# Statistical Inference:
# 600 Exercises, 100 Stories, and Solutions

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*Some technical stuff*

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The Kioskvelter Project

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1234-5678

*To my somebody*
– N.L.H.

*To my somebody*
– E.A.S.

# Preface

There are various books on various shelfs concerning the broader themes of *statistical inference*, both generally and with its many specialised themes. These include Bayesian analysis (parametric and nonparametric), testing theory, optimal estimation, confidence distributions, large-sample theory with the necessary probability on board, survival analysis and event history analysis, Markov chains and processes, optimal decision theory, time series, machine learning, processing of big data, image analysis, multivariate analysis, model selection and averaging, goodness of fit methods, stochastic processes, change-point identification, prediction, etc. This book is decidedly different, in several key respects.

### 600 Exercises

exercises, exercises

First of all, we change gears from the traditional narrative of 'theory development first, then a few exercises and illustrations', to 'crisp introduction text to a theme, then jump into exercises mode'. So when we come to Bayesian analysis in Ch. 7, for example, readers & doers are pushed rather quickly into working with and understanding from our generous lists of exercises. Thus theory is developed as we go, through problems and their solutions, from simpler statements and illustrations to the needed lemmas and theorems, as opposed to the more traditional 'theory first' approach. This is in line with what Aristotle wrote, in *Nicomachean Ethics*; 'for the things we have to learn before we can do them, we learn by doing them'; people learn to cook by cooking. Our approach is hence correlated with educational principles around Experimental Learning, and also the so-called Flipped Classroom.

The introductory Ch. 1 works through a generous list of parametric models, their construction and properties, from the more famous to various extensions and connections to yet other models. A healthy segment of our readers would know the basics of these models already, but may use the chapter for reference, as do we for the rest of the book. Ch. 2 then builds the apparatus for large-sample approximations, the uses of which truely abound in probability theory and both theoretical and applied statistics, with the Law of Large Numbers, the Central Limit Theorem with extensions, the delta method, moment-generating functions, etc. Chs. 3 and 4 then develop the fundamental concepts and methods for statistical estimation, confidence, and testing, with the required Neyman–Pearson theory and other excursions.

The important Ch. 5, on likelihoods and their ramifications, can of course be worked through and appreciated separately, but is also used in various guises in several later chapters, specifically for the Bayesian methods of Ch. 7, the survival and event history

Ch. 11, for model selection in Ch. 12, and for Markov chain models in Ch. 13. Ch. 6 goes into minimum divergence inference, generalising and in various situations robustifying likelihood inference. In Ch. 8 we build confidence, with the frequentist no-prior methodology for summarising inference in terms of distributions for focus parameters, with consequences also for meta-analyses, ways of combining information across diverse sources.

In various situations statisticians need for modelling and inference purposes to go beyond the finite-dimensional world of data points and vectors. Ch. 10 develops the appropriate tools and results for classes of stochastic processes, with Donsker theorems, Brownian motion and bridges, along with statistically relevant constructions of running t-tests, monitoring processes for models, log-likelihood maxima bridges, and more. Some of these are also used in the survival chapter Ch. 11.

Model selection themes have been part of statistical practice since the early models were invented, but have become more important and ubiqitous in the modern world of computing, access to an increasing volume of datasets, and available software to run through thousands of model candidates. Ch. 12 develops these themes, with emphasis on the established acronyms AIC, BIC, FIC (the Akaike, the Bayes, the Focused Information Criterion). Markov chains and processes are the themes of Ch. 13, where also the important Markov Chain Monte Carlo methods are tended to; the MCMC algorithms play an important role both in finite-dimensional and infinite-dimensional Bayesian work, as we see in Chs. 7 and 14, respectively. Our final Ch. 15 takes us through a sample of modern statistical and machine learning, with the ridge and the lasso for high-dimensional data, along with conformal prediction.

Notably, each chapter is rounded off with a 'Notes and Pointers' section, commenting on how key ideas have evolved historically, and with connections to other chapters and yet further themes.

### 100 Statistical Stories

Secondly, we, with the active participation of you, the readers & doers, illustrate and showcase concepts, modelling, theory development via a generous list of Statistical Stories. These range from the perhaps simple illustrations and applications to the rather more challenging terrain, where theory needs further development or new twists, in order to solve real-life real-data problems. Our stories are sorted into the broader categories of (i) Demography, Epidemiology, Medicine; (ii) Biology, Climate, Ecology; (iii) Art, History, Literature, Music; (iv) Economics, Political Science, Sociology; (v) Sport; (vi) Simulated stories; and a final bag of (vii) Miscellaneous stories. Significantly, also the stories are told in the spirit of exercises, from start observations concerning the context of and the problems at hand, to developing the right modelling and then the tools required for solving these. Students and readers are then meant to get involved in the problem solving process from the start, as opposed to listening to a polished final lecture by someone with struggle and efforts might finally having solved the problem. Ploughing through the pieces of these stories will also provide good training in the subtle tasks of statistical modelling, checking, and selection.

stories, stories

It is to be noted that most stories engage tools from not merely one chapter at a time, so to speak, but often via several themes from several chapters. So as opposed to illustrating a certain Bayesian method, for example, there might be alternative and equally relevant approaches, depending on the context, drawn from other chapters in our

book. Also, for some stories, theory needs further development as we go.

Importantly, we also provide *solutions* to all 600 Exerciss and 100 Statistical Stories. There is not space in the physical book to include these, however, so they are placed online in the GitHub site pointed to below. That site also has *all datasets* used through the book, with `R` and `Python` code when required.

Our methodological horizon is wide, so also in that sense our book is on the amibitious side of the spectrum. We have chapters on a fair range of methodological themes, from basic models to estimation and testing, on large-sample theory with central limit theorems and so on, empirical processes, parametric and nonparamtric Bayes, confidence distributions, survival analysis, Markov chains and processes, and modern methods for statistical learning. Though we do manage to develop several of these themes rather broadly, to the state of strong applicability also for challenging uses, the book is not intended to be a full-semester full-course book for each of its subthemes. Several types of courses can however easily be built from the book, as we offer advice on below.

### Appendixes A, B, C, D, E

At the end of the book are four appendixes with exercises that takes the reader through some of the required mathematical and probability background. Our list of exercises are selective, and we have strived to only include material that is strictly necesseary for solving and understanding the exercises in other chapters, and that hopefully also sheds some light on the short&crisp exercises as well as on the statistical stories.

Appendix A gives a concise and selective introduction to metric spaces. and includes results about such spaces that are used several places in the book, but in particular in Ch. 10. Statistics builds on probability, and probability is a branch of measures theory. Appendix B gives an introduction to measures theory. This includes exercises on $\sigma$-algebras, Caratheodory's extension theorem, Dynkin's lemma, the Lebesgue integral, the convergence theorems, the Radon–Nikodym theorem, and so on. Needless to say, the theory developed in this appendix underlies all of the statistical theory in the book, even though this theory is not actively present in each exercise. The measure theory appendix provides the background for Appendix C on probability theory. This appendix gives a measure theoretic introduction to probability theory, and develops such familiar concepts as the expectation, independence, and conditional expectation, to mention some, in a rigorous fashion. Appendix D contains exercises on martingale theory in discrete and continuous time. Some martingale theory is present in Ch. 10, and is used very actively in Ch. 11 on survival analysis. Martingale theory is a subject that deserves book length treatises (of which there are many, see Notes & Pointers), in our martingale appendix, however, we have attempted to keep the theory a the bare minimum required to work with survival analysis from the counting process point of view, without having to take too many mathematical results for granted.

Finally, and crucially, Appendix E goes through the datasets used for our Statistical Stories, with references and brief backgrounds and descriptions, along with pointers to where in our book the datasets are used.

### This book, and its accompanying GitHub site

As made clear above the book is broad-horizoned, and will have different strata of readers. The required start level for this book to be readable and useful is meant to be at end-

of-bachelor and above, as in Master's and PhD levels (and above!). Prerequisites, for the core of readers, should be basic mathematics, including calculus, linear algebra with matrices and so on; familiarity and experience with programming in or via suitable statistical or mathematical software systems, such as `R` and `Python`, and specifically with the ability to code things on one's own, outside the terrain of the application of already fine-tuned algorithms of packages; and having operating knowledge of classic statistical models and their parameters. Our start chapter 1 goes through such models, though, along with combinations, variations, extensions, so it is perfectly possible to start almost from scratch, with a certain uphill parameter. Otherwise the already flagged intention and spirit is 'come along', learning away as one proceeds, chapter by chapter, story by stories.

Our intended strata of readers are roughly as follows. (xx more polish here. xx) (i) The Linear Readers, who will benefit from having the stamina to work through chapter by chapter (ideally also exercise for exercise), and appropriate subsets of our stories. These readers will be at a Master's or PhD level. We even hope and trust that experienced scholars above these levels will find things to like, to be surprised and inspired by; some of the material, also in the collection of Stories, is advanced. (ii) The Chapter Samplers, who need to freshen up or learn more about certain themes, like confidence distributions or survival analysis, without necessarily having the time to or the interest for all the other chapters. Such readers will also be guided to and benefit from relevant subsets of our 100 Statistical Stories. (iii) The Statistical Stories Readers, for those who already know the basics on statistical models, parameter estimation and testing, some Bayes, etc. (iv) Our book is also for the specialists inside certain themes, who wish to learn even more and to see how these subfields are applied in new fashions, possibly in combination with methods and themes from other areas.

the Hjort-Stoltenberg book website

The GitHub site `Hjort-Stoltenberg-Book` (xx to be fixed), already pointed t above, is seen as a vital companion to the physical book. It has all datasets used; code in `R` and `Python`, i.e. scripts used to generate our figures, tables, and analyses; and, importantly, solutions to all our 600 Exercises and all our 100 Statistical Stories. The site will be actively maintained, and will with time also include follow-up exercises, snippets of course notes, etc.

### Road maps to courses to give based on our book

as many as ten courses can be taught from our book

Several types of courses can be taught from this book. Depending on the course parameters of ambition, time alotted, and background level, lecturers could supplement material with course notes and pointers to specialised literature. This is a list of such, reflecting both the book's general ambitions and our own teaching experiences at the Master and PhD level for statistics and data science at the Department of Mathematics, University of Oslo. Below we provide the skeletons of relevant Road Maps, for the different courses. For each of these, a reasonable going-through of the long list of statistical models of Ch. 1 would be required and useful, without necessarily working through all of them, so below we omit further mention of that chapter. The other chapters point back to that initial cavalcade of models when needed.

(xx give a version with modern statistical learning, 1, 5, 14, 15.) (xx could give our matrix of stories and methods, if nice. xx)

(xx Below there will be better and more precise Road Maps, where we point to the relevant subsection of chapters, course by course. xx)

A A basic course in probability models and statistical modelling: more of Ch. 1 than for the other courses, and most of 3. A suitable selection of these Stories (xx more, edit xx): #1, #2, #16, #22, #25, #43, #63, #70, #73, #75, #76, #77, #79, #83, #84, #85, #89. The lecturer and the students could then for these stories focus on the modelling and interpretation aspects, and perhaps leave the inference aspects for other rainy days.

B Hard-core statistical inference, with parametric models, etc.: half of Ch. 2, then most of 3, 4, half of 5, 7, 9. A suitable selection of these Stories: #1, #2, #5, #16, #20, #24, #25, #26, #31, #38, #40, #43, #46, #50; #73, #77, #78, #82, #88, #100.

C Modern Bayesian statistics, the essence of Ch. 3, 5, then the full 7, 14; this could also be coupled with addition emphasis on estimation and decision theory, risk functions, with Ch. 9. A suitable selection of these Stories: #1, #4, #17, #18, #21, #31, #54, #57, #62, #74, #89, #103.

D Confidence distributions, perhaps alongside classical Bayes: the essence of Ch.s. 3, 5, then the full 7, 8, 14. A suitable selection of these Stories: #1, #3, #4, #5, #7, #11, #12, #28, #30, #44, #45, #61, #62, #77, #81, #89, #101.

E Large-sample theory, the careful probablity theory leading to CLT and more, ideally coupled with parts of empirical processes, convergence, approximations, applications in statistics: most of Chs. 2, 5, 6, 10, perhaps even with parts of 11, 13 . A suitable selection of these Stories (xx more, edit xx): #14, #20, #22, #25, #32, #37, #43, #45, #49, #62, #77, #82, #83, #85, #92, #96, #98.

F Survival and event history analysis: the essence of Chs. 2, 3, 4, 5, then the full 11. A suitable selection of these Stories: #4, #8, #9, #14, #18, #19, #37, #47.

G Model selection and model averaging: the essence of Chs. 2, 3, 4, 5, then the full 12. A suitable selection of these Stories: #7, #15, #16, #20, #22, #24, #25, #26, #35, #36, #37, #44, #49, #50, #53, #59, #63, #73, #85, #95.

H Markov chains and processes, with applications: half of Ch. 5, then Chs. 10, 13. A suitable selection of these Stories: #3, #13, #14, #23, #45, #63, #84.

I Finally we point to the possibility of building a different type of course, on statistical modelling and applications, with emphasis on the stories, and less concern for all the technical details. This is also a way to learn about and appreciate modern statistics, concepts, modelling, computation, scientific context. The selection of stories, in volume and in detail, would of course depend on the time given and the interests of the lecturer and the audience.

(xx then exhortive words to this effect: Go forth! After having worked your way through significant chunks of this book, including both chapters and stories, you'll be fit to do wonders and be sought after on the job market. xx)

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Nils Lid Hjort and Emil Aas Stoltenberg
Blindern, some day in 2025

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# Part I

# 600 Exercises

# I.1

# Statistical models

In this chapter we study families of distributions and densities that we are to meet time and again in this book. A partial list includes the uniform, normal and multinormal, chi-squared, the t and the F, Gamma, exponential, Weibull, Beta, Dirichlet, Poisson, compound Poisson, binomial, multinomial, geometric, Pareto, Gumbel, logistic. These families have parameters, with values to be set for certain studies or illustrations, or for purposes of confidence setting and tests; more generally these parameters are estimated from data, as we return to in several later chapters. We also learn fruitful ways of extending and mixing given families of distributions; in such fashions the classical models can be building blocks for forming new ones. Mathematical techniques for deriving crucial properties include those of moment-generating functions, convolutions, and double expectation.

*Key words:* distributions, models, moments, moment-generating functions, parameters, quantiles, sums, transformations

The aim of this chapter is to go through a generous list of parametric statistical models, from the well-known distributions connected with the normal model, to the Beta and the Gamma, to the binomial, Poisson, and negative binomial for discrete data, etc., along with deriving their basic properties. These models turn up repeatedly in later chapters and in our Statistical Stories, with variations, as direct models for data, or as building blocks for more complicated constructions. The normal and multinormal distributions play important roles, also because these become fruitful simple-to-use approximations to sometimes much more complicated exact distributions.

These models, for probability theory and statistics, rely on deeper mathematical constructions and considerations, with random variables being measurable functions on probability spaces, measure and integration theory, etc. For this book it has been practical to organise that body of mathematical theory in Appendix B. For the present chapter on models we take certain notions and basic definitions for granted, with background and more detail in that appendix. Thus we deal here with classes of distributions, parameters, probability densities, cumulative distribution functions, conditional and marginal distributions, means and variances, quantiles, correlations, and so on. In particular, any nonnegative function $f(y)$ integrating to 1 over some interval is a probability density over that interval; if $Y$ is drawn from this $f$ it has a cumulative distribution function (c.d.f.) $F(y) = \Pr(Y \le y)$, which in the continuous case is the same as $\int_{-\infty}^{y} f(y')\,\mathrm{d}y'$; the mean of a random variable $Y$ drawn from this distribution is $\mathrm{E}\,Y = \int y f(y)\,\mathrm{d}y$; its median is $F^{-1}(\frac{1}{2})$; its variance $\mathrm{Var}\,Y = \mathrm{E}\,(Y - \mathrm{E}\,Y)^2$; the covariance between two random

variables $X$ and $Y$ with means $a$ and $b$ is $\operatorname{cov}(X,Y) = \mathrm{E}\,(X-a)(Y-b) = \mathrm{E}\,XY - ab$; etc. We also deal with sums of random variables, drawn from the same or different distributions. In one of its classical forms, the density $h(z)$ of $Z = X + Y$, where $X$ and $Y$ are independent with densities $f(x)$ and $g(y)$, is the convolution formula

$$h(z) = \int f(z-y)g(y)\,\mathrm{d}y = \int f(x)g(z-x)\,\mathrm{d}x, \tag{1.1}$$

see Ex. C.7 for more. There are also occasions in this chapter where the rules for double expectation and variance, i.e.

$$\mathrm{E}\,X = \mathrm{E}\,\mathrm{E}\,(X \mid Y) \quad \text{and} \quad \operatorname{Var} X = \mathrm{E}\operatorname{Var}(X \mid Y) + \operatorname{Var}\mathrm{E}\,(X \mid Y), \tag{1.2}$$

see Ex. C.10, come in handy.

In addition to defining and presenting a list of useful models, and diving into their properties and inter-connections, we develop certain tools, useful also in later chapters. These include transformations (see Ex. 1.12), moment-generating functions (see Ex. 1.32, with more in Ex. C.21), characteristic functions (see Ex. 1.36), conditional distributions, mixtures, and simulation. (xx make sure we have a little bit on simulation. xx) Also included is material on the general exponential family class, which has several of the classic models as special cases (see Ex. 1.56, with follow-up material in Ch. 4).

Importantly, several of the central models worked with in this introduction chapter find uses inside wider contexts, e.g. for regression situations, as we shall return to in later chapters. To indicate that direction of model building, below we learn about a random variable $Y$ having a gamma distribution with parameters $(a,b)$, which we write as $Y \sim \mathrm{Gam}(a,b)$; see Ex. 1.9. Now suppose there is a dataset consisting of measurements $(x_{i,1}, x_{i,2}, y_i)$ for individuals $i = 1,\ldots,n$, where the main outcome $Y_i$ is influenced by the covariates $x_{i,1}, x_{i,2}$. Then a gamma regression model could take the form $Y_i \mid (x_{i,1}, x_{i,2}) \sim \mathrm{Gam}(a_i, b)$, with $a_i = \exp(\beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2})$. The traditional multiple linear gression model is of a similar type, with $Y_i$ given the covariates having normal distributions, with mean function linear in the covariates. Tools developed in later chapters may then be applied to estimate parameters, with confidence intervals, testing, comparisons, prediction, etc.

(xx just a few pointers to later chapters. CLT. normal approximations. estimation, testing. calibrate with what's in the abstract. we may point to more complex models, making clear that these classic families of distributions are often used as stepping stones. could point to Markov chains etc., but not really touching these in this chapter. also: take care with mentions of limit distributions and CLT, which we may choose to touch here and there, but details come in Ch. 2. xx)

(xx as of 30-June-2025, we have a little fortellerproblem: we do mgf and characteristic functions in Appendix A, good, but then need just a bit of basic models there, complete with $M(t)$ formulae for the normal, the binomial, just a few more. but those models are more formally introduced here in Ch1. so how to deal with this. xx)

(xx one para to point to Main Stuff, also signalled with the fermat. xx)

### Normal, bi- and multinomial, exponential, gamma, mixing

**Ex. 1.1** *The normal distribution.* The perhaps most famous and broadly useful distribution in probability theory and statistics is the normal distribution, also called the Gaußian distribution. It is also a building block for various inferred and related models

and distributions, as we learn later in the chapter. In its standard form, before we add on two more parameters, the normal density is

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp(-\tfrac{1}{2}x^2) \quad \text{on the real line.}$$

We call this the standard normal distribution, and write $X \sim \mathrm{N}(0,1)$ to indicate this. It is standard in statistics and probability theory to use $\phi(x)$ for its density and $\Phi(x)$ for its cumulative distribution function (c.d.f.).

**(a)** There are myriad ways of demonstrating that $1/(2\pi)^{1/2}$ is the correct constant here, i.e. that $I = \int \exp(-\frac{1}{2}x^2)\,\mathrm{d}x = (2\pi)^{1/2}$. You are allowed to take this for granted, but attempt to show it via expressing $I^2$ as a double integral, featuring $\exp\{-\frac{1}{2}(x^2+y^2)\}$, and then substituting $x = r\cos\theta$ and $y = r\sin\theta$, followed by the use of double integration tools from calculus.

**(b)** Show that for $X$ a standard normal, its mean is zero and its variance is one.

**(c)** With $X$ a standard normal, consider $Y = \xi + \sigma X$, with $\xi$ any number and $\sigma$ positive. Show that its mean and standard deviation are $\mu$ and $\sigma$, and that its density can be written

Gauß

$$f(y) = \phi\Big(\frac{y-\xi}{\sigma}\Big)\frac{1}{\sigma} = \frac{1}{(2\pi)^{1/2}\sigma} \exp\Big\{-\tfrac{1}{2}\Big(\frac{y-\xi}{\sigma}\Big)^2\Big\}.$$

We write $Y \sim \mathrm{N}(\xi, \sigma^2)$ to indicate this distribution. Show that $\Pr(\xi - 1.96\,\sigma \le Y \le \xi + 1.96\,\sigma) = 0.95$. Find the $c$ such that $\Pr(|Y-\xi| \le c\sigma) = 0.50$.

**(d)** With $X$ a standard normal, consider $Z = X^2$. Find its c.d.f., and show that its density becomes $g(z) = (2\pi)^{-1/2}\exp(-\frac{1}{2}z)/\sqrt{z}$. We learn about the chi-squared distribution in Ex. 1.48; this $X^2$ has such a chi-squared distribution, with degrees of freedom equal to 1, which we write as $X^2 \sim \chi^2_1$.

**(e)** Consider $X_1, X_2, X_3$ being independent and standard normal. Work out the means and variances of $X_1^2$, $X_1^2 + X_2^2$, $X_1^2 + X_2^2 + X_3^2$. Simulate say $10^4$ realisations of these distributions, check their histograms, and describe their different behaviour close to zero.

**(f)** Consider the enigmatic density $f(x) = e^{-\pi x^2}$ on the real line, featuring and combining the eternal mathematical constants $e$ and $\pi$, integrating to 1. What is its standard deviation, and what is the probability that an $X$ with this distribution is inside $[-1,1]$?

**(g)** A statistician is helping clients to predict the future outcome $Y$ of some variable. The predictions $\widehat{Y}$ are such that $\widehat{Y} - Y$ is approximately a zero-mean normal with standard deviation 5.0. What is the probability that the prediction is inside $(-1,1)$ of the future outcome?

**(h)** (xx some pointers, placed here or elsewhere. point to mgf already, for the linear combination property. then a simple question to illustrate this. xx)

**Ex. 1.2** *Normal sums.* Sums of independent normals have themselves normal distributions. This is clearly easiest to demonstrate via moment-generating functions (m.g.f.s), see Ex. 1.33 below, but it is worth doing this via convolution formulae too.

**(a)** Let $X$ and $Y$ be independent standard normals. Show that $X+Y \sim \mathrm{N}(0,2)$, via the convolution formula (1.1). With a bit more algebraic work, show that if $X_1 \sim \mathrm{N}(\xi_1, \sigma_1^2)$ and $X_2 \sim \mathrm{N}(\xi_2, \sigma_2^2)$ are independent, then $X_1 + X_2 \sim \mathrm{N}(\xi_1+\xi_2, \sigma_1^2+\sigma_2^2)$.

**(b)** Generalise this: show that if $X_i \sim \mathrm{N}(\xi_i, \sigma_i^2)$ for $i = 1, \ldots, m$, and these are independent, then $Z = \sum_{i=1}^m a_i X_i$ is also normal, with mean $\sum_{i=1}^m a_i\xi_i$ and variance $\sum_{i=1}^m a_i^2\sigma_i^2$.

**(c)** Sometimes 2 + 2 might be 5, for extremely high values of 2. If independent $X$ and $Y$ are 2 and 2, but observed with with some Gaussian noise on top, of standard deviation level $\sigma$, find a formula $p(\sigma)$ for the probability that $X+Y$ is outside $[3.5, 4.5]$, i.e. 3 or 5 or even farther away from 4 when rounded off to the nearest integer. Plot this function.

**Ex. 1.3** *Binomial distribution.* One of the more old, classic, and deservedly famous distributions in probability and statistics is the binomial. If there is a fixed probability $p = P(A)$ of a certain event $A$ taking place, in a certain type of experiment, then the number $Y$ of times $A$ is seen, in $n$ independent experiments, is the binomial, which we write as $Y \sim \text{binom}(n, p)$.

the binomial distribution

**(a)** Show that

$$\Pr(Y = y) = \binom{n}{y} p^y (1-p)^{n-y} \quad \text{for } y = 0, 1, \ldots, n.$$

This involves the essential combinatorial fact that the number of ways one may place precisely $y$ 1s in a total of $n$ possible position is $\binom{n}{y} = n!/\{y!\,(n-y)!\}$. Explain that $Y$ can be expressed as $X_1 + \cdots + X_n$, where $X_i$ is a simple 0-1 variable, with $\Pr(X_i = 1) = p$, and where these are independent. Such $X_i$ are called *Bernoulli variables*. Use this to prove the classic formulae $np$ and $np(1-p)$ for mean and variance. Also, deduce the $\Pr(Y = y)$ formula from the $Y = \sum_{i=1}^n X_i$ description.

Bernoulli variables

**(b)** If the first question to ask concerning a distribution is about its centre (its mean, or perhaps its median), and the second is about its spread (its standard deviation, or perhaps a different measure, like its interquartile range), then the third question would be about its skewness, the degree of asymmetry. The classical skewness definition of a distribution, or equivalently of a random variable $Y$ having that distribution, is $\text{skew} = \mathrm{E}\, W^3$, where $W = (Y - \mathrm{E}\, Y)/(\mathrm{Var}\, Y)^{1/2}$ is the normalised version of $Y$, i.e. linearly transformed to have mean zero and standard deviation one. Show for the binomial $(n, p)$ case that its skewness is

the skewness

$$\text{skew} = \mathrm{E}\Big[\frac{Y - np}{\{np(1-p)\}^{1/2}}\Big]^3 = \frac{1-2p}{\{np(1-p)\}^{1/2}},$$

which for fixed $p$ goes to zero with rate $1/\sqrt{n}$ (i.e. $\sqrt{n}\,\text{skew}$ tends to a positive constant). Briefly discuss what this entails regarding the degree of asymmetry for the binomial distribution.

**(c)** After the skewness comes the so-called kurtosis, defined as $\text{kurt} = \mathrm{E}\, W^4 - 3$, with $W$ as in the previous point. The minus 3 is there in order for the kurtosis to be zero for the normal distribution; show that this is the case. Then show that

the kurtosis

$$\text{kurt} = (1/n)[1/\{p(1-p)\} - 6],$$

for the binomial, and comment.

**Ex. 1.4** *Trinomial probabilities.* (xx emil looks it over and checks if this is suitable here in Ch1, perhaps before Ex. 1.5. if not in App A. xx) Consider the so-called trinomial distribution for a random pair $(X, Y)$, with probability mass function

$$f(x,y) = \frac{n!}{x!\,y!\,(n-x-y)!} p^x q^y (1-p-q)^{n-x-y} \quad \text{for } x \geq 0,\ y \geq 0,\ x+y \leq n.$$

Here $n$ is the total count number, $p, q$ the probabilities of events of type One and Two in repeated experiments, with $p + q < 1$. With $Z = n - X - Y$ representing the number of

events of type Three (not One, not Two), this is a model for the numbes of events One, Two, Three in $n$ independent experiments; hence the trinomial name. See also Ex. 1.5.

**(a)** Verify that what is here called the probability mass function is the same as the density of the distribution with respect to counting measure on the set of $(x, y)$ with $x \geq 0$, $y \geq 0$, $x + y \leq n$ – or, for that matter, with respect to counting measure on the set of all pairs $(x, y)$ with $x \geq 0$, $y \geq 0$.

**(b)** Show by summing over the $y$ that the distribution of $X$ becomes a binom$(n, p)$ from Ex. 1.3.

**(c)** Show that $Y \,|\, (X = x) \sim \text{binom}(n - x, q/(1-p))$. Give a formula for $\mathrm{E}\,(Y \,|\, X = x)$, and deduce the formula for $\mathrm{E}\,X$ from this. Find also the covariance between $X$ and $Y$, using this scheme of conditioning with respect to $X = x$ first. Deduce that the correlation between them is $-\{p/(1-p)\}^{1/2}\{q/(1-q)\}^{1/2}$.

**(d)** Find a formula for $\Pr(X \leq x_0, Y \leq y_0)$, expressed as a sum over $x \in \{0, 1 \ldots, x_0\}$ (as opposed to a double sum over lots of $(x, y)$ pairs). For a setup with $n = 50$, $(p, q) = (0.22, 0.33)$, compute the probability $\Pr(X \leq 15, Y \leq 20)$.

**Ex. 1.5** *The multinomial model.* The binomial model, with basic properties treated in Ex. 1.3, is about sorting and counting events in two categories; if $Y \sim \text{binom}(n, p)$, then also $n - Y \sim \text{binom}(n, 1-p)$. The *multinomial model* is the natural extension of also Ex. 1.4 to more than two and three categories. Suppose there are $n$ independent experiments, where each time one (and only one) of the events $A_1, \ldots, A_k$ takes place, with the same probabilities $p_1, \ldots, p_k$ for each experiment. Let then $Y = (Y_1, \ldots, Y_k)$, with $Y_j$ counting the number of times $A_j$ occurred, for $j = 1, \ldots, k$. Of course $Y_1 + \cdots + Y_k = n$, and $p_1 + \cdots + p_k = 1$, so there are $k - 1$ free parameters in the model.

**(a)** Show that $Y_j \sim \text{binom}(n, p_j)$, and deduce that we already know $\mathrm{E}\,Y_j = np_j$ and $\mathrm{Var}\,Y_j = np_j(1-p_j)$, even before we start working on the joint distribution of $(Y_1, \ldots, Y_k)$.

the multinomial model

**(b)** Show that the joint probability distribution becomes

$$f(y_1, \ldots, y_k) = \Pr(Y_1 = y_1, \ldots, Y_k = y_k) = \frac{n!}{y_1! \cdots y_k!} p_1^{y_1} \cdots p_k^{y_k}$$

for nonnegative $(y_1, \ldots, y_k)$ with sum $n$. The first factor $n!/(y_1! \cdots y_k!)$ is a combinatorial one, the number of different ways one may place '1' in $y_1$ positions, '2' in $y_2$ positions, etc., up to '$k$' in $y_k$ positions. Note that this generalises the classic $n!/(y_1!\, y_2!) = \binom{n}{y_1}$ for the binomial case, the number of ways one may place '1' in $y_1$ ways (and hence '2' in $n - y_1$ ways) in a list $1, \ldots, n$.

**(c)** Show that each pair has a trinomial distribution, e.g.

$$\Pr(Y_1 = y_1, Y_2 = y_2) = \frac{n!}{y_1!\, y_2!\, (n - y_1 - y_2)!} p_1^{y_1} p_2^{y_2} (1 - p_1 - p_2)^{n - y_1 - y_2}$$

for $y_1 \geq 0, y_2 \geq 0, y_1 + y_2 \leq n$. Note that formulae from Ex. 1.4 therefore apply to pairs $(Y_i, Y_j)$ here. For $i \neq j$, show that $\mathrm{cov}(Y_i, Y_j) = -np_ip_j$, and find the correlation between $Y_i$ and $Y_j$.

**(d)** Among the most used acronyms of statistical parlance is *i.i.d.*, for idependent and identically distributed. Explain in the present setup that $Y = Z_1 + \cdots + Z_n$, where $Z_1, \ldots, Z_n$ are i.i.d., with $Z_i$ taking values $(1, 0, \ldots, 0), \ldots, (0, \ldots, 0, 1)$ with probabilities $p_1, \ldots, p_k$. Derive again the formulae for means, variances, covariances, starting with this representation.

i.i.d.

**Ex. 1.6** *Histograms.* Suppose data $Y_1, \ldots, Y_n$ are i.i.d. from some continuous density $f$ over some interval $[a, b]$. Create disjoint cells $C_1, \ldots, C_k$, with $C_j = (a_{j-1}, a_j]$, for $a = a_0 < \cdots < a_k = b$. Let $N_j$ count the number of data points in cell $j$. The *histogram*, associated with the chosen cells, is then

histogram

$$\widehat{f}(x) = \widehat{p}_j/h_j \quad \text{for } x \in C_j, \text{ with lengths } h_j = a_j - a_{j-1},$$

where $\widehat{p}_j = N_j/n$ estimates $p_j = \Pr(Y_i \in C_j)$. In most automatic histogram algorithms, the width is taken constant across cells. The notation $\widehat{f}(x)$ indicates that beyond being an effective way of showing the essential spread and shape of the data, it is *an estimate* of the underlying $f(x)$.

**(a)** Carry out some simulations, samling $n$ datapoints from the standard normal, creating histograms with $k$ cells. For small and big $n$, play with $k$ small, big, and about right. In `R`, you may use `hist(y,breaks=20,prob=T)`, etc.

**(b)** Show that $(N_1, \ldots, N_k)$ is multinomial. Taking a constant cell width $h$, for simplicity, explain that we may write $p_j = \int_{C_j} f \, \mathrm{d}y = f(\zeta_j)h$, for a suitable $\zeta_j$ inside cell $C_j$. From this derive that for $x \in C_j$, we have $\mathrm{E}\,\widehat{f}(x) = f(\zeta_j)$ and $\mathrm{Var}\,\widehat{f}(x) = f(\zeta_j)\{1 - f(\zeta_j)h\}/(nh)$. Argue that for the histogram to achieve $\mathrm{E}\,\widehat{f}(x) \to f(x)$ and $\mathrm{Var}\,\widehat{f}(x) \to 0$, as $n$ increases, we need $h \to 0$ and $nh \to \infty$.

**Ex. 1.7** *Hazard rates and survival functions.* Here and below we shall partly follow the implied tradition of using say $T$ and $f(t)$ and $F(t)$, for random variables with their densities and c.d.f.s, rather than say $Y$ and $f(y)$ and $F(y)$, when these relate to *time*. – Consider a random variable $T$ on the halfline $[0, \infty)$, with density $f$ and c.d.f. $F$. Classes of such distributions are sometimes most conveniently or fruitfully defined and discussed in terms of their hazard or cumulative hazard functions, as opposed to their densities and c.d.f.s, as we outline here; see also Ch. 11.

**(a)** For any nonnegative random variable $T$, the survival curve, or function, is $S(t) = \Pr(T \geq t)$. For $T$ continuous, $S(t) = 1 - F(t)$, but explain that in the general case, we need $S(t) = 1 - F(t-)$, with $F(t-)$ the left hand limit of $F$ at position $t$. Show that integrated survival is equal to mean survival:

$$\mathrm{E}\,T = \int_0^\infty S(t) \, \mathrm{d}t = \int_0^\infty t \, \mathrm{d}F(t). \tag{1.3}$$

(You may write the latter as $\int_0^\infty \int_0^\infty I(s < t) \, \mathrm{d}s \, \mathrm{d}t$ and then use the Fubini theorem.) Show from this that $\mathrm{E}\,T^p = \int_0^\infty \{1 - F(t^{1/p})\} \, \mathrm{d}t$, for any positive $p$.

**(b)** Show that

hazard rate function

$$\Pr(T \in [t, t + \varepsilon] \,|\, T \geq t) = h(t)\varepsilon + O(\varepsilon^2) \quad \text{for small } \varepsilon, \tag{1.4}$$

in terms of the so-called *hazard rate* function $h(t) = f(t)/\{1 - F(t)\}$. With $T$ interpreted as the time to a certain event, the function $h(t)$ describes the chance of this event taking place in the next instance, among those having survived up to $t$.

**(c)** So we may deduce the hazard rate from the density. Starting instead with $h(t)$, define first *the cumulative hazard* $H(t) = \int_0^t h(s) \, \mathrm{d}s$, and show that $F(t) = 1 - \exp\{-H(t)\}$. The function $S(t) = \Pr(T \geq t) = \exp\{-H(t)\}$ is important in its own right, and is called *the survival function*.

**(d)** Suppose an individual has survived up to time $t_0$. Show that

$$\Pr(T \geq t \,|\, T \geq t_0) = \frac{S(t)}{S(t_0)} = \exp[-\{H(t) - H(t_0)\}] \quad \text{for } t \geq t_0.$$

Show that the median lifetime, for such an invidual having lived up to $t_0$, is $t^* = H^{-1}(H(t_0) + \log 2)$.

**(e)** Suppose an event takes place at $T$ as soon as either one of two independent continuously distributed $T_1$ and $T_2$ lifetimes occur. With these having cumulative hazards $H_1$ and $H_2$, show that $\Pr(T \geq t) = \exp\{-H_1(t) - H_2(t)\}$. Explain that this entails additive hazards, with $h(t) = h_1(t) + h_2(t)$.

**Ex. 1.8** *The exponential distribution.* The *exponential distribution* is a simple but important one, in probability theory and statistics, which with positive parameter $\theta$ has the density $f(t, \theta) = \theta \exp(-\theta t)$ for $t > 0$. We write $T \sim \text{Expo}(\theta)$ to indicate this.

the exponential distribution

**(a)** Show that the c.d.f. becomes $F(t, \theta) = 1 - \exp(-\theta t)$, and find the median. Show also that we may write $T = T_0/\theta$, where $T_0$ has the unit exponential distribution with density $\exp(-t_0)$. Show that $T$ has mean and variance $1/\theta$ and $1/\theta^2$.

**(b)** Using Ex. 1.7, show that the hazard rate is constant, $h(t) = \theta$, and that the cumulative hazard rate is $H(t) = \theta t$. Show also that the exponential distribution is the only one where the hazard rate is constant.

**(c)** Show that the median survival time is $(\log 2)/\theta$. If an individual has survived up to time $t_0$, what is the median survival time?

**(d)** Assume certain light bulbs have a longevity distribution with the property that $\Pr(T \geq t_0 + t \,|\, T \geq t_0)$ does not depend on $t_0$. Argue that such light bulbs may be sold as if they were brand new, as long as they are still alive. Show that their distribution must be exponential.

the memoryless property

**(e)** Consider $T = \min(T_1, \ldots, T_k)$, with these being independent exponentials with parameters $\theta_1, \ldots, \theta_k$. Show that $T$ is also exponential, with hazard $\theta_1 + \cdots + \theta_k$.

**Ex. 1.9** *The Gamma distribution.* The gamma function is important in various branches in mathematics, probability theory, and statistics, and is defined as $\Gamma(a) = \int_0^\infty x^{a-1} \exp(-x)\,\mathrm{d}x$ for $a$ positive. We may hence define a family of probability densities via $g_0(t, a) = \Gamma(a)^{-1} t^{a-1} \exp(-t)$ for $t > 0$. This is called the *Gamma distribution* with shape parameter $a$.

the gamma function

**(a)** With $T_0$ having this density, and $b$ a positive scale parameter, show that $T = T_0/b$ has density

$$g(t, a, b) = \{b^a/\Gamma(a)\} t^{a-1} \exp(-bt) \quad \text{for } t > 0.$$

This is the two-parameter $\text{Gam}(a, b)$ distribution. Verify that $\Gamma(1) = 1$, that $\Gamma(a+1) = a\Gamma(a)$ for all $a > 0$, and that $\Gamma(m) = (m-1)!$ for $m = 1, 2, \ldots$.

the Gamma distribution

**(b)** When $T$ has the $\text{Gam}(a, b)$ distribution, show that the mean and variance are $a/b$ and $a/b^2$. Find also that $\mathrm{E}\, T^p = \{\Gamma(a+p)/\Gamma(a)\}/b^p$, valid for any $p$, as long as $p > -a$. Use this to show that the skewness and kurtosis become equal to skew $= 2/a^{1/2}$ and kurt $= 6/a$. Finally, regarding moments, find that the inverse gamma distributed variable $1/T$ has mean $b/(a-1)$ and finite variance $b^2/\{(a-1)^2(a-2)\}$, as long as $a > 2$.

**(c)** Verify that for $a = 1$ we have the exponential distribution, with density $b \exp(-bt)$ and cumulative $1 - \exp(-bt)$. Show that $a = 2$ gives density $b^2 t \exp(-bt)$ and cumulative

$1-\exp(-ty)(1+bt)$. More generally, show that the cumulative is

$$\int_0^t g(s,a,b)\,\mathrm{d}s = 1-\exp(-bt)\Big\{1+bt+\frac{(bt)^2}{2!}+\cdots+\frac{(bt)^{a-1}}{(a-1)!}\Big\}$$

for the case of $a$ being an integer.

**(d)** For $a$ an integer, give an explicit expression for the hazard function $h(t,a,b)$, as per (1.4), and show that it converges to $b$ as time increases. Show that this is the case also for any $a$, i.e. not only for integers; it increases from zero to $b$, if $a>1$, and decreases from infinity to $b$, if $a<1$.

**(e)** Let $T_1,T_2$ be independent and exponential with the same $\theta$. Show that $T_1+T_2\sim \mathrm{Gam}(2,\theta)$. With $T_1,\ldots,T_k$ seen as the independent waiting times between events, show that the time to event $k$ is a $\mathrm{Gam}(k,\theta)$.

**(f)** With $T_1\sim\mathrm{Gam}(a_1,b)$ and $T_2\sim\mathrm{Gam}(a_2,b)$ independent, show that $T_1+T_2\sim\mathrm{Gam}(a_1+a_2,b)$. Generalise. This may indeed be accomplished via the convolution formulae from Ex. C.7, but as for other instances it becomes easier to prove such statements via m.g.f.s; see Ex. 1.32–1.33.

**Ex. 1.10** *Mixing the exponential.* Sometimes waiting time type data do not follow an exact exponential distribution, but rather one characterised as a mixture of such; $T$ given $\theta$ has the $\mathrm{Expo}(\theta)$ distribution, but the values of $\theta$ vary from occasion to occasion, or from individual to individual.

**(a)** Suppose indeed that $T\,|\,\theta\sim\mathrm{Expo}(\theta)$ but that $\theta$ has some density $g(\theta)$. Show that the density of $T$ then becomes $f(t)=\int_0^\infty \theta\exp(-\theta t)g(\theta)\,\mathrm{d}\theta$.

**(b)** Suppose the distribution of $\theta$ is such that $1/\theta$ has mean value $1/\theta_0$ and a positive standard deviation $\tau$. Show, starting with $\mathrm{E}\,(T\,|\,\theta)=1/\theta$ and $\mathrm{Var}\,(T\,|\,\theta)=1/\theta^2$, that $\mathrm{E}\,T=1/\theta_0$ and $\mathrm{Var}\,T=1/\theta_0^2+2\tau^2$; see (1.2). The case of a very tight distribution for the $\theta$ corresponds to $\tau$ small, which again means the case of a constant rate $\theta_0$ for all.

**(c)** A convenient class of distributions for $\theta$ is the Gamma, with parameters $(a,b)$, from Ex. 1.9. Its mean and variance are $a/b$ and $a/b^2$; now find also the mean and variance of $1/\theta$. Show that the density of $T$ can be written

$$f(t,a,b)=\int_0^\infty \theta\exp(-\theta t)\frac{b^a}{\Gamma(a)}\theta^{a-1}\exp(-b\theta)\,\mathrm{d}\theta=\frac{ab^a}{(b+t)^{a+1}},$$

and also that its c.d.f. is

$$F(t,a,b)=1-\Big(\frac{b}{b+t}\Big)^a=1-\frac{1}{(1+t/b)^a}.$$

**(d)** (xx a bit more. the hazard rate function $h(t)=f(t)/\{1-F(t)\}$ is decreasing. expressions for quantiles, $t_0(p,a,b)=b\{1/(1-p)^{1/a}-1\}$, solution to $F(t)=p$, to be used for Story #56 for fitting the distribution to the 95 between-war-times. xx)

**(e)** Find an expression for the hazard rate function $h(t,a,b)=f(t,a,b)/\{1-F(t,a,b)\}$, and comment on its form, compared to the exponential case.

**(f)** Find the mean and variance of $T$, for the $g(t,a,b)$ distribution. This might be used to estimate $(a,b)$ from data. (xx could point to Story #56, perhaps with better calibration. xx)

**Ex. 1.11** *Gamma-mixing the gamma.* A given parametric distribution may sometimes be fruitfully extended by placing a separate distribution on one of its parameters. The following is an illustration.

**(a)** Consider a distribution which for given individuals is a gamma, but where the scale parameter varies between individuals. Specifically, suppose $Y \,|\, b \sim \text{Gam}(a_0, b)$ and that $b$ has a distribution with $\text{E}\, 1/b = 1/b_0$ and $\text{Var}\, 1/b = \tau^2$. Show that $Y$ has mean $a_0/b_0$ and variance $a_0/b_0^2 + (a_0 + a_0^2)\tau^2$.

**(b)** For the special case of $b \sim \text{Gam}(c, d)$, thus leading to a 3-parameter model, find the density $f(y, a, c, d)$ for $Y$. (xx work a bit with parametrisation here; big $(c, d)$ correspond to old gamma. the following to be cleaned and sent to solutions. xx)

$$\bar{f}(y) = \int_0^\infty \frac{b^a}{\Gamma(a)} y^{a-1} \exp(-by) \frac{d^c}{\Gamma(c)} b^{c-1} \exp(-db)\, \text{d}b = \frac{d^c}{\Gamma(c)} \frac{\Gamma(a+c)}{\Gamma(a)} \frac{y^{a-1}}{(d+y)^{a+c}}.$$

### Transformations, uniform, Pareto, Cauchy, Beta, Dirichlet

**Ex. 1.12** *Transformation from $X$ to $Y$.* We often encounter transformations, from one variable $X$ to another $Y$, also in the vector case. We need formulae for how the density $g(y)$ of the $Y$ can be found in terms of the density $f(x)$ for $X$.

**(a)** In the one-dimensional case, suppose $X = h(Y)$, equivalently $Y = h^{-1}(X)$, where $h$ is smooth and increasing. Show that $\Pr(Y \le y) = \Pr(X \le h(Y))$, with density formula

$$g(y) = f(h(y))h'(y).$$

Show also that if $x = h(y)$ is continuous and decreasing, the formula becomes $g(y) = f(h(y))|h'(y)|$. Write down density formulae for the variables $Y_1 = \exp(X)$, $Y_2 = 3.33 - 2.22\, X$, $Y_3 = \log X$ (assuming for that case that $X$ is positive).

**(b)** Show that if $X$ is normal, then a linearly transformed $Y = a + bX$ is also normal. Show that if $X \sim \text{Gam}(a, b)$, with density proportional to $x^{a-1} \exp(-bx)$, then $Y = bX \sim \text{Gam}(a, 1)$.

**(c)** Suppose then that $X = (X_1, \ldots, X_p)^{\text{t}}$ and $Y = (Y_1, \ldots, Y_p)^{\text{t}}$ are vectors, with transformations binding them together,

$$\begin{pmatrix} X_1 \\ \vdots \\ X_p \end{pmatrix} = \begin{pmatrix} h_1(Y_1, \ldots, Y_p) \\ \vdots \\ h_p(Y_1, \ldots, Y_p) \end{pmatrix}, \quad \begin{pmatrix} Y_1 \\ \vdots \\ Y_p \end{pmatrix} = \begin{pmatrix} h_1^{-1}(X_1, \ldots, X_p) \\ \vdots \\ h_p^{-1}(X_1, \ldots, X_p) \end{pmatrix}.$$

We write this as $X = h(Y)$ and $Y = h^{-1}(X)$, for short. It is assumed that these systems of equations have unique solutions, and that the transformations are smooth, with continuous partial derivatives. In particular, the so-called Jacobi matrix

$$J(y) = \frac{\partial h(y)}{\partial y} = \frac{\partial h(y_1, \ldots, y_p)}{\partial y_1 \cdots \partial y_p},$$

having $\partial h_i(y)/\partial y_j$ as its $(i, j)$ component, exists, and is continuous, with a non-zero determinant $\det(J(y))$. – Now, if $X$ has density $f(x)$, show that

$$\Pr(Y \in B) = \int_{h(B)} f(x)\, \text{d}x = \int_B f(h(y))|\det(J(y))|\, \text{d}y.$$

This shows that $Y$ has density $g(y) = f(h(y))|\det(J(y)))|$. This is essentially the multidimensional 'integration by substitution' formula of calculus.

**(d)** Let $X, Y$ be independent standard exponentials, and consider $(U, V) = (X+Y, X/Y)$. Show that these become independent, having densities $u\exp(-u)$ and $1/(v+1)^2$. More generally, suppose in this construction of $(U, V)$ that $X, Y$ are independent from the $\text{Gam}(a, 1)$. Show that $U \sim \text{Gam}(2a, 1)$, that $V$ has density $\{\Gamma(2a)/\Gamma(a)^2\}v^{a-1}/(v+1)^{2a}$, and that they are independent.

**Ex. 1.13** *Transformations to the uniform.* We say that a variable $U$ is uniform on the interval $[a, b]$ if its density is constant over that interval, i.e. $1/(b-a)$, and zero outside. In particular, we write $U \sim \text{unif}(0, 1)$ to indicate a variable with the uniform distribution on the unit interval.

**(a)** For such a $U \sim \text{unif}(0, 1)$, find its mean and variance. Find the probabilities that $U$ lands in $[0.03, 0.04]$, or in $[0.77, 0.78]$.

**(b)** Let $F$ be a continuous and increasing c.d.f. for a variable $X$. Show that $U = F(X)$ is uniform on the unit interval. Conversely, we may start with $U \sim \text{unif}(0, 1)$ and map to $X' = F^{-1}(U)$. Show that $X'$ has distribution $F$. This is hence a simulation recipe, generating realisations from a given and perhaps not standard distribution.

**(c)** Consider the c.d.f. $F(x) = (x/10)^{3.33}$ on $[0, 10]$. Simulate $10^4$ independent $X_i$ from this distribution, by transforming uniforms. Make a fine histogram, with the density $f(x)$ plotted alongside.

**Ex. 1.14** *Polar and spherical coordinates.* Points in the plane may be transformed from Cartesian to polar coordinates, and vice versa; similarly, but with more delicate details, points in the three-dimensional space can be represented in spherical coordinates.

**(a)** Suppose $X$ and $Y$ are independent and standard normal, and transform to polar coordinates, $X = R\cos A$ and $Y = R\sin A$. Find the density $g(r, a)$ for $(R, A)$, with $R$ positive and $A \in [0, 2\pi]$. Show in particular that length $R$ and angle $A$ become independent, with $A$ having a uniform distribution on $[0, 2\pi]$ (i.e. the flat density $1/(2\pi)$ over that interval). Find also the distribution of $Z = Y/X = \tan A$; see also Ex. 1.18.

**(b)** Explain that a somewhat idiosyncratic but nevertheless feasible way of simulating pairs of standard normals is to draw $R$ from the density $h_0(r) = r\exp(-\frac{1}{2}r^2)$ and $A$ from the uniform over $[0, 2\pi]$, followed by the polar transformation. Do this in your computer, generating say $10^4$ pairs $(X_i, Y_i)$. Then tweak your code, drawing $R$ from some density $h$ different from $h_0$. Show that this amounts to the joint density

$$f(x, y) = h((x^2+y^2)^{1/2})(x^2+y^2)^{-1/2}/(2\pi).$$

Take e.g. $h(r) = \frac{1}{2}\gamma r^{\gamma-1}\exp(-\frac{1}{2}r^\gamma)$, with $\gamma$ close to but not equal to 2. Generate and plot $10^4$ points $(x, y)$ via this recipe, perhaps via the inverse c.d.f. trick of Ex. 1.13; how can you spot that these data are *not* independent normal pairs?

**(c)** Then go from plane to space. Start with $(X, Y, Z)$ a triple of independent standard normals, and transform to $(R, A, B)$ via

$$X = R\sin A\cos B, \quad Y = R\sin A\sin B, \quad Z = R\cos A.$$

Set up the Jacobi matrix $\partial(x, y, z)/\partial(r, a, b)$ of partial derivatves, and show that its determinant becomes $r^2\sin a$. Use this to show that the joint density of $(R, A, B)$ can be written in product form $g_1(r)g_2(a)g_3(b)$, with $g_1$ the density proportional to $r^2\exp(-\frac{1}{2}r^2)$, $g_2(a) = \frac{1}{2}\sin(a)$ over $[0, \pi]$, and $g_3(b) = 1/(2\pi)$ over $[-\pi, \pi]$.

**(d)** Just to see operationally that transformations go two ways, simulate say $10^5$ standard normal triples $(X, Y, Z)$ via simulating triples $(R, A, B)$ from the densities found

away. Check that the resulting $(X, Y, Z)$ look adequate. As for the plane question, tweak your code to draw $R$ from a somewhat different distribution than proportional to $r^2 \exp(-\frac{1}{2}r^2)$, and explain that you have invented a generalisation of the triple independent normal.

**Ex. 1.15** *Ordering exponentials.* Let $Y_1, Y_2, Y_3$ be independent unit exponentials, with density $f(y)\exp(-y)$ for $y$ positive, and order them, to $Y_{(1)} < Y_{(2)} < Y_{(3)}$. Then define the so-called spacings between them, $Z_1 = Y_{(1)}$, $Z_2 = Y_{(2)} - Y_{(1)}$, $Z_3 = Y_{(3)} - Y_{(2)}$.

**(a)** Show first that the joint density of $(Y_{(1)}, Y_{(2)}, Y_{(3)})$ is $3!\, f(y_{(1)})f(y_{(2)})f(y_{(3)})$ on the set $y_{(1)} < y_{(2)} < y_{(3)}$. Find then the joint density for $(Z_1, Z_2, Z_3)$, and show that they are independent.

**(b)** Then generalise, considering i.i.d. unit exponentials $Y_1, \ldots, Y_n$, ordered into $Y_{(1)} < \cdots < Y_{(n)}$. Work with the scaled spacings $V_1 = nY_{(1)}$, $V_2 = (n-1)(Y_{(2)} - Y_{(1)})$, up to $V_{n-1} = 2(Y_{(n-1)} - Y_{(n-2)})$, $V_n = Y_{(n)} - Y_{(n-1)}$. Show that

$$Y_{(1)} = \frac{V_1}{n}, Y_{(2)} = \frac{V_1}{n} + \frac{V_2}{n-1}, \ldots, Y_{(n)} = \frac{V_1}{n} + \frac{V_2}{n-1} + \cdots + \frac{V_{n-1}}{2} + \frac{V_n}{1},$$

and then show that in fact $V_1, \ldots, V_n$ are i.i.d. unit exponentials.

**(c)** The Euler constant $\gamma_e = 0.5772...$ is defined as the limit of $1 + 1/2 + \cdots + 1/n - \log n$. Use the above to show that $Y_{(n)} = \max_{i \le n} Y_i$ has mean close to $\log n + \gamma_e$, and variance converging to $\pi^2/6$.

**Ex. 1.16** *The Pareto distribution.* Consider a variable $T$ defined on the range $T \ge t_0$, for some positive $t_0$, with c.d.f. $F(t) = 1 - (t_0/t)^\theta$ for $t \ge t_0$, for some positive parameter $\theta$. This is the Pareto distribution, and we may write $T \sim \text{pareto}(t_0, \theta)$ to indicate this.

the Pareto distribution

**(a)** Use the mean formula of (1.3) to show that the mean of a Pareto is $t_0\theta/(\theta - 1)$, for $\theta > 1$. Show furthermore that the variance is $t_0^2\theta/\{(\theta-1)^2(\theta-2)\}$, for $\theta > 2$, and that the median is $\text{med}(T) = t_0 2^{1/\theta}$.

**(b)** Show that the density is $f(t, \theta) = \theta t_0^\theta / t^{\theta+1}$ for $t \ge t_0$; you may use this to find the mean formula again. Furthermore, explain that the hazard rate is $h(t) = \theta/t$.

**(c)** Show that $\Pr(T \ge t \,|\, T \ge t_1) = (t_1/t)^\theta$, for $t \ge t_1$. So $T \,|\, (T \ge t_1)$ is Pareto $(t_1, \theta)$. Deduce from this that

$$\text{med}(T - t \,|\, T \ge t) = ct = (2^{1/\theta} - 1)t \quad \text{for all } t.$$

the Lindy Effect

This is linked to the so-called Lindy Effect: the longer the life is observed to be (of a company, an idea, a party), the longer is the remaining lifetime expected to be (note that $c > 1$). Show in fact that the Pareto distribution is the only one with the property that $\text{med}(T - t \,|\, T \ge t) = ct$ for all $t$.

**(d)** Show that $Y = \log(T/t_0)$ is exponential with parameter $\theta$. So a representation of the Pareto is $T = t_0 \exp(Y)$, with $Y \sim \text{Expo}(\theta)$. You may use this to find the mean formula once more.

**Ex. 1.17** *Maxima of i.i.d. samples.* To illustrate one of the many ways in which suitable start models may be generalised and extended, consider $Y_1, \ldots, Y_n$ i.i.d. from some distribution with c.d.f. $F$ and density $f$, and from these define $M_n = \max_{i \le n} Y_i$. (xx brief pointer to order statistics and sample quantiles in Chs 2, 3. xx)

**(a)** Show that $M_n$ has c.d.f. $G_n(z) = F(z)^n$ with density $g_n(z) = nF(z)^{n-1}f(z)$. Show that its median is $\mu_m = F^{-1}((\frac{1}{2})^{1/m})$. Compute and draw these, for say $n = 1, \ldots, 10$, for the case of the $Y_i$ being standard normal, tracking also the medians $\mu_m$.

**(b)** Before pursuing some maxima, consider a situation with $X_1$ and $X_2$ being independent, with c.d.f.s $H_1, H_2$ and densities $h_1, h_2$. Show that

$$\Pr(X_1 \le X_2) = \int H_1(x_2)h_2(x_2)\,\mathrm{d}x_2 = \int \{1 - H_2(x_1)\}h_1(x_1)\mathrm{d}x_1.$$

**(c)** Long jumpers A and B have jumps being $\mathrm{N}(7.90, 0.10^2)$ and $\mathrm{N}(7.80, 0.15^2)$, respectively (on the metres scale). What is the probability that A jumps longer than B, with one jump each? Consider also $p_m$, the probability that B after $m$ jumps is better than A after $m = 1, 2, 3, 4, 5, 6$ jumps (in the sense of 'best jump so far'), assuming that all jumps are independent. Show that $p_m = \int F_1(z)^m m F_2(z)^{m-1} f_2(z)\,\mathrm{d}z$, in terms of the two distributions. Compute $p_m$ both via numerical integration and via simulations.

**Ex. 1.18** *Ratios and the Cauchy.* If $(X, Y)$ has a certain distribution, what happens to the ratio $V = Y/X$?

**(a)** Suppose that $X$ and $Y$ are independent with the same density $f$ on $(0, \infty)$. Show that $V = Y/X$ has density $g(v) = \int_0^\infty x f(x) f(vx)\,\mathrm{d}x$. With $X$ and $Y$ independent from the same exponential distribution, show that $g(v) = 1/(1+v)^2$.

**(b)** With $X$ and $Y$ independent from the same Gamma $(a, b)$, show that $V = Y/X$ has density $\{\Gamma(2a)/\Gamma(a)^2\}\, v^{a-1}/(1+v)^{2a}$.

**(c)** Suppose now that $X$ and $Y$ are independent from the same density $f$, symmetric around zero. Show that $V$ has density $g(v) = 2\int_0^\infty x f(x) f(vx)\,\mathrm{d}x$. For the special case of a ratio of two independent standard normals, show that

the Cauchy

$$g(v) = (1/\pi)/(1+v^2) \quad \text{with c.d.f.} \quad G(v) = \tfrac{1}{2} + (1/\pi)\arctan v.$$

This is the Cauchy distribution (in its standard form). Show that it has no mean. Find its interquartile range. Deduce the $Y/X$ distribution also via Ex. 1.14(a).

**(d)** There is an infinitely long straight line ahead of you, with distance $r$ to the nearest point. You kick a ball towards the line, with some angle $A \in (-\pi/2, \pi/2)$. Show that it crosses the line at position $X = r\tan(A)$. Show that with $A$ having density $g$ and c.d.f. $G$, over $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, then $X$ has c.d.f. $F(x) = G(\arctan(x/r))$. When $A$ is uniform over that interval, show that $X/r$ is standard Cauchy. Investigate what happens if $A$ is uniform over a tighter interval, like $(-\pi/4, \pi/4)$.

**(e)** What distribution does the random angle $A$ need to have, in order for $X$ to be normal? For $r = 1$ unit away from the line, show that the peculiar density $g(a) = \phi(\tan(a))/\cos^2(a)$ leads to a standard normal $X$. Graph $g$ to see that it is symmetric and bimodal, with modes at $\pm$ 45 degrees.

**Ex. 1.19** *The Beta distribution.* An important class of distributions, over the unit interval $(0, 1)$, is the *Beta distribution*, with two positive parameters. We write $p \sim \mathrm{Beta}(a, b)$ if its density is

the Beta distribution

$$\mathrm{be}(p, a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} p^{a-1}(1-p)^{b-1} \quad \text{for } p \in (0, 1).$$

**(a)** As an introductory exercise, of separate interest, suppose $X$ and $Y$ are independent Gamma variables with parameters $(a, 1)$ and $(b, 1)$. Construct from these the sum $Z = X + Y$ and ratio $P = X/(X+Y)$. Finding the joint density for $(P, Z)$, demonstrate that $Z$ and $P$ are independent, with $Z \sim \mathrm{Gam}(a+b, 1)$ and $P$ having precisely the $\mathrm{be}(p, a, b)$ density. This in particular shows that the integration constant is the correct one, i.e. that $\int_0^1 p^{a-1}(1-p)^{b-1}\,\mathrm{d}p = \Gamma(a)\Gamma(b)/\Gamma(a+b)$.

**(b)** Compute and display a few of these densities, for $(a, b)$ of your choice. Note that the uniform is the special case of $(a, b) = (1, 1)$.

**(c)** Show that $\mathrm{E}\, p = p_0 = a/(a+b)$ and that $\mathrm{Var}\, p = p_0(1-p_0)/(a+b+1)$.

**(d)** Find a formula for $\mathrm{E}\, p^m$, for $m = 1, 2, \ldots$, in terms of $(a, b)$, and in terms of the reparametrisation $(cp_0, c(1-p_0))$. Use this to find

$$\mathrm{E}\,(p-p_0)^3 = \frac{2p_0(1-p_0)(1-2p_0)}{(c+1)(c+2)},$$

with a consequent formula for the skewness. For fixed mean $p_0$, show that the skewness tends to zero with increasing $c$.

**(e)** Examine the particular Beta$(\frac{1}{2}, \frac{1}{2})$ distribution. Show that its density and c.d.f. become $f(p) = (1/\pi)/\{p(1-p)\}^{1/2}$ and $F(p) = (2/\pi)\arcsin(\sqrt{p})$. Find is quantile $F^{-1}(q)$. Plot these functions.

**Ex. 1.20** *The Dirichlet distribution.* Let $G_1, \ldots, G_k$ be independent and Gamma distributed, with parameters $(a_1, 1), \ldots, (a_k, 1)$. With $G = G_1 + \cdots + G_k$ their sum, consider the random ratios

$$(X_1, \ldots, X_{k-1}) = (G_1/G, \ldots, G_{k-1}/G).$$

It inherits a distribution, with density $h(x_1, \ldots, x_{k-1})$, worked with below, in the simplex where each $x_i \geq 0$ and $x_1 + \cdots + x_{k-1} < 1$. Taking also $X_k = G_k/G = 1 - (X_1 + \cdots + X_{k-1})$ on board, we have a vector $(X_1, \ldots, X_k)$ of random probabilities summing to 1 over its $k$ categories. Its distribution has a name: it's the Dirichlet distribution, with $k$ categories, and parameters $(a_1, \ldots, a_k)$, which we write as $X \sim \mathrm{Dir}(a_1, \ldots, a_k)$.

the Dirichlet distribution

**(a)** Suppose $(X_1, X_2, X_3, X_4, X_5, X_6) \sim \mathrm{Dir}(a_1, a_2, a_3, a_4, a_5, a_6)$. Show that $(X_1 + X_4 + X_6, X_2, X_3 + X_5) \sim \mathrm{Dir}(a_1 + a_4 + a_6, a_2, a_3 + a_5)$. Generalise and formalise this summing-over-cells property of the Dirichlet distribution.

**(b)** With $X \sim \mathrm{Dir}(a_1, \ldots, a_k)$, show that each $X_i \sim \mathrm{Dir}(a_i, a - a_i)$, with $a = a_1 + \cdots + a_k$, and that this is the same as a Beta$(a_i, a - a_i)$. Show from this that

$$\mathrm{E}\, D_i = \xi_i = a_i/a, \quad \mathrm{Var}\, D_i = \xi_i(1-\xi_i)/(a+1),$$

Show also the $\mathrm{cov}(D_i, D_j) = -\xi_i\xi_j/(a+1)$ for $i \neq j$.

**(c)** We have been able to derive certain basic properties above, without really needing an expression for the density of a Dirichlet vector. We tend to this now, using the transformation machinery of Ex. 1.12. Show in fact, starting with $(G_1, \ldots, G_k)$ and transforming to $(X_1, \ldots, X_{k-1}, G)$, (i) that $(X_1, \ldots, X_{k-1})$ has the density

$$h(x_1, \ldots, x_{k-1}) = \frac{\Gamma(a)}{\Gamma(a_1)\cdots\Gamma(a_k)} x_1^{a_1-1} \cdots x_{k-1}^{a_{k-1}-1}(1 - x_1 - \cdots - x_{k-1})^{a_k-1}$$

over the simplex; (ii) that $G \sim \mathrm{Gam}(a, 1)$; (iii) that these are independent; and (iv) that this also verifies the implied formula for integrating $x_1^{a_1-1} \cdots x_{k-1}^{a_{k-1}-1}(1 - x_1 - \cdots - x_{k-1})^{a_k-1}$ over the simplex. Note that these efforts and results generalise the findings of Ex. 1.19(a).

**Ex. 1.21** *Dirichlets inside Dirichlets.* The summing-over-cells property of the Dirichlet, see Ex. 1.20, has other consequences and angles, and we shall learn here that long Dirichlet vectors might be split into Dirichlet parts via Dirichlet cuts.

**(a)** Start with $A, B, C$ independent gammas with parameters $(a,1)$, $(b,1)$, $(c,1)$. Form from these $X = A/(A+B)$, $Y = (A+B)/(A+B+C)$, $Z = A+B+C$. Explain that we already know that $X \sim \text{Beta}(a,b)$, $Y \sim \text{Beta}(a+b,c)$, $Z \sim \text{Gam}(a+b+c,1)$. Show in fact that $X, Y, Z$ also are independent. This means working out their joint distribution; establish first that the inverse transform is $A = XYZ$, $B = (1-X)YZ$, $C = (1-Y)Z$, and that the associated Jacobi determinant becomes $yz^2$. Explain that all of this leads to the product Beta representation

$$\frac{A}{A+B+C} = \frac{A}{A+B}\,\frac{A+B}{A+B+C} = \text{Beta}(a,b)\,\text{Beta}(a+b,c) = \text{Beta}(a,b+c).$$

**(b)** (xx nils rant, to be edited. link to pinned-down Dirichlet processes. xx) Suppose $(X_1,\ldots,X_k,Y_1,\ldots,Y_\ell)$ is $\text{Dir}(a_1,\ldots,a_k,b_1,\ldots,b_\ell)$, which we may represent as $X_i = G_i/(S+T)$, $Y_j = H_j/(S+T)$, with the $G_i$ and $H_j$ being independent gamma variables with the appropriate parameters, and sums $S = \sum_{i=1}^k G_i$ and $T = \sum_{j=1}^\ell H_j$. Explain that $W = \sum_{i=1}^k X_i = S/(S+T)$ is a $\text{Beta}(a,b)$, with $a = \sum_{i=1}^k a_i$ and $b = \sum_{j=1}^\ell b_j$, and show that

$$X_i = \frac{G_i}{S+T} = W\frac{G_i}{S} = WX_i' \quad\text{and}\quad Y_j = \frac{H_j}{S+T} = (1-W)\frac{H_j}{T} = (1-W)Y_j',$$

where $X' = (X_1',\ldots,X_k') \sim \text{Dir}(a_1,\ldots,a_k)$ and $Y' = (Y_1',\ldots,Y_\ell') \sim \text{Dir}(b_1,\ldots,b_\ell)$, independent of $W$. We learn that the long Dirichlet vector $(X,Y)$ may be split into two separate Dirichlet vectors $X' = X/w$ and $Y' = Y/(1-w)$, by conditioning on $W = w$.

**(c)** Now generalise to longer vectors. Let $X = (X_1,\ldots,X_k)$ be a $\text{Dir}(a_1,\ldots,a_k)$, where each $X_i$ is a vector $(X_{i,1},\ldots,X_{i,m_i})$, with corresponding $a_i = (a_{i,1},\ldots,a_{i,m_i})$. Write $W_i = \sum_{j=1}^{m_i} X_{i,j}$ for the sum over $X_i$. Explain first that $(W_1,\ldots,W_k) \sim \text{Dir}(b_1,\ldots,b_k)$, where $b_i = \sum_{j=1}^{m_i} a_{i,j}$ is the sum over $a_i$. Show next that with $X_i' = X_i/W_i$, then $X_i' \sim \text{Dir}(a_i)$, and that $X_1',\ldots,X_k'$ are independent of $(W_1,\ldots,W_k)$.

**(d)** The previous point implies that a long Dirichlet vector may be split into independent Dirichlet components, in several ways. In the setting above, starting with the long $(X_1,\ldots,X_k)$, suppose we condition on $(W_1,\ldots,W_k) = (w_1,\ldots,w_k)$, a given probability vector. With $X_i' = X_i/w_i$, show that $X_1',\ldots,X_k'$ are independent, with $X_i' \sim \text{Dir}(a_i)$. This is the sorcerer's apprentice property of the Dirichlet distribution; pin it down, into subsets with given sums, and see that each part is a scaled Dirichlet again.

**Ex. 1.22** *The Beta-binomial distribution.* Sometimes comparable binomial experiments may be modelled and analysed jointly, but where the success probability is not the same across studies. The $p = \Pr(\text{girl})$ may e.g. vary from family to family, see Story #2.

**(a)** Suppose in general terms that $Y \,|\, p \sim \text{binom}(n,p)$, and that $p$ has a distribution with mean $p_0$ and standard deviation $\tau_0$. Using the double expectation rule (1.2), show that

$$\text{E}\,Y = np_0 \quad\text{and}\quad \text{Var}\,Y = np_0(1-p_0) + n(n-1)\tau_0^2.$$

Hence the extra-binomial component of the variance, the $n(n-1)\tau_0^2$, becomes more noticeable with increasing $n$. The case of $\tau_0 = 0$ corresponds to the usual binomial.

**(b)** Suppose $Y \,|\, p \sim \text{binom}(n,p)$ and that $p \sim \text{Beta}(a,b)$. Show that this leads to the

distribution

$$\begin{aligned}\Pr(Y=y) &= \int_0^1 \binom{n}{y} p^y (1-p)^{n-y} g(p,a,b)\,\mathrm{d}p \\ &= \binom{n}{y} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \frac{\Gamma(a+y)\Gamma(b+n-y)}{\Gamma(n+a+b)} \quad \text{for } y=0,1,\ldots,n.\end{aligned}$$

Give formulae for the mean and variance of $Y$. For the special case of the uniform for $p$, show that all outcomes for $Y$ are equally likely.

**Ex. 1.23** *The Dirichlet-multinomial distribution.* Here we deal with the natural extension of the Beta-binomial setup of Ex. 1.22, from the case of two categories to more than two.

**(a)** Let $Y = (Y_1, \ldots, Y_k)$, for given probability vector $p = (p_1, \ldots, p_k)$, have a multinomial $(n, p_1, \ldots, p_k)$ model, as per Ex. 1.5. Assume then that the $p$ is not fixed, but with $p_i$ variances $\tau_{0,i}^2$ around mean $p_{0,i}$. Show that $Y_i$, marginally, has mean $np_{0,i}$ and variance $np_{0,i}(1-p_{0,i}) + n(n-1)\tau_{0,i}^2$.

**(b)** Let in particular $p \sim \mathrm{Dir}(cp_0)$, with parameters $cp_0 = (cp_{0,i}, \ldots, cp_{0,k})$. Show that

$$\mathrm{Var}\, Y_i = \{n + n(n-1)/(c+1)\} p_{0,i}(1-p_{0,i}) = \frac{c+n}{c+1} n p_{0,i}(1-p_{0,i}),$$

with a clear overdispersion factor with respect to multinomial variation.

**(c)** Show that the marginal distribution of $(Y_1, \ldots, Y_k)$, now overdispersed compared to the multinomial, becomes

$$\begin{aligned}\bar{f}(y_1,\ldots,y_k) &= \int \frac{n!}{y_1!\cdots y_k!} p_1^{y_1}\cdots p_{k-1}^{y_{k-1}} (1-p_1-\cdots-p_{k-1})^{y_k} \\ &\qquad g(p_1,\ldots,p_{k-1})\,\mathrm{d}p_1\cdots\mathrm{d}p_{k-1} \\ &= \frac{n!}{y_1!\cdots y_k!} \frac{\Gamma(c)}{\Gamma(cp_{0,1})\cdots\Gamma(cp_{0,k})} \frac{\Gamma(cp_{0,1}+y_1)\cdots\Gamma(cp_{0,k}+y_k)}{\Gamma(c+n)}.\end{aligned}$$

**(d)** For the case of Dirichlet parameters $cp_0 = (1, \ldots, 1)$, show that all outcomes $(y_1, \ldots, y_k)$ have the same probability, and find a formula for how many different outcomes there can be.

### Laws of small numbers, Poisson, geometric, negative binomial

**Ex. 1.24** *The Poisson distribution.* Counting a high number of events with small probabilities leads to the Poisson distribution, which we define here. A count variable $Y$ is said to have the Poisson distribution with parameter $\theta > 0$ if

$$\Pr(Y=y) = \exp(-\theta)\theta^y/y! \quad \text{for } y = 0,1,2,\ldots.$$

We write $Y \sim \mathrm{Pois}(\theta)$ to indicate this.

**(a)** Show that the probabilities indeed sum to 1. Verify next that $\mathrm{E}\, Y = \theta$, $\mathrm{E}\, Y(Y-1) = \theta^2$ and show from this that the variance is equal to the mean. Show further that $\mathrm{E}\, Y(Y-1)(Y-2) = \theta^3$, $\mathrm{E}\, Y(Y-1)(Y-2)(Y-3) = \theta^4$, and with further algebra that for $W = (Y-\theta)/\sqrt{\theta}$, we have skew $= \mathrm{E}\, W^3 = 1/\sqrt{\theta}$, kurt $= \mathrm{E}\, W^4 - 3 = 1/\theta$. Show also that $\mathrm{Var}\,(Y-\theta)^2 = 2\theta^2 + \theta$.

**(b)** With $Y \sim \mathrm{Pois}(\theta)$, what is the most probable outcome? What is the probability that $Y$ is odd?

**(c)** Show that the sum of two independent Poisson variables is Poisson, with parameter equal to the sum of the two parameters. Generalise.

**(d)** Consider $Y \sim \text{binom}(n, p)$, and assume that $n$ grows, while $p$ becomes small, in the fashion of $np \to \theta$. Show that $Y$ then tends to the Pois($\theta$) distribution, in the sense that the point probabilities converge. See also Ex. 2.8 for a fuller picture.

**(e)** In some event counting applications there are more zeros than predicted by the Poisson, leading naturally to a more general model with

$$\Pr(Y = 0) = p_0, \quad \Pr(Y = y) = (1 - p_0) \exp(-\theta)\theta^y / y! \, / \{1 - \exp(-\theta)\} \quad \text{for } y \geq 1.$$

Verify that these probabilities sum to 1, and find expressions for the mean and variance. This model is sometimes called the zero-inflated Poisson, since situations with $p_0 > \exp(-\theta)$ are prevalent, but also cases with $p_0 < \exp(-\theta)$ are allowed. Simulate say 1000 datapoints from the model with $\theta = 3.00$ and $p_0 = 0.25$, and check the histogram.

**Ex. 1.25** *The geometric distribution.* Suppose $Y$ has the distribution with point probabilities $f(y) = (1-p)^{y-1}p$ for $y = 1, 2, \ldots$. This is the geometric distribution, and we write $Y \sim \text{geom}(p)$ to indicate this.

**(a)** Show that the probabilities $f(y)$ indeed sum to 1. Suppose independent experiments are carried out, each time with probability $p$ that a certain event $A$ takes place. With $Y$ the first time $A$ happens, show that $Y \sim \text{geom}(p)$.

**(b)** Show that $Y$ has mean $1/p$ and variance $(1-p)/p^2$, via direct summation of $\sum_{y=1}^{\infty} yf(y)$ etc. If $Y$ is the number of times you need to roll a six-sided die until it shows a '6', find the mean and the standard deviation.

**(c)** Another way of finding the mean and variance is as follows. With probability $p$, $Y = 1$; with complementary probability $1 - p$, $Y = 1 + Y'$, with $Y'$ having the same distribution as $Y$. Show that this leads to $\text{E}\, Y = p + (1-p)(1 + \text{E}\, Y)$ and solve. Use this representation to also find the variance. Show also that $\text{E}\,(Y - 1/p)^3 = (1-p)(2-p)/p^3$.

**(d)** Show that $\Pr(Y \geq y) = (1-p)^{y-1}$, derive a formula for $\Pr(Y \geq y_0 + y \,|\, Y \geq y_0)$, and comment.

**(e)** A simple related distribution is when one starts counting at 0, not at 1, so to speak. Show that with $Y \sim \text{geom}(p)$, as defined above, the variable $Y_0 = Y - 1$ has point probabilities $\Pr(Y_0 = y) = q^y p$ for $y = 0, 1, \ldots$, writing $q = 1 - p$. Show that $Y_0$ has mean $(1-p)/p$ and variance $(1-p)/p^2$.

**(f)** Suppose $Y$ given $p$ has geometric probabilities $(1-p)^{y-1}p$ for $y = 1, 2, \ldots$, but that $p$ itself stems from a uniform distribution. Find the distribution for $Y$; note that this implies $1/2 + 1/6 + 1/12 + 1/20 + \cdots = 1$.

**Ex. 1.26** *Time to last event.* Suppose independent geometric experiments are carried out by $m$ individuals, say players throwing dice until they get the '6'. How long time does it take until the event in question has taken place, for all individuals?

**(a)** Let $Y_1, \ldots, Y_m$ be the time needed for the $m$ individuals to see the event in question. Show that $F(y) = \Pr(Y_i \leq y) = 1 - (1-p)^y$ for $y = 1, 2, \ldots$. For $Z_m = \max(Y_1, \ldots, Y_m)$, show that its c.d.f. is $H_m(z) = F(z)^m$, and use this to find a formula for the median time to final event: $z^*_m = \log(1 - (\frac{1}{2})^{1/m}) / \log(1-p)$ (which may be rounded to the nearest integer).

**(b)** For $p = 1/6$, compute also the mean $\text{E}\, Z_m$, for say $m = 1, \ldots, 100$. Plot this, along with the median times $z^*_m$.

**Ex. 1.27** *Mixing the Poisson.* Suppose observations come from Poisson mechanisms, but with different parameters, forming their own distribution. There are several versions and uses of such Poisson overdispersion models. (xx pointer to Poisson regression with overdispersion, perhaps in Ch5. xx)

**(a)** Suppose $Y \mid \theta \sim \text{Pois}(\theta)$ but that $\theta$ has a distribution with mean $\theta_0$ and variance $\tau_0^2$. Show that $Y$ has mean $\theta_0$ and variance $\theta_0 + \tau_0^2$.

**(b)** Specialise to the case of $\theta \sim \text{Gam}(a, b)$, see Ex. 1.9. Show that $\text{E}\, Y = \theta_0 = a/b$ and that $\text{Var}\, Y = \theta_0(1+1/b)$. Argue that with a large $b$ we come back to pure Poisson. Show also that the marginal distribution of $Y$ becomes

$$f(y, a, b) = \frac{\Gamma(a+y)}{\Gamma(a)\, y!} \frac{b^a}{(b+1)^{a+y}} = \frac{\Gamma(a+y)}{\Gamma(a)\, y!} \Big(\frac{b}{b+1}\Big)^a \Big(\frac{1}{b+1}\Big)^y \quad \text{for } y = 0, 1, 2, \ldots.$$

negative binomial

We are discovering the general *negative binomial* distribution in the process, of the form

$$g(y, a, p) = \frac{\Gamma(a+y)}{\Gamma(a)\, y!}(1-p)^y p^a \quad \text{for } y = 0, 1, \ldots, \tag{1.5}$$

for parameters $a > 0$, $p \in (0, 1)$; see Ex. 1.28 for more details.

**Ex. 1.28** *The negative binomial.* We met the negative binomial distribution in Ex. 1.27 and now point to other features and constructions.

**(a)** Let $X_1, X_2$ be independent from the geometric distribution $q^x p$ for $x = 0, 1, \ldots$, with $q = 1 - p$. Show that $Y = X_1 + X_2$ has distribution $\Pr(Y = y) = (y+1)q^y p^2$, for $y = 0, 1, \ldots$. For $Y = X_1 + X_2 + X_3$ a sum of three such independent geometric variables, show that $\Pr(Y = y) = \binom{y+2}{2} q^y p^3$ for $y = 0, 1, \ldots$.

**(b)** Generalise to the case of $Y = X_1 + \cdots + X_a$, the sum of $a$ independent geometric variables, each with $q^x p$ for $x = 0, 1, \ldots$. Show that

$$\Pr(Y = y) = \binom{y+a-1}{a-1} q^y p^a = \frac{\Gamma(y+a)}{\Gamma(a)\, y!}(1-p)^y p^a \quad \text{for } y = 0, 1, \ldots,$$

i.e. the negative binomial with parameters $(a, p)$. Deduce that the number of ways in which one may find nonnegative numbers $x_1, \ldots, x_a$ with a given sum $y$ is $\binom{y+a-1}{a-1} = (y+a-1)!/\{(a-1)!\, y!\}$. In how many ways may one find 5 nonnegative numbers with sum 100? And with 10 nonnegative numbers with sum 100?

**(c)** How do we know that the negative binomial probabilities (1.5) sum to one, also when the $a$ is a non-integer? Deduce from this that

$$\sum_{y=0}^{\infty} \frac{\Gamma(y+a)}{\Gamma(a)} \frac{u^y}{a!} = \frac{1}{(1-u)^a} \quad \text{for } u \in (0, 1).$$

Show that $\text{E}\, Y = aq/p$, $\text{Var}\, Y = aq/p^2$.

**(d)** (xx the step from $Y = X_1 + \cdots + X_a$, counting from zero, to $Y' = X_1' + \cdots + X_a'$, counting each from one, so that $Y' \geq a$. and just a bit more. reason for the negative binomial term. xx)

**(e)** In one of the episodes of the television series *Siffer* (NRK, 2011), programme leader Jo Røislien announced he would flip his coin and land 'krone' ten times in a row – which he then proceeded to do. He looked a bit tired, though; he had just kept on doing this, complete with his opening statement, until he had achieved the ten krone in a row event,

and then showed only this crowning minute on tv. About how many times did he need to flip his coin, in total, before he (and his camera man) could show that final string of crowns? Simulate the process, and give a histogram of say 1,000 realisations.

**Ex. 1.29** *Conditioning on Poisson sums and a generalised binomial.* We start with Poisson sums and see a connection to the binomial. Using generalised Poissons then lead to generalisations of the binomial.

**(a)** Let $X$ and $Y$ be independent Poissons with parameters $\theta_1, \theta_2$. Show that $X$ given $X+Y=n$ is a binomal $(n,p)$, with $p=\theta_1/(\theta_1+\theta_2)$. Generalise to the case of $X_1,\ldots,X_m$ being independent Poissons, with parameters $\theta_1,\ldots,\theta_m$. Show that their conditional distribution given $X_1+\cdots+X_m=n$ is a multinomial with count $n$ and probabilities $(p_1,\ldots,p_m)$, where $p_j=\theta_j/(\theta_1+\cdots+\theta_m)$.

**(b)** To generalise the above, and in its turn also the binomial distribution, consider the gamma-mixed Poissons of 1.27. Specifically, let $X\,|\,\theta_1 \sim \text{Pois}(\theta_1)$ and $Y\,|\,\theta_2 \sim \text{Pois}(\theta_2)$, with gamma distributions $(c\theta_{0,1}, c)$ and $(c\theta_{0,2}, c)$ for $\theta_1$ and $\theta_2$; when $c$ is large, this means tight concentration around $\theta_{1,0}, \theta_{0,2}$, and we're back to Poisson. Show that the conditional distribution of $X$ given $X+Y=n$ may be written

$$f(x\,|\,n) \propto \binom{n}{x}\Gamma(c\theta_{0,1}+x)\Gamma(c\theta_{0,2}+n-x) \quad \text{for } x=0,1,\ldots,n,$$

where '$\propto$' means 'proportional to'. For a particular case, explain that with $c\theta_{0,1}=c\theta_{0,2}=1$, $X$ has the uniform distribution on $0,1,\ldots,n$.

**(c)** It may not be easy to sum the terms above directly, to find the normalisation constant, but show via expressions in Ex. 1.22 that we in fact must have

$$f(x\,|\,n) = \binom{n}{x}\frac{\Gamma(c\theta_{0,1}+c\theta_{0,2})}{\Gamma(c\theta_{0,1})\Gamma(c\theta_{0,2})}\frac{\Gamma(c\theta_{0,1}+x)\Gamma(c\theta_{0,2}+n-x)}{\Gamma(c\theta_{0,1}+c\theta_{0,2}+n)}$$

for $x=0,1,\ldots,n$. In yet other words, we have reinvented the Beta-binomial distribution with gamma-mixing of Poisson parameters. Also, for large $c$, we are back to plain binomial $(n,p)$, with $p=\theta_{0,1}/(\theta_{0,1}+\theta_{0,2})$.

**(d)** To invent other generalisations of the binomial, therefore, we might attempt other extensions of the Poisson. One such is to let $X$ and $Y$ have point probabilities proportional to $\theta_1^x/(x!)^\gamma$ and $\theta_2^x/(y!)^\gamma$, for some $\gamma\neq 1$; we work with such models in Ex. 4.35 and Story #26. Show here that $X$ given $X+Y=n$ has the distribution

$$f(x\,|\,n) \propto \binom{n}{x}^\gamma p^x(1-p)^{n-x} \quad \text{for } x=0,1,\ldots,n,$$

where $p=\theta_1/(\theta_1+\theta_2)$. Compute these probabilities, for say $n=50$, $p=0.33$, and some values of $\gamma$ around 1. Check then numerically $np_0=\mathrm{E}\,X$ and the dispersion ratio $\rho=\mathrm{Var}\,X/\{np_0(1-p_0)\}$, to learn that there is overdispersion and underdispersion, compared to the binomial case, for $\gamma<1$ and $\gamma>1$, respectively.

### Moments, moment-generating functions, characteristic functions

**Ex. 1.30** *Jensen, Hölder, and Cauchy–Schwarz.* The three inequalites named after the four mathematicians in the title of this exercise are used time and again in probability and statistics.

**(a)** Recall that the function $g\colon \mathbb{R} \to \mathbb{R}$ is convex if $g(\lambda x + (1-\lambda)y) \le \lambda g(x) + (1-\lambda)g(y)$ for all $x, y$ and all $\lambda \in [0,1]$. Show that if $g$ is convex, then $g$ is continuous, and that

$$\lambda_1 g(x_1) + \cdots + \lambda_k g(x_k) \ge g(\lambda_1 x_1 + \cdots + \lambda_k x_k),$$

for any $\lambda_1, \ldots, \lambda_k \in [0,1]$ such that $\lambda_1 + \cdots + \lambda_k = 1$. Use this to show that for any simple random variable $X_m = \sum_{j=1}^m a_j I_{A_j}$ (see Ex. B.10 for the precise definition), $\mathrm{E}\, g(X_m) \ge g(\mathrm{E}\, X_m)$ for any convex function $g$. This is Jensen's inequality for simple random variables.

**(b)** For a convex function $g\colon \mathbb{R} \to \mathbb{R}$ we can for any $x_0$ choose $a$ and $b$ such that $a + bx_0 = g(x_0)$ and $a + bx \le g(x)$ for all $x$ (make a drawing!). Use this to show Jensen's inequality for general random variables $X$, namely that for any convex function $g\colon \mathbb{R} \to \mathbb{R}$

Jensen's inequality

$$\mathrm{E}\, g(X) \ge g(\mathrm{E}\, X).$$

To remember what way the inequality goes, recall the variance is always nonnegative, so $\mathrm{Var}(X) = \mathrm{E}\, X^2 - (\mathrm{E}\, X)^2 \ge 0$, and that $g(x) = x^2$ is a convex function.

**(c)** Let $X$ and $Y$ be random variables such that $\mathrm{E}\,|X|^p$ and $\mathrm{E}\,|Y|^q$ are finite, where $p, q > 1$ and $1/p + 1/q = 1$. Set $K = \mathrm{E}\,|X|^p$, so that $Q(A) = K^{-1}\mathrm{E}\, I_A |X|^p$ is a probability measure (see Ex. B.14(e)), and let $\mathrm{E}_Q(\cdot)$ be the expectation with respect to $Q$. Use Jensen's inequality to show that

$$\frac{1}{K^q}\big(\mathrm{E}\,|XY|\big)^q = \Big(\mathrm{E}_Q \frac{|Y|}{|X|^{p-1}}\Big)^q \le \frac{1}{K}\mathrm{E}\,|Y|^q,$$

Hölder's inequality

and from this, deduce Hölder's inequality,

$$\mathrm{E}\,|XY| \le \big(\mathrm{E}\,|X|^p\big)^{1/p}\big(\mathrm{E}\,|Y|^q\big)^{1/q}.$$

Cauchy–Schwarz inequality

**(d)** From the above, deduce the Cauchy–Schwarz inequality $|\mathrm{E}\, XY|^2 \le \mathrm{E}\, X^2\, \mathrm{E}\, Y^2$.

**(e)** Show that for any random variables $X$ and $Y$

$$\mathrm{cov}(X,Y)^2 \le \mathrm{Var}(X)\mathrm{Var}(Y),$$

entailing that the correlation of two random variables lies between $-1$ and 1.

**Ex. 1.31** *Moments.* Consider a random variable $X$ with c.d.f. $F$. Its mean is $\mathrm{E}\, X = \int x\, \mathrm{d}F(x)$, and we may of course define higher moments.

**(a)** Use results of Ex. C.3 to show that $\mathrm{E}\, X^k$, first seen as $\int y\, \mathrm{d}G_k(y)$, the mean of the variable $Y = X^k$ with distribution $G_k$ inherited from $F$, is also the same as $\int x^k\, \mathrm{d}F(x)$; thus there is no ambiguity there.

**(b)** For $r < s$, show that $(\mathrm{E}\,|X|^r)^{1/r} \le (\mathrm{E}\,|X|^s)^{1/s}$, i.e. $h(r) = (\mathrm{E}\,|X|^r)^{1/r}$ is a non-decreasing function in $r$. In particular, note that if $|X|$ has a finite $s$-moment, then all moments of smaller order are also finite. Illustrate by computing and graphing $h(r)$ and $\log h(r)$ for the case of $X \sim \mathrm{Expo}(1)$.

**(c)** For $X$ a standard normal, show that $\mathrm{E}\,|X|^p = 2^{p/2}\Gamma(\frac{1}{2}(p+1))/\sqrt{\pi}$ for $p \ge 0$, a formula that also can be written $(\frac{1}{2})^{p/2}\Gamma(p+1)/\Gamma(\frac{1}{2}p+1)$. Compute and graph the function $h(r)$ for this case. For $p$ an even integer, the formula simplifies to $(\frac{1}{2})^{p/2}p!/(\frac{1}{2}p)!$.

**(d)** For a random variable $X$ with finite fourth moment, we have defined its skewness skew and kurtosis kurt in Ex. 1.3. Give expressions skew $= h_3(\mu_1, \mu_2, \mu_3)$ and kurt $= h_4(\mu_1, \mu_2, \mu_3, \mu_4)$ in terms of the moments $\mu_j = \mathrm{E}\, X^j$, and also expressions skew $= h_3^*(\mu_2^*, \mu_3^*)$ and kurt $= h_4^*(\mu_2^*, \mu_3^*, \mu_4^*)$ in terms of the centralised moments $\mu_j^* = \mathrm{E}\,(X - \mu_1)^j$.

**Ex. 1.32** *Moment-generating functions.* For a random variable $Y$, with distribution $P$, its m.g.f. is

$$M(t) = \mathrm{E}\,\exp(tY) = \int \exp(ty)\,\mathrm{d}P(y),$$

defined for each $t$ at which the expectation exists. The moment-generating function is useful for finding and characterising distributions, for finding their moments, for handling the distributions of sums of variables, and in connection with distributional limits. When $Y$ has a density $f(y)$ (with respect to Lebesgue measure), we have $M(t) = \int \exp(ty)f(y)\,\mathrm{d}y$, and if it is discrete with pointmasses $f(y)$ for sample space $S$, say, then $M(t) = \sum_{y\in S} \exp(ty)f(y)$. The expectation operator is more general, however, and $M(t)$ is perfectly defined also for intermediate cases where $Y$ can have both discrete and continuous parts; see Ex. C.3.

**(a)** For a standard normal $Y \sim \mathrm{N}(0,1)$, show that $M(t) = \exp(\frac{1}{2}t^2)$. When $Y \sim \mathrm{N}(\xi,\sigma^2)$, derive $M(t) = \exp(\xi t + \frac{1}{2}\sigma^2 t^2)$.

**(b)** A moment-generating function has that name since it generates moments; we indeed have $M'(0) = \mathrm{E}\,Y$, $M''(0) = \mathrm{E}\,Y^2$, etc., under mild conditions, see Ex. C.21 for details. Use this to find the first four moments of the standard normal.

**(c)** For $Y \sim \mathrm{Expo}(\theta)$, show that $M(t) = 1/(1-t/\theta)$, for $t < \theta$.

**(d)** For $Y \sim \mathrm{Gam}(a,b)$, with density $\{b^a/\Gamma(a)\}y^{a-1}\exp(-by)$, show that $M(t) = \{b/(b-t)\}^a$, for $t < b$. In particular, $M(t) = 1/(1-t)^a$ for $\mathrm{Gam}(a,1)$.

**(e)** Suppose $Y$ is equal to zero with probability 0.90, but a standard normal with probability 0.10. Find the $M(t)$, and generalise.

**(f)** For the binomial $(n,p)$, show that $M(t) = \{1-p+p\exp(t)\}^n$.

**(g)** For $Y \sim \mathrm{Pois}(\theta)$, find $M(t) = \exp\{\theta(e^t-1)\}$. Use this, with Ex. 1.27, to find $M(t)$ also for the negative binomial $(a,p)$. (xx hm, should give the formula here. xx)

**(h)** Let $Y = \pm 1$ with probabilities $\frac{1}{2},\frac{1}{2}$. Show that

$$M(t) = \cosh(t) = \tfrac{1}{2}(e^t + e^{-t}) = 1 + (1/2)t^2 + (1/4!)t^4 + (1/6!)t^6 + \cdots.$$

**(i)** For the uniform distribution on the unit interval, show that $M(t) = \{\exp(t)-1\}/t$, for $t \neq 0$, and with $M(0) = 1$. For $Y$ having the uniform distribution on the $[-1,1]$ interval, show that

$$M(t) = \frac{\exp(t)-\exp(-t)}{2t} = \frac{\sinh t}{t},$$

and that this function may be written as the infinite sum $1 + (1/3!)t^2 + (1/5!)t^4 + \cdots$.

**Ex. 1.33** *Distribution of sums via moment-generating functions.* Importantly, the m.g.f. $M(t) = \mathrm{E}\,\exp(tY)$, if it exists in a neighbourhood around zero, characterises the distribution; variables whose m.g.f.s are identical in such a neighbourhood have identical distributions. See details in Ex. C.21.

**(a)** Suppose $M(t) = (\frac{1}{2})^{25}\{1+\exp(t)\}^{25}$. What is the underlying distribution? Similarly, if $M(t) = 0.99 + 0.01\exp(\frac{1}{2}t^2)$, what is the distribution?

**(b)** With $X$ and $Y$ being independent, with m.g.f.s $M_1$ and $M_2$, show that the sum $Z = X+Y$ has m.g.f. $M_X(t)M_Y(t)$.

**(c)** With $X$ and $Y$ independent and standard normal, show that $X+Y$ is $\mathrm{N}(0,2)$. Redo the questions of Ex. 1.2.

**(d)** Redo a part of Ex. 1.24, showing that sums of independent Poissons are Poisson. Similarly, if $X, Y, Z$ are independent binomials $(n_1, p), (n_2, p), (n_3, p)$, identify the distribution of $W = X + Y + Z$.

**(e)** If $X_1, \ldots, X_n$ are independent gamma variables, with parameters $(a_1, b), \ldots, (a_n, b)$, find the distribution of $X_1 + \cdots + X_n$.

the Laplace distribution

**Ex. 1.34** *The Laplace distribution.* The Laplace or double exponential distribution, in its simplest form, has density $f_0(y) = \frac{1}{2}\exp(-|y|)$, on the real line; note the cusp at its centre point zero.

**(a)** Let $V_1$ and $V_2$ be independent standard exponentials. Show that $Y = V_1 - V_2$ has this density $f_0(y)$. Deduce from this that its m.g.f. is $M_0(t) = 1/(1-t^2)$, for $|t| < 1$.

**(b)** More generally, consider $Y = V_1 - V_2$ where these two are independent and Expo$(\theta)$. Show that $Y$ has density $f(y) = \frac{1}{2}\theta\exp(-\theta\,|y|)$, with zero mean and variance $2/\theta^2$. Also, show that its m.g.f. is $M(t) = 1/\{1-(t/\theta)^2\}$ for $|t| < \theta$. The Laplace with variance 1 is hence that with $\theta = \sqrt{2}$.

**(c)** Suppose $X$ for given $\sigma$ is a $\mathrm{N}(0, \sigma^2)$, but that the variance $V = \sigma^2$ has some distribution. Show that the m.g.f. for such a normal scale mixture becomes $M(t) = \mathrm{E}\,\exp(tX) = M_V(\frac{1}{2}t^2)$, where $M_V(s)$ is the m.g.f. for $V$. In particular, show that if $X \,|\, \sigma \sim \mathrm{N}(0, \sigma^2)$ and $\sigma^2 \sim \mathrm{Expo}(1)$, then $X$ has the Laplace distribution with variance 1.

**(d)** If $X \,|\, V \sim \mathrm{N}(0, V)$, and $V$ has density $g(v)$, show that $X$ has density $f(x) = \int_0^\infty \phi(x/v^{1/2})(1/v^{1/2})g(v)\,\mathrm{d}v$. Translate the result above to the interesting formula

$$\begin{aligned}\int_0^\infty \frac{1}{(2\pi)^{1/2}}\exp\Big(-\frac{1}{2}\frac{x^2}{v}\Big)\frac{\exp(-v)}{v^{1/2}}\,\mathrm{d}v &= 2\int_0^\infty \frac{1}{(2\pi)^{1/2}}\exp\{-(\tfrac{1}{2}x^2/w^2 + w^2)\}\,\mathrm{d}w \\ &= \tfrac{1}{2}\sqrt{2}\exp(-\sqrt{2}|x|).\end{aligned}$$

Use this to find a formula for the integral $\int_0^\infty \exp\{-(av^2 + b/v^2)\}\,\mathrm{d}v$.

**Ex. 1.35** *Summing uniforms.* [xx to be polished. xx] can we simply invert $\{\exp(t) - 1\}^n/t^n$, for $n = 2, 3, \ldots$? $\mathrm{E}\,N = e$ for $N$, the number of uniforms to sum to reach 1. xx]

**(a)** Let $U_1, U_2, \ldots$ be i.i.d. uniforms on the unit interval. Show that the densities for $U_1 + U_2$ and $U_1 + U_2 + U_3$ may be written

$$f_2(x) = \begin{cases} x & \text{when } 0 \le x \le 1, \\ 2 - x & \text{when } 1 \le x \le 2, \end{cases}$$

and

$$f_3(x) = \begin{cases} \frac{1}{2}x^2 & \text{when } 0 \le x \le 1, \\ \frac{1}{2}(-2x^2 + 6x - 3) & \text{when } 1 \le x \le 2, \\ \frac{1}{2}(3-x)^2 & \text{when } 2 \le x \le 3. \end{cases}$$

Show that $f_2$ is continuous, but that its derivative has a jump at position $x = 1$. Show however that $f_3$ is smoother, with a continuous derivative.

**(b)** Generalise the above to the case of $X = U_1 + \cdots + U_n$; show that its density may be written

$$f_n(x) = \frac{1}{(n-1)!}\sum_{j=0}^{\lfloor x \rfloor}(-1)^j\binom{n}{j}(x-j)^{n-1}$$

for $0 \le x \le n$, where $\lfloor x \rfloor$ is the so-called floor function, the largest integer equal to or to the left of $x$ (so $\lfloor 2.99 \rfloor = 2$, $\lfloor 3.00 \rfloor = 3$, $\lfloor 3.01 \rfloor = 3$). In particular, show that $f_n(x) = x^{n-1}/(n-1)!$ for $0 \le x \le 1$. This is sometimes called the Irwin–Hall distribution, from the related and concurrently published papers Irwin (1927), Hall (1927).

**(c)** Draw the densities $f_1, f_2, f_3, f_4, f_5$ in a diagram, and comment on their forms.

**(d)** Let now $N$ be the number of uniforms $U_j$ needed in order for their sum to exceed 1. Show that

$$\Pr(N > n) = \Pr(U_1 + \cdots + U_n < 1) = F_n(1),$$

with $F_n$ the cumulative for $f_n$. Also show that in fact $F_n(1) = 1/n!$. Use this to deduce that $\Pr(N = n) = 1/\{n\,(n-2)!\}$ and that $\mathrm{E}\,N = e$. As a follow-up computational exercise, simulate a high number of such $N$ to calculate the mathematical constant $e$ to three decimal places. Check if you get Ibsen's and Tolstoy's birth years right.

**Ex. 1.36** *Characteristic functions.* Above we have found multiple uses for moment-generating functions, as in Ex. 1.32 and 1.33. Another very useful transform is the characteristic function, worked with in App. B, see Ex. C.24, which is also a crucial technical tool in the development of the large-sample theory of Ch. 2. For a variable with distribution $P$, its definition is $\varphi(t) = \mathrm{E}\,\exp(itX) = \int \{\cos(tx) + i\sin(tx)\}\,\mathrm{d}P(x)$, with $\mathrm{d}P(x)$ to be read as $f(x)\,\mathrm{d}x$ when $P$ has a density $f$. Notably, this $\varphi(t)$ always exists, also for distributions without means, etc. We note here the inversion theorem from Ex. C.34, that if $\int |\varphi(t)|\,\mathrm{d}t$ is finite, then there is a density $f(x) = (2\pi)^{-1} \int \exp(-itx)\varphi(t)\,\mathrm{d}t$, see Ex. C.34, which is also seen to be continuous.

**(a)** For the standard normal, show that $\varphi(t) = \exp(-\frac{1}{2}t^2)$, and that $\varphi(t) = \exp(-\frac{1}{2}\sigma^2 t^2)$ for the $\mathrm{N}(0, \sigma^2)$.

**(b)** Show that $\varphi(t)$ is real, and then equal to $\mathrm{E}\cos(tX)$, if and only if the distribution is symmetric around zero.

**(c)** For $U$ a unit exponential, show that $\varphi(t) = 1/(1-it)$. Deduce that $X = U - V$, with its Laplace distribution, has $\varphi(t) = 1/(1+t^2)$, and that

$$\tfrac{1}{2}\exp(-|x|) = (2\pi)^{-1} \int \cos(tx) \frac{1}{1+t^2}\,\mathrm{d}t.$$

**(d)** Show from this, essentially by changing from $(x,t)$ to $(t,x)$, that the standard Cauchy has $\varphi(t) = \exp(-|t|)$.

**(e)** In Ex. 1.18(d) we saw that if a ball is kicked from distance $r$ to an infinitely long straight line, with random angle $A$ being uniform on $(-\frac{1}{2}\pi, \frac{1}{2}\pi)$, then the position where the ball crosses the line is $X = rX_0$, where $X_0$ is Cauchy. Now assume $r$ is not fixed, but comes from a unit exponential distribution. Show that $X$, a scale mixture of Cauchys, has characteristic function $\phi(t) = 1/(1+|t|)$.

**(f)** For $V$ a uniform on $[-1,1]$, show $\varphi(t) = (\sin t)/t$. Prove from this that even though $\int (\sin t)/t\,\mathrm{d}t = \pi$, the integral $\int |(\sin t)/t|\,\mathrm{d}t$ must be infinite. Show that

$$f(x) = (2\pi)^{-1} \int \cos(tx) \Big(\frac{\sin t}{t}\Big)^2\,\mathrm{d}t$$

is equal to the triangular density on $[-2,2]$.

**(g)** Just as for m.g.f.s, sums of independent components are associated with products of characteristic functions. Show that if $X, Y, Z$ are independent, with $\varphi_1(t), \varphi_2(t), \varphi_3(t)$,

then $X+Y+Z$ has $\varphi_1(t)\varphi_2(t)\varphi_3(t)$ as its characteristic function. When $X_1,\ldots,X_n$ are i.i.d. with characteristic function $\varphi(t)$, show that $(X_1+\cdots+X_n)/\sqrt{n}$ has $\varphi_n(t) = \varphi(t/\sqrt{n})^n$. We learn in Ch. 2 that if the $X_i$ have mean zero and finite variance $\sigma^2$, then $\varphi_n(t)\to\exp(-\frac{1}{2}\sigma^2t^2)$. In this connection, you may numerically compute

$$f_n(x) = (2\pi)^{-1}\int \cos(tx)\Big(\frac{\sin t/\sqrt{n}}{t/\sqrt{n}}\Big)^n\,\mathrm{d}t$$

and verify that it tends to the $\mathrm{N}(0,1/12)$ density; cf. Ex. 1.35.

**(h)** Show that there cannot be two i.i.d. variables with sum having the uniform distribution. (xx but pointer to story with $U=V_1+V_2$, two independent but with different distributions. xx)

**Ex. 1.37** *Characterisation theorems.* (xx to be repaired carefully. start with symmetry, then remove that assumption. may push these two things to Ch2. xx)

**(a)** If $X,Y$ are independent with the same distribution, and $(X+Y)/\sqrt{2}\sim X$, show that $X$ must be a zero-mean normal. (xx pointer to CLT. xx)

**(b)** For another characterisation lemma, suppose that $(X+Y)/2\sim X$, and show by induction that this implies the curious property that with $X_1,\ldots,X_n$ i.i.d. from such a model, then the sample mean $\bar{X}_n$ has the same distribution for every $n$. Show that $X$ must be Cauchy. Why is this not contradicting the LLN?

**Ex. 1.38** *Cumulants and the cumulant-generating function.* For a variable $X$ with m.g.f. $M(t)$, assumed finite in an interval around zero, it is sometimes fruitful to work with the *cumulant-generating function* $K(t)=\log M(t)$. When expanded in a power series around zero, with

cumulants

$$K(t) = K'(0)t + \tfrac{1}{2}K''(0)t^2+\cdots = \sum_{j=1}^{\infty}\frac{\kappa_j}{j!}t^j,$$

we call the coefficients $\kappa_j = K^{(j)}(0)$ the *cumulants* of the distribution.

**(a)** For the $\mathrm{N}(\xi,\sigma^2)$ distribution, show that $K(t)=\xi t+\frac{1}{2}\sigma^2t^2$. For the unit exponential distribution, show $K(t)=t+t^2/2+t^3/3+\cdots$, with $\kappa_j=(j-1)!$ for $j\ge 1$.

**(b)** If $X$ has mean $\xi$, write $X=\xi+X_0$, Show that $K(t)=\xi t+K_0(t)$, where $K_0(t)=\log M_0(t)$, with $M_0$ the m.g.f. of the zero-mean variable $X_0$. Hence the cumulants $\kappa_j$ for $X$ are the same as the cumulants $\kappa_{j,0}$ for $X_0$, for $j\ge 2$.

**(c)** Via successive derivatives of $M(t)=\exp\{K(t)\}$, show that

$$\begin{aligned}
M' &= MK',\\
M'' &= M\{K''+(K')^2\},\\
M''' &= M\{K'''+3K''K'+(K')^3\},\\
M'''' &= M\{K''''+4K'''K'+6K''(K')^2+3(K'')^2+(K')^4\}.
\end{aligned}$$

Write $\xi$ and $\sigma^2$ for the mean and variance of $X$. From the equations, show first that $K'(0)=\kappa_1=\xi$ and that $K''(0)=\kappa_2=\sigma^2$, and next that

$$\mathrm{E}\,X^3 = \kappa_3+3\xi\kappa_2+\xi^3,\quad \mathrm{E}\,X^4 = \kappa_4+4\xi\kappa_3+6\xi^2\kappa_2+3\kappa_2^2+\xi^4.$$

**(d)** Skewness and kurtosis were defined in Ex. 1.3. We may now find useful expressions for these in terms of the cumulants. Show first via formulae above that

$$\mathrm{E}\,(X-\xi)^3=\kappa_3,\quad \mathrm{E}\,(X-\xi)^4=\kappa_4+3\sigma^4.$$

Explain that this leads to $\text{skew}(X) = \kappa_3/\sigma^3$, $\text{kurt}(X) = \kappa_4/\sigma^4$.

**Ex. 1.39** *Generating functions.* Moment-generating functions, studying distributions via the transformation $M(t) = \text{E}\exp(tX)$, have several close relatives, which might be more convenient for certain classes of distributions. It is e.g. common to use *Laplace transformations* $L(s) = \text{E}\exp(-sX)$ for distributions on $[0, \infty)$, then studied for $s \geq 0$. Here we work through the basic properties of *generating functions*, primarily used for distributions on the nonnegative integers. If $\Pr(Y = j) = p_j$, for $j = 0, 1, 2, \ldots$, define $G(s) = \text{E}\, s^Y = \sum_{j=0}^{\infty} p_j s^j = p_0 + p_1 s + p_2 s^2 + \cdots$, called the generating function for that distribution, or for variables having that distribution.

Laplace transforms

generating functions

**(a)** Show that $G(s) = M(\log s)$, for $s$ such that the latter exists. Demonstrate that $G(s)$ is finite, for $|s| < 1$, and also for $s = 1$. Find the generating functions for (i) the binomial $(n, p)$; (ii) the Poisson with parameter $\theta$; (iii) the geometric with $\Pr(Y = j) = q^{j-1}p$ for $j \geq 1$, with $q = 1 - p$; answers are

$$G_1(s) = (1 - p + ps)^n, \quad G_2(s) = \exp\{-\theta(1 - s)\}, \quad G_3(s) = \frac{ps}{1 - qs},$$

the latter valid for $|s| < 1/q$.

**(b)** Returning to the general case, give an expression for $G'(s)$, show that $G'(1) = \text{E}\, Y$, and that $G''(1) = \text{E}\, Y(Y - 1)$. Find the mean and variance for the Poisson using generating functions.

**(c)** Suppose $X$ and $Y$ are random variables taking on values in $\{0, 1, 2, \ldots\}$, and that their generating functions are equal, on an interval around zero. Show that $X$ and $Y$ must have identical distributions.

**(d)** Show that if $X, Y, Z$ are independent, with generating functions $G_1, G_2, G_3$, then the generating function for $X + Y + Z$ is $G_1(s)G_2(s)G_3(s)$. Show from this, and the previous point, yet again, that a sum of independent Poissons is a Poisson.

**Ex. 1.40** *Getting $Y$ from two copies of $X$.* Let $X_1, X_2$ be independently drawn from some distribution on the nonnegative integers, and from these draw $Y$ from the binomial $(X_1 + X_2, \frac{1}{2})$.

**(a)** Writing $\xi$ and $\sigma^2$ for the mean and variance for the $X$ distribution, set up formulae for the conditional mean and variance of $Y$, given $X_1, X_2$. From these show that $\text{E}\, Y = \xi$, $\text{Var}\, Y = \frac{1}{2}\xi + \frac{1}{2}\sigma^2$.

**(b)** Write $G_0(s) = \text{E}\, s^X$ for the generating function of the $X_i$. Show that the generating function for $Y$ may be written $G(s) = G_0(\frac{1}{2} + \frac{1}{2}s)^2$. When $X_i \sim \text{binom}(m, p)$, show from this that $Y \sim \text{binom}(2m, \frac{1}{2}p)$. Then show that if the $X_i$ are Poisson, then $Y$ reproduces the same distribution.

**Ex. 1.41** *Sums of random lengths and the compound Poisson.* Let $X_1, X_2, \ldots$ be i.i.d., from a distribution with mean $\xi$, variance $\sigma^2$, and m.g.f. $M_0(t)$. Consider then a random sum of these random elements; $Z = \sum_{i=1}^{N} X_i$, where $N$ has some distribution with mean $\lambda$, variance $\tau^2$, and generating function $G(s) = \text{E}\, s^N$. We define $Z$ as zero if $N = 0$.

**(a)** Show that $Z$ has m.g.f. $M(t) = G(M_0(t))$. Show that $Z$ has mean $\lambda\xi$ and variance $\lambda\sigma^2 + \xi^2\tau^2$.

**(b)** Consider the so-called compound Poisson variable $Z = \sum_{i=1}^{N} X_i$, where the $X_i$ are i.i.d. with m.g.f. $M_0(s)$ and $N \sim \text{Pois}(\lambda)$. Show that the m.g.f. of $Z$ may be written

compound Pòisson

$$\text{E}\exp(tZ) = \text{E}\, M_0(s)^N = \exp[\lambda\{M_0(t) - 1\}],$$

and that this leads to mean $\lambda\xi$ and variance $\lambda(\xi^2+\sigma^2)$. Find an expression for $\mathrm{E}\,\exp(tZ)$ for the particular case of $X_i \sim \mathrm{Gam}(a,b)$.

**(c)** Consider now $Z = \sum_{i=1}^N X_i$, where the $X_i$ are i.i.d. unit exponentials. For given $N$, the $Z$ is a $\mathrm{Gam}(N,1)$. With $N \sim \mathrm{geom}(p)$, show that $Z \sim \mathrm{Expo}(p)$. Generalise.

**(d)** Using terminology and results from Ex. 1.38, show that the cumulant-generating function for the compound Poisson is $K(t) = \lambda\{M_0(t)-1\}$, with cumulants $\kappa_j = \lambda\,\mathrm{E}\,X^j$. Show from this that $\mathrm{skew}(Z) = (1/\lambda)^{1/2}\gamma_3$ and $\mathrm{kurt}(Z) = (1/\lambda)\gamma_4$, with $\gamma_3 = \mathrm{E}\,X^3/(\mathrm{E}\,X^2)^{3/2}$ and $\gamma_4 = \mathrm{E}\,X^4/(\mathrm{E}\,X^2)^2$. For a Poisson sum $Z = \sum_{i=1}^N X_i$ of i.i.d. standard normals, so that $Z\,|\,(N=n) \sim \mathrm{N}(0,n)$, find the variance and kurtosis.

**Ex. 1.42** *The logarithmic distribution.* Consider a variable $X$ with point probabilities $\Pr(X=x) = c(p)^{-1}p^x/x$ for $x = 1,2,\ldots$, with $p$ a parameter in $(0,1)$. This distribution is sometimes called the logarithmic distribution.

**(a)** Show that we must have $c(p) = -\log(1-p)$. Find expressions for the m.g.f. $M(t)$, its mean, and its variance. Comment on the cases where $p$ is close to zero, or close to one. Show also for its generating function that $G(s) = \mathrm{E}\,s^X = c(ps)/c(p)$.

**(b)** Consider $Z = \sum_{i=1}^N X_i$, with the $X_i$ being i.i.d. with this logarithmic distribution, and $N$ is Poisson, with parameter expressed as $\lambda c(p)$. Find the mean and variance of $Z$, and show that its distribution is a negative binomial. We learn that the negative binomial is inside the class of compound Poissons.

### The multinormal, the t, the chi-squared, the F

**Ex. 1.43** *Mean and variance matrix for a random vector.* Consider a random vector of length $p$, say $Y = (Y_1,\ldots,Y_p)^{\mathrm{t}}$. Its mean vector is defined as the vector of means, i.e. $\xi = \mathrm{E}\,Y = (\mathrm{E}\,Y_1,\ldots,\mathrm{E}\,Y_p)^{\mathrm{t}}$, and its *variance matrix* $\Sigma$ (also often called the covariance matrix), of dimension $p \times p$, has the variances on the diagonal and the covariances outside.

Cauchy–Schwarz inequality

**(a)** For a random $(Z_1,Z_2)$, with finite variances $\sigma_1^2$ and $\sigma_2^2$, and correlation $\rho$, show from

$$\mathrm{Var}\,(Z_1/\sigma_1 + Z_2/\sigma_2) = 2+2\rho \quad \text{and} \quad \mathrm{Var}\,(Z_1/\sigma_1 - Z_2/\sigma_2) = 2-2\rho$$

that the correlation must be in $[-1,1]$. If $\rho = 1$, what can be said about $Z_1, Z_2$? Show also that $\mathrm{E}\,|Z_1Z_2| \le (\mathrm{E}\,Z_1^2)^{1/2}\,(\mathrm{E}\,Z_2^2)^{1/2}$, which is one of several ways to state the Cauchy–Schwarz inequality.

**(b)** Show that the elements of $\Sigma$ are $\mathrm{cov}(Y_i,Y_j)$, for $i,j = 1,\ldots,p$, and that we may write $\Sigma = \mathrm{E}\,(Y-\xi)(Y-\xi)^{\mathrm{t}}$. If we transform $Y$ to $Z = AY+b$, with $A$ a $m \times p$ matrix and $b$ a vector of length $m$, show that $\mathrm{E}\,Z = A\xi + b$ and that $\mathrm{Var}\,Z = A\Sigma A^{\mathrm{t}}$. Explain that this generalises the usual rule $\mathrm{Var}\,(aY) = a^2\mathrm{Var}\,Y$ for one-dimensional variables.

**(c)** Let $X_1,X_2,X_3,X_4$ be i.i.d. standard normals. Set up the variance matrix for $(X_1, X_1+X_2, X_1+X_2+X_3, X_1+X_2+X_3+X_4)^{\mathrm{t}}$.

**(d)** For the multinomial model studied in Ex. 1.5, with $Y = (Y_1,\ldots,Y_k)^{\mathrm{t}}$ counting the number of $n$ events that fall in categories $1,\ldots,k$, with probabilities $p_1,\ldots,p_k$, show that $\mathrm{E}\,Y = np$, where $p$ is the vector of $p_1,\ldots,p_k$. Show also that its variance matrix can be written $\Sigma = D - pp^{\mathrm{t}}$, where $D$ is diagonal with elements $p$.

**(e)** For the multinomial model, with $\mathbf{1}$ the vector $(1,\ldots,1)^{\mathrm{t}}$, show that $\mathbf{1}^{\mathrm{t}}\Sigma\mathbf{1} = 0$. Explain that this is related to the linear relationships $\sum_{j=1}^k p_j = 1$ and $\sum_{j=1}^k Y_j = n$. The matrix is hence not of full rank, and not invertible. Work however with the shorter

vector $Y_0 = (Y_1, \ldots, Y_{k-1})^{\mathrm{t}}$, and show that its variance matrix may be written $\Sigma_0 = D_0 - p_0 p_0^{\mathrm{t}}$, where $p_0 = (p_1, \ldots, p_{k-1})^{\mathrm{t}}$ and $D_0$ is diagonal with these elements. Show that $\Sigma_0^{-1} = D_0^{-1} + (1/p_k)\mathbf{1}_0\mathbf{1}_0^{\mathrm{t}}$, where $\mathbf{1}_0$ is the vector of $k-1$ 1s. For any vector $x = (x_1, \ldots, x_k)$ with sum zero, and with $x_0$ being the shortened version $(x_1, \ldots, x_{k-1})$, demonstrate that $x_0^{\mathrm{t}}\Sigma_0^{-1}x_0 = \sum_{j=1}^k x_j^2/p_j$.

**Ex. 1.44** *The multinormal distribution.* Let $Y = (Y_1, \ldots, Y_p)^{\mathrm{t}}$ be a random vector of length $p$. We say that it is multinormally distributed, with mean vector $\xi$ and variance matrix $\Sigma$, which needs to be positive definite, provided its joint density can be written

$$f(y) = (2\pi)^{-p/2}|\Sigma|^{-1/2}\exp\{-\tfrac{1}{2}(y-\xi)^{\mathrm{t}}\Sigma^{-1}(y-\xi)\},$$

where the domain for $y$ is all of $\mathbb{R}^p$. We write $Y \sim \mathrm{N}_p(\xi, \Sigma)$ to indicate this distributon.

**(a)** Show that $\int yf(y)\,\mathrm{d}y$ indeed is equal to $\xi$, so calling it the mean vector is appropriate. Show also that $\mathrm{E}\,(Y-\xi)(Y-\xi)^{\mathrm{t}}$, calculated from the density, is equal to $\Sigma$.

**(b)** Show that if $Y \sim \mathrm{N}_p(\xi, \Sigma)$, then $Y - \xi \sim \mathrm{N}_p(0, \Sigma)$.

**(c)** Assume now that $A$ is an invertible $p \times p$ matrix, and consider the transformation $Z = AY$. Show that if $Y \sim \mathrm{N}_p(\xi, \Sigma)$, then $Z = AY \sim \mathrm{N}_p(A\xi, A\Sigma A^{\mathrm{t}})$.

**(d)** By the spectral decomposition theorem of linear algebra, there is an orthonormal matrix $P$, with $PP^{\mathrm{t}} = I = P^{\mathrm{t}}P$, such that $P\Sigma P^{\mathrm{t}} = D = \mathrm{diag}(\lambda_1, \ldots, \lambda_p)$, with these values being the eigenvalues of $\Sigma$. Show that $Z = P(Y-\xi)$ has components $Z_1, \ldots, Z_p$ which are independent, with $Z_j \sim \mathrm{N}(0, \lambda_j)$.

**(e)** Show that a vector $Y$ is multinormal if and only if all linear combinations are normal. In particular, if $Y \sim \mathrm{N}_p(\xi, \Sigma)$, then $V = c^{\mathrm{t}}Y = c_1Y_1 + \cdots + c_pY_p$ is normal $\mathrm{N}(c^{\mathrm{t}}\xi, c^{\mathrm{t}}\Sigma c)$. [xx need to say something careful about allowing constants to be seen as normal, with zero variance. xx]

**(f)** Generalise point (c) to state that with any matrix $A$, of size say $q \times p$, the transformed $Z = AY$ is a multinormal $\mathrm{N}_q(A\xi, A\Sigma A^{\mathrm{t}})$.

**(g)** For the binormal case, with means $\xi_1, \xi_2$, standard deviations $\sigma_1, \sigma_2$, and correlation $\rho$, show that the density may be written

$$f(x,y) = \frac{1}{2\pi}\frac{1}{\sigma_1\sigma_2(1-\rho^2)^{1/2}}\exp\Big[-\tfrac{1}{2}\frac{1}{1-\rho^2}\Big\{\Big(\frac{x-\xi_1}{\sigma_1}\Big)^2 + \Big(\frac{y-\xi_2}{\sigma_2}\Big)^2 - 2\rho\Big(\frac{x-\xi_1}{\sigma_1}\Big)\Big(\frac{x-\xi_1}{\sigma_1}\Big)\Big\}\Big].$$

Show that $X$ and $Y$ are independent if and only if the correlation is zero.

**(h)** We learn that the situation is easy and clean for the multinormal case, where independence is equivalent to zero correlation. This is in general more complicated. (i) Let $X \sim \mathrm{N}(0,1)$ and $Y = DX$, with $D$ a random sign with equal probabilities for $-1, 1$. Show that $Y$ also is standard normal, that the correlation is zero, but that they are dependent. (ii) For another example, again with $X \sim \mathrm{N}(0,1)$, let $Y = X$ if $|X| \le a$ but $Y = -X$ if $|X| > a$. Show that $Y$ is standard normal. Show that the correlation $\rho(a)$ goes from $-1$ to 1 as $a$ goes from zero to infinity; thus there is an $a_0$ for which the correlation is zero. Also, find this $a_0$ numerically (answer: 1.537). In these zero-correlation constructions, with normal marginals, the point is that there is not joint binormality.

**Ex. 1.45** *The multinormal and conditional distributions.* Consider a multinormally distributed vector, of length $p+q$, blocked into subvectors of sizes $p$ and $q$. Let us write

this as

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim \mathrm{N}_{p+q}(\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}).$$

**(a)** By carrying out a linear transformation, and using results from Ex. 1.44, show that

$$Z = Y_1 - \Sigma_{12}\Sigma_{22}^{-1}Y_2 \sim \mathrm{N}_p(\xi_1 - \Sigma_{12}\Sigma_{22}^{-1}\xi_2, \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}),$$

and that this $Z$ is independent of $Y_2$.

**(b)** Show that the distribution of $Y_1$ given $Y_2 = y_2$ must be multinormal. Derive the important formulae for the conditional mean and variance,

$$\mathrm{E}(Y_1 \,|\, y_2) = \xi_1 + \Sigma_{12}\Sigma_{22}^{-1}(y_2 - \xi_2), \qquad \mathrm{Var}(Y_1 \,|\, y_2) = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.$$

Note that the conditional mean is a linear function in $y_2$ and that the conditional variance matrix is constant, not depending on $y_2$.

**(c)** Now study the simplest two-dimensional prototype case, with

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathrm{N}_2(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}).$$

Show that $Y \,|\, x \sim \mathrm{N}(\rho x, 1 - \rho^2)$ and that $X \,|\, y \sim \mathrm{N}(\rho y, 1 - \rho^2)$. Discuss implications for situations where an easy to measure $X$ might be a proxy for a harder to come by $Y$. How can you estimate $Y$ from $X$, and with what precision?

**(d)** In generalisation of this special binormal case, consider a binormal

$$\begin{pmatrix} Y \\ X \end{pmatrix} \sim \mathrm{N}_2(\begin{pmatrix} \xi \\ \xi_0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & \rho\sigma\sigma_0 \\ \rho\sigma\sigma_0 & \sigma_0^2 \end{pmatrix}).$$

In particular, the $X$ is seen as stemming from a $\mathrm{N}(\xi_0, \sigma_0^2)$ distribution. Show that $Y \,|\, (X = x)$ is normal, with constant variance $\sigma^2(1 - \rho^2)$ and linear mean function $\mathrm{E}(Y \,|\, x) = \xi + \rho(\sigma/\sigma_0)x$. This is essentially the linear regression model, for $Y \,|\, x$, here derived as a consequence of binormality. Without pretensions of having precise numbers, guess the binormal parameters in your population for $(X, Y)$ being height and weight, and deduce from this a model predicting a person's weight from his or her height.

**(e)** Generalise this to situations with a $(p + 1)$-dimensional normal distribution for $(Y, X_1, \ldots, X_p)$, with $Y$ seen as the main outcome, influenced by covariates $X_1, \ldots, X_p$. Show that $Y$ given these covariates is normal, with a constant variance, and find the linear conditional mean function $\mathrm{E}(Y \,|\, X_1, \ldots, X_p)$. This is essentially the linear multiple regression model; see Ex. 3.34.

**(f)** (xx a bit here on regression towards the mean. xx)

**Ex. 1.46** *How tall is Nils?* Assume that the heights of Norwegian men above the age of twenty follow the normal distribution $\mathrm{N}(\xi, \sigma^2)$ with $\xi = 180$ cm and $\sigma = 9$ cm.

**(a)** Given this information only, what is your point estimate of his height, and what is your 95 percent prediction interval?

**(b)** Assume now that you learn that his four brothers are actually 195 cm, 207 cm, 196 cm, 200 cm tall, and furthermore that correlations between brothers' heights in the population of Norwegian men is equal to $\rho = 0.80$. Use this information about his four brothers to revise your initial point estimate of his height, and provide the updated 95 percent prediction interval. Is Nils a statistical outlier in his family?

**(c)** Suppose that Nils has $n$ brothers and that you learn their heights. Give formulae for the updated normal parameters $\xi_n$ and $\sigma_n$, in the conditional distribution of his height given these extra pieces of information. Use this to clarify the following statistical point: Even if you get to know all facts concerning 99 brothers, there should be a limit to your confidence in what you may infer about Nils.

**Ex. 1.47** *Simulating from the multinormal.* For given $\xi$ and $\Sigma$, how can we simulate realisations $y$ from the $\mathrm{N}_p(\xi,\Sigma)$? We take it for granted here that yes, we can simulate from the one-dimensional $\mathrm{N}(\xi,\sigma^2)$, e.g. via `xi + sigma*rnorm(100)` in `R`. For simplicity we take $\xi=0$ here, without loss of generality.

there are simulation algorithms in packages, like `mvrnorm` om `R`, but the recipe here works better for large dimension

**(a)** From details of Ex. 1.44, there is a spectral decomposition, with $P\Sigma P^{\mathrm{t}}=D$, with the diagonal $D$ having the eigenvalues of $\Sigma$ and $P$ orthonormal, i.e. $PP^{\mathrm{t}}=I_n=P^{\mathrm{t}}P$, with $I_n$ the identity matrix. Show that with $D^{1/2}$ the diagonal with the square-roots of the eigenvalues, and with $\Sigma^{1/2}=P^{\mathrm{t}}D^{1/2}P$, this is a matrix which when squared becomes $\Sigma$.

**(b)** Explain from this that with $N=(N_1,\ldots,N_p)^{\mathrm{t}}$ a string of i.i.d. standard normals, $Y=\Sigma^{1/2}N\sim\mathrm{N}_p(0,\Sigma)$.

**(c)** Explain from Ex. 1.45 that a perhaps long multinormal vector $(Y_1,Y_2)$, with lengths $p$ and $q$, can be simulated in blocks; first $Y_1$, then $Y_2\,|\,Y_1$. This is sometimes practical in high dimensions.

**(d)** To attempt this in practice, simulate say $n=50$ copies $Y_i=(Y_{i,1},\ldots,Y_{i,p})^{\mathrm{t}}$, in dimension say $p=10$, with all variables being i.i.d. standard normal. From these compute the empirical variance matrix $\Sigma_n$, with empirical variances on the diagonal and covariances outside, i.e. $\sigma_{j,k}=(n-1)^{-1}\sum_{i=1}^n(Y_{i,j}-\bar Y_j)(Y_{i,k}-\bar Y_k)$ for $j,k=1,\ldots,p$. Then simulate from the multinormal $\mathrm{N}_p(0,\Sigma_n)$. Check to see if their empirical variance matrix is close to $\Sigma_n$ again.

**Ex. 1.48** *The chi-squared.* This exercise goes through some basic properties of the chi-squared; see also Ex. 1.53 for its eccentric cousin, the noncentral or eccentric chi-squared.

**(a)** We say that a nonnegative variable $X$ has the chi-squared distribution, with degrees of freedom $m$, and write $X\sim\chi^2_m$ for this, when its density takes the form

$$g_m(x)=\frac{1}{2^{m/2}\Gamma(m/2)}x^{m/2-1}\exp(-\tfrac12 x)\quad\text{for } x>0.$$

Show that its m.g.f. becomes $M(t)=(1-2t)^{-m/2}$, for $t<\frac12$. Show further that $\mathrm{E}\,X=m$, $\mathrm{Var}\,X=2m$, and that the skewness, i.e. $\mathrm{E}\,W^3$ with $W=(X-\mathrm{E}\,X)/(\mathrm{Var}\,X)^{1/2}=(X-m)/(2m)^{1/2}$, is $(8/m)^{1/2}$. From the density, show $\mathrm{E}\,(\chi^2_m)^p=2^p\Gamma(m/2+p)/\Gamma(m/2)$.

**(b)** Via the m.g.f., show the simple and basic convolution property for the chi-squared, that if $X_1,\ldots,X_n$ are independent and chi-squared distributed with degrees of freedom $m_1,\ldots,m_n$, then the sum $Z=\sum_{i=1}^n X_i$ is chi-squared too, with degrees of freedom $\sum_{i=1}^n m_i$. A generalisation is given in Ex. 1.53.

**(c)** If $N$ is standard normal, show that $X=N^2\sim\chi^2_1$. Establish that if $X\sim\chi^2_m$, with $m$ a natural number, then it may be represented as $X=N_1^2+\cdots+N_m^2$, in terms of independent standard normals $N_1,\ldots,N_m$. Note, however, that the $\chi^2_m$ with the density $g_m(x)$ above may be used also when $m$ is not a natural number.

**(d)** There are connections between the chi-squared and the Gamma distribution (see Ex. 1.9). Show that the $\mathrm{Gam}(\frac12 m,\frac12)$ is the $\chi^2_m$; that if $Y\sim\mathrm{Gam}(a,b)$, then $X=2bY\sim\chi^2_{2a}$; and that if $Z\sim\chi^2_m$, then $\frac12 Z\sim\mathrm{Gam}(\frac12 m,1)$.

**(e)** Consider independent $X \sim \chi^2_a$ and $Y \sim \chi^2_b$. Show, perhaps via Gamma distribution ratios in Ex. 1.12, that $R = X/(X+Y) \sim \text{Beta}(\frac{1}{2}a, \frac{1}{2}b)$.

**(f)** When $X \sim \chi^2_m$, show that $\text{E} \log X = \log 2 + \psi(\frac{1}{2}m)$, where $\psi(x) = \Gamma'(x)/\Gamma(x)$ is the digamma function.

**(g)** Consider $W = AB$, a product of independent standard normals. Use (1.2) to show that its m.g.f. is $M(t) = 1/(1-t^2)^{1/2}$, for $|t| < 1$. Deduce from this that $W$ has the same distribution as $\frac{1}{2}(K - L)$, where $K$ and $L$ are independent with $\chi^2_1$ distributions.

**(h)** Then consider $W = AB + CD$, where $A, B, C, D$ are independent standard normals. Show that $W$ has density $\frac{1}{2}\exp(-|w|)$, i.e. the Laplace distribution worked with in Ex. 1.34.

**(i)** Consider the second degree equation $x^2 + Bx + C = 0$ from school. If there are many such equations, with $B$ and $C$ being independent standard normal, how many of these equations will have both roots real?

**Ex. 1.49** *Tell me about $X$ and $X + Y$.* In exercises above we have seen that sums of independent normals, Poissons, chi-squares are respectively normal, Poisson, chi-square. Sometimes we meet questions going the other way.

**(a)** Suppose $X$ and $Y$ are independent, and that you learn the distributions of $X$ and $Z = X + Y$. Show that $Y$ must have m.g.f. $M_Y(t) = M_Z(t)/M_X(t)$. As an illustration, suppose $X \sim \text{N}(0,1)$ and $X + Y \sim \text{N}(0,2)$. Show that $Y \sim \text{N}(0,1)$.

**(b)** Explain similarly that if $X$ and $Y$ are independent, with $X$ and $X+Y$ being Poisson, then $Y$ is also Poisson.

**(c)** Suppose $X$ and $Y$ are independent, that $X \sim \chi^2_a$ and that $Z = X+Y \sim \chi^2_{a+b}$. Show that the only possibility is then the expected one, that $Y \sim \chi^2_b$.

**(d)** Consider $X_1, \ldots, X_n$ i.i.d. standard normal. Writing as usual $\bar{X}$ for the sample mean, we know that $\sqrt{n}\bar{X}$ is also standard normal. Show that the vector of $X_i - \bar{X}$ is independent of $\bar{X}$. Writing $Z_0 = \sum_{i=1}^n (X_i - \bar{X})^2$, show from $\sum_{i=1}^n X_i^2 = Z_0 + n\bar{X}^2$ that $Z_0 \sim \chi^2_{n-1}$. (xx calibrate with other things; this is giving a simpler proof of the lemma below, with no orthogonal transformations. xx)

**(e)** (xx for later, perhaps in Ch3: also nice for $p$-dimensional things with $Q(\theta) = \sum_{j=1}^k (y_j - \theta)^{\text{t}} \Sigma_j^{-1} (y_j - \theta)$, where $Q_{\min} = Q(\widehat{\theta}) \sim \chi^2_{(k-1)p}$, independent of $\widehat{\theta} = A^{-1} \sum_{j=1}^k \Sigma_j^{-1} y_j$. xx)

**Ex. 1.50** *Orthonormal transformations.* [xx check and calibrate with chi-squared things, to get the order right. restructure text. xx] We have seen in Ex. 1.44 that a multinormal vector can be sent via a linear transformation to independent one-dimensional normal components, and vice versa. This also leads to useful characterisation and representation theorems involving independence. In the present exercise we shall e.g. find a proof that the sample mean $\bar{Y}$ and the sample variance statistic $Z = \sum_{i=1}^n (Y_i - \bar{Y})^2$ are independent; this fact, which does not hold outside the normal family, is actively used in Ex. 3.4, and will also be utilised elsewhere, like in Ex. 3.6 xx).

**(a)** Suppose $X = (X_1, \ldots, X_n)$ is a vector with i.i.d. and standard normal components, and let $A$ be an orthonormal matrix, which means $AA^{\text{t}} = I = A^{\text{t}}A$. In yet other words, each row of $A$ and each column of $A$ has length 1, rows are ortogonal, as well as columns. Show that $Y = AX$ must have components $Y_1, \ldots, Y_n$ which are also i.i.d. and standard normal. – Here you may also use the general transformation formula of Ex. 1.12.

**(b)** To exemplify the above, show that if $X_1, X_2$ are independent and standard normal,

then also $Y_1, Y_2$, where

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} (X_1 + X_2)/\sqrt{2} \\ (X_1 - X_2)/\sqrt{2} \end{pmatrix},$$

must be independent and standard normal.

**(c)** When $A$ is orthonormal, show that it preserves length, so $\|Au\| = \|u\|$, for any vector $u$; here $\|u\|$ is Euclidean length, so $\|u\|^2 = u_1^2 + \cdots + u_n^2$.

**(d)** Let again $X_1, \ldots, X_n$ be i.i.d. standard normals. Construct an orthogonal matrix $A$ by letting its first row be $(1/\sqrt{n}, \ldots, 1/\sqrt{n})$, and define $Y = AX$. Then show that $Y_1 = \sqrt{n}\bar{X} = \sum_{i=1}^n X_i/\sqrt{n}$, and that $Z_0 = \sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=2}^n Y_i^2$.

**(e)** Conclude from this that (i) $\sqrt{n}\bar{X} \sim \mathrm{N}(0,1)$, (ii) $Z_0 \sim \chi^2_{n-1}$, and (iii) $\bar{X}$ and $Z_0$ are idependent. This was proven more directly in Ex. 1.49 above, via decompositions of chi-squares.

**(f)** Show that this implies the following classical and important properties, starting with an independent sample $Y_1, \ldots, Y_n$ from the $\mathrm{N}(\xi, \sigma^2)$: The statistics $\bar{Y}$ and $Z = \sum_{i=1}^n (Y_i - \bar{Y})^2$ are independent, with $\bar{Y} \sim \mathrm{N}(\xi, \sigma^2/n)$ and $Z \sim \sigma^2 \chi^2_{n-1}$. Show also from this that the classical empirical variance

$$\widehat{\sigma}^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \tag{1.6}$$

is unbiased for the population variance, i.e. $\mathrm{E}\,\widehat{\sigma}^2 = \sigma^2$. Construct an unbiased estimator for $\sigma$, of the type $c_n \widehat{\sigma}$. [xx point to generalisations for the linear regression model. xx]

**(g)** Consider the general multinormal distribution $Y \sim \mathrm{N}_p(\xi, \Sigma)$, with invertible $\Sigma$. Show that $K = (Y - \xi)^{\mathrm{t}} \Sigma^{-1} (Y - \xi) \sim \chi^2_p$. Suppose $Y = y_{\mathrm{obs}}$ is observed, that $\Sigma$ is known, but $\xi$ unknown. Give a confidence region $R$ such that $\xi \in R$ with probability 90 percent. How does this region shrink, if you observe 100 vectors from the multinormal, rather than merely 1?

**Ex. 1.51** *The t distribution.* Consider independent variables $X \sim \mathrm{N}(0,1)$ and $K \sim \chi^2_m$. The ratio $t = X/(K/m)^{1/2}$ is then said to have the t distribution, with $m$ degrees of freedom. We write $t \sim t_m$ to indicate this.

**(a)** Looking at the denominator first, show that $K/m$ has mean 1 and variance $2/m$, indicating that it will be close to 1 with high probability when $m$ is moderate or big. Explain that this implies that $t$ is not far from having a standard normal distribution for growing $m$. Finer details regarding this are in Ch. 2.

**(b)** Find the mean and variance of $t$.

**(c)** Show that its density can be written

$$g_m(x) = \frac{\Gamma((m+1)/2)}{\Gamma(m/2)} \frac{1}{\sqrt{m\pi}} \frac{1}{(1 + x^2/m)^{(m+1)/2}}.$$

Show that $g_m(x)$ tends to the standard normal density $\phi(x)$ as $m$ increases, matching the intuition above. Show also that the Cauchy distribution, see Ex. 1.18, is the $t_1$ distribution.

**(d)** Find also the skewness and kurtosis for the $t_m$ distribution. In particular, show that the latter is $\mathrm{kurt} = 6/(m-4)$ for $m > 4$.

**(e)** Assume $Y_1, \ldots, Y_n$ are i.i.d. $\mathrm{N}(\xi, \sigma^2)$. With $\widehat{\sigma}$ the empirical standard deviation, from (1.6), show that

$$t = (\bar{Y} - \xi)/(\widehat{\sigma}/\sqrt{n})$$

has the t-distribution with $n-1$ degrees of freedom. This is the classic t-statistic dating all the way back to Student (1908),

**Ex. 1.52** *The noncentral t distribution.* Consider $X \sim \mathrm{N}(\xi, 1)$, independent of $K \sim \chi^2_m$. The ratio $t = X/(K/m)^{1/2} = (\xi + \mathrm{N}(0,1))/(\chi^2_m/m)^{1/2}$ is then said to follow that noncentral t distribution, written $t \sim t_m(\xi)$, with degrees of freedom $m$ and noncentrality parameter $\xi$. Its distribution is a complex one, but both the density $g_m(t, \xi)$ and c.d.f. $G_m(t, \xi)$ are implemented in various software packages, as e.g. `dt(t,m,ksi)` and `pt(t,m,ksi)` in `R`.

**(a)** Explain that the classical t distribution of Ex. 1.51 corresponds to the special case of $\xi = 0$. Via arguments in that exercise, explain also that the $t_m(\xi)$ becomes close to a $\mathrm{N}(\xi, 1)$ for growing $m$.

**(b)** Find expressions for the mean and variance of the $t_m(\xi)$.

**(c)** Consider again an i.i.d. sample $Y_1, \ldots, Y_n$ from $\mathrm{N}(\xi, \sigma^2)$, with sample mean $\bar{Y}$ empirical standard deviation $\widehat{\sigma}$, as in Ex. 1.51. Show that $t = \sqrt{n}\bar{Y}/\widehat{\sigma}$ has the $t_{n-1}(\xi/\sigma)$ distribution. We shall need this for studying the power of the t test in Ex. 4.13.

**Ex. 1.53** *The noncentral chi-squared.* Consider also the so-called noncentral chi-squared distribution, say $K \sim \chi^2_m(\lambda)$, with $\lambda$ the excentre or eccentricity parameter; the case of $\lambda = 0$ corresponds to the ordinary $K \sim \chi^2_m$. It is the distribution of $Y_1^2 + \cdots + Y_m^2$, where the $Y_i$ are independent normals, with $Y_i \sim \mathrm{N}(\xi_i, 1)$, and $\lambda = \sum_{i=1}^m \xi_i^2$.

**(a)** Show that the m.g.f. of $K \sim \chi^2_m(\lambda)$ may be written

$$M(t) = \mathrm{E}\,\exp(tK) = \frac{\exp\{\lambda t/(1-2t)\}}{(1-2t)^{m/2}} \quad \text{for } t < \tfrac{1}{2}.$$

**(b)** Also the noncentral chi-squared distributions have convolution properties, generalising those of Ex. 1.48. If $K_i \sim \chi^2_{m_i}(\lambda_i)$, and these are independent, for $i = 1, \ldots, n$, show that $\sum_{i=1}^n K_i$ is another noncentral chi-squared, with degrees of freedom $\sum_{i=1}^n m_i$ and excentre parameter $\sum_{i=1}^n \lambda_i$.

**(c)** Another property which can be established, remarkbly without yet having seen the density of a $\chi^2_m(\lambda)$, is the following, using Ex. 1.49: if $X$ and $Y$ and independent, with $X \sim \chi^2_{m_1}(\lambda_1)$ and $X + Y \sim \chi^2_{m_1+m_2}(\lambda_1 + \lambda_2)$, then by necessity $Y \sim \chi^2_{m_2}(\lambda_2)$.

**(d)** Its density can be expressed in several ways; show that this is one such valid formula:

$$f(k, m, \lambda) = \sum_{j=0}^{\infty} \{\exp(-\tfrac{1}{2}\lambda)(\tfrac{1}{2}\lambda)^j/j!\}\, g_{m+2j}(k),$$

where $g_{m+2j}(k)$ is the $\chi^2_{m+2j}$ density. In other words, the noncentral chi-squared is a Poisson mixture of central chi-squared distributions. Show that this entails the representation $K \,|\, (J = j) \sim \chi^2_{m+2j}$, where $J \sim \mathrm{Pois}(\frac{1}{2}\lambda)$. Also non-integer values of $m$ are allowed here.

**(e)** Establish that for $K \sim \chi^2_m(\lambda)$, we have $\mathrm{E}\,K = m + \lambda$ and $\mathrm{Var}\,K = 2m + 4\lambda$. Show also that the skewness of $K$ becomes $2^{3/2}(m + 3\lambda)/(m + 2\lambda)^{3/2}$. What is required in order for this skewness to tend to zero?

**(f)** With $N$ a standard normal, let $K = (\lambda^{1/2} + N)^2$, which has the $\chi^2_1(\lambda)$ distribution. Consider the normalised variable

$$\frac{K - (1+\lambda)}{(2+4\lambda)^{1/2}} = \frac{N^2 + 2\lambda^{1/2}N - 1}{(2+4\lambda)^{1/2}}.$$

Work out its m.g.f. and show that it tends to $\exp(\frac{1}{2}t^2)$ for growing $\lambda$.

**(g)** More generally, with $K \sim \chi^2_m(\lambda)$, work out a formula for the m.g.f. $M(t)$ for $Z = \{K - (m+\lambda)\}/(2m+4\lambda)^{1/2}$. For any fixed $m$, show that $M(t) \to \exp(\frac{1}{2}t^2)$ as $\lambda$ grows, and comment on this finding.

**Ex. 1.54** *Noncentral chi-squared for empirical variances.* We saw in Ex. 1.50 that if $X_1, \ldots, X_n$ are i.i.d. $\mathrm{N}(a, 1)$, with common $a$, then $Z = \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi^2_{n-1}$, with consequences for the empirical variance estimator. Here are some fruitful generalisations.

**(a)** Let the $X_i$ have non-identical means, $X_i \sim \mathrm{N}(a_i, 1)$. Show that $Z \sim \chi^2_{n-1}(\lambda)$, with noncentrality parameter $\lambda = \sum_{i=1}^n (a_i - \bar{a})^2$.

**(b)** Assume now that $X_i \sim \mathrm{N}(a_i, 1/m_i)$ for $i = 1, \ldots, n$, perhaps reflecing sample sizes $m_i$ for different groups, and with $M = \sum_{i=1}^n m_i$. Consider $Z = \sum_{i=1}^n m_i (X_i - \widetilde{X})^2$, with $\widetilde{X} = \sum_{i=1}^n (m_i/M) X_i$. Show that $Z = \sum_{i=1}^n m_i X_i^2 - M\widetilde{X}^2$, and that its distribution is a $\chi^2_{n-1}(\lambda)$, with $\lambda = \sum_{i=1}^n m_i (a_i - \widetilde{a})^2$, where $\widetilde{a} = \sum_{i=1}^n (m_i/M) a_i$.

**(c)** Let $X \sim \mathrm{N}_p(\xi, \Sigma)$. Show that $Z = X^{\mathrm{t}}\Sigma^{-1}X \sim \chi^2_p(\xi^{\mathrm{t}}\Sigma^{-1}\xi)$.

**Ex. 1.55** *The F distribution.* As we have seen Ex. 1.50, with a normal sample $X_1, \ldots, X_n$, the distribution of the classical empirical variance estimator $\widehat{\sigma}^2 = (n-1)^{-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is governed by $\widehat{\sigma}^2 \sim \sigma^2 \chi^2_m / m$, where $m = n - 1$ is the degrees of freedom. Suppose there are two independent samples, from normal distributions $\mathrm{N}(\xi_1, \sigma_1^2)$ and $\mathrm{N}(\xi_2, \sigma_2^2)$, of sample sizes $n_1$ and $n_2$, with estimators $\widehat{\sigma}_1^2$ and $\widehat{\sigma}_2^2$.

**(a)** Let $\rho = \sigma_1/\sigma_2$, the ratio of standard deviations. For the ratio of the two empirical variances, show that

$$R^2 = \frac{\widehat{\sigma}_1^2}{\widehat{\sigma}_2^2} = \rho^2 F, \quad \text{where } F \sim \frac{\chi^2_{m_1}/m_1}{\chi^2_{m_2}/m_2},$$

with degrees of freedom $m_1 = n_1 - 1$ and $m_2 = n_2 - 1$, and with the two chi-squareds being independent. We say that $F$ has the F distribution, or Fisher distribution, with degrees of freedom $(m_1, m_2)$, and write $F \sim F(m_1, m_2)$.

**(b)** Show that when $F \sim F(m_1, m_2)$, then $1/F \sim F(m_2, m_1)$. Show furthermore that

$$\mathrm{E}\, F = \frac{m_2}{m_2 - 2}, \quad \mathrm{E}\, F^2 = \frac{m_1 + 2}{m_1} \frac{m_2^2}{(m_2 - 2)(m_2 - 4)},$$

these expressions being finite when $m_2 > 2$ and $m_2 > 4$, respectively. Find also an expression for the variance. Verify that both $\mathrm{E}\, F$ and $\mathrm{E}\, F^2$ tend to 1 as the degrees of freedom increase.

**(c)** The main aspects of the F distribution have been worked out, above, from the constructive definition given in (a), without actually needing any formula for its density; also, probabilities are found using software packages, like `pf(x,m1,m2)` in `R`. Once in a while one needs the density function, however. Show first that the cumulative function can be written

$$\Pr(F \le x) = \Pr(\chi^2_{m_1}/m_1 \le x\chi^2_{m_2}/m_2) = \int_0^\infty G(xy(m_1/m_2), m_1) g(y, m_2)\, \mathrm{d}y,$$

in terms of the cumulative $G(\cdot, m)$ and densiy $g(\cdot, m)$ of the $\chi^2_m$. Then take the derivative to get

$$h(x, m_1, m_2) = \int_0^\infty g(xy(m_1/m_2), m_1) y (m_1/m_2) g(y, m_2)\, \mathrm{d}y.$$

Complete the math to land at

$$h(x, m_1, m_2) = \frac{\Gamma(\frac{1}{2}(m_1+m_2))}{\Gamma(\frac{1}{2}m_1)\Gamma(\frac{1}{2}m_2)} (m_1/m_2)^{m_1/2} \frac{x^{m_1/2-1}}{\{1+(m_1/m_2)x\}^{(m_1+m_2)/2}}.$$

### The exponential family class

**Ex. 1.56** *The exponential family class.* Many parametric models fall under the wide umbrella of the exponential family class, which we treat in this and the folllowing exercises. This will be properly generalised and extended down the road, but we start with this definition: Suppose $Y$ has model density of the form

exponential family

$$\begin{aligned} f(y, \theta) &= \exp\{\theta_1 T_1(y) + \cdots \theta_p T_p(y) - k(\theta_1, \ldots, \theta_p)\} h(y) \\ &= \exp\{\theta^{\mathrm{t}} T(y) - k(\theta)\} h(y), \end{aligned} \tag{1.7}$$

for appropriate functions $T_1(y), \ldots, T_p(y)$ and $h(y)$; the $k(\theta)$ function is there to secure integration to one. We require that there is no fixed linear relationship among $T_1(y), \ldots, T_p(y)$, and also that the support of the distribution, the smallest closed set having probability 1, is the same, for all parameter values. We then say $Y$ is of the exponential family class, with *data functions* $T(y) = (T_1(y), \ldots, T_p(y))^{\mathrm{t}}$ and *natural parameters* $\theta = (\theta_1, \ldots, \theta_p)^{\mathrm{t}}$.

**(a)** Before we start developing the general theory for the full class, we verify that a few classic models are under its umbrella. For the following models, write the model density in a form matching (1.7). (i) $Y \sim \text{binom}(n, p)$. (ii) $Y \sim \text{Pois}(\theta)$. (iii) $Y \sim \text{Beta}(a, b)$. (iv) $Y \sim \text{Gam}(a, b)$. (v) $Y \sim \text{N}(\xi, \sigma^2)$, with known $\sigma$ (and see Ex. 1.57). (vi) Let $(X, Y, Z)$ be trinomial $(n, p, q, r)$, with $r = 1 - p - q$. First work with $f(x, y, p, q)$, and then with the submodel where $p = a^2$, $q = 2a(1-a)$, $r = (1-a)^2$.

**(b)** Show that we must have

$$k(\theta) = \log\Big( \int \exp\{\theta^{\mathrm{t}} T(y)\} h(y)\, \mathrm{d}\nu(y) \Big),$$

assumed to be finite for at least some $\theta$, and with $\mathrm{d}\nu(y)$ the dominating measure (often enough simply $\mathrm{d}y$, for models with continuous densities). Let in fact $\Theta$ be the set of $\theta$ such that $k(\theta)$ is finite, called the natural parameter region. Show that $\Theta$ is a convex set.

natural parameter region

**(c)** The score function of a model is $u(y, \theta) = \partial \log f(y, \theta)/\partial\theta$; see Ex. 5.6 for more on this. For the class of models studied here, show that $u(y, \theta) = T(y) - \xi(\theta)$, where

$$\xi(\theta) = \frac{\partial k(\theta)}{\partial \theta} = \frac{\int T(y) \exp\{\theta^{\mathrm{t}} T(y)\} h(y)\, \mathrm{d}y}{\int \exp\{\theta^{\mathrm{t}} T(y)\} h(y)\, \mathrm{d}y}.$$

Show that the score function must have mean zero, which here means $\mathrm{E}_\theta\, T(Y) = \xi(\theta)$. Show also that $\mathrm{Var}_\theta\, T(Y) = \partial^2 k(\theta)/\partial\theta\partial\theta^{\mathrm{t}}$, giving variances and covariances of the $T_j(Y)$ in one matrix formula.

**(d)** We come back to general likelihood theory in Ch. 5, but for now show that if $Y_1, \ldots, Y_n$ are i.i.d. from the exponential family density, then the logarithm of the joint density can be written

$$\ell_n(\theta) = \sum_{i=1}^n \log f(Y_i, \theta) = n\{\theta^{\rm t}\bar{T} - k(\theta)\},$$

with $\bar{T} = (1/n)\sum_{i=1}^n T(Y_i)$ the vector of averages $\bar{T}_j = (1/n)\sum_{i=1}^n T_j(y_i)$. If this is a family with say $p = 3$ parameters, and $n = 10000$, then the full relevant information required for computing the $\ell_n$ function is captured in the 3 averages $\bar{T}_1, \bar{T}_2, \bar{T}_3$. This is related to the sufficiency concept, returned to in Ch. 4. Show also that this $\ell_n(\theta)$ is a concave function.

**Ex. 1.57** *The normal and binormal exponential family members.* The normal and binormal (and multinormal) classes of distributions belong under the exponential family umbrella.

**(a)** Consider first the normal $(\xi, \sigma^2)$ model. Explain that its density may be written

$$f(y) = \frac{1}{(2\pi)^{1/2}} \exp\Big(-\tfrac{1}{2}\frac{y^2}{\sigma^2} + \frac{\xi y}{\sigma^2} - \tfrac{1}{2}\frac{\xi^2}{\sigma^2}\Big),$$

and argue from this that the normal is of exponential family type, with natural parameters $\xi/\sigma^2$ and $1/\sigma^2$, and data functions equivalent to $(y, y^2)$.

**(b)** Consider then $(X, Y)$ binormal, with its five parameters $\xi_1, \xi_2, \sigma_1, \sigma_2, \rho$, see Ex. 1.44. Show that the distribution is of the exponential family type, with natural parameters

$$\frac{\xi_1}{(1-\rho^2)\sigma_1^2}, \frac{1}{(1-\rho^2)\sigma_1^2}, \frac{\xi_2}{(1-\rho^2)\sigma_2^2}, \frac{1}{(1-\rho^2)\sigma_2^2}, \frac{\rho}{(1-\rho^2)\sigma_1\sigma_2}.$$

In the terminology of exponential families, identify also the associated $T_1(x, y), \ldots, T_5(x, y)$.

**(c)** Show that $(\xi_1, \xi_2, \sigma_1, \sigma_2, \rho)$ is in one-to-one correspondence with the five exponential class natural parameters above.

**(d)** Suppose now that $\xi_1 = \xi_2$ and $\sigma_1 = \sigma_2$, so this distribution for $(X, Y)$ has now three parameters. Show that it is inside the exponential family, and identify its natural parameters.

**Ex. 1.58** *A log-linear density model on the unit interval.* The machinery of the exponential family class makes it easy to construct new models, starting with relevant data functions $T_j$, in the language of Ex. 1.56.

**(a)** For densities on the unit interval, start with data function $T(y) = y - \frac{1}{2}$, to define $f(y, \theta) = \exp\{\theta T(y) - k(\theta)\}$ for $y \in [0, 1]$. Find the $k(\theta)$, a formula for the c.d.f. $F(y, \theta)$, and the mean and variance for $Y$ having this density.

**(b)** Then go on to

$$f(y, \theta_1, \theta_2) = \exp\{\theta_1(y - \tfrac{1}{2}) + \theta_2(y - \tfrac{1}{2})^2 - k(\theta_1, \theta_2)\},$$

i.e. using $T_j(y) = (y - \frac{1}{2})^j$ as data functions, for $j = 1, 2$. Set up an integral function for $k(\theta_1, \theta_2)$ and some algorithms for determining the means, variances, and covariance, for $T_1(Y), T_2(Y)$, for any given $(\theta_1, \theta_2)$.

**(c)** Spell out the necessary details for the third order log-linear density model, which uses $T_j(y) = (y - \frac{1}{2})^j$ for $j = 1, 2, 3$. Again, set up algorithms so that you may compute the means, variances, covariances for $T_1, T_2, T_3$, for any $(\theta_1, \theta_2, \theta_3)$.

## Yet other models: log-normal, Weibull, Gompertz, Gumbel, et al.

the log-normal distribution

**Ex. 1.59** *The log-normal distribution.* Starting with $X \sim \mathrm{N}(\xi, \sigma^2)$, the variable $Y = \exp(X)$ is said to be a *log-normal*, and we write $Y \sim \mathrm{logN}(\xi, \sigma^2)$ to indicate this.

**(a)** Consider the view that the distribution should or could have been named the exponormal instead – would you agree? Show that with $Y \sim \mathrm{logN}(\xi, \sigma^2)$, its mean and variance are $\exp(\xi + \frac{1}{2}\sigma^2)$ and $\{\exp(2\sigma^2) - \exp(\sigma^2)\}\exp(2\xi)$.

**(b)** Show that its density may be written $\phi_\sigma(\log y - \xi)/y = \sigma^{-1}\phi(\sigma^{-1}(\log y - \xi))/y$, for $y > 0$. Find its mode.

**(c)** Assume that $Y \,|\, \xi \sim \mathrm{logN}(\xi, \sigma^2)$, and that $\xi \sim \mathrm{N}(\xi_0, \tau^2)$. Show that marginally, $Y \sim \mathrm{logN}(\xi_0, \sigma^2 + \tau^2)$. Make explicit the connection to Ex. 1.65.

**(d)** Show that a product of independent log-normals is log-normal. Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the $\mathrm{logN}(\xi, \sigma^2)$ distribution. Explain what happens to their harmonic mean, $Z_n = (Y_1 \cdots Y_n)^{1/n}$.

**(e)** Assume a random time variable $T$ has the $\mathrm{logN}(0, 1)$ distribution. Find a formula for its hazard rate $h(t)$, and show that $h(t) \doteq (\log t)/t$ for growing $t$. Plot the exact hazard rate, along with its approximation, and comment.

**Ex. 1.60** *The Weibull distribution.* The Weibull distribution, with positive parameters $(a, b)$, has c.d.f. $F(t) = 1 - \exp\{-(t/a)^b\}$ for $t \geq 0$. The $b$ is called the shape parameter, with $a$ a scale parameter. The Weibull generalises the exponential distribution, which is the special case of $b = 1$. Other parametrisations are sometimes convenient, as with $1 - \exp(-ct^b)$.

Ernst Hjalmar Waloddi Weibull

**(a)** Find a formula for the median, and more generally for the $q$-quantile $F^{-1}(q)$. Show that the density can be written $f(t) = \exp\{-(t/a)^b\}bt^{b-1}/a^b$ for $t > 0$. Find also a formula for the hazard rate, and draw this in a diagram, for $b = 0.9, 1.0, 1.1$, say for $a = 1$.

**(b)** Use the general mean formula of (1.3) to work through the details of

$$\mathrm{E}\, T^p = \int_0^\infty \Pr(T^p \geq u)\, \mathrm{d}u = \int_0^\infty \exp\{-(u^{1/p}/a)^b\}\, \mathrm{d}u = a^p\, \Gamma(1 + p/b).$$

Show that this leads to mean $a\,\Gamma(1 + 1/b)$ and variance $a^2\,\{\Gamma(1 + 2/b) - \Gamma(1 + 1/b)\}^2$. With $T$ from the Weibull $(a, b)$, plot the function $\mathrm{sd}(T)/\mathrm{E}\, T$ as a function of $b$. (xx write out. also reparametrisation, with $1 - \exp(-ct^b)$. point to story. xx)

**(c)** Show that $V = (T/a)^b \sim \mathrm{Expo}(1)$, and use this to give a recipe for simulating outcomes from any Weibull.

**Ex. 1.61** *The Gompertz distribution.* The Gompertz distribution, with positiver parameters $(a, b)$, has hazard rate $h(t) = a\exp(bt)$.

**(a)** Find the cumulative hazard rate, the c.d.f., and the density. Find also a formula for the median, and more generally for the $q$ quantile, expressed via $(a, b)$. (xx pointer to Story #37. more; round off. xx)

**(b)** Suppose an individual has survived up to time $t_0$. Show that her cumulative hazard rate, for the remaining lifetime, is $H(t) - H(t_0) = (a/b)\{\exp(bt) - \exp(bt_0)\}$. Give a formula for $t^*(t_0)$, her median survival time. (xx then brief application of this, for Norwegian women, using perhaps rough estimates of $(a, b)$, via data from Human Mortality Index. give $(t_0, t^*(t_0))$ as a graph, for women born in perhaps 1900, 1960, 2020. can also be a Story. xx)

**Ex. 1.62** *The Gumbel distribution.* Here we work with the Gumbel distribution, useful e.g. in models for extreme values. the Gumbel distribution

**(a)** Let $X_1, \ldots, X_n$ be i.i.d. from the standard exponential distribution. Show that their maximum value $M_n$ has c.d.f. $\{1 - \exp(-m)\}^n$. Deduce that $M_n - \log n$ has c.d.f. $G_n(u) = \{1 - (1/n)\exp(-u)\}^n$, for all $u \geq -\log n$.

**(b)** Show that the limit c.d.f. for $M_n - \log n$ becomes $G(u) = \exp\{-\exp(-u)\}$, and that this defines a c.d.f. on the full line. This is called the *Gumbel distribution*. Find its median and interquartile range.

**(c)** Find its density $g(u) = \exp\{-u - \exp(-u)\}$, and draw it in a diagram, along with the densities $g_n$ for say $n = 10, 20, 30$, for $M_n - \log n$.

**(d)** With $U$ having the Gumbel distribution, show that its m.g.f. becomes $M(t) = \Gamma(1-t)$, for $t < 1$. To find the moments, show by taking derivatives of $\Gamma(a) = \int_0^\infty v^{a-1}\exp(-v)\,\mathrm{d}v$ at position $a = 1$ that

$$\mathrm{E}\,U^p = (-1)^p\Gamma^{(p)}(1) = (-1)^p\int_0^\infty (\log v)^p \exp(-v)\,\mathrm{d}v.$$

These may be found numerically. You may also work with the cumulant-generating function, where a certain connection can be made to the Riemann zeta function, namely

$$K(t) = \log\Gamma(1-t) = \gamma_e t + \sum_{j=2}^\infty \frac{\zeta(j)}{j}t^j$$

for $|t| < 1$. Here $\gamma_e = 0.5772...$ is the Euler constant, see Ex. 1.15, and $\zeta(j) = \sum_{n=1}^\infty 1/n^j$. In the terminology of Ex. 1.38, show that $\kappa_1 = \gamma_e$, $\kappa_2 = \zeta(2)$, $\kappa_3 = 2\,\zeta(3)$, $\kappa_4 = 6\,\zeta(4)$. With $\zeta(2) = \pi^2/6$, $\zeta(3) = \pi^3/25.7943$, $\zeta(4) = \pi^4/90$, establish that $\mathrm{E}\,U = \gamma_e$, $\mathrm{Var}\,U = \sigma^2 = \pi^2/6$, $\mathrm{skew}(U) = 2\,\zeta(3)/\sigma^3 \doteq 1.1395$, $\mathrm{kurt}(U) = 12/5 = 2.4$.

**(e)** To appreciate the perhaps strange-looking connection from Gumbel moments to the zeta function, show from efforts of Ex. 1.15 that

$$M_n - \log n = \sum_{i=1}^n W_i/i + a_n$$

where $W_i = V_i - 1$, for i.i.d. unit exponentials $V_i$, and $a_n \to \gamma_e$. Find formulae for the mean, variance, skewness, kurtosis of $M_n - \log n$ based on this, and take their limits; these will agree with formulae found above.

**Ex. 1.63** *The logistic distribution.* Consider the logistic distribution (in its standard form), with c.d.f. $H(x) = \exp(x)/\{1 + \exp(x)\}$, over the real line. the logistic distribution

**(a)** Show that $H$ indeed is a proper c.d.f., and that its density is $h(x) = \exp(x)/\{1 + \exp(x)\}^2 = H(x)\{1 - H(x)\}$, symmetric around zero. Find its interquartile range.

**(b)** With $X$ having the logistic distribution, show that its m.g.f. $M(t)$ becomes

$$\int \frac{\exp\{(t+1)x\}}{\{1+\exp(x)\}^2}\,\mathrm{d}x = \int_0^\infty \frac{u^t}{(1+u)^2}\,\mathrm{d}u = \int_0^1 v^t(1-v)^{1-t}\,\mathrm{d}t = \Gamma(1+t)\Gamma(1-t),$$

for $|t| < 1$. Show from this that actually $X = U - V$, in terms of independent Gumbel distributed $U$ and $V$, see Ex. 1.62. Demonstrate this fact also directly, via convolution. Use this representation to show that $\mathrm{Var}\,X = \pi^2/3$ and that its kurtosis is $\mathrm{kurt} = 6/5 = 1.2$.

**(c)** Here we managed to find moments of the logistic via the Gumbel difference representation. We may also use the intriguing formula $\Gamma(1+t)\Gamma(1-t) = \pi t/\sin(\pi t)$, a consequence of Euler's reflection formula. Take derivatives of this function to find the variance and kurtosis formulae.

**(d)** A scaled version of the logistic has c.d.f. $H(x,\tau) = H(x/\tau) = \exp(x/\tau)/\{1+\exp(x/\tau)\}^2$, for a suitable positive $\tau$. Show that the variance is $\tau^2\pi^2/3$, which is 1 for $\tau = \sqrt{3}/\pi$. Draw a figure with the density, alongside the standard normal, and comment.

**Ex. 1.64** *The logistic-normal distribution on the unit interval.* The most prominent model for distributions on the unit interval is the Beta, also useful in various guises and roles outside its direct use for fitting datasets; see Ex. 1.19. Here we work with a different class, useful for versions of logistic regression, which we meet in Ch. 5. Let $H(u) = \exp(u)/\{1+\exp(u)\}$ the logistic transform, worked with in Ex. 1.63, taking any real $u$ to the unit interval.

**(a)** Solve $H(u) = v$ to find the inverse transform $H^{-1}(v) = \log\{v/(1-v)\}$. Start with $X$ a standard normal and define $V = H(X)$. Show that its c.d.f. is $G(v) = \Pr(X \le H^{-1}(v)) = \Phi((H^{-1}(v)))$, with density

$$g(v) = \phi(H^{-1}(v))\frac{1}{v(1-v)} = \frac{1}{(2\pi)^{1/2}}\exp[-\tfrac{1}{2}\{\log v - \log(1-v)\}^2]\frac{1}{v(1-v)}.$$

**(b)** Generalise to the case where $X \sim \mathrm{N}(\xi,\sigma^2)$, where the density becomes

$$g(v,\xi,\sigma) = \phi((H^{-1}(v)-\xi)/\sigma)\frac{1}{\sigma}\frac{1}{v(1-v)}.$$

We call this the logistic-normal with parameters $(\xi,\sigma)$. Draw c.d.f.s and densities of this type, for some combinations of parameters. Explain that if $X_1,\ldots,X_p$ has a joint multinormal distribution, then the random probability

$$p(X_1,\ldots,X_p) = H(\beta_0+\beta_1X_1+\cdots+\beta_pX_p)$$

has the logistic-normal distribution, and identify its parameters. In this sense the logistic-normal class is closed under linear combinations of the underlying $H^{-1}(V)$, whereas the Beta class does not have similar properties.

**(c)** Suppose $Y \mid p$ is binom$(n,p)$, and that $p$ is logistic-normal. There are no closed-form expressions for the resulting probabilities

$$\bar f(y) = \binom{n}{y}\int_0^1 p^y(1-p)^{n-y}g(p,\xi,\sigma)\,\mathrm{d}p \quad \text{for } y = 0,1,\ldots,n,$$

but it may be worked with numerically. Display a few of these $\bar f(y)$, for say $n = 100$ and a few values of $(\xi,\sigma)$.

**(d)** We may also work the other way here: suppose $V = H(X)$ is Beta$(cp_0, c(1-p_0))$; what is then the distribution for $X$?

**Ex. 1.65** *A normal with a normal mean is normal.* (xx preliminary version; need just a few editorial decisions regarding where to place it, and how. xx) The normal distribution has a convenient coherence type property: if $X$ given its mean parameter is normal, and this mean parameter itself is normal, then $X$, marginally, is again normal. This is also related to what is found in Ex. 1.45.

**(a)** Consider first independent $X_1$ and $X_2$ with densities $f_1$ and $f_2$. Show that $\int f_1 f_2 \, \mathrm{d}x$ is the density of $X_1 - X_2$, evaluated at zero. Write then $\phi_\sigma(x-\xi) = \sigma^{-1}\phi(\sigma^{-1}(x-a))$ for the $\mathrm{N}(\xi,\sigma^2)$ density. Show that

$$\int \phi_{\sigma_1}(x-\xi_1)\phi_{\sigma_2}(x-\xi_2)\,\mathrm{d}x = \phi_{(\sigma_1^2+\sigma_2^2)^{1/2}}(\xi_1-\xi_2).$$

**(b)** Assume that $X$ given $\xi$ has mean $\xi$ and variance $\sigma^2$, and further that $\xi$ stems from its own distribution, with mean $\xi_0$ and variance $\tau^2$. Show that $X$, marginally, has mean $\xi_0$ and variance $\sigma^2+\tau^2$. Then specialise to the normal case, with $X \,|\, \xi$ and $\xi$ having normal distributions. Show that $X$ indeed also is normal, (i) by integrating out the $\xi$, with respect to its distribution, and also (ii) by arguing via $X = \xi + \varepsilon = \xi_0 + \delta + \varepsilon$, where $\varepsilon$ and $\delta$ are zero-mean normals with variances $\sigma^2$ and $\varepsilon^2$. (xx point to connected thing for logN. xx)

**(c)** Suppose $Y \,|\, (x_1,x_2)$ is normal $\mathrm{N}(a+b_1x_1+b_2x_2,\sigma^2)$, as in linear regression models we will study in later chapters; see e.g. Ex. 3.34. Assume then that $(x_1,x_2)$ themselves have a distribution, in its space of covariate pairs, and that this distribution is binormal. Show that $Y$, marginally, is normal, and give formulae for its mean and variance.

**Ex. 1.66** *Mixing the normal scale.* (xx at the moment nils thinks this exercise will go away, partly with material in Story #73 and partly elsewhere. point back to and calibrate with Ex. 1.34. xx) Suppose $X \sim \mathrm{N}(\xi,\sigma^2)$ for given parameters $(\xi,\sigma)$, but that there are background mechanisms producing these $(\xi,\sigma)$. In various settings this leads to good 'mixtures of normals' models for actually observed data.

**(a)** Suppose a given individual has his $\xi$ and that his associated $X$ is a $\mathrm{N}(\xi,\sigma^2)$. Assume next that in a population of such $X$, there is a distribution $\xi \sim \mathrm{N}(\xi_0,\sigma^2_{\mathrm{extra}})$ of their means. Show that an $X$ sampled from that population is a $\mathrm{N}(\xi_0,\sigma^2+\sigma^2_{\mathrm{extra}})$. From a statistical modelling viewpoint we have simply 'put in more in the $\sigma$', perhaps stretched its interpretation a little, without inventing or having to invent a new model for the observed $X$, per se. Also, without knowing more, or perhaps having a separate experiment, we cannot identify the components of the observed variance.

**(b)** Now turn attention to the scale. With the mean $\xi$ kept fixed, but $\sigma$ having some density $\pi(\sigma)$, show that $X$ has density $\bar f(x) = \int (1/\sigma)\phi((x-\xi)/\sigma)\pi(\sigma)\,\mathrm{d}\sigma$, and the variance of $X$ is the mean of the distribution of $\sigma^2$. Assume for simplicity of presentation that $\xi = 0$, and work with the case where the distribution of $\sigma$ is such that $1/\sigma^2 \sim \mathrm{Gam}(a,b)$ (it is common to express this by saying that $\sigma^2$ has an inverse gamma distribution). Work out that

$$\bar f(x) = \frac{1}{(2\pi)^{1/2}} \frac{b^a}{\Gamma(a)} \frac{\Gamma(a+\frac12)}{(b+\frac12 x^2)^{a+1/2}}.$$

It is useful to transform this to a member of the well-known distributions, to facilitate computations of probabilities etc. Show therefore that

$$V = (\tfrac12 X^2/b)/(1+\tfrac12 X^2/b) \sim \mathrm{Beta}(\tfrac12, a),$$

and express the c.d.f. of $X$ in terms of the c.d.f. of this Beta distribution: $\Pr(|X| \le x) = \mathrm{Be}((\frac12 x^2/b)/(1+\frac12 x^2/b), \frac12, a)$, with $\mathrm{Be}(\cdot,a,b)$ the c.d.f. of a $\mathrm{Beta}(a,b)$. Check these details via simulations.

**Ex. 1.67** *Normal mixtures.* [xx to come. mean, variance, skewness. xx] Suppose $Y$ is such that with probability $p_j$, it is a normal $(\xi_j,\sigma_j^2)$, with probabilities $p_1,\ldots,p_k$ summing

to 1. Its density may be written $f(y) = \sum_{j=1}^k p_j \phi_{\sigma_j}(y - \xi_j)$, where $\phi_\sigma(u) = \sigma^{-1}\phi(\sigma^{-1}u)$ is the density of a $\mathrm{N}(0, \sigma^2)$. Such distributions are called normal mixtures.

**(a)** With $J$ taking values $1, \ldots, k$, with probabilities $p_1, \ldots, p_k$, let $Y \,|\, (J = j) \sim \mathrm{N}(\xi_j, \sigma_j^2)$. Show that this $Y$ has the density above; this amounts to a way of representing and interpreting a normal mixture.

**(b)** From $\mathrm{E}\,(Y \,|\, J) = \xi_J$ and $\mathrm{Var}\,(Y \,|\, J) = \sigma_J^2$, show that

$$\mathrm{E}\,Y = \bar\xi = \sum_{j=1}^k p_j \xi_j, \quad \mathrm{Var}\,Y = \mathrm{E}\,(Y - \bar\mu)^2 = \sum_{j=1}^k p_j \sigma_j^2 + \sum_{j=1}^k p_j (\xi_j - \bar\xi)^2.$$

**(c)** (xx someting; display a few. xx)

**Ex. 1.68** *The hypergeometric distribution.* You draw a sample of $n$ items from a bag of $N$, which has $A$ of Type One and $B = N - A$ of type Two. Consider $X$, the number among the sampled $n$ which are of Type One.

**(a)** Show that $X$ has distribution

$$f(x) = \Pr(X = x) = \binom{A}{x}\binom{B}{n-x}\Big/\binom{N}{n}.$$

For which $x$ is this positive? Explain the identity $\sum_{x=0}^A \binom{A}{x}\binom{B}{n-x} = \binom{A+B}{n}$.

**(b)** Show that $\mathrm{E}\,X = nA/N = np$, with $p = A/N$ the proportion of Type One in the bag. This may be done work with $\sum_{x=0}^n x f(x)$, or by writing $X = J_1 + \cdots + J_n$, with $J_i$ and indicator for selected item $i$ being a Type One or not.

**(c)** Explain that if one samples one item at the time, followed by replacing the item, then the $J_1, \ldots, J_n$ above are independent Bernoulli variables with probability $p = A/N$, leading in that case to binomial variance $np(1-p)$. For the present hypergeometric setting, where $n$ items sampled in one go without replacement, the $J_i$ are dependent; show that $\mathrm{cov}(J_1, J_2) = p(A-1)/(N-1) - p^2 = -p(1-p)/(N-1)$. Deduce that the variance formula becomes $\mathrm{Var}\,X = c_n np(1-p)$, with $c_n$ being the shrinking factor $(N-n)/(N-1)$. This may be accomplished working algebraically with $\mathrm{E}\,X(X-1)$, or via the representation above.

**(d)** (xx just a bit more. comparing with binomial. xx)

a copula, coupling the binormal with other distributions

**Ex. 1.69** *Building bivariate dependence models.* (xx something here. need to have something beside the multinormal for dependence. xx) Let $X$ and $Y$ have densities $f_1(x)$ and $f_2(y)$, with c.d.f.s $F_1(x)$ and $F_2(y)$. To model dependence between them, the idea pursued here is mapping them to the normal scale, then using the binormal model, and mapping back.

**(a)** Write $X = F_1^{-1}(\Phi(U))$ and $Y = F_2^{-1}(\Phi(V))$, where $(U, V)$ is binormal with zero means, unit variances, and correlation $\rho$. Show that $X$ and $Y$ indeed have densities $f_1$ and $f_2$. Writing $g_\rho(u, v)$ for the binormal density, show that the joint density can be written

$$f(x, y) = f_1(x) f_2(y) R_\rho(\Phi^{-1}(F_1(x)), \Phi^{-1}(\Phi(y))), \quad \text{with } R_\rho(u, v) = \frac{g_\rho(u, v)}{\phi(u)\phi(v)}.$$

with the dependence factor $R_\rho$ of course being 1 when $\rho = 0$. Show in fact that

$$R_\rho(u, v) \,=\, \frac{1}{(1-\rho^2)^{1/2}} \exp\Big\{-\tfrac{1}{2}\frac{1}{1-\rho^2}(\rho^2 u^2 + \rho^2 v^2 - 2\rho uv)\Big\}.$$

**(b)** Now consider $f_1(x) = \exp(-x)$ and $f_2(y) = \exp(-y)$, i.e. two unit exponentials. Construct a bivariare pair $(X, Y)$ via the recipe above, with a density $f_\rho(x, y)$ having unit exponential marginals. Compute the correlations $\text{corr}(X, Y)$ as a function of $\rho$, which might be easiest via simulations. (xx simulate $(X, Y)$ for some values of $\rho$. note very different cases, $\rho = -0.99$ and $\rho = 0.99$. the range of correlations is $(-0.64, 1)$. the density formula is this. xx)

$$f(x, y) = \exp(-x - y) R_\rho\big(\Phi^{-1}(1 - \exp(-x)), \Phi^{-1}(1 - \exp(-y))\big).$$

**(c)** (xx one more example. for the uniform case, we have $\text{corr}(X, Y)$ quite close to $\rho$, but they remain different. xx)

**Ex. 1.70** *Alice and Bob correlate their binomials.* Here we show how correlated binomials may be constructed, leading also to correlated random walks.

**(a)** Alice flips her fair coin $n$ times, with i.i.d. 0-1 outcomes $A_1, \ldots, A_n$. Bob has two coins, and mixes between them depending on Alice's outcomes: if $A_i = 1$, he uses the plus-coin with probability $\frac{1}{2} + a$ for heads; and if $A_i = 0$ he uses his minus-coin with $\frac{1}{2} - a$ for heads. With $B_i$ his outcome, show that $\Pr(B_i = 1) = \frac{1}{2}$, and argue therefore that both $X_n$ and $Y_n$ are binomial $(n, \frac{1}{2})$ variables, where $X_n = \sum_{i=1}^n A_i$ and $Y_n = \sum_{i=1}^n B_i$ are the number of heads for Alice and for Bob. Show that the correlation between these two binomials is $2a$.

**(b)** In the little story-telling above, Bob observes Alice's outcomes, one by one, which then influence his choice between two coins; Alice doesn't even need to be aware of Bob's existence. Explain however that we from observed pairs of coin flips $(A_i, B_i)$ never can see the difference between that scenario and the alternative one, that Bob is the one flipping his fair coin, without caring for Alice, before she chooses between two biased ones. This is arguably an instance of what Breiman (2001) alludes to as the Rashomon Effect (from a Japanese movie in which different persons report very differently about something they have all observed): data alone cannot help us uncover which of the chains of action have been at work. See Story #8 for further discussion. Show indeed that as long as Alice and Bob have a joint scheme of producing outcomes $(0, 0), (0, 1), (1, 0), (1, 1)$, with probabilities respectively $\frac{1}{4}(1 + a), \frac{1}{4}(1 - a), \frac{1}{4}(1 - a), \frac{1}{4}(1 + a)$, then $(X_n, Y_n)$ have the correlated binomial distribution.

the Rashomon Effect: different models may offer equally good explanations

**(c)** Find a way to compute $f(x, y) = \Pr(X_n = x, Y_n = y)$, for $x, y = 0, 1, \ldots, n$.

**(d)** Leaving the Rashomon aspects to the side, generalise the first setup to the case of two correlated binomials $(n, p)$, where $p$ is not necessarily $\frac{1}{2}$. Take indeed $\Pr(A_i = 1) = p$ and then $\Pr(B_i = 1 \,|\, A_i = 1) = p + a$, $\Pr(B_i = 0 \,|\, A_i = 0) = 1 - p + ap/(1 - p)$, for $a < \min(p, 1 - p)$, and show that this works properly. What is the correlation between $X_n$ and $Y_n$?

**(e)** Show that $\sqrt{n}(X_n/n - p, Y_n/n - p)$ tends in distribution to a binormal zero-mean $(X, Y)$, with variances $p(1 - p)$ and covariance $ap$.

**(f)** (xx brief pointer to two correlated random walks, Ch9, with two correlated Brownian motions. also good ML exercise, finding $\widehat{a}$ based on having observed Alice and Bob random walks, easy $\sqrt{n}(\widehat{a} - a)$, test for $a = 0$, etc.)

## Notes and pointers

(xx to come. a bit old literature, but crisply, and not systematic. brief genesis of the normal, a few sentences on the chi-squared, Pearson (1900), the t, Student (1908), the F, the Dirichlet, more. we also point to essential things in later chapters. point out that the normal is famous and useful also because of a host of approximation methods. xx)

(xx mention things for Irwin (1927) and Hall (1927) and before. xx)

(xx where do we have a precise theorem on $M_X = M_Y$ implying $F = G$? inversion formula? Jensen is Johan Jensen. Vector form too?)

(xx find the little fun fact from Hjort (1994), regarding t and midway between cauchy and normal. xx)

The chi-squared is on the list over deservedly famous distributions in probability theory and statistics, and stems from Karl Pearson's famous 1900 paper, 'On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling'. [xx a bit more: he establishes the chi-square distribution, the test carrying the chi-squared name, and sets up a rigorous concepttual framework for hypothesis testing. xx]

(xx point to uses of expofamily in both Chs. 4 and 5. xx)

# I.2

# Large-sample theory

The broad themes of this chapter are the concepts, details, methods, results, applications pertaining to three modes of convergence for random variables: convergence in probability, convergence almost surely, convergence in distribution. The first two have to do with random variables $X_n$ coming close to some limit $X$, with increasing $n$, typically indexed by sample size; often the limit is merely a constant. The chief result here, with various extensions and uses, is the Law of Large Numbers, that the empirical average of a sequence of observations tends to the expected value of the underlying distribution. The third mode of convergence rather involves the distribution of $X_n$ coming close to the distribution of some limit $X$, with the Central Limit Theorem being a prime statement. These machineries also lead to practically useful approximations; the idea is that a complicated distribution may be approximated by something much simpler. This is in particular helped by a collection of approximation methods called the delta method. The theory is developed first for functions of i.i.d. sequences, involving tools of moment-generating functions and characteristic functions, along with various probability inequalities. It is then extended to cover cases of independent variables from non-equal distributions, culminating in the Lindeberg theorem, giving precise conditions under which a sum of independent components approaches normality. Methods and results from this chapter are crucial for developing the likelihood theory of Ch. 5, and also for several later chapters.

*Key words:* central limit theorems, characteristic functions, delta method, large-sample approximations, laws of large numbers, Lindeberg conditions, modes of convergence

In this chapter we study convergence of sequences $(X_n)_{n\geq 1}$ of random elements. In statistics, the index $n$ typically refers to the sample size, and $X_n$ is some function of the $n$ data points available, for example an estimator. The modes of convergence we study take place as $n$ grows without bounds; hence the name large-sample theory.

A random element $X$ is a function defined on a probability space $(\Omega, \mathcal{F}, \Pr)$, and taking values in some space $\mathcal{X}$. Throughout this chapter, and indeed throughout this book, $\mathcal{X}$ is a metric space equipped with its Borel-$\sigma$-algebra, i.e., the smallest $\sigma$-algebra containing all the open sets. When $\mathcal{X}$ is a subset of the real line, then $X$ is a random variable; when $\mathcal{X}$ is a subset of $\mathbb{R}^k$ for some $k \geq 1$, $X$ is a random vector, so a random vector of dimension one is a random variable; and when $\mathcal{X}$ is a set if functions, $X$ is called a random or stochastic process. In this chapter we concentrate mainly on convergence of random variables and vectors, with the more involved themes of convergence of stochastic

processes studied in Ch. 10. Several of the results of the present chapter are valid for stochastic processes, however; which ones will be clear from the context.

Applications of large-sample theory are plentiful in probability and statistics, partly to understand crucial phenomena better, and partly to provide fruitful and practical approximations; an estimator or a statistic might have a very complicated exact distribution, but have a simple to use and sometimes accurate large-sample approximation. The key convergence concepts, with ensuing approximations and applications, are as follows.

First, if $X_n = (X_{n,1}, \ldots, X_{n,k})$ is a sequence of random vectors, and $a = (a_1, \ldots, a_k)$ is some point, we say that $X_n$ *converges to* $a$ *in probability*, written $X_n \to_{\rm pr} a$, if for each positive $\varepsilon$,

$$\Pr(\|X_n - a\| \geq \varepsilon) \to 0, \quad \text{as } n \text{ tends to infinity.} \tag{2.1}$$

Here $\|x - a\| = (\sum_{j=1}^k (x_j - a_j)^2)^{1/2}$ is the Euclidean distance, and hence ordinary distance in the one-dimensional case. If $\widehat{\theta}_n$ is an estimator of the value $\theta_0$, we say that the estimator is *consistent* if $\widehat{\theta}_n \to_{\rm pr} \theta_0$. Various results from the present chapter will indeed have immediate applications in Ch. 3. consistency of an estimator

Second, a stronger version of convergence is $X_n$ converges to $a$ *almost surely*, that is, the convergence occurs with probability one. This means that

$$\Pr(\lim_n \|X_n - a\| = 0) = 1, \tag{2.2}$$

and we write $X_n \to_{\rm a.s.} a$. We say that an estimator $\widehat{\theta}_n$ is strongly consistent for $\theta_0$ if $\widehat{\theta}_n \to_{\rm a.s.} \theta_0$. A strong achievement indeed is the *strong Law of Large Numbers* (LLN), which says that the LLN

$$\text{if } X_1, X_2, \ldots \text{ are i.i.d. with finite mean } \xi, \text{ then } \bar{X}_n = n^{-1} \sum_{i=1}^n X_i \to_{\rm a.s.} \xi, \tag{2.3}$$

with no further assumptions required. One may readily prove weaker versions of the LLN, as in Ex. 2.10(a) and Ex. 2.15, but with the development of sharper tools, and separate valuable results along the way, we reach the strong LLN in Ex. 2.58–2.59. It immediately has many applications and uses, as we shall see.

We note that limits in probability and almost surely can easily be defined also when the limit is a random vector and not merely a constant, just by replacing $a$ with a random vector $X$ in (2.1)–(2.2). Most often, though, these two convergence concepts are used for cases when the limit is a constant.

The third and statistically speaking most important concept is that of *convergence in distribution*. If the random vectors $X_n$ and $X$ of dimension $k \geq 1$ have c.d.f.s $F_n$ and $F$, we say that $X_n$ converges in distribution to $X$ if $X_n \to_d X$ $F_n \Rightarrow F$

$$F_n(x) \to F(x), \quad \text{for all continuity points } x = (x_1, \ldots, x_k) \text{ of } F. \tag{2.4}$$

We write $X_n \to_d X$ or $F_n \Rightarrow F$ to indicate convergence in distribution, allowing for simplicity also statements like $X_n \to_d {\rm N}(0,1)$. This form of convergence is also called *weak convergence* and *convergence in law*. The bigger sibling of the LLN is the Central Limit Theorem (CLT): if $X_1, X_2, \ldots$ are i.i.d. random variables with variance $\sigma^2$, then the CLT

$$\sqrt{n}(\bar{X}_n - {\rm E}\, X_1) \to_d {\rm N}(0, \sigma^2), \tag{2.5}$$

again with no further assumptions needed beyond a finite variance. Below we go considerably further, however, in detail, in extensions, in applications. In particular, with additional tools and efforts we reach the Lindeberg theorem, with precise necessary conditions for a sum of independent variables from different distributions to approach normality. Such results are, for example, used to establish approximate normality for estimators in regression models.

Reaching the CLTs, with variations, requires hard mathematical work, with various technical details to sort out. When the main theorems have been established, however, along with further tools, their actual use for statistical applications might be relatively straightforward. In particular, functions of approximately normal variables are also approximately normal, as we learn in the subsection on delta methods.

### Modes of convergence

**Ex. 2.1** *Implications.* Of the three modes of convergence, convergence almost surely is the strongest, convergence in distribution the weakest, with convergence in probability lying somewhere in the middle. In the following, $X_n$ and $X$ are random vectors. In statistical applications, $(X_n)_{n\geq 1}$ will often be a sequence of estimators, and the limit a constant, e.g., the parameter of a model.

**(a)** Show that $X_n \to_{\text{a.s.}} X$ if and only if the event $\cap_{n\geq 1} \cup_{k\geq n} \{\|X_k - X\| \geq \varepsilon\}$ has probability zero for all $\varepsilon > 0$.

**(b)** Show that if $X_n \to_{\text{a.s.}} X$, then $X_n \to_{\text{pr}} X$.

**(c)** Show that if $X_n \to_{\text{pr}} X$ then $X_n \to_d X$.

**(d)** Let $Y$ be a nonnegative random variable. Show that $\Pr(Y \geq a) \leq \mathrm{E}\,(Y)/a$ for any $a > 0$, which is Markov's inequality. Deduce that if $\mathrm{E}\,\|X_n - X\| \to 0$, then $X_n \to_{\text{pr}} X$.

Markov's inequality

**(e)** Show that $\|X_n - X\| \to_{\text{a.s.}} 0$ is equivalent to $|X_{n,j} - X_j| \to_{\text{a.s.}} 0$ for each $j$, and that the same is true for convergence in probability.

**Ex. 2.2** *The Borel–Cantelli lemma and convergence almost surely.* Let $A_1, A_2, \ldots$ be a sequence of events. Consider $\cap_{n\geq 1} \cup_{m\geq n} A_m$, the full-sequence event corresponding to the $A_n$ occurring infinitely often.

**(a)** Let $A_1, A_2, \ldots$ be a sequence of events, and let $N$ be the total number of occurrences of the $A_i$. Show that $\mathrm{E}\,N = \sum_{i=1}^{\infty} \Pr(A_i)$.

**(b)** Show that if $\sum_{n=1}^{\infty} \Pr(A_n)$ is convergent, then $\Pr(\cap_{n\geq 1} \cup_{m\geq n} A_m) = 0$. So sooner or later there will be a finite (but random) $n$, such that none of the $A_m$ will ever occur, for $m > n$.

The Borel–Cantelli lemma

**(c)** Assume in addition that the $A_1, A_2, \ldots$ are independent events. Show that if $\sum_{n=1}^{\infty} \Pr(A_n)$ is divergent, then $\Pr(A_{\text{i.o.}}) = 1$. In particular, for the case of independent events, there can't be say a 50 percent chance that there will be infinitely many occurrences. To show this, you may prove and use the inequality $1 + x \leq \exp(x)$, valid for all $x \in \mathbb{R}$.

**(d)** Let $X_1, X_2, \ldots$ be i.i.d. from the unit exponential distribution. Will there be infinitely many cases with $X_i \geq 0.99 \log i$, with $X_i \geq \log i$, with $X_i \geq 1.01 \log i$?

**(e)** Let $X_1, X_2, \ldots$ be i.i.d. standard normal. Show that

$$\Pr(X_i \geq a) = \frac{\phi(a)}{a}\{1 + o(1)\}, \quad \text{as } a \to \infty,$$

and use this to show that there will be infinitely many cases with $|X_i| \geq (2 \log i)^{1/2}$, but

only finitely many cases with $|X_i| \geq (2.001 \log i)^{1/2}$. The quantity $\{1 - \Phi(a)\}/\{\phi(a)/a\}$ is sometimes called the Mills ratio.

**Ex. 2.3** *The converses do not hold.* The implications arrows in Ex. 2.1(b)–(c) only point in one direction, as we will now see.

**(a)** Let $X_1, X_2, \ldots$ be Bernoulli random variables with success probabilities $p_1, p_2, \ldots$. Show that $X_n \to_{\text{pr}} 0$ as long as $p_n \to 0$, but that $X_n \to_{\text{a.s.}} 0$ takes more. What is the probability for having infinitely many $X_n = 1$, for $p_n = 1/n^{0.99}$, for $p_n = 1/n$, for $p_n = 1/n^{1.01}$? Conclude that we can have convergence in probability, but not almost surely.

**(b)** Let $U$ be unif $(0, 1)$, and divide the unit interval in $2, 3, \ldots$ pieces: $A_1 = [0, 1/2]$, $A_2 = (1/2, 1]$, $A_3 = [0, 1/4]$, $A_4 = (1/4, 1/2]$, $A_5 = (1/2, 3/4]$, and so on. Define the random variables $X_n = I(U \in A_n)$, and show that $X_n \to_{\text{pr}} 0$, but not almost surely.

**(c)** Find an example where $X_n \to_d X$, but there is not convergence in probability.

**(d)** Let $X_n$ be a sequence of binary random variable with distributions $\Pr(X_n = a_n) = p_n$ and $\Pr(X_n = 0) = 1 - p_n$. Construct the $a_n$ and $p_n$ sequences such that $X_n \to_{\text{pr}} 0$, but $\mathrm{E}\,|X_n|$ does not converge to zero.

**(e)** Similarly, let $X_n = \theta + Z_n$, for $\theta \in \mathbb{R}$, where $Z_1, Z_2, \ldots$ are independent random variable with distribution $\Pr(Z_n = 2^n) = \Pr(Z_n = -2^n) = 1/(3n)$ and $\Pr(Z_n = 0) = 1 - 2/(3n)$. Show that $X_n \to_{\text{pr}} \theta$, but that $\mathrm{E}\,|X_n - \theta|$ does not converge to zero.

**Ex. 2.4** *Partial converses.* As we have just seen, the implication arrows of Ex. 2.1 only point in one direction. We do, however, have certain partial converses.

**(a)** As as partial converse to Ex. 2.1(b), show that if $X_n \to_{\text{pr}} X$, then there is a subsequence $(n_k)_{k \geq 1}$ such that $X_{n_k} \to_{\text{a.s.}} X$. To construct such a subsequence, you may use the Borel–Cantelli lemma.

**(b)** Prove the converse of (a), to obtain the lemma saying that $X_n \to_{\text{pr}} X$ if and only if any subsequence $(X_{n_k})$ has a further subsequence $(X_{n_{k_j}})$ such that $X_{n_{k_j}} \to_{\text{a.s.}} X$.

**(c)** As a partial converse to Ex. 2.1(c), show that if $X_n \to_d a$ for some constant $a \in \mathbb{R}$, then $X_n \to_{\text{pr}} a$. Generalise to higher dimensions.

**(d)** As a partial converse to Ex. 2.1(d), we have that if $X_n \to_{\text{pr}} X$ and $\|X_n\|$ is bounded by a random variable $Y$ such that $|Y|^p$ is integrable for some $p \geq 1$, then $\mathrm{E}\,\|X_n - X\|^p \to 0$. This is a version of Lebesgue's dominated convergence theorem, see Ex. B.13(e), and you may consult that exercise before solving this one.

**(e)** Another partial converse to Ex. 2.1(d) is provided by the following lemma. If $X_n$ is a sequence of nonnegative random vectors such that $X_n \to_{\text{pr}} X$ and $\mathrm{E}\,X_n \to \mathrm{E}\,X$, with $X$ integrable, then $\mathrm{E}\,\|X_n - X\| \to 0$. Prove it first in one dimension, where you may use that $|y| = y + 2\max(-y, 0)$, then generalise.

**Ex. 2.5** *Uniform integrability.* Yet another partial converse to Ex. 2.1(d) is obtained by introducing the concept of *uniform integrability*. A sequence $X_1, X_2, \ldots$ of random variables is uniformly integrable if

$$\lim_{M \to \infty} \sup_{n \geq 1} \mathrm{E}\,|X_n| I(|X_n| > M) = 0. \tag{2.6}$$

As we show in this exercise, if $(X_n)_{n \geq 1}$ is uniformly integrable and $X_n \to_{\text{pr}} X$, then $\mathrm{E}\,|X_n - X| \to 0$. Thus, compared to Ex. 2.4(d) (which is essentially Lebesgue's dominated convergence theorem), the uniform integrability assumption takes the place of the

domination assumption imposed in the dominated convergence theorem. Moreover, as we show in (c) and (i), uniformly integrability is a bit weaker than boundedness, and turns out to be necessary for convergence in mean.

**(a)** Show that $\mathrm{E}\,|X| < \infty$ is equivalent to $\lim_{M\to\infty} \mathrm{E}\,|X|I(|X| > M) = 0$. This explains why the property in (2.6) is called *uniform* integrability.

**(b)** Show that if $(X_n)_{n\geq 1}$ is uniformly integrable, then $\sup_n \mathrm{E}\,|X_n| < \infty$. Exhibit a sequence of random variables for which the converse does not hold.

**(c)** Show that if $(X_n)_{n\geq 1}$ is dominated by an integrable $Y$, then it is uniformly integrable. Construct a sequence of random variables that is uniformly integrable, but that is not dominated by an integrable random variable.

**(d)** Let $a, b \in \mathbb{R}$ and suppose that $(X_n)_{n\geq 1}$ and $(Y_n)_{n\geq 1}$ are uniformly integrable. Show that $aX_n + bY_n$ is uniformly integrable.

**(e)** Let $(X_n)_{n\geq 1}$ be a uniformly integrable sequence of random variables such that $X_n \to_{\text{a.s.}} X$. Show that the limit $X$ must be integrable, and that $\mathrm{E}\,|X_n - X| \to 0$. To show the integrability part, consult Fatou's lemma Ex. B.13(b). For the convergence in mean part, the trick is to truncate $Y_n = |X_n - X|$ and take it from there.

**(f)** Show that in (e), it is sufficient that $X_n$ converges in probability to $X$.

**(g)** Show that if $(X_n)_{n\geq 1}$ is such that $\sup_n \mathrm{E}\,|X_n|^{1+\delta} < \infty$ for some $\delta > 0$, then it is uniformly integrable.

**(h)** Show that $(X_n)_{n\geq 1}$ is uniformly integrable if and only if

$$\lim_{M\to\infty} \limsup_{n\to\infty} \mathrm{E}\,|X_n|I(|X_n| > M) = 0,$$

and $X_n$ is integrable for each $n$.

**(i)** The claim in (e) has a converse. Show that if $(X_n)_{n\geq 1}$ and $X$ are nonnegative and integrable, $X_n \to_{\text{a.s.}} X$ and $\mathrm{E}\,X_n \to \mathrm{E}\,X$, then $(X_n)_{n\geq 1}$ is uniformly integrable.

**(j)** Deduce from (i) that if $(X_n)_{n\geq 1}$ and $X$ are integrable (but not necessarily nonnegative), $X_n \to_{\text{a.s.}} X$ and $\mathrm{E}\,|X_n - X| \to 0$, then $(X_n)_{n\geq 1}$ is uniformly integrable.

**(k)** Show that if $X_n^2$ and $Y_n^2$ are uniformly integrable, then $X_nY_n$ is uniformly integrable. Show also that if $X_n^p$ is uniformly integrable, then so is $X_n^r$ for all $1 \leq r \leq q$.

**Ex. 2.6** *Scheffé's lemma.* If we replace the random vectors in Ex. 2.4(e) with the densities $(f_n)_{n\geq 1}$ and $f$, then the limit is automatically integrable, and we obtain the lemma typically known by the name of Scheffé.

Scheffé's lemma

**(a)** Let $Y_n$ and $Y$ be random vectors with densities $f_n$ and $f$ with respect to a dominating measure $\mu$. Show that if $f_n(y) \to f(y)$ for $\mu$-almost all $y$, then $\int |f_n - f|\,\mathrm{d}\mu \to 0$, and consequently $Y_n \to_d Y$.

**(b)** Suppose $X_n$ has the Gamma distribution with parameters $(a_n, b)$, where $a_n \to 1$. Show that $X_n$ tends in distribution to an exponential with mean $1/b$.

**(c)** Show that under the conditions of (a) we have $\sup_B |\Pr(Y_n \in B) - \Pr(Y \in B)| \to 0$, the supremum being over all Borel sets $B$. Since $Y_n \to_d Y$ only requires $\Pr(Y_n \in B) \to \Pr(Y \in B)$ for sets of the form $B = (-\infty, y]$ (or, more generally for all continuity sets $B$, see Ex. 2.18), convergence $\Pr(Y_n \in B) \to \Pr(Y \in B)$ uniformly in $B$ is stronger than what is required for convergence in distribution: For a classical example thereof, consider the sequence $Y_n = 1/n$. Clearly, $Y_n \to_d 0$, but show that $\sup_B |\Pr(Y_n \in B) - \Pr(Y \in B)| = 1$.

**(d)** For the case of $Y_n$ and $Y$ on the integers $0, 1, 2, \ldots$, show that $p_n(j) = \Pr(Y_n = j)$ converges to $p(j) = \Pr(Y = j)$ for each $j$, if and only if $Y_n \to_d Y$.

**(e)** If $Y_n$ and $Y$ have densities $f_n$ and $f$, and $Y_n \to_d Y$, we should expect $f_n \to f$. This is not alway happening, however. Consider the case of $F_n(y) = y + (n\pi)^{-1}\sin(n\pi y)$. Plot the $F_n$ and its density $f_n$, for some $n$. Show that $Y_n \to_d$ unif, but that $f_n(y)$ does not converge to 1 for all $y$. It is also instructive to transform to the approximate normal scale, via $X = \Phi^{-1}(Y)$. Show that $X$ then has density $g_n(x) = \phi(x)\{1 + \cos(n\pi\Phi(x))\}$, with very notable oscillations, but where the c.d.f. $G_n(x)$ nevertheless tends to the standard normal.

**Ex. 2.7** *From discrete to continuous.* Often enough discrete distributions have continuous limits.

**(a)** Let $X_n$ have distribution $\Pr(X_n = j/n) = 1/(n+1)$ for $j = 0, 1, \ldots, n$. Show that $X_n \to_d X$, where $X$ has the uniform distribution on the unit interval. Show also that $\sup_B |\Pr(X_n \in B) - \Pr(X \in B)|$ does *not* converge to zero (cf. Ex. 2.6(c)).

**(b)** With perhaps similar techniques as in (a), consider $X_n$ with distribution $\Pr(X_n = j/n) = j/\{n(n+1)/2\}$ for $j = 1, \ldots, n$. Find its limit distribution.

**(c)** Let $X_1, X_2, \ldots$ be independent Bernoulli random variables with success probability $p$. Only using (2.4), show that $\sqrt{n}(\bar{X}_n - p) \to_d \mathrm{N}(0, p(1-p))$. You ought to use Stirling's formula $n! \sim \sqrt{2\pi n}(n/e)^n$ here; we prove that formula via probabilistic tricks in Ex. 2.35.

**Ex. 2.8** *Many small probabilities give a Poisson.* The Law of Small Numbers says that if we sum a high number of 0-1 variables, with each having a small probability of 1, then we're close to a Poisson. This was touched upon in Ex. 1.24, but here we give a clearer picture, with more to come in Ex. 2.29, with natural setups yielding Poisson limits.

der Gesetz der kleinen Zahlen

**(a)** Suppose $Y_n \sim \mathrm{binom}(n, p_n)$, with $p_n$ becoming small with growing $n$ in a way which has $np_n \to \lambda$. Show that $Y_n \to_d \mathrm{Pois}(\lambda)$. As a small demonstration, take $n = 100$ and $p = 0.05$. Compare the exact and the limiting probabilities.

**(b)** Suppose $(X_n, Y_n)$ has the trinomial distribution, with parameters $(n, p_n, q_n)$, see Ex. 1.4, where we also found the correlation between them. Assume now that $p_n, q_n$ become small with $n$, such that $np_n \to \lambda_1$, $nq_n \to \lambda_2$. Show that the correlation betweem $X_n$ and $Y_n$ tends to zero, and that $(X_n, Y_n) \to_d (X, Y)$, where $X$ and $Y$ are independent and Poisson with parameters $\lambda_1, \lambda_2$. Generalise to the multinomial situation.

**Ex. 2.9** *Maximum of uniforms.* Let $X_1, \ldots, X_n$ be i.i.d. random variables with the uniform distribution on $(0, \theta)$, and set $M_n = \max_{i \le n} X_i$.

**(a)** Show that $M_n \to_{\mathrm{pr}} \theta$, that is, the maximum of the observations is consistent for the unknown endpoint.

**(b)** Find the limit distribution of $V_n = n(\theta - M_n)$, and use this result to find an approximate 90 percent confidence interval for $\theta$, i.e., a random interval $[L_n, U_n]$ such that $\Pr(L_n \le \theta \le U_n)$ is approximately 0.90.

### Convergence in probability and tail bounds

**Ex. 2.10** *Markov, Chebyshov, and the Law of Large Numbers.* In view of the definition of convergence in probability in (2.1), it is certainly useful to find mathematically manageable bounds for so-called tail probabilities, i.e., $\Pr(|X| \ge a)$ for a random variable $X$. There are several such, as we learn here, with more to come in Ex. 2.16. Their uses include assessing how likely it might be that an estimator is some distance off its target.

**(a)** Markov's inequality was proven in Ex. 2.1(d): If $X$ is nonnegative, then $\Pr(X \ge a) \le \mathrm{E}\,(X)/a$ for each $a > 0$. More generally, if $h(x)$ nonnegative and nondecreasing, with

the Chebyshov inequality

$h(a)$ positive, show that $\Pr(X \geq a) \leq \mathrm{E}\, h(X)/h(a)$. From this, deduce the Chebyshov inequality: if $X$ has finite variance (but is not necessarily nonnegative), then

$$\Pr(|X - \mathrm{E}\, X| \geq \varepsilon) \leq \mathrm{Var}\,(X)/\varepsilon^2, \quad \text{for } \varepsilon > 0.$$

the LLN via Chebyshov

**(b)** With $X_1, \ldots, X_n$ being i.i.d. from the same distribution as $X$, with mean $\xi$ and standard deviation $\sigma$, show that for the empirical mean $\bar{X}_n$ that $\Pr(|\bar{X}_n - \xi| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2)$. Since the right hand side tends to zero as $n$ tends to infinity, you have now proven the version of the *Law of Large Numbers* (LLN) that says that the empirical mean of $n$ i.i.d. random variables with finite second moments converges in probability to its expectation as $n$ tends to infinity.

**(c)** Let $Y_1, Y_2, \ldots$ be i.i.d. random variables with expectation $\theta$ and finite variance $\sigma^2$. Consider the weighted average $\widehat{\theta}_n = \sum_{i=1}^n w_i Y_i / \sum_{i=1}^n w_i$, for nonnegative and fixed weights $w_i$. Give a condition for consistency ensuring $\widehat{\theta}_n \to_{\mathrm{pr}} \theta$, in terms of these weights. What happens for $w_i = 1/i$, and for $w_i = 1/i^{1.5}$?

**(d)** If $X$ has mean $\xi$, and a finite fourth moment, show that $\Pr(|X - \xi| \geq \varepsilon) \leq \mathrm{E}\,|X - \xi|^4/\varepsilon^4$. For $X \sim \mathrm{N}(\xi, \sigma^2)$, deduce that $\Pr(|X - \xi| \geq \varepsilon) \leq 3\sigma^4/\varepsilon^4$.

**(e)** With $X_1, \ldots, X_n$ i.i.d. from a distribution with finite fourth moment, write $\gamma_4 = \mathrm{E}\,\{(X_i - \xi)/\sigma\}^4 - 3$ for its kurtosis. Show that

$$\mathrm{E}\,|\bar{X}_n - \xi|^4 = \frac{\sigma^4}{n^2}\{3 + (1/n)(\gamma_4 - 3)\}.$$

Show hence that $\Pr(|\bar{X}_n - \xi| \geq \varepsilon) \leq 3.01\,\sigma^4/(n^2\varepsilon^2)$, for all large enough $n$. When is this a sharper result than that of the Chebyshov inequality?

**Ex. 2.11** *The binomial and the empirical distribution function.* For i.i.d. observations $Y_1, \ldots, Y_n$, we form *the empirical cumulative distribution function*, the e.c.d.f., with $F_n(t) = n^{-1} \sum_{i=1}^n I(Y_i \leq t)$. Plotting $F_n$ for a given dataset is informative, also for comparing with given distributions.

**(a)** Before returning to $F_n$, consider $X_n \sim \mathrm{binom}(n, p)$ and the familiar ratio $\widehat{p}_n = X_n/n$. Show that indeed $\widehat{p}_n \to_{\mathrm{pr}} p$. Historically speaking, this is the earliest clear LLN statement.

**(b)** Coming back to the empirical c.d.f., explain that $F_n(t)$ for given $t$ is just a binomial proportion, and that $F_n(t) \to_{\mathrm{pr}} F(t)$ for each $t$. In particular, the events $F_n(t) \leq F(t) - \varepsilon$ and $F_n(t) \geq F(t) + \varepsilon$ both have probabilities going to zero, for positive $\varepsilon$.

**(c)** We come back to finer analysis of $F_n$ in Ex. 2.61 and 10.27, but find the variance of $F_n(t)$ and the covariance betweem $F_n(t_1)$ and $F_n(t_2)$.

**Ex. 2.12** *Continuous mapping for convergence in probability.* The theorem known as the continuous mapping theorem (which will be proved in its entirety during the course of this chapter) says that the three modes of convergence introduced above are preserved under a continuous mapping. In this exercise we look at the convergence in probability part of this theorem.

**(a)** Let $X_n$ be a sequence of random variables such that $X_n \to_{\mathrm{pr}} a$, with $a$ being a constant. Show that if $g$ is a function continuous at point $x = a$, then $g(X_n) \to_{\mathrm{pr}} g(a)$.

**(b)** Suppose more generally that $X_n \to_{\mathrm{pr}} X$, with the limit being a random variable such that $\Pr(X \in C) = 1$, for $C \subset \mathbb{R}$. Show that if $g$ is uniformly continuous on $C$, then $g(X_n) \to_{\mathrm{pr}} g(X)$.

**(c)** Show that in (b) it is sufficient that $g$ is continuous on $C$. There are several ways of showing this, one involves the subsequences result from Ex. 2.4(a).

**(d)** Generalise the above to sequences $X_n = (X_{n,1}, \ldots, X_{n,k})$ of random vectors and continuous functions $g\colon \mathbb{R}^k \to \mathbb{R}^m$.

**Ex. 2.13** *Toeplitz lemma and convergence in probability.* Toeplitz lemma says that if $a_1, a_2, \ldots$ is a nonnegative sequence such that $\sum_{j=1}^{\infty} a_j = \infty$ and $x_1, x_2, \ldots$ is a sequence such that $x_n \to x$, then $\sum_{j=1}^{n} a_j x_j / \sum_{j=1}^{n} a_j \to x$ as $n \to \infty$. If we set $a_j = 1$ for all $j$, the lemma says that $\bar{x}_n = n^{-1} \sum_{i=1}^{n} x_i \to x$ as long as $x_n \to x$. In this exercise we look into the conditions under which this carries over to the situation where $x_n$ is replaced by a sequence $X_n$ of random variables converging in probability to some random or constant limit. The example in (b) and the lemma in (c) are from Linero and Rosalsky (2013).

**(a)** Prove the Toeplitz lemma.

**(b)** Let $X_n$ be a sequence of independent random variables with distributions $\Pr(X_n = 0) = 1 - n^{-1}$ and $\Pr(X_n = n) = n^{-1}$. Show that $X_n \to_p 0$, and that the empirical means $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$ is such that its characteristic function

$$\mathrm{E}\, \exp(it\bar{X}_n) \to \exp\left( \int_0^1 \frac{e^{itx} - 1}{x} \, \mathrm{d}x \right),$$

which you may peek ahead to Ex. 2.36 to show. As a consequence of Ex. 2.28, the limit here is the charactertistic function of a nondegenerate random variable, say $Z$. We are thus in the situation that $X_n \to_p 0$, while $\bar{X}_n \to_d Z$, which shows that Toeplitz lemma does not translate directly to convergence in probability.

**(c)** Suppose that $X_n$ is a uniformly integrable sequence of random variables such that $X_n \to_p 0$, and let $\bar{X}_n = n^{-1} \sum_{i=1}^{n} X_i$. Show that $\bar{X}_n \to_p X$. You may look back at Ex. 2.5(f) before attempting to prove this.

**Ex. 2.14** *Quantiles.* Here we learn that the empirical quantiles are consistent for their population counterparts. Further informative analyses are in Ex. 3.21, including convergence in distribution, but here we limit attention to direct convergence in probability.

**(a)** Suppose first that $U_1, \ldots, U_n$ are i.i.d. from the uniform distribution on the unit interval, and let $M_n$ be the empirical median. With the empirical c.d.f. $F_n$ from Ex. 2.11, use the fact that $F_n(t) \to_{\mathrm{pr}} t$, for each $t$, to show that $M_n \to_{\mathrm{pr}} \frac{1}{2}$. Then generalise to the case of observations $Y_1, \ldots, Y_n$ from a distribution with continuous and strictly increasing c.d.f. $F$, showing for the empirical median that $M_n \to_{\mathrm{pr}} F^{-1}(\frac{1}{2})$.

**(b)** More generally, for a $q \in (0,1)$, let $Q_n = Q_n(q)$ be the empirical $q$-quantile, defined here to be $Y_{(\lfloor nq \rfloor)}$, the $\lfloor nq \rfloor$ order statistic, where $\lfloor x \rfloor = \max\{m \in \mathbb{Z}\colon m \le x\}$. Show that $Q_n$ is consistent for the population quantile $F^{-1}(q)$. floor function

**(c)** Show that the interquartile range, the 0.75 quantile minus the 0.25 quantile, is consistent for the population interquartile range $F^{-1}(0.75) - F^{-1}(0.25)$. Show more generally that if $\theta = g(F^{-1}(q_1), \ldots, F^{-1}(q_r))$ is any continuous function of a finite number of quantiles, then $\widehat{\theta} = g(Q_n(q_1), \ldots, Q_n(q_r))$ is consistent for $\theta$.

**Ex. 2.15** *Improving on the weak LLN.* We have seen in Ex. 2.10(a) that the LLN holds if the distribution has a finite variance. Here we get rid of the finite variance condition. For the almost sure version of the LLN see Ex. 2.58.

**(a)** Let $X_1, \ldots, X_n$ be i.i.d. with finite mean $\xi$. Truncate the random variables involved,

i.e., write $X_i = X_i I_{|X_i|\le M} + X_i I_{|X_i|>M}$ for some $M > 0$, and show that for any $\varepsilon > 0$,

$$\Pr(|\bar{X}_n - \xi| \ge \varepsilon) \le \frac{8M}{n\varepsilon^2}\,\mathrm{E}\,|X| + \frac{4}{\varepsilon}\,\mathrm{E}\,|X-\xi| I_{|X-\xi|>M},$$

law of large numbers

**(b)** Let $X_1, X_2, \ldots$ be i.i.d. random variables with finite mean $\xi$, i.e., no assumption is made about its second moment. Show that $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i \to_p \xi$.

**(c)** Let $X_1, X_2, \ldots$ be i.i.d. random variables with finite variance $\sigma^2$. Show that the empirical variance $n^{-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ is consistent for $\sigma^2$.

**Ex. 2.16** *Further tail bound inequalities.* In Ex. 2.10 we learned about the Markov and Chebyshov inequalities; here we work out further tail bounds for our toolboxes.

**(a)** Suppose $X$ has a finite m.g.f. $M(t) = \mathrm{E}\,\exp(tX)$, as per Ex. 1.32. Show that

$$\Pr(X \ge a) \le q(a) = \min\{t \ge 0\colon\ \exp(-ta)M(t)\}.$$

Writing $M(t) = \exp\{K(t)\}$, show that this leads to $q(a) = \exp\{K(t_a) - at_a\}$, where $t_a$ is the solution to $K'(t_a) = a$, if positive. Apply this to $X \sim \mathrm{N}(0,1)$ and $a$ positive, and show that $\Pr(X \ge a) \le \exp(-\frac{1}{2}a^2)$. Show that this is indeed sharper than the tail bound $1/a^2$, from the simpler Chebyshov inequality, for all $a > 0$.

**(b)** For $X \sim \mathrm{N}(0,1)$ and $a$ positive, show that

$$\Pr(|X| \ge a) \text{ is smaller than each of } \frac{1}{a^2}, \frac{3}{a^4}, \frac{15}{a^6}, 2\exp(-\tfrac{1}{2}a^2).$$

(xx a bit more here, rounding it off. more inequalities in Ch. 2. xx)

**(c)** Let $X \sim \chi^2_m$, which has mean and variance $m$ and $2m$. Consider $p_m(a) = \Pr(X \ge m + am^{1/2})$. Show that $p_m(a) \le q_m(a)$, where

$$\begin{aligned} q_m(a) &= \min\{t \ge 0\colon\ \exp(-ta - t/\sqrt{m})(1 - 2t/\sqrt{m})^{-m/2} \\ &= \exp(-\tfrac{1}{2}a\sqrt{m})(1 + a/\sqrt{m})^{m/2}. \end{aligned}$$

Compare this bound both with bounds from the Markov inequality, and with the exact limit of $p_m(a)$, as $m$ grows.

**(d)** Let $X_1, X_2, \ldots$ be i.i.d. with mean zero and variance one, so that $\sqrt{n}\bar{X}_n \to_d \mathrm{N}(0,1)$, but you don't need to use that here. Assume the m.g.f. $M(t) = \mathrm{E}\,\exp(tX_1) = \exp\{K(t)\}$ is finite. Show that

$$\Pr(\sqrt{n}\bar{X}_n \ge a) \le \frac{M(t/\sqrt{n})^n}{\exp(ta)} = \exp\{nK(t/\sqrt{n}) - ta\},$$

for each $t$. (xx then a bit more. tail inequality. not too far from good bound $\exp(-\frac{1}{2}a^2)$. briefly mention and point to large deviations theory. xx)

**Ex. 2.17** *Bernshteĭn and Weierstraß.* In c. 1885, Karl Weierstraß proved one of the fundamental and insightful results of approximation theory, that any given continuous function can be approximated uniformly well, on any finite interval, by polynomials. Here we give a constructive and relatively straightforward proof of the Weierstraß theorem, involving so-called Bernshteĭn polynomials. Let $g\colon [0,1] \to \mathbb{R}$ be continuous, and construct

$$B_n(p) = \mathrm{E}_p\, g(X_n/n) = \sum_{j=0}^n g(j/n)\binom{n}{j} p^j (1-p)^{n-j} \quad \text{for } p \in [0,1],$$

where $X_n \sim \mathrm{binom}(n,p)$. Note that $B_n(p)$ is a polynomial of degree $n$.

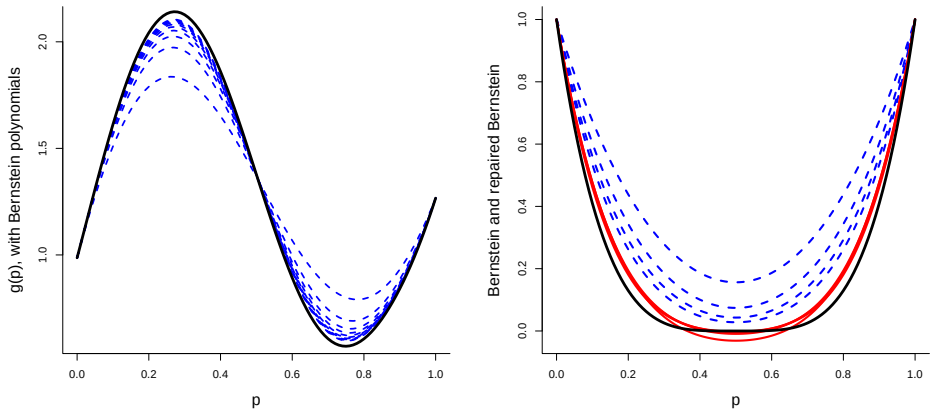



Figure 2.1: Left panel: the given non-polynomial function $g(p)$ (full black curve), along with approximating Bernshteĭn polynomials of order $10, 20, \ldots, 100$. Right panel: for the function $g_2(p) = (2p-1)^4$, the Bernshteĭn polynomials $B_n$ (dashed curves), along with the bias-modified ones $B_n^*$ (full curves), of order $4, 6, 8, 10$.

**(a)** Show that $B_n(p) \to_{\rm pr} g(p)$, for each $p$. Then show that the convergence is actually uniform. In some detail, for $\varepsilon > 0$, find $\delta > 0$ such that $|x-y| < \delta$ implies $|g(x)-g(y)| < \varepsilon$ (which is possible, as a continuous function on a compact interval is always uniformly continuous). Then show

$$|B_n(p) - g(p)| \le \varepsilon + 2M \Pr(|X_n/n - p| \ge \delta),$$

with $M$ a bound on $|g(p)|$. Show from this that indeed $\max_p |B_n(p) - g(p)| \to 0$.

**(b)** Consider the marvellous function

$$g_1(p) = \sin(2\pi p) + \exp(1.234 \sin^3 \sqrt{p}) - \exp(-4.321 \cos^5 p^2)$$

on the unit interval. Compute the Bernshteĭn polynomials of various orders, and display these in a diagram, alongside the curve of $g$. Construct a version of Figure 2.1, left panel, which does this for $n = 10, 20, \ldots, 90, 100$. How high $n$ is needed for the maximum absolute difference to creep below 0.01?

**(c)** Assuming smoothness of the $g$ function, and writing $\widehat{p} = X_n/n$ for the binomial fraction, use Taylor expansion $g(\widehat{p}) = g(p) + g'(p)(\widehat{p}-p) + \frac{1}{2}g''(p)(\widehat{p}-p)^2 + O_{\rm pr}(|\widehat{p}-p|^3)$ to show that $B_n(p) = g(p) + \frac{1}{2}g''(p)p(1-p)/n + o(1/n)$. This invites a repaired Bernshteĭn polynomial, of the form $B_n^*(p) = B_n(p) - \frac{1}{2}C_n(p)p(1-p)/n$, with $C_n(p) = {\rm E}_p g''(X_n/n)$. Explain that this is still a polynomial, now of order $n+2$, and show in examples that it often succeeds better than $B_n$ in coming close to the target $g(p)$. Make a version of Figure 2.1, right panel, which shows the $B_n$ and the $B_n^*$, of order $4, 6, 8, 10$, for the case of $g_2(p) = (2p-1)^4$. Check here that $\max_p |B_n^*(p) - g(p)|$ becomes rather smaller than $\max_p |B_n(p) - g(p)|$.

**(d)** Assuming $g$ has a derivative, construct a confidence band of the type $B_n(p) \pm 1.96\,\widehat{\sigma}(p)/\sqrt{n}$, for a certain $\widehat{\sigma}(p)$, with the property that it for each given $p$ covers the underlying $g(p)$ with probability tending to 0.95.

**(e)** Let now $g(x, y)$ be an arbitrary function on the unit simplex, $\{(x, y)\colon x \geq 0, y \geq 0, x + y \leq 1\}$. Construct a mixed polynomial $B_n(x, y)$ of degree $n$ such that it converges uniformly to $g$ on the simplex.

### Convergence in distribution

**Ex. 2.18** *The Portmanteau theorem.* So far we have taken as our definition of $X_n \to_d X$ that we have pointwise convergence $F_n(x) = \Pr(X_n \leq x) \to \Pr(X_n \leq x) = F(x)$ at all points $x$ where $F$ is continuous. A type of convergence we denote $F_n \Rightarrow F$. A limitation of this definition is that it only works for random vectors, and since we soon enough want to study convergence in distribution in spaces other than $\mathbb{R}^k$, we need a more general definition. Let $(\mathcal{X}, d)$ be a metric space equipped with its Borel-$\sigma$-algebra $\mathcal{B} = \mathcal{B}(\mathcal{X})$ (i.e., the smallest $\sigma$-algebra containing all the open sets), and let $P_n$ and $P$ be probability measures on the measurable space $(\mathcal{X}, \mathcal{B})$, typically the distributions of some random elements $X_n$ and $X$ taking values in $\mathcal{X}$. We say that $P_n$ *converges weakly* to $P$, denoted $P_n \Rightarrow P$, if

weak convergence $P_n \Rightarrow P$

$$\int g \,\mathrm{d}P_n \to \int g \,\mathrm{d}P, \tag{2.7}$$

for all bounded and continuous functions $g\colon \mathcal{X} \to \mathbb{R}$, a collection of functions we denote $C_b(\mathcal{X})$. When $P_n$ and $P$ are the distributions of some random elements $X_n$ and $X$, we might just write $\mathrm{E}\, g(X_n) \to \mathrm{E}\, g(X)$ for the convergence in (2.7). If $X_n$ and $X$ are random vectors, so that $\mathcal{X} = \mathbb{R}^k$ for some $k \geq 1$, then (2.7) is equivalent to $X_n \to_d X$ as defined in (2.4), a fact we prove in (e) of this exercise. As will become clear as we proceed, however, the definition in (2.7) is vastly more general than (2.4), in particular, it works when $\mathcal{X}$ is a function space, see Ch. 10. It is also often much easier to work with. In this exercise we first prove the 'bare bones', or classical, Portmanteau theorem, valid for all types of metric spaces, then proceed to proving its equivalence with $X_n \to_d X$. Ex. 2.19 presents additional equivalent statements. The classical Portmanteau theorem states that the following four statements are equivalent:

$C_b(\mathcal{X})$

(i) $P_n \Rightarrow P$;
(ii) $\limsup_n P_n(F) \leq P(F)$ for every closed set $F$;
(iii) $\liminf_n P_n(B) \geq P(B)$ for every open set $B$;
(iv) $P_n(C) \to P(C)$ for every set $C$ that is $P$-continuous, in the sense that $P(\partial C) = 0$, where $\partial C = \bar{C} \setminus C^\circ$ is the boundary of $C$ (the closure minus the interior);

We prove the Portmanteau theorem through a string of subexercises.

**(a)** For any set $A$ define $d(x, A) = \inf\{d(x, y)\colon y \in A\}$, i.e., the distance from $x$ to $A$, and define the function

$$\eta(t) = \begin{cases} 1, & \text{if } t \leq 0, \\ 1 - t, & \text{if } 0 \leq t \leq 1, \\ 0, & \text{if } t \geq 1. \end{cases}$$

For some $\varepsilon > 0$, the function $\eta(d(x, A)/\varepsilon)$ will be used to approximate the indicator function $I_A$. Let $F = [a, b]$ be a closed interval on the real line, and make a sketch $\eta(d(x, F)/\varepsilon)$. Show that for any closed set $F$, that $x \mapsto \eta(d(x, A)/\varepsilon)$ is bounded and continuous, and that for $F^\varepsilon = \{x\colon d(x, F) < \varepsilon\}$

$$I_F(x) \leq \eta(d(x, A)/\varepsilon) \leq I_{F^\varepsilon}(x).$$

**(b)** Use the inequalities from (a) to show that (i) implies (ii) and (iii).

**(c)** Show that (ii) and (iii) imply (iv).

**(d)** To show that (iv) implies (i), take $f$ to have its values inside $[0,1]$, without loss of generality, write

$$\mathrm{E}\, f(X_n) = \int \int_0^1 I\{y \le f(x)\}\, \mathrm{d}y\, \mathrm{d}P_n(x) = \int_0^1 \Pr\{f(X_n) \ge y\}\, \mathrm{d}y,$$

and finish up the argument. You may prove and use that for a continuous function $f$, we have the inclusions $\overline{f^{-1}(B)} \subset f^{-1}(\bar{B})$ and $f^{-1}(B^\circ) \subset (f^{-1}(B))^\circ$, which entails the inclusion $\partial f^{-1}(B) \subset f^{-1}(\partial B)$.

**(e)** Finally, consider the setting where have c.d.f.'s $F_n(x) = P_n(-\infty, x]$ and $F(x) = P(-\infty, x]$. Show that for $P_n \Rightarrow P$ it is necessary and sufficient that $F_n \Rightarrow F$. To show the sufficiency part, express the open set $B \subset \mathbb{R}$ as a union of the disjoint open intervals $(a_1, b_1), (a_2, b_2), \ldots$, and deduce that $F_n \Rightarrow F$ implies (iii) of the Portmanteau theorem.

**Ex. 2.19** *More in the Portmanteau.* It is often convenient to have a smaller class of functions to work with than the bounded and continuous one. In this exercise we show that $P_n \Rightarrow P$ is equivalent to $\int f\, \mathrm{d}P_n \to \int f\, \mathrm{d}P$ for all bounded and continuous functions with bounded and continuous derivatives of all orders. This equivalence can be shown in several ways, the trick we use here is from Pollard (1984).

**(a)** As a preliminary that will be useful below, for $k = 1, 2, \ldots$ consider the functions $g_k(x,u) = |x-u|^k \exp(-\frac{1}{2}(x-u)^2)$. For a given $\varepsilon > 0$ define

$$K_k(u) = \begin{cases} |u+\varepsilon|^k \exp(-\frac{1}{2}(u+\varepsilon)^2), & u < -k^{1/2} - \varepsilon, \\ k^{k/2} \exp(-\frac{1}{2}k), & -k^{1/2} - \varepsilon \le u < k^{1/2} + \varepsilon, \\ |u-\varepsilon|^k \exp(-\frac{1}{2}(u-\varepsilon)^2), & k^{1/2} + \varepsilon \le u, \end{cases}$$

and show that $g_k(x,u) \le K_k(u)$ for all $x \in (-\varepsilon, \varepsilon)$ and that $\int K_k(u)\, \mathrm{d}u < \infty$.

**(b)** Let $f\colon \mathbb{R} \to \mathbb{R}$ be a bounded and continuous function. With $X$ having any distribution, $Z \sim \mathrm{N}(0,1)$ independent of $X$, and $\sigma > 0$, show that $\mathrm{E}\, f(X + \sigma Z) = \mathrm{E}\, f_\sigma(X)$ where (see Ex. C.25)

$$f_\sigma(x) = \frac{1}{\sqrt{2\pi}\sigma} \int f(u) \exp(-\tfrac{1}{2}(x-u)^2/\sigma^2)\, \mathrm{d}u.$$

Show that $f_\sigma(x)$ is bounded and continuous, and has bounded and continuous derivatives of all orders.

**(c)** Let $Z_1, Z_2, \ldots$ be a standard normal random variables with $Z_i$ independent of $X_i$ for each $i \ge 1$. Show that $X_n \to_d X$ if and only if $X_n + \sigma Z \to_d X + \sigma Z$ for each $\sigma > 0$.

**(d)** From (b) and (c), conclude that $X_n \to_d X$ if and only if $\mathrm{E}\, f(X_n) \to \mathrm{E}\, f(X)$ for all bounded and continuous functions, having bounded and continuous derivatives of all orders. This version of the Portmanteau theorem will be of use in Ex. 2.31.

**Ex. 2.20** *Continuous mapping for convergence in distribution.* Let $X_1, X_2, \ldots$ and $X$ be $k$-dimensional random vectors, and $g\colon \mathbb{R}^k \to \mathbb{R}^m$ a function that is continuous on a set $C \subset \mathbb{R}^k$ such that $\Pr(X \in C) = 1$. Suppose that $X_n \to_d X$. continuous mapping

**(a)** Assume that $g$ is continuous on all of $\mathbb{R}^k$, and show that $g(X_n) \to_d g(X)$.

**(b)** Now suppose that $C$ is a proper subset of $\mathbb{R}^k$, and let $F$ be a closed set. The challenge here is that while the sets $\{g(X_n) \in F\}$ and $\{X_n \in g^{-1}(F)\}$ are identical, the

set $g^{-1}(F)$ need not be closed, as $g$ is only assumed continuous on $C$. Use the fact that a set is included in its closure followed by the Portmanteau theorem to establish that

$$\limsup_n \Pr(X_n \in g^{-1}(F)) \leq \Pr(X \in \overline{g^{-1}(F)}).$$

It then remains to prove that $\Pr(X \in \overline{g^{-1}(F)}) \leq \Pr(X \in g^{-1}(F))$ as the reverse inequality is immediate from the inclusion of a set in its closure. To show this inequality, start by combining the continuity of $g$ on $C$ and the fact that a closed set contains all its limits points to show that $\overline{g^{-1}(F)} \cap C \subset g^{-1}(F)$, and take it from there.

**Ex. 2.21** *Skorokhod's theorem.* If $X_n \to_d X$, we can construct random variables $Y_n$ and $Y$ on a common probability space such that $Y_n \sim X_n$ and $Y \sim X$ and $Y_n(\omega) \to Y(\omega)$ for all $\omega$. This result is interesting in its own right, but also leads to simplifications of certain proofs.

**(a)** Let $F$ be a c.d.f. in $\mathbb{R}$ and suppose that the inverse function $F^{-1}\colon [0,1] \to \mathbb{R}$ exists, and for $U \sim \text{unif}(0,1)$ define $Y = F^{-1}(U)$. Show that $Y \sim F$. A more formal way of presenting this result is to consider the probability space $([0,1], \mathcal{B}, \text{Leb})$, i.e., the unit interval equipped with the Borel-$\sigma$-algebra and Lebesgue measure, and define $Y(\omega) = F^{-1}(\omega)$ for each $\omega \in (0,1)$. Then that $\text{Leb}(Y \leq y) = \text{Leb}(\{\omega \in (0,1)\colon Y(\omega) \leq y\}) = F(y)$, which is to say that $Y$ has distribution $F$.

**(b)** If $F^{-1}$ does not exist, the results from (a) still holds. Define

$$Y(\omega) = \inf\{x\colon F(x) \geq \omega\}, \quad \text{for } \omega \in (0,1).$$

Argue that $\{x\colon F(x) \geq \omega\} = [Y(\omega), \infty)$ since $F$ is a c.d.f., in particular nondecreasing and right-continuous; and that consequently $F(x) \geq \omega$ if and only if $Y(\omega) \leq x$. Show that $\text{Leb}(Y \leq y) = F(y)$ yet again.

**(c)** Let $F_n$ and $F$ be c.d.f.s on $\mathbb{R}$ and suppose that $F_n \Rightarrow F$. Construct $Y_n(\omega) = \inf\{x\colon F_n(x) \geq \omega)$ and $Y(\omega) = \inf\{x\colon F(x) \geq \omega)$ as above. Given $\varepsilon > 0$, choose a continuity point $x$ of $F$ such that $Y(\omega) - \varepsilon < x < Y(\omega)$, and for $\omega' > \omega$ choose a continuity point $y$ of $F$ such that $Y(\omega) \leq Y(\omega') < y < Y(\omega') + \varepsilon$. Show that $Y(\omega) \leq \liminf_n Y_n(\omega) \leq \limsup_n Y_n(\omega) \leq Y(\omega')$, and use that $Y$ only has countably many points of discontinuity to conclude that $Y_n(\omega) \to Y(\omega)$ for Lebesgue almost all $\omega \in (0,1)$. Finally, modify $Y_n$ and $Y$ on the null set to get pointwise convergence.

**(d)** Let $(X_n)_{n\geq 1}$ be an uniformly integrable sequence of random variables such that $X_n \to_d X$. Combine Skorokhod's theorem with Ex. 2.5(e) to show that $X$ is integrable and $\mathrm{E}\, X_n \to \mathrm{E}\, X$.

**(e)** Prove the continuous mapping theorem (see Ex. 2.20(b)) for random variables, this time using Skorokhod's theorem.

**(f)** For random variables, use Skorokhod's theorem to prove that (2.4) implies (i) and (iv) of the Portmanteau theorem .

**Ex. 2.22** *The Cramér–Slutsky rules.* The usefulness of the three results in (c) below, together known as the Cramér–Slutksy rules, will become abundantly clear as we progress.

**(a)** Show that if $X_n \to_d X$, and $Y_n - X_n \to_{\text{pr}} 0$, then also $Y_n \to_d X$. This says that variables which are essentially close, for growing $n$, have identical limit distributions.

**(b)** Show that if $X_n$ and $Y_n$ are sequences of random vectors such that $X_n \to_d X$ and $Y_n \to_{\text{pr}} a$, for a random variable $X$ and a constant $a$, then $(X_n, Y_n) \to_d (X, a)$.

Cramér–Slutsky

**(c)** Show that if $X_n \to_d X$ and $Y_n \to_{\text{pr}} a$, as above, then

(i) $X_n + Y_n \to_d X + a$;
(ii) $X_n Y_n \to_d Xa$;
(iii) $X_n / Y_n \to_d X/a$, provided $a \neq 0$.

Explain why rules (i) and (ii) also hold when $Y_n$ and $a$ are matrices.

**(d)** Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli($p$). Looking back at Ex. 2.7(c) and Ex. 2.15(c), show that $\sqrt{n}(\bar{X}_n - p)/\widehat{\sigma}^2 \to_d \text{N}(0,1)$, where $\widehat{\sigma}^2 = \widehat{p}(1-\widehat{p})$.

**(e)** Suppose $X_n \to_d X$, in dimension $p$, and that $A_n \to_{\text{pr}} A$, a symmetric matrix, in the sense that $A_{n,i,j} \to_{\text{pr}} A_{i,j}$ for each component $(i,j)$. Show that $X_n^{\text{t}} A_n X_n \to_d X^{\text{t}} A X$.

**(f)** Assume $X_n \to_d \text{N}_p(0, \Sigma)$, with a positive definite $\Sigma$, and that there is a consistent estimator $\widehat{\Sigma}$ for this matrix. Show that $X_n^{\text{t}} \widehat{\Sigma}^{-1} X_n \to_d \chi^2_p$.

**(g)** The claim in (b) is in general not true if the constant limit in probability $a$ is replaced with a (nondegenerate) random variable. Construct an example where $X_n \to_d X$ and $Y_n \to_{\text{pr}} Y$, for a random variable $Y$, but $(X_n, Y_n)$ does not converge in distribution.

**Ex. 2.23** *Stochastic $O_{\text{pr}}$ and $o_{\text{pr}}$ symbols.* The symbols $O_{\text{pr}}(\cdot)$ and $o_{\text{pr}}(\cdot)$ are the stochastic analogues of $O(\cdot)$ and $o(\cdot)$. Let $X_n$ be a sequence of random variables. Then $X_n = o_{\text{pr}}(1)$ if $X_n \to_{\text{pr}} 0$; and $X_n = O_{\text{pr}}(1)$ if $X_n$ is bounded in probability, i.e., for any $\varepsilon > 0$ there is an $M$ such that $\Pr(|X_n| > M) < \varepsilon$ for all $n$. If $r_n$ is another sequence of random variables, then $X_n = o_p(r_n)$ if $X_n = r_n o_{\text{pr}}(1)$, and $X_n = O_{\text{pr}}(r_n)$ if $X_n = r_n O_{\text{pr}}(1)$. Typically, $r_n$ will be a sequence of constants, $r_n = n^{-1}$ and $r_n = n^{-1/2}$ for example, appear frequently in parametric statistics.

**(a)** Show that $X_n = o_{\text{pr}}(r_n)$ if and only if $\Pr(|X_n| \geq \varepsilon r_n) \to 0$ for each $\varepsilon > 0$.

**(b)** Show that $X_n = O_{\text{pr}}(r_n)$ if and only if for any $\varepsilon > 0$ there are $M$ and $n_0$ such that $\Pr(|X_n| \geq M r_n) < \varepsilon$ for all $n \geq n_0$.

**(c)** Show that if $\sqrt{n} X_n \to_d \text{N}(0, \sigma^2)$, then $X_n = O_{\text{pr}}(1/\sqrt{n})$. More generally, show that if $n^\alpha X_n \to_d Z$, then $X_n = O_{\text{pr}}(n^{-\alpha})$.

**(d)** Show that $o_{\text{pr}}(1) + o_{\text{pr}}(1) = o_{\text{pr}}(1)$; $o_{\text{pr}}(1) + O_{\text{pr}}(1) = O_{\text{pr}}(1)$; $o_{\text{pr}}(1) O_{\text{pr}}(1) = o_{\text{pr}}(1)$; $1/(1 + o_{\text{pr}}(1)) = O_{\text{pr}}(1)$; $o_{\text{pr}}(O_{\text{pr}}(1)) = o_{\text{pr}}(1)$; and $o_{\text{pr}}(1)/(1 + o_{\text{pr}}(1)) = o_{\text{pr}}(1)$.

**(e)** Let $r \colon \mathbb{R}^k \to \mathbb{R}$ be a function such that $g(0) = 0$, and $X_n$ a sequence of random vectors. Show that if $r(x) = o(\|x\|^p)$ as $x \to 0$, then $R(X_n) = o_p(\|X_n\|^p)$; and that if $r(x) = O(\|x\|^p)$ as $x \to 0$, then $r(X_n) = O_p(\|X_n\|^p)$.

**(f)** Let $g \colon \mathbb{R} \to \mathbb{R}$ be a function such that $g(\theta) - g(\theta_0) = g'(\theta_0)(\theta - \theta_0) + o(|\theta - \theta_0|)$ as $\theta \to \theta_0$, and let $\widehat{\theta}_n$ be an estimator sequence such that $\sqrt{n}(\widehat{\theta}_n - \theta_0)$ converges in distribution. Show that $\sqrt{n}(g(\widehat{\theta}_n) - g(\theta_0)) = g'(\theta_0)\sqrt{n}(\widehat{\theta}_n - \theta_0) + o_p(1)$ using the rules about $o_{\text{pr}}(\cdot)$ and $O_{\text{pr}}(\cdot)$ that you have derived above in the above exercises.

**(g)** Suppose that $X_n = o_{\text{pr}}(n^{-\alpha})$ for some $\alpha > 0$, and that $n^\alpha X_n$ is dominated by an integrable random variable $Y$, or is uniformly integrable (which is weaker, as we recall from Ex. 2.5). Show that $\text{E}\, X_n = o(n^{-\alpha})$.

**Ex. 2.24** *Tightness.* Consider the sequence $(X_n)_{n \geq 1}$ of random variables with distribution $\Pr(X_n = x) = 1/3$ for $x = 0, 1/2, n$. You may verify that the sequence $F_n(x) = \Pr(X_n \leq x)$ of distribution functions converges pointwise the limiting function,

$$G(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{3}, & 0 \leq x < \frac{1}{2}, \\ \frac{2}{3}, & x \geq \frac{1}{2}. \end{cases}$$

This function, albeit a limit of distributions functions, is *not* itself a distribution function. The problem is that $G(\infty) = 2/3$, meaning that one third of the probability mass of $F_n(x)$ has escaped to inifinity. Tightness is a condition ensuring that a limit of distribution functions is itself a distribution function. A sequence $X_1, X_2, \ldots$ of random variables is tight if for any $\varepsilon > 0$ there exists a constant $K$ so that

tightness

$$\Pr(|X_n| > K) < \varepsilon, \quad \text{for all } n.$$

A tight sequence of random variables is also said to be bounded in probability.

**(a)** Show that $(X_n)_{n\geq 1}$ is tight if and only if for any $\varepsilon > 0$ there are $a$ and $b$ such that

$$\Pr\{X_n \in (a, b]\} > 1 - \varepsilon, , \quad \text{for all } n,$$

and that this is the same as requiring that for any $\varepsilon > 0$ we can find $a$ and $b$ such that $F_n(a) < \varepsilon$ and $F_n(b) > 1 - \varepsilon$ for all $n$, in terms of the c.d.f.s. $F_n$ of $X_n$.

**(b)** Show that $(X_n)_{n\geq 1}$ is tight if and only if for any $\varepsilon > 0$ there is a $K > 0$ so that $\limsup_n \Pr(|X_n| > K) < \varepsilon$.

**(c)** Show that $(X_n)_{n\geq 1}$ is tight if and only if for any sequence of constants $a_1, a_2, \ldots \geq 0$ tending to zero, $a_n Y_n \to_p 0$.

**(d)** Suppose that for some $\delta > 0$ there is an $M > 0$ and an $n_0 \geq 1$ such that $\mathrm{E}\,|X_n|^\delta \leq M$ for all $n \geq n_0$. Show that $(X_n)_{n\geq 1}$ is tight.

**(e)** Show that if $(X_n)_{n\geq 1}$ is uniformly integrable, then it is tight.

**(f)** Let $(X_n)_{n\geq 1}$ be a tight sequence or random variables with c.d.f.s $(F_n)_{n\geq 1}$. Suppose that there is a non-decreasing and right-continuous function $F$ such that $F_n(x) \to F(x)$ at each continuity point of $F$. Show that $F$ must be a c.d.f., and hence $X_n \to_d X$, where the random variable $X$ has distribution $F$.

**Ex. 2.25** *Helly's theorem* ...says that for any sequence $(F_n)_{n\geq 1}$ of c.d.f.'s there is a non-decreasing and right-continuous function $F$ and a subsequence $(F_{n_k})_{k\geq 1}$ such that $F_{n_k}(x) \to F(x)$ at each $x$ where $F$ is continuous. The limit function $F$ need not be a c.d.f., as it may lack one of the three defining properties of a c.d.f. We know from Ex. 2.24(f), however, that if $(F_n)_{n\geq 1}$ is a tight sequence, then $F$ is a bona fide c.d.f. In this exercise we prove Helly's theorem for c.d.f.'s, first on the real line, then on $\mathbb{R}^k$.

**(a)** Let $\mathbb{Q} = \{q_1, q_2, \ldots\}$ be the rational numbers, and consider the infinite array

$$\begin{array}{ccccc} F_1(q_1) & F_2(q_1) & F_3(q_1) & \ldots \\ F_1(q_2) & F_2(q_2) & F_3(q_2) & \ldots \\ F_1(q_3) & F_2(q_3) & F_3(q_3) & \ldots \\ \vdots & \vdots & \vdots & \end{array}$$

Since $F_n(q_1)$ lies between zero and one for all $n$, the the Bolzano–Weierstrass theorem says that there is a convergent subsequence $(F_{n_{1,k}}(q_1))_{k\geq 1}$, say $F_{n_{1,k}}(q_1) \to G(q_1)$ as $k \to \infty$. The sequence $(F_{n_{1,k}}(q_2))_{k\geq 1}$ is also bounded between zero and one, so we can extract a further subsequence $(n_{2,k})_{k\geq 1} \subset (n_{1,k})_{k\geq 1}$ such that $F_{n_{2,k}}(q_2)$ converges to a limit $G(q_2)$ as $k \to \infty$. Continuing in this manner we get the array

the diagonal method

$$\begin{array}{ccccc} n_{1,1} & n_{1,2} & n_{1,3} & \ldots \\ n_{2,1} & n_{2,2} & n_{2,3} & \ldots \\ n_{3,1} & n_{3,2} & n_{3,3} & \ldots \\ \vdots & \vdots & \vdots & \end{array}$$

where each row is included in the row above, i.e., $(n_{j,k})_{k\geq 1} \subset (n_{j-1,k})_{k\geq 1}$ for $j = 2, 3, \ldots$. Argue that the diagonal sequence $n_k = n_{k,k}$ is such that $F_{n_k}(q) \to G(q)$ for each $q \in \mathbb{Q}$.

**(b)** Define the function

$$F(x) = \inf\{G(q) \colon q > x, q \in \mathbb{Q}\}.$$

Since $F(x)$ is an infinum over an set that shrinks with growing $x$, $F(x)$ is non-decreasing. Also, as $F(x)$ is an infinum we can for any $x$ and any $\varepsilon > 0$ find an $s \in \mathbb{Q}$ with $s > x$ such that $G(s) < F(x) + \varepsilon$. Use this to prove that $F(x)$ must be right-continuous.

**(c)** Let $x$ be a continuity point of $F$ (see Ex. C.2(c) for the why such points exist). Argue that for any $\varepsilon > 0$ we can find $y \in \mathbb{R}$ and $s, r \in \mathbb{Q}$ such that $y < r < x < s$ and

$$F(x) - \varepsilon \leq G(r) \leq G(s) \leq F(x) + \varepsilon.$$

Use these inequalities to prove that $\lim_k F_{n_k}(x) = F(x)$, proving Helly's theorem.

**(d)** Let $\mathbb{Q}^k = \{q_1, q_2, \ldots\}$ be an enumeration of the $k$-dimensional vectors with rational components. Mimic the one-dimensional proof and generalise Helly's theorem to sequences of c.d.f.'s on $\mathbb{R}^k$. The changes you need to make are essentially notational.

**Ex. 2.26** *Prokhorov's theorem...* comprises (a) and (b) of this exercise. With Helly's theorem at hand, part (a) was essentially proved in Ex. 2.24(f), but we include it here for completeness.

**(a)** If $(X_n)_{n\geq 1}$ is tight, then there is a subsequence $(n_k)_{k\geq 1}$ such that $X_{n_k} \to_d X$ for some random variable $X$.

**(b)** If $X_n \to_d X$ for some random variable $X$, then $(X_n)_{n\geq 1}$ is tight.

**(c)** The importance of Prokhorov's theorem stems from the following lemma, which you are to show: The sequence $(X_n)_{n\geq 1}$ converges in distribution to $X$ if and only if each subsequence $(X_{n_k})_{k\geq 1}$ contains a further subsequence $(X_{n_{k_j}})_{j\geq 1}$ that converges in distribution to $X$. The proof is similar to the analogous result from real analysis saying that a sequence $(a_n)_{n\geq 1} \subset \mathbb{R}$ converges to $a$ if and only if each subsequence of $(a_n)_{n\geq 1}$ contains a further subsequence converging to $a$. Here, if $a_n \to a$, then clearly every subsequence also converges to $a$. For the converse, assume that $a_n$ does not converge to $a$ and extract a subsequence with no convergent further subsequence.

**(d)** Suppose that $(X_n)_{n\geq 1}$ is tight, and that every subsequence of $X_{n_k}$ that converges weakly at all, converges to the same random variable $X$. Show that then $X_n \to_d X$.

**Ex. 2.27** *Characteristic functions converging to a characteristic function.* Let $(X_n)_{n\geq 1}$ be a sequence of random variables with characteristic functions $\varphi_1(t), \varphi_2(t), \ldots$. Suppose that $\varphi_n(t) \to \varphi(t)$, and that we recognise $\varphi(t)$ as the characteristic function of some random variable $X$. In such cases we can conclude that $X_n \to_d X$ without further ado.

**(a)** Suppose that $X_n$ and $X$ have characteristic functions $\varphi_n(t)$ and $\varphi(t)$, respectively, and that $\varphi_n(t) \to \varphi(t)$, where $\varphi(t)$ is a characteristic function. As an auxiliary assumption, suppose that $\varphi_n(t)$ is dominated by a function $g(t)$ such that $\int g(t)\,\mathrm{d}t < \infty$. By Ex. C.25(h) the $X_n$ then have densities

$$f_n(x) = \frac{1}{2\pi} \int \exp(-itx)\varphi_n(t)\,\mathrm{d}t.$$

Show that under these assumptions $X_n \to_d X$.

**(b)** Next, we reduce the general case to the special case studied in (a), by using the result from Ex. 2.19(c). Suppose that $X_n$ and $X$ have characteristic functions $\varphi_n(t)$ and $\varphi(t)$, respectively, and that $\varphi_n(t) \to \varphi(t)$. Let $Z$ be a standard normal random variable, and $\sigma$ some positive constant. Show that $X_n + \sigma Z \to_d X + \sigma Z$, and conclude that $\varphi_n(t) \to \varphi(t)$ implies $X_n \to_d X$.

**Ex. 2.28** *Lévy's continuity theorem and some more.* In Ex. 2.27 we made two crucial assumptions: that $\varphi_n(t) \to \varphi(t)$, *and* that the limiting function $\varphi$ is indeed a characteristic function. What if we drop the second assumption and only assume that the characteristic functions $\varphi_n$ converge to *some* function $\varphi$? Lévy's continuity theorem says that if this limiting function is continuous at zero, then it must be a characteristic function. In more terse prose, if $\varphi_n(t) = \mathrm{E}\exp(itX_n)$ and $\varphi_n(t) \to \varphi(t)$ with $\varphi$ some function continous at zero, then $\varphi$ is the characteristic function of some random variable $X$, and $X_n \to_d X$. The key to this theorem is that the continuity of $\varphi$ at zero entails that $(X_n)_{n\geq 1}$ is tight. In fact, given that $\varphi_n(t) \to \varphi(t)$ for some function $\varphi(t)$, there is equivalence between the following three statements:

(i) $\varphi(t)$ is a characteristic function,

(ii) the $X_n$ sequence is tight;

(iii) $\varphi(t)$ is continuous at zero;

and, as we know from Ex. 2.27, the convergence $\varphi_n(t) \to \varphi(t)$ combined with (iii) implies that convergence in distribution takes place.

The implication (i)⇒(iii) is immediate from the uniform continuity of characteristic functions, see Ex. C.24(a); and (i)⇒(ii) follows from combining Ex. 2.26(b) and Ex. 2.27. In (a) of this exercise we prove (ii)⇒(i); while (iii)⇒(i), which is Lévy's continuity theorem, is proven in (b)–(c).

**(a)** Suppose that $X_n$ has characteristic function $\varphi_n$ for $n = 1, 2, \ldots$, and that $(X_n)_{n\geq 1}$ is tight. Use the lemma in Ex. 2.26(c) to show that if $\varphi_n(t) \to \varphi(t)$ for *some* function $\varphi$, then $\varphi(t)$ must be a characteristic function.

**(b)** Start by applying Fubini's theorem (see Ex. B.18) to show that if $X$ has characteristic function $\varphi$ and cumulative distribution function $F$, then

$$\int_{-\varepsilon}^{\varepsilon} \{1 - \varphi(t)\}\,\mathrm{d}t = 2\varepsilon \int \big(1 - \frac{\sin(x\varepsilon)}{x\varepsilon}\big)\,\mathrm{d}F(x).$$

In particular, the integral of $\varphi(t)$ on a symmetric interval around zero is really a real number, that is, the complex component disappears. Deduce that

$$\frac{1}{\varepsilon}\int_{-\varepsilon}^{\varepsilon} \{1 - \varphi(t)\}\,\mathrm{d}t \geq 2 \int_{|x\varepsilon|\geq c} \big(1 - \frac{\sin(x\varepsilon)}{x\varepsilon}\big)\,\mathrm{d}F(x) \geq 2(1 - 1/c)\Pr\{|X| \geq c/\varepsilon\},$$

with the value $c = 2$ yielding the tail inequality

$$\Pr\{|X| \geq 2/\varepsilon\} \leq \frac{1}{\varepsilon}\int_{-\varepsilon}^{\varepsilon} \{1 - \varphi(t)\}\,\mathrm{d}t. \tag{2.8}$$

**(c)** If we now have a collection of random variables, where their characteristic functions have approximately the same level of smoothness around zero, then we should get tightness from (2.8). Assume that $X_1, X_2, \ldots$ have characteristic functions $\varphi_1, \varphi_2, \ldots$, and

that $\varphi_n(t)$ converges to *some* function $\varphi(t)$ that is continuous at zero. For a given $\varepsilon' > 0$, find $\varepsilon > 0$ such that $|1 - \varphi(t)| \leq \varepsilon'$ for $|t| \leq \varepsilon$. Show that

$$\limsup_{n\to\infty} \Pr\{|X_n| \geq 2/\varepsilon\} \leq \frac{1}{\varepsilon} \int_{-\varepsilon}^{\varepsilon} \{1 - \varphi(t)\}\, \mathrm{d}t \leq 2\varepsilon'.$$

We've hence found a broad interval, namely $[-2/\varepsilon, 2/\varepsilon]$, inside which each $X_n$ lies, with high enough probability, which means that $(X_n)_{n\geq 1}$ is tight, by Ex. 2.24(b).

**(d)** As a bonus, show that if $X_1, X_2, \ldots$ is a tight sequence of random variables with characteristic functions $\varphi_1, \varphi_2, \ldots$, then $(\varphi_n)_{n\geq 1}$ is uniformly equicontinuous. That is, for each $\varepsilon > 0$ there is a $\delta > 0$ such that $|t - s| < \delta$ implies $|\varphi_n(t) - \varphi_n(s)| < \varepsilon$ for all $n$.

**Ex. 2.29** *Showing convergence in distribution via moment generating functions.* Here we spell out an m.g.f. analogue of the Lévy continuity theorem, see Ex. 2.28. Let $(X_n)_{n\geq 1}$ have m.g.f.s $\{M_n(t)\}_{n\geq 1}$ and suppose that all these exist on the same interval $[-a, a]$.

**(a)** Expand on Ex. 2.16(a) to show that

$$\Pr(|X_n| \geq K) \leq \exp(-Ka)\{M_n(a) + M_n(-a)\}, \quad \text{for all } n.$$

Show that if $M_n(t)$ converges to *some* function $M(t)$ for all $t$ in some interval around zero that contains $[-a, a]$, then $(X_n)_{n\geq 1}$ is tight.

**(b)** Deduce from (a) that $X_n \to_d X$, where $X$ is a random variable with m.g.f. $M(t)$. Note that since $x \mapsto \exp(tx)$ is not bounded, this last part is more involved than in the characteristic function case, cf. Ex. 2.28(a). [xx bit is it correct? Perhaps must also assume that E $\exp(tX) < \infty$ in interval around zero. xx]

**(c)** For illustration, consider $X_1, \ldots, X_n$ being independent 0-1 Bernoulli variables with $p_i = \Pr(X_i = 1)$. Show that if $\max_{i\leq n} p_i \to 0$ and $\sum_{i=1}^n p_i \to \lambda$, then $\sum_{i=1}^n X_i \to_d \mathrm{Pois}(\lambda)$. Note that this generalises the classical binomial to Poisson theorem, see Ex. 2.8, and also that proving this result directly from the point probabilities would be difficult.

**(d)** Suppose $X_1, X_2, \ldots$ are independent Bernoulli with $p_i = i/n$, and consider $Y_n(t) = \sum_{i\leq t\sqrt{n}} X_i$. Show that $Y_n(t) \to_d \mathrm{Pois}(\frac{1}{2}t^2)$. The limit is actually a full Poisson process in $t$, with independent increments (see Ch. 10).

**Ex. 2.30** *Showing convergence in two steps.* Suppose one wishes to prove that $X_n \to_d X$, but that technical issues make it easier to first prove that an approximation to $X_n$ converges to an approximation to $X$. With a suitable extra condition this might suffice. Here we suppose that the random elements involved take values in a common (separable) metric space $(\mathcal{X}, d)$.

**(a)** For the approximations $A_{n,k}$ to $X_n$ and $A_k$ to $X$, suppose (i) that $A_{n,k} \to_d A_k$, for each $k$; (ii) that $A_k \to_d X$ as $k \to \infty$; and also (iii) that

$$\lim_{k\to\infty} a_k(\varepsilon) \to 0 \text{ for each } \varepsilon > 0, \quad \text{where } a_k(\varepsilon) = \limsup_{n\to\infty} \Pr(d(X_n, A_{n,k}) \geq \varepsilon) = 0.$$

Consider the closed sets $F$ and $F^\varepsilon = \{x\colon d(x, F) \leq \varepsilon\}$, and use the Portmanteau theorem to show that $X_n \to_d X$.

**(b)** Suppose somebody clever has managed to prove the CLT for bounded i.i.d. variables (for example using m.g.f.s): If $A_1, A_2, \ldots$ are i.i.d. with mean zero and variance one, and $|A_i| \leq k$, then $Z_n = \sqrt{n}\bar{A}_n \to_d \mathrm{N}(0, 1)$. How can you then prove that the CLT is valid also for unbounded i.i.d. random variables, as long at only their variance is finite?

## Central limit theorems

**Ex. 2.31** *The central limit theorem, Lindeberg's proof.* Let $X_1, X_2, \ldots$ be i.i.d. random variables with mean zero and unit variance, define $X_{n,i} = X_i/\sqrt{n}$, and let $Z \sim \mathrm{N}(0,1)$. We will show that $\mathrm{E}\, f(\sum_{i=1}^n X_{n,i}) \to \mathrm{E}\, f(Z)$ for all bounded and continuous functions with bounded and continuous derivatives of all orders, which, by Ex. 2.19 is equivalent to $\sum_{i=1}^n X_{n,i} \to_d \mathrm{N}(0,1)$. Introduce $Z_{n,i} = Z_i/\sqrt{n}$ for $i = 1, 2, \ldots, n$ and $n \geq 1$, where the $Z_1, Z_2, \ldots$ are i.i.d. standard normal random variables independent of the $X_1, X_2, \ldots$. In particular, $\sum_{i=1}^n Z_{n,i} \sim \mathrm{N}(0,1)$ by Ex. C.25, for example.

**(a)** What is in essence Lindeberg's idea, back in 1922, was to show that $\mathrm{E}\, f(\sum_{i=1}^n X_{n,i}) - \mathrm{E}\, f(\sum_{i=1}^n Z_{n,i})$ tends to zero by replacing the summands $X_{n,i}$ by the Gaussiand summands $Z_{n,i}$, one by one. Show that by so doing the difference $f(\sum_{i=1}^n X_{n,i}) - f(\sum_{i=1}^n Z_{n,i})$ can be expressed as the (telescoping) sum

$$\begin{aligned} f(\sum_{i=1}^n X_{n,i}) - f(\sum_{i=1}^n Z_{n,i}) &= \sum_{k=1}^n \{f(A_k) - f(A_{k-1})\} \\ &= \sum_{k=1}^n \{f(R_k + X_{n,k}) - f(R_k + Z_{n,k})\}, \end{aligned}$$

where $A_k = \sum_{i=1}^k X_{n,i} + \sum_{i=k+1}^n Z_{n,i}$ and $R_k = \sum_{i=1}^{k-1} X_{n,i} + \sum_{i=k+1}^n Z_{n,i}$, so that $A_n = \sum_{i=1}^n X_{n,i}$ and $A_0 = \sum_{i=1}^n Z_{n,i}$.

**(b)** Let $f$ be a bounded and continuous functions with bounded and continuous derivatives up to and including the third order. Show that for any $\varepsilon > 0$ there are $\delta > 0$ and $K$ such that

$$|f(x+y) - f(x) - f'(x)y - \tfrac{1}{2} f''(x) y^2| \leq \varepsilon y^2 I(|y| < \delta) + K y^2 I(|y| \geq \delta),$$

uniformly in $x$ and $y$.

**(c)** Verify that for each $k = 1, \ldots, n$ and $n \geq 1$, $X_{n,k}$ is independent of $R_k$ and that $Z_{n,k}$ is independent of $R_k$. Use this and the findings from (a) and (b) to show that given $\varepsilon > 0$ we can choose $\delta > 0$ and find $K$ such that

$$|\mathrm{E}\, f(\sum_{i=1}^n X_{n,i}) - \mathrm{E}\, f(\sum_{i=1}^n Z_{n,i})| \leq 2\varepsilon + K \sum_{k=1}^n \mathrm{E}\, X_{n,k}^2 I_{|X_{n,i}| \geq \delta} + K \sum_{k=1}^n \mathrm{E}\, Z_{n,k}^2 I_{|Z_{n,i}| \geq \delta}.$$

Please conclude, and you will have shown the CLT for i.i.d. random variables.

**(d)** Suppose $Y_1, Y_2, \ldots$ are i.i.d. with finite mean $\xi$ and standard deviation $\sigma$. Show that the random sum, normalised to have mean zero and variance one, i.e.,

$$\Big(\sum_{i=1}^n Y_i - n\xi\Big)/\sqrt{n\sigma^2} = \sqrt{n}(\bar{Y}_n - \xi)/\sigma,$$

tends to the $\mathrm{N}(0,1)$ in distribution.

**Ex. 2.32** *The Lindeberg theorem.* We now extend the central limit theorem to independent random variables that do not necessarily have the same distribution.

Lindeberg CLT

**(a)** Let $X_1, X_2, \ldots$ be independent mean zero random variables with variances $\sigma_1^2, \sigma_2^2 \ldots$, and form $X_{n,j} = X_j/B_n$, where $B_n^2 = \sigma_1^2 + \cdots + \sigma_n^2$. In view of the remainder term in Ex. 2.31(c), the Lindeberg condition is natural: Assume that

$$\sum_{j=1}^n \mathrm{E}\, X_{n,j}^2 I(|X_{n,j}| \geq \varepsilon) \to 0, \quad \text{as } n \to \infty, \text{ for each } \varepsilon > 0. \tag{2.9}$$

Show that the Lindeberg condition implies that $\max_{i\le n}\sigma_i^2/B_n^2 \to 0$ as $n\to\infty$; and proceed to show that $\sum_{j=1}^n X_{n,j} \to_d \mathrm{N}(0,1)$ by making the appropriate modifications to the proof for the i.i.d. case studied in Ex. 2.31.

**(b)** Suppose $X_1, X_2, \ldots$ are i.i.d. with mean zero and variance $\sigma^2$, then form from these $Z_n = \sum_{i=1}^n iX_i/(\sum_{i=1}^n i^2)^{1/2}$. Show that $Z_n \to_d \mathrm{N}(0,\sigma^2)$. Generalise.

**Ex. 2.33** *Convergence in distribution for bounded variables.* Suppose variables $X_n$ all live on a bounded interval, which we for simplicity and without loss of generality below take to be $[0,1]$.

**(a)** With $X$ a candidate limit variable, also on $[0,1]$, we know from the Portmanteau theorem of Ex. 2.18 that $X_n \to_d X$ if and only if $\mathrm{E}\,g(X_n)\to \mathrm{E}\,g(X)$ for each bounded continuous $g$. Show via the Weierstraß polynomial approximation theorem of Ex. 2.17 that this is also equivalent to $\mathrm{E}\,X_n^k \to \mathrm{E}\,X^k$ for each $k$.

**(b)** Then show that this again is equivalent to convergence of the m.g.f.s, i.e. $M_n(t)\to M(t)$ for each $t$. For an illustration with an infinite sum of Bernoullis, check Story #83.

**(c)** We saw in Ex. 2.7 that if $X_n$ is uniform on $1/n, 2/n, \ldots, n/n$, then $X_n \to_d U$, the uniform on the unit interval. Show from this that $s_n(k) = (1/n^{k+1})(1^k+2^k+\ldots+n^k) \to 1/(k+1)$ for each $k$.

**Ex. 2.34** *Proving the CLT with moment generating functions.* Here we prove a central limit theorem for i.i.d. random variables $X_1, X_2, \ldots$ whose m.g.f. $M(t) = \mathrm{E}\exp(tX_1)$ exist on an interval around zero. See Ex. C.21 for what this assumption entails about the random variables. Throughout, we assume that $\mathrm{E}\,X_1 = 0$ and $\mathrm{Var}\,X_1 = \sigma^2$. The CLT proof via characteristic functions is clearly more general, but in various cases it is simply more convenient to give proofs via m.g.f.s.

**(a)** Show that $M(t) = 1 + \frac12\sigma^2t^2 + o(t^2)$ as $t\to 0$.

**(b)** Show that $\sqrt{n}\bar X_n = n^{-1/2}\sum_{i=1}^n X_i$ has moment generating function of the form

$$M_n(t) = M(t/\sqrt{n})^n = \{1 + \tfrac12\sigma^2t^2/n + o(t^2/n)\}^n,$$

and conclude that $M_n(t)\to\exp(\frac12t^2\sigma^2)$ as $n\to\infty$. Explain, in view of Ex. 2.29, that this proves the CLT, under these m.g.f. conditions.

**Ex. 2.35** *Proving the Stirling formula via the CLT.* The approximation formula

the Stirling approximation

$$n! \doteq n^ne^{-n}\sqrt{2\pi n}, \quad \text{in the sense of} \quad \lim_{n\to\infty}\frac{n!}{n^n\exp(-n)(2\pi n)^{1/2}} = 1,$$

is a famous one. Here we shall prove this formula via the CLT, specifically for Poisson variables and gamma variables.

named after J. Stirling (1692–1770), though A. de Moivre stated the formula earlier

**(a)** If $X_n\sim\mathrm{Pois}(n)$, show that $Z_n = (X_n-n)/\sqrt{n}\to_d Z$, a standard normal. Show that $\exp(-n)(1+n/1+n^2/2!+\cdots+n^n/n!)\to\frac12$.

**(b)** Write $p_n(j) = \Pr(X_n=j)$ for the Poisson probabilities. Work out that $\mathrm{E}\max(X_n-n,0) = \sum_{j\ge n}(j-n)p_n(j) = np_n(n)$, perhaps by discovering a telescoping series. Show from this that

$$\mathrm{E}\max(0,Z_n) = \sum_{j\ge n}\frac{j-n}{\sqrt{n}}\exp(-n)\frac{n^j}{j!} = \sqrt{n}\exp(-n)n^n/n! \to \mathrm{E}\max(0,Z),$$

and that the right hand side is $1/(2\pi)^{1/2}$. Deduce Stirling from this.

**(c)** Whereas the arguments above lead to a clean proof of Stirling, attempt also the following. With $\varepsilon$ small, argue that

$$\sum_{n\le j\le n+\varepsilon\sqrt{n}} \exp(-n)\frac{n^j}{j!} \doteq \frac{1}{(2\pi)^{1/2}}\varepsilon.$$

Prove Stirling from this.

**(d)** To learn more, from the arguments above, show that

$$a_n(c) = \mathrm{E}\, Z_n\, I(0\le Z_n\le c) = \sum_{n\le j\le n+c\sqrt{n}} \frac{j-n}{\sqrt{n}}\exp(-n)\frac{n^j}{j!} \to a(c),$$

where $a(c) = \mathrm{E}\, Z\, I(0\le Z\le c)$; the previous case corresponds to $c=\infty$. For the limit, show that $a(c) = \phi(0)-\phi(c) = (2\pi)^{-1/2}\{1-\exp(-\frac12 c^2)\}$. For the left hand side, write $[x]$ for the integer value of $x$, and use the telescoping series trick to show that

$$a_n(c) = \frac{1}{\sqrt{n}}\{np_n(n) - np_n([n+c\sqrt{n}])\} = \sqrt{n}\exp(-n)\frac{n^n}{n!}\Big\{1-\frac{n^{[c\sqrt{n}]}}{(n+1)\cdots([n+c\sqrt{n}])}\Big\}.$$

Deduce from this that

$$\frac{n^{[c\sqrt{n}]}}{(n+1)\cdots([n+c\sqrt{n}])} \to \exp(-\tfrac12 c^2),$$

for any $c>0$. You may also try to show this directly, but here we get it from the Poisson and probability theory.

**(e)** Consider now gamma distributed variables $X_n\sim \mathrm{Gam}(n,1)$, with densities $g_n(x) = \Gamma(n)^{-1}x^{n-1}\exp(-x)$ for $x$ positive. Explain via the CLT that $Z_n = (X_n-n)/\sqrt{n}\to_d Z$, a standard normal. Show from this that we for each $(a,b)$ have

$$L_n(a,b) = \mathrm{E}\, Z_n\, I(Z_n\in[a,b]) = \int_{n+a\sqrt{n}}^{n+b\sqrt{n}} \frac{x-n}{\sqrt{n}} g_n(x)\,\mathrm{d}x \to L(a,b),$$

where $L(a,b) = \int_a^b z\phi(z)\,\mathrm{d}z = \phi(a)-\phi(b)$.

**(f)** Show that $xg_n(x) = ng_{n+1}(x)$, and that this leads to

$$L_n(a,b) = \sqrt{n}\int_{n+a\sqrt{n}}^{n+b\sqrt{n}} (g_{n+1}-g_n)\,\mathrm{d}x.$$

With $G_n$ the c.d.f. for $g_n$, show via expressions found in Ex. 1.9 that $G_n(t)-G_{n+1}(t) = \exp(-t)t^n/n!$. From this deduce that

$$L_n(0,\infty) = \sqrt{n}\int_n^\infty (g_{n+1}-g_n)\,\mathrm{d}x = \sqrt{n}\exp(-n)n^n/n! \to 1/(2\pi)^{1/2},$$

proving the Stirling again. Working with $L_n(0,b)$, deduce also that

$$\exp(-b\sqrt{n})(1+b/\sqrt{n})^n \to \exp(-\tfrac12 b^2).$$

This may be verified independently; the present point is that these formulae, including the Stirling, follow from the CLT and probability theory for sums of gammas.

**Ex. 2.36** *Two useful lemmas.* In proving the central limit theorems below, the following two lemmas will be useful.

**(a)** Demonstrate that with $z_{n,1}, z_{n,2}, \ldots$ a sequence of real numbers coming closer to zero, we have $\prod_{i=1}^n (1+z_{n,i}) \to \exp(z)$, provided (i) $\sum_{i=1}^n z_{n,i} \to z$; (ii) $\max_{i\le n} |z_{n,i}| \to 0$; and (iii) $\sum_{i=1}^n |z_{n,i}|$ stays bounded. It may be helpful to show first that

$$\log(1+x) = x - \tfrac{1}{2}x^2 + \tfrac{1}{3}x^3 - \cdots = x + x^2 K(x),$$

with $K(x)$ a continuous function such that $|K(x)| \le 1$ for $|x| \le \frac{1}{2}$, and $K(x) \to -\frac{1}{2}$ when $x \to 0$. These statements are valid also when the $z_{n,i}$ and the $x$ are complex numbers inside the unit ball, in which case the logarithm is the natural complex extension of the real logarithm.

**(b)** Let $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$ be complex numbers such that $|x_j|, |y_j| \le 1$ for all $j$. Show that $|x_1x_2 - y_1y_2| \le |x_1 - y_1| + |x_2 - y_2|$, and proceed by induction to show that

$$|\prod_{j=1}^n x_j - \prod_{j=1}^n y_j| \le \sum_{j=1}^n |x_j - y_j|, \quad \text{for all } n.$$

**Ex. 2.37** *The Lindeberg CLT via truncations.* Here we present a proof of the CLT for independent random variables using m.g.f.s., but without assuming that the m.g.f.s. of the random variables involved necessarily exist. Since the m.g.f. of a bounded random variable always exists (prove it), the trick is to truncate the random variables involved: Let $X_1, X_2, \ldots$ be independent random variables with mean zero and variances $\sigma_1^2, \sigma_2^2, \ldots$. Write $B_n^2 = \sigma_1^2 + \cdots + \sigma_n^2$ and $X_{n,j} = X_j/B_n$ for $j = 1, \ldots, n$. We are to show that $\sum_{j=1}^n X_{n,j}$ converges in distribution to a standard normal random variable, provided the Lindeberg condition (2.9) is satisfied, which we assume throughout.

**(a)** For some $\varepsilon > 0$, define $\mu_{n,j} = \mathrm{E}\, X_{n,j} I(|X_{n,j}| < \varepsilon)$, as well as the random variables $Y_{n,j} = X_{n,j} I(|X_{n,j}| < \varepsilon) - \mu_{n,j}$ and $v_{n,j} = X_{n,j} I(|X_{n,j}| \ge \varepsilon) + \mu_{n,j}$. Verify that

$$X_{n,j} = Y_{n,j} + v_{n,j},$$

and also $|Y_{n,j}| \le 2\varepsilon$, the latter implying that the m.g.f. of $Y_{n,j}$ exist for all $j$ and $n$. Show that

$$\mathrm{E}\, \exp(tY_{n,j}) = 1 + \tfrac{1}{2}t^2 \mathrm{E}\, Y_{n,j}^2 + r_{n,j}(t),$$

where the remainders are such that $|\sum_{j=1}^n r_{n,j}(t)| \le \varepsilon \exp(2\varepsilon)$ for $|t| \le 1$.

**(b)** By Ex. 2.36(a), showing that $\prod_{j=1}^n \mathrm{E}\, \exp(tY_{n,j}) \to \exp(\frac{1}{2}t^2)$ now comes down to verifying that (i) $\sum_{j=1}^n \mathrm{E}\, Y_{n,j}^2 \to 1$; (ii) $\max_{j\le n} \mathrm{E}\, Y_{n,j}^2 \to 0$ and $\max_{j\le n} |r_{n,j}| \to 0$ ; and (iii) $\limsup_{n\to\infty} |\mathrm{E}\, \exp(tY_{n,j}) - 1|$ is finite. Please verify these conditions, remembering that it is sufficient to consider $|t| \le 1$.

**(c)** Finally, show that $\sum_{j=1}^n v_{n,j} \to_{\mathrm{pr}} 0$ and conclude that $\sum_{j=1}^n X_{n,j} \to_d \mathrm{N}(0,1)$.

**Ex. 2.38** *Proving the CLT with characteristic functions.* It can be argued that the most elegant, unified, and general proof of the CLT is obtained by the use of characteristic functions. This is because, contrary to the m.g.f., the characteristic function of a random variable always exists.

**(a)** Show that if $X$ has finite mean $\xi$, then its characteristic function satisfies $\varphi(t) = 1 + i\xi t + o(t)$ as $t \to 0$. Also, its derivative exists, with $\varphi'(t) = \mathrm{E}\, iX \exp(itX)$, and in particular $\varphi'(0) = i\xi$ (see Ex. C.24 for more details).

**(b)** Show similarly that if $X$ has finite variance $\sigma^2$, then

$$\varphi(t) = 1 + i\xi t - \tfrac{1}{2}(\xi^2 + \sigma^2)t^2 + o(t^2) \quad \text{as } t \to 0.$$

**(c)** If $X_1, X_2, \ldots$ are i.i.d. with mean zero and finite variance $\sigma^2$, then show that $Z_n = \sqrt{n}\bar{X}_n = n^{-1/2}\sum_{i=1}^n X_i$ has characteristic function of the form

$$\varphi_n(t) = \{1 - \tfrac{1}{2}\sigma^2 t^2/n + o(1/n)\}^n.$$

Prove the CLT from this.

**Ex. 2.39** *The Lindeberg CLT, yet again.* Let $(X_{n,i})_{i\le n, n\ge 1}$ be a triangular array of mean zero random variables with variances $(\sigma^2_{n,i})_{i\le n, n\ge 1}$ such that $X_{n,1}, \ldots, X_{n,n}$ are independent for each $n$. We say that the triangular array is row wise independent.

**(a)** Let $(Y_{n,i})_{i\le n, n\ge 1}$ be a triangular array of row wise independent mean zero normally distributed random variables with variances $\mathrm{Var}(Y_{n,i}) = \sigma^2_{n,i}$ for all $i$ and $n$. Using the second lemma of Ex. 2.36, show that for any $\varepsilon > 0$

$$|\mathrm{E}\,\exp(it\sum_{i=1}^n X_{n,i}) - \mathrm{E}\,\exp(it\sum_{i=1}^n Y_{n,i})| \le \tfrac{1}{3}|t|^3\varepsilon\sum_{i=1}^n \sigma^2_{n,i} + L_n(\varepsilon) + L'_n(\varepsilon),$$

where $L_n(\varepsilon) = \sum_{i=1}^n \mathrm{E}\,X^2_{n,i} I(|X_{n,i}| \ge \varepsilon)$ and $L'_n(\varepsilon) = \sum_{i=1}^n \mathrm{E}\,Y^2_{n,i} I(|Y_{n,i}| \ge \varepsilon)$.

**(b)** Deduce that $\sum_{i=1}^n X_{n,i} \to_d \mathrm{N}(0,1)$ provided $\sum_{i=1}^n \sigma^2_{n,i} \to 1$ and the Lindeberg condition holds.

**Ex. 2.40** *Lindeberg and Liapunov and some more.* In this exercise we summarise various implications pertaining to the Lindeberg condition. Let $X_1, X_2, \ldots$ be independent mean zero random variables with variances $\sigma_1^2, \sigma_2^2, \ldots$, set $B_n^2 = \sigma_1^2 + \cdots + \sigma_n^2$, and define $X_{n,i} = X_i/B_n$. Recall that the Lindeberg condition is satisfied if for each $\varepsilon > 0$, we have $\sum_{i=1}^n \mathrm{E}\,X^2_{n,i} I(|X_{n,i}| \ge \varepsilon) \to 0$ as $n$ tends to infinity.

**(a)** Show that if the Lindeberg conditon holds, then $\max_{i\le n} \sigma_i^2/B_n^2 \to 0$.

**(b)** Show that if $B_n^2$ is convergent, then the Lindeberg condition fails.

**(c)** In view of (b), we assume that $B_n$ diverges. Show that the Lindeberg condition holds for uniformly bounded random variables, i.e., $|X_i| \le M$ for all $i$.

**(d)** Show that if $\mathrm{E}\,|X_i|^{2+\delta} < M$ for all $i$ and some $\delta > 0$, still assuming that $B_n$ diverges, of course.then the Lindeberg condition holds.

**(e)** Suppose that there is a sequence of constants $K_1, K_2, \ldots$ such that $|X_i| < K_i$ for each $i$, and that $K_n/B_n \to 0$. Show that the Lindeberg conditon holds.

**(f)** Show that the Liapunov condition, that is $\sum_{i=1}^n \mathrm{E}\,|X_{n,i}|^{2+\delta} \to 0$ for some $\delta > 0$, implies the Lindeberg condition.

**(g)** Suppose that $n^{-1}B_n^2 \to \sigma^2 > 0$, and show that then

$$\frac{1}{n}\sum_{i=1}^n \mathrm{E}\,X_i^2 I(|X_i| \ge \sqrt{n}\varepsilon) \to 0,$$

for each $\varepsilon > 0$ if and only if the Lindeberg condition holds; and that under these conditions $\sqrt{n}\bar{X}_n \to_d \mathrm{N}(0, \sigma^2)$, where $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$.

**(h)** Let $Y_1, Y_2, \ldots$ be i.i.d. random variables with finite second moments. Show that $\max(Y_1, \ldots, Y_n)/\sqrt{n}$ tends to zero in probability.

**(i)** Show that if $Y_1, Y_2, \ldots$ are i.i.d. with mean $\mu$ and variance $\sigma^2$, then the Lindeberg condition holds. Thus, the Lindeberg CLT contains the CLT for i.i.d. random variables.

**Ex. 2.41** *Limiting normality of linear combinations of i.i.d. variables.* Let $\varepsilon_1, \varepsilon_2, \ldots$ be i.i.d. from some distribution with mean zero and finite variance $\sigma^2$. For a sequence of multiplicative constants $a_1, a_2, \ldots$, consider

$$Z_n = \frac{\sum_{i=1}^n a_i \varepsilon_i}{B_n} = \sum_{i=1}^n (a_i/B_n)\varepsilon_i, \quad \text{with } B_n^2 = \sigma^2 \sum_{i=1}^n a_i^2,$$

which has mean zero and variance 1. The question is what should be demanded of the $a_i$ sequence, to ensure that $Z_n \to_d \mathrm{N}(0,1)$ (even if the $\varepsilon_i$ distribution might be looking say skewed and multimodal and strange).

**(a)** Let $D_n = \max_{i \le n} |a_i|/B_n$. Writing $G$ for the distribution of $\varepsilon_i$, show that

$$\sum_{i=1}^n \mathrm{E}\left|\frac{a_i\varepsilon_i}{B_n}\right|^2 I\Big(\Big|\frac{a_i\varepsilon_i}{B_n}\Big| \ge \delta\Big) \le \sum_{i=1}^n \frac{a_i^2}{B_n^2}\mathrm{E}\,\varepsilon_i^2\, I(D_n|\varepsilon_i| \ge \delta) \le \int_{|u|\ge\delta/D_n} u^2 \,\mathrm{d}G(u),$$

and conclude that $Z_n \to_d \mathrm{N}(0,1)$ provided $D_n \to 0$.

**(b)** Under a variety of setups, one actually has $D_n \to 0$, which is hence not at all a strict condition. Verify that the condition holds, and hence limiting normality, in the following cases: (i) $a_i = 1$ (which corresponds to the plain CLT); (ii) all $|a_i|$ inside some positive $[b,c]$ interval; (iii) $a_i = i$; (iv) $a_i = i^2$ (and generalise); (v) $a_i = 1/\sqrt{i}$. Show however that the condition does not hold for $a_i = 1/i$.

**(c)** A grasshopper sits at zero and then starts jumping, to the right and left with equal probability, and with jump sizes $1, 2, 3, \ldots$. With $S_n$ her position after $n$ jumps, show that $S_n/n^{3/2}$ has a normal limit. Though she keeps on passing zero, she will not be in that vicinity; show that $\Pr(|S_n| \le cn^{1.49}) \to 0$, for each $c$.

**(d)** [xx fikse xx]Another important case to understand well is when the $a_i$ can be considered an i.i.d. sequence, drawn from their own distribution. Show that $D_n \to_{\mathrm{pr}} 0$ if the $a_i$ distribution has finite variance. (xx nils thinks this is if and only if, actually. what happens with $Z_n$ if the $a_i$ are drawn from say the $1/|x|^2$ distribution, for $|x| \ge 1$? xx)

**Ex. 2.42** *Higher-order expansions of m.g.f.s.* (xx to be polished. the aim is to give more info for CLT, of the type $|F_n(t) - \Phi(t)| \le c/\sqrt{n}$ for some $c$. xx) [xx intro text here, perhaps by Nils. (xx nils pushes an earlier thing from Ch2 to this place; then we edit and prune and clean. xx) ]

**(a)** Consider a variable $Y$, with m.g.f. $M(t) = \mathrm{E}\exp(tY)$, assumed to be finite in at least a neighbourhood around zero. We have seen in Ex. C.21 that $\mathrm{E}\,Y^r = M^{(r)}(0)$. Write $\xi$ and $\sigma^2$ for the mean and variance of $Y$. Show that $M(t) = 1 + \xi t + o(h)$, for $|t|$ small. Taking a Taylor expansion to the next step, show that $M(t) = 1 + \xi t + \frac{1}{2}(\xi^2 + \sigma^2)t^2 + o(t^2)$. Deduce also that $\log M(t) = \xi t + \frac{1}{2}\sigma^2 t^2 + o(t^2)$.

**(b)** We may also take the expansion to the third order, but it is simpler and more insightful to proceed from $Y = \xi + Y_0$, with $Y_0$ having mean zero. Show that

$$M(t) = \exp(t\xi)\,\mathrm{E}\exp(tY_0) = \exp(t\xi)\{1 + \tfrac{1}{2}\sigma^2 t^2 + \tfrac{1}{6}\gamma_3 t^3 + o(|t|^3)\},$$

where $\gamma_3 = \mathrm{E}\,(Y - \xi)^3$.

**(c)** Consider $Y_1, \ldots, Y_n$ i.i.d. from a distribution with mean zero and m.g.f. $M(t)$ being finite around zero. Show that $Z_n = \sqrt{n}\bar{Y}$ has

$$\begin{aligned} M_n(t) &= \mathrm{E}\,\exp(tZ_n) = M(t/\sqrt{n})^n \\ &= \{1 + \tfrac{1}{2}\sigma^2 t^2/n + \tfrac{1}{6}\gamma_3 t^3/n^{3/2} + o(|t|^3/n^{3/2})\}^n. \end{aligned}$$

Show from this that under the assumptions given, $\log M_n(t) = \frac{1}{2}\sigma^2 t^2 + \frac{1}{6}\gamma_3 t^3/\sqrt{n} + o(1/\sqrt{n})$. Explain why this is a proof of the CLT.

**(d)** (xx round off, point to CLT, identify remainder term with skewness. xx)

### Joint convergence in distribution

**Ex. 2.43** *Characterisitc functions of random vectors and Cramér–Wold.* With $X = (X_1, \ldots, X_k)^{\mathrm{t}}$ a random vector, in dimension $p$, we define its characteristic functions as

$$\varphi(t_1, \ldots, t_k) = \mathrm{E}\,\exp(it^{\mathrm{t}}X) = \mathrm{E}\,\exp\{i(t_1X_1 + \cdots + t_kX_k)\}$$

for $t = (t_1, \ldots, t_k)^{\mathrm{t}}$. Let $F(x) = F(x_1, \ldots, x_k)$ be the c.d.f. of $X$, and $\varphi(t) = \varphi(t_1, \ldots, t_k)$ its characteristic function. From Ex. C.26 we have the inversion formula

$$F(b) - F(a) = \lim_{\sigma\to 0} \frac{1}{(2\pi)^k} \int_{\mathbb{R}^k} \left( \prod_{i=1}^{k} \frac{\exp(-it_jb_j) - \exp(-it_ja_j)}{-it_j} \right) \varphi(t) \exp(-\tfrac{1}{2}t^{\mathrm{t}}t\sigma^2)\, \mathrm{d}t,$$

valid for all continuity points $a = (a_1, \ldots, a_k)$ and $b = (b_1, \ldots, b_k)$ of $F$, using the shorthand notation $\mathrm{d}t = \mathrm{d}t_1 \cdots \mathrm{d}t_k$. Ex. C.26 also contains other results pertaining to the c.f. $\varphi(t_1, \ldots, t_p)$ that you may need in this exercise.

**(a)** Let $X_n = (X_{n,1}, \ldots, X_{n,p})^{\mathrm{t}}$ be a sequence of random vectors with characteristic functions $\varphi_n(t_1, \ldots, t_p)$. Suppose that $\varphi_n(t_1, \ldots, t_p) \to \varphi(t_1, \ldots, t_p)$, where $\varphi$ is the characteristic function of some random vector $X = (X_1, \ldots, X_p)^{\mathrm{t}}$. Mimic the steps taken in Ex. 2.27 to show that then $X_n \to_d X$.

Cramér–Wold device

**(b)** Show that $X_n \to_d X$ if and only if $a^{\mathrm{t}}X_n \to_d a^{\mathrm{t}}X$ for each $a \in \mathbb{R}^k$, i.e., if and only if all linear combinations converge. This is the Cramér–Wold theorem, also known as the Cramér–Wold device.

**(c)** Show that a random pair $(X_n, Y_n)$ converges in distribution to the binormal $\mathrm{N}_2(0, \Sigma)$, with $\Sigma$ having diagonal elements 1, 1, and correlation $\rho$, if and only if $aX_n + bY_n \to_d \mathrm{N}(0, a^2 + b^2 + 2\rho ab)$ for each $(a, b)$.

**Ex. 2.44** *Tightness, Prokhorov, and the continuity theorem in* $\mathbb{R}^k$. A sequence of random vectors $X_n = (X_{n,1}, \ldots, X_{n,k})^{\mathrm{t}}$ is tight if for any $\varepsilon > 0$ there is a $K$ such that

$$\Pr(\|X_n\| > K) < \varepsilon \quad \text{for all } n.$$

**(a)** Let $F_n$ is the c.d.f. of $X_n$, and show that $X_n$ being tight is equivalent to that for every $\varepsilon > 0$ there are $a = (a_1, \ldots, a_k)$ and $b = (b_1, \ldots, b_k)$ such that $F_n(b) - F_n(a) > 1 - \varepsilon$ for all $n$, where .

**(b)** Show that all the $X_{n,1}, \ldots, X_{n,k}$ are tight if and only if $X_n$ is tight. Deduce that $X_n$ is tight if and only if any linear combination $a^{\mathrm{t}}X_n = a_1X_{n,1} + \cdots + a_pX_{n,p}$ is tight.

**(c)** Show that if the sequence $(X_n)_{n\geq 1}$ of random vectors is tight, then the corresponding sequence of characteristic functions $\{\varphi_n(t_1, \ldots, t_p)\}_{n\geq 1}$ is uniformly equicontinuous.

Prokhorov's theorem in $\mathbb{R}^k$

**(d)** Prokohorov's theorem, proven for the one-dimensional case in Ex. 2.26, extends to the multi-dimensional case (and further, see Ex. 10.7): Demonstrate that (i) if $X_n \to_d X$, then $X_n$ is tight; and (ii) that if $(X_n)_{n\geq 1}$ is tight, then there is a subsequence $(n_k)_{k\geq 1}$ such that $(X_{n_k})_{k\geq 1}$ converges in distribution. Recall that Helly's theorem for random vectors was proven in Ex. 2.25(d)

**(e)** Let $\varphi_n$ and $F_n$ be, respectively, the characteristic function and the c.d.f. of the random vector $X_n = (X_{n,1}, \ldots, X_{n,k})$ for $n = 1, 2, \ldots$. Amend the derivations of Ex. 2.28 to show that for any $\varepsilon > 0$, and with $A_\varepsilon = (-\varepsilon, \varepsilon) \times \cdots \times (-\varepsilon, \varepsilon)$,

$$\begin{aligned}\int_{A_\varepsilon} \{1 - \varphi_n(t)\}\,\mathrm{d}t &\geq 2^k \varepsilon^k \int_{\cap_{j=1}^k \{|\varepsilon x_j| \geq c_k\}} \Big(1 - \prod_{j=1}^k \frac{\sin(x_j\varepsilon)}{x_j\varepsilon}\Big)\,\mathrm{d}F_n(x_1, \ldots, x_k) \\ &\geq 2^k \varepsilon^k (1 - 1/c_k^k)\,\Pr\big(\cap_{j=1}^k \{|X_{n,j}| \geq c_k/\varepsilon\}\big),\end{aligned}$$

so that with $c_k = 2/(2^k - 1)^{1/k}$ we get the inequality

Lévy's continuity theorem in dim. $k$

$$\Pr\big(\cap_{j=1}^k \{|X_{n,j}| \geq c_k/\varepsilon\}\big) \leq \frac{1}{\varepsilon^k} \int_{-\varepsilon}^{\varepsilon} \cdots \int_{-\varepsilon}^{\varepsilon} \{1 - \varphi_n(t_1, \ldots, t_k)\}\,\mathrm{d}t_1 \cdots \mathrm{d}t_k.$$

Combine this inequality with (b) to show that if $\phi_n(t_1, \ldots, t_k)$ converges to some limit function $\varphi(t_1, \ldots, t_p)$ that is continuous at $(t_1, \ldots, t_p) = (0, \ldots, 0)$, then $(X_n)_{n\geq 1}$ is tight, and that consequently the limit function $\varphi$ must be a c.f., which by Ex. 2.43(a) entails that $X_n \to_d X$, where $X$ is a random variable with c.f. $\varphi$.

**Ex. 2.45** *The multi-dimensional CLT.* In this exercise we prove two canonical central limit theorems for random vectors taking values in $\mathbb{R}^k$.

multivariate CLT

**(a)** Suppose $X_1, X_2, \ldots$ are i.i.d. random vectors with mean $\xi$ and variance matrix $\Sigma$. Invoke the Cramér–Wold device to show that $\sqrt{n}(\bar{X}_n - \xi) \to_d \mathrm{N}_k(0, \Sigma)$.

multivariate Lindeberg theorem

**(b)** Let $X_1, X_2, \ldots$ be independent random vectors of dimension $k$, with means $\xi_1, \xi_2, \ldots$, and with finite positive definite variance matrices $\Sigma_1, \Sigma_2, \ldots$. Let $B_n = (\Sigma_1 + \cdots + \Sigma_n)^{1/2}$, and write $Z_{n,i} = B_n^{-1}(X_i - \xi_i)$. Show that

$$Z_n = (\Sigma_1 + \cdots + \Sigma_n)^{-1/2} \sum_{i=1}^n (X_i - \xi_i) = \sum_{i=1}^n Z_{n,i} \to_d \mathrm{N}_k(0, I_k),$$

provided the multivariate Lindeberg condition holds, i.e.,

$$\sum_{i=1}^n \mathrm{E}\,\|Z_{n,i}\|^2 I(\|Z_{n,i}\| \geq \varepsilon) \to 0, \quad \text{for each } \varepsilon > 0. \tag{2.10}$$

**(c)** Show that the multi-dimensional Liapunov condition $\sum_{i=1}^n \mathrm{E}\,\|Z_{n,i}\|^{2+\delta} \to 0$ for some $\delta > 0$, implies the Lindeberg condition in (2.10).

**(d)** Suppose that $B_n^2/n$ converges to some positive definite $\Sigma$, and that the Lindeberg condition $n^{-1}\sum_{i=1}^n \mathrm{E}\,\|X_i - \xi_i\|^2 I(\|X_i - \xi_i\| \geq \sqrt{n}\varepsilon) \to 0$ for each $\varepsilon > 0$, holds. Show that $n^{-1/2}\sum_{i=1}^n (X_i - \xi_i) \to_d \mathrm{N}_p(0, \Sigma)$.

**(e)** Suppose that $B_n^2/n$ converges to some positive definite $\Sigma$, and that there is a $K$ and $\delta > 0$ such that $\mathrm{E}\,|X_{i,j} - \xi_{i,j}|^{2+\delta} \leq K$ for $j = 1, \ldots, k$ and for all $i \geq 1$. Show that $n^{-1/2}\sum_{i=1}^n (X_i - \xi_i) \to_d \mathrm{N}_p(0, \Sigma)$.

**Ex. 2.46** *Joint limiting normality for empirical mean and variance.* Suppose $X_1, X_2, \ldots$ are i.i.d. with mean $\xi$ and standard deviation $\sigma$. We assume that also the skewness and kurtosis are finite, $\gamma_3 = \mathrm{E}\,(X_i - \xi)^3/\sigma^3$ amd $\gamma_4 = \mathrm{E}\,(X_i - \xi)^4/\sigma^4 - 3$.

**(a)** Show from the two-dimensional CLT that

$$\begin{pmatrix} \sqrt{n}(\bar{X}_n - \xi) \\ \sqrt{n}(\widehat{\sigma}_0^2 - \sigma^2) \end{pmatrix} \to_d \mathrm{N}_2\Big(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & \gamma_3\sigma^3 \\ \gamma_3\sigma^3 & \sigma^4(2+\gamma_4) \end{pmatrix}\Big),$$

where $\widehat{\sigma}_0^2 = n^{-1}\sum_{i=1}^n (X_i - \xi)^2$.

**(b)** Then show that with $\widehat{\sigma}^2 = n^{-1}\sum_{i=1}^n (X_i - \bar{X})^2$, which is a 'real estimator', as opposed to $\widehat{\sigma}_0^2$, which uses $\xi$, then we have $\sqrt{n}(\widehat{\sigma}^2 - \widehat{\sigma}_0^2) \to_{\mathrm{pr}} 0$; see Ex. 2.22. Conclude that the two-dimensional limit distribution result above continues to hold with $\sqrt{n}(\widehat{\sigma}^2 - \sigma^2)$ replacing $\sqrt{n}(\widehat{\sigma}_0^2 - \sigma^2)$.

**(c)** Let in particular the distribution of the $X_i$ be normal, so $X_i \sim \mathrm{N}(\xi, \sigma^2)$. Show that $\gamma_3$ and $\gamma_4$ are equal to zero, so the general result above simplifies to

$$\begin{pmatrix} \sqrt{n}(\bar{X}_n - \xi) \\ \sqrt{n}(\widehat{\sigma}^2 - \sigma^2) \end{pmatrix} \to_d \mathrm{N}_2\Big(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma^2 & 0 \\ 0 & 2\sigma^4 \end{pmatrix}\Big).$$

For another instructive application, involving the Gamma distribution, see Ex. 3.27.

**(d)** Let $X_1, \ldots, X_n$ be independent Bernoulli variables with $\Pr(X_i = 1) = p$. With $\widehat{p} = \bar{X} = Y/n$ the familiar binomial ratio, explain first that the empirical variance here is $\widehat{\sigma}^2 = \widehat{p}(1-\widehat{p})$. Then use the results above to find the joint limiting distribution of $\sqrt{n}(\widehat{p} - p, \widehat{\sigma}^2 - p(1-p))$.

### The delta method

**Ex. 2.47** *The delta method: the basics.* There are important and often reasonably simple to use approximation methods, in probability theory and statistics, going by the name of *the delta method*. It is related to functions of variables often being approximately linear, if the variables in question are not too spread out, with consequences for approximate normality.

**(a)** We start out in dimension one. Suppose in generic terms that $\sqrt{n}(A_n - a) \to_d Z$, with random variables $A_n$ and a constant $a$. The limits are taken with respect to the index $n$ tending to infinity, where $n$ typically but not always is the size of an underlying sample. Show first that $A_n \to_{\mathrm{pr}} a$. Now consider a function $g$, defined in at least a neighbourhood around $a$, and assumed to have a continuous derivative there. Show via the mean value theorem that $g(A_n) - g(a) = g'(B_n)(A_n - a)$, for some $B_n$ between $A_n$ and $a$, and use this to show that $\sqrt{n}\{g(A_n) - g(a)\} \to_d g'(a)Z$.

**(b)** For the typical case where the limit is a zero-mean normal, say $Z \sim \mathrm{N}(0, \tau^2)$, explain that $\sqrt{n}\{g(A_n) - g(a)\} \to_d \mathrm{N}(0, g'(a)^2\tau^2)$.

**(c)** For the vector case, assume $\sqrt{n}(A_n - a) \to_d Z$ in dimension $p$, and consider a smooth function $g(a) = g(a_1, \ldots, a_p)$, assumed to have continuous first order derivatives in a neighbourhood around $a$. Explain that $A_n \to_{\mathrm{pr}} a$, so $A_n$ will be inside that neighbourhood with probability tending to one. With $c = g^*(a)$ the vector of derivatives $\partial g(a)/\partial a_1, \ldots, \partial g(a)/\partial a_p$, evaluated at $a$, show that $\sqrt{n}\{g(A_n) - g(a)\} \to_d c^{\mathrm{t}}Z = c_1Z_1 + \cdots + c_pZ_p$. In particular, if $Z \sim \mathrm{N}_p(0, \Sigma)$, explain that the limit $c^{\mathrm{t}}Z$ is a $\mathrm{N}(0, c^{\mathrm{t}}\Sigma c)$. The main point is that the convergence in distribution, from the previous points, holds also jointly.

**(d)** Sometimes one also needs the vector-to-vector extension of results above, leading to the joint limit distribution of several such $g$ functions. Let $g = (g_1, \ldots, g_q)^{\mathrm{t}} \colon \mathbb{R}^p \to \mathbb{R}^q$

be such a function, with component functions having first order derivatives at position $a$, yielding a Jacobi matrix $J = \partial g(a)/\partial a$ of dimension $q \times p$. Show that $\sqrt{n}\{g(A_n) - g(a)\} \to_d JZ$. In particular, if $Z \sim \mathrm{N}_p(0, \Sigma)$, then the limit is $\mathrm{N}_q(0, J\Sigma J^{\mathrm{t}})$.

**(e)** The delta method has been formulated here in terms of limit distributions, implying approximate normality for function of approximately normal variables. Explain that in the framework above, with $\sqrt{n}(A_n - a) \to_d \mathrm{N}_p(0, \Sigma)$, then $g(A_n)$ is approximately normal, with mean $g(a)$ and variance $c^{\mathrm{t}}\Sigma c/n$. The delta method hence gives approximations also to means, variances, and covariances of smooth functions of other variables.

**(f)** The $\sqrt{n}$ factor above comes from the most typical uses of these methods, where variances of estimators go to zero with speed $1/n$, in terms of sample size. Show however that the mathematics goes through for any increasing sequence; if $d_n(A_n - a) \to_d Z$, with $d_n$ tending to infinity, then $d_n\{g(A_n) - g(a)\} \to_d g'(a)Z$. There are indeed situations where the rate might be $n^{2/3}$ or $n^{1/3}$.

**Ex. 2.48** *Applying the delta method.* Here we exercise our delta method muscles, to see how the general recipes of Ex. 2.47 may be applied in a few situations. As is clear, once we have established limiting normality for certain quantities, e.g. via the CLT, there is a host of easy-to-harvest applications for functions of these start quantities.

**(a)** If $\sqrt{n}(A_n - a) \to_d \mathrm{N}(0, 1)$, find out what happens to $\sqrt{n}(A_n^3 - a^3)$, and to $\sqrt{n}\{\exp(kA_n) - \exp(ka)\}$, where $k$ is a constant.

**(b)** For $Y$ a binomial $(n, p)$, we have of course $\mathrm{Var}\,\widehat{p} = p(1-p)/n$ for the classic estimator $\widehat{p} = Y/n$. Use the delta method to find approximations to the means, variances, and distributions of (i) the estimated odds ratio $\widehat{p}/(1-\widehat{p})$; (ii) the estimated log-odds-ratio $\log\widehat{p} - \log(1-\widehat{p})$; (iii) the transformed estimator $\widehat{\gamma} = 2\arcsin(\widehat{p}^{1/2})$. In particular, for the latter, show $\sqrt{n}(\widehat{\gamma} - \gamma) \to_d \mathrm{N}(0, 1)$, with $\gamma = 2\arcsin(p^{1/2})$.

**(c)** Suppose $\widehat{p}_1 = Y_1/n$ and $\widehat{p}_2 = Y_2/n$ are two binomial estimates, with the same sample size $n$. Then $\sqrt{n}(\widehat{p}_1 - p_1) \to_d Z_1$ and $\sqrt{n}(\widehat{p}_2 - p_2) \to_d Z_2$, where $Z_j \sim \mathrm{N}(0, p_j(1-p_j))$. Find the approximate normal distribution of $\widehat{p}_1/\widehat{p}_2$, viewed as an estimator of $p_1/p_2$. Modify arguments appropriately to find a good approximation to the variance of $\widehat{p}_1/\widehat{p}_2$, and its approximate normal distribution, also in the case of unequal sample sizes, say $n_1$ and $n_2$. [xx pointer to Story #1. xx]

**(d)** Suppose $Y_1, \ldots, Y_n$ are independent from the geometric distribution with $\Pr(Y_i = y) = (1-p)^{y-1}p$ for $y = 1, 2, \ldots$. We learned in Ex. 1.25 that the mean and variance are $1/p$ and $(1-p)/p^2$. Find first the limiting distribution of $\sqrt{n}(\bar{Y} - 1/p)$ and then that of $\sqrt{n}(\widehat{p} - p)$, where $\widehat{p} = 1/\bar{Y}$.

**(e)** Suppose $(a, b)$ is a certain position on the map, where one only has estimates, say $A_n$ and $B_n$, for its x- and y-coordinates. Assume these are independent, approximately unbiased, and approximate normal, after $n$ measurements. We formalise a version of this as $\sqrt{n}(A_n - a) \to_d N_1$ and $\sqrt{n}(B_n - b) \to_d N_2$, the limit variables $N_1, N_2$ being independent and standard normal. Having observed $A_n$ and $B_n$, explain how you can put up 90 percent confidence intervals for $a$ and $b$ separately. Construct also a 90 percent confidence circle for $(a, b)$.

**(f)** Let us pass from Cartesian to polar coordinates, letting

$$R_n = \|(A_n, B_n)\| = (A_n^2 + B_n^2)^{1/2} \quad \text{and} \quad \widehat{\alpha}_n = \arctan(B_n/A_n),$$

seen as estimators of the length $r = \|(a, b)\|$ and angle $\alpha = \arctan(b/a)$. Find the limit distributions for $\sqrt{n}(R_n - r)$ and $\sqrt{n}(\widehat{\alpha}_n - \alpha)$, and show that these are independent in

the limit.

**(g)** Suppose one observes $(A_n, B_n) = (4.44, 2.22)$, with $n = 100$. Construct and display an approximate 90 percent confidence circle for $(a, b)$, and then approximate 90 percent confidence intervals for the length $r$ and angle $\alpha$. How can you construct confidence intervals for $r$ and $\alpha$ jointly, say $I_{r,n}$ and $I_{\alpha,n}$, such that the probability that $(r \in I_{r,n}) \cap (\alpha \in I_{\alpha,n})$ converges to 0.90?

**Ex. 2.49** *Limiting normality for multinomials.* Consider the multinomial setup of Ex. 1.5, with $(Y_1, \ldots, Y_k)$ counting the number of events of type $1, \ldots, k$ in $n$ independent experiments, each time with probabilities $p = (p_1, \ldots, p_k)^{\mathrm{t}}$. Here we sort out the basic large-sample behaviour of the relative frequencies $\widehat{p}_j = Y_j/n$.

**(a)** Show that these $\widehat{p}_j = Y_j/n$ are consistent, with $\sqrt{n}(\widehat{p}_j - p_j) \to_d \mathrm{N}(0, p_j(1-p_j))$. Show more generally that there is full joint convergence in distribution here; $X_n = \sqrt{n}(\widehat{p} - p) \to_d Z \sim \mathrm{N}_k(0, \Sigma)$, where $\Sigma$ is the matrix with elements $\sigma_{j,\ell} = p_j\delta_{j,\ell} - p_jp_\ell$. It may be written $\Sigma = D - pp^{\mathrm{t}}$ with $D$ diagonal with elements $p_j$. Verify that this is consistent with $\sum_{j=1}^k Z_j = 0$.

**(b)** For $\gamma = g(p_1, \ldots, p_k)$ any smooth function of the relative frequencies, with natural estimator $\widehat{\gamma} = g(\widehat{p}_1, \ldots, \widehat{p}_k)$, show that $\sqrt{n}(\widehat{\gamma} - \gamma) \to_d \mathrm{N}(0, \tau^2)$, with $\tau^2 = c^{\mathrm{t}}\Sigma c = c^{\mathrm{t}}Dc - (c^{\mathrm{t}}p)^2$, where $c = \partial g(p)/\partial p$. Check what this says, for the case of $\gamma = p_1 + \cdots + p_k$.

**(c)** Consider $(X, Y, Z)$ being trinomial $(n, p, q, r)$. With $\widehat{p} = X/n$, $\widehat{q} = Y/n$, $\widehat{r} = Z/n$, find the limit distribution for $\widehat{\gamma} = \widehat{p}/(\widehat{q}\widehat{r})^{1/2}$, as well as for $\widehat{\delta} = 2\arcsin(\widehat{p}^{1/2}) - 2\arcsin(\widehat{q}^{1/2})$.

**Ex. 2.50** *Smooth functions of means.* Suppose $Y_1, \ldots, Y_n$ are i.i.d. from some distribution. (xx point suitably to Ex. 3.21, with joint normality for empirical quantiles, after which methods of this exercise apply also to these; smooth functions of means and quantiles are approximately normal. xx)

**(a)** Consider sample averages $\bar{g}_j = n^{-1}\sum_{i=1}^n g_j(Y_i)$, for functions $g_1, \ldots, g_p$, with finite variances. With $\xi_j = \mathrm{E}\, g_j(Y_i)$ the means, show from the CLT that $\sqrt{n}(\bar{g}_1 - \xi_1, \ldots, \bar{g}_p - \xi_p)^{\mathrm{t}}$ tends to a zero-mean multinormal $(X_1, \ldots, X_p)$, with covariance matrix $\Sigma$, having elements $\sigma_{j,k} = \mathrm{cov}\{g_j(Y_i), g_k(Y_i)\}$. Let then $A(x_1, \ldots, x_p)$ be any smooth function of these means. With $\bar{A}_n = A(\bar{g}_1, \ldots, \bar{g}_p)$, show that

$$\sqrt{n}\{\bar{A}_n - A(\xi_1, \ldots, \xi_p)\} \to_d a_1X_1 + \cdots + a_pX_p \sim \mathrm{N}(0, \tau^2),$$

with $\tau^2 = a^{\mathrm{t}}\Sigma a$, in terms of $a_j = \partial A(\xi_1, \ldots, \xi_p)/\partial\xi_j$.

**(b)** For an illustration, let $Y_1, \ldots, Y_n$ be i.i.d. from the unit exponential distribution. Show that the skewness and kurtosis are skew $= 2$ and kurt $= 6$. Identify the multinormal limit of $\sqrt{n}(\bar{g}_1 - 1, \bar{g}_2 - 2, \bar{g}_3 - 6, \bar{g}_4 - 24)$, for $\bar{g}_j = n^{-1}\sum_{i=1}^n Y_i^j$. Find the limit distribution for $\widehat{\sigma} = (\bar{g}_2 - \bar{g}_1^2)^{1/2}$. Then find the limit distribution of $\sqrt{n}(\mathrm{skew}^* - 2)$ and $\sqrt{n}(\mathrm{kurt}^* - 6)$, with the empirical skew$^* = n^{-1}\sum_{i=1}^n W_i^3$ and kurt$^* = n^{-1}\sum_{i=1}^n W_i^4 - 3$, with $W_i = (Y_i - \bar{g}_1)/\widehat{\sigma}$. Find also the joint binormal limit for skew$^*$, kurt$^*$.

**Ex. 2.51** *Delta method calculus for the normal case.* Let $Y_1, \ldots, Y_n$ be i.i.d. from the normal $\mathrm{N}(\xi, \sigma^2)$, with standard estimators $\widehat{\xi} = \bar{Y}$ and $\widehat{\sigma}^2 = (n-1)^{-1}\sum_{i=1}^n (Y_i - \bar{Y})^2$. In various exercises in Ch. 1 we have worked with *exact finite-sample* calculus, for certain basic parameters, like the mean, variance, the quantile $\gamma_q = \xi + z_p\sigma$. Here we show how the delta method, starting with the basic limit distributions for the two parameters, can be used to put up large-sample normal approximations for *any* functions of the parameters, in cases where it might be too hard to carry out exact finite-sample calculus.

**(a)** Since the skewness and the kurtosis for the normal are zero, show that the general result from Ex. 2.45 implies that $(\sqrt{n}(\bar{Y}-\xi), \sqrt{n}(\widehat{\sigma}-\sigma))^{\mathrm{t}}$ tends to say $(A,B)^{\mathrm{t}}$, with these being independent and zero-mean normals with variances $\sigma^2$ and $\frac{1}{2}\sigma^2$. Show this directly, from the normality assumptions, as opposed to deriving it as a special case of the general statement. Note also that the $\sqrt{n}(\bar{Y}-\xi) \sim \mathrm{N}(0,\sigma^2)$ holds exactly, for each finite $n$.

**(b)** With $\alpha = g(\xi,\sigma)$, for any smooth function of the two parameters, the natural estimator is $\widehat{\alpha} = g(\bar{Y},\widehat{\sigma})$. Show that

$$\sqrt{n}(\widehat{\alpha}-\alpha) \to_d cA + dB \sim \mathrm{N}(0, (c^2+\tfrac{1}{2}d^2)\sigma^2),$$

where $c$ and $d$ are the partial derivatives of $g$, evaluated at the position $(\xi,\sigma)$. Show how this leads to construction of confidence intervals for the $\alpha$ parameter.

**(c)** Consider the probability $p = \Pr(Y \geq y_0) = 1-\Phi((y_0-\mu)/\sigma)$, for some given threshold $y_0$, and the associated estimator $\widehat{p} = 1-\Phi((y_0-\bar{Y})/\widehat{\sigma})$. Finr the limit distribution of $\sqrt{n}(\widehat{p}-p)$, and use this to put up a confidence interval for $p$, with coverage level converging to 0.90. Compare with the simpler estimator $p^* = n^{-1}\sum_{i=1}^n I(Y_i \geq y_0)$, the binomial proportion, which bypasses the normal assumption.

**(d)** Then consider the parameter $\kappa = \xi/\sigma$, the normalised mean (so its value is unchanged when one passes from say millimetres to metres). For $\widehat{\kappa} = \bar{Y}/\widehat{\sigma}$, show that $\sqrt{n}(\widehat{\kappa}-\kappa) \to_d \mathrm{N}(0, 1+\frac{1}{2}\kappa^2)$, and construct an approximate 90 percent confidence interval. For this particular parameter, check also Ex. 3.39 for an exact method. (xx check if $g(\kappa) = \sqrt{2}\log(\kappa+(2+\kappa^2)^{1/2})$ is an ok variance stabilising transformation; $\sqrt{n}\{g(\widehat{\kappa})-g(\kappa)\}$ should tend to $\mathrm{N}(0,1)$. xx)

**(e)** Assume $\xi$ is positive, so that the so-called coefficient of variation $\gamma = \sigma/\xi$ has a natural interpretation. With $\widehat{\gamma} = \widehat{\sigma}/\widehat{\xi}$ the plug-in estimator, show that $\sqrt{n}(\widehat{\gamma}-\gamma)$ tends to a normal with variance $\tau^2 = \frac{1}{2}\gamma^2(1+2\gamma^2)$. Find also a variance stabilising transformation, from $\gamma$ to $\gamma^* = h(\gamma)$, with the property that $\sqrt{n}(\widehat{\gamma}^*-\gamma^*) \to_d \mathrm{N}(0,1)$: one such is $h(\gamma) = \log\gamma - \log(1+(1+2\gamma^2)^{1/2})$. Explain how confidence intervals can be set for $\gamma$ via this.

**Ex. 2.52** *Estimating mean and standard deviation outside normality.* Let $Y_1,\ldots,Y_n$ be i.i.d. from a distribution with finite fourth moment, and consider the usual mean $\bar{Y}_n$ and empirical standard deviation $S_n$. Under normality we have precise finite-sample results regarding their distributions, see Ex. 1.50, but here we investigate behaviour outside normality.

**(a)** Let as on previous occasions $\gamma_3$ and $\gamma_4$ be the skewness and kurtotis of the distribution. Use the delta method, with previous results from Ex. 2.45, to show that

$$\begin{pmatrix}\sqrt{n}(\bar{Y}_n-\xi)\\ \sqrt{n}(\widehat{\sigma}_n-\sigma)\end{pmatrix} \to_d \mathrm{N}_2\Big(\begin{pmatrix}0\\0\end{pmatrix}, \sigma^2\begin{pmatrix}1 & \frac{1}{2}\gamma_3\\ \frac{1}{2}\gamma_3 & \frac{1}{2}+\frac{1}{4}\gamma_4\end{pmatrix}\Big).$$

Explain that this implies $\Pr(|\bar{Y}_n-\xi| \leq 1.96\,\widehat{\sigma}) \to 0.95$, even outside normality; the normality-based standard recipes regarding inference for the mean parameter is hence not muich hampered by that modelling assumption.

**(b)** The situation is different when it comes to inference for the standard deviation, however. Show that $\sqrt{n}(\widehat{\sigma}-\sigma)/\widehat{\kappa} \to_d \mathrm{N}(0,1)$, if $\widehat{\kappa}$ is a consistent estimator for $\kappa = (\frac{1}{2}+\frac{1}{4}\gamma_4)^{1/2}$. Construct indeed such an estimator; see also Ex. 3.12. For an application of this, with sample size up to a million, see Story #92.

**Ex. 2.53** *Approximate variances and the delta method.* We have built a well-working apparatus around the delta method and seen various applications. Statements reached are in terms of precise limit distributions, though without going into the quality of the resulting approximations. The present exercise goes into the details of variances and covariances for functions of sample averages.

**(a)** Suppose $X_1, \ldots, X_n$ are i.i.d. with mean zero, variance $\sigma^2$, and finite skewness $\gamma_3 = \mathrm{E}\,(X_i/\sigma)^3$ and kurtosis $\gamma_4 = \mathrm{E}\,(X_i/\sigma)^4 - 3$. With $\bar{X}_n$ as usual being the average, show then that

$$\begin{aligned} \mathrm{E}\,\bar{X}_n^2 &= \sigma^2/n, \\ \mathrm{E}\,\bar{X}_n^3 &= (\sigma^3/n^2)\gamma_3, \\ \mathrm{E}\,\bar{X}_n^4 &= (\sigma^4/n^2)(3+\gamma_4/n), \\ \mathrm{Var}\,\bar{X}_n^2 &= (\sigma^4/n^2)(2+\gamma_4/n). \end{aligned}$$

**(b)** Let then $Y_1, \ldots, Y_n$ be i.i.d., with finite mean $\xi$, standard deviation $\sigma$, skewness $\gamma_3$, kurtosis $\gamma_4$. With $\bar{Y}_n$ the sample average, consider then the variable

$$Z_n = a_0 + a_1(\bar{Y}_n - \xi) + \tfrac{1}{2}a_2(\bar{Y}_n - \xi)^2.$$

Show that $\mathrm{E}\,Z_n = a_0 + \frac{1}{2}a_2\sigma^2/n$, and that

$$\mathrm{Var}\,Z_n = a_1^2\sigma^2/n + (1/n^2)\{\tfrac{1}{4}a_2^2\sigma^4(2+\gamma_4/n) + a_1a_2\sigma^3\gamma_3\}.$$

**(c)** Consider any smooth function $Z_n = g(\bar{Y}_n)$. Since $\bar{Y}_n$ is close to $\xi$ with high probability, it makes sense to carry out a Taylor expansion,

$$Z_n = g(\xi) + g'(\xi)(\bar{Y} - \xi) + \tfrac{1}{2}g''(\xi)(\bar{Y} - \xi)^2 + \delta_n,$$

where $\delta_n$ is a smaller-sized remainder term; you may prove that $n^{3/2}\delta_n$ is bounded in probability, provided $g$ has three derivatives in a neighbourhood around $\xi$. Show from the above that

$$\mathrm{Var}\,Z_n = g'(\xi)^2\sigma^2/n + (1/n^2)\{\tfrac{1}{4}g''(\xi)^2\sigma^4(2+\gamma_4/n) + g'(\xi)g''(\xi)\sigma^3\gamma_3\} + o(1/n^2).$$

There is hence a clear leading $O(1/n)$ term for the variance, with other terms being of size $O(1/n^2)$. Explain how this relates to the basics of the delta method from Ex. 2.47.

**(d)** Suppose $A$ is a random variable with mean $a$ and finite variance, and that $g(y)$ is smooth in a neighbourhood around $a$. Use the Taylor approximation

$$g(y) = g(a) + g'(a)(y-a) + \tfrac{1}{2}g''(a)(y-a)^2 + O(|y-a|^3),$$

valid for $y$ close to $a$, to show that

$$\mathrm{E}\,g(A) \doteq g(a) + \tfrac{1}{2}g''(a)\mathrm{Var}\,Y, \quad \mathrm{Var}\,g(Y) \doteq g'(a)^2\,\mathrm{Var}\,Y,$$

and indicate the sizes of the error terms involved.

**Ex. 2.54** *The empirical correlation coefficient under binormality.* For i.i.d. data pairs $(X_i, Y_i)$, the classical empirical correlation coefficient is

$$R_n = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\{\sum_{i=1}^n (X_i - \bar{X})^2\}^{1/2}\{\sum_{i=1}^n (Y_i - \bar{Y})^2\}^{1/2}} = n^{-1}\sum_{i=1}^n \frac{X_i - \bar{X}}{\widehat{\sigma}_1}\frac{Y_i - \bar{Y}}{\widehat{\sigma}_2}, \quad (2.11)$$

with $\widehat{\sigma}_1^2$ and $\widehat{\sigma}_2^2$ the empirical variances $n^{-1}\sum_{i=1}^n (X_i - \bar{X})^2$ and $n^{-1}\sum_{i=1}^n (Y_i - \bar{Y})^2$. Here we find the the limit distribution of $R_n$ under binormality.

**(a)** Assume first that the $(X_i, Y_i)$ pairs are from a zero-mean binormal with variances 1 and correlation $\rho \in (-1, 1)$; see Ex. 1.44. Use results from Ex. 1.45, including $Y_i \,|\, X_i \sim \mathrm{N}(\rho X_i, 1-\rho^2)$, to derive expressions for $\mathrm{E}\, X_i^2 Y_i^2$, $\mathrm{E}\, X_i^3 Y_i$, $\mathrm{E}\, X_i Y_i^3$. Use these to show that

$$\Sigma = \mathrm{Var} \begin{pmatrix} X_i^2 \\ Y_i^2 \\ X_i Y_i \end{pmatrix} = \begin{pmatrix} 2 & 2\rho^2 & 2\rho \\ 2\rho^2 & 2 & 2\rho \\ 2\rho & 2\rho & 1+\rho^2 \end{pmatrix}.$$

Use the CLT to argue that

$$\begin{pmatrix} A_n \\ B_n \\ C_n \end{pmatrix} = \sqrt{n} \begin{pmatrix} n^{-1}\sum_{i=1}^n X_i^2 - 1 \\ n^{-1}\sum_{i=1}^n Y_i^2 - 1 \\ n^{-1}\sum_{i=1}^n X_i Y_i - \rho \end{pmatrix} \to_d \begin{pmatrix} A \\ B \\ C \end{pmatrix} \sim \mathrm{N}_3(0, \Sigma).$$

With $R_{n,0} = C_n/(A_n B_n)^{1/2}$, use the delta method to show that $\sqrt{n}(R_{n,0} - \rho) \to_d Z = -\frac{1}{2}\rho A - \frac{1}{2}\rho B + C$, and that in fact $Z \sim \mathrm{N}(0, (1-\rho^2)^2)$.

**(b)** Then generalise to the situation where the $(X_i, Y_i)$ pairs are i.i.d. from a zero-mean binormal, with standard deviations $\sigma_1, \sigma_2$ and correlation $\rho$. Show that we still have $\sqrt{n}(R_{n,0} - \rho) \to_d \mathrm{N}(0, (1-\rho^2)^2)$.

**(c)** Then go one step further, to the full five-parameter binormal situation, with unknown means $\xi_1, \xi_2$, standard deviations $\sigma_1, \sigma_2$, and correlation $\rho$. Argue first that we must have $\sqrt{n}(R_{n,1} - \rho) \to_d \mathrm{N}(0, (1-\rho^2)^2)$, where $R_{n,1}$ is as in (2.11) but using $\xi_1, \xi_2$ instead of $(\bar{X}, \bar{Y})$. Then, finally, show what we really are after, that $\sqrt{n}(R_n - \rho)$ must have the same limit distribution.

**(d)** Explain that if $h(\rho)$ is a smooth function, then $\sqrt{n}\{h(R_n) - h(\rho)\} \to_d h'(\rho)(1-\rho^2)\,\mathrm{N}(0,1)$. Show that with the particular choice $\zeta = \frac{1}{2}\log\{(1+\rho)/(1-\rho)\}$ the variance is being stabilised, and $\sqrt{n}(\widehat{\zeta} - \zeta) \to_d \mathrm{N}(0,1)$, where $\widehat{\zeta} = \frac{1}{2}\log\{(1+R_n)/(1-R_n)\}$. This is called Fisher's zeta. Show that $\widehat{\zeta} \pm 1.645/\sqrt{n}$ becomes an approximate 90 percent confidence interval for $\zeta$, and transform this to an approximate 90 percent confidence interval for $\rho$. Show in fact that this inverse transformation is $\rho = \{\exp(2\zeta) - 1\}/\{\exp(2\zeta) + 1\}$.

variance stabilising transformation

Fisher's zeta, with inverse

**Ex. 2.55** *The empirical correlation coefficient, general case.* Here we use some of the arguments of Ex. 2.54 to find the limit distribution of the empirical correlation $R_n$ of (2.11) also outside binormality. Assume $(X_1, Y_1), \ldots, (X_n, Y_n)$ are i.i.d. pairs from a distribution with means $\xi_1, \xi_2$, standard deviations $\sigma_1, \sigma_2$, and correlation $\rho$. Write $a_{j,k} = \mathrm{E}\, U_i^j V_j^k$ for cross moments of the standardised $U_i = (X_i - \xi_1)/\sigma_1$ and $V_i = (Y_i - \xi_2)/\sigma_2$, where it is assumed that fourth order moments $a_{4,0}$ and $a_{0,4}$ are finite.

**(a)** Show that $\sqrt{n}(R_n - R_{n,0}) \to_{\mathrm{pr}} 0$, where $R_{n,0}$ is as $R_n$, but using the real $\xi_1, \xi_2$ instead of their estimators $\bar{X}, \bar{Y}$. Show also that the distribution of $R_n$ and $R_{n,0}$ must depend on $\rho$ but not on $\xi_1, \xi_2, \sigma_1, \sigma_2$. We may hence carry out our large-sample investigation with the standardised $(U_i, V_i)$ rather than the $(X_i, Y_i)$. Work with

$$\begin{pmatrix} A_n \\ B_n \\ C_n \end{pmatrix} = \begin{pmatrix} \sqrt{n}(n^{-1}\sum_{i=1}^n U_i^2 - 1) \\ \sqrt{n}(n^{-1}\sum_{i=1}^n V_i^2 - 1) \\ \sqrt{n}(n^{-1}\sum_{i=1}^n U_i V_i - \rho) \end{pmatrix},$$

and show that $(A_n, B_n, C_n)^{\mathrm{t}} \to_d (A, B, C)^{\mathrm{t}} \sim \mathrm{N}_3(0, \Sigma)$, for the variance matrix $\Sigma$ of $(U_i^2, V_i^2, U_i V_i)^{\mathrm{t}}$. Spell out the elements of this matrix, using the $a_{j,k}$. Check that this agrees with the $\Sigma$ of Ex. 2.54 under binormality.

**(b)** Then show that $\sqrt{n}(R_n - \rho) \to_d Z = -\frac{1}{2}\rho A - \frac{1}{2}\rho B + C$, and give an expression for the limit distribution variance $\tau^2$. Explain how $\tau$ may be estimated from the data, and how this leads to confidence intervals of the type $R_n \pm 1.96\,\widehat{\tau}/\sqrt{n}$ for $\rho$.

**(c)** For a concrete illustration, consider the joint density $f(x,y) = 1 + a(x - \frac{1}{2})(y - \frac{1}{2})$ for $(x,y)$ in the unit square. Find the allowed parameter range for $a$, and a formula for the correlation coefficient $\rho$ in terms of $a$. Then apply the above to find the limit distribution of $\sqrt{n}(R_n - \rho)$.

**Ex. 2.56** *The delta method outside root-n terrain.* (xx to come. not always $\sqrt{n}(X_n - a) \to_d \mathrm{N}(0,\tau^2)$ terrain. different limits, different speeds. xx)

**Ex. 2.57** *Stretching the delta method.* (xx to be filled in. with $\sqrt{n}(X_n - a) \to_d V$, we have $Z_n = \sqrt{n}\{g(X_n) - g(a)\} \to_d g'(a)V$ for a fixed $g(x)$. here we consider $Z_n = \sqrt{n}\{g_n(X_n) - g_n(a)\}$. with $g_n''(a) = o(\sqrt{n})$ we may still have the right approximation. example: $Z_n = \sqrt{n}\{\exp(c_n X_n) - 1\}$. xx)

### The strong law of large numbers

**Ex. 2.58** *The Strong Law of Large Numbers: the basics.* (xx to be cleaned. xx) Suppose $X_1, X_2, \ldots$ are i.i.d. from a distribution with finite $\mathrm{E}\,|X_i|$. Then the mean $\xi = \mathrm{E}\,X_i$ exists, and we are aiming to prove the strong LLN of (2.3), that the event

$$A = \{\bar{X}_n \to \xi\} = \cap_{\varepsilon>0} \cup_{n_0 \geq 1} \cap_{n \geq n_0} \{|\bar{X}_n| \leq \varepsilon\}$$

has probability equal to one hundred percent. We may for simplicity and without loss of generality take $\xi = 0$ below.

**(a)** Show that $A$ is the same as $\cap_{N\geq 1} \cup_{n_0 \geq 1} \cap_{n \geq n_0} \{|\bar{X}_n| \leq 1/N\}$, and deduce in particular from this that $A$ is actually measurable – so it does make well-defined sense to work with its probability.

**(b)** Show that if $\Pr(A_N) = 1$ for all $N$, then $\Pr(\cap_{N\geq 1} A_N) = 1$ – if you're fully certain about a countable number of events, then you're also fully certain about all of them, jointly. This is actually not true with a bigger index set: if $X \sim \mathrm{N}(0,1)$, then you're 100 percent sure that $B_x = \{X \text{ is not } x\}$ takes place, for each single $x$, but from this does it *not* follow that you should be sure about $\cap_{\text{all } x} B_x$. Explain why.

**(c)** Show that $\Pr(A) = 1$ if and only if $\Pr(B_{n_0}) \to 0$, for each $\varepsilon > 0$, where $B_{n_0} = \cup_{n \geq n_0} \{|\bar{X}_n| \geq \varepsilon\}$. In words: for a given $\varepsilon$, the probability should be very low that there is *any* $n \geq n_0$ with $|\bar{X}_n| \geq \varepsilon$.

**(d)** A simple bound is of course $\Pr(B_{n_0}) \leq \sum_{n \geq n_0} \Pr\{|\bar{X}_n| \geq \varepsilon\}$, so it suffices to show, if possible, under appropriate conditions, that $\sum_{n\geq 1} \Pr\{|\bar{X}_n| \geq \varepsilon\}$ is a convergent series. With finite variance $\sigma^2$, explain that the classic simple Chebyshov bound, see Ex. 2.10, does *not* solve any problem here.

**(e)** (xx calibrate better with Ex. 2.10. xx) Show, however, that if the fourth moment is finite, then

$$\Pr\{|\bar{X}_n| \geq \varepsilon\} \leq \frac{1}{\varepsilon^4} \mathrm{E}\,|\bar{X}_n|^4 \leq \frac{c}{\varepsilon^4} \frac{1}{n^2},$$

for a suitable $c$. So under this condition, which is moderately hard, we've proven the strong LLN.

**(f)** One may squeeze more out of the chain of arguments here, which we now indicate, without full details. Assume $\mathrm{E}\,|X_i|^r$ is finite, for some $r > 2$, like $r = 2.02$. Then one may show, via arguments in von Bahr (1965), that the sequence $\mathrm{E}\,|\sqrt{n}\bar{X}_n|^r$ is bounded. This leads to the bound

$$\Pr\{|\bar{X}_n| \geq \varepsilon\} \leq \frac{1}{(\sqrt{n}\varepsilon)^r} \mathrm{E}\,|\sqrt{n}\bar{X}_n|^r,$$

and these form a convergent series. We have hence proven (modulo the von Bahr thing) that the strong LLN holds for finite $\mathrm{E}\,|X_i|^{2+\varepsilon}$, an improvement over the finite $\mathrm{E}\,|X_i|^4$ condition. – To get further, trimming away on the conditions until we are at the Kolmogorovian position of only requiring finite mean, we need more technicalities; see the following Ex. 2.59.

**Ex. 2.59** *The Strong Law of Large Numbers: nitty-gritty details.* This exercise goes through the required extra technical details, along with a few intermediate lemmas, to secure a full proof of the full LLN theorem: as long as $\mathrm{E}\,|X_i|$ is finite, the infinite sequence of sample means $\bar{X}_n$ will with probability equal to one hundred percent converge to $\xi = \mathrm{E}\,X_i$.

**(a)** We start with Kolmogorov's inequality: Consider independent zero-mean variables $X_1, \ldots, X_n$ with variances $\sigma_1^2, \ldots, \sigma_n^2$, and with partial sums $S_i = X_1 + \cdots + X_i$. Then

$$\Pr\{\max_{i \leq n} |S_i| \geq \varepsilon\} \leq \frac{\mathrm{Var}\,S_n}{\varepsilon^2} = \frac{1}{\varepsilon^2} \sum_{i=1}^{n} \sigma_i^2.$$

Note that this is a much stronger result than the special case of caring only about $|S_n|$, with $\Pr\{|S_n| \geq \varepsilon\} \leq \mathrm{Var}\,S_n/\varepsilon^2$, which is the Chebyshov inequality. To prove it, work with the disjoint decomposition

$$A_i = \{|S_1| < \varepsilon, \ldots, |S_{i-1}| < \varepsilon, |S_i| \geq \varepsilon\} \quad \text{and} \quad A = \cup_{i=1}^{n} A_i = \{\max_{i \leq n} |S_i| \geq \varepsilon\}.$$

Show that $\mathrm{E}\,S_n^2 \geq \mathrm{E}\,S_n^2 I(A) = \sum_{i=1}^{n} \mathrm{E}\,S_n^2 I(A_i)$, that

$$\mathrm{E}\,S_n^2 I(A_i) = \mathrm{E}\,(S_i + S_n - S_i)^2 I(A_i) \geq \varepsilon^2 \Pr(A_i),$$

and that this leads to the inequality asked for.

**(b)** Consider a sequence of independent $X_1, X_2, \ldots$ with means zero and variances $\sigma_1^2, \sigma_2^2, \ldots$. Show that if $\sum_{i=1}^{\infty} \sigma_i^2$ is convergent, then $\sum_{i=1}^{\infty} X_i$ is convergent with probability 1. – It suffices to show that the sequence of partial sums $S_n = X_1 + \cdots + X_n$ is Cauchy with probability 1. Show that this is the same as

$$\lim_{n \to \infty} \Pr\big[\cup_{i,j \geq n} \{|S_i - S_j| \geq \varepsilon\}\big] = 0 \quad \text{for each } \varepsilon > 0.$$

Use the Kolmogorov inequality to show this.

**(c)** A quick example to illustrate this result is as follows. Consider $X = X_1/10 + X_2/100 + X_3/1000 + \cdots$, a random number in the unit interval, with the $X_i$ independent, and with no further assumptions. Show that $X$ exists with probability 1.

**(d)** Prove that if $\sum_{i=1}^{\infty} a_i/i$ converges, then $\bar{a}_n = (1/n) \sum_{i=1}^{n} a_i \to 0$. To show this, consider $b_n = \sum_{i=1}^{n} a_i/i$, so that $b_n \to b$ for some $b$. Show $a_n = n(b_n - n_{n-1})$, valid also for $n = 1$ if we set $b_0 = 0$, and which leads to $\sum_{i=1}^{n} a_i = nb_n - b_0 - b_1 - \cdots - b_{n-1}$.

**(e)** From the above, deduce that if $X_1, X_2, \ldots$ are independent with means $\xi_1, \xi_2, \ldots$ and variances $\sigma_1^2, \sigma_2^2, \ldots$, and $\sum_{i=1}^\infty \sigma_i^2/i^2$ converges, then $\bar X_n - \bar\xi_n \to_{\text{a.s.}} 0$. Here $\bar\xi_n = (1/n)\sum_{i=1}^n \xi_i$.

**(f)** Use the above to show that if $X_1, X_2, \ldots$ are independent with zero means, and all variances are bounded, then indeed $\bar X_n \to_{\text{a.s.}} 0$. Note that this is a solid generalisation of what we managed to show in (xx calibrate xx) – first, the distributions are allowed to be different (not identical); second, we have landed at a.s. convergence with the mild assumption of finite and bounded variances, whereas we there needed the harsher conditions of finite fourth moments.

**(g)** We're close to the Pole. For i.i.d. zero mean variables $X_1, X_2, \ldots$, split them up with the little trick

$$X_i = Y_i + Z_i, \quad \text{with} \quad Y_i = X_i I(|X_i| < i), \quad Z_i = X_i I(|X_i| \ge i).$$

We have $\bar X_n = \bar Y_n + \bar Z_n$, so it suffices to demonstrate that $\bar Y_n \to_{\text{a.s.}} 0$ and $\bar Z_n \to_{\text{a.s.}} 0$ (since an intersection of two sure events is sure). Use the Borel–Cantelli lemma, in concert with $\mathrm{E}\,|X_i| = \int_0^\infty \Pr(|X_i| \ge x)\,\mathrm{d}x$, to show that only finitely many $Z_i$ are non-zero. Then use previous results to demonstrate $\bar Y_n - \bar\xi_n \to_{\text{a.s.}} \to 0$ and $\bar\xi_n \to 0$, where $\bar\xi_n$ is the average of $\xi_i = \mathrm{E}\,Y_i$.

**(h)** So we've managed to prove the Strong LLN; good. Attempt also to prove the interesting converse that if $\mathrm{E}\,|X_i| = \infty$, then the sequence of sample means is pretty erratic indeed:

$$\Pr\{\limsup_{n\to\infty} \bar X_n = \infty\} = 1, \quad \Pr\{\liminf_{n\to\infty} \bar X_n = -\infty\} = 1.$$

Simulate a million realisations from the density $f(x) = 1/x^2$, for $x \ge 1$, in your nearest computer, display the sequence of $\bar X_n$ on your screen, and comment.

**Ex. 2.60** *Yes, we converge with probability 1.* We've proven that the sequence of empirical means converges almost surely to the population mean, under the sole condition that this mean is finite. This half-automatically secures almost sure convergence of various other natural quantities, almost without further efforts.

**(a)** Suppose $X_1, X_2, \ldots$ are i.i.d. with finite variance $\sigma^2$. Show that the classical empirical standard deviation $\widehat\sigma = \{\sum_{i=1}^n (X_i - \bar X_n)^2/(n-1)\}^{1/2}$ converges a.s. to $\sigma$. Note again that nothing more is required than a finite second moment.

**(b)** Suppose the third moment is finite, such that the skewness $\gamma_3 = \mathrm{E}\,\{(X-\xi)/\sigma\}^3$ is finite. Show that $\widehat\gamma_{3,n} = (1/n)\sum_{i=1}^n (X_i - \bar X_n)^3/\widehat\sigma^3$ is strongly consistent for $\gamma_3$.

**(c)** Then suppose the fourth moment is finite, such that the kurtosis $\gamma_4 = \mathrm{E}\,\{(X-\xi)/\sigma\}^4 - 3$ is finite. Construct a strongly consistent estimator for this kurtosis.

**(d)** Assume that $(X_1, Y_1), (X_2, Y_2), \ldots$ is an i.i.d. sequence of random pairs, with finite variances, and define the population correlation coefficient in the usual fashion, as $\rho = \mathrm{cov}(X,Y)/(\sigma_1\sigma_2)$. Show that the usual empirical correlation coefficient $R_n$ of (2.11) converges with probability one to $\rho$.

**(e)** Formulate and prove a suitable statement regarding almost sure convergence of smooth functions of means.

**(f)** Let $X_1, X_2, \ldots$ be an i.i.d. sequence of nonnegative variables such that $\mathrm{E}\,\log X_i = \xi$ is finite. Show that the harmonic means $H_n = (X_1 \cdots X_n)^{1/n}$ converge with probability 1 to $\exp(\xi)$.

**Ex. 2.61** *Glivenko–Cantelli theorem.* For i.i.d. observations $Y_1, \ldots, Y_n$, we form the empirical c.d.f., as in Ex. 2.11, with $F_n(t) = n^{-1} \sum_{i=1}^n I(Y_i \leq t)$. We have seen that since $F_n(t)$ is just a binomial ratio, $F_n(t) \to_{\text{a.s.}} F(t)$, for each $t$. It is a remarkable fact that this convergence also takes place uniformly, with probability 1. This is the Glivenko–Cantelli theorem: with $D_n = \max_t |F_n(t) - F(t)|$, the max taken over all $t$ in the domain in question, we have $\Pr(D_n \to 0) = 1$. This means that regardless of any strange or complicated aspects of the distribution $F$, with enough data one will be able to learn these. See also Ex. 3.9 and 10.27 for more information, regarding the speed with which $D_n \to 0$.

Glivenko–Cantelli

**(a)** Choose $t_1 < \cdots < t_m$, creating a finite number of cells $[t_j, t_{j+1})$, where we take $t_0 = -\infty$ and $t_{m+1} = \infty$. With $A_{m,j}$ the event that $F_n(t_j) \to F(t_j)$, argue that $\Pr(\cap_{j=1}^m A_{m,j}) = 1$.

**(b)** Consider any $t$ in the cell $[t_j, t_{j+1})$. Writing $D_n(t) = F_n(t) - F(t)$, use monotinicity of $F_n$ and $F$ to show that

$$D_n(t_j) - \{F(t_{j+1}) - F(t_j)\} \leq F_n(t) - F(t) \leq D_n(t_{j+1}) + F(t_{j+1} - F(t_j).$$

Deduce that

$$\max_{t_j \leq t < t_{j+1}} |D_n(t)| \leq B_m + C_m,$$

where $B_m = \max_{1 \leq j \leq m} |D_n(t_j)|$ and $C_m = \max_{1 \leq j \leq m} \{F(t_{j+1}) - F(t_j)\}$.

**(c)** Show that $\Pr(\limsup D_n \leq C_m) = 1$.

**(d)** For each $\varepsilon > 0$, show that a partition into cells can be arranged, with high $m$ if required, so that $C_m \leq \varepsilon$. Conclude that $\Pr(D_n \to 0) = 1$.

**(e)** Choose some moderately complicated normal mixture, of the type $f = \sum_{j=1}^k p_j \mathrm{N}(\mu_j, \sigma_j^2)$; see Ex. 1.67. Then simulate a high number $n$ of data from this distribution, and read off $D_n = \max_t |F_n(t) - F(t)|$. Check out how high $n$ needs to be to have $D_n \leq 0.01$, say, with high probability, in a few situations.

### Contiguity

[xx couple of exercises xx]

### Stable and conditional convergence, and nonnormal limits

**Ex. 2.62** *Stable convergence.* This exercise introduces the notion of stable convergence of random variables, which is a form of convergence lying between convergence in probability and convergence in distribution (to paraphrase Jacod and Mémin (1981), an early article on the topic). On a probability space $(\Omega, \mathcal{F}, \Pr)$, let $(X_n)_{n \geq 1}$ be a sequence of random variables, and let $\mathcal{G} \subset \mathcal{F}$. The sequence $X_n$ converges $\mathcal{G}$-stably to the random variable $X$, where $X$ is defined on an extension $(\Omega \times \mathbb{R}, \mathcal{G} \otimes \mathcal{B}(\mathbb{R}), \Pr')$, if

$$\mathrm{E}\, \xi f(X_n) \to \mathrm{E}'\, \xi f(X), \tag{2.12}$$

for every bounded $\mathcal{G}$-measurable random variable $\xi$, and every bounded and continuous function $f$. See Ex. C.19 for extensions of probability spaces. Here $\mathrm{E}'(\cdot)$ denotes the expectation with respect to $\Pr'$. We write $X_n \to_{\mathcal{G}\text{-st.}} X$ to indicate this form of convergence.

**(a)** Show that if $X_n \to_{\mathcal{G}\text{-st.}} X$ then $X_n \to_d X$.

**(b)** To see that the converse of (a) is not true, consider the sequence $X_n = I(n \text{ odd})Y + I(n \text{ even})Y'$, where $Y$ and $Y'$ are i.i.d. random variables with common distribution $F$. Show that $X_n \to_d F$, but that $X_n$ fails to converge $\mathcal{G} = \sigma(Y)$ stably, for example. This example is from Aldous and Eagleson (1978).

**(c)** Suppose that $(X_n, Y) \to_d (X, Y)$ for every $\mathcal{G}$-measurable $Y$. We'll soon see that this is equivalent to $X_n \to_{\mathcal{G}\text{-st.}} X$. For now, assume that the limit $X$ is also $\mathcal{G}$-measurable, and show that then $X_n \to_p X$. – This illustrates that to have $\mathcal{G}$-stable convergence of $X_n$ to $X$ without also having convergence in probability, $X$ must be realised in a fashion that does not render it $\mathcal{G}$-measurable, hence the extension of the original probability space.

**(d)** Show that the following are equivalent:

(i) $X_n \to_{\mathcal{G}\text{-st.}} X$;
(ii) $(X_n, Y) \to_d (X, Y)$ for every $\mathcal{G}$-measurable random vector $Y$;
(iii) $(X_n, Y) \to_{\mathcal{G}\text{-st.}} (X, Y)$ for every $\mathcal{G}$-measurable random vector $Y$;
(iv) $(X_n, Y_n) \to_d (X, Y)$ for every sequence $(Y_n)_{n\geq 1}$ of random variables and every $\mathcal{G}$-measurable $Y$ such that $Y_n \to_p Y$;
(v) $\mathrm{E}\, I_G f(X_n) \to \mathrm{E}'\, I_G f(X)$ for every $G \in \mathcal{G}$ and every bounded and continuous $f$;
(vi) $\mathrm{E}\,\{f(X_n)\,|\,G\} \to \mathrm{E}'\,\{f(X)\,|\,G\}$, for every $G \in \mathcal{G}$ with $\Pr(G) > 0$, and every bounded and continuous $f$;
(vii) $\mathrm{E}\, I_G \exp(itX_n) \to \mathrm{E}'\, I_G \exp(itX)$ for every $G \in \mathcal{G}$.

To prove this, Ex. 2.19 and Ex. C.32(c) is of help.

**(e)** [xx rewrite xx] Let $Q_n$ and $Q$ be versions of the conditional distributions of $X_n$ and $X$ given $\mathcal{G}$, respectively, and let $Q_n f = \int_{\mathbb{R}} f(x) Q_n(\cdot, \mathrm{d}x)$, with $Qf$ similarly defined. Show that $X_n \to_{\mathcal{G}\text{-st.}} X$ is equivalent to $\mathrm{E}\,\xi Q_n f \to \mathrm{E}\,\xi Q f$, for every $\xi$ and $f$ as above.

**(f)** [xx cramer slutsky for stable convergence here xx]

**Ex. 2.63** *Conditional convergence.*[xx introtext here xx]

**(a)** Suppose that $(X_n, Y_n) \to_d (X, Y)$. Show that we then also have marginal convergence, that is $X_n \to_d X$ and $Y_n \to_d Y$.

**(b)** Show that $(X_n, Y_n) \to_d (X, Y)$ is equivalent to

$$\mathrm{E}\,\{f(X_n)\,|\,Y_n \in B\} \to \mathrm{E}\,\{f(X_n)\,|\,Y_n \in B\},$$

for all bounded and continuous $f$ and all continuity sets $B$ of the distribution of $Y$ such that $\Pr(Y \in B) > 0$ and $\Pr(Y_n \in B) > 0$ for all $n$.

**(c)** Suppose $(X_n, Y_n) \to_d (X, Y)$, a binormal zero-mean limit, say $\mathrm{N}_2(0, \Sigma)$, with

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}.$$

Deduce from (b) that $X_n\,|\,(|Y_n| \leq \varepsilon) \to_d X\,|\,(|Y| \leq \varepsilon)$ for each $\varepsilon > 0$, i.e., that the conditional distribution of $X_n$ given $|Y_n| \leq \varepsilon$ converges in distribution to that of $X$ given $|Y| \leq \varepsilon$. Show that $X\,|\,(|Y| \leq \varepsilon) \to_d X\,|\,(Y = 0) \sim \mathrm{N}(0, (1-\rho^2)\sigma_X^2)$ as $\varepsilon \to 0$.

**(d)** Consider a distribution for $X_i$ with mean zero and variance $\sigma^2$, where we indeed know that $\sqrt{n}\bar{X}_n \to_d \mathrm{N}(0, \sigma^2)$. Suppose $X_i$ also has an integrable characteristic function

$\varphi(t)$, implying by Ex. C.25 the existence of a smooth density $f$ for $X_i$ and also a density $f_n$ for $\sqrt{n}\bar{X}_n$. Show that

$$f_n(z) = 1/(2\pi) \int \exp(-itz)\varphi(t/\sqrt{n})^n \, \mathrm{d}t \to \sigma^{-1}\phi(\sigma^{-1}z),$$

i.e., that there is convergence not merely for the cumulatives, but for the densities too. To do so, you may split the domain of integration in two parts $|t| > \varepsilon\sqrt{n}$ and $|t| \le \varepsilon\sqrt{n}$, for some $\varepsilon > 0$, and use Ex. C.27.

**(e)** Suppose next that $X_i$ is discrete, without a continuous density; for a concrete example, consider $X_i = \pm 1$ with equal probabilities, and for which $\varphi(t) = \cos t$. Then $Z_n = \sqrt{n}\bar{X}_n$ does not have a density, but we may add a little Gaussian noise, to form $Z_n^* = \sqrt{n}\bar{X}_n + \xi_n$, with $\xi_n \sim \mathrm{N}(0, \varepsilon_n^2)$. Show that $Z_n^*$ has density

$$f_n^*(z^*) = 1/(2\pi) \int \exp(-itz^*)\varphi(t/\sqrt{n})^n \exp(-\tfrac{1}{2}\varepsilon_n^2 t^2) \, \mathrm{d}t,$$

and that this again converges to the normal density $\sigma^{-1}\phi(\sigma^{-1}z^*)$ provided merely that $\varepsilon_n \to 0$.

**(f)** We may use the same trick also in the vector case. Specifically, with a $p$-dimensional $X_i$ having zero mean and covariance matrix $\Sigma$, show (i) that if $X_i$ has an integrable characteristic function, then the density for $Z_n = \sqrt{n}\bar{X}_n$ tends to the $\mathrm{N}_p(0, \Sigma)$ density; and (ii) that if $X_i$ is discrete, without a density, then $Z_n^* = \sqrt{n}\bar{X}_n + \xi_n$, with some small Gaussian added noise $\xi_n \sim \mathrm{N}_p(0, \varepsilon_n^2\Sigma)$ with $\varepsilon_n \to 0$, has a density $f_n^*(z^*)$ which tends to the same $\mathrm{N}_p(0, \Sigma)$ density.

**(g)** Suppose $(X_n, Y_n) \to_d (X, Y)$, a zero-mean binormal. If there is also density convergence, with $(X_n, Y_n)$ having a density $f_n(x, y)$ tending to the appropriate binormal density $f(x, y)$; and $f_n(x, y)$ as well as the marginals $f_{Y_n}(y)$ are continuous, show that $X_n \,|\, (|Y_n| \le \delta_n)$ tends to $X \,|\, (Y = 0)$, for an appropriate sequence $\delta_n \to 0$.

**(h)** Show that the same limiting distribution statement as in (g) holds also when $(X_n, Y_n)$ has a discrete distribution, or a noncontinuous density, using the 'adding small Gaussian noise to get densities' trick.

**Ex. 2.64** *Limiting normality of rank sums statistics.* (xx to be edited and polished; nils rant so far. we put it in if it looks smooth enough, and with a brief pointer to Story #76. point to Swensen (1983). xx) In a population of $n$ individuals, followed on some continuous scale, a subgroup of interest, of size $m$, has ranks $X_1, \ldots, X_m$. These form a randomly selected subset of size $m$ from $\{1, \ldots, n\}$, with all such $\binom{n}{m}$ subsets equally likely. The rank sum $Z_n = X_1 + \cdots + X_m$ is the Wilcoxon statistic.

**(a)** Explain that one may write $Z_n = \sum_{i=1}^n iJ_i$, where the 0-1 variables $J_i$ are such that precisely $m$ of them are 1, and with all $\binom{n}{m}$ subsets of such 1s being equally likely. Find $\mathrm{E}\, J_i$, $\mathrm{Var}\, J_i$, $\mathrm{cov}(J_i, J_j)$ for $j \ne i$. Writing $p = m/n$ for the sample ratio, show using either of the representations $\sum_{i=1}^m X_i$ or $\sum_{i=1}^n iJ_i$ that

$$\mathrm{E}\, Z_n = \tfrac{1}{2}m(n+1) \doteq \tfrac{1}{2}n^2 p, \quad \mathrm{Var}\, Z_n = (1/12)(n+1)m(n-m) \doteq (1/12)n^3 p(1-p).$$

**(b)** We aim indeed at showing limiting normality of $Z_n$ here, with both $n$ and $m$ becoming larger, with $m/n \to p$. Explain that this must mean

$$(Z_n - \tfrac{1}{2}n^2 p)/n^{3/2} \to_d \mathrm{N}(0, (1/12)p(1-p)).$$

We cannot use CLT or Lindeberg for studying $Z_n$, since the $X_i$ are dependent, as are the $J_i$. Consider however a different parallel setup, involving independent Bernoulli variables $J_1^*, \ldots, J_n^*$ with $\Pr(J_i^* = 1) = p = m/n$. Explain that the distribution of $Z_n$ is the same as the distribution of $Z_n^* = \sum_{i=1}^n i J_i^*$ given $\sum_{i=1}^n J_i^* = m$. Show now that

$$\begin{pmatrix} A_n \\ B_n \end{pmatrix} = \begin{pmatrix} (1/\sqrt{n}) \sum_{i=1}^n (i/n)(J_i^* - p) \\ (1/\sqrt{n}) \sum_{i=1}^n (J_i^* - p) \end{pmatrix} \to_d \begin{pmatrix} A \\ B \end{pmatrix} \sim \mathrm{N}_2(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1/3 & 1/2 \\ 1/2 & 1 \end{pmatrix}).$$

**(c)** Find the distribution of $A \,|\, (B = b)$, and show in particular that $A \,|\, (B = 0) \sim \mathrm{N}(0, 1/12)$. This gives a clear limiting normality statement for $Z_n^*$, conditional on $\sum_{i=1}^n J_i^* = m$, and nicely solves our Wilcoxon problem; show that it corresponds precisely to the $(Z_n - \frac{1}{2}n^2 p)/n^{3/2}$ limiting statement above. (xx some extra care needed. but an easy and instructive way to show normality for Wilcoxon, and also for other related variables. to illustrate, find limiting normality for $X_1^{1/2} + \cdots + X_m^{1/2} = \sum_{i=1}^n i^{1/2} J_i$. xx)

**(d)** (xx if we manage: also a link via the uniform order statistic process, and to integrals of salt-and-pepper processes, with $W_n = \int_0^1 ([ns]/n)\, \mathrm{d}C_n(s)$, with the $\mathrm{d}C_n(s)$ being $\mathrm{d}s$ or 0 with probabilities $p$ and $1-p$, but conditional on the random region $\int_0^1 \mathrm{d}C_n(s)$ being $p = m/n$. if we're lucky there is a limit expressible as integral or Brownian bridge, for Ch9. nils will attempt to fix this and at least make the idea more precise. xx)

**Ex. 2.65** *Nonnormal limits.* (xx polish, and calibrate with Story #96. xx) Normally limits are normal, but not always. Here we shall indeed work with variables with mean zero and variance one, where the sample averages have nonnormal limits. The basic construction is as follows. Let $U_1, U_2, \ldots$ be i.i.d., with mean zero and variance one, and with m.g.f. $M_0(s) = \mathrm{E}\exp(sU_i)$ finite in a neighbourhood around zero; in particular, all moments for the $U_i$ are finite. Let independently of these $J_1, J_2, \ldots$ be independent Bernoulli variables with $\Pr(J_i = 1) = 1/i$, $\Pr(J_i = 0) = 1 - 1/i$. Then form

$$Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n J_i \sqrt{i}\, U_i = \sum_{i=1}^n J_i \sqrt{i/n}\, U_i.$$

A picture to have in mind is that most of the terms will be zero, with non-zero contributions becoming both more rare and more big as time proceeds.

**(a)** Show that there will with probability one be infinitely many $J_i = 1$, i.e. non-zero terms in the $Z_n$ sum as $n$ grows.

**(b)** Show that the terms $J_i \sqrt{i} U_i$ have mean zero and variance one; hence also the normalised sample average $Z_n$ has mean zero and variance one. Find also an expression for the kurtosis $\kappa_n = \mathrm{E}\, Z_n^4 - 3$ of $Z_n$, and show that $\kappa_n \to \frac{1}{2} a_4$, where $a_4 = \mathrm{E}\, U_i^4$. Compare this to what we are 'used to' from the Lindeberg theorem.

**(c)** We already know that if $Z_n$ has a limit distribution, it can't be normal. Working with the m.g.f., show that

$$M_n(t) = \mathrm{E}\exp(tZ_n) = \prod_{i=1}^n \Big[1 + \frac{1}{i}\{M_0(t\sqrt{i/n}) - 1\}\Big],$$

for all $t$ around zero for which $M_0(t)$ is finite.

**(d)** Show that

$$\prod_{i=1}^n \Big[1 + \frac{1}{i}\{M_0(t\sqrt{i/n}) - 1\}\Big] \to \exp\Big\{\int_0^1 \frac{M_0(t\sqrt{x}) - 1}{x}\, \mathrm{d}x\Big\}.$$

Work first with Special Case One, where we let $U_i$ have the simple symmetric two-point distribution $\Pr(U_i = 1) = \Pr(U_i = -1) = \frac{1}{2}$. Find the limiting kurtosis for $Z_n$ in this case. Show that $M_0(s) = \frac{1}{2}e^s + \frac{1}{2}e^{-s} = 1 + (1/2!)s^2 + (1/4!)s^4 + \cdots$, and use this to find an infinite-sum expression for the limit of $M_n(t)$. Have you now proved that $Z_n$ has a limit distribution?

**(e)** Then work with Special Case Two, where the $U_i$ have a double exponential distribution, of the form $f(u) = \frac{1}{2}\sqrt{2}\exp(-\sqrt{2}|u|)$ on the real line (the $\sqrt{2}$ factor is there to ensure variance one). Find the m.g.f. $M_0(s)$ for the $U_i$, and then the m.g.f. $M(t)$ for the limit distribution of $Z_n$.

**(f)** For most cases, regarding the distribution for the $U_i$, it is hard to learn the explicit distribution for $Z_n$ (even in cases where there might be a clear distribution for its limit). For Special Case Two, however, find the explicit distribution for $Z_n$, for any given $n$.

### CLTs for dependent random variables

**Ex. 2.66** *A CLT for 1-dependent variables.* (xx decide later if these few should be pushed to Ch. 13. xx) Consider a stationary sequence $Y_1, Y_2, \ldots$, with mean zero and variance one, being 1-dependent. Stationarity means $(Y_1, \ldots, Y_r)$ having the same distribution as $(Y_{i+1}, \ldots, Y_{i+r})$, for any $i$ and block lengths $r$, and 1-dependence means that $Y_i, Y_{i+1}$ may be dependent, but $Y_1, \ldots, Y_i$ is independent of $Y_{i+2}, Y_{i+3}, \ldots$. This exercise reaches a CLT for $\sum_{i=1}^n Y_i$, representing a genuine extension of the usual CLT and Lindeberg theorems from independence.

**(a)** Writing $\rho = \mathrm{corr}(Y_i, Y_{i+1})$, show that $(1/k)\mathrm{Var}(Y_1 + \cdots + Y_k) = 1 + 2(1 - 1/k)\rho$ which then goes to $1 + 2\rho$ for increasing $k$.

**(b)** For a given block length $k$, split $Y_1 + \cdots + Y_n$ into $[n/k]$ blocks, and write block $j$ of these as $U_j + V_j$, with $U_j$ as sum of $k-1$ consecutive observations and $V_j$ the last one of that block. Write then

$$Z_n = (1/\sqrt{n})\sum_{i=1}^n Y_i = (1/\sqrt{n})\Big(\sum_{j=1}^{[n/k]} U_j + \sum_{j=1}^{[n/k]} V_j + E_n\Big),$$

with $E_n$ any extra left after the $k[n/k]$ variables captured in these first $[n/k]$ blocks.

**(c)** Explain why $U_1, U_2, \ldots$ are independent, so that the usual CLT applies to these. Show that $(1/\sqrt{n})\sum_{j=1}^{[n/k]} U_j \to_d \mathrm{N}(0, \tau_k^2)$, with $\tau_k^2 = (1/k)\mathrm{Var}\,(Y_1 + \cdots + Y_{k-1})$.

**(d)** Then use Ex. 2.30 to prove that $(1/\sqrt{n})\sum_{i=1}^n Y_i \to_d \mathrm{N}(0, 1 + 2\rho)$, i.e. a CLT for 1-dependent variables.

**(e)** Assume $X_1, X_2, \ldots$ are i.i.d. with mean zero and variance one. Consider $Z_n = (1/\sqrt{n})(X_1X_2 + X_2X_3 + \cdots + X_{n-1}X_n)$. Show that $Z_n \to_d \mathrm{N}(0,1)$. Show also that

$$Z_n' = (1/\sqrt{n})\sum_{i=1}^{n-1}(X_i - \bar{X}_n)(X_{i+1} - \bar{X}_n)$$

has the same limit distribution, where $\bar{X}_n$ as usual is the sample mean.

**Ex. 2.67** *A CLT for m-dependent variables.* In natural generalisation of Ex. 2.66, consider a stationary $m$-dependent sequence $Y_1, Y_2, \ldots$, with mean zero and variance $\sigma^2$. There is accordingly potential dependence among $Y_1, \ldots, Y_m$, but for any $i$, $(Y_1, \ldots, Y_i)$ is independent of $(Y_{i+m+1}, \ldots, Y_n)$.

**(a)** Writing $\operatorname{cov}(Y_i, Y_j) = \sigma^2\rho(|j-i|)$, with the autocorrelation function $\rho(\cdot)$, show first that in general terms,

$$(1/n)\operatorname{Var}\Big(\sum_{i=1}^n Y_i\Big) = \sigma^2\Big\{1 + 2\sum_{j=1}^n (1 - j/n)\rho(j)\Big\}.$$

Then show that for the case of $m$-dependence, for any $k \geq m$, we have $(1/k)\operatorname{Var}(Y_1 + \cdots + Y_k) \to \sigma^2\{1 + 2\sum_{j=1}^m \rho(j)\}$,

**(b)** Extend arguments and techniques from Ex. 2.66 to show that $(1/\sqrt{n})\sum_{i=1}^n Y_i$ tends to a zero-mean normal with variance $\sigma^2\{1 + 2\rho(1) + \cdots + 2\rho(m)\}$.

**(c)** (xx a bit on how the acf works for an i.i.d. sequence: $\sqrt{n}\widehat{\rho}(j) \to_d \mathrm{N}(0,1)$, for each $j$. xx)

### Further material and leftovers, as of 30-June-2025

**Ex. 2.68** *Summing geometrically many terms.* Suppose $Y_1, Y_2, \ldots$ are i.i.d. with mean zero, variance $\sigma^2$, and m.g.f. $M_0(t)$. With $\Pr(N = n) = (1-p)^{n-1}p$ for $n \geq 1$, i.e. a geometric distribution, consider $Z_p = p^{1/2}(Y_1 + \cdots + Y_N)$, with the $Y_i$ being independent of $N$.

**(a)** Show first that the generating function for $N$ is $\mathrm{E}\, s^N = ps/\{1 - (1-p)s\}$ for $|s| < 1/(1-p)$; see Ex. 1.39. Show that $Z_p$ has variance $\sigma^2$, and that its m.g.f. may be written

$$K_p(t) = \mathrm{E}\,\exp(tZ_p) = pM_0(p^{1/2}t)/\{1 - (1-p)M_0(p^{1/2}t)\}.$$

**(b)** Then use $M_0(t) = 1 + \frac{1}{2}\sigma^2t^2 + o(|t|^2)$ to demonstrate that as $p \to 0$, with increasing number of terms $\mathrm{E}\, N = 1/p$, we have $K_p(t) \to 1/(1 - \frac{1}{2}\sigma^2t^2)$ for $|t| < \sqrt{2}/\sigma$. This shows that $Z_p \to_d L_\sigma$, the Laplace distribution with standard deviation $\sigma$, see Ex. 1.34.

**(c)** For the particular case of a sum of randomly many normal terms, let the $Y_i$ be i.i.d. standard normal. Show what $Z_p \,|\, N \sim \mathrm{N}(0, pN)$, and that $pN \to_d \mathrm{Expo}(1)$ as $p \to 0$. Explain how this matches Ex. 1.34.(c).

**(d)** (xx one or two further illustrations, with Laplace limit of $p^{1/2}(Y_1 + \cdots + Y_N)$. with $Y_i = Q_i - 1$, Poisson, we learn $p^{1/2}(V_N - N) \to_d L$, where $V_N \sim \mathrm{Pois}(N)$. similarly with $p^{1/2}(W_N - N)$, where $W_N \,|\, N \sim \chi^2_N$, randomly many degrees of freedom. xx)

**(e)** (xx something to simulate, to illustrate the cusp behaviour at the centre. with the $Y_i$ having mean $\xi$ and variance 1, we have

$$p^{1/2}(Y_1 - \xi + \cdots + Y_N - \xi) = p^{1/2}N(\bar{Y} - \xi) = (pN)^{1/2}\, N^{1/2}(\bar{Y} - \xi) \to_d L_1.$$

this gives different inference for $\xi$, and different predictions for $\bar{Y}$, than what we're used to from normal terrain. we're also in scale mixtures of normal terrain, with random variance tending to a unit exponential. i'll look for variations. i do like the $pN \to_d \mathrm{Expo}(1)$, since it gives the cool cusp in the limit for $Z_p$, but other variations for $pN \to_d V$ are ok too. xx)

**(f)** (xx just ranting away a bit until things settle. xx) More generally, with $Y_1, Y_2, \ldots$ being i.i.d. with zero mean and unit variance, consider

$$Z_p = p^{1/2}(Y_1 + \cdots + Y_N) = (pN)^{1/2}N^{1/2}\bar{Y}_N,$$

with $N$ having a distribution such that $pN \to_d V$, say, as $p \to 0$. From the CLT, $N^{1/2}\bar{Y}_N \to_d \mathrm{N}(0,1)$, so this amounts to a situation with a normal limit, but a random variance, as in $X \,|\, V \sim \mathrm{N}(0,V)$ but $V$ random. Here $\mathrm{E}\exp(tX) = \mathrm{E}\exp(\frac{1}{2}t^2V) = M_0(\frac{1}{2}t^2)$, where $M_0(u) = \mathrm{E}\exp(uV)$ is the mgf for $V$. Also, $X$ has density

$$\bar{f}(x) = \int \phi(x/v^{1/2})(1/v^{1/2})\,\mathrm{d}H(v),$$

with $H$ the distribution of $V$. – The special case above amounts to $N \sim \mathrm{geom}(1/p)$, where $pN$ tends to the unit exponential, and where $X$ gets the Laplace distribution. For another case, consider $N \,|\, \lambda \sim \mathrm{Pois}(\lambda/p)$, and with $\lambda$ having its own distribution with mean and variance $\lambda_0$ and $\tau_0^2$, say. Show that $\mathrm{E}\,pN = \lambda_0$ and that $\mathrm{Var}\,pN \to \tau_0^2$. Also consider the special case for this setup where $\lambda \sim \mathrm{Gam}(a,b)$. From

$$\mathrm{E}\,\{\exp(-spN) \,|\, \lambda\} = \exp[-(\lambda/p)\{1-\exp(-sp)\}]$$

deduce

$$\mathrm{E}\exp(-spN) = \frac{1}{[1+(1/b)(1/p)\{1-\exp(-sp)\}]^a} \to \frac{1}{(1+s/b)^a},$$

and that $pN \to_d V_{a,b}$, another $\mathrm{Gam}(a,b)$. With the construction above, the normal scale mixture variable $X$ has mgf $1/(1-\frac{1}{2}t^2/b)^a$, and density

$$\bar{f}(x) = \int \phi(x/v^{1/2})(1/v^{1/2})\frac{b^a}{\Gamma(a)}v^{a-1}\exp(-bv)\,\mathrm{d}v.$$

This is a Laplace, for $a = 1$. (xx but different, and interesting, for other $a$. round this off. xx)

**(g)** (xx i think we can use these tools to form a full Laplace process, for BNP use or otherwise. we should tune in to a $Z_p(t) = p^{1/2}(Y_1 + \cdots + Y_{N(t)})$, with a clever $N(t)$. will look at $N(t)$ being a negative binomial process with mean $t/p$. nils thinks this works: (i) $\lambda \sim \mathrm{Gam}(1,b)$, with mean $\lambda_0 = 1/n$. (ii) $N \,|\, \lambda \sim \mathrm{Pois}(\lambda/p)$. write down $\mathrm{E}\,(pN \,|\, \lambda)$ and $\mathrm{Var}\,(pN \,|\, \lambda)$, then unconditional mean and variance for $pN$. (iii) find limit in distribution of $pN$. (iv) study $Z_p = p^{1/2}\sum_{i=1}^N Y_i$, close to $(pN)^{1/2}$ times a normal, etc. (v) make it into a full Laplace process, by having $\lambda_t \sim \mathrm{Gam}(1,b_t)$. xx)

**Ex. 2.69** *How many records are set?* Consider i.i.d. observations $X_1, X_2, \ldots$ from some continuous distribution on the line. Define $R_n = 1$ if $X_n$ is bigger than all previous datapoints, i.e. a new record has been set.

**(a)** Explain first that the number of records set, in the course of the first $n$ occasions, is $Z_n = \sum_{i=1}^n R_i$. Show that $\Pr(R_n = 1) = 1/n$, so that $\mathrm{E}\,Z_n = H_n = 1+1/2+\cdots+1/n$, the partial sum of the harmonic series. As we have noted earlier, $H_n - \log n \to \gamma_e = 0.5772...$, the Euler constant, so the number of records set is approximately $\log n + \gamma_e$.

**(b)** Show that $R_1, \ldots, R_n$ are independent, and deduce from this that $\mathrm{Var}\,Z_n = H_n - \sum_{i=1}^n 1/i^2 \doteq \log n + \gamma_e - \pi^2/6$. Use the Lindeberg theorem to show that $(Z_n - H_n)/H_n^{1/2} \to_d \mathrm{N}(0,1)$. Show also that $(Z_n - \log n)/(\log n)^{1/2} \to_d \mathrm{N}(0,1)$, which gives a simpler but somewhat more crude approximation to $Z_n$ probabilities. Use these normal limits to give a prediction band, with $\Pr(a_n \le Z_n \le b_n) \to 0.95$.

**(c)** Now consider $A_n = Z_{2n} - Z_n$, the number of records set during $n, n+1, \ldots, 2n$. Show that $A_n$ tends to the Poisson with mean $\log 2$. Generalise.

**Ex. 2.70** *Characterisations of the normal and the Cauchy.* There are many characterisation theorems in probability theory, results saying that appropriate assumptions or properties fully characterise certain distributions- Here we give a few such, using characteristic functions. (xx check with care. xx)

**(a)** Suppose $F$ is a distribution, symmetric around zero, such that if $X$ and $Y$ are independent from this distribution, then $(X+Y)/\sqrt{2}$ has the same distribution. Prove that the distribution is normal.

**(b)** Suppose $F$ is a distribution, symmetric around zero, such that if $X$ and $Y$ are independent from this distribution, then $(X+Y)/2$ has the same distribution. Prove that the distribution is a Cauchy.

**(c)** (xx one more. xx)

## Notes and pointers

(xx to come. we point to certain famous things from the past: Kolmogorov (1933b), Lindeberg (1922), Borel and Cantelli. tail bounds. emil's extension of the Inlow (2010) paper, from CLT to Lindeberg. more on Lindeberg and the history of CLT developments in Cramér (1976), also see Schweder (1980, 1999). mention the classical text Billingsley (1968). xx)

(xx Weierstraß, Hveberg (2019), A generation or so later, such results had been generalised to so-called Stone–Weierstraß theorems, stating, in various forms, that certain classes of functions are rich enough to deliver uniform approximations to bigger classes of functions. This is useful also in branches of probability theory. xx)

(xx the first of the two lemmas: Nils beta 1990. Weierstrass: see also Hveberg (2019). our result xxxx if an extension of one by Inlow (2010). xx)

(xx include also something on Fra Preface i *Life and Times of CLT*: "For those who teach a course in probability whose objective is to prove the Central limit theorem of interest [...] is commentary on the characteristic function approach employed by Lyapunov versus the 'very simple' proof, as Le Cam describes it, given by Lindeberg". also, LeCam (1986). xx)

[xx for Scheffé: see Scheffé (1947), but also Kusolitsch (2010), who explains that the result is a special case of results published by F. Riesz in 1928. see also what Scheffe says in his paper about comments he got from Morse. xx]

(xx push this to Notes. xx) Inlow (2010) has shown how one can prove the usual CLT without the technical use of characteristic and hence complex functions. Essentially, he writes the $X_i$ in question as $Y_i + Z_i$ with $Y_i = X_i\, I\{|X_i| \le \varepsilon\sqrt{n}\}$ and $Z_i = X_i\, \{|X_i| > \varepsilon\sqrt{n}\}$, after which 'ordinary' m.g.f.s may be used for the part involving the $Y_i$, yielding the normal limit, supplemented with analysis to show that the part involving the $Z_i$ tends to zero in probability. – It is a non-trivial matter to extend Inlow's arguments, from the CLT to the Lindeberg theorem, but this is precisely what is done in Stoltenberg (2019). Check that note, on the book website, and make sure you understand its main tricks and steps.

(xx When Jarl Waldemar Lindeberg was reproached for not being sufficiently active in his scientific work, he said, 'Well, I am really a farmer'. And if somebody happened to say that his farm was not properly cultivated, his answer was, 'Of course my real job is to be a mathematics professor'. Hundred years ago!, i.e. in 1920, he published his first paper on the CLT, and in 1922 he generalised his findings to the classical Lindeberg

Theorem, with the famous Lindeberg Condition, securing limiting normality of a sum of independent but not identically distributed random variables. He did not know about Ляпунов's earlier work, and therefore not about условие Ляпунова, the Lyapunov condition, which we treat below as a simpler-to-reach condition than the more general one of Lindeberg. Other luminaries whose work touch on these themes around the 1920ies and beyond include Paul Lévy, Harald Cramér, William Feller, and, intriguingly, Alan Turing who (allegedly) won the war and invented computers etc. xx)

(xx point to a couple of characterisation theorems books, kagan linnik rao, one more. xx)

(xx for Notes: The little $\log(1+x)$ lemma is stated, proven, and used in Hjort (1990b, Appendix). xx)

(xx the material is from Hjort and Pollard (1993) and Hjort (1986a). xx)

[xx For notes and pointers: Ex. 2.19(b) is from Jacod and Protter (2004, p. 166), they have it from Pollard (1984) xx].

ToDo notes, of 30-June-2025.

Clean and calibrate. Include a couple of classic nonparametric test procedures, like Wilcoxon, the sign test, more, to showcase the use of Lindeberg things to show limiting normality of such statistics too. point to Hjort and Pollard (1993) and Hjort (1986a).

Include some non-normal limits. Can take $\widehat{\mu}^* = \{1-c(D_n)\}\widehat{\mu}_{\rm narr} + c(D_n)\widehat{\mu}_{\rm wide}$ from model selection. And point to $n^{2/5}$ rates for $f$ estimation.

# I.3

## Parameters, estimators, precision, confidence

With data observed from a statistical model, the theme of this chapter is that of constructing estimators for unknown statistical parameters, along with assessing their precision. Basic concepts include the bias, the variance, the mean squared error of estimators. The development also naturally leads to the important notion and basic machinery of confidence intervals. General estimation methods covered here include the method of moments and the method of quantiles; these can also be combined. For regression setups, with response variables influenced by covariates, we go through the method of least squares. To understand and utilise the properties of classes of estimators in general models, we utilise the machinery of large-sample normal approximations, from Ch. 2. This also enables one to assess precision and to compare different competing estimators. Similar remarks apply also for the more versatile method of maximum likelihood, treated in Ch. 5.

*Key words:* approximations, confidence, estimators, linear regression, model parameters, moment matching, quantile matching, risk

Most statistical models have *parameters*, as we learn from the generous variety of models in Ch. 1. Parameters may then be fine-tuned, or estimated, from data, which is the grand theme of the present chapter. In generic terms, if a model has density $f(y, \theta)$, with $\theta = (\theta_1, \ldots, \theta_p)^{\mathrm{t}}$ its parameter vector, we use data $\mathcal{D}$ to construct *an estimator* $\widehat{\theta} = \widehat{\theta}(\mathcal{D})$. Thus $f(y, \widehat{\theta})$ is the fitted model, which we use for interpretation and inference, themes we return to in more detail in later chapters. The data $\mathcal{D}$ can often be in the form of direct independent observations $y_1, \ldots, y_n$ from the model, but can be in vector form, and also be different in character, involving censoring mechanisms, or measurement error.

A sensible minimum demand for an estimator is that it should tend to the right value, with increasing data volume. Formally, if $\widehat{\theta} = \widehat{\theta}_n$ is the estimator for some parameter whose true value is $\theta_0$, based on a sample of size $n$, then we say that the estimator (or, more pedantically, the sequence of estimators) is *consistent* provided $\widehat{\theta}_n$ converges in probability to $\theta_0$, i.e. $\widehat{\theta}_n \to_{\mathrm{pr}} \theta_0$ in the terminology of Ch. 2.

consistency of an estimator

One often needs estimates and inference methods for *focus parameters*, those of particular and context-driven interest, which are one-dimensional functions $\mu = \mu(\theta_1, \ldots, \theta_p)$ of the underlying model parameter vector. If $\widehat{\mu}$ is an estimator for this parameter, we often care about its mean, represented here as

focus parameter

$$\mathrm{E}_\theta\, \widehat{\mu} = \mu + b(\theta). \tag{3.1}$$

The footscript signals that the expectation operator is at work at the parameter position $\theta$. The $b(\theta)$ is termed *the bias*; in various cases it is a function of $\mu$ only, but in general it depends on the full parameter vector $\theta$. If $\mathrm{E}_\theta\,\widehat{\mu} = \mu$, at all positions $\theta$, we say the estimator is *unbiased*. In addition to wishing for estimators with small bias, we care about its variability, and often about its *mean squared error*

unbiased estimator

mean squared error

$$\mathrm{mse}(\widehat{\mu},\theta) = \mathrm{E}_\theta\,(\widehat{\mu}-\mu)^2 = \mathrm{Var}_\theta\,\widehat{\mu} + b(\theta)^2, \tag{3.2}$$

the classic variance plus squared bias. This is a function of the unknown parameter, and gives a way of understanding and comparing performance for competing estimation schemes. When we can sort out the mathematics properly, depending on the situation at hand, we then choose estimators with smaller mse than those of competitors.

The $\mathrm{mse}(\widehat{\mu},\theta)$ of (3.2) is sometimes called the risk, or risk function, and relates to having chosen squared error $(\widehat{\mu}-\mu)^2$ as the underlying measure of quality. Other ways in which to compare and rank performance, involving also different quality functions and risk functions, are dealt with in Ch. 9.

Fundamental and conspicuous instruments in the statistical toolkits, when summarising and reporting findings of investigations, are *confidence intervals* and *testing of null hypotheses*. The development in the present chapter deals with the former, regarding interpretation, construction, properties, performance, whereas the following chapter handles the latter, along with further connections.

Consider in general terms data $y$, perhaps a long vector or a data matrix, from a model with parameter $\theta = (\theta_1,\ldots,\theta_p)$, with $\mu = \mu(\theta_1,\ldots,\theta_p)$ a parameter of interest. Then $[L(y),U(y)]$ is a *confidence interval*, with confidence level $\alpha$, like 0.90 or 0.95 or an even higher 0.99, provided

confidence interval

$$\Pr_\theta\{L(Y) \le \mu \le U(Y)\} = \alpha \quad \text{for all } \theta. \tag{3.3}$$

Thus $[L(Y),U(Y)]$ is a random interval, and with a high number of repeated situations it will capture the underlying $\mu$ a fraction $\alpha$ of the times. The reported confidence interval is $[L_{\mathrm{obs}},U_{\mathrm{obs}}] = [L(y_{\mathrm{obs}}),U(y_{\mathrm{obs}})]$, computed based on the actually observed data $y_{\mathrm{obs}}$. Occasionally one also wants to construct *confidence regions* for a parameter vector, as opposed to confidence intervals for each of its components. Thus if $(a,b)$ is a parameter pair in a model, a random region $R$, based on data, is a 95 percent confidence region provided $\Pr_\theta((a,b)\in R) = 0.95$ for all model parameters $\theta$.

In various setups one can study distributions, biases, variances, confidence etc. quite accurately, as will be seen in many exercises below. Often enough this might be too complicated, however, and one relies instead on good approximations. There is indeed a host of normal approximations, sometimes with additional tools for finetuning these, as studied in Ch. 2, with yet more to come in Ch. 5. These methods may be understood, appreciated, seen in action, and used for new situations, without necessarily having been through each $\delta$ and $\varepsilon$ of Ch. 2.

In this chapter we learn certain estimation principles, including those associated with the method of moments and the method of quantiles. There is also room for combining such methods, or for coming up with new estimators in unfamiliar waters. We go on to more advanced models and hence estimation methods in later chapters (and in several of our stories), but included below is the basics of linear regression and the least sum of squares methods. The more versatile and often well-performing method of *maximum likelihood* will be studied with care in Ch. 5.

[xx In this brief intro there should be a figure, conveying some basic ideas. We may snikinnføre confidence intervals, but that comes with more weight in Ch. 4, along with testing and power and p-values. we do mention a few key concepts here in intro, like unbiasedness, low variance, etc. xx]

### Precise estimation in a few classical models

**Ex. 3.1** *Mean squared error.* Suppose data lead to an estimator $\widehat{\mu}$ for a focus parameter $\mu = \mu(\theta)$, in a model with parameter $\theta$.

**(a)** Verify the mse formula (3.2), and note its Pythagorean character. Make a little right-angle triangle figure with absolute bias and standard deviation for the two short sides and root-mse, rmse $=$ mse$^{1/2}$, on the long side. These root operations bring the risk components down to the original scale of the measurements.

**(b)** For a simple situation, let $Y \sim \mathrm{N}(\theta, 1)$, with $\theta$ to be estimated. Find formulae for the mean squared errors of the three estimators $0.9\,Y$, $Y$, $1.1\,Y$. Note the interplay between bias and variance.

**(c)** Generalise to the case of $Y_1, \ldots, Y_n$ being i.i.d. from $\mathrm{N}(\theta, 1)$. Find $\mathrm{mse}(\widehat{\theta}, \theta)$ for the three estimators $0.99\,\bar{Y}$, $\bar{Y}$, $1.01\,\bar{Y}$, with $\bar{Y}$ the sample average. Comment on what you find.

**(d)** In somewhat more general terms, consider an i.i.d. sample $Y_1, \ldots, Y_n$ from a distribution with unknown mean $\mu$ and variance $\sigma^2$. Show that $\bar{Y}$ is unbiased with variance $\sigma^2/n$. If your estimator for $\mu$ is $c_n\bar{Y}$, what is required of $c_n$, in order for the mean squared error to go to zero with growing $n$?

**Ex. 3.2** *Binomial estimation.* Consider $Y$ being binomial $(n, p)$, as in Ex. 1.3.

**(a)** To estimate $p$ the canonical choice is $\widehat{p} = Y/n$. Find its mean and variance, and a formula for $\mathrm{mse}(\widehat{p}, p) = \mathrm{E}_p\,(\widehat{p} - p)^2$.

Bayes estimator

**(b)** Then compare the simple binomial unbiased proportion with the estimator $\widehat{p}_B = (Y+1)/(n+2)$; this is arguably the very first Bayes estimator. Find its bias and variance, and a formula for $\mathrm{mse}(\widehat{p}_B, p)$. Draw the two mse functions in a diagram, for say $n = 10$. When is the Bayes estimator better than the $Y/n$, according to this criterion?

**Ex. 3.3** *Estimating the normal mean.* Consider independent observations $Y_1, \ldots, Y_n$ from the normal distribution $\mathrm{N}(\xi, \sigma^2)$.

**(a)** Prove that the sample average $\bar{Y} = n^{-1}\sum_{i=1}^n Y_i$ has the $\mathrm{N}(\xi, \sigma^2/n)$ distribution. This $\bar{Y}$ is the canonical estimator for $\mu$. Find also a clear formula for its risk, or mean squared error, namely $\mathrm{mse}(\bar{Y}, \mu, \sigma) = \mathrm{E}_{\mu,\sigma}(\bar{Y} - \mu)^2$. The subscript indicates that the mean operator is with respect to the probability mechanism dictated by $(\mu, \sigma)$.

**(b)** Then generalise the above somewhat, by finding the mean and variance also for the estimator $\widehat{\mu} = b\bar{Y}$, with $b$ a constant (which might be close to 1). Use this to put up a clear expression for

$$\mathrm{mse}(b\bar{Y}, \mu, \sigma) = \mathrm{E}_{\mu,\sigma}(b\bar{Y} - \mu)^2.$$

Illustrate this, for values $b = 0.98, 1.00, 1.02$, and comment. For what values of the parameters $(\mu, \sigma)$ will the estimator $0.98\,\bar{Y}$ be better than the classic $\bar{Y}$? Are there values of the parameters where $1.02\,\bar{Y}$ is better than the plain $1.00\,\bar{Y}$?

**(c)** Suppose the starting assumptions about the data at hand are changed to merely saying that the $Y_i$ are i.i.d. with mean $\mu$ and standard deviation $\sigma$, i.e. we avoid saying

that the distribution of the error terms $\varepsilon_i = (Y_i - \mu)/\sigma$ needs to be exactly standard normal. How does this affect your findings and claims for the previous points?

**Ex. 3.4** *Estimating the normal variance.* As in Ex. 3.3, suppose there are i.i.d. data $Y_1, \ldots, Y_n$ from the $\mathrm{N}(\xi, \sigma^2)$. Here we care about the standard deviation parameter $\sigma$. As we saw in Ex. 1.50, $Z = \sum_{i=1}^n (Y_i - \bar{Y})^2 \sim \sigma^2 \chi^2_m$, where $m = n - 1$. Also, the $Z$ is stochastically independent of the sample mean $\bar{Y}$.

**(a)** Use the statement above to find the mean and variance of $\widehat{\sigma}^2 = cZ$ (where $c$ ought to be about $1/n$). Find the mean squared error $\mathrm{mse}(cZ, \sigma) = \mathrm{E}_\sigma\,(cZ - \sigma^2)^2$. Check in particular the result for $c = 1/(n-1)$, the classical factor to make the estimator unbiased; for $c = 1/n$, which comes out of the maximum likelihood paradigm (see Ch. 5); and for $c = 1/(n+1)$. Find the best possible constant $c$ for estimators of this type $cZ$, using the mean squared error on the $\sigma^2$ scale as criterion.

**(b)** Find also the mean and variance of $dZ^{1/2}$, seen as an estimator of $\sigma$, i.e. on the standard deviation scale, not that of the variance. Find an expression for $\mathrm{mse}(dZ^{1/2}, \sigma) = \mathrm{E}_\sigma\,(dZ^{1/2} - \sigma)^2$. Find the best $d$, according to this criterion.

**(c)** We may also finetune $\sigma$ estimation on the log-scale. Examine the risk function $\mathrm{mse}(kZ^{1/2}, \sigma) = \mathrm{E}_\sigma\,\{\log(kZ^{1/2}) - \log\sigma\}^2$, and find the best value of $k$.

**(d)** A different solution to the issue of determining 'the best constant' when estimating $\sigma$, disregarding tradition and mathematical convenience, might be as follows. With $\widehat{\sigma}^2 = Z/(n-1)$ being the traditional sample variance, with $1/(n-1)$ selected to achieve unbiasedness on the $\sigma^2$ scale, consider $\sigma^* = c_n\widehat{\sigma}$, with $c_n$ to be fine-tweaked perhaps a little bit away from 1. Find the $c_n$ that makes $\mathrm{risk}(c_n\widehat{\sigma}, \sigma) = \mathrm{E}_\sigma\,|c_n\widehat{\sigma} - \sigma|$ smallest. This means relying on absolute error as loss function, and the solution needs numerical minimisation of a function which needs numerical integration. Give a table with these optimal $c_n$ for say $n = 10, \ldots, 30$. Show that $c_n \to 1$ as $n$ grows.

**Ex. 3.5** *Confidence interval for an exponential rate.* Choose a sample size $n$, and simulate i.i.d. variables $Y_1, \ldots, Y_n$ from the exponential distribution, see Ex. 1.8, with parameter $\theta$ equal to say $\theta_0 = 3.33$.

**(a)** Construct a 90 percent confidence interval $[L, U]$ for $\theta$. Check if $\theta_0$ is contained in this interval, for the data you generated. Repeat the experiment say 100 times, and record how often the intervals contain $\theta_0$. What is in fact the distribution of $N$, the number of the 100 intervals which cover the truth?

**(b)** In addition to checking whether the intervals cover the truth, compute the length $D = U - L$, and give a histogram of its distribution. Find $\mathrm{E}\,D$. Repeat also these experiments with a couple of other sample sizes, and comment.

**Ex. 3.6** *Confidence interval for a normal variance.* Let $Y_1, \ldots, Y_n$ be i.i.d. from a normal distribution. How can we set up confidence intervals for the standard deviation $\sigma$ (or, equivalently, for the variance $\sigma^2$)? Writing $m = n - 1$, the sample variance is $\widehat{\sigma}^2 = Z/m$, with $Z = \sum_{i=1}^n (Y_i - \bar{Y})^2 \sim \sigma^2 \chi^2_m$, from Ex. 1.50.

**(a)** Start with $[a, b] = [\Gamma_m^{-1}(0.05), \Gamma_m^{-1}(0.95)]$, an interval covering the $\chi^2_m$ with probability 0.90. Transform $a \le m\widehat{\sigma}^2/\sigma^2 \le b$ to the confidence interval $\mathrm{ci} = [(m/b)^{1/2}\widehat{\sigma}, (m/a)^{1/2}\widehat{\sigma}]$. Show in detail that indeed $\Pr_\sigma(\sigma \in \mathrm{ci}) = 0.90$.

**(b)** Most often one wishes to estimate and assess the $\sigma$ parameter directly, being on the same scale as the measurements, but once in a while it would be more natural to communicate and interpret results on the variance scale. Show in suitable detail that

$\Pr_\sigma((m/b)\widehat\sigma^2 \le \sigma^2 \le (m/a)\widehat\sigma^2) = 0.90$; confidence intervals can in this fashion be easily transformed, say from $\theta$ to $g(\theta)$, as here, from $\sigma^2$ to $\sigma$, or the other way around.

**(c)** The construction above is 'equitailed', starting with 0.05 probability to the left of $a$ and 0.05 probability to the right of $b$. One might somewhat more generally use any $[a, b]$ with 0.90 probability for the $\chi^2_m$, needing $\Gamma_m(b) - \Gamma_m(a) = 0.90$. The length of the 90 percent $\sigma$ interval above is proportional to $1/a^{1/2} - 1/b^{1/2}$. Minimise this function, say for $m = 10, 20, 30, 40$. Compare these length-minimising 0.90 intervals with the simpler ones, and comment.

**(d)** Go through the analogue exercise, this time working with confidence intervals on the $\sigma^2$ scale, where the length is proportional to $1/a - 1/b$; minimise this, under the constraint $\Gamma_m(b) - \Gamma_m(a) = 0.90$, and compare with the above, for say $m = 10, 20, 30, 40$. The moral is that such delicate stretching doesn't matter so much, and in statistical practice we are typically content with the equitailed construction. See however the more principled construction of confidence distributions in Ch. 8.

**Ex. 3.7** *Confidence interval for a normal mean.* Here we go through the basics for constructing confidence intervals for normal means. Since approximations to normality abound in statistical theory and practice, what we learn here quickly finds use also outside the strict normality assumptions.

**(a)** Start with the simplest prototype setup, a single $Y$ from the $\mathrm{N}(\xi, 1)$ model. Show that the random interval $[Y - 1.96, Y + 1.96]$ captures $\xi$ with probability 0.95.

**(b)** Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the $\mathrm{N}(\xi, \sigma^2)$ model, at the moment with standard deviation parameter $\sigma$ taken known. With $\bar Y$ as usual being the data average, show that $\sqrt{n}(\bar Y - \xi)/\sigma$ is standard normal, and deduce from this that $\bar Y \pm 1.96\,\sigma/\sqrt{n}$ is a 95 percent confidence interval for $\xi$.

**(c)** In most cases also the $\sigma$ is unknown, however. Let $\widehat\sigma$ be the usual empirical standard deviation, see e.g. Ex. 3.4. Show that the natural construction

$$t = \frac{\bar Y - \xi}{\widehat\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar Y - \xi)}{\widehat\sigma} = \frac{\sqrt{n}(\bar Y - \xi)/\sigma}{\widehat\sigma/\sigma} \tag{3.4}$$

has a distribution not depending on the two parameters, and that this distribution, call it $G_n$, is symmetric around zero. Deduce also that with $t_{0,n} = G_n^{-1}(0.975)$, the random interval $\bar Y \pm t_{0,n}\widehat\sigma/\sqrt{n}$ covers $\xi$ with probability 0.95.

**(d)** It is then a matter of finding and perhaps tabulating the distribution $G_n$ of $t$. It is in fact the celebrated $t$ distribution, with $\mathrm{df} = n - 1$ degrees of freedom, see Ex. 1.51. But even without that specific knowledge detail, we could easily have simulated a high number for $t$, from (3.4), and read off the required quantile.

**(e)** In more generality, suppose $\beta$ is a model parameter for which there is an estimator $\widehat\beta$ with distribution $\mathrm{N}(\beta, c_n^2\sigma^2)$, say, with a known factor $c_n$. Suppose also that there is a statistically independent estimator $\widehat\sigma$ for $\sigma$, with the property that $\widehat\sigma^2/\sigma^2 \sim \chi^2_m/m$, for a known $m$. Then show that $t = (\widehat\beta - \beta)/(c_n\widehat\sigma) \sim t_m$. Put up a 0.99 confidence interval for $\beta$ based on this.

**(f)** Suppose $X_1, \ldots, X_{n_1}$ and $Y_1, \ldots, Y_{n_2}$ are random samples from two normal groups with equal variance, say $\mathrm{N}(\xi_1, \sigma^2)$ and $\mathrm{N}(\xi_2, \sigma^2)$. Find the distribution of the difference of sample means $D_n = \bar X - \bar Y$, and construct a confidence interval for $\delta = \xi_1 - \xi_2$. This is related to the classic two-sample t-test, see Ex. 4.4.

**Ex. 3.8** *Normal quantiles: estimation and confidence.* Consider again the setup of Ex. 3.3, with a sample of $Y_i$ from the normal $\mathrm{N}(\xi, \sigma^2)$ model. In the present exercise we care about quantiles, as opposed to 'only' the mean or the standard deviation.

**(a)** Show that the c.d.f. of $Y_i$ may be written $F(y) = \Pr(Y_i \leq y) = \Phi((y-\xi)/\sigma)$, with $\Phi(x)$ the c.d.f. for the standard normal. Show that the $q$ quantile $F^{-1}(q)$ is equal to $\gamma_q = \xi + z_q\sigma$, with $z_q = \Phi^{-1}(q)$. Thus the 0.95 quantile is $\gamma_{0.95} = \xi + 1.645\,\sigma$, etc.

e.g. the `pnorm` function in `R`

**(b)** Find the mean and variance of the natural estimator $\widehat{\gamma}_q = \bar{Y} + z_q\,\widehat{\sigma}$, where $\widehat{\sigma} = (Z/m)^{1/2}$, with $Z$ as in Ex. 3.4 and $m = n-1$.

**(c)** Show that we may write

$$\begin{aligned}\widehat{\gamma}_q - \gamma_q &\sim \xi + (\sigma/\sqrt{n})N + z_q\sigma(K_m/m)^{1/2} - \xi - z_q\sigma \\ &= \sigma[(1/\sqrt{n})N + z_q\{(K_m/m)^{1/2} - 1\}],\end{aligned}$$

in terms of $N \sim \mathrm{N}(0,1)$ and $K_m \sim \chi^2_m$, with these being independent. Verify that the pivot

$$W_{n,q} = \frac{\sqrt{n}(\widehat{\gamma}_q - \gamma_q)}{\widehat{\sigma}} \sim \frac{N + z_q\sqrt{n}\{(K_m/m)^{1/2} - 1\}}{(K_m/m)^{1/2}}$$

has a distribution not depending on $\xi, \sigma$. It can be simulated, for any given sample size $n$ and quantile level $q$. With $a_n$ such that $\Pr(-a_n \leq W_{n,q} \leq a_n) = 0.95$, convert this to a 95 percent confidence interval for $\gamma_q$. Note that for $q = 0.50$, the median case, the distribution of the $W_{n,q}$ is the $t_m$, the t with degrees of freedom $m = n-1$.

**(d)** Use the delta method of Ex. 2.47 to show that $\sqrt{n}\{(K_m/m)^{1/2} - 1\} \to_d \mathrm{N}(0, \frac{1}{2})$, and explain that $W_{n,q} \to_d \mathrm{N}(0, 1 + \frac{1}{2}z_q^2)$. Construct an approximate 95 percent confidence interval for $\gamma_q$ based on this. Simulate a dataset of size e.g. $n = 50$, use both methods from this exercise to find 95 percent confidence intervals for the $\gamma_{0.75}$ quantile. Note that the delta method of Ex. 2.51 leads to approximate confidence intervals for *any* smooth $\gamma = g(\xi, \sigma)$; the point here is that a more exact analysis works for the quantile.

**Ex. 3.9** *The empirical distribution function.* Assume there is an i.i.d. dataset $Y_1, \ldots, Y_n$ from an unknown distribution, with c.d.f. $F(t) = \Pr(Y_i \leq t)$. We have learned about the empirical distribution function in Ex. 2.11 and 2.61, with $F_n(t) = n^{-1}\sum_{i=1}^n I(Y_i \leq t)$ converging uniformly to the underlying $F$, with probability one; here we learn about how fast this happens, via convergence in distribution.

more on the empirical c.d.f.

**(a)** Explain that the empirical c.d.f. is the cumulative of the probability measure that puts probability mass $1/n$ at each data point. This is the natural nonparametric estimator of the unknown $F$.

**(b)** Construct a version of Figure 3.1, left panel, where $n = 100$ datapoints are simulated from the distribution $f = 0.50\,\mathrm{Expo}(r_1) + 0.50\,\mathrm{Expo}(r_2)$, with rates $r_1 = 2.00$ and $r_2 = 4.00$. Check how well the nonparametric estimator $F_n$ succeeds in coming close to $F$.

**(c)** Since we know so much about the binomial, we quickly learn a few basic properties of the $F_n$. Show that $F_n(t)$ is unbiased for $F(t)$, and that its variance is $F(t)\{1-F(t)\}/n$.

**(d)** Consider the process $Z_n(t) = \sqrt{n}\{F_n(t) - F(t)\}$. Show that it has mean zero, and that $Z_n(t) \to_d Z(t)$, say, where $Z(t)$ is a zero-mean normal with variance $F(t)\{1-F(t)\}$. Show also that

$$\mathrm{cov}\{Z_n(t), Z_n(t')\} = F(t)\{1 - F(t')\} \quad \text{for } t \leq t'.$$

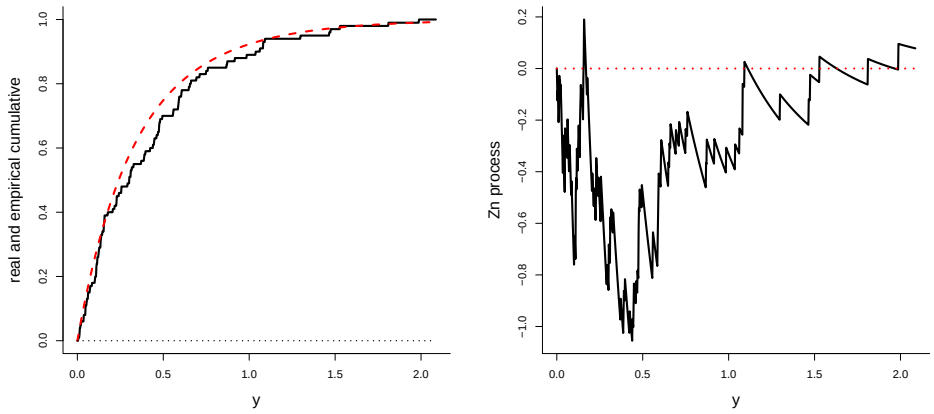



Figure 3.1: *Left panel: the real underlying data-generating $F(t)$ (dashed, red), with the empirical distribution function $F_n(t)$ (full line, black), computed from a sample of $n = 100$ data points from $F$. Right panel: the process $Z_n(t)$, computed for the data used for the same data. In 95 percent of such cases, the maximum absolute value of the $Z_n$ process will be below 1.358.*

Compute and display the $Z_n$ plot, using the same data values as for the previous figure; in other words, construct a version of Figure 3.1, right panel. Such plots may e.g. be used to check model adequacy – if the data come from a distribution not close to the $F$ used to construct the plot, then the $Z_n$ plot will deviate aignificantly from the zero line. To understand what might qualify as 'significantly different from the zero line' means we need theory for the behaviour of the full $Z_n$ process, not merely the pointwise result that $Z_n(t) \to_d \mathrm{N}(0, F(t)\{1 - F(t)\})$. We return to such questions in Ch. 10.

**Ex. 3.10** *Estimating the normal density.* Most often the statistical interest lies in estimating some parameter related to, or expressed through, the normal distribution, like the mean, spread, or quantile, as illustrated above. In some situations one wishes to estimate the density itself. Consider once again a sample $Y_1, \ldots, Y_n$ from the normal $\mathrm{N}(\xi, \sigma^2)$.

**(a)** For the parameter $\sigma$, we shall again use $Z = \sum_{i=1}^n (Y_i - \bar{Y})^2 \sim \sigma^2 \chi^2_m$, with $m = n-1$, as in several previous exercises. With the traditional default estimator $\widehat{\sigma}^2 = Z/(n-1)$ of (1.6), find formulae for the mean of $1/\widehat{\sigma}^2$ and $\log \widehat{\sigma}$.

**(b)** In Ex. 9.34 we construct an unbiased estimator of the normal density $f(y, \xi, \sigma)$. For the present exercise, construct unbiased estimators for $1/\sigma^2$ and for $\log \sigma$, and then for the log-density function $\log f(y, \xi, \sigma) = -\log \sigma - \frac{1}{2}(y-\xi)^2/\sigma^2 - \frac{1}{2}\log(2\pi)$. For this task, show in fact that $\mathrm{E} \log \widehat{\sigma} = \log \sigma + a_m$, with $a_m = \mathrm{E} \log(\chi^2_m/m) = \{\log 2 + \psi(\frac{1}{2}m)\}/(2m)$, and that

$$\mathrm{E} \frac{m}{m-2} \frac{(y-\widehat{\xi})^2}{\widehat{\sigma}^2} = \frac{(y-\xi)^2 + \sigma^2/n}{\sigma^2}.$$

**(c)** For independent samples $X_1, \ldots, X_{n_1}$ and $Y_1, \ldots, Y_{n_2}$ from normal populations $\mathrm{N}(\xi_1, \sigma_1^2)$ and $\mathrm{N}(\xi_2, \sigma_2^2)$, construct an unbiased estimator for $d(y) = \log\{f_2(y)/f_1(y)\}$.

### Confidence via normal approximations

**Ex. 3.11** *Confidence intervals via normal approximations.* (xx make connection to Wald tests to come in Ex. 4.5. xx) As we've already seen in various situations of Ch. 2, there are often estimators for interest parameters for which there is approximate normality. Then various recipes under strict normality can still be used, but as approximations.

**(a)** Suppose $\mu$ is such a parameter of interest, for which there is an estimator $\widehat{\mu}$, being approximately normal, in the mathematical sense of $\sqrt{n}(\widehat{\mu}-\mu) \to_d \mathrm{N}(0,\tau^2)$, for some appropriate limiting variance $\tau^2$. Suppose also that there is a consistent estimator $\widehat{\tau}$ of $\tau$, with $\widehat{\tau}/\tau \to_{\mathrm{pr}} 1$. Show that $Z_n = \sqrt{n}(\widehat{\mu}-\mu)/\widehat{\tau} \to_d \mathrm{N}(0,1)$; you may check with Ex. 2.22.

**(b)** Show under these mild and very frequently met assumptions that

$$\Pr(\mu \in \widehat{\mu} \pm 1.96\,\widehat{\tau}/\sqrt{n}) \to 0.95.$$

In other words, the $[\widehat{\mu}-1.96\,\widehat{\tau}/\sqrt{n}, \widehat{\mu}+1.96\,\widehat{\tau}/\sqrt{n}]$ is an *approximate* or *asymptotic* 95 percent interval for $\mu$. Note the grand generality here; this simple construction works in a large variety of situations, also in nonparametric setups, cases with dependent data, etc.

**(c)** The simplest interesting application of this standard recipe is for the unknown mean $\xi$ of a population. Verify via the CLT of Ch. 2 that $\sqrt{n}(\bar{Y}-\xi)/\widehat{\sigma} \to_d \mathrm{N}(0,1)$. Hence the t-based interval $\bar{Y} \pm 1.96\,\widehat{\sigma}/\sqrt{n}$, for which we have very precise probability computations under normality, is large-sample correct even if the data are not at all normal.

**(d)** Suppose $(X,Y,Z)$ is trinomial $(n,p,q,r)$, with $p+q+r=1$. Construct an approximate 90 percent confidence interval for $d=q-p$, then for $\gamma = r/(p+q)$. Check that you see how similar and not complicated tasks can be tended to in the examples of Ex. 2.48.

**Ex. 3.12** *Confidence intervals for the standard deviation, outside normality.* Consider i.i.d. data $Y_1,\ldots,Y_n$, from a distribution with mean $\xi$ and standard deviation $\sigma$, from which we compute the classical sample mean $\widehat{\xi}=\bar{Y}$ and empirical variance $\widehat{\sigma}^2$ of (1.6). Here we illustrate the general large-sample methods by building confidence intervals for $\sigma$, with no assumptions on the distribution of the data, like normality. The only mild assumption we make is a finite fourth moment, in order for $\widehat{\sigma}$ to have a clear limit distribution. See Figure 3.2 for 100 simulated confidence intervals, all attempting to capture the true value, here $\sigma=1$, for two different sample sizes.

**(a)** Make sure you understand and can show that $\widehat{\xi}$ and $\widehat{\sigma}$ are consistent for $\xi$ and $\sigma$, from the LLN theorems of Ch. 2.

**(b)** Via results from Ex. 2.46 and the delta method of Ch. 2, show that

$$\begin{pmatrix} \sqrt{n}(\widehat{\xi}-\xi) \\ \sqrt{n}(\widehat{\sigma}-\sigma) \end{pmatrix} \to_d \mathrm{N}_2\Big(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \sigma^2 \begin{pmatrix} 1 & \frac{1}{2}\gamma_3 \\ \frac{1}{2}\gamma_3 & \frac{1}{2}+\frac{1}{4}\gamma_4 \end{pmatrix}\Big),$$

in terms of skewness $\gamma_3 = \mathrm{E}\,\{(Y-\xi)/\sigma\}^3$ and kurtosis $\gamma_4 = \mathrm{E}\,\{(Y-\xi)/\sigma\}^3 - 3$.

**(c)** Show that $\widehat{\gamma}_4 = (1/n)\sum_{i=1}^n \{(Y_i-\bar{Y})/\widehat{\sigma}\}^4 - 3$ is consistent for $\gamma_4$, and use this to construct an approximate 90 percent confidence interval for $\sigma$. Note that this is a nonparametric procedure, totally free of other distributional assumptions, like normality; *if* one assumes normality, as an extra condition, one may do more, of course.

**(d)** For an illustration, consider the unit exponential distribution; show that the standard deviation is 1 and that the kurtosis is $\gamma_4 = 6$. Simulate a suitably high number of datasets of size $n=333$ from this distribution. For each simulated dataset, compute

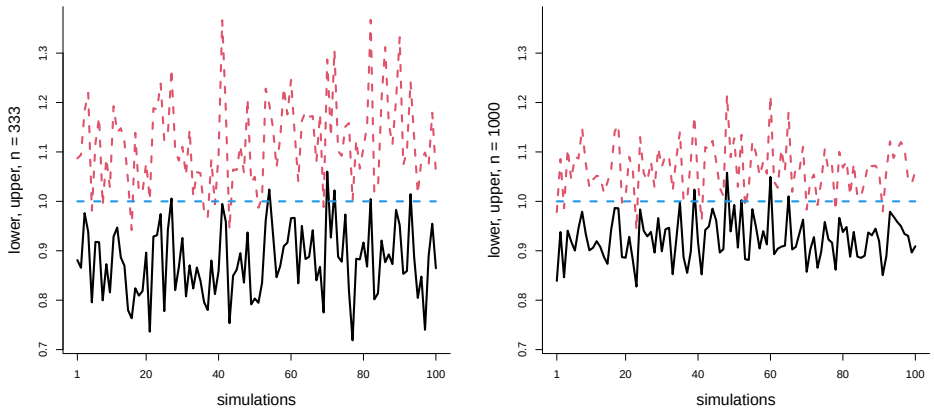



Figure 3.2: *Simulations, with datasets from the unit exponential, displaying lower and upper confidence points for 90 percent intervals; the intervals attempt to cover the true value $\sigma = 1$, and will succeed about 90 percent of the time. Left panel: with $n = 333$; right panel: with $n = 1000$.*

$\widehat{\gamma}_4$, to check how close it is to $\gamma_4$, along with the approximate 90 percent confidence interval for $\sigma$. Construct a version of Figure 3.2 (left panel for $n = 333$ and right panel for $n = 1000$). Examine in particular the coverage of your intervals: how often do they contain the correct $\sigma$? Use simulations to check how big $n$ must be, in order for $\widehat{\gamma}_4$ to be inside $[5.5, 6.5]$ with probability at least 95 percent.

**(e)** Generate a dataset of size $n = 333$ from the unit exponential, and construct an approximate 90 percent confidence ellipsoid on your screen for $(\xi, \sigma)$. Check if it contains the true values.

**Ex. 3.13** *Inference for the normal density, via large-sample methods.* Consider again i.i.d. data $Y_1, \ldots, Y_n$ from a normal density $f(y, \xi, \sigma)$. In Ex. 3.10 we constructed an estimator $\widehat{f}(y)$ with the property that $\log \widehat{f}(y)$ is exactly unbiased for $\log f(y, \xi, \sigma)$, for any given $n$. Here we instead work out the large-sample approximations for the direct estimator $f^*(y) = f(y, \widehat{\xi}, \widehat{\sigma})$, having plugged in the usual empirical mean and empirical standard deviation.

**(a)** We have seen in Ex. 2.51 that $(\sqrt{n}(\widehat{\xi} - \xi), \sqrt{n}(\widehat{\sigma} - \sigma))$ tends to $(A, B)$, independent zero-mean normals with variances $\sigma^2$ and $\frac{1}{2}\sigma^2$. For fixed $y$, wite $x = (y - \xi)/\sigma$. Use the delta method to explain that (i) $\sqrt{n}(\log \widehat{\sigma} - \log \sigma) \to_d (1/\sigma)B$; (ii) $\sqrt{n}((y - \widehat{\xi})/\widehat{\sigma} - (y - \xi)/\sigma) \to_d (1/\sigma)(-A - xB)$; (iii) $\sqrt{n}\{(y - \widehat{\xi})^2/\widehat{\sigma}^2 - x^2\} \to_d (1/\sigma)(-2xA - x^2 B)$. Then combine these to show that

$$\sqrt{n}\{\log f^*(y) - f(y, \xi, \sigma)\} \to_d (1/\sigma)(-B + xA + \tfrac{1}{2}x^2 B),$$

a zero-mean normal with variance $\tau^2 = x^2 + (\frac{1}{2}x^2 - 1)^2 = \frac{1}{2}(1 + x^4)$.

**(b)** Show that the log-unbiased estimator $\log \widehat{f}(y)$ is close enough to this $\log f^*(y)$ to make the scaled difference $\sqrt{n}\{\log f^*(y) - \log \widehat{f}(y)\}$ go to zero in probability; explain that the two estimators therefore have the same limit distribution.

**(c)** Explain how a pointwise 90 percent confidence band can be constructed for the log-density, of the type $\log f^*(y) \pm 1.645\,\widehat{\tau}(y)$, which then may be transformed back to the

density scale.

**Ex. 3.14** *Variance of the variance estimator.* Let $Y_1, \ldots, Y_n$ be i.i.d., with mean $\xi$, variance $\sigma^2$, and finite kurtosis $\gamma_4 = \mathrm{E}\,(Y_i - \xi)^4/\sigma^4 - 3$.

**(a)** With $A = \sum_{i=1}^n (Y_i - \bar{Y})^2$, show that $\mathrm{E}\,A = (n-1)\sigma^2$. This says that $\widehat{\sigma}^2 = A/(n-1)$ is unbiased for the variance, regardless of the underlying distribution.

**(b)** If the $Y_i$ are actually normal, then $A \sim \sigma^2 \chi^2_m$, with $m = n-1$. Show that $\mathrm{Var}\,\widehat{\sigma}^2 = 2\sigma^4/(n-1)$.

**(c)** Outside normality, work out an expression for $\mathrm{Var}\,A$, and show that

$$\mathrm{Var}\,\widehat{\sigma}^2 = \Big(3 + \gamma_4 - \frac{n-3}{n-1}\Big)\frac{\sigma^4}{n} = \frac{2\sigma^4}{n-1} + \frac{\gamma_4 \sigma^4}{n}.$$

simulate a high number of samples of size $n = 12$ from the t distribution $t_m$, with say $m = 6$, and 'verify' the formula.

**(d)** Check that the finite-sample formula above matches the large-sample limit for $\sqrt{n}(\widehat{\sigma}^2 - \sigma^2)$.

**Ex. 3.15** *The binomial, the normal approximation, and confidence intervals.* Consider $Y_n$, a binomial $(n, p)$. To invoke the CLT it is practical to use the $Y_n = X_1 + \cdots + X_n$ representation, in terms of $X_i$ being i.i.d. Bernoullis.

**(a)** Use the CLT of Ch. 2 to deduce that the normalised variable

$$W_n = \frac{\sum_{i=1}^n (X_i - p)}{\{\mathrm{Var} \sum_{i=1}^n (X_i - p)\}^{1/2}} = \frac{Y_n - np}{\{np(1-p)\}^{1/2}} = \frac{\widehat{p} - p}{\{p(1-p)/n\}^{1/2}}$$

converges to the standard normal $\mathrm{N}(0,1)$ with increasing $n$. Discuss briefly the relevance of the skewness result from Ex. 1.3 in light of the limiting normality.

**(b)** With $\widehat{p}_B = (Y+1)/(n+2)$, show that the difference between $W_n$ and $W_{n,B} = \sqrt{n}(\widehat{p}_B - p)$ is so small, for large $n$, that $W_{n,B}$ must have the same normal limit. The confidence intervals we construct below, based on $\widehat{p}$, can therefore alternatively be based on $\widehat{p}_B$.

**(c)** Verify from the above that $\Pr(-1.96 \le W_n \le 1.96) \to 0.95$ as sample size increases, and use this to construct an interval, based on having observed $Y_n = y$ in a given experiment with known $n$, which covers the true $p$ with probability approximately 95 percent. binomial confidence

**(d)** There are actually several constructions of such confidence intervals, with this property. Considering the basic estimator $\widehat{p} = Y/n$ again, write $\sigma_n^2 = p(1-p)/n$ for its variance, and $\widehat{\sigma}_n^2 = \widehat{p}(1-\widehat{p})/n$ for its estimated variance. Verify that both asymptotic equivalence

$$W_n = (\widehat{p} - p)/\sigma_n \quad \text{and} \quad W_n' = (\widehat{p} - p)/\widehat{\sigma}_n$$

tend to the standard normal in distribution. Now show that the arguments above, used for $W_n'$ in lieu of $W_n$, lead to the confidence interval $\widehat{p} \pm 1.96\,\widehat{\sigma}_n$ instead. Exemplify, with $n = 100$, for the three cases $y = 22$, $y = 55$, $y = 77$, where you compute both versions of the 95 percent confidence interval for $p$.

**(e)** Suppose certain details related to your applied research project require that you compute the probability $p$ that $L \le 1.33\,R$, where

$$L = \{(G_1/G)(G_2/G)(G_3/G)(G_4/G)\}^{1/4}, \quad R = \{(G_5/G)(G_6/G)(G_7/G)\}^{1/3},$$

in terms of $G_1, \ldots, G_8$ being i.i.d. from the $\chi^2_{12}$ distribution (the chi-squared with degrees of freedom equal to 12), and $G = \sum_{i=1}^8 G_i$. Since it's hard to find an exact formula, or an exact answer in other ways, you *simulate* a high number sim of such vectors $(G_1, \ldots, G_8)$, and check for each simulation whether the event just described takes place or not. How large should sim be, in order for your simulation based estimate of $p$ to be correct to three decimal places? Carry out such simulations and thus find $p$. Display also a histogram of simulated $L/R$.

**Ex. 3.16** *Confidence intervals via the delta method.* The delta method of Ex. 2.47 can be used to derive approximate normality of various estimators, after which the generic recipe of Ex. 3.11 may be used to form confidence intervals. Once limiting normality has been established, there are automatically limiting normality statements and uses for transformated parameters.

**(a)** Suppose $\sqrt{n}(\widehat{\mu} - \mu) \to_d \mathrm{N}(0, \kappa^2)$, for some parameter $\mu$ of interest. Confidence intervals then take the form $\widehat{\mu} \pm (c/\sqrt{n})\widehat{\kappa}$. With a smooth transformation to $\gamma = g(\mu)$ and $\widehat{\gamma} = g(\widehat{\mu})$, explain that we have $\sqrt{n}(\widehat{\gamma} - \gamma) \to_d g'(\mu)\kappa\,\mathrm{N}(0,1)$. If $\gamma = g(\mu)$ is monotone, with an inverse $\mu = g^{-1}(\gamma)$, explain that one via back-transformation can form the alternative interval

$$g^{-1}\big(\widehat{\gamma} \pm (c/\sqrt{n})g'(\widehat{\mu})\widehat{\kappa}\big).$$

Explain also how this simplifies for variance stabilising transformations, where $g'(\mu)\kappa = 1$.

**(b)** Illustrate the above for a binomial $(n, p)$ situation, where we know from Ex. 2.48 that $\sqrt{n}(\widehat{\gamma} - \gamma) \to_d \mathrm{N}(0,1)$, with $\gamma = g(p) = 2\arcsin(p^{1/2})$: construct the interval on the scale of $\gamma$, and then back-transform to the $p$ scale. Check numerically how close such intervals are to the standard ones, for a few combinations of $n$ and binomial outcome $y$.

**(c)** Show via Taylor expansion that $g^{-1}(\widehat{\gamma} + \varepsilon) = \widehat{\mu} + (g^{-1})'(\widehat{\gamma})\varepsilon + O(\varepsilon^2)$ for small $\varepsilon$. Deduce that that the transformation intervals above become large-sample equivalent to the $\widehat{\mu} \pm (c/\sqrt{n})\widehat{\kappa}$ we started out with.

**(d)** (xx ranting. xx) So which $g$ should one perhaps use? Suppose $g_a$ is one such transformation, with a tweaking parameter $a$. Argue that the best choices of $a$ are those for which $g_a(\widehat{\mu}) - g_a(\mu)$ are close to having zero-mean normal distributions. Play a bit with this idea, for the case of having normal approximation confidence intervals for $\sigma$ in the $\mathrm{N}(\xi, \sigma^2)$ situation. The starting point here would be $\sqrt{n}(\widehat{\sigma} - \sigma) \to_d \mathrm{N}(0, \frac{1}{2}\sigma^2)$. Attempt the transformation $g_a(\sigma) = \sigma^a$; the task is to check accurate normality for the appopriate $\sqrt{n}(\widehat{\sigma}^a - \sigma^a)/\widehat{\kappa}_a$.

### Quantiles and sample quantiles

**Ex. 3.17** *Uniform ordering.* Consider $U_1, \ldots, U_n$ i.i.d. from the uniform distribution on the unit interval. Order these, to $U_{(1)} < \cdots < U_{(n)}$.

**(a)** Explain that in order to have $U_{(i)} \in [u, u + \mathrm{d}u]$, one needs $i - 1$ points in $[0, u]$, 1 point in $[u, u + \mathrm{d}u]$, and $n - 1$ points in $(u + \mathrm{d}u, 1]$. Show from this trinomial setup that $U_{(i)}$ has density

connection to Beta distributions

$$g_i(u) = \frac{n!}{(i-1)!\,1!\,(n-i)!} u^{i-1}(1-u)^{n-i} \quad \text{for } u \in (0,1).$$

Explain that $U_{(i)} \sim \mathrm{Beta}(i, n - i + 1)$, see Ex. 1.22, and explain from this that

$$\mathrm{E}\,U_{(i)} = p_i = i/(n+1) \text{ and that } \mathrm{Var}\,U_{(i)} = p_i(1-p_i)/(n+2).$$

**(b)** Explain further that $U_{(i)} \leq p$ is equivalent to the binomial count $Z$ of datapoints falling in $[0,p]$ being at least $i$, and that this gives rise to the following equation:

Beta and binomial connection

$$\frac{n!}{(i-1)!\,(n-i)!}\int_0^p u^{i-1}(1-u)^{n-i}\,\mathrm{d}u = \sum_{j=i}^n \binom{n}{j} p^j(1-p)^{n-j}.$$

**(c)** With $i<j$, show that $(U_{(i)},U_{(j)})$ has density

$$g_{i,j}(u,v) = \frac{n!}{(i-1)!\,1!\,(j-i-1)!\,1!\,(n-j)!}u^{i-1}(v-u)^{j-i-1}(1-v)^{n-j}$$

for $u<v$. The idea behind the reasoning, and the ensuing notation, is that in order to see $U_{(i)} \in [u,u+\mathrm{d}u]$ and $U_{(j)} \in [v,v+\mathrm{d}v]$, there is a multinomial situation, with five boxes $[0,u],[u,u+\mathrm{d}u],[u+\mathrm{d}u,v],[v,v+\mathrm{d}v],[v+\mathrm{d}v,1]$, inside which we need to find $i-1,1,j-i-1,1,n-j$ datapoints.

**(d)** For an i.i.d. uniform sample $U_1,\ldots,U_n$ on $[0,1]$, consider the uniform range $R_n = U_{(n)}-U_{(1)}$, where we know that $U_{(1)} \sim \mathrm{Be}(1,n)$. Show that given $U_{(1)}=u$, $U_{(n)}$ can be represented as $u+Z$, where $Z$ is the maximum of another uniform sample, of size $n-1$, on $[0,1-u]$. Use this to show that the c.d.f. of $R_n$ can be expressed as $H_n(r) = nr^{n-1}(1-r)+r^n = nr^{n-1}-(n-1)r^n$, and show that this is the $\mathrm{Be}(n-1,2)$ distribution.

**(e)** In general, if $Y_1,\ldots,Y_n$ are i.i.d. from a density $f$, show that the joint density for the full order statistic vector $(Y_{(1)},\ldots,Y_{(n)})$ is $n!\,f(y_{(1)})\cdots f(y_{(n)})$, on the set where $y_{(1)}<\cdots<y_{(n)}$. In particular, for order statistics from the uniform distribution, show that the joint density of $(U_{(1)},\ldots,U_{(n)})$ is flat and equal to $n!$ on the set $u_{(1)}<\cdots<u_{(n)}$.

**(f)** Use this, in conjunction with Ex. 1.20, to demonstrate that

connection to exponential and Dirichlet

$$\begin{aligned}(U_{(1)},U_{(2)},\ldots,U_{(n)}) &=_d (D_1, D_1+D_2,\ldots,D_1+\cdots+D_n)\\ &=_d (V_1/S,(V_1+V_2)/S,\ldots,(V_1+\cdots+V_n)/S),\end{aligned}$$

with $V_1,\ldots,V_n,V_{n+1}$ being i.i.d. from the unit exponential, with sum $S=V_1+\cdots+V_{n+1}$, and $(D_1,\ldots,D_n,D_{n+1})$ is a flat Dirichlet $(1,\ldots,1,1)$. The differences $D_i = U_{(i)}-U_{(i-1)}$ are called the *spacings*. We use '$=_d$' to signal equality in distributions. Show that this leads to the representation of the order statistics process as

$$U_{([nq])} = \sum_{i=1}^{[nq]} V_i \Big/ \sum_{i=1}^{n+1} V_i \quad \text{for } 0\leq q\leq 1. \tag{3.5}$$

Here $[nq]$ is the largest integer less than or equal to $nq$. We have seen in Ex. 2.14 that $U_{([nq])} \to_{\mathrm{pr}} q$, via consistency properties of the empirical c.d.f. Now prove this result again, via the representation (3.5) and the law of large numbers.

**Ex. 3.18** *Ratios of ordered uniforms.* (xx again, need checing and calibration, regarding what is told where. xx) Let $U_1,\ldots,U_n$ be an i.i.d. sample from the uniform distribution on the unit interval, and order these into $U_{(1)}<\cdots<U_{(n)}$. From these form the ratios

$$V_1 = U_{(1)}/U_{(2)},\ V_2 = U_{(2)}/U_{(3)},\ \ldots, V_{n-1} = U_{(n-1)}/U_{(n)},\ V_n = U_{(n)}/1.$$

**(a)** Show that the inverse transformation leads to the representation

$$U_{(n)} = V_n,\ U_{(n-1)} = V_nV_{n-1},\ \ldots, U_{(2)} = V_nV_{n-1}\cdots V_2,\ U_{(1)} = V_nV_{n-1}\cdots V_2V_1.$$

**(b)** Find the joint probability density for $(V_1, \ldots, V_n)$, and show in fact that these are independent, with

$$V_1 \sim \text{Beta}(1,1),\ V_2 \sim \text{Beta}(2,1),\ \ldots, V_{n-1} \sim \text{Beta}(n-1,1),\ V_n \sim \text{Beta}(n,1).$$

**(c)** Independently of details above, we know from Ex. 3.17 that $U_{(i)}$ has a Beta$(i, n-i+1)$ distribution. Explain that this Beta$(i, n-i+1)$ can be represented as a product of different independent Beta variables.

**(d)** It is of course a somewhat cumbersome simulation recipe for generating a uniform sample, but it is a useful exercise, opening doors & minds to fruitful generalisations: For $n = 10$, say, generate ordered uniform samples of size $n$ in your computer via the representation above, in terms of products of Beta variables. Carry out some checks to see that each single $U_{(i)}$ then has the right distribution, i.e. as described in (c).

**(e)** Work with the following generalisation of the construction above: Let $X_1, \ldots, X_n$ be an i.i.d. sample from the distribution with density $f(x) = ax^{a-1}$, i.e. a Beta$(a, 1)$. Again form the ratios $V_i = X_{(i)}/X_{(i+1)}$ as above, leading to $X_{(i)} = V_i V_{i+1} \cdots V_n$. Show that the $V_i$ are again independent, now with $V_i \sim \text{Beta}(ai, 1)$.

**Ex. 3.19** *Min and max of two uniforms.* Suppose $Y_1, Y_2$ are i.i.d. from a density $f(y)$, and order them, to $V_1 < V_2$. (xx ask per august and martin why this particular probability calculation was of value. also check relevance for window prior for even. xx)

**(a)** Show that $(V_1, V_2)$ has joint density $2f(v_1)f(v_2)$, on the set where $v_1 < v_2$.

**(b)** Then consider the special case of two datapoints from the uniform distribution on the unit interval, ordered to $V_1 < V_2$. Show that $R = V_1/V_2$ is another uniform on the unit interval, and that $W = V_2 - V_1$ is a Beta$(1, 2)$. Show that $\Pr(V_2 - V_1 \le c) = \Pr(V_2 - V_1 > c) = \frac{1}{2}$, for $c = 1 - 1/\sqrt{2} = 0.2929$.

**(c)** Find also the joint distribution for $(R, W)$ here.

**Ex. 3.20** *The sample median.* Let $Y_1, \ldots, Y_n$ be i.i.d. from a positive density $f$ with true median $\theta = F^{-1}(\frac{1}{2})$.

**(a)** Suppose for simplicity that $n$ is odd, say $n = 2m + 1$. Show that $M_n$ has density of the form

$$g_n(y) = \frac{(2m+1)!}{m!\,m!} F(y)^m \{1 - F(y)\}^m f(y).$$

**(b)** Show then that the density of $Z_n = \sqrt{n}(M_n - \theta)$ can be written in the form $h_n(z) = g_n(\theta + z/\sqrt{n})/\sqrt{n}$. Prove that

$$h_n(z) \to (2\pi)^{-1/2} 2f(\theta) \exp\{-\tfrac{1}{2} 4 f(\theta)^2 z^2\},$$

where the Stirling approximation formula of Ex. 2.35 may be of use. The limit is the density $h(z)$ of the normal $\text{N}(0, \tau^2)$, with $\tau = \frac{1}{2}/f(\theta)$. We have hence proved $Z_n \to_d \text{N}(0, \tau^2)$, by Scheffé's lemma; see Ex. 2.6.

**(c)** So when is the sample mean best, and when might the sample median be the better estimator, when it comes to estimating the centre point $\theta$ of a symmetric density? Explain that this for large $n$ is a matter of the ratio

$$\rho = \frac{\sigma}{\frac{1}{2}/f(\theta)} = 2\sigma f(\theta),$$

where $\sigma$ is the standard deviation for $f$. Argue that if $\rho < 1$, then the sample mean is best, and that if $\rho > 1$, then the sample median is the best.

**(d)** Compare the limiting distributions for the sample mean and the sample median for the normal density, the double exponential density $\frac12\exp(-|y|)$, and the Cauchy density $(1/\pi)/(1+y^2)$.

**(e)** Consider the t distribution, with degrees of freedom $\nu$, see Ex. 1.51. Find an expression for the ratio $\rho=\rho(\nu)$, plot $(\nu,\rho(\nu))$ in a diagram, and comment. Show that $\rho(\nu)$ approaches $(2/\pi)^{1/2}=0.7979$ for large $\nu$. Show that for $\nu<4.678$, there is roughness at the top, and the median is best; whereas for $\nu>4.678$, there is a smoother density at the top, and the mean is best.

**(f)** Carry out a similar analysis for the binormal symmetric mixture model $f=\frac12\,\mathrm{N}(-a,1)+\frac12\,\mathrm{N}(a,1)$. For which values of $a$ is the sample median a better estimator of the centre point then the sample mean? [xx later on, another chapter: the estimator which says $\widehat\theta$ is sample median if $A_n$ and sample mean if $A_n^c$, where $A_n$ is the event that $\frac12/\widehat f(\widehat\theta_0)<\widehat\sigma$. xx]

**Ex. 3.21** *Sample quantiles.* Suppose $Y_1,\ldots,Y_n$ are i.i.d. from some distribution with positive density $f$ and c.d.f. $F$. The sample median estimates the population median $F^{-1}(0.50)$, and similarly the sample quantile $Q_n(q)$, at any prescribed level $q\in(0,1)$, estimates the population quantiles $F^{-1}(q)$. Built-in functions like `quantile(data,0.33)` in `R` find such sample quantiles directly, so users do not need the cumbersome linear interpolation fiddling between the two ordered observations coming closest to $nq$, or to care too much about ties in the data due to rounding-off errors. This exercise finds limit distributions for $\sqrt n\{Q_n(q)-F^{-1}(q)\}$. sample quantiles

**(a)** Suppose $U\sim$ unif and let $Y=F^{-1}(U)$. Show that $Y$ has distribution $F$, and hence density $f$.

**(b)** Explain that the full order statistic vector $Y_{(1)}<\cdots<Y_{(n)}$ may be represented via a correspondingly ordered sample of the uniform, as $F^{-1}(U_{(1)})<\cdots<F^{-1}(U_{(n)})$, with $U_{(i)}$ being the $i$th ordered observations in an i.i.d. sample $U_1,\ldots,U_n$ from the uniform, studied in Ex. 3.17. In particular, $Y_{(i)}$ has the same distribution as $F^{-1}(U_{(i)})$.

**(c)** This also means that if we work out basic approximation results for the order statistics from the uniform, we are a modest delta method step away from similar results for the general case of a density $f$. In particular, suppose we manage to show $\sqrt n(U_{([nq])}-q)\to Z_q$, for some $Z_q$. Show that this implies $\sqrt n\{Q_n(q)-F^{-1}(q)\}\to_d c(q)Z_q$, with $c(q)=(F^{-1})'(q)=1/f(\mu_q)$, writing $\mu_q=F^{-1}(q)$.

**(d)** To illustrate this point in a simple case first, show from what we already know in Ex. 3.20 that $\sqrt n(U_{([0.50\,n])}-0.50)\to_d\mathrm{N}(0,0.50^2)$, for the uniform median. Then show for a general density $f$ that $\sqrt n\{Q_n(0.50)-\mu\}\to_d\mathrm{N}(0,0.50^2/f(\mu)^2)$, with $\mu=F^{-1}(0.50)$ the population median. Here we used an exact expression for the density of the median.

**(e)** There are actually several other ways of proving this median of uniforms result. Such an alternative approach is to use the representation (3.5). Explain that $U_{([0.50\,n])}=A_n/(A_n+B_n)$, with $A_n$ and $B_n$ the averages of the first and second half of i.i.d. variables $V_1,\ldots,V_{n+1}$ from the unit exponential. Use the CLT for the joint limit distributions of $\sqrt n(A_n-\frac12)$ and $\sqrt n(B_n-\frac12)$, and then use the delta method to land the $\mathrm{N}(0,0.50^2)$ limit.

**(f)** Then generalise to the case of any given quantile level $q$. Show first that $\sqrt n(U_{[nq]}-q)\to_d\mathrm{N}(0,q(1-q))$, and then that the limit distribution is $\mathrm{N}(0,q(1-q)/f(\mu_q)^2)$ for $\sqrt n\{Q_n(q)-\mu_q\}$.

**(g)** For the case of two quantiles jointly, like the lower and upper sample quartiles, show for the uniform case that with $q_1 < q_2$,

$$\begin{pmatrix} \sqrt{n}\{Q_n(q_1) - q_1\} \\ \sqrt{n}\{Q_n(q_2) - q_2\} \end{pmatrix} \to_d \mathrm{N}_2(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} q_1(1-q_1) & q_1(1-q_2) \\ q_1(1-q_2) & q_2(1-q_2) \end{pmatrix}).$$

Prove this via the explicit density for $(U_{(i)}, U_{(j)})$, given in Ex. 3.17, using the Scheffé Lemma of Ex. 2.6.

**(h)** It is also instructive to use representation (3.5) via i.i.d. unit exponentials. Prove the joint binormal limit distribution statement from this. Then generalise to the case of $r$ quantiles, for levels $q_1 < \cdots < q_r$. Show for the uniform case that there is a joint multivariate normal limit, with variances $q_j(1-q_j)$ and covariances $-q_j q_\ell$ for $j < l$. Then carry out the transformation arguments needed to prove that for the case of an underlying positive density $f$, there is limiting joint normality for $\sqrt{n}(Q_{n,j} - \mu_j)$, where the limit has variances $q_j(1-q_j)/f(\mu_j)^2$ and covariances $-q_j - q_k/\{f(\mu_j)f(\mu_k)\}$ for $j < k$. See Ex. 10.39 for convergence of a full quantile process.

**(i)** So how variable are sample quantiles, how much bigger are variances out in the tails, compared to in the middle? Explain that the limiting standard deviation at quantile level $q$ is $\mathrm{sd}(q) = c(q)\{q(1-q)\}^{1/2} = \{q(1-q)\}^{1/2}/f(\mu_q)$. For a few densities, including the uniform, the normal, the exponential, compute and plot this $\mathrm{sd}(q)$.

**Ex. 3.22** *Good and bad estimators.* Suppose $X_1, \ldots, X_n$ are i.i.d. from the density $f(x, \theta) = \exp\{-(x-\theta)\}$ for $y \geq \theta$, i.e. a unit exponential starting at parameter $\theta$.

**(a)** Explain that we have $X_i = \theta + Y_i$, with the $Y_i$ being i.i.d. from the unit exponential, and hene that the order statistics can be represented as $X_{(i)} = \theta + Y_{(i)}$, cf. Ex. 1.15.

**(b)** For the smallest and largest observations, show that $\widehat{\theta}_A = X_{(1)} - 1/n$ and $\widehat{\theta}_B = X_{(n)} - H_n$ are unbiased estimators of $\theta$, with $H_n = 1 + 1/2 + \cdots + 1/n$ the partial sum of the harmonic series. Find their variances.

**(c)** (xx a bit more. spell out that $\widehat{\theta}_B$ is not consistent. a bit on $X_{(i)} - c_i$ too, where $c_i = 1/n + \cdots + 1/(n-i+1) = H_n - H_{n-i}$. median is ok. xx)

**Ex. 3.23** *Smooth functions of quantiles.* (xx point to Ex. 2.50 for similar method and similar results. xx) Suppose $Y_1, \ldots, Y_n$ are i.i.d. from some distribution with continuous and positive density $f$. The inverse c.d.f. $F^{-1}$ is then well-defined.

**(a)** Consider some smooth function $B(\mu_1, \ldots, \mu_r)$ of the quantiles $\mu_j = F^{-1}(q_j)$. Via Ex. 3.21, identify first the multinormal limit $X \sim \mathrm{N}_r(0, \Sigma)$ of the vector with components $\sqrt{n}(Q_{n,j} - \mu_j)$, with $Q_{n,j} = F_n^{-1}(\mu_j)$ the sample quantiles. Then establish that with $\widehat{B} = B(Q_{n,1}, \ldots, Q_{n,r})$, we have $\sqrt{n}(\widehat{B} - B) \to_d c^{\mathrm{t}} X \sim \mathrm{N}(0, \tau^2)$, with $\tau^2 = c^{\mathrm{t}}\Sigma c$. Here $c$ is the vector of $c_j = \partial B(\mu_1, \ldots, \mu_r)/\partial \mu_j$.

**(b)** Suppose $Y_1, \ldots, Y_n$ are independent unit exponentials. With $p < q$, find the limit distribution of the ratio $\widehat{\gamma} = F_n^{-1}(q)/F_n^{-1}(p)$. (xx check: we do have an exercise with quartiles and $F_n^{-1}(1-\varepsilon) - F_n^{-1}(\varepsilon)$ for symmetric distributions? xx)

**(c)** Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the Weibull distribution with c.d.f. $F(y) = 1 - \exp(-y^b)$. Show that the quantile function is $\mu_p = F^{-1}(p) = d(p)^{1/b}$, with $d(p) = -\log(1-p)$. With $p < q$, find the limit distribution of

$$\sqrt{n}\Big[\frac{Q_{n,q}}{Q_{n,p}} - \Big\{\frac{d(q)}{d(p)}\Big\}^{1/b}\Big].$$

Argue that with $R_n = Q_{n,q}/Q_{n,p}$, the solution $\widehat{b}$ to $\log R_n = (1/b)\log\{d(q)/d(p)\}$ is a consistent estimator. Find the limit distribution of $\sqrt{n}(\widehat{b}-b)$.

**(d)** (xx one more, with three or four quantiles at work. perhaps point to quantile pyramid things. xx)

**(e)** (xx might wait until a Ch9 exercise with processes. xx) Consider quantiles levels $p<q<r$ for i.i.d. observations $U_1,\ldots,U_n$ from the uniform distribution on the unit interval. For empirical quantiles $Q_{n,p}, Q_{n,q}, Q_{n,r}$, explain via Ex. 3.21 that

$$\begin{pmatrix}\sqrt{n}(Q_{n,p}-p)\\ \sqrt{n}(Q_{n,q}-q)\\ \sqrt{n}(Q_{n,r}-r)\end{pmatrix} \to_d \begin{pmatrix}A\\B\\C\end{pmatrix} \sim \mathrm{N}_3(0,\Sigma),$$

and identify the elements of $\Sigma$. From the empirical quantiles, form the two successive ratios $R_n = Q_{n,q}/Q_{n,p}$ and $S_n = Q_{n,r}/Q_{n,q}$. Show that

$$\begin{pmatrix}\sqrt{n}(R_n-q/p)\\ \sqrt{n}(S_n-r/q)\end{pmatrix} \to_d \begin{pmatrix}R\\S\end{pmatrix} = \begin{pmatrix}(1/p)\{-(q/p)A+B\}\\ (1/q)\{-(r/q)B+C\}\end{pmatrix}.$$

Do the algebra to verify that $R$ and $S$ have variances $(1/p)^2\{p(1-q/p)^2+q-p\}$ and $(1/q)^2\{q((1-r/q)^2+r-q\}$, and that they are in fact independent.

**Ex. 3.24** *Exercises with sample quantiles.* [xx various things, using the general results above. interquartile range $R_n = Q_n(0.75)-Q_n(0.25)$, e.g. for the normal. for the Cauchy, cool factoid: $\sqrt{n}(R_n-2) \to_d \mathrm{N}(0,\pi^2)$. Limit distribution of sample median, given the two 0.25 and 0.75 quartiles. a little link to the nonparametric quantile processes of Hjort and Petrone (2007) and the more general quantile pyramids of Hjort and Walker (2009). also pointer to fuller process result in Ch. 10. the limit is $(F^{-1})'(q)W^0(q)$. xx]

**(a)** (xx perhaps pushed to BNP chapter. xx) Consider building a model for $V_1<V_2<V_3$ as follows: (i) $V_2 \sim \mathrm{unif}(0,1)$; (ii) given $V_2=v_2$, let independently $V_1 \sim \mathrm{unif}(0,v_2)$ and $V_3 \sim \mathrm{unif}(v_2,1)$. Show that $(V_1,V_2,V_3)$ has mean $(1/4,2/4,3/4)$.

**(b)** Show that the densities of $V_1$ and $V_3$ become $g_1(v_1) = -\log v_1$ and $g_3(v_3) = -\log(1-v_3)$ on the unit interval.

**(c)** Generalise to the case of $V_1<\cdots<V_7$, thought of as random versions of the seven octiles: (i) $V_4 \sim \mathrm{unif}(0,1)$; (ii) given $V_4$, let $V_2 \sim \mathrm{unif}(0,V_4)$ and $V_6 \sim (V_4,1)$, independently; (iii) given $V_2,V_4,V_6$, let $V_1,V_3,V_5,V_7$ be independent and uniform on the intervals $(0,V_2),(V_2,V_4),(V_4,V_6),(V_6,1)$, respectively. Show that $V_j$ has mean $j/8$, and find the densities for each individual $V_j$.

**Ex. 3.25** *Which order statistics interval contains the true median?* Let $Y_1,\ldots,Y_n$ be i.i.d. from a positive and smooth density $f$, with cumulative $F$. With $Y_{(1)}<\cdots<Y_{(n)}$ the order statistics, which of the subintervals $(Y_{(i)},Y_{(i+1)})$ will contain the true median, $\mu = F^{-1}(\frac12)$?

**(a)** Show that $p_i = \Pr\{\mu\in(Y_{(i)},Y_{(i+1)})\} = \Pr\{\frac12\in(U_{(i)},U_{(i+1)})\}$, in terms of the order statistics from a uniform sample. We allow $i=0,1,\ldots,n$, here, for the $n+1$ possibilities for which interval shall contain $\mu$, writing $u_{(0)}=0$ and $u_{(n+1)}=1$.

**(b)** Given $U_{(i)}=u$, show that the distribution of $U_{(i+1)}$ is the same as that of $u+(1-u)W$, where $W$ is the smallest of $n-i$ observations from the uniforom in the unit interval.

Use this to show that

$$p_i = \Pr\{U_{(i)} < \tfrac12 < U_{(i+1)}\} = \int_0^{1/2} \Pr\{U_{(i+1)} > \tfrac12 \,|\, U_{(i)} = u\} g_i(u)\,\mathrm{d}u$$
$$= \int_0^{1/2} \Big(\frac{\frac12}{1-u}\Big)^{n-i} \mathrm{be}(u, i, n-i+1)\,\mathrm{d}u,$$

involving a Beta density, as per Ex. 3.21. Show that this indeed leads to the explicit probability

$$p_i = (\tfrac12)^{n-i} \frac{n!}{(i-1)!\,(n-i)!} \int_0^{1/2} u^{i-1}\,\mathrm{d}u = \binom{n}{i} (\tfrac12)^n.$$

Hence we've reached the binomial probabilities, for a binom$(n, \frac12)$, via direct probability calculations. Try also to give a direct argument.

**(c)** (xx generalise to general quantile $\mu_p = F^{-1}(p)$. xx)

### Moment matching methods

**Ex. 3.26** *Moment matching estimators.* Suppose $Y_1, \ldots, Y_n$ are i.i.d. from some model $f(y, \theta)$, where $\theta = (\theta_1, \ldots, \theta_p)^{\mathrm{t}}$ is of dimension $p$. The *method of moments* consists in fitting the first $p$ empirical moments to the theoretical ones. In detail, one computes

method of moments

$$M_1 = \bar Y = n^{-1} \sum_{i=1}^n Y_i,\ M_2 = n^{-1} \sum_{i=1}^n (Y_i - \bar Y)^2,\ \ldots,\ M_p = n^{-1} \sum_{i=1}^n (Y_i - \bar Y)^p,$$

and solves the $p$ equations $M_1 = g_1(\theta), \ldots, M_p = g_p(\theta)$, where $g_1(\theta) = \mathrm{E}_\theta\, Y$, $g_2(\theta) = \mathrm{E}_\theta\, \{Y - g_1(\theta)\}^2$, up to $g_p(\theta) = \mathrm{E}_\theta\, \{Y - g_1(\theta)\}^p$.

**(a)** For one-parameter models, explain that this amounts to fitting the empirical and theoretical mean. If $Y_1, \ldots, Y_n$ are i.i.d. geom$(p)$, see Ex. 1.25, use $\mathrm{E}\, Y_i = 1/p$ to find the method of moments estimator for $p$. For another application, assume $Y_1, \ldots, Y_n$ follow the distribution with c.d.f. $y^\theta$ on $[0, 1]$. Find the method of moments estimator for $\theta$.

**(b)** For two-parameter models, explain that the method of moments means fitting the empirical mean and variance to the theoretical ones. If $Y_1, \ldots, Y_n$ are i.i.d. Beta$(a, b)$, see Ex. 1.19, find expressions for the method of moments estimators for $a, b$.

**(c)** Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the uniform distribution on $[0, b]$. Find the moment estimator for $b$. Then assume that the data are from a uniform om $[a, b]$, with both parameters unknown. Find the moment estimators.

**(d)** Define now $h_j(\theta) = \mathrm{E}_\theta\, Y^j$ and $N_j = n^{-1} \sum_{i=1}^n Y_i^j$, for $j = 1, \ldots, p$. The 'method of direct moments' is to solve the equations $h_j(\theta) = N_j$ for $j = 1, \ldots, p$. For $p = 3$, set up the two systems of three equations with three unknowns, i.e. $g_j(\theta) = M_j$ and $h_j(\theta) = N_j$ for $j = 1, 2, 3$, and show that the solutions $\widehat\theta = (\widehat\theta_1, \widehat\theta_2, \widehat\theta_3)$ are the same. Show in fact that the two methods, fitting centralised or direct methods, are identical, in the general case, for $p \geq 2$.

**(e)** For a given application one may choose one's method based on convenience and practicality. Sometimes formulae for the direct moments are more easily found that those for the centralised moments. The case for using centralised moments is partly numerical safety; the $M_j$ numbers may be much smaller than the $N_j$. Exemplify this by seeting up the two equations, with two unknowns, both for the centralised moments and for the direct moments, for the case of the Beta$(a, b)$ distribution.

**(f)** Generate $n = 100$ data points via the equation $y_i = [\exp\{a(\xi + \sigma N_i)\} - 1]/a$, for say $(a, \xi, \sigma) = (0.33, 0.55, 0.77)$, with the $N_i$ being standard normal. This is a skewed extension of the usual normal model, which corresponds to $a \to 0$ here. Find formulae for the first three moments for this distribution. From your data, use the method of moments to estimate the three parameters.

**Ex. 3.27** *Moment fitting estimators for the Gamma distribution.* We now apply the moment matching principle of Ex. 3.26 to the Gamma model with parameters $(a, b)$, with density proportional to $y^{a-1}\exp(-by)$ for $y$ positive; see Ex. 1.9, where we also give the mean, variance, skewness, kurtosis.

**(a)** With $\bar{Y}$ and $V_n$ the usual sample mean and sample variance, find explicit formulae for the moment estimators $\widehat{a}, \widehat{b}$; show that $\widehat{a} = \bar{Y}^2/V_n$ and $\widehat{b} = \widehat{Y}/V_n$.

**(b)** Use skewness and kurtosis formulae, in combination with Ex. 2.45, to show that

$$\begin{pmatrix} \sqrt{n}(\bar{Y} - a/b) \\ \sqrt{n}(V_n - a/b^2) \end{pmatrix} \to_d \begin{pmatrix} U \\ W \end{pmatrix} \sim \mathrm{N}_2\Big(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} a/b^2 & 2a/b^3 \\ 2a/b^3 & (2+6/a)a^2/b^4 \end{pmatrix}\Big).$$

Transform this, via the delta method, to find the limit distribution for the moment estimators.

**Ex. 3.28** *Moment method estimation in a symmetric Beta.* Consider an i.i.d. dataset $Y_1, \ldots, Y_n$ from the symmetric Beta$(a, a)$, with sample mean $\bar{Y}$ and sample variance $V_n = (n-1)^{-1}\sum_{i=1}^n (Y_i - \bar{Y})^2$.

**(a)** Show that $Y_i$ has mean $\frac{1}{2}$ and variance $\sigma^2 = \frac{1}{4}/(2a+1)$. Explain that the moment method with the sample mean does not work in this situation, but that the moment method using the sample variance leads to $\widehat{a} = 1/(8V_n) - \frac{1}{2}$. Show that it is consistent.

**(b)** Via patient integration, show that the kurtosis is

$$\mathrm{kurt} = \int_0^1 \frac{(x-\frac{1}{2})^4}{\sigma^4}\frac{\Gamma(2a)}{\Gamma(a)^2}\{x(1-x)\}^{a-1}\,\mathrm{d}x - 3 = -\frac{6}{2a+3}.$$

Then use Ex. 2.46 to show that $\sqrt{n}(V_n - \sigma^2) \to_d \mathrm{N}(0, \tau^2)$, with

$$\tau^2 = \sigma^4(2 + \mathrm{kurt}) = \frac{1/16}{(2a+1)^2}\frac{4a}{2a+3}.$$

**(c)** Use the delta method to find the limiting distribution of $\sqrt{n}(\widehat{a} - a)$, and explain how this may be used to construct a 90 percent confidence interval for $a$. (xx later: compare variance to that of the ML method, with efficiency loss. check if the following is ok. xx)

$$\kappa^2 = (2a+1)^2 a/(2a+3).$$

**Ex. 3.29** *Moment method estimators for the exponential family.* Consider an exponential family type model, studied in Ex. 1.56, with density of the form $f(y, \theta) = \exp\{\theta^{\mathrm{t}} T(y) - k(\theta)\}h(y)$. Here $T(y) = (T_1(y), \ldots, T_p(y))^{\mathrm{t}}$ is a collection of data functions. As we saw in the exercise pointed to, many classical models are special cases. Now suppose $Y_1, \ldots, Y_n$ are i.i.d. from such a model.

**(a)** Spell out the basic moment matching method, fitting empirical and model based moments for $Y_i, Y_i^2, \ldots, Y_i^p$.

**(b)** The moment matching idea is however flexible enough to allow us to choose other data functions than those associated with $Y_i, \ldots, Y_i^p$. Show that matching the moments

of $T_1(Y_i), \ldots, T_p(Y_i)$ corresponds to solving the equations

$$\bar{T}_j = \xi_j(\theta) = \partial k(\theta)/\partial\theta_j \quad \text{for } j = 1, \ldots, p,$$

with $\bar{T}_j = n^{-1}\sum_{i=1}^n T_j(Y_i)$.

**(c)** Consider the setup of Ex. 3.27, with the Gam$(a, b)$ distribution. There we studied moment matching estimators based on sample mean $\bar{Y}$ and sample variance $V_n$. Show that the principle above, using the exponential family structure, leads to fitting $\bar{Y} = a/b$ and $n^{-1}\sum_{i=1}^n \log Y_i = \psi(a) - \log b$. We shall learn via likelihood theory of Ch. 5 that these are more precise than the moment based ones, under model conditions.

**Ex. 3.30** *Mean and variance for background distribution for random sums.* Suppose there is a hidden background machine drawing first the number of dice $N$ and then reporting the sum $Z = X_1 + \cdots + X_N$ of outcomes from having thrown those $N$ dice (without reporting $N$).

**(a)** For the individual random $X_i$, show that the mean and variance are $\xi = 3.50$ and $\sigma^2 = 35/12 = 2.9167$. Use results from Ex. 1.41 to find formulae for the mean and variance of $Z$.

**(b)** If $Z_1, \ldots, Z_m$ are observed, from such a two-layer-random machine, with mean 33.33 and standard deviation 12.12, estimate the mean and standard deviation for $N$.

### Quantile matching methods

**Ex. 3.31** *Quantile fitting estimators.* An alternative to the method of moments, described in Ex. 3.26, is to fit empirical and theoretical quantiles (actually in several ways). If $Y_1, \ldots, Y_n$ are i.i.d. from a density $f(y, \theta)$, for a parameter vector of length $p$, with quantiles $Q(r, \theta) = F^{-1}(r, \theta)$, we choose quantile levels $r_1 < \cdots < r_p$, and solve the $p$ equations $Q_n(r_j) = Q(r_j, \theta)$ with respect to the $p$ unknown parameters, where $Q_n(r) = F_n^{-1}(r)$ is the empirical quantile.

method of quantiles

**(a)** Suppose the distribution to be fitted has c.d.f. $F(y) = y^\theta$ on the unit interval. Find the estimator corresponding to fitting the empirical to the theoretical median. Starting with the limit distribution for the median, see Ex. 3.20, find the limit distribution for $\sqrt{n}(\widehat{\theta} - \theta)$. More generally, find the estimator $\widehat{\theta}_r$ corresponding to fitting the $r$ level quantile, and then the limit distribution for $\sqrt{n}(\widehat{\theta}_r - \theta)$. For large $n$, which value of $r$ gives the most precise estimator?

**(b)** Consider $Y_1, \ldots, Y_n$ from the location Cauchy density $f_0(y - \theta)$, with $f_0(x) = (1/\pi)/(1+x^2)$ the standard Cauchy. Its c.d.f. is $F_0(x) = \frac{1}{2} + (1/\pi)\arctan x$, see Ex. 1.18. Show that the $r$ level quantile is $\mu + F_0^{-1}(r)$, and that this leads to the estimator $\widehat{\mu}_r = Q_n(r) - F_0^{-1}(r)$. Find the limit distribution for $\sqrt{n}(\widehat{\mu}_r - \mu)$. What quantile level $r$ leads to the sharpest estimator?

**(c)** For data $Y_1, \ldots, Y_n$ from a normal $(\xi, \sigma^2)$, find estimators for the two parametrs using the median and interquartile range. Find the limit distributions. More generally, construct an estimator $\hat{\sigma}_r$ for $\sigma$ based on the quantile difference $Q_n(1-r) - Q_n(r)$. Find the limit distribution and identify the best value of $r$.

**Ex. 3.32** *Moment fitting and quantile fitting for the Weibull.* As a general illustration of moment and quantile fitting estimation methods, consider the Weibull distribution with c.d.f. $F(t) = 1 - \exp\{-(t/a)^b\}$ for $t \geq 0$, see Ex. 1.60.

**(a)** Take e.g. $(a, b) = (3.33, 1.44)$, and simulate $n = 100$ realisations. (i) Compute average and standard deviation for these, and compute estimates $(\widehat{a}_m, \widehat{b}_m)$. (ii) From 0.25 and 0.75 quantiles, fit the two relevant equations to compute $(\widehat{a}_q, \widehat{b}_q)$. Take the trouble to display three curves, the correct underlying cumulative hazard function $A(t) = (t/a)^b$ along with the two estimated versions.

**(b)** Repeat the experiment many times, to see how close the two $(\widehat{a}, \widehat{b})$ is to $(a, b)$. Also, as an instance of a focused question, check how close the two median estimates $\widehat{m} = \widehat{a}(\log 2)^{1/\widehat{b}}$ are to the real median? Which of the two estimating schemes is best? We should point here to the likelihood methodologies of Ch. 5; the maximum likelihood method will be the winning strategy, beating both moment and quantile fitting, under model conditions.

### Minimum sum of squares and linear regression

**Ex. 3.33** *Linear regression and least squares estimation.* Consider observed pairs $(x_i, Y_i)$ for $i = 1, \ldots, n$, where $Y_i$ conditionally on covariate $x_i$ is modelled to have mean $a_0 + bx_i$ and common variance $\sigma^2$. This is classical linear regression, widely used in theoretical and applied statistics, most often analysed and used with the additional assumption that the $Y_i$ are normally distributed. Importantly, the model is extended to the case of multiple covariates below, see Ex. 3.34, 3.35, with more to come regarding statistical testing (xx in Ex. 4.37 and one more xx). To see some of these classical methods in action, check out Stories #53 and #33.

**(a)** It is helpful to reparametrise the regression line from $a_0 + bx_i$ to $a + b(x_i - \bar{x})$. Show that minimising the sum of squares $Q(a, b) = \sum_{i=1}^n \{y_i - a - b(x_i - \bar{x})\}^2$ leads to

$$\widehat{a} = (1/n)\sum_{i=1}^n y_i = \bar{y}, \quad \widehat{b} = \sum_{i=1}^n (x_i - \bar{x})y_i/M_n, \quad \text{with } M_n = \sum_{i=1}^n (x_i - \bar{x})^2.$$

**(b)** Show that $\widehat{a}$ and $\widehat{b}$ are unbiased, with zero covariance, and variances $\sigma^2/n$ and $\sigma^2/M_n$.

**(c)** Let $Q_0 = \min_{a,b} Q(a, b) = \sum_{i=1}^n \{y_i - \widehat{a} - \widehat{b}(x_i - \bar{x})\}^2$ be the minimum sum of squares. Show that

$$Q(a, b) = \sum_{i=1}^n \{y_i - a - b(x_i - x)\}^2 = Q_0 + n(\widehat{a} - a)^2 + M_n(\widehat{b} - b)^2.$$

Use this to show that $\widehat{\sigma}^2 = Q_0/(n-2)$ is an unbiased estimator of $\sigma^2$.

**(d)** We have found natural and unbiased estimators for $a, b, \sigma^2$, without yet making assumptions of the underlying distributions for $Y_i$, beyond means and variance. Assume now, however, that the distributions are normal, so that $Y_i \sim \mathrm{N}(a + b(x_i - \bar{x}), \sigma^2)$. Show that (i) $\widehat{a} \sim \mathrm{N}(a, \sigma^2/n)$; (ii) $\widehat{b} \sim \mathrm{N}(b, \sigma^2/M_n)$; (iii) $\widehat{\sigma}^2 \sim \sigma^2\chi^2_m/m$, with $m = n - 2$; and (iv) that these three statistics are mutually independent. For this last point, it may be helpful to follow the line of proof for independence of sample mean and sample variance for normal data, used in Ex. 1.50, now with an orthonormal matrix $A$ with first row $(1/\sqrt{n}, \ldots, 1/\sqrt{n})$ and second row $((x_1 - \bar{x})/M_n^{1/2}, \ldots, (x_n - \bar{x})/M_n^{1/2})$.

**(e)** Construct confidence intervals for $b$, for $\sigma$, and for $\mathrm{E}(Y \mid x_0) = a + b(x_0 - \bar{x})$, the mean value at a given position $x_0$.

**Ex. 3.34** *Linear multiple regression and least squares.* The celebrated linear multiple regression model remains a cornerstone success story of theoretical and applied statistics. It is a bag of tools for investigating the extent to which covariates $x$ influence the outcomes of certain interest variables $Y$. The standard formulation of the model is as follows. The data collected can be organised into $(x_i, Y_i)$, for individuals or objects $i = 1, \ldots, n$, where $x_i = (x_{i,1}, \ldots, x_{i,p})^{\mathrm{t}}$ is of dimension $p$ and $y_i$ of dimension one. The model then postulates that

$$Y_i = x_i^{\mathrm{t}}\beta + \varepsilon_i = x_{i,1}\beta_1 + \cdots + x_{i,p}\beta_p + \varepsilon_i \quad \text{for } i = 1, \ldots, n,$$

where the $\varepsilon_i$ are i.i.d. from the normal $\mathrm{N}(0, \sigma^2)$. Thus there are $p+1$ parameters at work here, the regression coefficients $\beta = (\beta_1, \ldots, \beta_p)^{\mathrm{t}}$ and the error distributiuon standard deviation $\sigma$. – Note that the very classical case of $y_i = a + bx_i + \varepsilon_i$, associated with a scatterplot of $(x_i, Y_i)$, is a special case; see Ex. 3.33.

**(a)** With $Y$ the vector of $Y_i$, $\varepsilon$ the vector of $\varepsilon_i$, and $X$ the $n \times p$ matrix having $(x_{i,1}, \ldots, x_{i,p})$ as its row $i$, show that

$$Y = X\beta + \varepsilon \sim \mathrm{N}_n(X\beta, \sigma^2 I),$$

with $I$ the $n \times n$ identity matrix. This is a practical and compact linear algebra version of the model formulation. We do assume that $X$ is of full rank $p$, so that the symmetric matrix $X^{\mathrm{t}}X$ is invertible. This amounts to there being at least $p$ linearly independent covariate vectors in the $X$ matrix; in particular, we must have $n \geq p$ to identify the $\beta_j$ coefficients directly from data.

**(b)** The least squares estimator $\widehat{\beta}$ is the minimiser of $Q(\beta) = \|Y - X\beta\|^2 = \sum_{i=1}^n (Y_i - x_i^{\mathrm{t}}\beta)^2$. Show that $\sum_{i=1}^n (Y_i - x_i^{\mathrm{t}}\widehat{\beta})x_i = 0$. With $\Sigma_n$ the $p \times p$ matrix $n^{-1}\sum_{i=1}^n x_i x_i^{\mathrm{t}} = n^{-1}X^{\mathrm{t}}X$, show that

$$\widehat{\beta} = (X^{\mathrm{t}}X)^{-1}X^{\mathrm{t}}Y = \Sigma_n^{-1} n^{-1} \sum_{i=1}^n x_i Y_i.$$

Prove also that $Q(\beta) = Q_0 + n(\widehat{\beta} - \beta)^{\mathrm{t}}\Sigma_n(\widehat{\beta} - \beta)$, with $Q_0 = \min_{\mathrm{all}\ \beta} Q(\beta) = \sum_{i=1}^n \widehat{\varepsilon}_i^2$, writing $\widehat{\varepsilon}_i = Y_i - x_i^{\mathrm{t}}\widehat{\beta}$ for the estimated residuals.

**(c)** Show that $\widehat{\beta}$ is unbiased and that its variance matrix can be written $\sigma^2(X^{\mathrm{t}}X)^{-1} = (\sigma^2/n)\Sigma_n^{-1}$.

**(d)** Show also that $\widehat{\beta}$ has a multinormal distribution, so that in fact $\widehat{\beta} \sim \mathrm{N}_p(\beta, (\sigma^2/n)\Sigma_n^{-1})$. This is the key result about the least squares estimators. We also need precise information for estimating $\sigma$; see Ex. 3.35.

**Ex. 3.35** *The residuals and their variances.* The setup is an in the previous Ex. 3.34, the $Y \sim \mathrm{N}_n(X\beta, \sigma^2 I)$ linear regression model. Above we focused on the least squared method and the ensuing properties for the estimators of the regression coefficients, and found $\widehat{\beta} \sim \mathrm{N}_p(\beta, (\sigma^2/n)\Sigma_n^{-1})$. We also need to deal carefully with estimators of $\sigma$, the residual standard deviation, also since we encounter statistics of the type $(\widehat{\beta}_j - \beta_j)/\widehat{\sigma}$.

**(a)** From the basic $Y = X\beta + \varepsilon$ we may define the estimated residuals as

$$\widehat{\varepsilon} = Y - X\widehat{\beta} = (I - H)Y, \quad \text{where} \quad H = X(X^{\mathrm{t}}X)^{-1}X^{\mathrm{t}},$$

the so-called hat matrix, of size $n \times n$. Show that $H$ is symmetric and idempotent, which means that $H^{\mathrm{t}} = H$ and $H^2 = H$. This also implies $(I - H)H = 0$.

**(b)** Now consider the random minimum achieved by the $Q(\beta)$ which was used in the least squares operation,

$$Q_0 = \min\{Q(\beta)\colon \text{all } \beta\} = Q(\widehat{\beta}) = \|Y - X\widehat{\beta}\|^2 = \sum_{i=1}^{n}(Y_i - x_i^{\mathrm{t}}\widehat{\beta})^2.$$

The main result, arrived at below, is that $Q_0/\sigma^2 \sim \chi^2_m$, with degrees of freedom $m = n-p$, and that $Q_0$ is independent of $\widehat{\beta}$. Show first that

$$\begin{pmatrix} X\widehat{\beta} \\ \widehat{\varepsilon} \end{pmatrix} = \begin{pmatrix} HY \\ (I-H)Y \end{pmatrix} \sim \mathrm{N}_{2n}(\begin{pmatrix} X\beta \\ 0 \end{pmatrix}, \sigma^2 \begin{pmatrix} H & 0 \\ 0 & I-H \end{pmatrix}).$$

In particular, these two random vectors are independent; also, $Q_0 = \|\widehat{\varepsilon}\|^2 = Y^{\mathrm{t}}(I-H)Y$ is consequently independent of $X\widehat{\beta}$.

**(c)** Show that $(I-H)X = 0$, which implies $\widehat{\varepsilon} = (I-H)Y = (I-H)(Y-X\beta) = (I-H)\varepsilon$ and hence $Q_0 = \varepsilon^{\mathrm{t}}(I-H)\varepsilon$. We also reach the simple idenity

$$\|\varepsilon\|^2 = \varepsilon^{\mathrm{t}}H\varepsilon + \varepsilon^{\mathrm{t}}(I-H)\varepsilon,$$

where the left-hand side is a $\sigma^2\chi^2_n$ and the two terms on the right-hand side being independent. Show that the first term on the right-hand side is a $\sigma^2\chi^2_p$. Via independence and a moment-generating function argument show then that $Q_0 \sim \sigma^2\chi^2_{n-p}$ (this is essentially the basic trick of Ex. 1.49).

**(d)** With $Q_0 \sim \sigma^2\chi^2_m$, with degrees of freedom $m = n-p$, the natural estimator is $\widehat{\sigma}^2 = Q_0/m$. Show that it is unbiased and find its variance. Note that things we have learned in Ex. 1.50 and other exercises related to the empirical variance for a normal sample carry over to the the present linear regression model, simply with $n-p$ instead of $n-1$.

**Ex. 3.36** *Separate and joint confidence intervals.* Suppose a vector $Y$ is observed from some $\mathrm{N}_p(\theta, \Sigma)$, with unknown $\theta$ and known positive definite $\Sigma$. How can we set confidence intervals for the individual $\theta_j$, for given linear combinations, and then for all linear combinations jointly?

**(a)** Consider a given linear combination $\gamma = c^{\mathrm{t}}\theta = c_1\theta_1 + \cdots + c_p\theta_p$. Explain that $\widehat{\gamma} = c^{\mathrm{t}}Y \sim \mathrm{N}(\gamma, c^{\mathrm{t}}\Sigma c)$, and that the canonical 95 percent confidence interval for $\gamma$ is $\widehat{\gamma} \pm 1.96\,\kappa(c)$, with $\kappa(c) = (c^{\mathrm{t}}\Sigma c)^{1/2}$. Explain what this entails for the components $\theta_1, \ldots, \theta_p$.

**(b)** Leaving the situation above aside for a minute, prove the following mathematical fact: for given $x$, and a given symmetric positive definite $p \times p$ matrix $A$, the maximum of $r(c) = c^{\mathrm{t}}x/(c^{\mathrm{t}}Ac)^{1/2}$, taken over all $c$, is equal to $(x^{\mathrm{t}}A^{-1}x)^{1/2}$. Do this by first seeing that the ratio $r(c)$ does not depend on the length of $c$, so the problem is equivalent to finding the maximum of $c^{\mathrm{t}}x$ over all $c$ such that $c^{\mathrm{t}}Ac = 1$; you may proceed using Lagrange multipliers and a bit of linear algebra.

**(c)** Suppose now that one wishes to have intervals for *all* linear combinations, with confidence level 95 percent to hold jointly. To what extent does one need to stretch the $c^{\mathrm{t}}Y \pm 1.96\,\kappa(c)$ intervals, to secure such a joint-for-all 95 percent confidence? Show that

$$V = \max_{\text{all } c} \frac{|c^{\mathrm{t}}Y - c^{\mathrm{t}}\theta|}{(c^{\mathrm{t}}\Sigma c)^{1/2}} = \{(Y-\theta)^{\mathrm{t}}\Sigma^{-1}(Y-\theta)\}^{1/2},$$

and that $V^2 \sim \chi^2_p$. Explain then that with $z_p = \Gamma_p^{-1}(0.95)^{1/2}$, involving the upper 0.05 quantile of the $\chi^2_p$, the intervals $I(c) = c^{\rm t}Y \pm z_p\,\kappa(c)$ have the desired property, that $\Pr\{c^{\rm t}\theta \in I(c) \text{ for all } c\} = 0.95$. Check how this $z_p$ factor grows with $p$. For $p = 5$, for example, the separate confidence intervals need to be made 1.697 times wider, in order to secure that the confidence level is 95 percent for all linear combinations.

**(d)** (xx can point to Bonferroni method. in notes: Hjort (1988). xx)

**Ex. 3.37** *Confidence intervals for key parameters in linear regression models.* Consider the general linear regression setup of Ex. 3.34–3.35, where $Y_i = x_i^{\rm t}\beta + \varepsilon_i$ for $i = 1, \ldots, n$, and the $\varepsilon_i$ being i.i.d. $\mathrm{N}(0, \sigma^2)$. The compact version of this is $Y \sim \mathrm{N}_n(X\beta, \sigma^2 I)$. We use notation from these exercises below, with least squares estimator $\widehat\beta$, $\widehat\sigma^2 = Q_0/(n-p)$, etc.

**(a)** Construct a 90 percent confidence interval for $\sigma$.

**(b)** From the general results obtained above, show that $\widehat\beta_j \sim \mathrm{N}(\beta_j, \sigma^2 r_j/n)$, with $r_j$ the diagonal $(j,j)$ element of $\Sigma_n^{-1}$. Show from this that $t_j = \sqrt{n}(\widehat\beta_j - \beta_j)/(r_j^{1/2}\widehat\sigma)$ has a $t_m$ distribution, with degrees of freedom $m = n - p$, and construct a confidence interval for $\beta_j$ from this. If a 99 percent confidence interval for $\beta_j$ is outside zero, how can this be interpreted and used?

**(c)** More generally, consider $\gamma = c^{\rm t}\beta = c_1\beta_1 + \cdots + c_p\beta_p$, a linear combnination of the regression coefficients. Find the distribution of $\widehat\gamma = c^{\rm t}\widehat\beta$, and show that the interval $c^{\rm t}\widehat\beta \pm t_{m,0.975}\,\widehat\kappa(c)/\sqrt{n}$ covers the true $c^{\rm t}\beta$ with probability 95 percent, when $\widehat\kappa(c) = (c^{\rm t}\Sigma_n^{-1}c)^{1/2}\widehat\sigma$, with $t_{m,0.975}$ the upper 0.025 quantile of the $t_m$, with $m = n-p$ degrees of freedom. This may be used to assess the size of $\beta_1 - \beta_2$, of $\beta_1 - \frac12(\beta_2 + \beta_3)$, and similar constrasts.

**(d)** The situation is different if one wishes to have confidence intervals for *all* linear combinations, jointly, with a guarantee that the overall coverage is e.g. 95 percent. Show using Ex. 3.36 that

$$V_n = \max_{\text{all } c} \frac{\sqrt{n}|c^{\rm t}\widehat\beta - c^{\rm t}\beta|}{(c^{\rm t}\Sigma_n^{-1}c)^{1/2}\widehat\sigma} = \{n(\widehat\beta - \beta)^{\rm t}\Sigma_n(\widehat\beta - \beta)\}^{1/2}/\widehat\sigma,$$

and that $V_n^2 \sim F_{p,m}$. Use this to show that the intervals $I_n(c) = c^{\rm t}\widehat\beta \pm z_{p,m}\,\widehat\kappa(c)/\sqrt{n}$, with $z_{p,m}$ the 0.975 quantile of the $F_{p,m}$ distribution, have the property that $\Pr\{c^{\rm t}\beta \in I_n(c) \text{ for all } c\} = 0.95$.

**(e)** Use the result about $V_n$ to construct a confidence region, actually a confidence ellipsoid, for the full $\beta$ vector.

**Ex. 3.38** *Predicting the next y.* Suppose linear regression analysis has been carried out, for a given dataset $(x_1, y_1), \ldots, (x_n, y_n)$. How can we predict what happens with $Y_0$, associated with another covariate vector $x_0$? This could be for an individual outside the dataset, or in a time context, speculating about the next datapoint in a sequence. An illustration of methods given below is in Story #33.

**(a)** Assume the regression data are of the form $y_i \sim \mathrm{N}(x_i^{\rm t}\beta, \sigma^2)$ for $i = 1, \ldots, n$, as with Ex. 3.34, and consider a new $x_0$, for which the not yet unobserved $Y_0$ is independent of the other data, and with distribution $\mathrm{N}(x_0^{\rm t}\beta, \sigma^2)$. Show that $\widehat y_0 = x_0^{\rm t}\widehat\beta \sim \mathrm{N}(x_0^{\rm t}\beta, x_0^{\rm t}\Sigma_n^{-1}x_0\,\sigma^2/n)$, and use this to form a confidence interval for the mean of $Y_0$, as opposed to for $Y_0$ itself.

**(b)** Then show that

$$Y_0 - \widehat{y}_0 \sim \mathrm{N}(0, \sigma^2(1 + n^{-1}x_0^{\mathrm{t}}\Sigma_n^{-1}x_0/n)).$$

Construct a prediction confidence interval for $Y_0$ based on this. Comment on the situation with a large $n$, and on the difference between confidence for $Y_0$ and its mean.

**(c)** Then consider the classic linear regression case with only one variable studied in Ex. 3.33, with $Y_i \sim \mathrm{N}(a + b(x_i - \bar{x}), \sigma^2)$. For predicting a not yet observed $Y_0 \sim \mathrm{N}(a + b(x_0 - \bar{x}), \sigma^2)$, demonstrate that

$$Y_0 - \{\widehat{a} + \widehat{b}(x_0 - \bar{x})\} \sim \mathrm{N}(0, \sigma^2\{1 + 1/n + (x_0 - \bar{x})^2/M_n\}),$$

with $M_n = \sum_{i=1}^n (x_i - \bar{x})^2$. Show that this leads to the 95 percent confidence band

$$Y(x_0) \in \widehat{a} + \widehat{b}(x_0 - \bar{x}) \pm c\,\widehat{\sigma}\{1 + 1/n + (x_0 - \bar{x})^2/M_n\}^{1/2},$$

for an appropriate range of $x_0$ values, with $c = t_{0.975,n-2}$ the $t$ upper quantile. Illustrate this with a simulated dataset, to see both where the band is tight and where it becomes very broad. One learns that it is easier to predict outcomes related to $x_0$ closer to the middle of the covariate distribution.

**Ex. 3.39** *Inference for mean by standard deviation.* (xx point to Ex. 8.4 with clean CD. xx) Consider $Y_1, \ldots, Y_n$ being i.i.d. from $\mathrm{N}(\xi, \sigma^2)$. Estimation and precise inference for $\xi$ and for $\sigma$ separately is classic terrain, via the t and the chi-squared distribution, and we have seen in Ex. 3.8 how we can also deal with quantiles $\xi + z_q\sigma$ in a precise fashion. For more complicated parameters $g(\xi, \sigma)$ we might turn to large-sample approximations, as per Ex. 2.51. This exercise concerns however precise inference for the parameter $\kappa = \xi/\sigma$.

**(a)** We start with the following cleanly stated problem: suppose $T \sim t_m(a)$, with the noncentral t, as per Ex. 1.52, with known $m$ but unknown $a$. With $t_{\mathrm{obs}}$ the observed value of $T$, we form a confidence interval by collecting together all values $a$ for which $t_{\mathrm{obs}}$ is not surprising, defined by $I = \{a\colon 0.05 \le G_m(t_{\mathrm{obs}}, a) \le 0.95\}$. Show that indeed $\Pr_a(a \in I) = 0.90$. Again, the $G_m(t, a)$ is a somewhat complicated function, but is numerically available in packages.

**(b)** Coming back to the normal sample, write as usual $\widehat{\xi} = \bar{Y}$ and $\widehat{\sigma}^2 = (n-1)^{-1}\sum_{i=1}^n (Y_i - \bar{Y})^2$, and form $\widehat{\kappa} = \widehat{\xi}/\widehat{\sigma}$. Show that $X = \sqrt{n}\widehat{\kappa} \sim t_m(\sqrt{n}\kappa)$, with $m = n - 1$ degrees of freedom. Show that

$$I = \{\kappa\colon 0.05 \le G_m(\sqrt{n}\widehat{\kappa}, \sqrt{n}\kappa) \le 0.95\}$$

is a 90 percent confidence interval for $\kappa$. In practice we find the interval by computing and monitoring the $G_m(\sqrt{n}\widehat{\kappa}, \sqrt{n}\kappa)$, as a function of $\kappa$.

**(c)** Choose your favourite $(\xi, \sigma)$, generate a normal sample of size $n = 100$, and compute (i) the interval $\widehat{\kappa} \pm 1.645\,(1 + \widehat{\kappa}^2)^{1/2}/\sqrt{n}$, from Ex. 2.51, and (ii) the exact interval from the recipe above. Compare and comment.

**(d)** Consider now a linear regression setup, as in Ex. 3.34–3.35, with independent $y_i = x_i^{\mathrm{t}}\beta + \varepsilon_i$ for $i = 1, \ldots, n$. With some given linear combination $c^{\mathrm{t}}\beta$ of the regression components, consider $\kappa = c^{\mathrm{t}}\beta/\sigma$. In the standard notation of the exercises pointed to, show that $\widehat{\kappa} = c^{\mathrm{t}}\widehat{\beta}/\widehat{\sigma}$ is such that $t = \sqrt{n}\widehat{\kappa} \sim t_m(\sqrt{n}\kappa)$. Explain how a precise 90 percent confidence interval can be set for $\kappa$.

**(e)** In such a linear regression setup, suppose $x_a$ and $x_b$ are certain covariate values, outside of $x_1, \ldots, x_n$, associated with unknown or not yet observed $Y_a = x_a^{\rm t}\beta + \varepsilon_a$ and $Y_b = x_b^{\rm t}\beta + \varepsilon_b$. Show that

$$p = \Pr(Y_b > Y_a) = \Phi(\kappa/\sqrt{2}), \quad \text{with } \kappa = (x_b - x_a)^{\rm t}\beta/\sigma,$$

Explain that you via a clear 90 percent confidence interval for $\kappa$ also have a 90 percent confidence interval for $p$. (xx brief pointer to New Haven Story #33, where nils will put in estimate and interval for $p$, the probability that 1976 is warmer than 1907, based on data from 1912 to 1971. more on the details there, prediction vs. the mean change. xx)

**Ex. 3.40** *Linear regression outside normality.* The aim here is to show and appreciate that the classical coefficient estimators in linear regression setups are still approximately normal, even when the error terms distribution is not normal. That this is so is essentially thanks to basic large-sample theory, as partly summarised in Ex. 3.11, and specifically to the Lindeberg type theorems of Ch. 2; see in particular Ex. 2.41. (xx pointer to other regression large-sample results in Ch. 5. xx)

**(a)** We first deal with a simple setup with a single regression coefficient. Suppose $y_i = x_i\beta + \varepsilon_i$ for $i = 1, \ldots, n$, with covariates $x_i$ and error terms $\varepsilon_i$ being i.i.d. from a zero-mean distribution with finite variance $\sigma^2$. Show that the estimator minimising $Q_n(\beta) = \sum_{i=1}^n (y_i - x_i\beta)^2$ is $\widehat\beta = \sum_{i=1}^n x_i y_i / M_n$, where $M_n = \sum_{i=1}^n x_i^2$. Show further that $\widehat\beta$ is unbiased with variance $\sigma^2/M_n$.

**(b)** Then consider $Z_n = M_n^{1/2}(\widehat\beta - \beta)$. Show that it has zero mean and variance $\sigma^2$, and that it can be written $\sum_{i=1}^n (x_i/M_n^{1/2})\varepsilon_i$. Deduce that $\widehat\beta$ is approximately normal, even if the $\varepsilon_i$ are not normal, provided merely that $D_n = \max_{i\le n} |x_i|/M_n^{1/2} \to 0$. If in particular $(1/n)\sum_{i=1}^n x_i^2$ stays bounded, then the natural condition is $(1/\sqrt{n})\max_{i\le n}|x_i| \to 0$.

**(c)** Then consider the general linear regression model $Y_i = x_i^{\rm t}\beta + \varepsilon_i$ of Ex. 3.34, with the $x_i$ being $p$-dimensional covariate vectors and $\beta$ a $p$-dimensional vector of regression coefficients. We take the $\varepsilon_i$ to be i.i.d. with mean zero and finite variance $\sigma^2$, but do not stipulate normality. The least squares estimator is $\widehat\beta = \Sigma_n^{-1}(1/n)\sum_{i=1}^n x_i Y_i$ with $\Sigma_n = n^{-1}\sum_{i=1}^n x_i x_i^{\rm t}$. It is unbiased with variance matrix $(\sigma^2/n)\Sigma_n^{-1}$, assumed to have full rank. Assume $\Sigma_n \to \Sigma$, a full rank matrix. Show that $Z_n = \sqrt{n}(\widehat\beta - \beta) \to_d {\rm N}_p(0, \Sigma)$, under the condition $R_n = (1/\sqrt{n})\max_{i\le n}\|x_i\| \to 0$.

**(d)** Assume now that the $x_i$ are drawn i.i.d. from a distribution over the covariate space, with finite variance matrix. Show that $R_n \to_{\rm pr} 0$. (xx calibrate with what we have in 2.41. xx)

**(e)** Argue that the fine-tuned finite-sample confidence methods, developed in Ex. 3.34–3.35 under exact normality, continue to hold in the large-sample sense, even if the distributions are not normal. Give conditions for such results to hold.

**(f)** Suppose the same type of phenomenon is studied in both Denmark and Sweden, with the same meaning for $Y$ and covariates $x_1$ and $x_2$. This leads to regression estimators $\widehat\beta_D$ and $\widehat\beta_S$ for the two analyses. Construct confidence intervals for $d_j = \beta_{D,j} - \beta_{S,j}$, for $j = 1, 2$.

**Ex. 3.41** *Least squares estimation in other setups.* (xx to come. point is that minimising $Q(\theta) = \sum_{i=1}^n \{y_i - \xi_i(\theta)\}^2$ is a general principle, also outside linear regression. xx)

**(a)** (xx some easy cases. binomial. normal mean. poisson. xx)

**(b)** Assume $y_1, \ldots, y_n$ are i.i.d. and gamma distributed $(a, b)$, with known $a$. Find the least squares estimator $\widehat b$. Then find the mean and variance of this estimator.

**(c)** (xx regression, with mean $a + bx_i$, then mean $a \exp(bx_i)$. xx)

**Ex. 3.42** *Estimating with autocorrelated data.* Situations dealt with above have involved different models and questions, but have assumed independence between observations. In various setups, including those of time series, there is correlation, however, with consequences for estimators and their precision. The brief discussion below has implications for reading off estimates for key quantities in Markov Chain Monte Carlo settings in Bayesian analysis; see Ch. 7.

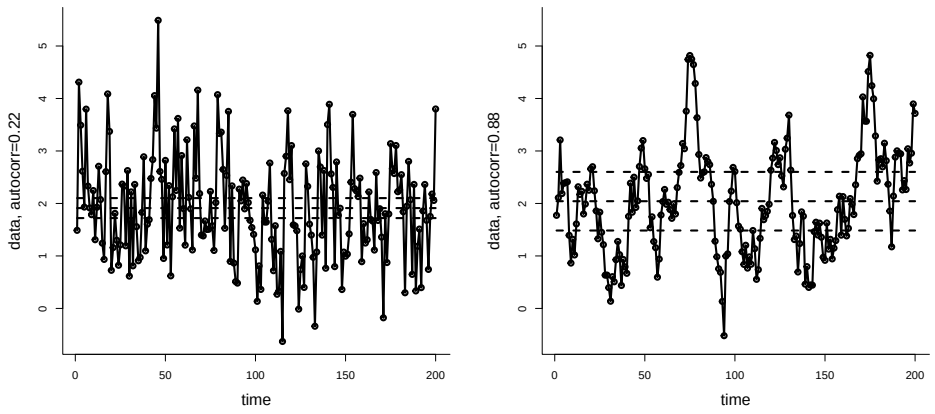



Figure 3.3: *Data with autocorrelation, with underlying* $\xi = 2.0$, $\sigma = 1.0$, *and then* $\rho = 0.22$ *(left panel) and* $\rho = 0.88$ *(right panel). The horizontal lines are for the estimate and the 95 percent confidence intervals for* $\xi$. *With higher autocorrelation, data are more connected, and estimates have larger variances.*

**(a)** Consider $V_1, \ldots, V_n$ i.i.d. with mean zero and variance one. From these we form a new series, with autocorrelation $\rho \in (-1, 1)$, as follows. Let $W_1 = V_1$ and then $W_i = \rho W_{i-1} + (1 - \rho^2)^{1/2} V_i$ for $i = 2, \ldots, n$. Show that these $W_i$ have mean zero and variance one, but $\mathrm{cov}(W_i, W_j) = \rho^{j-i}$ for $i < j$. A time series, with three crucial parameters mean $\xi$, variance $\sigma^2$, and autocorrelation $\rho$, is then formed as $Y_i = \xi + \sigma W_i$.

**(b)** With such an autocorrelated time series $Y_1, \ldots, Y_n$, the natural estimator of $\xi$ is still the sample average $\widehat{\xi} = \bar{Y}_n$, but its variance is affected. Show first in general terms that one for any $a \in (-1, 1)$ has

$$n^{-1} \sum_{1 \le i < j \le n} a^{j-i} = \frac{a}{1-a} \Big\{ 1 - \frac{1}{n} - \frac{1}{n} \frac{a}{1-a} (1 - a^{n-1}) \Big\} = \frac{a}{1-a} \{ 1 + O(1/n) \}.$$

Show that this leads to

$$\mathrm{Var}\, \bar{Y}_n = \frac{1}{n^2} \sum_{i,j} \mathrm{cov}(Y_i, Y_j) = \frac{\sigma^2}{n} \Big\{ 1 + 2 \frac{\rho}{1-\rho} + O(1/n) \Big\} = \frac{\sigma^2}{n} \Big\{ \frac{1+\rho}{1-\rho} + O(1/n) \Big\}.$$

Conclude that $\bar{Y}_n$ is unbiased and consistent for $\xi$, but with this adjusted variance.

**(c)** We're not aiming here for maximum generality (which would take us into the terrain of stationary time series); the main point to convey is that positively autocorrelated data still allow consistent estimation and inference for key quantities, but with bigger

variances, wider confidence intervals, etc. Assume therefore that the $V_i$ above are normal, implying that $(W_1, \ldots, W_n)$ and $(Y_1, \ldots, Y_n)$ have multinormal distributions. In addition to the basic moment structure, with $\mathrm{E}\, Y_i = \xi$, $\mathrm{Var}\, Y_i = \sigma^2$, $\mathrm{corr}(Y_i, Y_j) = \rho^{|j-i|}$, we can read off further variances and correlations, for $Y_i^2$, for $Y_i Y_{i+1}$, etc. Show that $\widehat{\sigma}^2 = n^{-1} \sum_{i=1}^n (Y_i - \bar{Y}_n)^2$ is consistent for $\sigma^2$, and that

$$\widehat{\rho} = \frac{1}{n} \sum_{i=1}^{n-1} \frac{Y_i - \widehat{\xi}}{\widehat{\sigma}} \frac{Y_{i+1} - \widehat{\xi}}{\widehat{\sigma}}$$

is consistent for $\rho$. Show that this implies $\sqrt{n}(\bar{Y}_n - \xi)/\widehat{g} \to_d \mathrm{N}(0, \sigma^2)$, with factor $\widehat{g}^2 = (1 + \widehat{\rho})/(1 - \widehat{\rho})$. Explain that this leads to these confidence intervals for $\xi$, adjusted for autocorrelation: $\widehat{\xi} \pm 1.96\, \widehat{g}\widehat{\sigma}/\sqrt{n}$. Construct a version of Figure 3.3, with $n = 200$ autocorrelated observations, with higher autocorrelation in the right panel, associated with wider confidence interval for the mean.

**(d)** Statisticians A and B need to estimate a mean parameter $\xi$ with a certain precision, from observations $Y_i$ with this mean and variance $\sigma^2$. A manages to get hold of i.i.d. data, and finds an estimate with a confidence interval after $n_A = 1000$ observations. B utilises a different setup for generating the data, and has to make do with a chain with autocorrelation $\rho = 0.90$. How much bigger does his sample size $n_B$ need to be, in order to have the same precision as that obtained by his colleague?

### Estimating a density

**Ex. 3.43** *The kernel density estimator.* Suppose $X_1, \ldots, X_n$ are i.i.d. from some smooth positive density $f$ on the real line, with c.d.f. $F$. To estimate $f$, introduce a kernel function $K$, a density symmetric around zero, and for any bandwidth $h$ write $K_h(u) = h^{-1} K(h^{-1}u)$. The kernel density estimator is

$$f_n(x) = n^{-1} \sum_{i=1}^n K_h(X_i - x).$$

Below we write $k_2 = \int u^2 K(u)\, \mathrm{d}u$ and $R(K) = \int K(u)^2\, \mathrm{d}u$, both taken to be finite. (xx point to Ex. 5.19 for parametric comparison. xx)

**(a)** Assuming that $f$ has two smooth derivatives, show that the mean of $f_n(x)$ is

$$g_n(x) = \int K_h(y - x)\, f(y)\, \mathrm{d}y = \int K(u) f(x + hu)\, \mathrm{d}u = f(x) + \tfrac{1}{2} k_2 f''(x) h^2 + o(h^3).$$

**(b)** For the variance, show that

$$\begin{aligned} \mathrm{Var}\, f_n(x) &= n^{-1} \mathrm{Var}\, K_h(X_i - x) = n^{-1} \Big[ \int K_h(y - x)^2\, f(y)\, \mathrm{d}y - g_n(x)^2 \Big] \\ &= (nh)^{-1} \int K(u)^2 f(x + hu)\, \mathrm{d}u - n^{-1} g_n(x)^2, \end{aligned}$$

with leading terms $R(K) f(x)/(nh) - f(x)^2/n$. Show that $f_n(x) \to_{\mathrm{pr}} f(x)$ provided the bandwidths satisfy $h \to 0$ and $nh \to \infty$. If $h = c/n^\alpha$, what is the allowed range of $\alpha$?

**(c)** For the integrated squared error, show that

$$\mathrm{mise}_n = \tfrac{1}{4} k_2 h^2 R(f'') + (nh)^{-1} R(K) + \text{smaller terms}.$$

Show next that the function $\frac{1}{4}ah^4 + b/(nh)$ is minimised at $h_0 = (b/a)^{1/5}n^{-1/5}$, with minimum value $(5/4)a^{1/5}b^{4/5}n^{-4/5}$. Hence deduce that the best bandwidth, in the sense of minimising the approximate mise, is

$$h_{0,n} = \{R(K)/k_2^2\}^{1/5}R(f'')^{-1/5}n^{-1/5},$$

with minimum mise

$$\min_h \text{mise} = (5/4)k_2^{2/5}R(K)^{4/5}R(f'')^{1/5}n^{-4/5} - R(f)/n + \text{smaller terms}.$$

**(d)** For the case of $f$ being a normal $\mathrm{N}(\xi, \sigma^2)$ density, using $K = \phi$ as the kernel function, show that

$$h_{0,n} = (4/3)^{1/5}\sigma\, n^{-1/5} = 1.0592\,\sigma\, n^{-1/5}.$$

How well one can estimate the unknown $f$ with the kernel method, for the case of a normal density, is seen through

$$\begin{aligned}\min_h \text{mise} &= (5/4)(3/4)^{1/5}(1/\sqrt{2})\phi(0)(1/\sigma)n^{-4/5} + O(1/n)\\ &= 0.3329\,(1/\sigma)n^{-4/5} + O(1/n).\end{aligned}$$

**(e)** The rule-of-thumb method is then to use the standard normal kernel, with bandwidth $h_n = 1.0592\,\widehat{\sigma}/n^{1/5}$, with $\widehat{\sigma}$ the usual empirical standard deviation, or perhaps a robust version. One often sees that this is a reasonably clever and reasonably robust bandwidth, even though its precise formula involves an auxiliary normality assumption which is not meant to be trusted. For an illustration, reproduce a version of Figure 16.5 of Story #5, producing density estimates for the birthweights from $n_0 = 115$ nonsmoking and the $n_1 = 74$ smoking mothers.

**(f)** (xx rounding off; we keep it short. could do parametric precision with ML after seeing Ch5 stuff. point to uses in our book. xx)

### More material, leftovers (as of 30-June-2025)

**Ex. 3.44** *Comparing estimators for a skewed distribution on the unit interval.* Consider i.i.d. observations $Y_1, \ldots, Y_n$ from the model with c.d.f. $F(y, \theta) = y^\theta$ on the unit interval, with $\theta$ an unknown positive parameter. Various estimators may be constructed and then compared in terms of precision.

**(a)** We start with the moment estimator. Show that the density is $f(y, \theta) = \theta y^{\theta-1}$, and that its mean is $\theta/(\theta + 1)$. Show that the moment method estimator becomes $\widehat{\theta}_m = \bar{Y}/(1 - \bar{Y})$, in terms of the sample average $\bar{Y}$.

**(b)** Explain that the $q$ level quantile is $y_q = q^{1/\theta}$, and that quantile matching leads to the estimator $\widehat{\theta}_q = \log q/\log Q_{n,q}$, with $Q_{n,q}$ the empirical quantile; a special case is the median matching estimator $\widehat{\theta}_{0.50} = -\log 2/M_n$, in terms of the median.

**(c)** Show that $Z_i = -\log Y_i \sim \text{Expo}(\theta)$. Argue that moment matching on this scale leads to the estimator $\widehat{\theta}_e = 1/\bar{Z}$, with $\bar{Z} = n^{-1}\sum_{i=1}^n Z_i$.

**(d)** To compare the three estimators we find their limit distributions. Use the CLT to identify the limit of $\sqrt{n}\{\bar{Y} - \theta/(\theta + 1)\}$, and then the delta method to establish that

$$\sqrt{n}(\widehat{\theta}_m - \theta) \to_d \mathrm{N}(0, \tau_m^2), \quad \text{where } \tau_m^2 = (1 + \theta)^4[\theta/(\theta + 2) - \{\theta/(\theta + 1)\}^2].$$

**(e)** For the quantiles, show first that $\sqrt{n}(Q_{n,q} - q^{1/\theta}) \to_d V$, where $V$ has variance $q(1-q)/\theta^2 q^{2-2/\theta}$. Again via the delta method, establish that

$$\sqrt{n}(\widehat{\theta}_q - \theta) \to_d -\frac{\theta^2}{\log q}\frac{1}{q^{1/\theta}}V \sim \mathrm{N}(0, \tau_q^2), \quad \text{where } \tau_q^2 = \frac{\theta^2}{(\log q)^2}\frac{1-q}{q}.$$

Examine this function numerically and find that the best value for $q$, leading to the smallest variance, is $q^* = 0.203$, with variance $1.544\,\theta^2$.

**(f)** For the transformed data estimator, show $\sqrt{n}(\widehat{\theta}_e - \theta) \to_d \mathrm{N}(0, \theta^2)$. Now compare the precision of the three estimators, and comment.

## Notes and pointers

(xx to come. we point to various matters, genesis of crucial concepts, and also point to chapters ahead. explain that yes, we've touched and used CLT and delta method and a bit more here, but with details and more material to come in Ch 4. xx)

[xx CLT for binomials: associated with the famous names de Moivre (who showed a version of this in 1733) and Laplace (who had a clearer and more general proof in 1812). xx]

Briefly genesis of неравенцво Маркова, the Markov, and the неравенцво Чебышёва, the Chebyshov (often anglisised to Chebyshev, but his name was really Чебышёв). mention Kahneman et al. (2020).

# I.4

## Testing, sufficiency, power

In previous chapters we have learned about classes of distributions, their parameters, ways of estimating these from data, along with assessment of precision and confidence intervals. The present chapter goes on to the fundamental statistical reporting tool of hypothesis testing. Statistical testing of hypotheses are data-based rules for when to reject (and hence, when not to reject) a hypothesis about the parameters of a model. Theory is developed to construct such tests in quite general setups. These are constructed to have a certain significance level, like 0.05, the intended low probability of rejecting the hypothesis if it is in fact true. It also has a power function, the probability of rejection as a function of how far the model parameters might be from the hypothesis. We learn the basics of the Neyman–Pearson theory for optimal testing, and see it panned out for many situations, including the general setting of exponential families. The fruitful concept of sufficiency, the notion that a lower-dimensional vector of summaries contains all statistical information about a model, is also developed in this chapter, with implications for both estimation and testing. Parts of the theory of this chapter are generalised and extended in Chapters 5, 8, 9.

*Key words:* ancillarity, completeness, conditional tests, exponential family, factorisation theorem, Neyman–Pearson optimality, power, p-value, sufficiency, testing, Wald ratios

In the broad context of analysing data from models, the previous chapter dealt generally speaking with estimators of relevant parameters, their precision, and comparisons, leading in particular to confidence intervals. The present chapter develops the methods and applications further, in the direction of *statistical testing of null hypotheses*. Studying such tests involves their interpretation, construction, properties, performance, along with connections to confidence and yet other themes.

Consider in general terms data $y$ stemming from a model with a parameter vector $\theta = (\theta_1, \ldots, \theta_p)$, and suppose one wishes to test the null hypothesis $H_0$ that $\theta$ is inside a well-defined subset $\Theta_0$ of the full parameter region $\Theta$. Precisely what constitutes a null hypothesis is a matter of scientific and statistical context, often reflecting the intentions and the overall aims of the data collection and its analysis. The null hypothesis is typically a statement, concerning the nature of the mechanisms studied, the incorrect rejection of which one attempts to avoid. A simple illustration is testing whether a particular regression coefficient is equal to zero (so $H_0$ could be '$\beta_3 = 0$'), or testing whether two parameters, perhaps for two groups, are equal (so $H_0$ would be '$\theta_A = \theta_B$').

A *test* for such a hypothesis is a rule saying '$H_0$ is to be rejected if data $y$ fall in the set $R$', along with the complementary statement '$H_0$ is not rejected if data $y$ fall in the set $R^c$'. We talk here of *the rejection region* $R$ and *the acceptance region* $R^c$. A fundamental aspect of such a test to look for is its *significance level*, typically meant to be a relatively small probability, like 0.05 or 0.01 or even smaller. We say that the test has level $\alpha$ provided

the level of a test

$$\Pr_\theta(\text{reject } H_0) \le \alpha \quad \text{for all } \theta \in \Theta_0. \tag{4.1}$$

With level 0.01 one is guaranteed such a low chance of falsely claiming that $H_0$ is wrong, if it is indeed correct. So for the illustrations briefly alluded to above, when statisticians after careful analyses reject '$\beta_3 = 0$' or '$\theta_A = \theta_B$', therefore going on to claim that their alternatives are valid, with '$\beta_3 \neq 0$' and '$\theta_A \neq \theta_B$', these claims are seen as trustworthy (and may make into publications), since the probability of these claims being false is so low.

Often tests are carried out via appropriate *test statistics*, as when constructing a $T = T(y)$, a function of the data $y$, with the property that $T$ ought to be inside some normal and well-understood range under $H_0$ conditions (we shall meet many cases of determining the *null distribution* of such test statistics), but bigger, if $H_0$ is wrong. In such cases, the rejection region takes the form $R = \{y\colon T(y) \ge t_0\}$, with $t_0$ the rejection threshold, chosen to have $\Pr_\theta(T(Y) \ge t_0) \le \alpha$ for all $\theta$ congruent with $H_0$.

test statistic, rejection threshold

Conceptually and operationally, there is a certain direction implied when setting up an hypothesis $H_0$ and its alternative. Rejecting $H_0$, with a test with low level testing level $\alpha$, leads to a positive claim about the alternative being true; the observed data have landed in a region which if $H_0$ were true has low probability. On the other hand, *not rejecting* the null hypothesis should not be seen as 'verifying' $H_0$; one needs to be content with the not so bold statement 'the observed data do not provide sufficient evidence for claiming that $H_0$ does not hold'.

We are also keenly interested in *the power* of a test, which is the detection chance

$$\text{pow}(\theta) = \Pr_\theta(\text{reject } H_0) \quad \text{as a function of } \theta. \tag{4.2}$$

So for $\theta \in \Theta_0$, the power is low, say bounded by the traditional 0.05, but interest is then behaviour of the power function in the alternative parameter domain $\Theta - \Theta_0$. Thus some tests are *stronger* than other tests with the same level, and we learn recipes for constructing such in the exercises below. The power of a test clearly depends on the quality of data, and typically of the sample size, as we shall see in exercises. Thus detecing that $\theta_A \neq \theta_B$, for parameters of two groups, in a setup where there really is a difference, becomes more likely with more data.

tests have power

Below we also define, discuss, and use *p-values*, which are commonly quoted in most branches of applied statistics work, typically to indicate how clear a potential finding is. The idea is to quantify how unlikely it is, to observe what is actually observed, if some relevant null hypothesis $H_0$ is actually true. If the test is set up to reject the null if an appropriate test statistic $T$ is sufficiently large, we're after $\Pr_\theta(T \ge t_{\text{obs}})$ under $H_0$ conditions, with $t_{\text{obs}}$ the observed $T$ for the given dataset. Some care is needed since that probability might depend on parameters under $H_0$. The more careful version is

the p-value

$$p = \max\{\Pr_\theta(T \ge t_{\text{obs}})\colon \theta \in \Theta_0\}. \tag{4.3}$$

A small p-value, like $p \leq 0.01$, casts serious doubt on $H_0$, since the observed $t_{\rm obs}$ is so unlikely. A rephrasing of the testing scheme, with significance level say 0.01, is to reject $H_0$ if $p \leq 0.01$.

A classic result for testing theory is the Neyman–Pearson Lemma, which in an idealised setup with just two possible densities identifies the most powerful method for testing one density against the other; see Ex. 4.8–4.10. This sharpens further questions for more general setups, and we manage to find optimal tests in a variety of setups, including for the broad exponential family class. This also necessitates exploring and developing certain themes of independent interest and use, those of sufficiency, ancillarity, completeness, conditional testing; see the string of exercises starting with Ex. 4.19. A core idea in that terrain is that of data compression; for various models, a low-dimensional vector of data summaries contains all relevant statistical information.

(xx then one paragraph with pointers to other chapters and perhaps to a few of the stories. xx)

### Testing, testing

**Ex. 4.1** *Testing a null hypothesis.* Here we introduce the notion of null hypotheses and their testing in a few simple setups.

**(a)** The probability $p = \Pr(A)$ of a certain event is meant to be $p_0 = 0.33$, if the machinery around it works as it should. To test this one carries out the relevant experiment $n = 100$ times, and the event takes place $y = 44$ times. Should you reject the 0.33 hypothesis? Show that with $Y \sim \text{binom}(n, p)$, the statistic $T = (Y - np_0)^2/\{np_0(1-p_0)\}$ is approximately a $\chi^2_1$, under the null assumption that $p = p_0$. Show also that $W$ tends to be bigger than a $\chi^2_1$, if $p \neq p_0$. Show that when using $T$ as test statistic, the p-value of (4.3) becomes $p = \Pr_{p_0}(T \geq t_{\rm obs})$, with $t_{\rm obs}$ the value of $T$ seen with $y_{\rm obs} = 44$. Compute this p-value, and decide whether the $p = 0.33$ hypothesis should be rejected at the 0.05 level.

**(b)** Suppose $(X, Y, Z)$ is trinomial with sum $n$ and probabilities $(p, q, r)$; see Ex. 1.5. For concreteness, suppose a theory holds that $(p, q, r) = (0.20, 0.30, 0.50)$, and that $(X, Y, Z) = (42, 47, 111)$ is observed. Is this enough to reject the theory in question? Show that

$$W = (X - np)^2/(np) + (Y - nq)^2/(nq) + (Z - nr)^2/(nr)$$

has mean equal to 2, and that it is approximately a $\chi^2_2$, using the multidimensional CLT for $(X, Y, Z)$. Explain how this may be used to test the theory mentioned, and carry out the test. This is actually the classic Pearson chi-squared test for multinomials; see Ex. 5.24 for details, generalisations, and discussion.

**(c)** You're rolling your die, but it takes you as many as $y_{\rm obs} = 15$ rolls to get your first '6'. Does this make you suspect that the probability $p$ of a '6' is not 1/6, but lower? With testing level $\alpha = 0.05$, what is the set of suspicious outcomes of $Y$, the number of throws to get the first '6'?

**Ex. 4.2** *Connections from confidence intervals to testing.* Though confidence intervals and testing are two different reporting tools, when summarising inference, there are clear connections. Suppose $\mu$ is a parameter of inference, perhaps a function of model parameters, for which we can build both confidence intervals and tests.

**(a)** Suppose one needs to test the one-point null hypothesis that $\mu = \mu_0$, a given value, and that $[L, U]$ is a 99 percent confidence interval. Show that the test consisting in rejecting, if $\mu_0$ is outside this interval, has level 0.01.

**(b)** Suppose on the other hand that there is a well-defined 0.01 level test procedure for testing $\mu = \mu_0$, against $\mu \neq \mu_0$, for each candidate value $\mu_0$. Gather together in a set $A$ all the $\mu_0$ values which are not rejected by the corresponding 0.01 level test. Show that $\Pr_\mu(\mu \in A) = 0.99$, making $A$ a 99 percent confidence region.

**(c)** Go through the relevant details, from confidence interval to test and vice versa, for the simple prototype case of the observation being $Y \sim \mathrm{N}(\theta, 1)$.

**Ex. 4.3** *Confidence intervals for quantiles.* Let $Y_1, \ldots, Y_n$ be i.i.d. from a continuous density, positive on its sample space. How can we construct confidence intervals for the median $\mu = F^{-1}(\frac{1}{2})$, and more generally for quantiles $\mu_q = F^{-1}(q)$? There are several approaches here, but here we give the basic method via the empirical c.d.f.

**(a)** Let $F_n$ be the empirical c.d.f. for the data, see Ex. 3.9. If $\mu_0$ is the true median, show that $F_n(\mu_0)$ is a simple $B_n/n$, with $B_n \sim \mathrm{binom}(n, \frac{1}{2})$, and that this implies $W_n(\mu_0) = \sqrt{n}\{F_n(\mu_0) - \frac{1}{2}\}/\frac{1}{2}$ being approximately a standard normal. Argue that a natural 0.05 level test for $\mu = \mu_0$ is to accept the hypothesis provided $|W_n(\mu_0)| \leq 1.96$.

**(b)** Following the general testing-to-confidence connection of Ex. 4.2, show that the associated 95 percent confidence interval becomes $\mathrm{ci}_n = \{\mu\colon |F_n(\mu) - \frac{1}{2}| \leq 1.96/(2\sqrt{n})\}$. In other words, we may read off the interval from a plot of $F_n$, without knowing or taking on board the details of the exact or approximate distribution of sample quantiles, as with Ex. 3.21.

**(c)** Generalise to the case of any quantile $\mu_q = F^{-1}(q)$. Show that the recipe above leads to the confidence interval $\mathrm{ci}_n = \{\mu_q\colon |F_n(\mu_q) - q| \leq z_\alpha\{q(1-q)\}^{1/2}/\sqrt{n}\}$, where $\Pr(|\mathrm{N}(0,1)| \leq z_\alpha) = \alpha$, the confidence level. For an illustration of these methods, check Story #5, where we plot the empirical c.d.f. and read off confidence intervals for quantiles $F^{-1}(q)$ at levels 0.10, 0.50, 0.90, for the weight of mothers pre pregnancy.

**(d)** Discuss ways in which more accurate confidence intervals can be constructed, using the exact binomial distribution; for the median, for example, $nF_n(\mu) \sim \mathrm{binom}(n, \frac{1}{2})$. This leads to slight modification of the horizontal bands when reading off intervals from the empirical c.d.f.

**Ex. 4.4** *t testing, one and two samples.* Testing the mean based on a sample of normal observations is a recurring problem, in several guises, and with the famous t test being the canonical procedure; details are given below. We also go through the basics for testing the difference of means for two normals samples. Due to the connections to confidence intervals discussed in Ex. 4.2 we also find accurate confidence intervals for the key parameters. Beyond their concrete relevance and repeated use in these standard setups, the t testing procedures are important since similar constructions can be worked with in large classes of more complicated setups, but then typically with approximations to key distributions, rather than the exact solutions found under these classic strict modelling assumptions.

**(a)** Suppose $X_1, \ldots, X_n$ are i.i.d. $\mathrm{N}(\xi, \sigma^2)$, so far assuming $\sigma$ to be known, and that one wishes to test $H_0\colon \xi = \xi_0$ against the alternative that $\xi \neq \xi_0$, where $\xi_0$ is some appropriate given value, like zero. Using the exact distribution of $\bar{X}$, cf. Ex. 1.2, show that $Z = (\bar{X} - \xi_0)/(\sigma/\sqrt{n}) = \sqrt{n}(\bar{X} - \xi_0)/\sigma$ is standard normal under $H_0$, and that $|Z|$ will tend to be bigger than a normal if $H_0$ is not true. Explain that the test which rejects

$H_0$ when $|Z| > 1.96$ has level 0.05.

**(b)** For the more realistic case of $\sigma$ not being known, the natural construction is the t statistic $t = \sqrt{n}(\bar{X} - \xi_0)/\widehat{\sigma}$, with $\widehat{\sigma}$ the empirical standard deviation. Show from Ex. 1.51 that $t \sim t_{n-1}$ under $H_0$, and write down a precise 0.05 level test.

**(c)** Suppose now that the mean $\xi$ for the population of $X_i$ is to be compared with the mean $\eta$ for another population, where we have i.i.d. data $Y_1, \ldots, Y_m \sim \mathrm{N}(\eta, \sigma^2)$. So the task is to test $H_0$ that $\xi = \eta$. Show first that $\bar{Y} - \bar{X} \sim \mathrm{N}(\eta - \xi, \sigma^2(1/m + 1/n))$. To build a t statistic we need an estimator for the variance. Writing $\widehat{\sigma}_1$ and $\widehat{\sigma}_2$ for the empirical deviances for samples 1 and 2, show that with

$$\widehat{\sigma}^2 = c_1\widehat{\sigma}_1^2 + c_2\widehat{\sigma}_2^2, \quad \text{using } c_1 = (n-1)/(n+m-2),\ c_2 = (m-1)/(n+m-2),$$

we have $\widehat{\sigma}^2 \sim \sigma^2\chi^2_{n+m-2}/(n+m-2)$, independent of $\bar{X} - \bar{Y}$. Conclude that $t = (\bar{X} - \bar{Y})/R \sim t_{n+m-2}$ under $H_0$, where $R = \widehat{\sigma}(1/m + 1/n)^{1/2}$.

**(d)** Use these building blocks to also construct a 90 percent confidence interval for $d = \xi - \eta$.

**Ex. 4.5** *Wald tests.* Here we present the basics of so-called Wald tests, a general and practical way of forming tests via approximate normality. Such tests are used very routinely when looking for and reporting findings about regression coeffiecients in all standard regression models; see Story #5 for an illustration of this. The Wald tests can be constructed almost immediately, from the confidence interval construction of Ex. 3.11, via the interval-to-test connection of Ex. 4.2, but we tend to a few details to allow for potential simplifications of assumptions. – Assume $Y_1, \ldots, Y_n$ comprise the data (not necessarily i.i.d.), from a suitable model with vector parameter $\theta$. Suppose further that $\mu = \mu(\theta)$ is a focus parameter, for which we need to test $\mu = \mu_0$, for a given null value $\mu_0$.

**(a)** Suppose there is an estimator $\widehat{\mu}$ with the property that $\sqrt{n}(\widehat{\mu} - \mu) \to_d \mathrm{N}(0, \tau^2)$, and that there is a consistent estimator $\widehat{\tau}$ for this limit spread $\tau$. Show as with Ex. 3.11 that $W_n = \sqrt{n}(\widehat{\mu} - \mu)/\widehat{\tau} \to_d \mathrm{N}(0,1)$. Show that the arguments go through, with a limiting standard normal, for $W_{n,0} = \sqrt{n}(\widehat{\mu} - \mu_0)/\widehat{\tau}_0$, at the null hypothesis, as long as $\widehat{\tau}_0$ is consistent for $\tau$ at this null position; also, technically speaking, we only need to establish $\sqrt{n}(\widehat{\mu} - \mu_0) \to_d \mathrm{N}(0, \tau^2)$ at the null hypothesis. Conclude that the test which rejects $\mu = \mu_0$ when $|W_{n,0}| \geq 1.96$ has level approaching 0.05 (and of course similarly with other chosen testing levels; level 0.01 corresponds to threshold 2.576, etc.).

**(b)** Explain that $p = \Pr(|N(0,1)| \geq W_{n,0,\mathrm{obs}})$ is an approximation to the exact p-value.

**(c)** Suppose $X \sim \mathrm{binom}(100, p)$, with a need for testing $p = 0.33$. Set up a Wald test, and compute the p-value, if $x_{\mathrm{obs}} = 44$. Suppose then that there is an additional $Y \sim \mathrm{binom}(100, q)$, and that one wishes to test $p = q$. Set up a Wald test, and compute the p-value, if $(X, Y)$ are observed to be $(44, 55)$.

**Ex. 4.6** *When confidence intervals for two parameters overlap.* Suppose confidence intervals for parameters $a$ and $b$ overlap. It might sound plausible that the null hypothesis $a = b$ will then not be rejected. To check aspects of this and related questions, consider a prototype setup where $\widehat{a} \sim \mathrm{N}(a, 1)$ and $\widehat{b} \sim \mathrm{N}(b, 1)$ are independent.

**(a)** With confidence level $1 - \alpha$, show that the canonical intervals are $\widehat{a} \pm z_0$ and $\widehat{b} \pm z_0$, with $z_0 = \Phi^{-1}(1 - \alpha/2)$, e.g. the standard 1.96 for level 0.95. Show that overlapping intervals corresponds to $|\widehat{d}| \leq 2z_0$, where $\widehat{d} = \widehat{b} - \widehat{a} \sim \mathrm{N}(b - a, 2)$

**(b)** Consider the test of $a = b$ which rejects when the intervals for $a$ and $b$ are disjoint. Show that this test has considerably lower significance level than $\alpha$; for 95 percent intervals this level is 0.0056.

**(c)** Yes, it is possible to have overlapping 95 percent intervals and yet reject $a = b$ at level 5 percent. Identify the range of $\widehat{d}$ where this happens.

**Ex. 4.7** *Consider the alternative.* If your null hypothesis $H_0$ is wrong, and therefore ought to be rejected by a clever test procedure, in which ways might it be wrong, and to what extent?

**(a)** For a simple illustration, suppose $\bar{x} = 2.22$ and $\bar{y} = 2.66$, sample means from $\mathrm{N}(\xi_1, \sigma^2)$ and $\mathrm{N}(\xi_2, \sigma)$, for sample sizes 100, for simplicity with known $\sigma = 1$. (i) Test the $H_0$, that the two means are equal; (ii) give a 95 percent confidence interval for $\delta = \xi_2 - \xi_1$. The point is that (ii) is more informative than the yes-no answer to the testing question.

**(b)** Suppose $X$ and $Y$ are binomial, of size $n = 100$, with probabilities $p$ and $q$. You observed $(x_{\mathrm{obs}}, y_{\mathrm{obs}}) = (33, 55)$. Test $H_0 \colon p = q$, and, in addition, find a confidence interval for $\rho = q/p$ (use the delta method), and comment.

**(c)** Suppose $X_1, \ldots, X_{10}$ are independent, with $X_j \sim \mathrm{N}(\xi_j, 1)$, and with $H_0$ equality of means is to be tested. Give first an interpretation of $\lambda$, where $\lambda^2 = \sum_{j=1}^{10}(\xi_j - \bar{\xi})^2$, where $\bar{\xi}$ is the average of the $\xi_j$. Explain that rejecting $H_0$ if $Z = \sum_{j=1}^{10}(X_j - \bar{X})^2 > \gamma_{0.95,9} = 16.919$ is a 0.05 level test. If the data deliver $z_{\mathrm{obs}} = 22.222$, you reject equality; what can you say about $\lambda$? (xx point to Ch. 8, a full CD for $\lambda$. xx)

**(d)** The alternatives to $H_0$ should be taken into account also for constructing a test that might be better than others. For the little scenario above, suppose one knows a priori that if $H_0$ is wrong, it must be because one of the means is different from the rest. Explain that if $H_0$ is wrong in this fashion, then

$$W = \max_{j \le 10} |X_j - \bar{X}_{(-j)}|$$

ought to be able to detect is, with $\bar{X}_{(-j)}$ the mean of the other nine $X_\ell$. Carry out simulations to find the upper 0.05 point in its null distribution. Then compare the power functions of the $Z$ and the $W$ test, against alternatives of the form $(\xi, \ldots, \xi + d, \ldots_{10})$, with $d$ the amount with which the single $\xi_j$ in question is different from the rest. You should find an explicit formula for the power function $\pi_Z(d)$ for $Z$, whereas $\pi_W(d)$ needs to be simulated.

**(e)** For the testing of equal means situation above, play with your code for other number of groups $m$ than 10. You should find that for $m$ moderate or big, the alternative-conscious $W$ test solidly outperforms the overall-modus $Z$. This type of alternatives, where $H_0$ can be wrong only in terms of a few percent of parameters studied can be significantly different from the rest, or perhaps from a common null value, are called *sparse*.

sparse alternatives

### Neyman–Pearson optimal testing

**Ex. 4.8** *The Neyman–Pearson Lemma: the basics.* Suppose data $y$ come from a density $f$, where there are just two possibilities: either $f = f_0$, which is the null hypothesis to be tested, or $f = f_1$, the alternative. Here there's an optimal strategy, made clear by the so-called Neyman–Pearson Lemma, part of the 49-page landmark paper Neyman and Pearson (1933). For simplicity of presentation we consider the continuous case, where $f_0$

and $f_1$ are densities on the relevant sample space $\mathcal{Y}$ (which can be multidimensional). – A *test function* is a $T\colon \mathcal{Y} \to [0,1]$, with $T(y)$ the probability of rejecting $f_0$ if the the data take on value $y$. This setup even allows the possibility of an element of randomisation, as in 'if $y$ turns out be 3.33 I throw some coins and I reject $H_0$ with probability 0.77'. Once in a blue while this might be of relevance, with discrete data, but in practice such a test function $T(y)$ takes on only values 1, for a rejection set $R$, and 0, for the complementary acceptance set $R^c$.

**(a)** Show that the probability of rejecting the null, if the null is true, can be written $\Pr_{f_0}(\text{reject}) = \int f_0 T \,\mathrm{d}y$.

**(b)** For a given testing level $\alpha$, like 0.01, let $T^*$ be the test which rejects when $f_1(y)/f_0(y) \geq c$, with $c$ tuned such that

$$\Pr_{f_0}(T^* \text{ rejects}) = \int_{y:\ f_1(y)/f_0(y)\geq c} f_0(y)\,\mathrm{d}y = \alpha.$$

Let $T$ be any other test function with the same level $\alpha$. Show that the power difference at $f_1$ can be written

$$\pi_{T^*}(f_1) - \pi_T(f_1) = \int f_1(T^* - T)\,\mathrm{d}y = \int (f_1 - cf_0)(T^* - T)\,\mathrm{d}y.$$

the Neyman–Pearson Lemma

**(c)** Show that among all possible tests, with level $\alpha$, the $T^*$ has the strongest detection power at position $f_1$.

**(d)** (xx a bit more, to cover discrete case; and we may also allow competitors with $\int f_0 T < \alpha$. xx)

**Ex. 4.9** *The Neyman–Pearson Lemma: more details.* Here we tend to some further details, related to the Neyman–Pearson Lemma and its proof, in Ex. 4.8.

**(a)** For two positive densities $f_0$ and $f_1$ defined on the same sample space, as with the Neyman–Pearson Lemma, consider the event $A_c = \{f_1(Y)/f_0(Y) \geq c\}$. Show that the function $p(c) = \Pr_{f_0}(A_c) = \int_{f_1(y)\geq cf_0(y)} f_0(y)\,\mathrm{d}y$ is a continuous and monotone function, starting at 1 and ending at 0, when $c$ travels through $[0,\infty)$. Hence deduce that there for given $\alpha$ really is a unique $c$ in the Neyman–Pearson recipe.

**(b)** Illustrate the $p(c) = \Pr_{f_0}(A_c)$ in a few concrete situations, including (i) $f_0 \sim \mathrm{N}(0,1)$ and $f_1 \sim \mathrm{N}(1,1)$; (ii) $f_0 \sim \mathrm{Gam}(2.2, 3.3)$ and $f_1 \sim \mathrm{Gam}(3.3, 2.2)$. (iii) $f_0 \sim \mathrm{Gam}(3.3, 2.2)$ and $f_1 \sim \mathrm{Gam}(2.2, 3.3)$. For these situations, find the rejection threshold $c$ such that the test rejecting when $f_1(y)/f_0(y) \geq c$ has significance level 0.05.

**(c)** For the three cases above, determine also the power at $f_1$. Note that the power at $f_1$ when testing $f_0$ is typically different from the power at $f_0$ when testing $f_1$.

**Ex. 4.10** *The Neyman–Pearson Lemma: applications.* For the simple two-possibilities setup we learn from the Neyman–Pearson lemma that there is a clear recipe for setting up the optimal test for $f = f_0$ against $f = f_1$. Here are some examples.

**(a)** Suppose $Y \sim \mathrm{N}(\theta, 1)$. Show that the optimal test of level $\alpha = 0.01$, for testing $\theta = 0$ vs. $\theta = 1.234$, is to reject if $Y \geq z_{0.99} = 2.326$, the upper 0.01 point of the standard normal.

**(b)** In this situation, verify that one finds the very same optimal 0.01 level test, for any alternative point $\theta_1 > 0$. Hence the $Y \geq z_{0.99}$ test is *uniformly most powerful*, against all positive alternative.

UMP, uniformly most powerful

**(c)** Generalise this to the case of data $Y_1, \ldots, Y_n$ being i.i.d. from the normal $\mathrm{N}(\theta, \sigma^2)$, with known $\sigma$. Show that the test which rejects $\theta = 0$ against $\theta > 0$ when $Z_n = \sqrt{n}\overline{Y}/\sigma > z_{1-\alpha}$ is uniformly most powerful, among all tets with level $\alpha$; here $z_{1-\alpha} = \Phi^{-1}(1-\alpha)$. Find its power function, and draw it in a diagram, for $\theta_0 = 1.234$, $\sigma = 1$, and for $n = 10, 20, 30$.

**(d)** Let $Y_1, \ldots, Y_n$ be i.i.d. from the $\mathrm{N}(\theta_0, \sigma^2)$ distribution, with $\theta_0$ known. Consider the problem of testing $\sigma = \sigma_0$ against $\sigma > \sigma_0$, at level say 0.01, where $\sigma_0$ is a prescribed null value. Show that the test which rejects when $V_n = \sum_{i=1}^n (Y_i - \theta_0)^2/\sigma_0^2 \geq \gamma_{n,0.99}$, the 0.99 quantile of the $\chi^2_n$, is uniformly optimal. Find its power function.

**(e)** Consider $f_0$, the standard normal, and $f_1(y) = \frac{1}{2}\sqrt{2}\exp(-\sqrt{2}|y|)$; they have both zero mean and unit variance. Find the optimal test for $f_0$ against $f_1$, with level 0.05, and find its detection power at $f_1$. Then do the opposite, constructing the best test at level 0.05 for $f_1$ against $f_0$, and find the power at $f_0$.

**(f)** Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the exponential $\theta\exp(-\theta y)$. Find the strongest 0.10 level test for $\theta = \theta_0$ against an alternative $\theta_1 > \theta_0$. Your test will not depend on the $\theta_1$, as long as it is to the right of $\theta_0$; hence this test is uniformly most powerful against these alternatives.

**Ex. 4.11** *Setting up the test via null simulations.* Consider the simple two-possibilities Neyman–Pearson setup, with data $Y$ from a density $f$, where one tests $f = f_0$ vs. $f = f_1$, for given densities $f_0$ and $f_1$.

**(a)** Writing $y_{\mathrm{obs}}$ for the observed data, the strongest test consists in rejecting when $T(y_{\mathrm{obs}}) = f_1(y_{\mathrm{obs}})/f_0(y_{\mathrm{obs}}) \geq c$, with $c$ finetuned to have level say $\alpha = 0.05$. Explain that to actually find the $c$, in a perhaps complicated situation, one may simulate a high number of $T(y^*) = f_1(y^*)/f_0(y^*)$, with $y^*$ drawn from $f_0$, and then set $c$ equal to the 0.95 quantile in this distribution.

**(b)** Explain that the Neyman–Pearson test may be defined and constructed using any monotone smooth transformation of the density ratio. In the simulation recipe above one may e.g. use simulations of $V(y) = \log f_1(y) - \log f_0(y)$, which might be numerically easier and safer, and then reject if the observed $V(y_{\mathrm{obs}})$ is larger than then 0.95 quantile point among the simulated $V(y^*)$.

**(c)** As an illustration, suppose the data are $Y_1, \ldots, Y_n$, with $n = 12$, and that one needs to test that their density $f$ is standard normal, vs. the alternative that $f$ is a $t_\nu$, with $\nu = 10.0$ degrees of freedom. Show that the strongest test, with level 0.05, is to reject when $V = \sum_{i=1}^n \{\log f_\nu(Y_i) - \log \phi(Y_i)\} \geq c$, with $c = 0.636$.

**(d)** Explain that this null distribution simulation scheme also works in each case where one tests a specified $H_0\colon f = f_0$, also when the alternative hypothesis might be a bigger space than merely a given $f_1$, as long as the test statistic has been decided upon.

**Ex. 4.12** *Density ratios and optimal testing: the normal and the Cauchy.* The Neyman–Pearson recipe is to reject when the density ratio $f_1(y)/f_0(y)$ is sufficiently big. This pans out differently in different situations, as illustrated here.

**(a)** For a single observation $Y$, consider testing $f_0 = \mathrm{N}(0,1)$ against $f_1 = \mathrm{N}(\theta_1, 1)$, with $\theta_1$ positive. Show that

$$\frac{f_1(y)}{f_0(y)} = \frac{\exp\{-\frac{1}{2}(y-\theta_1)^2\}}{\exp(-\frac{1}{2}y^2)} = \exp(\theta_1 y - \tfrac{1}{2}\theta_1^2).$$

Verify that this is a monotone function in $y$, regardless of the value of $\theta_1 > 0$. Argue

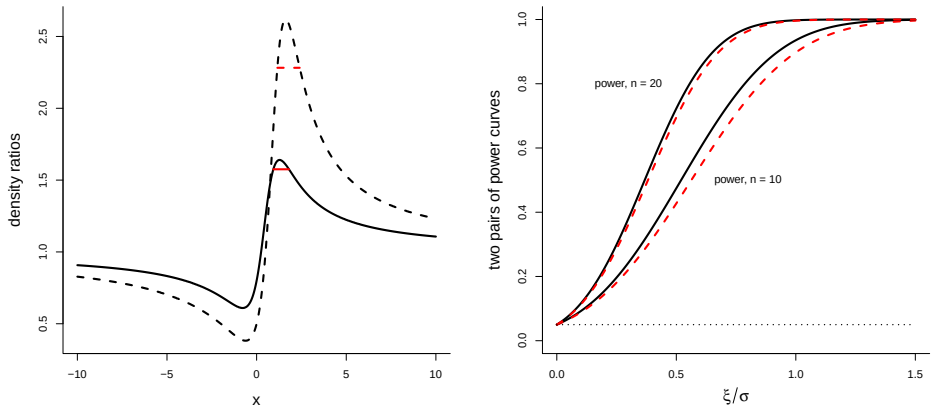



Figure 4.1: *Left panel: for the Cauchy density model, see Ex. 4.12, ratios $f(y,\theta_1)/f(y,0)$ (full line) and $f(y,\theta_2)/f(y,0)$ (slanted) are shown, for alternative values $\theta_1 = 0.50$ and $\theta_2 = 1.00$ to the null. Also indicated are the rejection intervals $[a_1, b_1]$ and $[a_2, b_2]$, for the optimal tests against $\theta_1$ and $\theta_2$. Right panel: two pairs of power. Testing $\xi = 0$ against $\xi > 0$, see Ex. 4.13, for a normal sample of size 10 (lower pair) and 20 (upper pair), at test level 0.05, as a function of $\xi/\sigma$. The lower power is for the t test (slanted, red); the upper power if for the normal based test (full, black), which requires that $\sigma$ is known.*

that 'reject $f_0$ provided $Y$ is big enough' becomes the uniformly optimal test. Exhibit the rejection threshold in $Y \geq c$, if the significance level is to be 0.10.

**(b)** The situation is rather different for the case of the Cauchy density $f(y,\theta) = (1/\pi)\{1+(y-\theta)^2\}^{-1}$. Suppose we wish to test $\theta = 0$ versus a positive $\theta_1$. Show that

$$\frac{f_1(y)}{f_0(y)} = \frac{f(y,\theta_1)}{f(y,0)} = \frac{1+y^2}{1+(y-\theta_1)^2},$$

and draw this function in a diagram, for a few values of $\theta_1$.

**(c)** For a concrete illustration, work through the alternative cases $\theta_1 = 0.50$ and $\theta_2 = 1.00$, for each case finding the rejection interval, say $[a_1, b_1]$ for the first and $[a_2, b_2]$ for the second, to give the optimal test, of level 0.10. (xx answers: $[0.933, 1.804]$ for $\theta_1$; $[1.161, 2.388]$ for $\theta_2$. construct a version of Figure 4.1, left panel. xx) – The point is that these rejection regions are different; the optimal test depends on the specific alternative, and there can be no uniformly optimal test.

**(d)** Again for the sake of concreteness, compute the optimal possible power, for any 0.10 level test, at $\theta_1 = 0.50$ and at $\theta_2 = 1.00$. Compare these powers with that of the simple test $Y \geq 3.078$.

**(e)** (xx there are two drastic differences here, between the simple normal and the non-simple Cauchy. the first is that the log-density-ratio $R(y,\theta_0,\theta_1) = \log f(y,\theta_1) - \log f(y,\theta_0)$ is monotone, for the normal, and not at all monotone for the Cauchy. the second is that of there being a simple one-dimensional sufficient statistic, in the case of $Y_1, \ldots, Y_n$ from the normal, whereas no such statistic exists for the Cauchy. where is sufficiency in kiosk? xx)

**(f)** (xx something about more regularity with $n$ data points; above we just did $n = 1$. xx)

**Ex. 4.13** *The t test and its power.* Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the $\mathrm{N}(\xi, \sigma^2)$, with testing of $\xi = 0$ required against $\xi > 0$. This is simple and standard, in the case of $\sigma$ being known, but needs the t test in the case of $\sigma$ being unknown and estimated from the data, as we have seen in Ex. 4.4, Setting up the test uses the relevant t distribution, from $t = \sqrt{n}\bar{Y}/\widehat{\sigma} \sim t_{n-1}$, but for studying the power function also the noncentral t distribution is required.

**(a)** Consider first the case of $\sigma$ being known. Show that $z = \sqrt{n}\bar{Y}/\sigma$ is standard normal under the null, and that the 0.05 level test becomes that of rejecting when $z \geq z_0 = \Phi^{-1}(0.95) = 1.645$.

**(b)** Show that at a given $\xi > 0$, we have $z \sim \mathrm{N}(\sqrt{n}\xi/\sigma, 1)$, and that this leads to the power function $\pi_{n,N}(\xi/\sigma) = \Phi(\sqrt{n}\xi/\sigma - z_0)$. Compute and display this power function for the case of $n = 10$ and $n = 20$, as with the black full curves of Figure 4.1 (right panel).

**(c)** Then consider the more complex situation where $\sigma$ is not known, needing the empirical standard deviation $\widehat{\sigma}$ of (1.6). We have seen in Ex. 1.51 that

$$t = \frac{\sqrt{n}\bar{Y}}{\widehat{\sigma}} = \frac{\sqrt{n}\bar{Y}/\sigma}{\widehat{\sigma}/\sigma} \sim \frac{\mathrm{N}(0,1)}{(\chi^2_{\mathrm{df}}/\mathrm{df})^{1/2}},$$

with nominator and denominator being independent, and where the degrees of freedom is $\mathrm{df} = n - 1$. The probability density $g_{\mathrm{df}}(x)$ and cumulative distribution $G_{\mathrm{df}}(x)$ of this $t_{\mathrm{df}}$ distribution are moderately complicated, see the exercise mentioned, but that does not concern us much, as long as we can consult a table or run an algorithm to find associated quantiles and probabilities. Show hence that the t test, with level 0.05, must consist of rejecting when $t \geq t_0 = G^{-1}_{\mathrm{df}}(0.95)$. Using `qt(0.95,df)` in `R`, we find 1.833 and 1.729 for $n = 10$ and $n = 20$. Check that `qt(0.95,df)` becomes close to 1.645 as df increases, and explain why.

**(d)** For the power of the t test, show that $\pi_{n,t}(\xi/\sigma) = \Pr_{\xi/\sigma}(t \geq t_0)$, where

$$t = \frac{\sqrt{n}\bar{Y}}{\widehat{\sigma}} \sim \frac{\mathrm{N}(\sqrt{n}\xi/\sigma, 1)}{(\chi^2_{\mathrm{df}}/\mathrm{df})^{1/2}},$$

again with nominator and denominator independent. Explain that the power function therefore can be written

$$\pi_{n,t}(\xi/\sigma) = \Pr_{\xi/\sigma}(t \geq t_{0.m}) = 1 - G_m(t_{0,m}, \sqrt{n}\xi/\sigma),$$

with $G_m(x, \lambda)$ denoting the c.d.f. for this noncentral $t$ with degrees of freedom $m$ and noncentrality parameter $\lambda$; see Ex. 1.52. This function is complicated, but can easily be found numerically via e.g. `pt(x,m,lambda)` in `R`. Construct a version of Figure 4.1, right panel, perhaps with other sample sizes than 10, 20. Comment on your findings.

**(e)** Describe how the two-sided tests and power functions pan out here, whem $\xi = 0$ is to be tested against $\xi \neq 0$. Make a corresponding version of Figure 4.1, right panel, with the relevant two-sided power functions.

**Ex. 4.14** *Establishing that a parameter is close to zero.* Suppose $\delta$ is some effect parameter, where the scientific context relates to establishing that it is close to zero. In

such situations it might be natural to test $H_0 : |\delta| \geq \varepsilon$, for some known small threshold $\varepsilon$, against the alternative that $\delta \in (-\varepsilon, \varepsilon)$. In some applications the $\delta$ might be the difference between two effect parameters, aiming to infer that these are close. This is turning the tables, somehow, compared to the more traditional setups, where the hypothesis to be tested is that a parameter is close to zero, against the alternative that it is some distance away.

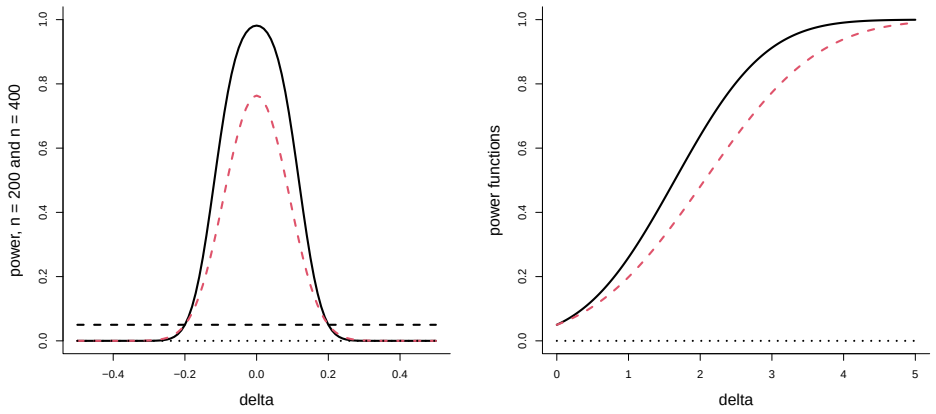



Figure 4.2: *Left panel: power functions for the bioequivalence tests of Ex. 4.14, with* $n = 200$ *(slanted curve) and* $n = 400$ *(full curve), and threshold* $\varepsilon = 0.20$. *Right panel: limiting local power functions for the two tests of Ex. 4.15, for* $\theta \leq \theta_0$ *against* $\theta > \theta_0$, *in terms of the local alternatives parameter* $\delta$, *with* $\theta_n = \theta_0 + \delta/\sqrt{n}$. *The more powerful test is based on* $\bar{Y}$, *the other on the median* $M_n$.

**(a)** To study the main features of such a situation, consider a prototype setup with i.i.d. observations $Y_1, \ldots, Y_n$ being $\mathrm{N}(\delta, 1)$. Show that $\sqrt{n}\bar{Y} \sim \mathrm{N}(\sqrt{n}\delta, 1)$, which implies $Z_n = n\bar{Y}^2 \sim \chi^2_1(n\delta^2)$.

**(b)** Explain that it is natural to reject the null, and hence claim that $|\theta|$ is small, if $|\sqrt{n}\bar{Y}| \leq c$, or equivalently $Z_n \leq c^2$, for $c$ calibrated to reach testing level e.g. $\alpha = 0.05$. This is equivalent to $Z_n \leq c^2$. Show that $\Pr_\delta(|\sqrt{n}\bar{Y}| \leq c) = \Gamma_1(c^2, n\delta^2)$, in terms of the c.d.f. for the noncentral $\chi^2_1$.

**(c)** Explain that to have significance level 0.05, we need $\Gamma_1(c^2, n\varepsilon^2) = 0.05$. Find the full power curve $\pi_n(\delta) = \Pr_\delta(Z_n \leq c^2)$, and comment on the level of its maximum. Construct a version of Figure 4.2, left panel, for threshold $\varepsilon = 0.20$, for sample sizes $n = 200$ and $n = 400$.

**Ex. 4.15** *Power and local power: a particular case.* This exercise studies a prototype situation in some detail; the type of calculations and results will be seen to be rather similar in a long range of different situations. Let $Y_1, \ldots, Y_n$ be i.i.d. data from $\mathrm{N}(\theta, \sigma^2)$. One wishes to test $H_0 : \theta = \theta_0$ vs. the alternative that $\theta > \theta_0$, where $\theta_0$ is a known value (e.g. 3.14). Two tests will be considered, based on respectively $\bar{Y}_n = n^{-1}\sum_{i=1}^n Y_i$ and the median $M_n$.

**(a)** For a given value of $\theta$, prove that $\sqrt{n}(\bar{Y}_n - \theta) \to_d \mathrm{N}(0, \sigma^2)$ and $\sqrt{n}(M_n - \theta) \to_d \mathrm{N}(0, (\pi/2)\sigma^2)$. Note that the first result is immediate and actually holds with exactness

for each $n$; the second result requires more care and follows from Ex. 3.20.

**(b)** Working under the null hypothesis $\theta = \theta_0$, show that

$$Z_n = \frac{\sqrt{n}(\bar{Y}_n - \theta_0)}{\widehat{\sigma}} \to_d \mathrm{N}(0,1), \quad Z_n^* = \frac{\sqrt{n}(M_n - \theta_0)}{(\pi/2)^{1/2}\widehat{\sigma}} \to_d \mathrm{N}(0,1),$$

where $\widehat{\sigma}$ is any consistent estimator of $\sigma$.

**(c)** With $z_{0.95} = \Phi^{-1}(0.95) = 1.645$, conclude from the above that the two tests that reject $H_0$ provided respectively

$$\bar{X}_n > \theta_0 + z_{0.95}\widehat{\sigma}/\sqrt{n} \quad \text{and} \quad M_n > \theta_0 + z_{0.95}(\pi/2)^{1/2}\widehat{\sigma}/\sqrt{n},$$

have the required asymptotic significance level 0.05; $\alpha_n = \Pr\{\text{reject } H_0 \,|\, \theta = \theta_0\} \to 0.05$. (There is one such $\alpha_n$ for the first test, and one for the other; both converge however to 0.05.)

**(d)** Our object is then to study the *local power*, the chance of rejecting the null hypothesis under alternatives of the type $\theta_n = \theta_0 + \delta/\sqrt{n}$, i.e. close to the null value. In generalisation of (b), show that

$$Z_n = \frac{\sqrt{n}(\bar{Y}_n - \theta_0)}{\widehat{\sigma}} \to_d \mathrm{N}(\delta/\sigma, 1), \quad Z_n^* = \frac{\sqrt{n}(M_n - \theta_0)}{(\pi/2)^{1/2}\widehat{\sigma}} \to_d \mathrm{N}((\pi/2)^{1/2}\delta/\sigma, 1),$$

where the convergence in question takes place under the indicated $\theta_0 + \delta/\sqrt{n}$ parameter values. Here we need to generalise the result of Ex. 3.20, to the case where $\theta_n$ moves with $n$. Writing $F_n$ for the distribution of $Y_i$, you may do this by first showing that $M_n$ has the same distribution as $F_n^{-1}(M_n^0) = \theta_0 + \delta/\sqrt{n} + \sigma\Phi^{-1}(M_n^0)$, where $M_n^0$ is the median for an i.i.d. sample from the uniform, and then using the delta method.

**(e)** Use these results to show that

$$\begin{aligned}
\pi_n(\delta) &= \Pr\{\text{reject with } \bar{Y}_n \text{ test} \,|\, \theta_0 + \delta/\sqrt{n}\} \to \Phi(\delta/\sigma - z_{0.95}),\\
\pi_n^*(\delta) &= \Pr\{\text{reject with } M_n \text{ test} \,|\, \theta_0 + \delta/\sqrt{n}\} \to \Phi((2/\pi)^{1/2}\delta/\sigma - z_{0.95}),
\end{aligned}$$

for the two power functions. Draw these in a diagram, and compare, as with Figure 4.2, right panel.

**(f)** Assume one wishes $n$ to be large enough to secure that the power function is at least at level $\beta$ for a certain alternative point $\theta_1$. Using the local power approximation, show that the required sample sizes are respectively

$$n_A \doteq \frac{\sigma^2}{(\theta_1 - \theta_0)^2}(z_{1-\alpha} + z_\beta)^2 \quad \text{and} \quad n_B \doteq \frac{\sigma^2/c^2}{(\theta_1 - \theta_0)^2}(z_{1-\alpha} + z_\beta)^2$$

for tests A (based on the mean) and B (based on the median), with $c = \sqrt{2/\pi}$. With level $\alpha = 0.05$, compute these sample sizes for the case of $\beta = 0.95$ and $\theta_1 = \theta_0 + \frac{1}{2}\sigma$.

**(g)** One sometimes defines the ARE, the asymptotic relative efficiency of test B with respect to test A, as

$$\mathrm{ARE} = \lim n_A(\theta_1, \beta)/n_B(\theta_1, \beta),$$

the limit in question in the sense of alternatives $\theta_1$ coming closer to the null hypothesis at speed $1/\sqrt{n}$. Show that the ARE in this particular situation becomes $c^2 = 2/\pi = 0.6366$; test A needs only ca. 64 percent as many data points to reach the same detection power as B needs.

**Ex. 4.16** *Power and local power: general results.* In Ex. 4.15 we examined two particular estimators and tests, for the mean parameter in the normal distribution, and found precise limit distributions and local power functions, for alternatives of order $1/\sqrt{n}$ from the null hypothesis. Here we go through the appropriate generalities. The setup is that of data, of sample size $n$, informing us about a key parameter $\theta$, where the null hypothesis $H_0\colon \theta = \theta_0$ is to be tested against the alternative $\theta > \theta_0$, with $\theta_0$ some specified value. We assume the data leads to one or more estimators $\widehat{\theta}$, with the basic premise that at the null parameter $\theta_0$, we have $A_n = \sqrt{n}(\widehat{\theta} - \theta_0) \to_d \mathrm{N}(0, \kappa^2)$, for the appropriate $\kappa$.

**(a)** With $\widehat{\kappa}$ an estimator for $\kappa$, under $H_0$ conditions, show that $Z_n = \sqrt{n}(\widehat{\theta} - \theta_0)/\widehat{\kappa}_0 \to_d \mathrm{N}(0,1)$ under $H_0$. Explain that the test which rejects when $Z_n \geq z_{1-\alpha} = \Phi^{-1}(1-\alpha)$ has asymptotic testing level $\alpha$, i.e. $\alpha_n = \Pr_0(Z_n \geq z_{1-\alpha}) \to \alpha$.

**(b)** We now make two more assumptions, concerning natural smoothness in a $O(1/\sqrt{n})$ size local neighbourhood around $\theta_0$: (i) if the true parameter behind the data is $\theta_n = \theta_0 + \delta/\sqrt{n}$, then the same limit $\mathrm{N}(0, \kappa^2)$ obtains for $A_n' = \sqrt{n}\{\widehat{\theta} - (\theta_0 + \delta/\sqrt{n})\}$; (ii) the estimator $\widehat{\kappa}$ still converges to the same $\kappa$ in probability. Show that $Z_n \to_d \mathrm{N}(\delta/\kappa, 1)$, and that this gives a limiting local power function,

$$\pi_n = \Pr(\text{reject} \,|\, \theta_0 + \delta/\sqrt{n}) \to \pi(\delta) = \Phi(\delta/\kappa - z_{1-\alpha}).$$

Show that this limit power function has derivative $\phi(z_{1-\alpha})/\kappa$ at zero.

**(c)** The above result is useful in its own right, providing an easy approximation to the power function for alternatives not far from the null hypothesis value. It may also be used for comparing different tests, built as above, for different estimators, say $\widehat{\theta}_j$. Assume the conditions above hold for these, with $\sqrt{n}(\widehat{\theta}_j - \theta_0) \to_d \mathrm{N}(0, \kappa_j^2)$, leading to estimators $\widehat{\kappa}_j$ for $\kappa_j$ and to test statistics $Z_{n,j} = \sqrt{n}(\widehat{\theta}_j - \theta_0)/\widehat{\kappa}_j$. Explain that the best estimators lead to the best tests, with derivatives $\phi(z_{1-\alpha})/\kappa_j$ at zero.

**(d)** (xx briefly about ARE, as above. xx)

**Ex. 4.17** *Optimal average power.* Supppose $Y$ is observed, perhaps a full vector, from a density $f$, where one wishes to test the null hypothesis $f = f_0$, a given density. As we saw in Ex. 4.12, there are cases where there is no uniformly optimal test, against all or a subset of alternatives; the optimal test at $f_1$ might be different from the one at $f_2$. In one-parameter models, this is caused by the log-density-ratio not being monotone.

**(a)** Consider alternative densities $f_1, \ldots, f_m$, given nonnegative weights of importance $w_1, \ldots, w_m$. These may be taken to have sum 1. The *weighted average power* of a test $T$, at these points and with these weights, is

$$\bar{\pi}_T = \sum_{j=1}^m w_j \pi_T(f_j) = \sum_{j=1}^m w_j \int f_j(y) T(y)\,\mathrm{d}y,$$

with $\pi_T(f_j)$ the power at $f_j$. Let $\bar{f}(y) = \sum_{j=1}^m w_j f_j$. Show that this average power is maximised, among all test functions $T(y)$ with level $\alpha$ at the null, by the $T^*(y)$ which rejects $H_0$ when $\bar{f}(y)/f_0(y) \geq c$, with $c$ tuned to give rejection probability $\alpha$ at the null.

**(b)** For a one-parameter model $f(y,\theta)$, consider testing of $\theta = \theta_0$ against $\theta > \theta_0$. For any test function $T(y)$ with rejection level $\alpha$ at the null, consider in general terms the weighted average power

$$\bar{\pi}_T = \int_{\theta > \theta_0} \pi_T(\theta)\,\mathrm{d}w(\theta),$$

with $\pi_T(\theta) = \int f(y,\theta)T(y)\,\mathrm{d}y$ the power of the test at position $\theta$. Show that $\bar\pi_T = \int_{\theta>\theta_0} \bar f(y)T(y)\,\mathrm{d}y$, featuring the density $\bar f(y) = \int_{\theta>\theta_0} f(y,\theta)\,\mathrm{d}w(\theta)$. This is the model density averaged over all alternatives to the null, as weighted by the $\mathrm{d}w(\theta)$ measure.

**(c)** Show that the test maximising this weighted average power is rejecting the null if $\bar f(y)/f(y,\theta_0) \geq c$, with $c$ tuned to have null level $\alpha$.

**(d)** Suppose $\theta$ is one-dimensional and that the log-density $\log f(y,\theta)$ is monotone in $\theta$. Explain that there then is a de facto uniformly optimal test for $\theta = \theta_0$ vs. $\theta > \theta_0$, and that the average power optimal test worked with here will be identical to this one, as long as the weight function is positive only for $\theta > 0$.

**(e)** (xx a little more. the marginal density, or predictive density, with a link to Bayes, but specifically with a 'prior' over the alternative space. can take Cauchy with $a\exp(-a\theta)$ over the halfline. xx)

**(f)** Consider the $\mathrm{N}(\xi,\sigma^2)$ model, and assume there is a need to test $f = \mathrm{N}(0,1)$ against the alternative that $\xi > 0$, or $\sigma > 1$, or both. show first that the NP test against alternatives $(1.1, 2.2)$ and $(2.2, 3.3)$ are indeed different, so there is no uniformly most powerful test. Then if a weight function $w(\xi,\sigma)$ is specified, in the domain $\xi > 0, \sigma > 1$, show that the average power optimal test consists in rejecting if

$$V(y_1,\ldots,y_n) = \frac{\bar f(y_1,\ldots,y_n)}{f_0(y_1)\cdots f_0(y_n)} = \int_{\xi>0,\sigma>1} \Big\{\prod_{i=1}^n \frac{f(y_i,\xi,\sigma)}{f(y_i,0,1)}\Big\}\,\mathrm{d}w(\xi,\sigma)$$

is larger than the treshold $c$, with this finetuned to be the upper quantile of its null distributon. As an illustration, suppose $n = 10$ and that the weight function is flat on $[0,4]\times[1,6]$. Simulate from $W(y_1,\ldots,y_n) = \log V(y_1,\ldots,y_n)$ under the null hypothesis, as per Ex. 4.11, to find the rejection threshold, to have significance level 0.01.

**(g)** (xx connect this to Wilks things in Ch5, which will work as a good approximation when testing an inner point of the parameter region. xx)

### Sufficiency, factorisation theorem, completeness

**Ex. 4.18** *Sufficiency.* Flip a coin ten times and record the number of heads. It is intuitively clear that the number of heads in the ten tosses is as informative about the unknown $\theta = \Pr(\text{heads})$ of the coin, as also knowing the exact ordering in which the heads and tails occurred. In fact, knowing the ordering in which the heads and tails occurred appears redundant for making inference about $\theta$. This leads us to the notion of a sufficient statistic. Let $X$ be data with a distribution from a family $\mathcal{P} = \{P_\theta\colon \theta\in\Theta\}$ of distributions on the measurable space $(\mathcal{X},\mathcal{B})$. In colloquial terms, the statistic $T = T(X)$ is sufficient for $\theta$ if for every $t$ and $\theta$, the conditional distribution of $X$ given $T = t$ is the same for all values of $\theta$. More formally, let $T\colon \mathcal{X}\to\mathcal{Y}$ be a statistic, then $T$ is sufficient for $\mathcal{P}$ (or for $\theta$) if there is a function $Q_t(B)$ of $B\in\mathcal{B}$ and $t\in\mathcal{Y}$, such that for each $B$ and $\theta\in\Theta$, we have $Q_t(B) = P_\theta(B\,|\,T=t)$ for $P_\theta T^{-1}$ almost all $t$, that is

sufficient statistic

$$\int_A Q_t(B)\,\mathrm{d}P_\theta T^{-1}(t) = \int_A P_\theta(B\,|\,T=t)\,\mathrm{d}P_\theta T^{-1}(t) = P_\theta(B\cap T^{-1}(A)),$$

for each $\theta$ and for all $A$ in the Borel-$\sigma$-algebra $\mathcal{B}(\mathcal{Y})$ on $\mathcal{Y}$.

**(a)** Let $X_1,\ldots,X_n$ be i.i.d. Bernoulli$(\theta)$, and set $T = \sum_{i=1}^n X_i$. Show that for each $t$, $\Pr_\theta(X_1 = x_1,\ldots,X_n = x_n\,|\,T=t) = 1/\binom{n}{t}$ on $\{(x_1,\ldots,x_n)\colon x_i = 0,1, \sum_{i=1}^n x_i = t\}$, and conclude that $T$ is sufficient for $\theta$.

**(b)** Let $Y_1, \ldots, Y_n$ be i.i.d. Pois($\theta$) and set $T = \sum_{i=1}^n Y_i$. Show that for each $t$

$$\Pr_\theta(Y_1 = y_1, \ldots, Y_n = y_n \,|\, T = t) = \frac{t!}{n^t} \prod_{i=1}^n \frac{1}{y_i!},$$

on $\{(y_1, \ldots, y_n)\colon y_i = 0, 1, 2, \ldots, \sum_{i=1}^n y_i = t\}$, and conclude that $T$ is sufficient for $\theta$.

**(c)** Let $X_1, \ldots, X_n$ be i.i.d. uniform on $(0, \theta)$ and set $T = \max_{i \le n} X_i$. Show that the conditional distribution of $X_1, \ldots, X_n$ given $T = t$ has density

$$f_\theta(x_1, \ldots, x_n \,|\, t) = \frac{1}{t^n},$$

on $\{(x_1, \ldots, x_n)\colon 0 \le x_i \le t \text{ for all } i\}$, and conclude that $T$ is sufficient for $\theta$.

**(d)** Suppose that $Y \sim \mathrm{N}(0, \sigma^2)$ and let $Z = \pm 1$ with equal probability be independent of $Y$. Show that $Y' = Z|Y|$ has the same distribution as $Y$. Consider the function

$$Q_t(B) = \tfrac{1}{2} I_B(t) + \tfrac{1}{2} I_B(-t), \quad B \in \mathcal{B},\, t \ge 0,$$

and show, using that $Y' \sim Y$, that $|Y|$ is sufficent for $\sigma^2$.

**(e)** Let $X$ and $Y$ be independent Expo($\theta$), and set $T = X + Y$. Let $U \sim \mathrm{unif}(0, 1)$ be independent of $X$ and $Y$, and define $X' = UT$ and $Y' = (1 - U)T$. Show that $(X, Y) \sim (X', Y')$. From this, show that $T$ is sufficient for $\theta$.

**(f)** Let $X$ be a random vector with a distribution from $\mathcal{P} = \{P_\theta\colon \theta \in \Theta\}$. Assume that $T$ is sufficient for $\theta$, and let $Q_t(B)$ be the function from the definition of sufficiency. Since $Q_t(B)$ is a version of $P_\theta(B \,|\, T = t)$, and $X$ is a random vector, there exists a version of $Q_t$ under $P_\theta T^{-1}$, $Q_t^\star$, such that $B \mapsto Q_t^\star(B)$ is a probability measure for all $t$ [xx (see Notes and Pointers in the appendix) xx]. Given $T = t$, let $X'$ be a random vector drawn from $Q_t^\star$. Show that $X' \sim X$.

**Ex. 4.19** *The factorisation theorem.* In Ex. 4.18 we found sufficient statistics through qualified guesswork. The factorisation theorem provides a recipe for finding sufficient statistics through (minor) manipulations of the density of a sample. Suppose that the family $\mathcal{P} = \{P_\theta\colon \theta \in \Theta\}$ of distributions on $(\mathcal{X}, \mathcal{B})$ is dominated by a $\sigma$-finite measure $\mu$. For each $\theta$, let $f_\theta$ be the density of $P_\theta$ with respect to $\mu$. The factorisation theorem says that the statistic $T\colon \mathcal{X} \to \mathcal{Y}$ is sufficient for $\mathcal{P}$ if and only if there exist nonnegative functions $h$ and $g_\theta$ such that

$$f_\theta(x) = g_\theta(T(x))h(x),$$

for all $\theta \in \Theta$. We prove this theorem step by step.

**(a)** Assume that $\mathcal{P}$ is dominated by a probability measure $P$, and that for each $\theta \in \Theta$, the density $f_\theta = \mathrm{d}P_\theta/\mathrm{d}P$ is such that $f_\theta(x) = g_\theta(T(x))$, i.e., $h(x) = 1$ for all $x \in \mathcal{X}$. Show that $T$ is sufficient for $\mathcal{P}$, and indeed also for the slightly larger family $\mathcal{P} \cup \{P\}$.

**(b)** For the converse of (a), assume that $\mathcal{P}$ is dominated by a probability measure $P$ and that $T$ is sufficient for $\mathcal{P} \cup \{P\}$. Let $f_\theta$ be a density of $P_\theta$ with respect to $P$, and show that $g_\theta(T(\cdot)) = \mathrm{E}_P(f_\theta \,|\, T)(\cdot)$ is also a density of $P_\theta$ with respect to $P$, i.e., that $\int_B g_\theta(T(x))\,\mathrm{d}P(x) = P_\theta(B)$ for all $B \in \mathcal{B}$. –Taken together, (a) and (b) prove the factorisation theorem in the case that $\mathcal{P}$ is dominated by (at least) one of its members, which is not wholly unusual, consider the family $\mathcal{P} = \{\mathrm{N}(\theta, 1)\colon \theta \in \mathbb{R}\}$, for example.

**(c)** To prove the factorisation theorem in full generality, we use the result from Ex. [xx abc xx] in the appendix, which says that if $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ is dominated by a $\sigma$-finite measure $\mu$, then there is a probability measure of the form $P = \sum_{j=1}^{\infty} a_j P_{\theta_j}$ equivalent to $\mathcal{P}$, i.e., $P_\theta(A) = 0$ for all $\theta$ if and only if $P(A) = 0$. Now, assume that for each $\theta \in \Theta$, $P_\theta$ has density $g_\theta(T(x))h(x)$ with respect to $\mu$, and show that $T$ is sufficient.

**(d)** Conversely, assume that $\mathcal{P}$ is dominated by $\mu$ and that $T$ is sufficient. Show that $P_\theta$ has density $g_\theta(T(x))h(x)$ with respect to $\mu$.

**(e)** Try to make sense of the colloquial definition of sufficiency given in Ex. 4.18, and show that it says the same thing as the formal definition of sufficiency.

**Ex. 4.20** *Some more sufficiency.* In this exercise we employ the factorisation theorem on some concrete examples, and touch on some important facts about sufficiency.

**(a)** Use the factorisation theorem to find sufficient statistics for i.i.d. samples from, respectively, the Bernoulli($\theta$) distribution; the N($\mu, 1$) distribution; the N($\mu, \sigma^2$) distribution with both parameters unknown; the unif$(0, \theta)$ distribution; and, lastly, the unif$(\theta, \theta + 1)$ distribution. What is particular about this latter?

**(b)** Let $X, Y$ be i.i.d. from $\mathcal{P} = \{\text{unif}(0, \theta) : \theta \geq 1\} \cup \{\text{unif}\{1, 2, \ldots, \theta\} : \theta \in \mathbb{N}\}$. Show that $T = \max(X, Y)$ is sufficient for $\mathcal{P}$. Find also an expression for the function $Q_t(B)$ from the definition of sufficiency (see Ex. 4.18), and show that this function may be chosen so that $B \mapsto Q_t(B)$ is not a probability measure for all $t$.

**(c)** A sufficient statistic is not unique, and different sufficient statistics may provide varying degress of data compression. At one extreme are sufficient statistics not providing any compression of the data. Show that the full sample is sufficient.

**(d)** Let $X_1, \ldots, X_n$ be i.i.d. from an unknown distribution with c.d.f. belonging to a family of continuous c.d.f.s. Let $T = (X_{(1)}, \ldots, X_{(n)})$ be the order statistics, and show that it is sufficient.

**(e)** Show that any one-to-one transformation of a sufficient statistic is sufficient.

**(f)** Let $X_1, \ldots, X_n$ be i.i.d. from a member of $\mathcal{P} = \{\text{N}(0, \sigma^2) : \sigma > 0\}$ and consider the statistics $T_1 = (X_1, \ldots, X_n)$, $T_2 = (X_1^2, \ldots, X_n^2)$, $T_3 = (X_1^2 + \cdots + X_k^2, X_{k+1}^2 + \cdots + X_n^2)$, and $T_4 = X_1^2 + \cdots + X_n^2$. Here, the data compression is increasing in the indices. Use the factorisation theorem to prove that all these statistics are all sufficient.

**Ex. 4.21** *Sufficiency in exponential families* (xx need intro some technical but important things for the expo family. calibrate how and when we do sufficiency. xx)

**(a)** Suppose that $Y$ stem from a member of the family $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, where each $P_\theta$ has density of the exponential family form $f_\theta(y) = \exp(\theta T(y) - k(\theta))h(y)$ with respect to some dominating measure $\mu$. Show that $T = T(Y)$ is sufficient for the natural parameter $\theta$; and that the distribution of $T$ has a density of the exponential family form, say $g_\theta(t) = \exp(\theta t - k(\theta))$, with respect to $\nu T^{-1}$, where $\mathrm{d}\nu(y) = h(y)\,\mathrm{d}\mu(y)$.

**(b)** Let $Y \sim \text{N}(0, \sigma^2)$. Write this normal density on exponential family form, and check the formula from (a) for the density $g_\theta(t)$ using the transformation methods of Ex. 1.12.

**(c)** Generalise (a) to the case where $Y_1, \ldots, Y_n$ are i.i.d. from a distribution in $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$, where each $P_\theta$ has density $f_\theta(y) = \exp\{\theta^{\mathrm{t}} T(y) - k(\theta)\}h(y)$ with respect to some dominating measure $\mu$. Show that $\bar{T} = (\bar{T}_1, \ldots, \bar{T}_p)^{\mathrm{t}}$, with $\bar{T}_j = (1/n)\sum_{i=1}^n T_j(Y_i)$ for $j = 1, \ldots, p$, is sufficient for the natural parameter $\theta = (\theta_1, \ldots, \theta_p)$; and that $\bar{T}$ has a distribution with density following the same exponential form with respect to an appropriate dominating measure.

**(d)** Here is a lemma that is useful for the following exercises. Let $\Pr_0$ and $\Pr_1$ be probability measures on the same measurable space $(\Omega, \mathcal{F})$, and let $X$ and $Y$ be random vectors with values in the measurable spaces $(\mathcal{X}, \mathcal{A})$ and $(\mathcal{Y}, \mathcal{B})$. Suppose that there are functions $a\colon \mathcal{X} \to (0, \infty)$ and $b\colon \mathcal{Y} \to [0, \infty)$ such that $\Pr_1$ has density $a(X)b(Y)$ with respect to $\Pr_0$. Show that the marginal distribution of $Y$ under $\Pr_1$ is

$$\Pr_1 Y^{-1}(B) = \int_B b(y) \mathrm{E}_0 \{a(X) \,|\, Y = y\} \, \mathrm{d}\Pr_0 Y^{-1}(y), \quad B \in \mathcal{B},$$

and that for each $A \in \mathcal{A}$,

$$\Pr_1(X \in A \,|\, Y = y) = \frac{\mathrm{E}_0\{I_A(X)a(X) \,|\, Y = y\}}{\mathrm{E}_0\{a(X) \,|\, Y = y\}}, \quad \Pr_1 Y^{-1} \text{ a.e.},$$

that is, under $\Pr_1 Y^{-1}$ the right hand side is a version of the conditional distribution of $X$ given $Y = y$.

**(e)** Let $\{P_{a,b}\colon (a, b) \in \Theta = \Theta_a \times \Theta_b\}$ be a family of distributions having densities of the exponential family form $f_{a,b}(y) = \exp\{a^{\mathrm{t}} U(y) + b^{\mathrm{t}} V(y) - k(a, b)\}h(y)$ with respect to $\mu$. Use (d) to show that the marginal distribution of $V = V(Y)$ has density

$$\exp(b^{\mathrm{t}} v - k(a, b)),$$

with respect to the measure

$$\mathrm{d}\lambda_a(v) = \exp(-b_0^{\mathrm{t}} v + k(a_0, b_0)) \mathrm{E}_{a_0,b_0}\{\exp((a - a_0)^{\mathrm{t}} U) \,|\, V = v\} \, \mathrm{d}P_{a_0,b_0} V^{-1}(v),$$

where $(a_0, b_0)$ is some fixed point in $\Theta$. The important thing here is that $V$ has a density belonging to an exponential family, not the exact form of the measure with respect to which this is a density.

**(f)** Use (d) to show that the conditional distribution of $U$ given $V = v$ has density,

$$\exp(a^{\mathrm{t}} u - k_v(a)), \quad \text{with } k_v(a) = \log \mathrm{E}_{a_0,b_0}\{\exp((a - a_0)^{\mathrm{t}} U) \,|\, V = v\},$$

with respect to the measure $\mathrm{d}\nu_v(u) = \exp(-a_0^{\mathrm{t}} u) \, \mathrm{d}P_{a_0,b_0}(U^{-1}(u) \,|\, V = v)$. In particular, the conditional density of $U$ given $V = v$ does not depend on $b$.

conditional sufficiency

**(g)** Show that, conditionally on $V = v$, $U$ is sufficient for the family $\{P_a^{U \,|\, V=v} \,|\, a \in \Theta_a\}$. where $P_a^{U \,|\, V=v}$ is the conditonal distribution of $U$ given $V = v$.

**(h)** Let $X \sim \mathrm{Expo}(b)$ and $U \sim \mathrm{Expo}(b + a)$ be independent, and define $V = X + Z$. By direct methods, for example the convolution formula in (1.1), find the density of the distribution of $V$, and show that the distribution of $U$ given $V = v$ has density

$$f_a(u \,|\, v) = \frac{a \exp(-au)}{1 - \exp(-av)}, \quad 0 \le u \le v.$$

Verify that the expression from (e) and (f) gives these same densities (with respect to Lebesgue measure), as they should.

**Ex. 4.22** *Minimal sufficiency.* In Ex. 4.20 we saw that for any model there are many different sufficient statistics, often with some providing more compression of the data than others. Since the purpose of a sufficient statistic is to compress the data, this naturally leads to a search for a sufficient statistic providing the maximum amount of data compression.If such a statistic exits, it is said to be minimal sufficient. Here is the

minimal sufficient

definition: Let $T$ be sufficient for $\mathcal{P} = \{P_\theta \colon \theta \in \Theta\}$. Then $T$ is minimal sufficient if for any other sufficient statistic $S$, there is a measurable function $g$ so that $P_\theta\{T = g(S)\} = 1$ for all $\theta \in \Theta$. Another way of saying this is that if $T$ is such $S(x) = S(y)$ implies $T(x) = T(y)$ for any sufficient statistic $S$, then $T$ is minimal sufficient.

**(a)** Show that any one-to-one function of a minimal sufficient statistic is a minimal sufficient statistic.

**(b)** Suppose that $T$ is minimal sufficient, and let $S$ be some sufficient statistic. Show that the $\sigma$-algebra generated by $T$ must be contained in the $\sigma$-algebra generated by $S$.

**(c)** Show that if $T(x) = T(y)$ if and only if

$$\theta \mapsto \frac{f_\theta(x)}{f_\theta(y)} \text{ is constant in } \theta,$$

then $T$ is minimal sufficient. Here we assume, for simplicity, that $f_\theta(y) > 0$ for all $y$. If this is not the case, then instead consider the case where $T(x) = T(y)$ if and only if $y \in \{y \colon f_\theta(y) = f_\theta(x)h(x,y)$ for all $\theta$ and some $h > 0\}$.

**(d)** With $X \sim \mathrm{N}(0, \sigma^2)$, show that the absolute value $|X|$ is minimal sufficient. Let $Y_1, \ldots, Y_n$ be i.i.d. $\mathrm{N}(\mu, \sigma^2)$, and show that $(\bar{Y}_n, S_n)$ is minimal sufficient, where $\bar{Y}_n = n^{-1}\sum_{i=1}^n X_i$ and $S_n = \sum_{i=1}^n (Y_i - \bar{Y})^2$. Let $Z_1, \ldots, Z_n$ be i.i.d. from a distribution with density $f_\theta(z) = \exp(-(z-\theta))$ for $z > \theta$ and $\theta \in \mathbb{R}$. Find a minimal sufficient statistic for $\theta$.

**(e)** Let $g(x)$ be a positive and integrable function on $\mathbb{R}$, set $c(a,b)^{-1} = \int_a^b g(x)\,\mathrm{d}x$, and define $f_{a,b}(x) = c(a,b)g(x)I_{(a,b)}(x)$, and suppose that $X_1, \ldots, X_n$ are i.i.d. from the distribution with density $f_{a,b}(x)$. Find a minimal sufficient statistic for $(a,b)$.

**(f)** Let $X_1, \ldots, X_n$ be i.i.d. from a distribution with density $f_\theta(x) = \frac{1}{2}\exp(-|x-\theta|)$ with $x, \theta \in \mathbb{R}$. Show that the order statistics are minimal sufficient.

**Ex. 4.23** *Ancillary statistics.* Ancillarity is, in a sense, the opposite of sufficiency. If $X$ stem from a member of $\mathcal{P} = \{P_\theta \colon \theta \in \Theta\}$, a statistic $V = V(X)$ is ancillary if its distribution is the same for all $\theta$. In other words, $V$ by itself does not provide any information about $\theta$. This does not, however, mean that $V$ should be disregarded when making inference on $\theta$. It just means that if you only learn $V = v$, you have not learned anything about $\theta$.

ancillary statistic

**(a)** Let $Y \,|\, (V = v) \sim \mathrm{N}(\mu, \sigma_V^2)$ where $V$ is Bernoulli $(\frac{1}{2})$. Why does it make the most sense to condition on $V = v$ when deriving a confidence interval for $\mu$?

**(b)** Let $X$ and $Y$ be independent $\mathrm{unif}(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, and set $T = \max(X,Y)$ and $V = |X - Y|$. Show that Show that $(T, V)$ is minimal sufficient, and that $V$ is ancillary.

**(c)** With $T$ and $V$ as in (b), compute the probability $\Pr(T - V \leq \theta \leq T)$ and the conditonal probability given $V = v$, $\Pr(T - \frac{1}{4}(1+v) \leq \theta \leq T + \frac{1}{4} - \frac{3}{4}v \,|\, V = v)$. Which of these two confidence intervals makes the most sense to you?

**(d)** Compute the probability $\Pr(T - V \leq \theta \leq T)$ and the conditonal probability given $V = v$, $\Pr(T - \frac{1}{4}(1+v) \leq \theta \leq T + \frac{1}{4} - \frac{3}{4}v \,|\, V = v)$. Which of these two confidence intervals makes the most sense to you?

**(e)** Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be i.i.d. with density $f_\theta(x,y) = \exp(-\theta x - y/\theta)$ for $x, y > 0$ and $\theta > 0$. Find the maximum likelihood estimator $\widehat{\theta}_n$ for $\theta$. Show that $(\widehat{\theta}_n, V)$ is minimal sufficient, where $V = (\sum_{i=1}^n X_i)^{1/2}(\sum_{i=1}^n Y_i)^{1/2}$; and that $V$ is ancillary.

**(f)** In many examples, the maximum likelihood estiamator is a minimal sufficient statistic. In (e) it is not. One way of measuring the information loss in only knowing $\widehat{\theta}_n$ compared to the full minimal sufficient statistic $(\widehat{\theta}_n, V)$, is by way of the Fisher information. Let $g_{\theta,n}$ be the density of $\widehat{\theta}_n$, and compute the difference $-\mathrm{E}\,\{\mathrm{d}^2/\mathrm{d}\theta^2 \sum_{i=1}^n \log f_\theta(X_i, Y_i)\} + \mathrm{E}\,\{\mathrm{d}^2/\mathrm{d}\theta^2 \sum_{i=1}^n \log g_{\theta,n}(\widehat{\theta}_n)\}$.

**Ex. 4.24** *Completeness.* Often, models are so harmoniously constructed that there are clear one-to-one connections between estimators (perhaps based on a set of summary statistics) and estimands, in the sense that there for each estimand is only one unbiased estimator. Clarifying such regularity leads to the concept of *completeness*, which turns out to be useful also when coming to conditional testing and optimal power in exercises below. Technically, suppose some vector $T = (T_1, \ldots, T_p)^{\mathrm{t}}$ has a distribution $f(t, \theta)$, with the property that $\mathrm{E}_\theta\, h(T) = 0$ for all $\theta \in \Theta$ implies $\Pr_\theta\{h(T) = 0\} = 1$ for all $\theta$, i.e. $h(t) = 0$ almost everywhere. We then say that $T$, or more formally its distribution, over the relevant parameter region, is complete.

complete

**(a)** Let $X \sim \mathrm{binom}(n, \theta)$, with $\theta \in (0, 1)$. Show that $X$ is complete; zero is the only unbiased estimator of zero. (You may appeal to properties of power series.) Show that $X$ is complete as long as the parameter region contains an open interval. Show similarly that if $X \sim \mathrm{geom}(p)$, see Ex. 1.25, then $X$ is complete, again requiring only that the parameter range for $p$ contains an open interval.

**(b)** With $Y_1, \ldots, Y_n$ i.i.d. from the uniform on $[0, \theta]$, consider $M = \max_{i \le n} Y_i$. Show that if the parameter region is the full halfline $\theta > 0$, then $M$ is sufficient and complete, but that $M$ is not complete if it is a priori known that $\theta \ge 1.234$, i.e. with a restricted parameter range.

**(c)** Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the uniform distribution over $[\theta - 1, \theta + 1]$. Show that $(Y_{(1)}, Y_{(n)})$, the smallest and largest, is sufficient, but not complete.

**(d)** Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the uniform distribution over $[a, b]$. Show that $(Y_{(1)}, Y_{(n)})$ is sufficient and complete. [xx check this xx]

**(e)** Consider $Y$ being uniform on $\{1, \ldots, \theta\}$, where $\theta \in \{1, 2, \ldots\}$ is an unknown parameter. Show that $Y$ is complete. When the parameter region is e.g. $\{4, 5, \ldots\}$, however, show that $Y$ is not complete.

**(f)** If $T$ is complete, show that any one-to-one transformation variable $T' = a(T)$ is also complete.

**(g)** Consider $Y_1, \ldots, Y_n$ an i.i.d. sample from the double exponential with density $f(y, \theta) = \frac{1}{2}\exp(-|y - \theta|)$. Show that the full set of order statistics $(Y_{(1)}, \ldots, Y_{(n)})$ is sufficient, but not complete; do this, by exhibiting two different unbiased estimators of $\theta$.

**Ex. 4.25** *Completeness for the exponential family.* For the large class of exponential family models, see Ex. 1.56 (xx and follow-up exercises above xx), there is a completeness lemma, as follows. Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the model $f(y, \theta) = \exp\{\theta^{\mathrm{t}} T(y) - k(\theta)\}h(y)$, with $T = (T_1, \ldots, T_p)^{\mathrm{t}}$ and $\theta = (\theta_1, \ldots, \theta_p)^{\mathrm{t}}$ varying in an open set, then the vector of sample averages $(\bar{T}_1, \ldots, \bar{T}_p)^{\mathrm{t}}$ is not merely sufficient, as seen in Ex. 4.22, but also complete. We shall freely use this lemma. (xx but point to proof, check BickelDoksum or Johansen or Brown or Schervish. perhaps requiring analyting continuation arguments. point is that $a(\theta) = \mathrm{E}_\theta\, h(\bar{T})$ is a super smooth functions with all derivatives smooth. xx)

completeness lemma for exponential family

**(a)** Let $Y_1, \ldots, Y_n$ be i.i.d. from the $\mathrm{N}(\xi, 1)$. Show that the full set $(Y_1, \ldots, Y_n)$ is not complete, but that the sample mean $\bar{Y}$ is.

**(b)** Consider $Y_1, \ldots, Y_n$ i.i.d. from the $\text{Gam}(a, b)$ model. Show that $(\sum_{i=1}^n Y_i, \sum_{i=1}^n \log Y_i)$ is sufficient and complete. Identify similarly a sufficient and complete pair of statistics for a sample from the $\text{Beta}(a, b)$.

**(c)** Consider an i.i.d. sample $Y_1, \ldots, Y_n$ from the $\text{N}(\xi, \sigma^2)$. Show that $(\sum_{i=1}^n Y_i, \sum_{i=1}^n Y_i^2)$ is sufficient and complete, and also that $(\bar{Y}, \widehat{\sigma})$ is sufficient and complete. Suppose then that the variance is postulated to be equal to the squared mean, so that the sample is from $\text{N}(\theta, \theta^2)$. Construct two different unbiased estimators of $\theta$, and show that this means that $(\bar{Y}, \widehat{\sigma})$ is not complete. You may similarly construct two different unbiased estimators of $\theta^2$.

**(d)** Consider the linear regression model with $Y_i \sim \text{N}(a + bx_i, \sigma^2)$, studied in Ex. 3.33. With $\widehat{a}, \widehat{b}$ the least squares estimators, and $Q_0 = \sum_{i=1}^n \{Y_i - \widehat{a} - \widehat{b}(x_i - \bar{x})\}^2$, show that the log-likelihood can be written $-n \log \sigma - \frac{1}{2}\{Q_0 + n(\widehat{a} - a)^2 + M_n(\widehat{b} - b)^2\}^2/\sigma^2$, with $M_n = \sum_{i=1}^n (x_i - \bar{x})^2$. Write this in the exponential family fashion, with natural parameters $1/\sigma^2, a/\sigma^2, b/\sigma^2$. Argue via the general exponential family results that $(Q_0, \widehat{a}, \widehat{b})$ is both sufficient and complete. Extend this, arguments and results, to the general multiple linear regression model of Ex. 3.34.

**Ex. 4.26** *Basu's Lemma.* Consider a setting with data $Y$ from some parametric family, indexed by $\theta \in \Theta$, where a suitable $T$ is sufficient and complete.

**(a)** Assume that a suitable statistic $Z = Z(Y)$ has a distribution not depending on the $\theta$. Show that $Z$ is independent of $T$. This is called Basu's Lemma. You may follow this path: start writing $\Pr_\theta(Z \in A) = p$, by assumption not depending on $\theta$. Show that $p = \text{E}_\theta\, h(T)$, for all $\theta$, where $h(t) = \Pr_\theta(Z \in A \,|\, T = t)$, another function not depending on $\theta$. Explain that this implies $\Pr_\theta(Z \in A \,|\, T = t) = p$ for all $\theta$. Basu's lemma

**(b)** Consider the familiar setup with $Y_1, \ldots, Y_n$ being i.i.d. from $\text{N}(\mu, \sigma^2)$. For $\sigma$ fixed, show that the mean $\bar{Y}$ is sufficient and complete for $\mu$; explain that all statistics $Z$ with distribution not depending on $\mu$ must be independent of $\bar{Y}$. Explain hence in particular that $\sum_{i=1}^n (Y_i - \bar{Y})^2$ is independent of $\bar{Y}$. This has been demonstrated in different ways in Ex. 1.49 and 1.50.

**(c)** In this normal sample setting, argue that the classic t ratio $t = \sqrt{n}(\bar{Y} - \mu)/\widehat{\sigma}$, with $\widehat{\sigma}$ the sample standard deviation, is independent of $(\bar{Y}, \widehat{\sigma})$.

**(d)** We have seen in Ex. 1.57 that the five-parameter binormal distribution is inside the exponential family class. With $(X_1, Y_1), \ldots, (X_n, Y_n)$ i.i.d. from the binormal, show first that $(\bar{X}, \bar{Y}, \widehat{\sigma}_1, \widehat{\sigma}_2, R_n)$ is complete and sufficient, with $R_n$ the empirical correlation coefficient. Show that $R_n$ is independent of $\bar{X}, \bar{Y}, \widehat{\sigma}_1, \widehat{\sigma}_2$ (xx check this with care xx).

### Optimal conditional testing

**Ex. 4.27** *Conditional tests.* Suppose in general terms that data $y$ are observed from a model with parameter $\theta$, where the null hypothesis $H_0 \colon \theta \in \Theta_0$ is to be tested, against the alternative that $\theta \notin \Theta_0$. Assume one computes $U = U(y)$ and $V = V(y)$. A *conditional test*, with respect to $V$, with level $\alpha$, is then to find a rejection region $R(v)$, using the distribution of $U$ given $V(y) = v$, with conditional tests

$$\Pr_\theta\{U(Y) \in R(v) \,|\, V(Y) = v\} \le \alpha \quad \text{for all } \theta \in \Theta_0.$$

Such tests are natural and important in multiparameter setups, as we shall see, and various constructions succeed in 'reducing the dimensionality' down to the analysis of a one-parameter family, where e.g. Neyman–Pearson more readily applies.

**(a)** Even when such a test has been constructed in a conditional modus, 'what is unlikely null behaviour of $U$ given that $V = v$', it may of course be translated or paraphrased without the conditioning: one rejects if $U(Y) \in R(V)$. Show that the test also has unconditional level $\alpha$.

**(b)** From unconditional to conditional: Conditional tests as above have the form $T(U, V) = I(U \in R(V))$, built to have $\mathrm{E}_\theta\{I(U \in R(v)) \,|\, v\} = \alpha$. Assume now that $V$ is complete at the boundary $\partial\Theta_0$ of the null hypothesis parameter region; see Ex. 4.24. Show that if *any* test $T(U, V)$ has constant level $\alpha$ at this boundary, then it is a conditional test, with this level; $\mathrm{E}_\theta\{T(U, V) \,|\, V = v\} = \alpha$ for all $\theta \in \bar{\Theta}_0$. (It might be useful to check the function $h(v) = \mathrm{E}_\theta\{T(U, V) \,|\, V = v\} - \alpha$.)

**Ex. 4.28** *Conditional tests: pairs of exponentials.* Suppose $X \sim \mathrm{Expo}(a)$ and $Y \sim \mathrm{Expo}(a + \delta)$, and that one wishes to test $\delta = 0$, i.e. equal distributions, against $\delta > 0$.

**(a)** Show that the joint density may be written $a(a + \delta)\exp(-az - \delta y)$, with $z = x + y$. Find the distribution of $Z = X + Y$, and show that the distribution of $Y$ given $Z = z$ has the density

$$g_\delta(y \,|\, z) = \frac{\delta \exp(-\delta y)}{\int_0^z \delta \exp(-\delta y')\,\mathrm{d}y'} = \frac{\delta \exp(-\delta y)}{1 - \exp(-\delta z)} \quad \text{for } 0 \le y \le z.$$

In particular, it does not depend on the $\theta$. For the null hypothesis case of $\delta = 0$, show that $Y \,|\, z$ is uniform on $[0, z]$.

**(b)** The natural conditional 0.05 level test is then to first compute $z$, and then to reject if $y \le 0.05\, z$. Show that it indeed has level 0.05, and that is the power optimal test among all conditional tests, using $Y$ given $z$. Verify that this conditional test is the same as the unconditional test of rejecting when $R = Y/(X + Y) \le 0.05$. Compute the power function of the $T^* = I(Y \le 0.05\, Z)$ test (in the testing function parlance of Ex. 4.8), conditional on $z$, and unconditionally.

**(c)** At the boundary of the null, where $\delta = 0$, show that $Z$ is complete. Show hence that *any* test with level 0.05 also must be a conditional on $Z$ 0.05 level test, via Ex. 4.28.

**(d)** We know that $T^*(y, z) = I(y \le 0.05\, z)$ is the most powerful conditional test with level 0.05; we now wish to extend this statement to $T^*$ actually being the most powerful among *all* tests with level 0.05. For any competing test $T(Y, Z)$ with level 0.05, show, since it must be a $z$-conditional 0.05 level test, where it cannot beat $T^*$, that

$$\mathrm{E}_{a,\delta}\{T^*(Y, Z) \,|\, z\} \ge \mathrm{E}_{a,\delta}\{T(Y, Z) \,|\, z\} \quad \text{for all } \delta > 0, z > 0.$$

There is equality, to 0.05, at $\delta = 0$. Show from this that $T^*$ is more powerful than such $T$, unconditionally; in suitable power function symbols, $\pi_{T^*}(a, \delta) \ge \pi_T(a, \delta)$ for all $\delta > 0$.

**(e)** Suppose now that there are $m$ independent pairs, $X_i \sim \mathrm{Expo}(a_i)$ and $Y_i \sim \mathrm{Expo}(a_i + \delta)$, with sums $Z_i = X_i + Y_i$; there are hence $m + 1$ parameters with $2m$ data points. Show that the optimal test is to reject when $U_m = Y_1 + \cdots + Y_m$ is small, given $z_1, \ldots, z_m$. Explain how the null distribution of $U_m$ can be evaluated via simulations. For an illustration, suppose three pairs $(x_i, y_i)$ are observed: $(0.927, 0.819)$, $(1.479, 0.408)$, $(3.780, 1.311)$. Carry out the test of $\delta$, and compute the p-value.

**Ex. 4.29** *Conditional tests: normal.* (xx various situations with distribution of $U \,|\, (V = v)$, followed by natural conditional test. xx) Consider a pair of normals, where interest lies in assessing their difference in means. This may of course be parametrised in different

ways, but one natural way is $x \sim \mathrm{N}(\theta, 1)$ and $y \sim \mathrm{N}(\theta + \delta, 1)$. One wishes to test $\delta = 0$ vs. $\delta > 0$, equivalent, of course, to testing equality of the means vs. $\mathrm{E}\, y > \mathrm{E}\, x$.

**(a)** Show that the joint likelihood can be written

$$\begin{aligned} f(x, y, \theta, \delta) &= (2\pi)^{-1} \exp[-\tfrac{1}{2}\{(x-\theta)^2 + (y-\theta-\delta)^2\}] \\ &= (2\pi)^{-1} \exp\{\theta z + \delta y - \tfrac{1}{2}x^2 - \tfrac{1}{2}y^2 - \tfrac{1}{2}\theta^2 - \tfrac{1}{2}(\theta+\delta)^2\}, \end{aligned}$$

where $z = x + y$, and with the main interaction between parameters and data being in the $\theta z + \delta y$ part.

**(b)** Show that $(y, z)$ is a binormal, and set up its mean vector and variance matrix. Then use Ex. 1.45, or other algebraic methods, to show that $y \,|\, z \sim \mathrm{N}(\frac{1}{2}(z+\delta), \frac{1}{2})$; in particular, its conditional distribution does not depend on $\theta$.

**(c)** Through the conditioning on $z$ the testing problem has been reduced from a two-parameter to a one-parameter situation. For $y \,|\, z \sim \mathrm{N}(\frac{1}{2}(z+\delta), \frac{1}{2})$, show that the optimal test is to reject when $y - \frac{1}{2}z > (1/\sqrt{2})c$, with $c = \Phi^{-1}(1-\alpha)$ the standard normal quantile.

**(d)** Show that the above test, constructed to be optimal in the model for $y \,|\, z$, is equivalent to that of rejecting whem $D = y - x > \sqrt{2}c$. (xx so the conditional test is an ordinary unconditional test in disguise, or vice versa, in this particular situation. the point is the general principle. xx)

**(e)** (xx Consider $m$ pairs of normal data, of the form $x_i \sim \mathrm{N}(\theta_i, 1)$ and $y_i \sim \mathrm{N}(\theta_i + \delta, 1)$. do the math, with the steps above. log joint density $\sum_{i=1}^m (\theta_i z_i + \delta y_i)$, with $z_i = x_i + y_i$. conditional test, $\sum_{i=1}^k y_i$ big given $z_1, \ldots, z_k$. xx)

**(f)** (xx something re power. xx)

**Ex. 4.30** *Conditional tests: Poisson.* (xx various situations with distribution of $U \,|\, (V = v)$, followed by natural conditional test. xx)

**(a)** We start with a single pair of Poissons, $x$ with mean $\theta$, $y$ with mean $\theta\gamma$. Show that the joint distribution becomes $\exp(-\theta - \theta\gamma)\theta^{x+y}\gamma^y/(x!\, y!)$. This inspires inspecting the distribution of $y$ given $z = x + y$. Show that $y \,|\, z \sim \mathrm{binom}(z, \gamma/(\gamma+1))$.

**(b)** To test $\gamma = 1$ against $\gamma > 1$, describe in details the natural conditional test which rejects when $y$ is big, given $z = x + y$.

**(c)** Next consider independent Poisson pairs $x_i, y_i$ for $i = 1, \ldots, m$, where $x_i$ has mean $\theta_i$ and $y_i$ mean $\theta_i\gamma$. The model hence has $m+1$ parameters for the $2m$ observations, with $\gamma$ the common multiplicative factor. Show that the joint distribution may be written

$$f = \exp\Big[-\sum_{i=1}^m (\theta_i + \theta_i\gamma) + \sum_{i=1}^m \{(x_i + y_i)\log\theta_i + y_i \log\gamma\}\Big] \frac{1}{x_1!\, y_1! \cdots x_m!\, y_m!}.$$

With $z_i = x_i + y_i$, find the distribution of $y_i \,|\, z_i$, and also the distribution of $S = \sum_{i=1}^m y_i$ given $z_1, \ldots, z_m$.

**(d)** Find the power optimal test for $\gamma = 1$ against $\gamma > 1$, among all those based on $S$ given $z_1, \ldots, z_m$.

**(e)** (xx more, rounding off. something with limit. point to Ch7 optimal CD. also do $Y \sim \mathrm{Pois}(m_0\theta_0)$ and $Y_1 \sim \mathrm{Pois}(m_1\theta_1)$, with $m_0$ and $m_1$ exposure time. With interest being in the ratio parameter $\gamma = \theta_1/\theta_0$, show that $Y_1 \,|\, (Z = z)$ is binomial $(z, m_1\gamma/(m_0 + m_1\gamma))$. xx)

**Ex. 4.31** *Conditional tests:* $2 \times 2$ *tables.* (xx various situations with distribution of $U \,|\, (V = v)$, followed by natural conditional test. xx)

**(a)** Consider two binomials $y_0 \sim \text{binom}(m_0, p_0)$ and $y_1 \sim \text{binom}(m_1, p_1)$. The outcomes in such situations are often presented as a two-by-two table,

$$\begin{matrix} y_0, & m_0 - y_0 \\ y_1, & m_1 - y_1 \end{matrix}$$

Consider the so-called logistic parametrisation

$$p_0 = H(\theta) = \frac{\exp(\theta)}{1+\exp(\theta)} \quad \text{and} \quad p_1 = H(\theta+\gamma) = \frac{\exp(\theta+\gamma)}{1+\exp(\theta+\gamma)}.$$

Show that $\theta = \log\{p_0/(1-p_0)\}$ and $\theta+\gamma = \log\{p_1/(1-p_1)\}$ in terms of the so-called log-odds. Show that the joint distribution can be written

$$f = \binom{m_0}{y_0}\binom{m_1}{y_1}\frac{\exp(\theta(y_0+y_1))}{\{1+\exp(\theta)\}^{m_0}}\frac{\exp(\gamma y_1)}{\{1+\exp(\theta+\gamma)\}^{m_1}}.$$

**(b)** (xx in view of ... check, calibrate. xx) This inspires reaching inference for $\gamma$ via the conditional distribution of $y_1$ given $z = y_0 + y_1$. Show that this distribution becomes

$$g_\gamma(y_1 \mid z) = \binom{m_0}{z-y_1}\binom{m_1}{y_1}\exp(\gamma y_1)\Big/\sum_{y_1' \le \min(m_1,z)}\binom{m_0}{z-y_1'}\binom{m_1}{y_1'}\exp(\gamma y_1')$$

for $y_1 = 0, 1, \ldots, \min(m_1, z)$. In particular, this so-called excentric hypergeometric distribution depends on $\gamma$ but not $\theta$. We recognise the ordinary hypergeometric for $\gamma = 0$; see Ex. 1.68.

**(c)** Show that the optimal 0.05 level conditional test for the null hypothesis of equality, $p_0 = p_1$, is to reject when $y_1 > c(z)$, with $c(z)$ the highest number with $\sum_{0 \le y_1 \le c(z)} g_0(y_1 \mid z) \le 0.95$.

**(d)** (xx the power. xx)

**(e)** (xx to $k$ two-by-two tables, $p_{i,0} = H(\theta_i)$ and $p_{i,1} = H(\theta_i+\gamma)$, $k+1$ parameters. optimal conditional test for $\sum_{i=1}^k y_{1,i}$ given $z_1, \ldots, z_k$, with $z_i = y_{i,0} + y_{i,1}$. point to Story #11. xx)

**Ex. 4.32** *The t test as an optimal conditional test.* Let $Y_1, \ldots, Y_n$ be i.i.d. from the normal $(\xi, \sigma^2)$, where we wish to test $\xi = 0$ against $\xi > 0$. The canonical classical test, of level say 0.05, is based on $t = \sqrt{n}\bar{Y}/\sigma$, rejecting if $t \ge t_{n-1,0.95}$, the upper 0.05 point of the $t_{n-1}$ distribution; see also Ex. 3.7 and (3.4). We cannot use the Neyman–Pearson lemma directly to demonstrate optimality of the t test, however. One of several optimality properties may be derived via conditioning.

**(a)** Write $U = \sqrt{n}\bar{Y}$ and $V = \sum_{i=1}^n Y_i^2$, so that in particular $W = \sum_{i=1}^n (Y_i - \bar{Y})^2 = V - U^2$. Show that the joint density of the data can be written

$$\begin{aligned} f &= \frac{1}{(2\pi)^{n/2}}\frac{1}{\sigma^n}\exp\Big\{-\tfrac{1}{2}\frac{1}{\sigma^2}\sum_{i=1}^n (y_i-\xi)^2\Big\} \\ &= \frac{1}{(2\pi)^{n/2}}\frac{1}{\sigma^n}\exp\Big\{\frac{\xi}{\sigma^2}\sqrt{n}U - \tfrac{1}{2}\frac{1}{\sigma^2}V - \tfrac{1}{2}\frac{\xi^2}{\sigma^2}\Big\}. \end{aligned}$$

Note that the testing problem is equivalent to testing $\lambda = 0$ against $\lambda > 0$, with $\lambda = \xi/\sigma^2$, the mathematics indicating that this is a parameter easier to work with than $\xi$.

**(b)** Find the distribution of $U \,|\, (V = v)$, and show in particular that it depends on the parameters only via $\lambda = \xi/\sigma^2$. It is convenient here to work with

$$T = \frac{U}{\{W/(n-1)\}^{1/2}} = (n-1)^{1/2} \frac{U}{(V - U^2)^{1/2}}.$$

**(c)** Show that the power optimal test, among all tests based on $U \,|\, (V = v)$, is to reject when $U$ is big, say $U \geq c(v)$, with $\Pr_0\{U \geq c(v) \,|\, V = v\} = 0.05$. Then show that this is actually the same as the t test.

**(d)** Show via arguments used in Ex. 4.28 and in other places above that the t test is not merely optimal among all conditional tests, for given level $\alpha$, i.e. given $V$, but among *all* tests with level $\alpha$.

**Ex. 4.33** *Conditional tests: multiparameter exponential models.* In the rather simple situation of Ex. 4.28, with the exponential pair $X \sim \text{Expo}(a)$ and $Y \sim \text{Expo}(a + \delta)$, with sum $Z = X + Y$, we learned that (i) there is a clear level $\alpha$ conditional test for $\delta = 0$ vs. $\delta > 0$, in terms of $Y \,|\, Z$; (ii) that test is uniformly most powerful against all $\delta > 0$, among all conditional tests; and (iii) all other level $\alpha$ competitors are in fact also $Z$-conditional. Hence the winning test, reject if $Y \leq \alpha Z$, is the uniformly most powerful level $\alpha$ test. – We shall see now that the same arguments essentially go through for the wide class of all exponential families. Consider data $Y$ from a density of the form $f(y, a, b) = \exp\{aU(y) + b^{\text{t}}V(y) - k(a, b)\}h(y)$, as in Ex. 4.21, with one-dimensional $U$ and $p$-dimensional $V$. Suppose we need to test $a = a_0$ against $a > a_0$, for some given null hypothesis value $a_0$.

**(a)** We have seen in the exercise pointed to that $U \,|\, (V = v)$ has a density depending on $a$ but not $b$, and that it has an exponential form. Assume for simplicity that the distribution of $U$ is continuous; mild formalistic additional arguments are required if the distribution is discrete. Deduce that there is a most powerful conditional level $\alpha$ test, say $T^*(y) = I\{U > c(V)\}$, with $c(v)$ determined from $\Pr_{a_0}(U > c(v) \,|\, V = v) = \alpha$, and with consequent power function $\pi(a, b) = \Pr_{a,b}\{U > c(V)\}$.

**(b)** Then consider *any* competing test $T(U, V)$ with level $\alpha$. From $\text{E}_{a_0,b}\, T(U(Y), V(Y)) = \alpha$, for all $b$, use completeness in $b$ of $V(Y)$ for fixed $a_0$ to prove that

$$\text{E}_{a_0,b}\{T(U(Y), V(Y)) \,|\, V(y) = v)\} = \alpha$$

for all $v$ (except perhaps in a region of probability zero), and for all $b$. Thus the $T$ competitor is also a level $\alpha$ $V$-conditional test, and we have proved that the conditional test is uniformly most powerful among *all* level $\alpha$ tests.

**(c)** The theory extends fruitfully to the case of testing $H_0 \colon a \leq a_0$ against $a > a_0$. Show that the test $T^* = I\{U > c(V)\}$ above, with $c(v)$ determined from $\Pr_{a_0}(U > c(v) \,|\, V = v) = \alpha$ at the boundary, is still of level $\alpha$. Then show that this test is uniformly most powerful against all competing tests with constant level $\alpha$ at the boundary $a = a_0$; one says that such tests are *unbiased*. This latter very mild limitation is in order for the completeness argument to go through.

**(d)** (xx briefly about two-sided tests. still based on $U \,|\, (V = v)$. xx)

**(e)** When $U \,|\, (V = v)$ has a discrete distribution, the arguments still go through, but one cannot expect to find $c(v)$ with e.g. $\Pr_{a_0}\{U > c(v) \,|\, V = v\} = 0.05$. There are two ways out of this mild quandary. The first is to be satisfied with level 0.042, say, if that is how close one comes to 0.050, by appropriate choice of $c(v)$. The other, if one pedantically

insists on 0.05, is to finetune $c(v)$ such that $\Pr_{a_0}\{U > c(v) \,|\, V = v\}$ is just below 0.05, and then identify the probability $r$ such that

$$\Pr_{a_0}\{U > c(v) \,|\, V = v\} + r \Pr_{a_0}\{U = c(v) \,|\, V = v\} = 0.05,$$

So one rejects if $U > c(V)$, or, but then with probability $r$, if $U = v(C)$.

**(f)** (xx go through the previous exercises about conditional tests, once more, make sure that the tests found there are really uniformly most powerful among all unbiased tests. xx)

**Ex. 4.34** *Optimal t testing in linear regression.* Consider the classic linear regression setup with $Y_i \sim \mathrm{N}(a + bx_i, \sigma^2)$ for $i = 1, \ldots, n$, see Ex. 3.33, and for simplicity of presentation here we assume the $x_i$ have been centred, so $\bar{x} = 0$. The three estimators are then $\widehat{a} = \bar{Y}$, $\widehat{b} = \sum_{i=1}^n x_i Y_i / M_n$, with $M_n = \sum_{i=1}^n x_i^2$, and $\widehat{\sigma}^2 = Q_0/(n-2)$, with $Q_0 = \sum_{i=1}^n (Y_i - \widehat{a} - \widehat{b}x_i)^2$.

**(a)** With $V = \sum_{i=1}^n Y_i^2$, show that $Q_0 = V - n\widehat{a}^2 - M_n\widehat{b}^2$, and that the likelihood may be written

$$\begin{aligned} L &= \frac{1}{\sigma^n} \exp\Big\{-\tfrac{1}{2}\frac{1}{\sigma^2}\sum_{i=1}^n (Y_i - a - bx_i)^2\Big\} \\ &= \frac{1}{\sigma^n} \exp\Big\{-\tfrac{1}{2}\frac{V}{\sigma^2} + \frac{a}{\sigma^2} n\widehat{a} + \frac{b}{\sigma^2} M_n \widehat{b} - \tfrac{1}{2}\frac{na^2 + M_n b^2}{\sigma^2}\Big\}, \end{aligned}$$

which is in the exponential family form, with natural parameters $1/\sigma^2$, $a/\sigma^2$, $b/\sigma^2$.

**(b)** Explain from the general exponential family testing theory that the uniformly most powerful test for $b = 0$ against $b > 0$, among all level $\alpha$ tests is to reject when $\widehat{b}$ is large enough, in its conditional distribution given $(V, \widehat{a})$. We need to find $c(V, \widehat{a})$ such that

$$\Pr_{a,\sigma}(\widehat{b} > c(V, \widehat{a}) \,|\, V, \widehat{a}) = \alpha$$

under $b = 0$. But this may be transformed to

$$W = \frac{\widehat{b}}{(V - n\widehat{a}^2)^{1/2}} = \frac{\widehat{b}}{(Q_0 + M_n\widehat{b}^2)^{1/2}} = \frac{\widehat{b}/Q_0^{1/2}}{(1 + M_n\widehat{b}^2/Q_0)^{1/2}} > \frac{c(V, \widehat{a})}{(V - n\widehat{a}^2)^{1/2}}.$$

But $W$ is a smooth increasing function of $t = \widehat{b}/(\widehat{\sigma}/M_n)$, the classic t ratio, which has the $t_{n-2}$ distribution under $b = 0$. Argue that $W$ is therefore independent of $(V, \widehat{a})$, and that the optimal test is to reject when $t > t_0$, the upper $\alpha$ in the $t_{n-2}$ distribtion. You have now shown that the traditional t test is optimal.

**(c)** Generalise the above to general linear multiple regression models, giving the result that the traditional t tests, for each of the $\beta_j$ coefficients, are optimal.

**Ex. 4.35** *A generalised Poisson distribution.* For a count variable $Y$, consider the model with point probabilities

$$f(y, \lambda, \gamma) = k(\lambda, \gamma)^{-1} \lambda^y / (y!)^\gamma \quad \text{for } y = 0, 1, 2, \ldots,$$

where $k(\lambda, \gamma)$ is the normalisation constant $\sum_{y=0}^\infty \lambda^y/(y!)^\gamma$. For $\gamma = 1$ we're back to ordinary Pois($\lambda$), with $k(\lambda, 1) = \exp(\lambda)$. This two-parameter generalised Poisson model is from Schweder and Hjort (2016, Examples 4.18, 8.16). A regression version model of this type is used in Story #26, to assess potential overdispersion in Poisson counts.

**(a)** Pick some $\lambda$, and compute and display curves of the mean $\xi(\lambda,\gamma)$ and the variance-to-mean ratio $\rho(\lambda,\gamma)$, for an interval of $\gamma$ around 1. Show that this ratio is decreasing in $\gamma$; hence $\gamma<1$ indicates overdispersion and $\gamma>1$ underdispersion, relative to the Poisson. Also show that the mean of $\log(Y!)$ is decreasing in $\gamma$.

**(b)** Show that the distribution is of the exponential family form, and that the sufficient statistics, after having observed a sample $Y_1,\ldots,Y_n$, is $T=\sum_{i=1}^n Y_i$ and $U=\sum_{i=1}^n \log(Y_i!)$. Show also that the joint distribution of these two must take the form

$$g_n(t,u)=\exp\{t\log\lambda-u\gamma-r_n(\lambda,\gamma)\}h_n(y_1,\ldots,y_n),$$

for appropriate functions $r_n$ and $h_n$.

**(c)** For an observed sample $Y_1,\ldots,Y_n$, to test the Poisson assumption, against overdispersion, show that the optimal test is to reject when $U$ is sufficiently small, given $T=t$. In other words, with level the classic 0.05, for example, we reject when $U\le u_0(t)$, where $u_0(t)$ is the 0.95 quantile of the distribution of $U$ given $T=t$, computed at $\gamma=1$, i.e. under Poisson conditions. (xx this needs more care; distribution of $U\,|\,(T=t)$ needs a formula or two, so we see that $U$ significantly small indicates $\gamma<1$. xx)

**(d)** There is no table or simple formula for the distribution of $U\,|\,(T=t)$, but show that it depends on $\gamma$, but not $\lambda$. Show that under $\gamma=1$, $(Y_1,\ldots,Y_n)\,|\,(T=t)$ is a multinomial with count $t$ and probabilities $(1/n,\ldots,1/n)$. Explain then how the distribution of $U\,|\,(T=t)$ may be simulated under Poisson conditions.

**(e)** (xx give them a dataset. nils checks the football matches dataset. decide later if this is a Story or an exercise. xx)

**Ex. 4.36** *Testing for correlation.* Consider binormal i.i.d. pairs $(X_i,Y_i)$, with parameters $\xi_1,\xi_2,\sigma_1,\sigma_2,\rho$, as in Ex. 1.57. Suppose we need to test $\rho=0$ vs. $\rho>0$.

**(a)** Argue as in the exercise pointed to that the binormal distribution is inside the exponential family class, with natural parameters

$$\frac{\xi_1}{(1-\rho^2)\sigma_1^2},\ \frac{1}{(1-\rho^2)\sigma_1^2},\ \frac{\xi_2}{(1-\rho^2)\sigma_2^2},\ \frac{1}{(1-\rho^2)\sigma_2^2},\ \frac{\rho}{(1-\rho^2)\sigma_1\sigma_2},$$

associated with data functions $X_i,X_i^2,Y_i,Y_i^2,X_iY_i$. Explain that the testing problem is equivalent to testing $\lambda=0$ vs. $\lambda>0$, where $\lambda=\rho/\{(1-\rho^2)\sigma_1\sigma_2\}$.

**(b)** Then argue from the general testing theory for the exponential class that there for given testing level $\alpha$ is a uniformly most powerful test, consisting in rejecting the null provided $A_n=n^{-1}\sum_{i=1}^n X_iY_i$ is big enough, in its conditional null distribution given $(\bar X,n^{-1}\sum_{i=1}^n X_i^2,\bar Y,n^{-1}\sum_{i=1}^n Y_i^2)$. Formally, the rejection threshold $c$, which depends on these variables, is determined by

$$\Pr_0\{A_n>c\,|\,(\bar X,n^{-1}\sum_{i=1}^n X_i^2,\bar Y,n^{-1}\sum_{i=1}^n Y_i^2)\}=\alpha,$$

with the footscript 0 indicating probability under $\rho=0$.

**(c)** Write $\widehat\sigma_1^2=n^{-1}\sum_{i=1}^n X_i^2-\bar X^2$ and $\widehat\sigma_2^2=n^{-1}\sum_{i=1}^n Y_i^2-\bar Y^2$ for the empirical variances. Explain that in the conditional situation, given values for $\bar X,\bar Y,\widehat\sigma_1,\widehat\sigma_2$, the requirement above can be transformed to $\Pr_0\{R_n>d\,|\,(\bar X,\bar Y,\widehat\sigma_1,\widehat\sigma_2)\}=\alpha$, with $R_n$ the empirical correlation coefficient, see (2.11).

**(d)** Then explain that $R_n$ is actually independent of $\bar X,\bar Y,\widehat\sigma_1,\widehat\sigma_2$, under the null, so that the optimal test becomes the simpler one, of rejecting if $R_n>d$, where $\Pr_0(R_n>d)=\alpha$.

**(e)** For this optimal test regime to be specified fully, it remains to find the null distribution of $R_n$. Explain that this null distribution does not depend on the actual values of $(\xi_1, \xi_2, \sigma_1, \sigma_2)$, which we therefore may take to be $(0, 0, 1, 1)$. For practical purposes one may simulate a high number of $R_n$ and then read off the $d$ quantile; explain also that $\sqrt{n}R_n \to_d \mathrm{N}(0,1)$, from Ex. 2.54, so that $z_{1-\alpha}/\sqrt{n}$ is an approximation to the upper $\alpha$ point. We show however in the following point that $T_n = m^{1/2}R_n/(1-R_n^2)^{1/2}$ has a $t_m$ distribution, under the null, with $m = n-2$ degrees of freedom. With e.g. $n = 25$, find the 0.95 quantile of the $R_n$ distribution.

**(f)** So let us find the null distribution for $R_n$. Write $s_x^2 = \sum_{i=1}^n (X_i - \bar{X})^2$, $s_y^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2$, $s_{xy} = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$, so that $R_n = s_{xy}/(s_x s_y)$. We may now use results from linear regression of the $Y_i$ with respect to the $X_i$. Study in particular the least squares estimators $\widehat{a}, \widehat{b}$, minimising $Q(a,b) = \sum_{i=1}^n \{Y_i - a - b(X_i - \bar{X})\}^2$, as with Ex. 3.33. Using results from that exercise, show (i) that $\widehat{a} = \bar{Y}$ and $\widehat{b} = s_{xy}/s_x^2$; (ii) that $Q_0 = Q(\widehat{a}, \widehat{b}) = s_y^2 - s_x^2\widehat{b}^2$; and (iii) that $T = \widehat{b}/\{(Q_0/m)^{1/2}/s_x\} = m^{1/2}s_x\widehat{b}/Q_0^{1/2}$ has the $t_m$ distribution, with $m = n-2$ degrees of freedom. Show that this implies

$$R_n = \widehat{b}\frac{s_x}{s_y} = \frac{s_x\widehat{b}}{(Q_0 + s_x^2\widehat{b}^2)^{1/2}} = \frac{T/m^{1/2}}{(1 + T^2/m)^{1/2}}.$$

Solve for $T$ to get the $T = m^{1/2}R_n/(1-R_n^2)^{1/2}$, pointed to and used in the previous point. Go on to work out the density of $R_n$, via the density $g_m$ for the $t_m$; show that it becomes

$$h_n(r) = g_m\Big(\frac{m^{1/2}r}{(1-r^2)^{1/2}}\Big)\frac{m^{1/2}}{(1-r^2)^{3/2}} = \frac{\Gamma(\frac{1}{2}(m+1))}{\Gamma(\frac{1}{2}m)\sqrt{\pi}}(1-r^2)^{(m-2)/2}$$

for $r \in (-1,1)$. As a curiosum, check that it is U-shaped for $n = 3$, uniform for $n = 4$, and then bell-shaped for $n \geq 5$.

**Ex. 4.37** *Inference for linear multiple regression.* [xx to be done. point back to Ex. 3.34, 3.35, 3.37, and point to one or two stories. xx] in this exercise we show typical and not so typical inference methods for the linear multiple regression model, using the key results reached in the previous exercise. confidence intervals, tests, also for $\sigma$, for a $p$ quantile $F^{-1}(q \,|\, x_0) = x_0^{\mathrm{t}}\beta + z_q\sigma$, and delta method for things like $\Pr(Y \leq y_0 \,|\, x_0)$. and prediction. xx]

**(a)** (xx typical things first. show that $\widehat{\beta}_j \sim \mathrm{N}(\beta_j, k_j^2\sigma^2/n)$, where $k_j^2 = \sigma_n^{j,j}$ the diagonal elements of $\Sigma_n^{-1}$. from this show $t_j = (\widehat{\beta}_j - \beta_j)/(k_j\widehat{\sigma}/\sqrt{n})$ is a $t_{n-p}$. then ci for each $\beta_j$. and test of $\beta_j = 0$. also ok for $\beta_j - \beta_k$ etc. xx)

**(b)** (xx inference for $\sigma$. xx)

**(c)** For a given individual, with covariate vector $x_0$, the outcome $Y_0$ has the distribution $\mathrm{N}(x_0^{\mathrm{t}}\beta, \sigma^2)$. Consider the inference task for a quantile in this distribution. Show that the $q$-quantile becomes $\gamma_q = x_0^{\mathrm{t}}\beta + z_q\sigma$, with $\Phi(z_q) = q$. With estimator $\widehat{\gamma}_q = x_0^{\mathrm{t}}\widehat{\beta} + z_q\widehat{\sigma}$, show that

$$W_q = \frac{\widehat{\gamma}_q - \gamma_q}{\widehat{\sigma}} = \frac{x_0^{\mathrm{t}}(\widehat{\beta} - \beta) + z_q(\widehat{\sigma} - \sigma)}{\widehat{\sigma}} = \frac{\mathrm{N}(0, x_0^{\mathrm{t}}\Sigma_n^{-1}x_0/n) + z_q\{(\chi_m^2/m)^{1/2} - 1\}}{(\chi_m^2/m)^{1/2}}.$$

(xx round off. the point is that $W_q$ can be simulated. also approximated with a normal. give data example. xx)

**(d)** (xx prediction, what will $Y_0$ be, at position $x_0$. also $\Pr(Y \leq y_0 \,|\, x_0)$. xx)

**Ex. 4.38** *How much of the variance is explained?* (xx the below to be polished and illustrated, and with a clearer link to $R^2$. exact cc($\rho$) to come in Ch. 8. point to illustration in Story #6. xx) In the linear regression model, the extent to which the covariates influence the outcomes may be assessed in several ways, one of which is to decompose the variance of the outcomes into a covariate part and a 'remaining variability' part. Such assessments relate also to 'signal plus noise' viewpoints; how strong is the signal?

**(a)** We start out writing the regression model as

$$Y_i = \beta_0 + \beta_1 x_{i,1} + \cdots + \beta_p x_{i,p} + \varepsilon_i = \beta_0 + x_i^{\mathrm{t}}\beta + \varepsilon_i \quad \text{for } i = 1, \ldots, n,$$

again with the $\varepsilon_i$ seen as i.i.d. $\mathrm{N}(0, \sigma^2)$, and further assume that the covariates have been centred, having their means subtracted, so that $\sum_{i=1}^n x_{i,j} = 0$ for $j = 1, \ldots, p$. This gives $\beta_0$ the interpretation as the overall mean of the $Y_i$. With $\Sigma_n = (1/n)\sum_{i=1}^n x_i x_i^{\mathrm{t}}$ the empirical $p \times p$ variance matrix for the $x_i$, show that the least squares estimators become

$$\widehat{\beta}_0 = \bar{Y} \sim \mathrm{N}(\beta_0, \sigma^2/n), \quad \widehat{\beta} = \Sigma_n^{-1}(1/n)\sum_{i=1}^n x_i Y_i \sim \mathrm{N}_p(\beta, (\sigma^2/n)\Sigma_n^{-1}),$$

and that these two are independent.

**(b)** Write $\widehat{\mu}_i = \widehat{\beta}_0 + x_i^{\mathrm{t}}\widehat{\beta}$ for the model based estimate of the outcome at $x_i$. For the sum of squared residuals, show that $Q_0 = \sum_{i=1}^n (Y_i - \widehat{\mu}_i)^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2 - n\widehat{\beta}^{\mathrm{t}}\Sigma_n\widehat{\beta}$. In other words,

$$V_n = \sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (Y_i - \widehat{\mu}_i)^2 + n\widehat{\beta}^{\mathrm{t}}\Sigma_n\widehat{\beta},$$

a neat decomposition of the full variability of outcomes as a sum of squared residuals and the covariate part $n\widehat{\beta}^{\mathrm{t}}\Sigma_n\widehat{\beta}$.

**(c)** (xx place a little caveat below, regarding interpretation; we need to think about the population being sampled from. xx) Standard themes for the linear regression model are developed with analyses carried out conditional on the covariates. Allow now a change in this narrative, where the $x_i$ are seen as having their own covariate distribution, with mean zero and variance matrix $\Sigma_n$. Show that a randomly selected outcome $Y_i$ then has variance $\beta^{\mathrm{t}}\Sigma_n\beta + \sigma^2$. Show that the covariate part of the full variability becomes

$$\rho = \frac{\beta^{\mathrm{t}}\Sigma_n\beta}{\beta^{\mathrm{t}}\Sigma_n\beta + \sigma^2} = \frac{\lambda}{\lambda + 1}, \quad \text{with } \lambda = \beta^{\mathrm{t}}\Sigma_n\beta/\sigma^2.$$

With $\widetilde{\sigma}^2 = Q_0/n$ (rather than the unbiased $\widehat{\sigma}^2 = Q_0/(n-p-1)$), show that this leads to

$$\widetilde{\rho} = \frac{\widehat{\beta}^{\mathrm{t}}\Sigma_n\widehat{\beta}}{\widehat{\beta}^{\mathrm{t}}\Sigma_n\widehat{\beta} + \widetilde{\sigma}^2} = \frac{V_n - \sum_{i=1}^n (Y_i - \widehat{\mu}_i)^2}{V_n} = 1 - \frac{\sum_{i=1}^n (Y_i - \widehat{\mu}_i)^2}{\sum_{i=1}^n (Y_i - \bar{Y})^2}.$$

This is often called the coefficient of determination, or $R^2$.

**(d)** To carry out precise inference for $\rho$, show first that $n\widehat{\beta}^{\mathrm{t}}\Sigma_n\widehat{\beta}/\sigma^2 \sim \chi^2_p(n\lambda)$; check with Ex. 1.54. Then show that with $\widehat{\lambda} = \widehat{\beta}^{\mathrm{t}}\Sigma_n\widehat{\beta}/\widehat{\sigma}^2$, we have

$$F = n\widehat{\lambda}/p = n\widehat{\beta}^{\mathrm{t}}\Sigma_n\widehat{\beta}/(p\widehat{\sigma}^2) \sim F(p, m, n\lambda),$$

the noncentral $F$ with $m = n - (p+1)$ the degrees of freedom for $\widehat{\sigma}^2$.

**(e)** Explain how this may be used to set confidence intervals for $\lambda$ and hence for $\rho$. You may check with Ex. 8.18 how this may be used to construct full confidence distributions for the fraction $\rho$ of variation explained by the covariates; for an illustration, see Story #6.

**Ex. 4.39** *Testing and confidence for linear combinations of regression parameters.* (xx nils rant, so far; will be cleaned and related to other themes. xx) linear regression, $\widehat{\beta} \sim \mathrm{N}_p(\beta, (\sigma^2/n)\Sigma_n^{-1})$.

**(a)** A useful little lemma is as follows. Let $x$ be a vector of length $p$, with $A$ a symmetric positive definite $p \times p$ matrix. Show that the maximum of $c^{\mathrm{t}}x$, under the condition $c^{\mathrm{t}}Ac = 1$, is $x^{\mathrm{t}}A^{-1}x$. You may show this via Lagrange multipliers and a bit of linear algebra.

**(b)** In the regression problem, show that we for each $c$ have $c^{\mathrm{t}}\widehat{\beta} \sim \mathrm{N}(c^{\mathrm{t}}\beta, (\sigma^2/n)c^{\mathrm{t}}\Sigma_n^{-1}c)$, and that

$$V_n(c) = \frac{\sqrt{n}(c^{\mathrm{t}}\widehat{\beta} - c^{\mathrm{t}}\beta)}{\widehat{\sigma}\, c^{\mathrm{t}}\Sigma_n^{-1}c} \sim t_m,$$

with $m = n - p$ degrees of freedom. Explain how this can be used to test $\gamma(c) = c^{\mathrm{t}}\beta = 0$, for any given $c$, and construct a confidence interval for such a $\gamma(c)$.

**(c)** Will the hypothesis $\gamma(c) = 0$ be rejected, for any linear combination? Show that

$$V_n^* = \max_{\text{all } c} V_n(c) = (\sqrt{n}/\widehat{\sigma})\{(\widehat{\beta} - \beta)^{\mathrm{t}}\Sigma_n(\widehat{\beta} - \beta)\}^{1/2},$$

and that $(V_n^*)^2$ has a Fisher distribution with degrees of freedom $(p, n-p)$. (xx then round off. joint confidence for all linear combinations. testing $\beta = 0$ via an F test. can also do contrasts, all $c$ with $\sum_{j=1}^p c_j = 0$. xx)

**Ex. 4.40** *Inference for ratios of standard deviations.* Suppose two independent samples, of sizes $n_1$ and $n_2$, come from two populations, with standard deviations $\sigma_1$ and $\sigma_2$. From the empirical standard deviations $\widehat{\sigma}_1$ and $\widehat{\sigma}_2$, form the ratio $R = \widehat{\sigma}_1/\widehat{\sigma}_2$, to be used for inference about the underlying ratio $\rho = \sigma_1/\sigma_2$.

**(a)** Suppose first that the two distributions are normal. We then saw in Ex. 1.55 that $R^2 = \rho^2 F$, where $F \sim F_{m_1,m_2}$, an F distribution with degrees of freedom $(m_1, m_2) = (n_1 - 1, n_2 - 1)$. Construct a 95 percent confidence interval for $\rho$ based on this. Also give a 0.05 level test for the equality hypothesis $\sigma_1 = \sigma_2$. this gives c.i. for $\rho$, and tests for $\rho = 1$. (xx answer: with $\Pr(a \le F \le b) = 0.95$, with $(a, b)$ found from quantiles of the F, we find $[R/b^{1/2}, R/a^{1/2}]$. test: accept if $a < R^2 < b$. xx)

**(b)** Then consider inference for the ratio $\rho$ outside the assumption of normally distributed data. From Ex. 3.12, find the representation

$$R = \frac{\widehat{\sigma}_1}{\widehat{\sigma}_2} \doteq \frac{\sigma_1}{\sigma_2} \frac{1 + (1/n_1)^{1/2}(\frac{1}{2} + \frac{1}{4}\gamma_{4,1})^{1/2} N_{n_1}}{1 + (1/n_2)^{1/2}(\frac{1}{2} + \frac{1}{4}\gamma_{4,2})^{1/2} N_{n_2}},$$

in terms of the kurtoses $\gamma_{4,1}$ and $\gamma_{4,2}$, where $N_{n_1}$ and $N_{n_2}$ are independent variables tending to standard normals as sample sizes increase. Use the delta method to deduce that $R/\rho \approx_d \mathrm{N}(1, \tau^2)$, with $\tau^2 = (1/n_1)(\frac{1}{2} + \frac{1}{4}\gamma_{4,1}) + (1/n_2)(\frac{1}{2} + \frac{1}{4}\gamma_{4,2})$. For normal data, show that this matches the distributional approximation $F^{1/2} \approx_d \mathrm{N}(1, 1/(2n_1) + 1/(2n_2))$.

**(c)** Construct an approximate 95 percent confidence interval for $\sigma_1/\sigma_2$, valid also outside normal data. For an application, see the Bach and Reger Story #50.

**(d)** (xx one more point. xx)

**Ex. 4.41** *Testing equality across groups.* (xx mention below that $k = 2$ is simple, with pairwise comparisons, t type things. xx) Suppose a parameter has the same interpretation across groups, say $\theta_j$ for groups $j = 1, \ldots, k$. associated with estimators, perhaps of different precision. How can we test the null hypothesis of no difference, i.e. $H_0 \colon \theta_1 = \cdots = \theta_k$?

**(a)** Suppose first that $Y_j \sim \mathrm{N}(\theta_j, \sigma^2)$, with independence, and the same known precision, i.e. $\sigma$. Explain from classic results of Ex. 1.49–1.50, that $Z = \sum_{j=1}^k (Y_j - \bar{Y})^2/\sigma^2 \sim \chi^2_{k-1}$, with $\bar{Y} = (1/k)\sum_{j=1}^k Y_j$ the natural estimator for the common parameter under $H_0$. Explain that in the case of $\sigma$ not being known, we still have $Z^* = \sum_{j=1}^k (Y_j - \bar{Y})^2/\widehat{\sigma}^2 \to_d \chi^2_{k-1}$, as long as $\widehat{\sigma}$ is consistent, with growing information for each group.

**(b)** A useful generalisation of the basic results for i.i.d. normal data is as follows. Suppose $Y_j \sim \mathrm{N}(0, 1/a_j)$, with the $a_j$ positive and $a = \sum_{j=1}^k a_j$. Define $Y^* = \sum_{j=1}^k (a_j/a) Y_j$ and then $Z = \sum_{j=1}^k a_j (Y_j - Y^*)^2$. Show that $Z = \sum_{j=1}^k a_j Y_j^2 - a(Y^*)^2$, that it is independent of $Y^*$, and that its distribution is a $\chi^2_{k-1}$. The classical case, leading to the distribution of the empirical variance and then to the t test, corresponds to all $a_j = 1$.

**(c)** A natural variation of the previous point is as follows. Assume in general terms that $Y_j \sim \mathrm{N}(\theta, \sigma_j^2)$, with independence, for $j = 1, \ldots, k$. Show that the minimiser of $Q(\theta) = \sum_{j=1}^k (Y_j - \theta)^2/\sigma_j^2$ is $\widehat{\theta} = (\sum_{j=1}^k Y_j/\sigma_j^2)/(\sum_{j=1}^k 1/\sigma_j^2)$, and that $Q_{\min} = Q(\widehat{\theta})$ is $\chi^2_{k-1}$, independent of $\widehat{\theta}$. Demonstrate also that of all unbiased estimators $\theta^* = \sum_{j=1}^k c_j Y_j$ of $\theta$, $\widehat{\theta}$, with weights proportionao to inverse variances, has the smallest variance. Again, if the $\sigma_j$ are consistently estimated, as opposed to known, explain that with $\theta^* = (\sum_{j=1}^k Y_j/\widehat{\sigma}_j^2)/(\sum_{j=1}^k 1/\widehat{\sigma}_j^2)$, we still have $Q^* = \sum_{j=1}^k (Y_j - \theta^*)^2/\widehat{\sigma}_j^2$ tending to $\chi^2_{k-1}$.

**(d)** Consider correlations $\rho_1, \ldots, \rho_k$ for $k$ groups of binormal data, with sample sizes $n_1, \ldots, n_k$, summing to the total $n$. Via Ex. 2.54, explain that $n_j^{1/2}(\widehat{\zeta}_j - \zeta_j) \to_d \mathrm{N}(0, 1)$, where $\widehat{\zeta}_j = \frac{1}{2}\log\{(1 + \widehat{\rho}_j)/(1 - \widehat{\rho}_j)\}$ is Fisher's zeta transform. Deduce that $Q = \sum_{j=1}^k n_j(\widehat{\zeta}_j - \zeta^*)^2 \to_d \chi^2_{k-1}$, with $\zeta^* = \sum_{j=1}^k (n_j/n)\widehat{\zeta}_j$. This is accordingly a natural test statistic for $\rho_1 = \cdots = \rho_k$.

**(e)** Situations similar to that of the previous point abound in applied statistics, when needing to compare more groups than two. Suppose $\widehat{p}_j = Y_j/n_j$ is estimating a binomial $p_j$, for independent experiments, for $j = 1, \ldots, k$. Then form $\widehat{p} = \sum_{j=1}^k (n_j/n)\widehat{p}_j$, and show that under $p_1 = \cdots = p_k$ conditions, $Q = \sum_{j=1}^k n_j(\widehat{p}_j - \widehat{p})^2 \to_d p(1-p)\chi^2_{k-1}$, using $p$ to denote the common value of the $p_j$. Construct a test for equality of the $p_j$.

**(f)** (xx one more. could test equality of medians in $k$ groups. xx)

**Ex. 4.42** *Testing equality across groups, vector parameter case.* In Ex. 4.41 we developed a recipe for testing equality across groups for any one-dimensional parameter. Here we lift those methods to the case where the $\theta_1, \ldots, \theta_k$ in question have dimension $p \geq 2$. We wish again to test equality of these. (xx to be used in Story #60. xx)

**(a)** Assume $Y_j \sim \mathrm{N}_p(\theta_j, \Sigma_j)$ for $j = 1, \ldots, k$, with known positive definite variance matrices. Under $H_0$ of $\theta_1 = \cdots = \theta_k$, show that $Q(\theta) = \sum_{j=1}^k (Y_j - \theta)^{\mathrm{t}} \Sigma_j^{-1} (Y_j - \theta)$ is a $\chi^2_{kp}$, and that it is minimised by $\widehat{\theta} = A^{-1} \sum_{j=1}^k \Sigma_j^{-1} Y_j$, with $A = \sum_{j=1}^k \Sigma_j^{-1}$. Show further (i) that $\widehat{\theta} \sim \mathrm{N}_p(\theta, A^{-1})$; (ii) that the full ensemble of $Y_j - \theta$ is independent of $\widehat{\theta}$; and (iii) that the natural test statistic $Q_0 = Q(\widehat{\theta}) = \sum_{j=1}^k (Y_j - \widehat{\theta})^{\mathrm{t}} \Sigma_j^{-1} (Y_j - \widehat{\theta})$ is a

$\chi^2_{(k-1)p}$.

**(b)** This $Q_0$ may be used as a test statistic for $\theta_1 = \cdots = \theta_k$, with an exact chi-square null distribution, provided the $\Sigma_j$ matrices are known. In various applications these are estimated from data, with $\widehat{\Sigma}_j$ using data for group $j$. This leads to $\widehat{\theta} = \widehat{A}^{-1}\sum_{j=1}^k \widehat{\Sigma}_j^{-1} Y_j$, with $\widehat{A} = \sum_{j=1}^k \widehat{\Sigma}_j^{-1}$. Show that if these $\widehat{\Sigma}_j$ are consistent, then $Q^* = Q^*(\widehat{\theta}) = \sum_{j=1}^k (Y_j - \widehat{\theta})^{\rm t}\widehat{\Sigma}_j^{-1}(Y_j - \widehat{\theta})$ tends to the $\chi^2_{(k-1)p}$, under the null.

**(c)** Suppose we have normal datasets with sample sizes $n_j$, yielding the usual parameter estimates $(\widehat{\xi}_j, \widehat{\sigma}_j)$, for groups $j = 1, \ldots, k$, From these compute $\widehat{\xi} = \sum_{j=1}^k (n_j/n)\widehat{\xi}_j$, the overall mean, and $\widehat{\sigma} = \prod_{j=1}^k \widehat{\sigma}_j^{n_j/n}$. Show that

$$Q = \sum_{j=1}^k \{n_j(\widehat{\xi}_j - \widehat{\xi})^2 + 2n_j(\log\widehat{\sigma}_j - \log\widehat{\sigma})^2\}$$

tends to $\chi^2_{2(k-1)}$ under the hypothesis of common $(\xi, \sigma)$. Write also down the Wilks test, based on attainde log-likelihood maxima. (xx do both, for the mothers and babies story, for the three ethnic groups. xx)

**(d)** Consider a probability distribution $b = (b_1, \ldots, b_s)$ for suitable outcomes $1, \ldots, s$, and suppose we have multinomial estimators $\widehat{b}_j = (\widehat{b}_{j,1}, \ldots, \widehat{b}_{j,s})$ for each of groups $j = 1, \ldots, r$, resulting from sample sizes $n_1, \ldots, n_k$, with total sum $n$. In other words, $\widehat{b}_j = (N_{j,1}, \ldots, N_{j,s})/n_j$, for the multinomial experiment for group $j$. The hypothesis to be tested is that of equal probability distribution across groups. Explain that $\widehat{b}_j \approx_d {\rm N}_s(b, \Sigma/n_j)$, where $\Sigma_j$ has $b_j(1-b_j)$ on the diagonal and $-b_j b_\ell$ outside. Write $\Sigma_0$ for the $(s-1)\times(s-1)$ submatrix of $\Sigma$, needed since $\Sigma$ does not have full rank, with $\widehat{b}_0$ and $\widehat{b}_{j,0}$ similarly being the shorter $(s-1)$-length vectors, so that the general method above can be applied. Then show that the test statistic $Q_0$ from the general recipe above can be written

$$\sum_{j=1}^k n_j(\widehat{b}_{j,0} - \widehat{b}_0)^{\rm t}\widehat{\Sigma}_0^{-1}(\widehat{b}_{j,0} - \widehat{b}_0) = \sum_{j=1}^k n_j \sum_{r=1}^s \frac{(\widehat{b}_{j,r} - \widehat{b}_r)^2}{\widehat{b}_r} = \sum_{j=1}^k \sum_{r=1}^s \frac{(N_{j,r} - E_{j,r})^2}{E_{j,r}},$$

which then tends to $\chi^2_{(k-1)(s-1)}$ under the null. You may check the Pearson statistic from Story #59, and check that this precisely identical to the test above, for testing independence between the two factors 'group' and 'bin'.

**Ex. 4.43** *One-way layout.* (xx a point about $k = 2$ at the end, where $F$ i suppose becomes simply $t^2$, with usual t test. xx) The most prominent special case of the setup from Ex. 4.41 is that with an i.i.d. normal sample for each of the groups in question, with the same variance for all data. Consider therefore $Y_{j,1}, \ldots, Y_{j,n_j}$ i.i.d. from ${\rm N}(\xi_j, \sigma^2)$, for group $j$, and with overall sample size $n = \sum_{j=1}^k n_j$. This is the so-called normal one-way layout, with the chief hypothesis to be tested is $H_0\colon \xi_1 = \cdots = \xi_k$.

**(a)** Let $\bar{Y}_j = (1/n_j)\sum_{r=1}^{n_j} Y_{j,r}$ be the group averages. Explain that we for the overall average have $\bar{Y} = (1/n)\sum_{j=1}^k \sum_{r=1}^{n_j} Y_{j,r} = \sum_{j=1}^k (n_j/n)\bar{Y}_j$. Show then that $Q = \sum_{j=1}^k n_j(\bar{Y}_j - \bar{Y})^2$ has ${\rm E}\, Q = (k-1)\sigma^2 + \sum_{j=1}^k n_j(\xi_j - \bar{\xi})^2$, with $\bar{\xi} = \sum_{j=1}^k (n_j/n)\xi_j$. Explain that $Q \sim \sigma^2\chi^2_{k-1}$, under equality of means, so that the test statistic $W = Q/\widehat{\sigma}^2$ approximately has the $\chi^2_{k-1}$ distribution under $H_0$, as long as $\widehat{\sigma}$ is consistent. This is already a satisfactory answer to the one-way layout testing problem, without further fine-tuning.

**(b)** In this setup further fine-tuning is available, however. Show that $Q_0 = \sum_{j=1}^k \sum_{r=1}^{n_j} (Y_{j,r} - \bar{Y}_j)^2 \sim \sigma^2 \chi^2_{n-k}$, making $\widehat{\sigma}^2 = Q_0/(n-k)$ the natural unbiased estimator of $\sigma^2$. Explain also that $Q_0$ is independent of $Q$. Under the null of equal means, then, deduce that

$$F = \frac{Q/(k-1)}{Q_0/(n-k)} = \frac{Q/(k-1)}{\widehat{\sigma}^2} = \frac{\sum_{j=1}^k n_j(\bar{Y}_j - \bar{Y})^2/(k-1)}{\sum_{j=1}^k \sum_{r=1}^{n_j}(Y_{j,r} - \bar{Y}_j)^2/(n-k)}$$

has the F distribution $F_{k-1,n-k}$. Checking whether an observed $F$ is too big, compared to this null distribution, is then a way to assess the hypothesis $\xi_1 = \cdots = \xi_k$. For an illustration, see Story #5.

**(c)** (xx a short thing estimating contrasts. xx)

**Ex. 4.44** *Testing equality of multinormal means.* (xx a brief thing, used in Story #44, can point to ML theory too. establish the following, then apply in two-three settings. xx) Suppose $A \sim \mathrm{N}_p(a, \Sigma_1)$ and $B \sim \mathrm{N}_p(b, \Sigma_2)$ are independent multinormal data vectors, with known variance matrices. How can we test $a = b$? Show that $W = (B-A)^{\mathrm{t}}(\Sigma_1 + \Sigma_2)^{-1}(A-B) \sim \chi^2_p$ under the null.

**Ex. 4.45** *The Behrens–Fisher problem.* (xx to come here, making it reasonable simple. one point is that problems go away with reasonable sample sizes. xx)

**Further material and leftovers (as of 30-June-2025)**

## Notes and pointers

(xx confidence intervals. testing. connections. power. Neyman–Pearson. point to Lehmann. and to later chapters, Ch. 8 for CDs. point to interplay between modelling, probability calculus, thinking, a bit of philosophy, and practice. xx)

(xx one-way layout: primarily to test equality of means, so 'anova' is arguably a slight misnomer. xx)

Basu's lemma: from Basu (1955), see also Ghosh (2002).

ToDo notes, as of 30-June-2025:

Lots, though the chapter is shaping up. There are lots of 'test $\theta = \theta_0$ against $\theta > \theta_0$ prose in exercises, since it's easiest and cleanest, with NP etc. But we need to say a couple of times that all of this generalises to $\theta \neq \theta_0$ etc.

the Lindqvist and Taraldsen things, for simulating from $U(x) \,|\, \{V(x) = v\}$. do the handball model from CLP.

Do the bioequivalence: test the $H_0$ that $\theta$ is outside $[-\varepsilon, \varepsilon]$ against the alternative that it is inside. Diffent type of tests, and different looking power function.

Do sample size things, and efficiency; $\pi_n(\theta) \doteq \Phi(\sqrt{n}(\theta - \theta_0)/\sigma - z_0)$, with informative derivative $\phi(0)\sqrt{n}/\sigma$ at $\theta_0$. Efficiency things.

Make an example or two, perhaps with Cauchy again, to see that the confidence region might not be an interval.

Point to Cox (1958) and also interviews with him regarding conditional stuff.

# I.5

# Maximum likelihood

Consider the joint density of a dataset $Y$ from a parametric model, say $f_{\text{full}}(y, \theta)$. The likelihood function, a fundamental concept in parametric inference, is just this density, but seen as a function of $\theta$, with $y$ fixed at the observed dataset $y_{\text{obs}}$. In this chapter we go through the fundamental likelihood inference methods, in particular with results for the maximum likelihood estimator, the maximiser of the likelihood, and for the attained maximum value itself. It is useful to see this general likelihood theory as a special case of the more flexible machinery of minimum divergence methods, which we come back to in Ch. 6. The divergene associated with maximum likelihood theory is Kullback–Leibler. The likelihood methods are practical and versatile, as is demonstrated in several exercises and stories, also for say non-standard regression models. With models outside the most familiar ones, inference analysis essentially flows from being able to programme the log-likelihood function. Further material connected to likelihood theory are Cramér–Rao information inequalities, Wilks theorems. Crucially, the theory developed does not in general presuppose that the parametric model worked with is correct, as results are established both under and outside the precise model conditions. (xx perhaps there is room for empirical likelihood. xx)

*Key words:* Cramér–Rao lower bounds, Fisher information matrix, Kullback–Leibler, least false parameters, log-likelihood, maximum likelihood, regression models, sandwich matrix, score function, Wilks tests

In earlier chapters we have met and worked with several methods for estimation parameters in different settings. Sometimes one estimates population parameters directly from data, like the mean, the median, the standard deviation, the skewness, the correlation, the median difference between groups, etc., without necessarily using parametric models. Very frequently, however, the most fruitful data analyses involve fitting some parametric model to the data, as in regression models, where regression coefficients are estimated and assessed to learn how covariates influence the main outcomes. In Chs. 3-4 we have already worked with methods associated with moment fitting, quantile fitting, and least squares for regression models, but the present chapter has a wider aim and applicability, specifically developing methodology for *minimum divergence function estimators*, and its primary special case, *maximum likelihood estimators*. There is extensive use of likelihood theory in Chs. 7, 8, 11, 12, 13.

In general terms, suppose data $y$ have been modelled via a parametric model, leading to a joint probability density $f_{\text{full}}(y, \theta)$, with $\theta$ in some relevant parameter region $\Theta$. Then

the *likelihood function* is $L(\theta) = f_{\text{full}}(y_{\text{obs}}, \theta)$, studied as a function of the parameters, with $y$ held fixed at the observed $y_{\text{obs}}$. The maximum likelihood (ML) estimator is the value $\widehat{\theta}$ maximising the likelihood, or equivalently the *log-likelihood function* $\ell(\theta) = \log f_{\text{full}}(y_{\text{obs}}, \theta)$. The simplest setup is that of i.i.d. observations $Y_1, \ldots, Y_n$ to be fitted to some parametric $f(y, \theta)$, with $\theta$ a parameter vector inside its relevant parameter region. The log-likelihood function is then

the ML estimator

log-likelihood function

$$\ell_n(\theta) = \sum_{i=1}^{n} \log f(Y_i, \theta). \tag{5.1}$$

The apparatus of minimum divergence function and likelihood estimation carries over to regression models too, where the density of $Y_i$ given a covariate vector $x_i$ is modelled via some $f(y_i \,|\, x_i, \theta)$.

Generally speaking there are two valid viewpoints when developing the required theory. The first takes *the model to be correct*, so there is a true parameter $\theta_0$ to be estimated, with associated inference. The second takes the parametric model to be a *sensible approximation* to the real and unknown data-generating mechanism. With i.i.d. data, the model density $f(y, \theta)$ is consequently seen as an approximation to the real underlying density $g(y)$. Estimation and inference then involves the *least false parameter* $\theta_0$, in a suitable sense making $f(y, \theta_0)$ coming as close to the real $g(y)$ as possible.

the agnostic view, with least false parameter values

A generic and powerful statistical idea is to estimate parameters by minimising a relevant distance or divergence from the true mechanism to the data. The arguably most important estimation method, the *maximum likelihood method*, is the special case associated with the *Kullback–Leibler divergence* $\text{KL}(g, f_\theta)$. Apart from clarifying the concepts, defining parameters and estimators, in such ways, developing the theory amounts to establishing clear results for the behaviour and performance of these estimators and related data-driven tests, confidence methods, etc. This again involves limiting normality, assessing and estimating variability, and so on. All of this leads to a fruitful and indeed practical general theory, which also in new situations, with models constructed for new purposes, allows the statistician to estimate, to reach relevant confidence statements, test and compare, predict, etc. In brief, this chapter has these connected parts, with associated correlated groups of exercises.

more on this in Ch. 6

(i) Motivating and building an apparatus for defining parameters and their estimators via maximised likelihoods, initially for i.i.d. setups.

(ii) Analysing ML methods involves studying score functions, information functions, and the Fisher information matrix, for general parametric models. This again relates to 'information inequalities', specifically of the Cramér–Rao type, establishing lower bounds for how small variances can be, with a given model and a given sample size $n$. The ML methods achieve these lower bounds, under model conditions, for growing $n$.

(iii) Methods are developed for deriving the basic properties, concerning limiting normality of estimators, chi-squared type results for profiled versions for focus parameters, and quadratic forms for classes of test criteria, via the mechanics of minimisers of random functions.

(iv) Importantly, the i.i.d. setup is then lifted to classes of regression models. Intriguingly, with attention to certain crucial details, these extensions turn out to be not too strenuous, partly with Lindeberg theorem arguments replacing CLT details for the simpler case. The general likelihood theory makes it relatively straightforward to define and work with not only multiple linear regression, but logistic regression, Poisson regression,

gamma type regressions, and more.

Carrying out data analyses using tools from this chapter is often even surprisingly straightforward, even in new situations with new models. This is partly due the general well-working theory, but also to the modern conveniences of software packages for numerical optimisation, easy calculation of derivatives and second derivatives, and so on, after having programmed the basic empirical distance function to be used. For maximum likelihood methodology, in particular, this is showcased in Story #5, with a logistic full-data model for how covariates influences birthweights, in Story #26, with regression models for the number of bird species on islands outside Ecuador, and in Story #95, concerning models for time-to-failure of machine components, These applications are also meant to inspire the invention of new parametric models in new situations, perhaps along with context relevant distance functions.

### Kullback–Leibler, likelihood functions

**Ex. 5.1** *Maximising the log-likelihood.* Here we work through some examples of setting up the log-likelihood function and finding the ML estimator.

**(a)** Suppose $Y \sim \text{binom}(100, \theta)$ and that you observe $y_{\text{obs}} = 22$. Set up the log-likelihood function $\ell(\theta)$ and plot it. Show that its maximiser is $\widehat{\theta} = 0.22$.

**(b)** Simulate ten values of $Y$ from the binomial $(100, 0.25)$, and plot the resulting ten log-likelihood functions. Note how they vary, giving different ML estimates of the underlying true $\theta_0 = 0.25$. With your code, experiment a bit with different sizes of the binomial $n$. For bigger $n$, the peak is sharper; show this mathematically.

**(c)** With i.i.d. observations $Y_1, \ldots, Y_n$ from some parametric $f(y, \theta)$, show that the log-likelihood function becomes $\ell_n(\theta) = \sum_{i=1}^n \log f(Y_i, \theta)$. With such data points from the two-parameter normal $\text{N}(\xi, \sigma^2)$ model, show that the log-likelihood function may be written

$$\begin{aligned} \ell_n(\xi, \sigma) &= \sum_{i=1}^n \{-\log \sigma - \tfrac{1}{2}(y_i - \xi)^2/\sigma^2 - \tfrac{1}{2}\log(2\pi)\} \\ &= -n \log \sigma - \tfrac{1}{2}\{Q_0 + n(\xi - \bar{Y})^2\}/\sigma^2 - \tfrac{1}{2} n \log(2\pi), \end{aligned} \tag{5.2}$$

with $\bar{Y}$ the sample mean and $Q_0 = \sum_{i=1}^n (Y_i - \bar{Y})^2$. Show that the ML estimators are $\widehat{\xi} = \bar{Y}$ and $\widehat{\sigma}^2 = Q_0/n$. Verify also that the log-likelihood maximum value is $\ell_{n,\max} = -n \log \widehat{\sigma} - \frac{1}{2} n\{1 + \log(2\pi)\}$. Compared to the classical empirical variance $Q_0/(n-1)$, see Ex. 1.50, we learn that the ML estimator has a small negative bias, but that the difference is small for moderate to large $n$.

**(d)** With data $y_1, \ldots, y_{100}$ from the exponential density $\theta \exp(-\theta y)$, set up the log-likelihood function and find a formula for its maximiser $\widehat{\theta}$.

**(e)** Simulate 50 points from the uniform $[0, \theta]$, with true value $\theta_0 = 1$. Find and plot the log-likelihood function, and read off the ML estimate $\widehat{\theta}$.

**(f)** Simulate 100 values $y_i$ from the $\text{Gam}(a_0, 1)$ distribution (see Ex. 1.9), with e.g. $a_0 = 3.33$. Set up and plot the log-likelihood function. Set up a little experiment where you keep track of the ML estimators $\widehat{a}$ in repeated experiments from the samme $\text{Gam}(a_0, 1)$.

**Ex. 5.2** *The likelihood and log-likelihood functions.* Consider the one-parameter model with density $f(y, \theta) = \exp(-\theta y^{1/2})\theta/(2y^{1/2})$ for $y > 0$, and assume $n = 12$ data points have been observed:

```
0.233 0.334 0.067 0.148 0.007 0.639 0.017 0.298 0.030 0.120 0.061 0.063
```

**(a)** Show that $f(y,\theta)$ indeed is a density, write down the log-likelihood function $\ell_n(\theta)$, and show that it is maximised at $\widehat{\theta} = 1/W_n$, with $W_n = n^{-1}\sum_{i=1}^n y_i^{1/2}$. Find also a formula for the Hessian at the maximum position, $\widehat{J}_{\rm obs} = -\ell_n''(\widehat{\theta})$.

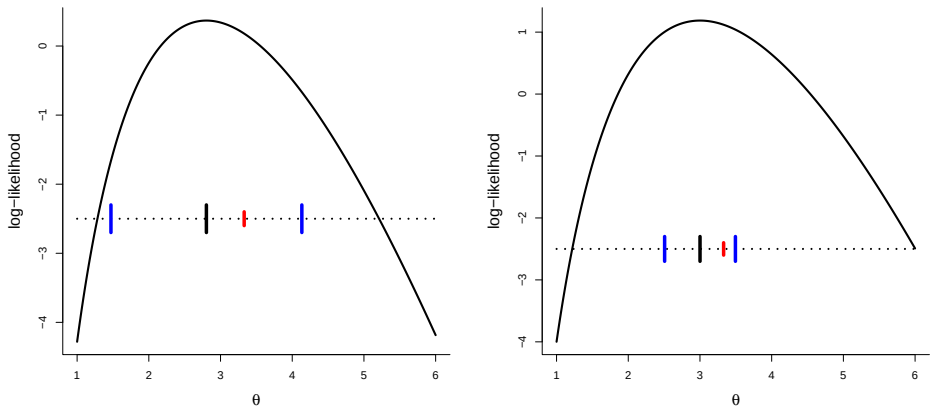



Figure 5.1: *Left panel: the log-likelihood function* $\ell_n(\theta)$ *for the simple* $n = 12$ *dataset of Ex. 5.2. The maximum is attained at* $\widehat{\theta} = 2.803$. *The blue bars indicate the 90 percent confidence interval coming from standard ML estimation theory, see Ex. 5.10, and is* $[1.472, 4.134]$. *The true value behind the data,* $\theta_0 = 3.33$, *is indicated as the red bar. Right panel: as for the left, but now with the bigger data set of* $n = 100$ *datapoints, from the same distribution, where the first 12 are as above. The minus second derivative* $\widehat{J}$ *at the ML position has increased from 1.527 to 11.103, causing the ML based confidence interval to become considerably tighter, and is* $[2.507, 3.494]$.

**(b)** Find the limit distribution of $\sqrt{n}(W_n - 1/\theta)$, using the CLT of Ch. 2, and use the delta method to find that the limit distribution of $\sqrt{n}(\widehat{\theta} - \theta)$ is $\mathrm{N}(0, \theta^2)$. You may verify already that this is what comes out of the general ML theory developed below, see e.g. Ex. 5.10.

**(c)** Anticipating general ML theory to come, check again with Ex. 5.10, it will be seen that $\widehat{\theta} \approx_d \mathrm{N}(\theta, 1/\widehat{J})$, under model conditions. Explain that this leads to approximate confidence intervals of the type $\widehat{\theta} \pm z_0/\widehat{J}^{1/2} = \widehat{\theta}(1 \pm z_0/\sqrt{n})$, with $z_0$ the appropriate normal quantile. We've actually simulated these few data points above from the model, with true parameter $\theta_0 = 3.33$. Construct a version of Figure 5.1, left panel.

**(d)** In general the ML estimator might have a complicated distribution (though it is approximately normal, as we have seen here). In this particular model its precise distribution may be worked out, however; show that $\widehat{\theta} \sim \theta\,(2n)/\chi^2_{2n}$. Use this to find a precise 90 confidence interval for $\theta$, and compare to the approximation given above.

**(e)** To simulate data from this model, show that $Y_i$ is equal in distribution to $(V_i/\theta_0)^2$, with the $V_i$ i.i.d. from the unit exponential. Make a computer programme to simulate $n$ points from the $f(y,\theta_0)$ model, and which then finds the log-likelihood function, the ML estimator, and the approximate 90 percent confidence interval, as above. Run such a programme for say 88 more points, forming a bigger dataset with $n = 100$ datapoints, and

comment on what you find. Produce a version of Figure 5.1, right panel. Comment on the main features here, including that $\widehat{J}$ becomes bigger with more data, yielding sharper confidence intervals. We would usually plot the log-likelihood and related aspects, as the confidence curves of Ch. 8 for a shorter range of parameter values than in this right panel, but we here choose to plot using the same range for both $n = 12$ and $n = 100$.

**Ex. 5.3** *The Kullback–Leibler divergence and the maximum likelihood method.* With i.i.d. data $Y_1, \ldots, Y_n$ from a density $g$, to be approximated with a parametric $f_\theta$, the particularly important *maximum likelihood estimation method* is worked with here, with $\widehat{\theta}$ the maximiser of $\ell_n(\theta) = \sum_{i=1}^n \log f(Y_i, \theta)$. It will be seen as the natural cousin to the Kullback–Leibler divergence as a measure of distance from the true density to the parametric approximation. Development in exercises below give a precise description of how the method actually behaves.

maximum likelihood

**(a)** For two densities $g$ and $f$, defined on a common support, the Kullback–Leibler distance, interpreted to be 'from the first density to the second', is

the Kullback–Leibler distance

$$\mathrm{KL}(g, f) = \int g \log \frac{g}{f} \, \mathrm{d}y. \tag{5.3}$$

It is an important concept and tool for communication and information theory, as for probability theory and statistics. The $\log(g/f)$ term will be both positive and negative, in different parts of the domain. Show nevertheless that indeed $\mathrm{KL}(g, f) \geq 0$, perhaps via the Jensen inequality, and that $\mathrm{KL}(g, f) = 0$ only when the two densities are equal a.e. In Ex. 5.4 we learn more details about the KL distance, and look into illustrations, but here the main point is to see its close connection to ML estimation.

**(b)** To get to know the KL distance, work with normal distributions. (i) For $g = \mathrm{N}(a, \sigma^2)$ and $f = \mathrm{N}(b, \sigma^2)$, show that $\mathrm{KL}(g, f) = \frac{1}{2}(b-a)^2/\sigma^2$, symmetric and a function of $|b-a|/\sigma$. (ii) For $g = \mathrm{N}(0, \sigma_1^2)$ and $f = \mathrm{N}(0, \sigma_2^2)$, show that $\mathrm{KL}(g, f) = \frac{1}{2}h(\sigma_1^2/\sigma_2^2)$, with $h(u) = u - \log u - 1$ a nonnegative function, but not symmetric around 1. So KL is not in general symmetric, and it is fruitful to view it as the distance *from $g$ to $f$*.

**(c)** Explain that the normalised log-likelihood $n^{-1}\ell_n(\theta)$ for each $\theta$ tends a.s. to the function $\mathrm{E}_G \log f(Y, \theta) = \int g \log f_\theta \, \mathrm{d}y$. When regularity conditions are in place to imply convergence of maximisers (we tend to such in exercises belos), then, explain that $\widehat{\theta} \to_{\mathrm{pr}} \theta_0 = \theta_0(G)$, the minimiser of $\mathrm{KL}(g, f_\theta)$. Explain that if the model is correct, then $\theta_0$ is simply the true parameter; when the model is not perfect, we call the $\theta_0$ the *least false parameter value*.

least false parameter value

**(d)** As a simple illustration (to be returned to in Ex. 6.4), generate say $n = 100$ points from the uniform distribution, and use the parametric density $f(y, \theta) = \theta y^{\theta-1}$ on the unit interval. Write down and plot the log-likelihood function, and find the ML estimate. Also show that the implied best parameter value, if the data stem from some $g$, rather than from the model, is $\theta_0 = 1/\mathrm{E}_g \log(1/Y)$.

**Ex. 5.4** *The Kullback–Leibler distance: details and illustrations.* (xx repair here. xx) In Ex. 5.3 we have seen that the machinery of maximum likelihood is intimately related to the KL distance $\mathrm{KL}(g, f) = \int g \log(g/f) \, \mathrm{d}y$. Here we work on illustrations to learn more.

**(a)** Find the KL distance from one Poisson to another.

**(b)** The KL distance is also perfectly well-defined and meaningful in higher dimension. Show that the KL distance from $\mathrm{N}_p(\xi_1, \Sigma)$ to $\mathrm{N}_p(\xi_2, \Sigma)$ can be expressed as $\frac{1}{2}\delta^2$, where

the Mahalanobis distance

$\delta = \{(\xi_2-\xi_1)^{\rm t}\Sigma^{-1}(\xi_2-\xi_1)\}^{1/2}$ is the so-called Mahalanobis distance between the two populations.

**(c)** Start from $d(g,f) = -\int g\log\{1+(f/g-1)\}\,{\rm d}y$, for densities which are not far from each other, and use Taylor expansion to find

$$\mathrm{KL}(g,f) \approx \tfrac12\int g(f/g-1)^2\,{\rm d}y = \tfrac12\Big(\int f^2/g\,{\rm d}y - 1\Big).$$

(xx some words indicating that the root-KL might have an easier interpretation. xx)

**(d)** (xx a bit of text, more than a question. xx) As noted the KL distance is not symmetric, so 'distance' has a direction. In various statistical setups it makes sens to interpret $d(g,f)$ as the the distance from 'home density $g$' to 'approximation candidate $f$'. As also becoming clear from examples above, it's somehow quadratic in nature, so when numbers are involved, measuring the KL distances, it would typically make more sense to give their square roots, as with $\{d(g,f_\theta)\}^{1/2}$, the degree of closeness of the parametric approximant $f_\theta$ to the ground truth $g$.

**Ex. 5.5** *KL approximation.* (xx to be edited. xx) For the following cases the point is to set up a data generating density $g$, and then check how well a certain parametric family $f(y,\theta)$ does the approximation job. For each case, this tells us how well the ML can do its job, with enough data. For the various cases, find the minimiser, i.e. the best approximation; find the minimum square-root distance $d(g,f(\cdot,\theta_0))^{1/2}$ (since this gives a better picture than on the KL scale itself); and plot the true $g$ alongside the parametric approximant.

**(a)** Let $g = 0.33\,\mathrm{N}(-1,1) + 0.67\,\mathrm{N}(1,1)$. Find the best normal approximation.

**(b)** Let $g$ be a Gamma with parameters $(2.22, 3.33)$. Find the best Weibull approximant, and also the best log-normal approximant. Similarly, start with a Weibull distribution, with parameters say $(3.33, 2.22)$, and find the best Gamma distribution approximation.

**(c)** Let $g = 0.95\,\mathrm{Expo}(1) + 0.05\,\mathrm{Expo}(0.01)$, which roughly means that about five percent of the data come from a distribution which much higher mean than the mainstream exponential data. Find the best exponential model approximation, and also the best Gamma and Weibull approximations. Display the true $g$ and these three best parametric approximations in the same diagram.

**(d)** Suppose data really come from $\mathrm{N}(0.333,\sigma_1^2)$, with $\sigma_1 = 1.111$, where a statistician fits the simpler $\mathrm{N}(0,\sigma^2)$ model. First, find out what happens to the ML estimator. Secondly, illustrate 'what goes on' by drawing e.g. ten samples of size $n = 50$ from the true density, and then display the ten versions of $n^{-1}\ell_n(\sigma)$, along with its limit $C(\sigma)$. Comment on your findings.

### Score function, Fisher information matrix, Cramér–Rao lower bounds

**Ex. 5.6** *Score functions and the Fisher information matrix.* Consider a parametric model with density $f(y,\theta)$ with respect to some measure $\mu$, where $\theta = (\theta_1,\ldots,\theta_p)^{\rm t}$, the parameter of the model is contained in some open parameter space $\Theta \subset \mathbb{R}^p$. Introduce score function

$$u(y,\theta) = \partial\log f(y,\theta)/\partial\theta \quad\text{and}\quad i(y,\theta) = \partial^2\log f(y,\theta)/\partial\theta\,\partial\theta^{\rm t},$$

called the *score function*, and is a $p$-dimensional column vector, and the *information function*, a $p\times p$ matrix. For the theory developed in this exercise, these partial derivatives

are assumed to exist and, they must be continuous in $\theta$. We also assume that the *support* for the distribution, the smallest closed set for which the density is positive, does not depend on $\theta$. Cases falling outside such assumptions are, for example, the uniform distribution on an unknown interval $[0, \theta]$.

first Bartlett identity

**(a)** Assume that $(\partial/\partial\theta)\int f(y,\theta)\,\mathrm{d}\mu(y) = \int(\partial/\partial\theta)f(y,\theta)\,\mathrm{d}\mu(y)$. Show that the score function has mean zero,

$$\mathrm{E}_\theta\, u(Y,\theta) = \int f(y,\theta)u(y,\theta)\,\mathrm{d}\mu(y) = 0.$$

Let next

$$K(\theta) = \mathrm{Var}_\theta\, u(Y,\theta) \quad \text{and} \quad J(\theta) = -\mathrm{E}_\theta\, i(Y,\theta),$$

second Bartlett identity

Assume also that $(\partial^2/\partial\theta\partial\theta^{\mathrm{t}})\int f(y,\theta)\,\mathrm{d}\mu(y) = \int(\partial^2/\partial\theta\partial\theta^{\mathrm{t}})f(y,\theta)\,\mathrm{d}\mu(y)$, and show that indeed $K(\theta) = J(\theta)$. This matrix is *the Fisher information matrix* for the model. It provides a measure of how much information about a parameter a data set provides. Note that the calculation of both $J(\theta)$ and $K(\theta)$ is taking place under the assumption that the model is actually correct.

**(b)** Consider a parametric density $f(y,\theta)$, with score function $u(y,\theta) = \partial \log f(y,\theta)$ and information matrix $J(\theta) = \mathrm{Var}_\theta\, u(Y,\theta)$; see Ex. 5.6 for more on these. Show here that

$$\mathrm{KL}(f(\cdot,\theta), f(\cdot,\theta+\varepsilon)) = \tfrac{1}{2}\varepsilon^{\mathrm{t}} J(\theta)\varepsilon + O(\varepsilon^3).$$

**(c)** Show that if $\theta \mapsto (\partial/\partial\theta)f(y,\theta)$ is continuous for each $y$, and each element of $(\partial/\mathrm{d}\theta)f(y,\theta)$ is bounded by an integrable function not depending on $\theta$, then the first partial derivative can be passed under the integral sign in $\int f(y,\theta)\,\mathrm{d}\mu(y)$. Deduce conditions under which we may pass the second partial derivative under the same integral sign.

**(d)** For the exponential model, with density $\theta\exp(-\theta y)$, find the score function, and compute the Fisher information function in two ways. The second (derivative) way of computing the Fisher information here was quite simple.

**(e)** Consider the general exponential family, with its natural parametrisation $f(y,\theta) = \exp\{\theta^{\mathrm{t}}T(y) - k(\theta)\}h(y)$, see Ex. 1.56. Explain that the score function becomes $T(y) - k'(\theta)$, with Fisher matrix $J(\theta) = k''(\theta)$, that of the second order derivatives of $k(\theta)$.

**(f)** For the normal $\mathrm{N}(\xi,\sigma^2)$ model, show that the score function can be expressed as

$$u(y,\xi,\sigma) = \begin{pmatrix} \frac{1}{\sigma}(y-\xi)/\sigma \\ \frac{1}{\sigma}\{(y-\xi)^2/\sigma^2 - 1\} \end{pmatrix} = \frac{1}{\sigma}\begin{pmatrix} z \\ z^2-1 \end{pmatrix},$$

writing $z = (y-\xi)/\sigma$, which is a standard normal when $y$ comes from the model. Demonstrate that the Fisher information matrix becomes

$$J(\xi,\sigma) = \mathrm{Var}_{\xi,\sigma}\, u(Y,\xi,\sigma) = \begin{pmatrix} 1/\sigma^2 & 0 \\ 0 & 2/\sigma^2 \end{pmatrix}.$$

**(g)** (xx Check with a few more of your favourite parametric models, where you find the score function and the information function, and where then formulae for both $J(\theta)$ and the variance matrix $K(\theta)$ of the score function, verifying that they are the same. ask for poisson, for binomial with parametrisation $p = \exp(\theta)/\{1+\exp(\theta)\}$, geometric. xx)

**(h)** If $Y$ has the uniform distribution on $[0, \theta]$, which of the regularity conditions listed above fail?

**Ex. 5.7** *Cramér–Rao lower bounds for estimators.* A certain basic and classic inequality provides a lower bound for the variance of any unbiased estimator of a given parameter. There are various versions and generalisations, some of which we go through here. Inequalities of the type encountered here are sometimes called 'information inequalities', as they may be used to define and analyse how much information there can be in a finite set of data. We also discover clear links to log-likelihoods and ML estimation; the behaviour of ML estimators for growing sample size, described in Ex. 5.10, exactly matches the Cramér–Rao lower bound for variances.

**(a)** To begin simply, suppose $Y$ is an observation from the density $f(y, \theta)$, assumed smooth in its one-dimensional parameter. Let $u(y, \theta) = \partial \log f(y, \theta)/\partial\theta$ be the score function, with finite variance, by definition equal to the Fisher information, $J(\theta) = \int f(y, \theta)u(y, \theta)^2 \, \mathrm{d}y$. Let $T = T(Y)$ be any estimator unbiased for $\theta$, and assume that $f(y, \theta)$ satisfies the conditions of Ex. B.14(a). From $\mathrm{E}_\theta T = \int T(y)f(y, \theta) \, \mathrm{d}y = \theta$, use the conditions on $f(y, \theta)$ to deduce that $(\mathrm{d}/\mathrm{d}\theta) \int T(y)f(y, \theta)u(y, \theta) \, \mathrm{d}y = 1$. Show that $\mathrm{cov}_\theta \, (T, u(Y, \theta)) = 1$, and, consequently $1 \leq \mathrm{Var}_\theta \, T \, \mathrm{Var} \, u(Y, \theta)$, from which the we get one-dimensional classic Cramér–Rao inequality

$$\mathrm{Var}_\theta \, T \geq 1/J(\theta).$$

**(b)** It is easy to generalise the above to the more interesting case of having more observation than one. Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the parametric model $f(y, \theta)$. Show that the arguments above still hold, essentially since $Y = (Y_1, \ldots, Y_n)$ can be considered a single datum from the model with joint density $f(y_1, \theta) \cdots f(y_n, \theta)$. Show that the score function now becomes $u(y_1, \ldots, y_n, \theta) = \sum_{i=1}^n u_0(y_i, \theta)$, writing for emphasis $u_0(y, \theta) = \partial \log f(y, \theta)/\partial\theta$ for the score function for a single observation. Deduce that the combined Fisher information for the full sample is $J_n = \mathrm{Var}_\theta \, u(Y_1, \ldots, Y_n) = nJ_0$, with $J_0 = \mathrm{Var}_\theta \, u_0(Y_i, \theta)$ the information in a single observation.

**(c)** Show from this that if $T = T(Y_1, \ldots, Y_n)$ is an unbiased estimator for $\theta$, then

Cramér–Rao lower bound

$$\mathrm{Var}_\theta \, T \geq \frac{1}{nJ_0(\theta)} = \frac{1/J_0(\theta)}{n}.$$

This says that there is a clear limit to how well one might estimate a parameter in a model, with $n$ observations. If you're not entirely satisfied with $\mathrm{Var}_\theta \, T = 0.10$, say, and wish for variance 0.05 instead, then shell out more money to get twice as many observations.

**(d)** Show more generally that if $T = T(Y_1, \ldots, Y_n)$ is an estimator for $\theta$, with mean $\mathrm{E}_\theta \, T = \theta + b(\theta)$, i.e. with a certain bias $b(\theta)$, then

$$\mathrm{Var}_\theta \, T \geq \frac{1}{n} \frac{\{1 + b'(\theta)\}^2}{J_0(\theta)}.$$

In particular, show that there's a lower bound on the mean squared error for *any* estimator (i.e. not merely the unbiased ones):

$$\mathrm{mse}(\theta) = \mathrm{E} \, \{T(Y_1, \ldots, Y_n) - \theta\}^2 \geq n^{-1}\{1 + b'(\theta)\}^2/J_0(\theta) + b(\theta)^2.$$

**(e)** Go through the following examples, in each case finding the score function, the information $J_0(\theta)$, and the lower bound for any unbiased estimator of the model parameter.

(i) $y$ is binomial $(n, \theta)$. (ii) $y$ is Poisson $\theta$. (iii) $y$ is normal $(\theta, \sigma^2)$, with $\sigma$ known. (iv) $y$ is normal $(\theta, \sigma^2)$, with $\theta$ known, and $\sigma$ to be estimated. Comment on the implications of your findings.

**Ex. 5.8** *Cramér–Rao bounds for the multidimensional case.* In generalisation of the above situation to the case of multiparameter models, assume first that $y$ is a single observation from the model $f(y, \theta)$, with $\theta = (\theta_1, \ldots, \theta_p)$ of dimension $p$. Let $u_0(y, \theta) = \partial \log f(y, \theta)/\partial\theta$ be the score function, for such a single $y$, with the $p \times p$ Fisher information matrix $J_0(\theta) = \mathrm{Var}_\theta\, u_0(Y, \theta)$ assumed positive definite.

**(a)** For a symmetric $p \times p$ matrix $A$ we write $A \geq 0$ provided it is nonnegative definite, i.e. that $c^{\mathrm{t}} A c \geq 0$ for all $c$. Show that the covariance matrix of a random vector is necessarily nonnegative, and that $A \geq 0$ is is equivalent to its eigenvalues being nonnegative. Explain that $a_{ii} \geq 0$, for the diagonal elements, but that we may still have $a_{i,j} < 0$ for some off-diagonal elements. With two symmetric matrices, we write $A \geq B$ if $A - B \geq 0$, which is the ordering of variance matrices we use below.

**(b)** Assume that $T = T(Y)$ is an unbiased estimator of $\theta$, which also means that $\mathrm{E}_\theta\, T_j(Y) = \theta_j$ for each component $j$. With $I_p$ the identity matrix of size $p \times p$, deduce from $\mathrm{E}_\theta\, T = \int T(y) f(y, \theta)\, \mathrm{d}y$ that

$$(\partial/\partial\theta)\, \mathrm{E}_\theta\, T = \int T(y) f(y, \theta) u_0(y, \theta)^{\mathrm{t}}\, \mathrm{d}y = I_p.$$

**(c)** Then work out that

$$\mathrm{Var}_\theta\, \{T - J_0(\theta)^{-1} u_0(Y, \theta)\} = \mathrm{E}_\theta\, \{T - \theta - J_0(\theta)^{-1} u_0(Y, \theta)\}\{T - \theta - J_0(\theta)^{-1} u_0(Y, \theta)\}^{\mathrm{t}}$$

can be expressed as $\mathrm{Var}_\theta\, T - J_0(\theta)^{-1}$. We have then shown a multidimensional version of the Cramér–Rao inequality, that $\mathrm{Var}_\theta\, T \geq J_0(\theta)^{-1}$.

**(d)** Generalise the above to the case of $n$ i.i.d. observations $Y_1, \ldots, Y_n$ from the model. Show that the information matrix for the full data set becomes

$$J_n(\theta) = \mathrm{Var}_\theta\, \frac{\partial \log\{f(Y_1, \theta) \cdots f(Y_n, \theta)\}}{\partial\theta} = n J_0(\theta),$$

Cramér–Rao lower bound, matrix case

and that for *any* unbiased estimator $T = T(Y_1, \ldots, Y_n)$ of $\theta$, we must have

$$\mathrm{Var}_\theta\, T \geq \{n J_0(\theta)\}^{-1} = n^{-1} J_0(\theta)^{-1}.$$

More generally, if independent observations $Y_1, \ldots, Y_n$ come from densities $f_1(y, \theta), \ldots, f_n(y, \theta)$, with Fisher information matrices $J_1(\theta), \ldots, J_n(\theta)$, show that any unbiased estimator $T$ of $\theta$ has $\mathrm{Var}_\theta\, T \geq \{J_1(\theta) + \cdots + J_n(\theta)\}^{-1}$.

**(e)** Suppose now that $\mu = \mu(\theta_1, \ldots, \theta_p)$ is a one-dimensional parameter in focus, and that $T = T(Y)$ is an unbiased estimator. With $c(\theta) = \partial\mu(\theta)/\partial\theta$ the $p \times 1$ gradient, show that

$$\mathrm{Var}_\theta\, \{T - c(\theta)^{\mathrm{t}} J(\theta)^{-1} u_0(Y)\} = \mathrm{Var}_\theta\, T - c(\theta)^{\mathrm{t}} J(\theta)^{-1} c(\theta),$$

and conclude the lower bound $\mathrm{Var}_\theta\, T \geq \lambda(\theta) = c(\theta)^{\mathrm{t}} J(\theta)^{-1} c(\theta)$. Generalise to the case of $Y_1, \ldots, Y_n$ being i.i.d. from $f(y, \theta)$: if $T_n = T(Y_1, \ldots, Y_n)$ is unbiased for $\mu$, show that $\mathrm{Var}_\theta\, T \geq \lambda(\theta)/n$. We learn also a bit more from these arguments. Show that

$$\mathrm{Var}_\theta \Big\{ T_n - n^{-1} \sum_{i=1}^n c(\theta)^{\mathrm{t}} J(\theta)^{-1} u_0(Y_i, \theta) \Big\} = \mathrm{Var}_\theta\, T_n - \lambda(\theta)/n,$$

so $T_n$ having a variance coming close to the lower bound means $T_n - \mu$ being equal to or close to the random variable $n^{-1}\sum_{i=1}^n c(\theta)^{\rm t} J(\theta)^{-1} u_0(Y_i,\theta)$.

**(f)** (xx something here, tying these matters to ML. consider $T_n^* = \theta + n^{-1} J(\theta)^{-1} \sum_{i=1}^n u_0(Y_i,\theta)$. show that this $T_n^*$ is unbiased and achieves the Cramér–Rao lower bound. this happens in exponential families, give pointer. the point is then that ML in the general case is close to this with growing $n$. xx)

**(g)** Also other estimators for $\theta$ deserve to be studied, even when they are not exactly unbiased. We start with a single observation $Y$ from $f(y,\theta)$, with score function $u_0(y,\theta)$ as above, and then generalise to $n$ observations afterwards. Assume therefore that $T = T(Y)$ is such that

$$\mathrm{E}_\theta\, T = \int T(y) f(y,\theta)\,{\rm d}y = \theta + b(\theta),$$

with components $\theta_j + b_j(\theta)$, for suitable bias functions $b_1(\theta),\ldots,b_p(\theta)$, perhaps not far from zero. Show that

$$(\partial/\partial\theta)\,\mathrm{E}_\theta\, T = \begin{pmatrix} 1+\partial b_1(\theta)/\partial\theta \\ \vdots \\ 1+\partial b_p(\theta)/\partial\theta \end{pmatrix} = \int T(y) f(y,\theta) u_0(y,\theta)^{\rm t}\,{\rm d}y.$$

Then work with $\mathrm{Var}_\theta\,[T - \{I_p + b'(\theta)\}J_0(\theta)^{-1}u_0(Y,\theta)]$ to demonstrate that

$$\mathrm{Var}_\theta\, T \geq \{I_p + b'(\theta)\}J_0(\theta)^{-1}\{I + b'(\theta)\}^{\rm t}.$$

**(h)** Generalise to the case of $n$ i.i.d. observations, to reach

$$\mathrm{Var}_\theta\, T \geq n^{-1}\{I_p + b'(\theta)\}J_0(\theta)^{-1}\{I + b'(\theta)\}^{\rm t}.$$

**(i)** (xx a bit more to round it off. an example or two. CR lower bound not always attained, in some models only for growing $n$, but that's ok. i make a separate point that the arguments also lead to bounds of type

$$\mathrm{Var}_\theta\, T \geq \{J_1(\theta) + \cdots + J_n(\theta)\}^{-1},$$

in cases with different situations or types of information sources, for the same $\theta$. tie it all to the large-sample ML results. xx)

**Ex. 5.9** *Fisher information and parameter transformations.* Consider a parametric model with density $f(y,\theta)$, with $\theta$ of length $p$, say. We have seen the central role the Fisher information matrix $J(\theta)$ plays, with more to come for the likelihood machinery below. Here we look into reparametrisation, going from $\theta = (\theta_1,\ldots,\theta_p)$ to say $\phi = (\phi_1,\ldots,\phi_p)$, with smooth 1-1 transformations $\theta = h(\phi)$, or $\phi = h^{-1}(\theta)$. It is assumed that the $p\times p$ matrix $G(\phi) = \partial h(\phi)/\partial\phi$, with rows $\partial h_j(\phi)/\partial\phi$, exists and is non-zero throughout the parameter space.

**(a)** Let $u(y,\theta) = \partial \log f(y,\theta)/\partial\theta$ be the score function with the $\theta$ parametrisation. Show that for the $\phi$ parametrisation, we have

$$u^*(y,\phi) = \frac{\partial \log f(y,h(\phi))}{\partial\phi} = \begin{pmatrix} u_1(y,h(\phi))^{\rm t} g_1(\phi) \\ \vdots \\ u_p(y,h(\phi))^{\rm t} g_p(\phi) \end{pmatrix} = G(\phi)u(y,h(\phi)).$$

**(b)** Show from this that the Fisher information matrix for the $\phi$ parametrisation becomes $J^*(\phi) = G(\phi)J(h(\theta))G(\phi)^{\rm t}$. Deduce also that the important inverse can be written $J^*(\phi)^{-1} = (G(\phi)^{\rm t})^{-1}J(h(\theta))^{-1}G(\phi)^{-1}$.

**(c)** As illustrations, consider the binomial $(n,\theta)$, with reparametrisations (i) $\phi_1 = \theta^{1/2}$; (ii) $\phi_2 = \log(\theta/(1-\theta))$, so that $\theta = \exp(\phi)/\{1+\exp(\theta)\}$; (iii) $\phi = 2\arcsin(\theta^{1/2})$. For each of these, find expressions for the Fisher information $J^*(\phi)$.

**(d)** Consider the normal $(\xi,\sigma^2)$. Find first the Fisher information matrix for the $(\xi,\sigma)$ parametrisation, then for the $(\xi,\lambda)$ parametrisation, where $\lambda = \sigma^2$.

### Maximum likelihood, profiled likelihood, deviance theorems

**Ex. 5.10** *Maximum likelihood estimators: a limit process and main heuristics.* After having motivated and worked through particular instances of the maximum likelihood function estimators, we now return to the general case, aiming to demonstrate both limiting normality and other associated results, finding recipes for large-sample approximations in the process. The aim now is to go through the basic ideas, involving also two main heuristics. Similar arguments will work also for rather more general estimation methods in Ch. 6. To make these heuristics precise we need more conditions and details, to which we return in further exercises below.

Setting the scene, we have $Y_1,\ldots,Y_n$ i.i.d. from some $G$, and we fit a parametric model $f(y,\theta)$ smooth in its parameters. Let $\theta_0 = \theta_0(G)$ be the least false parameter value, the maximiser of $\mathrm{E}_G \log(Y,\theta)$, and $\widehat\theta$ is its estimator, the maximiser of $\ell_n(\theta) = \sum_{i=1}^n \log f(Y_i,\theta)$. To present the basic ideas and main heuristics we shall start with these regularity conditions: (i) The true $\theta_0 = \theta_0(G)$ is an inner point in its parameter space inside $\mathbb{R}^p$. (ii) The $\log f(y,\theta)$ is smooth in $\theta$, in a neighbourhood around $\theta_0$, with at least two derivatives, i.e. the score function $u(y,\theta)$ and information function $i(y,\theta)$. (iii) The matrix $J = -\mathrm{E}_G\, i(Y,\theta_0)$ is finite and positive definite. (iv) The score function $u(Y,\theta_0)$ has finite variance matrix $K$.

regularity conditions

**(a)** A key idea is to work with the following random function. Write

$$A_n(s) = \ell_n(\theta_0 + s/\sqrt{n}) - \ell_n(\theta_0) = U_n^{\rm t}s - \tfrac12 s^{\rm t}J_n s + r_n(s),$$

with $U_n = (1/\sqrt{n})\sum_{i=1}^n u(Y_i,\theta_0)$, $J_n = -n^{-1}\sum_{i=1}^n i(Y_i,\theta_0)$. Show that $A_n(s)$ is properly defined for all large enough $n$, that $U_n$ has mean zero and tends to $U \sim \mathrm{N}_p(0,K)$, and that $J_n \to_{\rm pr} J$.

**(b)** It is also clear from the Taylor expansion argument that with moderate further regularity, the remainder term $r_n(s)$ will go to zero in probability, for each $s$. There is then convergence

$$A_n(s) \to_d A(s) = U^{\rm t}s - \tfrac12 s^{\rm t}Js, \tag{5.4}$$

for each $s$. Heuristic One is then to go from $A_n \to_d A$ to $\mathrm{argmax}(A_n) \to_d \mathrm{argmax}(A)$. Explain that this then leads to

$$\sqrt{n}(\widehat\theta - \theta_0) \to_d J^{-1}U, \quad \text{which is a } \mathrm{N}_p(0, J^{-1}KJ^{-1}).$$

We call this limit distribution variance matrix $J^{-1}KJ^{-1}$ the sandwich matrix.

the sandwich matrix

**(c)** Additionally, Heuristic Two is to go from $A_n \to_d A$ to $\max(A_n) \to_d \max(A)$. Argue that this entails

$$W_n = 2\{\ell_{n,\max} - \ell_n(\theta_0)\} \to_d W = U^{\rm t}J^{-1}U.$$

Show for this quadratic form that its mean and variance are $\mathrm{Tr}(J^{-1}K)$ and $2\,\mathrm{Tr}(J^{-1}KJ^{-1}K)$.

**(d)** For a simple illustration, consider data from a density $g$ on the positive half-line, with true mean and variance $\xi$ and $\sigma^2$. Fitting the data to the exponential $f(y,\theta) = \theta\exp(-\theta y)$, show first that the ML estimator aims at $\theta_0 = 1/\xi$. Then show $J = 1/\theta_0^2$ and $K = \sigma^2$, with sandwich $\Sigma = J^{-1}KJ^{-1} = \sigma^2/\xi^2$. Explain that if the model is correct, then $\sigma^2 = 1/\theta_0^2$, with $\Sigma = 1/\theta_0^2$.

**(e)** Back to the general narrative, assume the model holds, with some true parameter $\theta_0$, and Fisher information matrix $J = J(\theta_0)$, as per Ex. 5.6. Derive the celebrated and exceedingly useful results, for maximum likelihood, that

$$\sqrt{n}(\widehat{\theta}-\theta_0) \to_d \mathrm{N}_p(0,J^{-1}) \quad \text{and} \quad W_n = 2\{\ell_{n,\max}-\ell_n(\theta_0)\} \to_d \chi^2_p. \tag{5.5}$$

Explain that the limit distribution variance matrix precisely matches the best one could have hoped for, from the Cramér–Rao information inequalities of Ex. 5.7-5.8.

but beware Hodges

**Ex. 5.11** *Maximum likelihood: from heuristics to proofs.* (xx nils and emil check details carefully. repair and edit. xx) The general setup is as with the previous Ex. 5.10, with ML estimator $\widehat{\theta}$ maximising $\ell_n(\theta) = \sum_{i=1}^n \log f(Y_i,\theta)$, with observations coming from $G$. In addition to regulatity conditions (i)–(iv) given there, we postulate (v), that also third derivatives of $\log f(y,\theta)$ exist, and that these have finite means in a neighbourhood around $\theta_0$. We shall use the process $A_n(s) = \ell_n(\theta_0+s/\sqrt{n})-\ell_n(\theta_0)$ from the previous exercise. Another set of sufficient regularity conditions, in various cases simpler and cleaner, is worked with in Ex. 5.29–5.30 below.

**(a)** Explain first that the maximiser of $A_n$ is $\alpha_n = \sqrt{n}(\widehat{\theta}-\theta_0)$, where we shall also study the overall maximum $A_{n,\max} = A_n(\alpha_n)$ below. Let $B_n(s) = U_n^{\mathrm{t}}s - \frac{1}{2}s^{\mathrm{t}}J_n s$ be the quadratic approximation to $A_n$, with maximiser $\beta_n$ and overall maximum $B_{n,\max} = \max\{B_n(s)\colon \text{all } s\}$. Show that

$$\begin{aligned}\beta_n &= J_n^{-1}U_n \to_d J^{-1}U \sim \mathrm{N}_p(0,J^{-1}KJ^{-1}),\\ B_{n,\max} &= \tfrac{1}{2}U_n^{\mathrm{t}}J_n^{-1}U_n \to_d \tfrac{1}{2}U^{\mathrm{t}}J^{-1}U.\end{aligned}$$

So things are simple and clear for the quadratic approximation $B_n$; we need to show that the same results obtain for the real thing, the $A_n$.

**(b)** Supposing $|r_n(s)| \le \delta$ for all $s$ in a subset $S$, show from $A_n = B_n + r_n$ that

$$|\min_{s\in S} A_n(s) - \min_{s\in S} B_n(s)| \le \delta.$$

We next establish that $\alpha_n$ cannot be far away. Show that when $\|s\| \ge cn^{1/8}$, $B_n(s) \le D_n n^{1/4}$, with $D_n$ positive and bounded in probability; and that when $\|s\| \le cn^{1/8}$, then $|r_n(s)| \le E_n/n^{1/8}$, with $E_n$ also bounded in probability. Show from this that $\alpha_n = O_{\mathrm{pr}}(n^{1/8})$, and that $A_{n,\min} - B_{n,\min} \to_{\mathrm{pr}} 0$.

**(c)** Then consider $B_n$ a certain distance away from the maximum. For given small $\varepsilon$, show that for $v$ with $\|v\| \ge \varepsilon$, we have

$$B_n(\beta_n+v) = B_{n,\max} - \tfrac{1}{2}v^{\mathrm{t}}J_n v \le B_{n,\max} - \tfrac{1}{2}j_n\varepsilon^2,$$

where $j_n$ is the smallest eigenvalue of $J_n$. Show then that the event $\Omega_n$, where $A_n(\beta_n+v) \ge B_{n,\max} + \frac{1}{4}j_n\varepsilon^2$ for all $v$ with $\|v\| \ge \varepsilon$, must have $\Pr(\Omega_n) \to 1$. Prove from these established statements that $\alpha_n - \beta_n$ must tend to zero in probability. Conclude that

$$\begin{aligned}\sqrt{n}(\widehat{\theta}-\theta_0) &\to_d J^{-1}U \sim \mathrm{N}_p(0,J^{-1}KJ^{-1}),\\ 2\{\ell_{n,\max}-\ell_n(\theta_0)\} &\to_d W = U^{\mathrm{t}}J^{-1}U.\end{aligned} \tag{5.6}$$

These are the statements heuristically arrived at in Ex. 5.10, but now with full proofs.

**Ex. 5.12** *Profiling quadratic functions.* Consider some quadratic function $A(s) = \frac{1}{2}s^{\rm t}Js$, with $J$ symmetric and positive definite of dimension $p \times p$. It is useful to sort out minima of $A$ under different types of side constraints.

Lagrange multipliers for minimisation with constraints

**(a)** We start out examining the minimum of $A(s)$ over all $s$ with $c^{\rm t}s = x$, for some given vector $c$ and level $x$. The Lagrange multiplier way of solving such a problem is to minimise the function $\frac{1}{2}s^{\rm t}Js - \lambda(c^{\rm t}s - x)$, with no constraint on $s$, and in the process find the $\lambda$ agreeing with the constraint. Taking derivatives, show that minimum occurs for $s_0 = \lambda J^{-1}c$, leading to $c^{\rm t}s_0 = \lambda c^{\rm t}J^{-1}c$, which should then equal $x$. Explain that this leads to minimiser $s_0 = x/(c^{\rm t}J^{-1}c)$ and attained minimum $A_{\min} = \frac{1}{2}x^2/(c^{\rm t}J^{-1}c)$.

**(b)** Generalising the above, from a scalar to a vector, consider a $p_0 \times p$ matrix $C$ and an $x$ of dimension $p_0$, where it is assumed that $CJ^{-1}C^{\rm t}$ has full rank. The task is to work out the minimiser and minimum of $\frac{1}{2}s^{\rm t}Js$ over all $s$ with $Cs = x$. Work out that the minimiser is

$$s_0 = J^{-1}C^{\rm t}\lambda = J^{-1}C^{\rm t}(CJ^{-1}C^{\rm t})^{-1}x \text{ with } A_{\min} = \tfrac{1}{2}x^{\rm t}(CJ^{-1}C^{\rm t})^{-1}x.$$

**Ex. 5.13** *Profiling a log-likelihood function, I.* For data $Y_1, \ldots, Y_n$ from some distribution $G$, we are using the log-likelihood function $\ell_n(\theta)$ to find its maximiser $\widehat{\theta}$, seen as the canonical estimator for the least false parameter $\theta_0$, the ${\rm KL}(g, f_\theta)$ minimiser. For a focus parameter $\mu = k(\theta)$, a smooth function of $\theta = (\theta_1, \ldots, \theta_p)$, it is very useful to work with the associated profile function

log-likelihood profile

$$\ell_{n,{\rm prof}}(\mu) = \max\{\ell_n(\theta)\colon k(\theta) = \mu\}.$$

**(a)** As an introductory illustration, simulate $n = 100$ datapoints from some ${\rm Gam}(a_0, b_0)$, perhaps with $(a_0, b_0) = (2.22, 3.33)$. From the log-likelihood function $\ell_n(a, b)$, compute first the ML estimators, and then compute and display the profiled $\ell_{n,{\rm prof}}(\sigma)$, for the focus parameter $\sigma = a^{1/2}/b$, the standard deviation.

**(b)** Show that the consequent $\widehat{\mu} = k(\widehat{\theta})$ becomes normal for growing $n$. With setup and notation as in Ex. 5.10, prove indeed that $\sqrt{n}(\widehat{\mu} - \mu_0) \to_d c^{\rm t}J^{-1}U$, with $c = \partial k(\theta_0)/\partial\theta$, and that the limit is a zero-mean normal with variance $c^{\rm t}J^{-1}KJ^{-1}c$. Show also that this $\widehat{\mu}$ is identical to the maximiser of the profile function.

**(c)** In other words we already know the basic story for any focus parameter estimator $\widehat{\mu}$, thanks to the delta method. It is however fruiful to work with representations and approximations stemming from examining the associated profile function. With methods from Ex. 5.11, show that

$$\ell_n(\widehat{\theta} + s/\sqrt{n}) - \ell_n(\widehat{\theta}) = -\tfrac{1}{2}s^{\rm t}J_n s + r_n(s), \quad \text{with } r_n(s) = O_{\rm pr}(\|s\|^3/\sqrt{n}).$$

For the profiling, therefore, we must maximise this expression over all $s$ such that $k(\theta) = k(\widehat{\theta} + s/\sqrt{n}) = \mu$. With $k(\theta) = \widehat{\mu} + c_n^{\rm t}s/\sqrt{n} + O_{\rm pr}(\|s\|^2/n)$, here writing $c_n = \partial k(\widehat{\theta})/\partial\theta$, the essence is to minimise $\frac{1}{2}s^{\rm t}J_n s$ under $c_n^{\rm t}s = \sqrt{n}(\mu - \widehat{\mu}) = x_n$, say. Appealing to Ex. 5.12, show that this minimum becomes $\frac{1}{2}x_n^2/c_n^{\rm t}J_n^{-1}c_n = n(\widehat{\mu} - \mu)^2/c_n^{\rm t}J_n^{-1}c_n$. Fill in more details to prove that with $\mu_0 = k(\theta_0)$ the true parameter in question,

the deviance function

$$D_n(\mu_0) = 2\{\ell_{n,{\rm prof}}(\widehat{\mu}) - \ell_{n,{\rm prof}}(\mu_0)\} = \frac{n(\widehat{\mu} - \mu_0)^2}{c_n^{\rm t}J_n^{-1}c_n} + o_{\rm pr}(1) \to_d \frac{(c^{\rm t}J^{-1}U)^2}{c^{\rm t}J^{-1}c} \sim \kappa\chi^2_1,$$

with $\kappa = c^{\rm t}J^{-1}KJ^{-1}c/c^{\rm t}J^{-1}c$. The $D_n(\mu)$ is called *the deviance function* for the focus parameter in question.

**(d)** Verify, under the traditional circumstances that the model is adequate, that Wilks theorem

$$D_n(\mu_0) \to_d \chi^2_1 \quad \text{at the correct parameter value } \mu_0 = k(\theta_0). \tag{5.7}$$

We call this the focus parameter version of the Wilks thorem; see Ex. 5.14 for a vector generalisation. Explain how this can be used to construct confidence intervals.

**(e)** (xx explain that this often leads to better approximations than the direct limiting normality thing. then an illustration of this. and pointer to Wilks. and pointer to regression versions of these methods and results; the $Y_i$ need not at all be i.i.d. xx)

**Ex. 5.14** *Profiling a log-likelihood function, II: Wilks theorems.* We have developed maximum likelihood methods for estimating the least false model parameter vector $\alpha_0 = \alpha_0(G)$, defined as the minimiser of $\mathrm{KL}(g, f_\alpha)$. We now consider situations where some of the components of the $\alpha$ parameter are specified. Such occur when one tests for lower-dimensional structure, etc. This invites setting up the following framework, with a wide model having $\alpha = (\theta, \gamma)$ of length $p+q$, and the narrow model considered has $(\theta, \gamma_0)$, with $\theta$ unknown but $\gamma = \gamma_0$ fixed. Estimators $(\widehat\theta, \widehat\gamma)$ in the wide model maximise $\ell_n(\theta, \gamma)$, whereas $\widetilde\theta$ for the narrow model maximises $\ell_n(\theta, \gamma_0)$.

The theory developed in the previous exercises mentioned holds for the wide and the narrow models, separately, and below we postulate that the regularity conditions (i)–(v) put up in Ex. 5.10–5.11 are in force. Efforts of linear and matrix algebra are required in order to handle these models jointly, however. Define therefore

$$J_{\mathrm{wide}} = -\mathrm{E}_G \frac{\partial^2 \log f(Y, \theta_0, \gamma_0)}{\partial\alpha\partial\alpha^{\mathrm{t}}} = \begin{pmatrix} J_{00} & J_{01} \\ J_{10} & J_{11} \end{pmatrix} \quad \text{with inverse} \quad J^{-1}_{\mathrm{wide}} = \begin{pmatrix} J^{00} & J^{01} \\ J^{10} & J^{11} \end{pmatrix},$$

where $J_{00} = J_{\mathrm{narr}}$ is of size $p \times p$, etc. There is similarly a $(p+q) \times (p+q)$ matrix $K_{\mathrm{wide}}$, with submatrices $K_{00}, K_{01}, K_{10}, K_{11}$, the variance matrix of $U = (U_0^{\mathrm{t}}, U_1^{\mathrm{t}})^{\mathrm{t}}$, the score function $\partial \log f(Y, \theta_0, \gamma_0)/\partial\alpha$.

**(a)** The following developements are under the $\gamma = \gamma_0$ constraint, so $\alpha_0 = (\theta_0, \gamma_0)$ is the true parameter, determined by the distribution $G$. We let $J_{\mathrm{wide}}$ and $K_{\mathrm{wide}}$ be determined accordingly. The $q \times q$ matrix $J^{11}$ has an important role, here and on later occasions (as with the FIC in Ch. 12). Show that the $Q$ matrix

$$Q = J^{11} = (J_{11} - J_{10} J_{00}^{-1} J_{01})^{-1}. \tag{5.8}$$

Similarly, we have $J^{00} = J_{00}^{-1} + J_{00}^{-1} J_{01} Q J_{10} J_{00}^{-1}$. Show also that $J^{00} - J_{00}^{-1} = J^{01} J_{10} J_{00}^{-1}$, $J^{10} = -Q J_{10} J_{00}^{-1}$.

**(b)** Argue, under the $\gamma = \gamma_0$ constraint, that

$$\begin{pmatrix} \sqrt{n}(\widehat\theta - \theta_0) \\ \sqrt{n}(\widehat\gamma - \gamma_0) \end{pmatrix} \to_d J^{-1}_{\mathrm{wide}} \begin{pmatrix} U_0 \\ U_1 \end{pmatrix}, \quad \sqrt{n}(\widetilde\theta - \theta_0) \to_d J_{00}^{-1} U_0.$$

Show also, again using results reached earlier, that

$$\begin{aligned} \ell_{\mathrm{wide,max}} - \ell_n(\theta_0, \gamma_0) &\to_d \tfrac12 U^{\mathrm{t}} J^{-1}_{\mathrm{wide}} U, \\ \ell_{\mathrm{narr,max}} - \ell_n(\theta_0, \gamma_0) &\to_d \tfrac12 U_0^{\mathrm{t}} J_{00}^{-1} U_0, \end{aligned}$$

with $\ell_{\mathrm{wide}} = \ell_n(\widehat\theta, \widehat\gamma)$ and $\ell_{\mathrm{narr}} = \ell_n(\widetilde\theta, \gamma_0)$. Deduce that

$$W_n = 2(\ell_{\mathrm{wide,max}} - \ell_{\mathrm{narr,max}}) \to_d W = U^{\mathrm{t}} J^{-1}_{\mathrm{wide}} U - U_0^{\mathrm{t}} J_{00}^{-1} U_0. \tag{5.9}$$

Show that this limit variable has mean $\mathrm{Tr}(J^{-1}_{\mathrm{wide}} K_{\mathrm{wide}}) - \mathrm{Tr}(J_{00}^{-1} K_{00})$.

**(c)** Under model conditions, we have the Bartlett identity $J = K$, simplyfing matters. Show first from the above that $\mathrm{E}\, W = (p+q) - p = q$. (xx clean all of this. xx) Show that

$$\sqrt{n}(\widehat{\gamma} - \gamma_0) \to_d Z = J^{10}U_0 + J^{11}U_1 \sim \mathrm{N}_q(0, Q).$$

Use then linear and matrix algebra to show that $W$ above is in fact equal to $W' = Z^{\mathrm{t}}Q^{-1}Z$. Explain that we have reached the important result that under the narrow model, where $\gamma = \gamma_0$,

$$W_n = 2(\ell_{\mathrm{wide,max}} - \ell_{\mathrm{narr,max}}) \to_d \chi^2_q.$$

the Wilks theorem for testing a submodel of a bigger model

This is called *the Wilks theorem*, or, rather, one of several related Wilks theorems. Explain that the (5.7) reached in Ex. 5.13 is the special case of $q = 1$.

### ML theory and Wilks theorems: examples

**Ex. 5.15** *ML estimators in some models.* Here we work through some examples, where the task is to set up the log-likelihood function and if feasible also explicit formulae for the ML estimators. Importantly, the general theory developed above applies nicely, often with easy implentation, also when there is no explicit formula for the ML estimator.

**(a)** With $Y_1, \ldots, Y_n$ from the normal model $\mathrm{N}(\mu, \sigma^2)$, write down the log-likelihood function. Find the ML estimator for $\sigma$ when $\mu$ is a known value, and find also the ML estimators $(\widehat{\mu}, \widehat{\sigma})$ in the case where both parameters are unknown.

**(b)** Suppose $Y_i \sim \mathrm{Pois}(w_i\theta)$, with known exposure times $w_i$, and that the observations are independent, for $i = 1, \ldots, n$. Write down the log-likelihood function, find the ML estimator, and find its mean and variance.

**(c)** (xx one or two more, with explicit formulae for ML. xx)

**(d)** Let $Y_1, \ldots, Y_n$ be i.i.d. from the uniform model on $[0, \theta]$, with $\theta$ the unknown endpoint. Set up the likelihood function and find the ML estimator.

**Ex. 5.16** *Maximum likelihood for the Beta and Gamma models.* Consider the Beta and Gamma two-parameter models, with densities

$$\mathrm{be}(y, a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} y^{a-1}(1-y)^{b-1} \text{ and } g(y, a, b) = \frac{b^a}{\Gamma(a)} y^{a-1}\exp(-by),$$

for $y \in (0,1)$ and $y > 0$, respectively. The task is in each case to estimate the parameters based on an i.i.d. sample $Y_1, \ldots, Y_n$.

**(a)** We start with the Beta distribution, see Ex. 1.19, where we in particular have found formulae for the mean and variance in terms of $(a, b)$. With empirical mean and variance $\bar{y}$ and $\widehat{\sigma}^2$, show how $(a, b)$ can be fitted by solving the two equations $\bar{y} = \mathrm{E}\, Y$ and $\widehat{\sigma}^2 = \mathrm{Var}\, Y$. With solutions $(\widehat{a}_m, \widehat{b}_m)$ for these moment estimators, and assuming the Beta model is correct, explain how you can find limit distributions of $\sqrt{n}(\widehat{a}_m - a, \widehat{b}_m - b)$.

**(b)** Write down the log-likelihood function, say $\ell_n(a, b)$. Show that the ML estimators $(\widehat{a}, \widehat{b})$ are the solutions to the two equations

$$n^{-1}\sum_{i=1}^n \log Y_i = \psi(a) - \psi(a+b), \quad n^{-1}\sum_{i=1}^n \log(1-Y_i) = \psi(b) - \psi(a+b),$$

digamma function

where $\psi(x) = \partial \log\Gamma(x)/\partial x$ is the so-called digamma function. There are no explicit

formulae here, but the two equations may be easily solved numerically. Explain how the limit distribution for $\sqrt{n}(\widehat{a}-a,\widehat{b}-b)$ may be found, via the two-dimensional central limit theorem. – Here it turns out (i) that the ML estimators are more precise than the moment estimators, and (ii) that finding the limit distribution is rather easier via the general results about ML behaviour, already sorted out in Ex. 5.10.

**(c)** Then turn attention to the two-parameter Gamma model, where arguments and results will be similar. Show that the mean and variance are $a/b$ and $a/b^2$, and find moment estimators $\widehat{a}_m, \widehat{b}_m$ based on this. Find the limit distribution of $\sqrt{n}(\widehat{a}_m-a,\widehat{b}_m-b)$.

**(d)** Then write down the log-likelihood function, take derivatives, and show that the ML estimators $(\widehat{a},\widehat{b})$ are the solutions to the two equations

$$\bar{Y} = a/b, \quad n^{-1}\sum_{i=1}^{n} \log Y_i = \psi(a) - \log b.$$

Explain from Ex. 5.10 how the limit distribution of $\sqrt{n}(\widehat{a}-a,\widehat{b}-b)$ may be obtained. (xx also, we shall again find that ML estimators under model conditions are more precise than the moment estimators. xx)

**Ex. 5.17** *Maximum likelihood for the Gamma model.* (xx clean and fix regarding previous exercises and this one. xx) When the Gamma distribution is parametrised as $Y \sim \mathrm{Gam}(c\xi, c)$, we have mean $\xi$ and variance $\xi/c$, and hence a clean interpretation for the $c$. How will the ML estimators $(\widehat{\xi},\widehat{c})$ behave?

**(a)** Show that the log-density becomes

$$\ell = c\xi \log c - \log\Gamma(c\xi) + (c\xi-1)\log y - cy,$$

with score functions

$$\begin{aligned} \partial\ell/\partial\xi &= c\log c - c\psi(c\xi) + c\log y, \\ \partial\ell/\partial c &= \xi\log c + \xi - \xi\psi(c\xi) + \xi\log y - y. \end{aligned}$$

Show from this that equations determining the ML estimators are

$$\log c - \psi(c\xi) + A_n = 0, \quad \log c + 1 - \psi(c\xi) + A_n = \bar{Y}/c,$$

with $A_n = n^{-1}\sum_{i=1}^{n}\log Y_i$. From this again, deduce $\widehat{\xi} = \bar{Y}$, and that $\widehat{c}$ can be found from the single equation $\log c - \psi(c\bar{Y}) + \bar{Y} = 0$.

**(b)** Via taking 2nd order derivatives, show that the Fisher information matrix becomes

with $\psi'$ the digamma function

$$J(\xi,c) = \psi'(c\xi)\begin{pmatrix} c^2 & c^2\xi \\ c^2\xi & \xi^2 \end{pmatrix} - \begin{pmatrix} 0 & 0 \\ 0 & \xi/c \end{pmatrix}.$$

Find the limit distribution for $\sqrt{n}(\widehat{c}-c)$ from this.

**Ex. 5.18** *Log-linear mixing of densities.* How far have we moved, from A to B? Suppose i.i.d. data $Y_1,\ldots,Y_n$ come from a density which is percieved of as being 'betweeen' given densities $f_A$ and $f_B$. There are various ways of creating probabilistic bridges from A to B, via which one then can estimate the position of the current density, e.g. to assess whether it is close to A or to B. One such model takes $f(y,\lambda) = f_A(y)^{1-\lambda}f_B(y)^{\lambda}/R(\lambda)$, with $R(\lambda)$ the normalisation constant $\int f_A^{1-\lambda}f_B^{\lambda}\,\mathrm{d}y$. This creates such a bridge, with $\lambda\in[0,1]$, and with A and B corresponding to $\lambda=0$ and $\lambda=1$.

**(a)** Explain that $R(0) = R(1) = 1$, and that the derivative of $R(\lambda)$, at the endpoints 0 and 1, can be written

$$R'(0) = -\mathrm{KL}(f_A, f_B), \quad R'(1) = \mathrm{KL}(f_B, f_A).$$

in terms of the Kullback–Leibler distances discussed in Ex. 5.4 and later on. In particular, the $R(\lambda)$ has negative derivative at the start and positive derivative at the end.

**(b)** For a concrete illustration, suppose $f_A$ is $\mathrm{N}(0,1)$ and $f_B$ is $\mathrm{N}(c,1)$, with $c$ positive. Show that $R(\lambda) = \exp\{-\frac{1}{2}c^2\lambda(1-\lambda)\}$. From this explain that the score function is $u(y,\lambda) = cy - \lambda c^2$ and that the Fisher information becomes the constant $c^2$. Show indeed that $f(y,\lambda)$ is $\mathrm{N}(\lambda c, 1)$, that the ML estimator becomes $\widehat{\lambda} = \bar{Y}/c$, truncated to $[0,1]$. From general ML theory, explain that if the real $\lambda$ is inside $(0,1)$, then $\sqrt{n}(\widehat{\lambda} - \lambda) \to_d \mathrm{N}(0, 1/c^2)$. For $c$ small the variance will be very big, signifying that it is a hard task to estimate the balance parameter when $f_A$ and $f_B$ are close.

**(c)** Returning to the general setup, write $S(y) = \log f_B(y) - \log f_A(y)$. Show that the log-likelihood function becomes

$$\ell_n(\lambda) = n\{\lambda \bar{S}_n - \log R(\lambda)\}, \quad \text{with } \bar{S}_n = n^{-1}\sum_{i=1}^n S(y_i).$$

Explain that the ML estimator $\widehat{\lambda}$ hence is the solution to $\bar{S}_n = R'(\lambda)/R(\lambda)$. (xx a bit more. and example. limiting normality. evaluate for $\lambda = 0$ and $\lambda = 1$. xx)

**(d)** Bayes posterior: $\pi(\lambda)\exp[n\{\lambda S_n - \log R(\lambda)\}]$.

**(e)** (xx there is perhaps hope for the semiparametric construction $\widehat{f}(y) = f_0(y)^{1-\lambda} f_n(y)^{\lambda}/R_n(\lambda)$, resembling the Hjort–Glad estimator, see Hjort and Glad (1995). xx)

**Ex. 5.19** *Parametric precision when estimating densities.* (xx this exercise to be pushed in after basic ML. xx) Assume $X_1, \ldots, X_n$ are i.i.d. on the real line. In Ex. 3.43 we briefly investigated the kernel density estimator $f_n$, and found that under smoothness conditions, the $\mathrm{mise}_n = \int (f_n - f)^2\,\mathrm{d}x$ is of order $O(n^{-4/5})$ for the best bandwidths. Parametric precision is however better, provided the model can be trusted.

**(a)** Find useful expressions for the point-wise mean squared error $\mathrm{mse}(x) = \mathrm{E}\,\{\widehat{f}(x) - f(x)\}^2$ in the following cases: (i) $f$ the p.d.f. of $\mathrm{N}(\mu,\sigma^2)$ with $\sigma$ known; (ii) $f$ as in (i), but with $\mu$ known; (iii) $f$ as in (i), but with both parameters known; (iv) the exponential $f(x,\theta) = \theta\exp(-\theta x)$.

**(b)** Then find expressions for the integrated mean squared error $\mathrm{mise} = \mathrm{E}\int(\widehat{f} - f)^2\,\mathrm{d}x$, in cases (i)–(iv).

**(c)** Assume in general terms that data follow a parametric model $f(x,\theta)$, with $\theta_0$ the true parameter, of dimension say $p$. Writing $u(x,\theta) = \partial \log f(x,\theta)/\partial\theta$ for the score runction, use the general ML theory of Ch. 5 to show that

$$\begin{aligned}\sqrt{n}\{f(x,\widehat{\theta}) - f(x,\theta_0)\} &\to_d f(x,\theta_0)u(x,\theta_0)^{\mathrm{t}}Z,\\ \sqrt{n}f(x,\theta_0)^{1/2}\{\log f(x,\widehat{\theta}) - \log f(x,\theta)\} &\to_d f(x,\theta_0)^{1/2}u(x,\theta_0)^{\mathrm{t}}Z,\end{aligned}$$

where $Z \sim \mathrm{N}_p(0, J^{-1})$, with $J = J(\theta_0)$ the Fisher information matrix. Then use this to show that with

$$A_n = \int \frac{\{f(x,\widehat{\theta}) - f(x,\theta_0)\}^2}{f(x,\theta_0)}\,\mathrm{d}x \quad \text{and} \quad B_n = \int f(x,\theta_0)\{\log f(x,\widehat{\theta}) - \log f(x,\theta_0)\}^2\,\mathrm{d}x,$$

we have both $nA_n$ and $nB_n$ tending in distribution to $W = Z^{\rm t}JZ \sim \chi^2_p$. So various versions of weighted MISE go to zero with speed $p/n$, for parametrically estimated densities, under model conditions, in contrast to the slower $O(n^{4/5})$ speed for nonparametric estimators.

**(d)** (xx round off. other weight functions, perhaps. xx)

**Ex. 5.20** *Completing increasingly simpler tasks.* In a certain game of learning a player needs to complete tasks $1, 2, \ldots, n$, which become increasingly simpler with each passing of a new level. Assume that the time needed to complete these tasks are $V_1, \ldots, V_n$, with these being independent with $V_i \sim \text{Expo}(i/\theta)$, where $\theta$ is an unknown parameter. – For the following questions, you may encounter the partial sums

$$a_n = 1 + 1/2 + 1/3 + \cdots + 1/n, \quad b_n = 1 + 1/2^2 + 1/3^2 + \cdots + 1/n^2.$$

Here the first is slowly divergent, with $a_n \doteq \log n + 0.5772$, and the second is convergent, with $b_n \to \pi^2/6$, as shown by Euler in 1734, bringing him instant world fame.

**(a)** Find expressions for the mean and variance of $T_n = V_1 + \cdots + V_n$, the time it takes the player to complete all tasks. In particular, show that $T_n$ has mean $a_n\theta$. Put up the unbiased estimator based on $T_n$, say $\widehat{\theta}$. Find its variance, and show that the estimator is consistent.

**(b)** Then work out a formula for the log-likelihood function, based on having observed not merely the total time $T_n$, but the individual waiting times $V_1, \ldots, V_n$. Find the maximum likelihood estimator, say $\theta^*$. Show that also this estimator $\theta^*$ is unbiased, and compare its variance to that of $\widehat{\theta}$. Find also the Cramér–Rao lower bound for variances of unbiased estimators for $\theta$, and comment.

**(c)** Assume the game goes on, up to level $2n$, and consider the time a player needs to pass the last half of these levels, i.e. $T^*_n = T_{2n} - T_n$. Show that $T^*_n$ tends in probability to a certain limit as $n$ grows.

**Ex. 5.21** *Wilks theorems for the normal model.* Consider the familiar normal situation with $Y_1, \ldots, Y_n$ independent from $\text{N}(\xi, \sigma^2)$. The log-likelihood function is $\ell_n(\xi, \sigma)$, as in (5.2). Here we work with profiled consequences, also to illustrate the Wilks theorem of Ex. 5.13.

**(a)** For $\sigma$ in focus, show that

$$\ell_{n,\text{prof}}(\sigma) = -n \log \sigma - \tfrac{1}{2} Q_0/\sigma^2 - \tfrac{1}{2} n \log(2\pi),$$

with $Q_0 = \sum_{i=1}^n (Y_i - \bar{Y})^2$. From this show that the deviance function may be written

$$D_n(\sigma) = 2\{\ell_{n,\max} - \ell_{n,\text{prof}}(\sigma)\} = nh(\widehat{\sigma}^2/\sigma^2),$$

with $h(x) = x - \log x - 1$. The Wilks theorem implies that $D_n(\sigma) \to_d \chi^2_1$, under model conditions; also verify this via a two-term Taylor expansion for $h(x)$ around $x = 1$.

**(b)** Then with attention to $\xi$, show that

$$\ell_{n,\text{prof}}(\xi) = -n \log \widehat{\sigma}(\xi) - \tfrac{1}{2} n\{1 + \log(2\pi)\},$$

with $\widehat{\sigma}(\xi)^2 = n^{-1}\{Q_0 + n(\xi - \bar{Y})^2\}$. From this show that

$$D_n(\xi) = 2\{\ell_{n,\max} - \ell_{n,\text{prof}}(\xi)\} = n \log(1 + n^{-1} K_n(\xi)), \quad \text{with } K_n(\xi) = n(\bar{Y} - \xi)^2/\widehat{\sigma}^2.$$

Verify directly what the Wilks theorem says here, that $D_n(\xi) \to_d \chi^2_1$.

**(c)** The Wilks theorem gives in general terms the recipe $\{\mu\colon D_n(\mu) \le 1.96^2\}$ for an approximate 95 percent confidence interval. With some simulated data, with sample sizes say $10, 30, 50$, check what this recipe gives for forming intervals for $\xi$ and for $\sigma$, compared to the exact ones. (xx can also say, do this, exact and Wilks, for the $x_{i,E}$ data of Story #38, $n = 64$ word-length averages. xx)

**(d)** Importantly, the Wilks theorem with consequent confidence recipe can be used for *any* smooth function of the parameters here, i.e. not merely for $\xi$ and for $\sigma$. With some simulated data, illustrate this, by computing the log-likelihood profile function and the deviance function, for parameters $\gamma = \xi + 1.96\,\sigma$ and $p = 1 - \Phi((y_0 - \xi)/\sigma)$, for suitable threshold $y_0$.

**(e)** (xx can put in Behrens–Fisher problem here. Wilks will work well for $\ell_{\rm prof}(\delta)$, with $\delta = \xi_2 - \xi_1$, even with different variances. xx)

### Likelihood machinery for multinomial data

**Ex. 5.22** *log-likelihood and ML for the binomial, trinomial, multinomial.* Working through the likelihood mechanics of the binomial, trinomial, multinomial models provide good illustrations of the developed methodology. We already know the basic large-sample behaviour for the natural estimators in these models, via Ex. 2.49, but here we connect such results to the general likelihood theory.

**(a)** For $X \sim \text{binom}(n, p)$, a sum of independent Bernoulli variables, show that its log-likelihood function is $\ell_n(p) = X \log p + (n - X) \log(1 - p)$, with maximiser $\widehat{p} = X/n$. Show that $nJ = n/\{p(1-p)\}$, here with $J$ defined relative to a single Bernoulli variable. Explain that we may hence read off the limit distribution of $\sqrt{n}(\widehat{p}-p)$ being $\text{N}(0, p(1-p))$, without necessarily even knowing the $X/n$ formula, or about the CLT for binomials. One may even argue the other way, starting with the $X/n$ formula: since we know the limit is $\text{N}(0, p(1-p))$, via the CLT, we must have $J^{-1} = p(1-p)$, hence $J$ must be $1/\{p(1-p)\}$.

**(b)** For further illustration of the binomial likelihood mechanics, draw the log-likelihood function $\ell_n(p)$, computing both its maximiser, its maximum, and its second derivative at the maximum, for cases (i) $n = 20, y = 4$, (ii) $n = 40, y = 8$, (iii) $n = 200, y = 40$. Note how $\widehat{J}_{\rm obs} = -\ell_n''(\widehat{p})$ becomes bigger and the implied estimated standard deviance $1/\widehat{J}_{\rm obs}^{1/2}$ becomes smaller.

**(c)** Let $(X, Y)$ be trinomial $(n, p, q)$, and use $r = 1 - p - q$ and $Z = n - X - Y$. Show that the log-likelihood function becomes $X \log p + Y \log q + Z \log(1 - p - q)$, and for the Fisher information matrix that

$$J = \begin{pmatrix} 1/p + 1/r & 1/r \\ 1/r & 1/q + 1/r \end{pmatrix} \quad \text{with} \quad J^{-1} = \begin{pmatrix} p(1-p) & -pq \\ -pq & q(1-q) \end{pmatrix}.$$

Conclude, even before finding or perhaps without caring that there are clear formulae $\widehat{p} = X/n$ and $\widehat{q} = Y/n$ for the ML estimators, that $(\sqrt{n}(\widehat{p} - p), \sqrt{n}(\widehat{q} - q))$ tends to the binormal zero-mean distribution with the $J^{-1}$ covariance matrix. This may of course also be shown, for $(X/n, Y/n)$, without knowing that these are ML estimators.

**Ex. 5.23** *Maximum likelihood and Wilks testing for multinomials.* (xx check against Karl Pearson story, which might not be needed after all, and other uses of minimum chi-squared. xx) In generalisation of Ex. 5.22, consider a multinomial counts $(N_1, \ldots, N_k)$, with sum $n$ and probabilities $p = (p_1, \ldots, p_k)$ for the $k$ categories in question.

**(a)** Before we come to parametrised models, i.e. no constaints are made on the $p_j$ apart from their sum being 1, show that the log-likelihood function may be written $\ell_n(p) = \sum_{j=1}^k N_j \log p_j$, and that this is maximised at $\widehat{p}$, with $\widehat{p}_j = N_j/n$. Explain that each of these are binomial proportions, where we e.g. know that $\sqrt{n}(\widehat{p}_j - p_j) \to_d X_j \sim \mathrm{N}(0, p_j(1-p_j))$.

**(b)** We need joint convergence here, to understand other properties of the multinomial counts. Show that the vector of $X_{n,j} = \sqrt{n}(\widehat{p}_j - p_j)$ tends to the vector $X = (X_1, \ldots, X_k) \sim \mathrm{N}_k(0, \Sigma)$ in distribution, where $\Sigma$ is the $k \times k$ matrix with elements $p_j(\delta_{j,\ell} - p_\ell)$. Show that $\mathbf{1}^{\mathrm{t}}\Sigma\mathbf{1} = 0$, with $\mathbf{1}$ the vector of 1s, so $\Sigma$ is not invertible. Explain that we from general ML theory know that $(X_{n,1}, \ldots, X_{n,k-1}) \to_d (X_1, \ldots, X_{k-1}) \to_d \mathrm{N}_{k-1}(0, \Sigma_0)$, with $\Sigma_0 = J^{-1}$, in terms of the Fisher matrix $J$ for $(p_1, \ldots, p_{k-1})$. Argue therefore that we already know the form of $J^{-1}$, and that $J$ must be equal to

$$J = \Sigma_0^{-1} = \mathrm{diag}(1/p_1, \ldots, 1/p_{k-1}) + (1/p_k)\mathbf{1}_0\mathbf{1}_0^{\mathrm{t}},$$

writing $\mathbf{1}_0$ for the $(k-1)$-vector of 1s. Derive indeed the $J$ matrix expression from $-n^{-1}\partial^2 \ell_n(\widehat{p})/\partial p_0\, \partial p_0^{\mathrm{t}}$. Note that this generalises results of Ex. 5.22.

**(c)** Suppose one needs to test $p = p_{\mathrm{fix}}$, a fixed probability vector. Show via the log-likelihood ratio machinery that $W_n = 2n\sum_{j=1}^k \widehat{p}_j \log(\widehat{p}_j/p_{\mathrm{fix},j}) \to_d \chi^2_{\mathrm{df}}$, under the null model condition, with $\mathrm{df} = k-1$ degrees of freedom. Give a confidence region $R_n$ for the full probability vector $p$, such that $\Pr_p(p \in R_n) \to 0.95$.

**(d)** Often in multinomial models there are parametric theories concerning their probabilities. Suppose $p_j = p_j(\theta)$, for $j = 1, \ldots, k$, smooth functions of $\theta$, a parameter of dimension $r < k-1$. Write $u_j(\theta)$ and $\psi_j(\theta)$ for the 1st and 2nd derivatives of $\log p_j(\theta)$, where we below write $p_j, u_j, \psi_j$, for short, when these are evaluated at the true parameter $\theta_0$. Show from $\sum_{j=1}^k p_j(\theta) = 1$ that we have the Bartlett type identities $\sum_{j=1}^k p_j u_j = 0$ and $J = \sum_{j=1}^k p_j u_j u_j^{\mathrm{t}} = -\sum_{j=1}^k p_j \psi_j$. Explain next that the log-likelihood function is $\ell_n(\theta) = n\sum_{j=1}^k \widehat{p}_j \log p_j(\theta)$, with

$$\partial \ell_n(\theta)/\partial\theta = n\sum_{j=1}^k \widehat{p}_j u_j(\theta), \quad \partial^2 \ell_n(\theta)/\partial\theta\,\partial\theta = n\sum_{j=1}^k \widehat{p}_j \psi_j(\theta).$$

With $\widehat{\theta}$ the ML estimator, and following the general results of Ex. 5.3, or arguing directly in the present mutlinomial setup, show that

$$A_n(s) = \ell_n(\theta_0 + s/\sqrt{n}) - \ell_n(\theta_0) = U_n^{\mathrm{t}} s - \tfrac{1}{2} s^{\mathrm{t}} J_n s + o_{\mathrm{pr}}(1),$$

in terms of $U_n = \sum_{j=1}^k X_{n,j} u_j \to_d U \sim \mathrm{N}_r(0, J)$ and $J_n = -n^{-1}\sum_{j=1}^k \widehat{p}_j \psi_j \psi_j \to_{\mathrm{pr}} J$. From this explain that we also have

$$\sqrt{n}(\widehat{\theta} - \theta_0) \to_d \mathrm{N}_r(0, J^{-1}) \quad \text{and} \quad 2\{\ell_{n,\max} - \ell_n(\theta_0)\} \to_d \chi^2_{\mathrm{df}}, \text{ with } \mathrm{df} = k - 1 - r.$$

**(e)** For an illustration, suppose $(X, Y, Z)$ is trinomial probabilities $a^2, 2a(1-a), (1-a)^2$, as e.g. with so-called Hardy–Weinberg equilibtrium. Find the ML estimator of $a$ and its limit distribution, and construct a 90 percent confidence interval. (xx point to Stories. xx)

**Ex. 5.24** *Pearson and minimum chi-squared for multinomials.* Consider again the setup of Ex. 5.23. The Karl Pearson chi-squared, dating all the way back to his far-reaching

1900 paper, is the sum of normalised squares

$$Q_n = \sum_{j=1}^k \frac{(N_j - np_j)^2}{np_j} = n\sum_{j=1}^k \frac{(\widehat{p}_j - p_j)^2}{p_j},$$

measuring distance from the vector of $p_j$ to the vector of $\widehat{p}_j = N_j/n$ in a different fashion than with the log-likelihood ratio statistics worked with in the earlier exercies.

**(a)** With notation $X_{n,j} = \sqrt{n}(\widehat{p}_j - p_j) \to_d X_j$, as in the previous exercise, show that $Q_n \to_d Q = \sum_{j=1}^k X_j^2/p_j$. Show that both $Q_n$ and $Q$ have means $k-1$. The $Q$ is a sum of squared but dependencet normals, and $\sum_{j=1}^k X_j = 0$. Show that $Q_n$ may be expressed as $X_{n,0}^{\mathrm{t}}\Sigma_0^{-1}X_{n,0}$, and that its limit must be the quadratic form $X_0^{\mathrm{t}}\Sigma_0^{-1}X_0$. Conclude that $Q_n \to_d \chi^2_{k-1}$, which is the crowning achievement of Pearson (1900).

**(b)** Show that

$$Q_n - Q_n' = n\sum_{j=1}^k \frac{(\widehat{p}_j - p_j)^2}{p_j} - n\sum_{j=1}^k \frac{(\widehat{p}_j - p_j)^2}{\widehat{p}_j} \to_{\mathrm{pr}} 0,$$

so the two have the same large-sample behaviour; both may be used to test some $p = p_{\mathrm{fix}}$.

**(c)** Then consider a parametric model $p_j = p_j(\theta)$, as in Ex. 5.23. An estimation method different from the ML is to minimise

$$Q_n(\theta) = \sum_{j=1}^k \frac{\{N_j - p_j(\theta)\}^2}{N_j} = n\sum_{j=1}^k \frac{\{\widehat{p}_j - p_j(\theta)\}^2}{\widehat{p}_j} = n\sum_{j=1}^k \frac{p_j(\theta)^2}{\widehat{p}_j} - n.$$

Verify this last identity.

**(d)** It turns out that this alternative estimation method is close to the ML one, with the difference vanishing for growing $n$. The essentially lies with the first two derivatives of $\ell_n(\theta)$ and $Q_n(\theta)$ being closely related. To work with this, show first that the two first derivatives of $p_j(\theta)^2$, evaluated at $\theta_0$, become $2p_j^2u_j$ and $2(2p_j^2u_ju_j^{\mathrm{t}} + p_j^2\psi_j)$. Verify then that we may write

$$B_n(s) = Q_n(\theta_0 + s/\sqrt{n}) - Q_n(\theta_0) = V_n^{\mathrm{t}}s + \tfrac{1}{2}s^{\mathrm{t}}K_ns + o_{\mathrm{pr}}(1),$$

with $V_n = \sum_{j=1}^k 2X_{n,j}p_ju_j \to_d V = 2U$ and $K_n = \sum_{j=1}^k 2[2p_ju_ju_j^{\mathrm{t}} + p_j\psi_j] \to_{\mathrm{pr}} K = 4J - 2J = 2J$. Deduce, then, that $B_n(s) \to_d V^{\mathrm{t}}s + \frac{1}{2}s^{\mathrm{t}}Ks$, and compare with the result above for $A_n(s)$.

**(e)** Show from these efforts that $\sqrt{n}(\widehat{\theta} - \theta_0)$ and $\sqrt{n}(\widetilde{\theta} - \theta_0)$ have identical limit distributions, under model conditions. Show also that

$$Q_{n,\min} = \min \sum_{j=1}^k \frac{\{N_j - np_j(\theta)\}^2}{N_j} \quad \text{and} \quad \min Q'_{n,\min} = \min \sum_{j=1}^k \frac{\{N_j - np_j(\theta)\}^2}{np_j(\theta)}$$

are close enough for these two to have the same limit distribution. (xx be careful with the last step argument, via profiling of approximate quadratic functions, that $W_n$ and $Q_{\min}$ both have the $\chi^2_{\mathrm{df}}$ limit, under the model. xx)

**(f)** In addition to computing the full Pearson test statistic, to decide on acceptance vs. rejection of a model, typically along with finding the p-value in question, it is often useful to examine the individual terms in the chi-squared sum. Verify that with

Pearson residuals

$$\text{pear}_j = \{N_j - np_j(\widehat{\theta})\}/(np_j(\widehat{\theta})^{1/2} \quad \text{and} \quad \text{pear}'_j = \{N_j - np_j(\widehat{\theta})\}/N_j^{1/2},$$

which we call Pearson residuals, we have $\sum_{j=1}^k \text{pear}_j^2 = Q_{n,\min}$ and $\sum_{j=1}^k (\text{pear}'_j)^2 = Q'_{n,\min}$. If the model is good, the Pearson resdiduals ought to be approximately standard normal, so we may easily spot the boxes $j$ where there is significant discrepancy between observed $N_j$ and predicted $np_j(\widehat{\theta})$.

**(g)** Suppose Pearson's $Q_n = \sum_{j=1}^k (N_j - np_{j,0})^2/(np_{j,0})$ is used for testing the hypothesis $H_0$ that $p = p_0$, for a given relevant $(p_{0,1}, \ldots, p_{k,0})$. At the null, $Q_n \to_d \chi^2_{k-1}$, but happens outside $H_0$? Suppose the real state of affairs is $p_j = p_{j,0} + \delta_j/\sqrt{n}$, with $\sum_{j=1}^k \delta_j = 0$. With the notation above, show that $X_{n,j} = \sqrt{n}(\widehat{p}_j - p_{j,0}) \to_d X_j + \delta_j$. From this, demonstrate that $Q_n \to_d Q = \sum_{j=1}^k (X_j + \delta_j)^2/p_j$, and that this has a noncentral $\chi^2_{k-1}(\lambda^2)$ distribution, with excentre parameter $\lambda^2 = \sum_{j=1}^k \delta_j^2/p_{j,0}$. Give an interpretation for $\lambda$. What we have found is useful for studying power functions, but also for assessing the distance from the null hypothesis to reality, i.e. the size of $\lambda$. Give an estimator for this $\lambda$, based on having observed $Q_n$. local power

**(h)** (xx if we wish to: might hammer out limit distributions outside model, with two different sandwich matrices; the $W_n$ and $Q_n$ give the same at the model and up to $O(1/\sqrt{n})$ neighbourhoods only. xx)

**Ex. 5.25** *Contingency tables, and more on Pearson vs. Wilks.* In Exercises 5.23–5.24 we have seen how inference can be carried out in multinomial models, via either Pearson type minimum chi-squared or Wilks type log-likelihood. We learn a bit more here, about their large-sample equivalence, under model conditions. (xx check and calibrate with Story #59. we need one point with local power, for this story. xx)

**(a)** Consider again the multinomial setting of these two previous exercises, and with a smooth parametric model $p_j = p_j(\theta)$, with ensuing log-likelihood function $\ell_n(\theta) = \sum_{j=1}^k N_j \log p_j(\theta)$. Show in general terms that for $p$ close to $\widehat{p}$, we have

$$\widehat{p}\log p = \widehat{p}\log(\widehat{p} + p - \widehat{p}) = \widehat{p}\log p + (1/\widehat{p})(p - \widehat{p}) - \tfrac{1}{2}(1/\widehat{p}^2)(p - \widehat{p})^2 + O_{\rm pr}(|\widehat{p} - p|^3).$$

Explain that this implies

$$\sum_j \widehat{p}_j \log p_j(\theta) = \sum_j \widehat{p}_j \log \widehat{p}_j - \tfrac{1}{2}\sum_j \{\widehat{p}_j - p_j(\theta)\}^2/\widehat{p}_j + O_{\rm pr}(\max_j |\widehat{p}_j - p_j(\theta)|^3).$$

The implication for the log-likelihood is that

$$\ell_n(\theta) = \ell_{\text{wide,max}} - \tfrac{1}{2}Q_n(\theta) + \varepsilon_n(\theta), \quad \text{with } Q_n(\theta) = n\sum_j \{\widehat{p}_j - p_j(\theta)\}^2/\widehat{p}_j,$$

the Karl Pearson type weighted sum of squares. If the model holds, for some $\theta_0$, show that the remainder term goes to zero in probability inside neighbourhoods $\|\theta - \theta_0\| \le c/\sqrt{n}$. Argue from this that ML estimation for large samples is equivalent to minimum chi-squared, with $K_n = Q_n(\widetilde{\theta}) = \min Q_n(\theta)$. Deduce also for the Wilks statistic (xx pointer xx) that

$$W_n = 2(\ell_{\text{wide,max}} - \ell_{0,\max}) = K_n + o_{\rm pr}(1).$$

from Wilks theory of Ch5 we can infer that both $D_n$ and $K_n$ tend to $\chi^2_{\rm df}$, with df $= k - 1 - \dim(\theta)$, under model conditions. This extends the classical Karl Pearson 1900 work, where he developed the chi-squared for a fixed set of probabilities.

**(b)** We now consider a contingency table, with a multinomial $N_{i,j}$, the number of $(A = i, B = j)$, for an $r \times s$ table, and probabilities $p_{i,j}$, summing to 1. Let $\Pr(A = i) = p_{i,\cdot} = a_i$, $\Pr(B = j) = p_{\cdot,j} = b_j$, with the '$\cdot$' indicating summation over the index in question. Independence corresponds to $p_{i,j} = a_i b_j$ for all pairs. Letting $\widehat{p}_{i,j} = N_{i,j}/n$ be the direct estimates, show that the log-likelihood function for the independence model is

contingency table

$$\ell(a, b) = n \sum_{i,j} \widehat{p}_{i,j}(\log a_i + \log b_j) = n \sum_i \widehat{p}_{i,\cdot} \log a_i + n \sum_j \widehat{p}_{\cdot,j} \log b_j$$

with ML estimators $\widehat{a}_i = \widehat{p}_{i,\cdot}$ and $\widehat{b}_j = \widehat{p}_{\cdot,j}$. Deduce that the Wilks deviance statistic, see with (5.5), is

$$W_n = 2n\Big(\sum_{i,j} \widehat{p}_{i,j} \log \widehat{p}_{i,j} - \sum_i \widehat{a}_i \log \widehat{a}_i - \sum_j \widehat{b}_j \log \widehat{b}_j\Big).$$

Use results above to learn that under the null model of independene, $W_n$ and the classic

$$K_n = n \sum_{i,j} \frac{(\widehat{p}_{i,j} - \widehat{a}_i \widehat{b}_j)^2}{\widehat{p}_{i,j}} = \sum_{i,j} \frac{(N_{i,j} - E_{i,j})^2}{E_{i,j}}$$

have the same limit distribution $\chi^2_{\rm df}$, with df $= (r-1)(s-1)$, under the null. Here $E_{i,j} = n\widehat{a}_i \widehat{b}_j$ are the expected numbers in the cells, under the null; these are given to the right of the table of observed numbers. Show also that the same chi-squared limit obtains, under independence, whether one uses $E_{i,j}$ or $N_{i,j}$ in the denominator.

**(c)** (xx nice if we could complete the below things. relies on a lemma saying $X \sim \chi^2_a$ and $X + Y \sim \chi^2_{a+b}$ implies $Y$ independent of $X$ and being $\chi^2_b$. the below would then be a new little proof of chi-squared limit for $K_n$. at any rate, make clear from the prose that we have different proofs for the chis-square limit. xx) we have found the $\chi^2_{\rm df}$ limit in other ways, with df $= (r-1)(s-1)$, but it is interesting to understand another path too. we use $\sqrt{n}(\widehat{p}_{i,j} - p_{i,j}) \to_d A_{i,j}$, a big zero-mean multinormal. then work under independence with

$$\sqrt{n}(\widehat{p}_{i,j} - \widehat{a}_i \widehat{b}_j) \to_d B_{i,j} = A_{i,j} - a_i A_{\cdot,j} - b_j A_{i,\cdot}.$$

So $K_n = n \sum_{i,j} (\widehat{p}_{i,j} - \widehat{a}_i \widehat{b}_j)^2 / \widehat{p}_{i,j}$ tends to $K = \sum_{i,j} B^2_{i,j}/(a_i b_j)$. Multiply out and simplify to get

$$K = \sum_{i,j} \frac{(A_{i,j} - a_i A_{\cdot,j} - b_j A_{i,\cdot})^2}{a_i b_j} = \sum_{i,j} \frac{A^2_{i,j}}{a_i b_j} - \sum_i \frac{A^2_{i,\cdot}}{a_i} - \sum_j \frac{A^2_{\cdot,j}}{b_j}.$$

Intriguingly, the $A_{i,\cdot}$ and $A_{\cdot,j}$ are all independent, under the null, so $K = K_0 - K_a - K_b$, or $K_0 = K_a + K_b + K$, where $K_a$ and $K_b$ are independent and $\chi^2_{r-1}$, $\chi^2_{s-1}$, and we also know by KP 1900 story that $K_0 \sim \chi^2_{rs-1}$. but it is not clear how to derive that $K \sim \chi^2_{(r-1)(s-1)}$ from this. might involve some clever algebraic rewriting and Cochran's theorem.

**(d)** Often enough the two factors have dependencies, of course, and one needs models to assess such. Explain that for any score matrix $h_{i,j}$, the model

$$p_{i,j} = a_i b_j \exp(\gamma h_{i,j}) / \sum_{i',j'} a_{i'} b_{j'} \exp(\gamma h_{i',j'}).$$

is a bona fide 1-parameter extension of independence, How many free parameters are there in such a model? Check Story #63 for an illustration with fathers and sons and social mobility.

**(e)** (xx perhaps here: Pearson and Wilks have different limits outside the model, with two sandwiches. xx)

**Ex. 5.26** *Comparing two multinomials.* (xx check and point to Story #44. xx) Consider two multinomial vectors $M = (M_1, \ldots, M_k)$ and $N = (N_1, \ldots, N_k)$, with comparable probability vectors $p = (p_1, \ldots, p_k)$ and $q = (q_1, \ldots, q_k)$. How can we naturally test the hypothesis $H_0$ that $p = q$?

**(a)** Writing $n_1$ and $n_2$ for the two sample sizes, with sum $n$, write $\widehat{p}_j = M_j/n_1$ and $\widehat{q}_j = N_j/n_2$. Explain first that

$$\sqrt{n}(\widehat{p} - p) \to_d (1/c_1^{1/2})X \sim \mathrm{N}_k(0, \Sigma_1), \quad \sqrt{n}(\widehat{q} - q) \to_d (1/c_2^{1/2})Y \sim \mathrm{N}_k(0, \Sigma_2),$$

where $\Sigma_1 = D_1 - pp^{\mathrm{t}}$, $\Sigma_2 = D_2 - qq^{\mathrm{t}}$, with diagonal matrices $D_1$ and $D_2$ having $p$ and $q$ on their diagonals. Here $c_1 = n_1/n$ and $c_2 = n_2/n$, meant to stay stable for growing sample sizes.

**(b)** Under $H_0$, then, show that $\sqrt{n}(\widehat{p}-\widehat{q}) \to_d (1/c_1+1/c_2)^{1/2}Z$, with $Z \sim \mathrm{N}_k(0, D-pp^{\mathrm{t}})$. Deduce that

$$K_{n_1,n_2} = \frac{n_1 n_2}{n_1 + n_2} \sum_{j=1}^{k} \frac{(\widehat{p}_j - \widehat{q}_j)^2}{\widehat{r}_j} \to_d \chi^2_{k-1},$$

as long as $\widehat{r}_j \to_{\mathrm{pr}} p_j$ for each $j$; it is natural to take $\widehat{r}_j = c_1\widehat{p}_j + c_2\widehat{q}_j = (n_1X_j + n_2Y_j)/(n_1 + n_2)$.

### Maximum likelihood machinery in practice

**Ex. 5.27** *ML analysis via optimisers, gradients, and Hessians.* As long as you can programme your log-likelihood function, as with a suitable `logL <- function(theta)`, there are typically well-working optimisation algorithms that with a modicum of extra work will provide you with (i) the ML estimator $\widehat{\theta}$ itself, the maximiser; (ii) the $\ell_{\max} = \ell(\widehat{\theta})$, the maximum; (iii) the hessian matrix $\widehat{J}_{\mathrm{full}} = -\partial^2\ell(\widehat{\theta})/\partial\theta\,\partial\theta^{\mathrm{t}}$, the observed Fisher information; (iv) estimated score function at the data values, $\widehat{u}_i = u(y_i, \widehat{\theta})$, qua gradients; (v) the gradient vector $\widehat{c} = \partial\mu(\widehat{\theta})/\partial\theta$ associated with any focus parameter $\mu(\theta)$, required for delta method analysis; (vi) deviance functions $D(\mu)$ for parameters of interest; (vii) Wilks type differences $\ell_{\max,\mathrm{wide}} - \ell_{\max,\mathrm{narr}}$; and yet other quantities for certain further uses.

**(a)** To illustrate the general features, and how the crucial quantities flow from the log-likelihood function, imagine that you have never heard about the normal distribution and that you have not carefully read Chs. 1-4. Get hold of any dataset $y_1, \ldots, y_n$ under your fingertips. You are given the formula

counterfactually, you may even imagine that you haven't been through all the theorems and proofs

$$f(y, \xi, \sigma) = \frac{1}{(2\pi)^{1/2}} \frac{1}{\sigma} \exp\Big\{-\tfrac{1}{2}\frac{(y-\xi)^2}{\sigma^2}\Big\},$$

and proceed to programme the log-likelihood function, more or less as below, in your favourite software enviroment. You throw the `minuslogL` to a minimiser, like `nlm` in `R`, which then crucially gives you `ML`, the minimiser, and `Jhatfull`, the hessian matrix. Do this.

```
logL <- function(para)
{ xi = para[1]
```

```
  sigma = para[2]
  aux = -log(sigma) - 0.5*(yy - xi)^2/sigma^2 - 0.5*log(2*pi)
  sum(aux) }
  then
minuslogL <- function(para)
{ -logL(para) }
```

**(b)** Read off `sdhat = sqrt(diag(solve(Jhatfull)))`, the square-roots of the diagonal of the inverse Fisher information matrix $\widehat{J}_{\rm full}$. Get 95 percent confidence intervals from `ML-1.96*sdhat` to `ML+1.96*sdhat`.

**(c)** Under model circumstances the above is fine, but outside model conditions we also need the following. Do the programming required to obtain $\widehat{u}_i = \partial \log f(y_i, \widehat{\theta})$, without matematics, so to speak, but with a `grad` operation. From these, compute both $2 \times 2$ matrices $\widehat{J} = n^{-1} \widehat{J}_{\rm full}$ and $\widehat{K} = n^{-1} \sum_{i=1}^n \widehat{u}_i \widehat{u}_i^{\rm t}$, and comment regarding their similarity. Do also the sandwich matrix $\widehat{\Sigma} = \widehat{J}^{-1} \widehat{K} \widehat{J}^{-1}$. Explain that model robust confidence intervals are as above, but now with `sdhat2 = sqrt(diag(Sigmahat))`.

**(d)** Consider the probability $p = p(\xi, \sigma) = \Pr(Y \geq y_0) = 1 - \Phi((y_0 - \xi)/\sigma)$. Compute the ML estimate $\widehat{p} = p(\widehat{\xi}, \widehat{\sigma})$. Without doing the math, but just via programming, compute the 2-vector $\widehat{c} = \partial p(\widehat{\xi}, \widehat{\sigma})/\partial \theta$. Explain that $\widehat{c}^{\rm t} \widehat{\Sigma} \widehat{c}$ is the estimated variance of $\widehat{p}$, and compute a confidence interval.

**(e)** Via some extra programming, get hold of the profile functions $\ell_{\rm prof}(\xi)$ and $\ell_{\rm prof}(\sigma)$. From these compute and display deviance curves $D_1(\xi)$ and $D_2(\sigma)$. Do this also for the focus parameter $p = p(\xi, \sigma)$ above.

**(f)** Choose a suitable $\sigma_0$ value to be interested in. Redo the programming above, now as a function of only $\xi$, to find the log-likelihood maximum $\ell_{\rm max,narr}$. Then do Wilks to test the hypothesis $\sigma = \sigma_0$ against $\sigma \neq \sigma_0$.

**(g)** (xx point to and do the Cramér–Slutsky Wald test detail regarding $(\widehat{\mu} - \mu)/\widehat{\rm sd}$ being as ok as $(\widehat{\mu} - \mu)/{\rm sd}$. xx)

fix this, place it earlier

**Ex. 5.28** *We can do things.* The spirit of this exercise is to see that the log-likelihood machinery is useful, versatile, flexible, and not too hard to use in new situations, outside ordinary textbook terrain.

just do it

**(a)** One records the number of a certain event, say per week, over a time period. Suppose most of these counts are Poisson-like, with some parameter $\theta$, but that a fraction comes from another Poisson with a higher parameter. A model for such data is that $Y_i$ stems from the mixture distribution $(1-p)\,{\rm Pois}(\theta) + p\,{\rm Pois}(c\theta)$, with $c > 1$. (i) Find the mean and variance for this distribution. (ii) Generate such a dataset, say $y_1, \ldots, y_n$ with $n = 250$, $\theta = 10$, $c\theta = 20$, $p = 0.10$. Taking first $c = 2$ known, estimate the parameters $(p, \theta)$, along with confidence intervals. (iii) Using the same data, now with $c$ unknown, estimate all three parameters, with confidence intervals. Briefly investigate how much is earned in precision for estimating $(p, \theta)$ when $c$ is known compared to it being unknown.

**(b)** Suppose data $Y_i^0$ are generated according to some normal $(\xi, \sigma^2)$, but that we are only able to see those falling inside some observation window $[a, b]$. Write down the log-likelihood function, for the observed data $Y_i$. To see how it works, generate say $n_0 = 1000$ points from the standard normal, keep the $n$ amongst these that fall inside $[1.0, 3.0]$, and estimate the two parameters from these. Find also approximate confidence intervals for the two parameters.

**(c)** For two distributions with densities $f_1, f_2$ and c.d.f.s $F_1, F_2$, consider the bivariate

density

$$f(x,y,a) = f_1(x)f_2(y)[1 + a\{F_1(x) - \tfrac{1}{2}\}\{F_2(y) - \tfrac{1}{2}\}].$$

What is the range of the dependence parameter $a$? Show that the marginal distributions are $f_1, f_2$. Generate first binormal data with $\mathrm{N}(0,1)$ marginals, and estimate $a$, with confidence interval for $a$. Then generate another binormal dataset, and estimate the five parameters $(\xi_1, \xi_2, \sigma_1, \sigma_2, a)$, along with confidence intervals.

**(d)** Tired of normality? Then generalise it. With $\mathrm{Be}_{a,b}(x)$ the c.d.f. for the $\mathrm{Beta}(a,b)$ distribution, consider the four-parameter model with c.d.f. $F(y) = \mathrm{Be}_{a,b}(\Phi((y-\xi)/\sigma))$. Write up an expression for the log-likelihood function. Generate first $n = 500$ datapoints from the standard normal, and check if the ML metod succeeds in coming close to the true parameters $(0,1,1,1)$. Argue that you here have a test for normality, and spell out the ingredients. Then simulate another dataset, from $(a,b) = (1.23, 2.34)$, compute ML estimates and their estimated standard deviations.

### Convex criteria, log-concave likelihoods, differentiability in q.m.

**Ex. 5.29** *Minimisers of convex processes, I.* We have seen useful constructions, methods, and results for ML estimators in Ex. 5.10, 5.11, 5.13. There are issues worth refining and generalising, however. The regularity conditions required for the Taylor expansion based arguments to go fully through are a bit cumbersome, and there are important constructions where the $\mathrm{E}_G \log f(Y,\theta)$ is not smooth. Here we give the basics for how matters simplify, with weaker conditions, if the log-likelihood is concave, or more generally when the estimator is defined as the minimiser of a convex criterion function.

**(a)** From pointwise to uniform: Suppose $A_n(s)$ is a sequence of convex random functions defined on an open convex set $\mathcal{S}$ of $\mathbb{R}^p$, which convergences in probability to some $A(s)$, for each $s \in \mathcal{S}$. Show that the convergence is automatically uniform; $\max_{s\in\mathcal{S}} |A_n(s) - A(s)| \to_{\mathrm{pr}} 0$.

**(b)** Nearness of argmins: Suppose $A_n(s)$ is convex and is approximated by $B_n(s)$. Let $\alpha_n$ and $\beta_n$ be the argmins of $A_n$ and $B_n$. Then there is a probabilistic bound on how far these minimisers can be from each other: show that

$$\Pr(\|\alpha_n - \beta_n\| \geq \delta) \leq \Pr\{\Delta_n(\delta) \geq \tfrac{1}{2} h_n(\delta)\},$$

in which

$$\Delta_n(\delta) = \sup_{\|s-\beta_n\|\leq\delta} |A_n(s) - B_n(s)| \quad \text{and} \quad h_n(\delta) = \inf_{\|s-\beta_n\|=\delta} B_n(s) - B_n(\beta_n).$$

**(c)** Basic corollary: Suppose $A_n(s)$ is convex and can be represented as $\frac{1}{2}s^{\mathrm{t}}Js + U_n^{\mathrm{t}}s + C_n + r_n(s)$, where $J$ is symmetric and positive definite, $U_n$ is stochastically bounded, $C_n$ is arbitrary, and $r_n(s) \to_{\mathrm{pr}} 0$ for each $s$. For the approximation $B_n(s) = \frac{1}{2}s^{\mathrm{t}}Js + U_n^{\mathrm{t}}s + C_n$, show that $\beta_n = -J^{-1}U_n$ is its argmin. Then demonstrate that their minimisers as well as their minima are close. Specifically, show (i) that $\alpha_n - \beta_n \to_{\mathrm{pr}} 0$; and (ii) that $A_{n,\min} - B_{n,\min} \to_{\mathrm{pr}} 0$.

**(d)** Show that if in addition $U_n \to_d U$, then $\alpha_n \to_d -J^{-1}U$, and that $B_{n,\min} - C_n$ as well as $A_{n,\min} - C_n$ tend to $-\frac{1}{2}U^{\mathrm{t}}J^{-1}U$. These two statements are what we worked hard for in Ex. 6.9, 5.14 see (6.4)–(5.9), now obtained in a simpler fashion and with weaker smoothness assumptions, though bought with the extra convexity condition.

**(e)** Prove the following modest but useful generalisation of the above: the statements continue to hold if a random matrix $J_n$ replaces $V$, provided $J_n \to_{\rm pr} J$.

**Ex. 5.30** *Minimisers of convex processes, II.* The framework worked with now is as in Ex. 6.3 and 5.13, with $Y_1, \ldots, Y_n$ being i.i.d. from some $G$, a possibly multidimensional parameter $\theta_0$ defined as the minimiser of $H(\theta) = \int h(y,\theta)\,{\rm d}G(y)$, with estimator $\widehat\theta$ the minimiser of the distance function $H_n(\theta) = \int h(y,\theta)\,{\rm d}G_n(y) = n^{-1}\sum_{i=1}^n h(Y_i,\theta)$. Here we put in one more condition, however, that $h(y,\theta)$ is convex in $\theta$. The point is that this both simplifies various technical arguments, via methods of Ex. 5.29, and allows for nonsmooth distance functions.

**(a)** With $h(y,\mu) = |y-\mu|$, show that $\mu_0 = {\rm med}(G)$, the median, with $\widehat\mu = M_n$, the sample median. More generally, for some $q \in (0,1)$, consider

$$h_q(y,\mu) = q(y-\mu)_+ + (1-q)(\mu-y)_+ = \begin{cases} q(y-\mu) & \text{if } y \ge \mu, \\ (1-q)(\mu-y) & \text{if } y \le \mu. \end{cases}$$

Show that $\mu_0 = G^{-1}(q)$, the $q$ quantile.

**(b)** For an $\alpha \ge 1$, consider the parameter $\theta_0$ being the minimiser of ${\rm E}_G\,|Y-\theta|^\alpha$. Show that the distance function indeed is convex, and that the special cases $\alpha = 1$ and $\alpha = 2$ correspond to the median and the mean, respectively.

**(c)** We now work through regularity conditions ensuring control over the behaviour of such estimators. Part of the point is that we avoid needing smooth derivatives in $\theta$. Suppose that

$$h(y,\theta_0+\varepsilon) - h(y,\theta_0) = D(y)^{\rm t}\varepsilon + R(y,\varepsilon),$$

for a $D(y)$ with mean zero under $G$, and that

$${\rm E}\,\{h(Y,\theta_0+\varepsilon) - h(Y,\theta_0)\} = {\rm E}\,R(Y,\varepsilon) = \tfrac12\varepsilon^{\rm t}J\varepsilon + o(\|\varepsilon\|^2) \quad \text{as } \varepsilon \to 0$$

for a positive definite $J$. Assume furthermore that the variance matrix $K = {\rm Var}_G\,D(Y)$ is finite and that ${\rm Var}\,R(Y,\varepsilon) = o(\|\varepsilon\|^2)$. Show that $\sqrt n(\widehat\theta-\theta_0) = -J^{-1}(1/\sqrt n)\sum_{i=1}^n D(Y_i) + o_{\rm pr}(1)$. In particular, it tends to ${\rm N}_p(0, J^{-1}KJ^{-1})$. Show also that

$$W_n(\theta_0) = 2n\{H_n(\theta_0) - H_n(\widehat\theta)\} \to_d U^{\rm t}J^{-1}U,$$

and explain how this may be used to find a confidence region for $\theta_0$.

**(d)** The median: Suppose $Y_1, \ldots, Y_n$ are i.i.d. from a distribution $G$ with a density $g$ positive at the median $\mu$. For the median distance function $|y-\mu|$, show that

$$|y-(\mu+\varepsilon)| - |y-\mu| = D(y)\varepsilon + R(y,\varepsilon),$$

with $D(y) = I(y \le \mu) - I(y > \mu)$ and

$$R(y,\varepsilon) = \begin{cases} 2\{\varepsilon - (y-\mu)\}\,I(\mu \le y \le \mu+\varepsilon) & \text{if } \varepsilon > 0, \\ 2\{(y-\mu) - \varepsilon\}\,I(\mu+\varepsilon \le y \le \mu) & \text{if } \varepsilon < 0, \end{cases}$$

with $R(y,0) = 0$. Verify from this that ${\rm E}\,R(Y,\varepsilon) = g(\mu)\varepsilon^2 + o(\varepsilon^2)$ and ${\rm E}\,R(Y,\varepsilon)^2 = (4/3)g(\mu)|\varepsilon|^3 + o(|\varepsilon|^3)$. Deduce that $\sqrt n(M_n-\mu) \to_d {\rm N}(0, 1/\{4g(\mu)^2\})$.

**(e)** Generalise to the case of $\mu_q = G^{-1}(q)$, with $Q_{n,q} = G_n^{-1}(q)$ the empirical $q$ quantile. Show in fact that

$$Z_n(q) = \sqrt{n}(Q_{n,q} - \mu_q) = -g(\mu_q)^{-1}\sqrt{n}\{G_n(\mu_q) - q\} + \varepsilon_n(q),$$

where $\varepsilon_n(q) \to_{\rm pr} 0$ for each $q$. Derive from this that the limit is a $\mathrm{N}(0, q(1-q)/g(\mu_q)^2)$. We saw this in Ex. 3.21, but here the technique is distinctly different, and we find an interesting representation in terms of the empirical $G_n(\mu_q)$.

**(f)** Let $\xi_\alpha$ be the minimiser of $\mathrm{E}\,|Y_i - \xi|^\alpha$, with estimator $M_{n,\alpha}$ minimising $\sum_{i=1}^n |Y_i - \xi|^\alpha$. Show that

$$\sqrt{n}(M_{n,\alpha} - \xi_\alpha) \to_d \mathrm{N}(0, \tau_\alpha^2) \quad \text{with } \tau_\alpha^2 = \frac{\mathrm{E}\,|Y - \xi_\alpha|^{2(\alpha-1)}}{\{(\alpha-1)\,\mathrm{E}\,|Y-\xi_\alpha|^{\alpha-2}\}^2}.$$

Explain that if the distribution is symmetric, then the $\xi_\alpha$ is the same for each $\alpha$. Compute and display $\tau_\alpha$, for $\alpha \in [1,2]$, i.e. the range from median to mean, for the normal and for the Laplace,

**Ex. 5.31** *Maximum likelihood asymptotics with log-concave likelihood.* (xx edit with care. regularity condition on $R$. point to expofamily in later exercise. xx) Suppose $Y_1, \ldots, Y_n$ are i.i.d. from a density $g$, modelled as $f(y,\theta)$, where $\log f(y,\theta)$ is concave in $\theta$ for each $y$; the $\theta = (\theta_1, \ldots, \theta_p)$ is allowed to be multidimensional. Let $\ell_n(\theta)$ be the log-likelihood, with ML estimator $\widehat\theta$. From efforts of Ex. 5.10 we know $\widehat\theta$ is consistent for the least false parameter $\theta_0$, the KL minimiser, assumed here to be an inner point in the parameter space. log-concave density

**(a)** Show first that $\ell_n$ is concave. To start with mild conditions, assume merely that

$$\log f(y, \theta_0 + h) - \log f(y,\theta_0) = h^{\rm t} D(y) + R(y,h), \tag{5.10}$$

for a $D(y)$ with mean zero, and that $\mathrm{E}\,R(Y,h) = \frac12 h^{\rm t} J h + o(\|h\|^2)$ as $h \to 0$ for some positive definite $J$, along with $\mathrm{Var}\,R(Y,\varepsilon) = o(\|h\|^2)$. Show via the convexity driven methods of Ex. 5.29 and 5.30 that $\sqrt{n}(\widehat\theta - \theta_0) = J^{-1}U_n + o_{\rm pr}(1)$, where $U_n = (1/\sqrt{n})\sum_{i=1}^n u(Y_i,\theta_0)$. Here $U_n \to_d U \sim \mathrm{N}_p(0,K)$, with $K = \mathrm{Var}_g\, u(Y,\theta_0)$ assumed finite. Deduce the two fundamental results

$$\begin{aligned} Z_n &= \sqrt{n}(\widehat\theta - \theta_0) \to_d \mathrm{N}_p(0, J^{-1}KJ^{-1}), \\ W_n &= 2\{\ell_{n,\max} - \ell_n(\theta_0)\} \to_d U^{\rm t} J^{-1} U, \end{aligned} \tag{5.11}$$

with simplifications under model conditions. Show also that consistency of the ML estimator for $\theta_0$ is a simple consequence of the first statement. We have seen such results before, as in (5.5); the point is that they have now been derived more simply and under weaker conditions, as long as there is log-density concavity.

**(b)** Work through the details for the case of the Laplace distribution with $f(y,\theta) = \frac12 \exp(-|y-\theta|)$. The point is that with log-concavity, we reach the required results of (5.11) even without needing full smoothness in the parameter; here the score function is not defined for all values, etc. Usually, though, the $D(y)$ is the score function $u(y,\theta_0) = \partial \log f(y,\theta_0)/\partial\theta$ and $J = J(\theta_0)$ is the variance matrix of this score function, i.e. the Fisher information matrix, evaluated at the true position in the parameter space.

**(c)** (xx a couple of illustrations. for each, find the limit distribution of $\sqrt{n}(\widehat\theta - \theta_0)$. the poisson. gamma. beta. normal. also something like $Y_i \sim \mathrm{Pois}(e_i\theta)$, with different exposures $w_i$, the point is non-i.i.d. nils does this after mild extra editing for Ex. 5.29 and 5.30, with a bit of Lindeberg too. xx)

**(d)** (xx to polish. xx) something nontrivial. can do

$$f(y,\theta) = (1/k)\exp\{(y-\theta)\arctan(y-\theta)\}/\{1+(y-\theta)^2\}^{1/2},$$

which has log-density $(y-\theta)\arctan(y-\theta) - \frac{1}{2}\log\{1+(y-\theta)^2\}$ and nice score function $\arctan(y-\theta)$. something nontrivial II. and $f = (1/k)\exp(-|y-\theta|^{1.5})$.

**Ex. 5.32** *Differentiability in quadratic mean.* The key to proving asymptotic normality of the ML estimator for the log-concave densities in Ex. 5.31 was the assumption that $\log f(y,\theta_0+h) - \log f(y,\theta_0) = h^{\mathrm{t}}D(y) + R(y,h)$, where $D(y)$ and $R(y,h)$ satisfy the conditions specified in (a) of that exercise. This raises the question of what conditions are needed for such an expansion of the log-likelihood ratio to hold. As above $u(y,\theta) = \partial \log f(y,\theta)/\partial\theta$, and $i(y,\theta) = -\partial^2 \log f(y,\theta)/\partial\theta\partial\theta^{\mathrm{t}}$, and $J = -\mathrm{E}\, i(Y,\theta_0)$ and $K = \operatorname{Var} u(Y,\theta_0)$, where $\theta_0$ denotes the true parameter value.

**(a)** Assume that $\theta \mapsto \log f(y,\theta)$ is two times continuously differentiable for every $y$; that $\mathrm{E}\, u(Y,\theta_0) = 0$ and that every element of $-\partial^2 \log f(y,\theta)/\partial\theta\partial\theta^{\mathrm{t}}$ is bounded by an integreable function $k(y)$ not depending on $\theta$. Show that $\log f(y,\theta_0+h) - \log f(y,\theta_0) = h^{\mathrm{t}}u(y,\theta_0) + R(y,h)$, where $R(y,h)$ satisfies the conditons of Ex. 5.31(a) .

**(b)** With $Y_1,\dots,Y_n$ an i.i.d. sample from $f(y,\theta_0)$, show that the conditions in (a) entails the $A_n(h) = \sum_{i=1}^n \log f(Y_i,\theta_0+h/\sqrt{n})/f(Y_i,\theta_0) = \ell_n(\theta_0+h/\sqrt{n}) - \ell_n(\theta_0)$ can be expressed as

$$A_n(h) = h^{\mathrm{t}}U_n - \tfrac{1}{2}h^{\mathrm{t}}Jh + o_{\mathrm{pr}}(1). \tag{5.12}$$

where $U_n = n^{-1/2}\sum_{i=1}^n u(Y_i,\theta_0)$

**(c)** The representation in (5.12) can be derived under a weaker condition that only involves the first derivative of $\log f(y,\theta)$, namely differentiability in quadratic mean. Let $\mathcal{P} = \{P_\theta\colon \theta\in\Theta\}$ be a family of distributions, all with densities $f(y,\theta)$ with respect to some dominating measure $\mu$. We say that $\mathcal{P}$ is differentiable in quadratic mean at $\theta_0$ if

differentiability in quadratic mean

$$\int \big(f(y,\theta_0+h)^{1/2} - f(y,\theta_0)^{1/2} - \tfrac{1}{2}h^{\mathrm{t}}u(y,\theta_0)f(y,\theta_0)^{1/2}\big)^2\,\mathrm{d}\mu(y) = o(\|h\|^2), \tag{5.13}$$

as $h\to 0$. In this and the following five exercises we will see that this differentiability in quadratic mean assumption entails that the log-likelihood ratio $A_n(h)$ has the representation given in (5.12). Show that $\mathrm{E}\, u(Y,\theta_0) = 0$ and that $K$ is finite.

**(d)** Define $W_{n,i} = 2\big\{f(y,\theta+n^{-1/2}h)^{1/2}/f(y,\theta)^{1/2} - 1\big)$, then verify that $A_n(h) = 2\sum_{i=1}^n \log(\frac{1}{2}W_{n,i}+1)$, and show that by a Taylor expansion with the remainder on mean value form

$$A_n(h) = \sum_{i=1}^n W_{n,i} - \tfrac{1}{4}\sum_{i=1}^n W_{n,i}^2 + \tfrac{1}{4}\sum_{i=1}^n W_{n,i}^2 R(W_{n,i}),$$

where $R(2w) = 1-\{1+\tilde{w}(w)\}^{-2}$, and $\tilde{w}(w)$ is a point between 0 and $w$.

**(e)** We start by establishing some useful facts. Show that $n\,\mathrm{E}\, W_{n,i} = -\frac{1}{4}h^{\mathrm{t}}Kh$, that $\mathrm{E}\, W_{n,i}^2 = -4\,\mathrm{E}\, W_{n,i}$, and that $\mathrm{E}\,(W_{n,i} - n^{-1/2}h^{\mathrm{t}}u(Y_i,\theta_0))^2 \le 4D(h/\sqrt{n})$, where $D(h)$ is the expression on the left in (5.13). Deduce that

$$\mathrm{E}\,|W_{n,i}^2 - n^{-1}(h^{\mathrm{t}}u(Y_i,\theta_0))^2| \le n^{-1/2}4\{D(h/\sqrt{n})h^{\mathrm{t}}Kh\}^{1/2}.$$

**(f)** Define the mean zero random variables $\xi_{n,i} = W_{n,i} - n^{-1/2}h^{\mathrm{t}}u(Y_i,\theta_0) - \mathrm{E}\, W_{n,i}$, and verify, using the results from (e), that

$$\sum_{i=1}^n W_{n,i} = h^{\mathrm{t}}U_n - \tfrac{1}{4}h^{\mathrm{t}}Kh + \sum_{i=1}^n \xi_{n,i}.$$

Show also that $\sum_{i=1}^n \xi_{n,i} = o_{\rm pr}(1)$.

**(g)** Define the random variables $\eta_{n,i} = W_{n,i}^2 - (n^{-1/2}h^{\rm t}u(Y_i,\theta_0))^2$, as well as the mean zero random variables $\zeta_{n,i} = h^{\rm t}Kh - (h^{\rm t}u(Y_i,\theta_0))^2$. Use the facts deduced in (e), along with the law of large numbers, to show that

$$\sum_{i=1}^n W_{n,i}^2 = h^{\rm t}Kh - \sum_{i=1}^n \eta_{n,i} + n^{-1}\sum_{i=1}^n \zeta_{n,i} = h^{\rm t}Kh + o_{\rm pr}(1).$$

This means that at this point we have $A_n(h) = h^{\rm t}U_n - \frac12 h^{\rm t}Kh + \sum_{i=1}^n W_{n,i}^2 R(W_{n,i}) + o_{\rm pr}(1)$. It remains to show that $\sum_{i=1}^n W_{n,i}^2 R(W_{n,i}) = o_{\rm pr}(1)$.

**(h)** Add and subtract $n^{-1}(h^{\rm t}u(Y_1,\theta_0))^2$ from $W_{n,1}^2$ to show that

$$\Pr(W_{n,1}^2 \geq 2\varepsilon^2) \leq \frac{\|h\|}{n\varepsilon^2}{\rm E}\,\|u(Y_1,\theta_0)\|^2 I(\|u(Y_1,\theta_0)\| \geq \frac{\sqrt{n}\varepsilon}{\|h\|}) + \frac{4}{\varepsilon^2\sqrt{n}}\{D(h/\sqrt{n})h^{\rm t}Kh\}^{1/2},$$

for each $\varepsilon > 0$; and deduce from this expression that $\max_{1\leq i\leq n}|W_{n,i}| \to_{\rm pr} 0$ as $n \to \infty$. Finally, show that $\max_{1\leq i\leq n}|W_{n,i}| = o_{\rm pr}(1)$ implies that $\sum_{i=1}^n W_{n,i}^2 R(W_{n,i}) = o_{\rm pr}(1)$.

**(i)** [xx an exercise with conditions ensuring that a family of distributions is dqm xx]

**Ex. 5.33** *The exponential family class and ML.* The ML and log-likelihood machinery works particularly well for the exponential family class of models, see Ex. 1.56, 4.21. This is due the log-linear structure for parameters and sufficient statistics, and also to the consequent log-concavity of densities. Consider i.i.d. data $Y_1,\ldots,Y_n$ from a density $g$, modelled with the generic parametric $f(y,\theta) = \exp\{\theta^{\rm t}T(y) - k(\theta)\}h(y)$, with $\theta^{\rm t}T(y) = \theta_1 T_1(y) + \cdots + \theta_p T_p(y)$ involving basic data functions $T_1,\ldots,T_p$.

**(a)** Show that the score function is $u(y,\theta) = T(y) - \xi(\theta)$, with $\xi(\theta) = \partial k(\theta)/\partial\theta$. Also, the information function becomes $i(y,\theta) = -J(\theta)$, with $J(\theta) = \partial^2 k(\theta)/\partial\theta\,\partial\theta^{\rm t}$.

**(b)** Under model conditions, show that ${\rm E}_\theta\, T(Y) = \xi(\theta)$ and ${\rm Var}_\theta\, T(Y) = J(\theta)$. Outside model conditions, with $g$ not belonging to the $f_\theta$, assume merely that $T(Y)$ has true mean $\xi_0$ and true variance $K$. Show that the least false parameter $\theta_0$ minimising ${\rm KL}(g, f(\cdot,\theta))$ is characterised by $\xi_0 = \xi(\theta_0)$. This may also be written $\theta_0 = M(\xi_0)$, with $M$ the inverse map. Write also $J = J(\theta_0)$, the Fisher information matrix computed at the least false parameter position.

**(c)** Show that the log-likelihood function becomes $\ell_n(\theta) = n\{\theta^{\rm t}\bar T - k(\theta)\}$, in terms of sample averages $\bar T = (\bar T_1,\ldots,\bar T_p)^{\rm t}$, and that it is concave. Deduce that the ML estimator has $\xi(\widehat\theta) = \bar T$, of $\widehat\theta = M(\bar T)$.

**(d)** We have $U_n = \sqrt{n}(\bar T - \xi_0) \to_d U \sim {\rm N}_p(0,K)$, by the CLT. For the basic log-likelihood function process, show that

$$\begin{aligned} A_n(s) &= \ell_n(\theta_0 + s/\sqrt{n}) - \ell_n(\theta_0) = \sqrt{n}\bar T^{\rm t}s - n\{k(\theta_0 + s/\sqrt{n}) - k(\theta_0)\} \\ &= U_n^{\rm t}s - \tfrac12 s^{\rm t}J(\theta_0)s + o(\|s\|^3/\sqrt{n}) \to_d A(s) = U^{\rm t}s - \tfrac12 s^{\rm t}Js. \end{aligned}$$

This is as with Ex. 5.10-5.11, but now established rather simply and directly, with a minimum of regularity conditions. By appealing to Ex. 6.8-6.9, deduce also that

$$\sqrt{n}(\widehat\theta - \theta_0) \to_d {\rm N}_p(0, J^{-1}KJ^{-1}), \quad 2\{\ell_{n,\max} - \ell_n(\theta_0)\} \to_d W = U^{\rm t}J^{-1}U.$$

Under model conditions, the sandwich matrix is $J(\theta_0)^{-1}$ and $W \sim \chi_p^2$. Also, the Wilks theorems of Ex. 5.10 hold, for focus parameters and for testing submodels.

**(e)** (xx a bit more. xx) From $\widehat{\theta}_j = M_j(\bar{T})$, for components $M_1, \ldots, M_p$ of the mapping $\theta_0 = M(\xi_0)$, write

$$\widehat{\theta}_j = M_j(\xi_0) + M_j'(\xi_0)^{\mathrm{t}}(\bar{T} - \xi_0) + \tfrac{1}{2}(\bar{T} - \xi_0)^{\mathrm{t}} M_j''(\xi_0)(\bar{T} - \xi_0) + O_{\mathrm{pr}}(1/n^{3/2}).$$

Show that this implies $\mathrm{E}\,\widehat{\theta}_j = \theta_{0,j} + \frac{1}{2}c_j/n + o(1/n)$, with $c_j = \mathrm{Tr}(M_j''(\xi_0)K)$.

**(f)** (xx a simple example here. xx)

**Ex. 5.34** *ML asymptotics under model conditions: applications.* (xx cleaning required, and more of the directly useful up front; we estimate parameters and have confidence, almost automatically, via normality and delta method. xx) Results reached in Ex. 5.10 are central in applied statistics. The versatile ML machinery allows the statistician to construct good estimators for even complicated functions of parameters in new models, and to supplement these estimators with confidence intervals, tests, etc. We will also see that the general ML asymptotics results may be used to verify what we already knew, so to speak, regarding estimators in the more familiar models. (xx check all this to make sure that we don't become repetitive. xx)

**(a)** Let $Y$ be binomial $(n, p)$. Even before you find a formula for the ML estimator for $p$, show that $\sqrt{n}(\widehat{p} - p) \to \mathrm{N}(0, p(1-p))$. By all means, show also that $\widehat{p} = Y/n$.

**(b)** In a similar vein, study the classic case of $Y_1, \ldots, Y_n$ being i.i.d. from the normal $(\xi, \sigma^2)$. Using the Fisher information matrix found in Ex. 5.6, show that the ML estimators $\widehat{\xi}$ and $\widehat{\sigma}$ must be independent, in the limit, with approximate distributions $\mathrm{N}(\xi, \sigma^2/n)$ and $\mathrm{N}(\sigma, \frac{1}{2}\sigma^2)$. Remarkably, these results follow from the ML apparatus even without or before knowing any formulae for the estimators, and without or before knowing any finite-sample theory for these. As we know (xx crossref here xx) there is *exact* independence here, and the distribution for $\widehat{\xi}$ is exactly correct, for each $n$.

**(c)** (xx something with $\mathrm{Gam}(a, b)$, and approximate distribution for $\widehat{\mu}$, estimator for the median $\mu = \mu(a, b)$. illustrate also Wilks. which can be used without explicit formulae. xx)

**(d)** (xx the Weibull $F(t) = 1 - \exp\{-(t/a)^b\}$. perhaps an earlier exercise where we find $J(a, b)$. xx)

**(e)** Consider random i.i.d. pairs $(X_i, Y_i)$ from the standardised binormal distribution with zero means, unit variances, and correlation $\rho$. Set up the log-likelihood function $\ell_n(\rho)$, show that the Fisher information becomes $J(\rho) = (1 + \rho^2)/(1 - \rho^2)^2$, and find the limiting distribution of the ML estimator. How much better is the ML estimator compared to the usual empirical correlation coefficient $R_n$? Show that the ratio of variances, for the 1-parameter divided by that of the 5-parameter model, converges to $1/(1 + \rho^2)$.

**Ex. 5.35** *Examples of agnostic ML operations.* It is useful to go through a list of special cases, to see how the agnostic ML theory pans out in practice. Note that convergence to the normal $\mathrm{N}_p(0, J^{-1}KJ^{-1})$ takes place in general, model after model after model (including those you might invent next week), without any need for working with explicit formulae for the ML estimators etc.

**(a)** For the exponential model $\theta \exp(-\theta y)$, show that the score function is $u(y, \theta) = 1/\theta - y$, that its least false parameter value is $\theta_0 = 1/\xi_0$, in termas of the true mean $\xi_0 = \mathrm{E}\,Y$. Show that $\sqrt{n}(\widehat{\theta} - \theta_0)$ has limit distribution $\mathrm{N}(0, \sigma_0^2\theta_0^4)$, where $\sigma_0^2$ is the true variance. Show that this generalises the 'usual result' derived under model conditions.

**(b)** Then do the normal: assume data follow some density $g$, and the normal $\mathrm{N}(\xi, \sigma^2)$ model is used. We already know that the least false parameters are $\xi_0$ and $\sigma_0$, the true mean and standard deviation (i.e. even if $g$ is far from the normal). Assume that the fourth moment is finite, so that skew $= \mathrm{E}\, Z^3$ and kurt $= \mathrm{E}\, Z^4 - 3$ are finite, with $Z = (Y - \mathrm{E}\, Y)/\mathrm{sd}(Y) = (Y - \xi_0)/\sigma_0$. Working with the score function, and the second order derivatives, show that

$$J = \frac{1}{\sigma_0^2}\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{and} \quad K = \frac{1}{\sigma_0^2}\begin{pmatrix} 1 & \gamma_3 \\ \gamma_3 & 2+\gamma_4 \end{pmatrix}.$$

**(c)** For the ML estimators $\widehat{\xi}$ and $\widehat{\sigma}$, show from this that

$$\begin{pmatrix} \sqrt{n}(\widehat{\xi} - \xi) \\ \sqrt{n}(\widehat{\sigma} - \sigma) \end{pmatrix} \to_d \mathrm{N}_2\Big(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \sigma^2 \begin{pmatrix} 1 & \frac{1}{2}\gamma_3 \\ \frac{1}{2}\gamma_3 & \frac{1}{2} + \frac{1}{4}\gamma_4 \end{pmatrix}\Big).$$

Note that this is a 'rediscovery' of what we found in 2.52, but here we managed to find the limit distribution fully without knowing (or caring) about the exact expressions for the ML estimators.

**(d)** (xx one more case to come here. xx)

**Ex. 5.36** *An average power optimality property.* (xx we shall see how this pans out, and how it can be best told. make connection to BIC of Ch. 12. xx) Suppose $Y_1, \ldots, Y_n$ are i.i.d. from a smooth parametric model $f(y, \theta)$, where we need to test $\theta = \theta_0$, a given value, against $\theta \neq \theta_0$. In general there is no uniformly most powerful test. We have seen in Ex. 4.17, however, that there is a well-defined test maximising the weighted average power $\bar{\pi}_n = \int \pi_n(\theta)\, \mathrm{d}w(\theta)$, with $\pi_n(\theta)$ the power at position $\theta$ and $\mathrm{d}w(\theta)$ a given probability measure on the alternative region, here $\theta \neq \theta_0$. This optimal strategy is to use the Neyman–Pearson Lemma for the marginal density $\bar{f}(y_1, \ldots, y_n) = \int \prod_{i=1}^n f(y_i, \theta)\, \mathrm{d}w(\theta)$.

**(a)** Let $\ell_n(\theta)$ be the log-likelihood function, with $\widehat{\theta}$ the ML estimator, and write also $f_0$ for the model at the null value $\theta_0$. Show that the Neyman–Pearson ratio can be expressed as

$$\begin{aligned} R_n = \frac{\bar{f}(y_1, \ldots, y_n)}{f_0(y_1, \ldots, y_n)} &= \int \exp\{\ell_n(\theta) - \ell_n(\theta_0)\}\, \mathrm{d}w(\theta) \\ &= \exp\{\ell_n(\widehat{\theta}) - \ell_n(\theta_0)\} \int \exp\{\ell_n(\theta) - \ell_n(\widehat{\theta})\}\, \mathrm{d}w(\theta). \end{aligned}$$

**(b)** For $\theta$ close to $\widehat{\theta}$, use Taylor expansion to get

$$\ell_n(\theta) - \ell_n(\widehat{\theta}) \doteq -\tfrac{1}{2} n (\theta - \widehat{\theta})^{\mathrm{t}} J_n (\theta - \widehat{\theta}),$$

with $J_n = -n^{-1} \partial^2 \ell_n(\widehat{\theta}) / \partial\theta\, \partial\theta^{\mathrm{t}}$ the normalised Hessian matrix at the max point.

**(c)** The optimal test consists in rejecting when $R_n$ is above its null distribution threshold. Show that the above leads to

$$\begin{aligned} R_n &\doteq \exp(\tfrac{1}{2} D_n)(2\pi)^{p/2} |n J_n|^{1/2} w(\widehat{\theta}), \\ 2 \log R_n &\doteq D_n - p \log n + \log |J_n| + p \log(2\pi) + 2 \log w(\widehat{\theta}). \end{aligned}$$

Here $D_n = 2\{\ell_{n,\max} - \ell_n(\theta_0)\}$ is the Wilks or log-likelihood-ratio test statistic, with its $\chi^2_p$ limiting null distribution.

**(d)** Conclude that the $D_n$ test is an approximation to the maximum averaged power test, almost regardless of the weighting measure.

**(e)** (xx round this off. example. xx)

## Likelihoods in regression models

**Ex. 5.37** *Extending theory and methods to regression setups, I.* CHECK: Above we have dealt with likelihood methods, involving ML estimation, limit distributions under and outside model conditions, the Wilks theorem for profiled log-likelihoods, broadly valid for all smooth parametric models, etc. – but after all under simple i.i.d. conditions. Crucially, most of these concepts, methods, and results extend to classes of general regression models. Here we go through the various steps to see how the scene broadens and to learn the appropriate extensions for concepts, techniques, and results.

Consider in general terms regression data of the form $(x_i, Y_i)$, with $x_i$ a covariate vector, of length say $p$, thought to influence the main outcome $Y_i$. We assume here that the $Y_i$ are independent given the covariates. Let $f(y_i \,|\, x_i, \theta)$ be a suitable density for $y_i$ given $x_i$, with score function $u(y_i \,|\, x_i, \theta) = \partial \log f(y_i \,|\, x_i, \theta)/\partial\theta$ and information function $i(y_i \,|\, x_i, \theta) = \partial^2 \log f(y_i \,|\, x_i, \theta)/\partial\theta\,\partial\theta^{\rm t}$. The $\theta$ could comprise both regression coefficients and parameters describing the shape of the distributions. In this exercise we assume that the model holds, with $\theta_0$ denoting the true parameter, an inner point in the parameter spoace. We also postulate *the ergodic condition*, that averages over covariates stabilise, with increasing sample size; formally, for each bounded $h(x)$, there is a well-defined limit $h_0 = \int h(x)\,{\rm d}R(x)$ for $n^{-1}\sum_{i=1}^n h(x_i)$, for an appropriate distribution $R$ on the covariate space. For theory and applications, we do not need to model this $R$, or take it explicitly into account, beyond postulating its existence.

ergodic conditions

**(a)** First of all, there is a log-likelihood function, also in these regression setups, $\ell_n(\theta) = \sum_{i=1}^n \log f(y_i \,|\, x_i, \theta)$. The ML estimator $\widehat\theta$ is its maximiser, satisfying also $U_n(\widehat\theta) = 0$, with $U_n(\theta) = \sum_{i=1}^n u(y_i \,|\, x_i, \theta)$. Secondly, to extend theory and results for the i.i.d. case, see Ex. 5.10, we need to understand $U_n = n^{-1/2}U_n(\theta_0) = n^{-1/2}\sum_{i=1}^n u(Y_i \,|\, x_i, \theta_0)$. Show that it has mean zero and variance matrix $J_n = n^{-1}\sum_{i=1}^n J(x_i)$, where $J(x_i) = {\rm Var}_{\theta_0} u(Y_i \,|\, x_i, \theta_0)$. Under ergodic assumptions, there is convergence $J_n \to J$, say. Give Lindeberg type conditions under which $U_n \to_d U \sim {\rm N}_p(0, J)$.

**(b)** Extend techniques from Ex. 5.10 to deduce the natural parallel to $A_n \to_d A$ convergence in this general regression setting, that

$$A_n(s) = \ell_n(\theta_0 + s/\sqrt{n}) - \ell_n(\theta_0) = U_n^{\rm t} s - \tfrac12 J_n s + r_n(s) \to_d A(s) = U^{\rm t} s - \tfrac12 s^{\rm t} J s,$$

Set up clear but mild regularity conditions, as in Ex. 6.9, to secure the required $r_n(s) \to_{\rm pr} 0$ for each $s$. Explain as in Ex. 5.10 that this leads to the two fundamental results

$$\sqrt{n}(\widehat\theta - \theta_0) \to_d {\rm N}_p(0, J^{-1}) \quad \text{and} \quad 2\{\ell_{n,\max} - \ell_n(\theta_0)\} \to_d \chi^2_p,$$

seen before in (5.5), but now under the more general Lindeberg type conditions. Show also that the observed Fisher information matrix

observed Fisher information matrix

$$\widehat J_{n,\rm full} = -\partial^2 \ell_n(\widehat\theta)/\partial\theta\,\partial\theta^{\rm t}, \tag{5.14}$$

i.e. the Hessian mastrix associated with the maximisation of the log-likelihood, satisfies $\widehat J_n = n^{-1}\widehat J_{\rm full} \to_{\rm pr} J$. Deduce from this that $\widehat\theta \approx_d {\rm N}_p(\theta_0, \widehat J_{\rm full}^{-1})$. (xx need to point to Ch2 thing with weak LLN for averages of non-i.i.d. xx)

**(c)** (xx rounding this off. also log-likelihood profiling and deviance and $\chi^2_1$ limit and CDs. the point is that i.i.d. results all extend to the broad regression cases. point also to exercise below with outside the model. xx)

**Ex. 5.38** *Wald tests.* (xx something here, re Wald tests, used e.g. in regression models. uses abound. p-value. more on power for the two variations with two nevnere. point back to Ex. 4.5, and to a couple of stories where we use this tool. xx)

**Ex. 5.39** *Linear regression revisited.* (xx edit and clean. xx) Consider the linear regression model of Ex. 3.34, with $Y_i \,|\, x_i \sim \mathrm{N}(x_i^{\mathrm{t}}\beta, \sigma^2)$, for which exact finite-sample theory has been well developed. We now take another look at this classical model, with the general likelihood tools.

**(a)** For the log-likelihood, show that $\ell_n(\beta,\sigma) = -n\log\sigma - \frac{1}{2}Q(\beta)/\sigma^2 - \frac{1}{2}n\log(2\pi)$, with $Q(\beta) = \sum_{i=1}^n (y_i - x_i^{\mathrm{t}}\beta)^2$ Show that the ML estimator for $\beta$ is the least squares estimator $\widehat\beta = \Sigma_n^{-1} n^{-1} \sum_{i=1}^n x_i Y_i$, with $\Sigma_n = n^{-1}\sum_{i=1}^n x_i x_i^{\mathrm{t}}$, see the exercise mentioned, and that $\widehat\sigma = (Q_0/n)^{1/2}$, with $Q_0 = Q(\widehat\beta)$ the minimum of $Q(\beta)$.

**(b)** Show that the score function becomes

$$u(y_i \,|\, x_i, \beta, \sigma) = \begin{pmatrix} (1/\sigma^2)(y_i - x_i^{\mathrm{t}}\beta)x_i \\ -1/\sigma + (1/\sigma^3)(y_i - x_i^{\mathrm{t}}\beta)^2 \end{pmatrix} = \begin{pmatrix} (1/\sigma)\varepsilon_i x_i \\ (1/\sigma)(\varepsilon_i^2 - 1) \end{pmatrix}$$

in terms of $\varepsilon_i = (y_i - x_i^{\mathrm{t}}\beta)/\sigma$, which are independent standard normals under the model. With $(\beta_0, \sigma_0)$ the true parameters, deduce that the $(p+1)\times(p+1)$ Fisher information matrix becomes

$$J_n = n^{-1}\sum_{i=1}^n \mathrm{Var}_{\beta_0,\sigma_0}\, u(Y_i \,|\, x_i, \beta_0, \sigma_0) = (1/\sigma_0^2)\,\mathrm{diag}(\Sigma_n, 2),$$

with $\Sigma_n = n^{-1}\sum_{i=1}^n x_i x_i^{\mathrm{t}} = n^{-1}X^{\mathrm{t}}X$. Show also that the observed Fisher information matrix becomes $\widehat J_{n,\mathrm{full}} = (n/\widehat\sigma^2)\,\mathrm{diag}(\Sigma_n, 2)$.

**(c)** (xx then on to what likelihood theory implies for $\widehat\beta$ and $\widehat\sigma$. the $J_n$ and $\widehat J_n$. a $t_{n-p}$ vs. approximate normality. we reproduce the $\widehat\beta$ distribution, and come close for $\widehat\sigma^2$. xx)

**Ex. 5.40** *Logistic regression.* Consider binary outcome data, where the values 0-1 for $Y_i$ are influenced by a covariate vector $x_i$, of dimension say $p$. The logistic regression model takes the probabilities to be

logistic regression

$$p_i = \Pr(Y_i = 1 \,|\, x_i) = H(x_i^{\mathrm{t}}\beta) = \frac{\exp(x_i^{\mathrm{t}}\beta)}{1+\exp(x_i^{\mathrm{t}}\beta)} \quad \text{for } i = 1,\ldots,n, \tag{5.15}$$

with $H(u) = \exp(u)/\{1+\exp(u)\}$ the logistic transform, studied in Ex. 1.63. One may interpret the model via underlying or latent i.i.d. variables $Z_i$, having the logistic distribution, with the individuals having different thresholds; $p_i = \Pr(Z_i \le x_i^{\mathrm{t}}\beta)$ says that outcomes are '1' for those individuals whose $Z_i$ fall to the left of the threshold.

**(a)** Show that $H(u) = p$ means $u = H^{-1}(p) = \log\{p/(1-p)\}$, so that the model can be represented as $\log\{p_i/(1-p_i)\} = x_i^{\mathrm{t}}\beta$.

**(b)** Show that $\Pr(Y_i = y \,|\, x_i) = p_i^y(1-p_i)^{1-y}$, for the two outcomes, and deduce that the log-likelihood function can be written

$$\ell_n(\beta) = \sum_{i=1}^n \{y_i \log p_i + (1-y_i)\log(1-p_i)\} = \sum_{i=1}^n \big[y_i x_i^{\mathrm{t}}\beta - \log\{1+\exp(x_i^{\mathrm{t}}\beta)\}\big].$$

Show from this that the estimation equation, giving rise to the ML estimator $\widehat\beta$, is $\sum_{i=1}^n (y_i - p_i)x_i = 0$, and that

$$J_{n,\mathrm{full}}(\beta) = -\frac{\partial^2 \ell_n(\beta)}{\partial\beta\partial\beta^{\mathrm{t}}} = \sum_{i=1}^n p_i(1-p_i)x_i x_i^{\mathrm{t}} = \sum_{i=1}^n H(x_i^{\mathrm{t}}\beta)\{1-H(x_i^{\mathrm{t}}\beta)\}x_i x_i^{\mathrm{t}}.$$

This matrix is assumed here to be positive definite, which in particular requires $n \geq p$. Explain that the log-likelihood function is concave, with $\widehat{\beta}$ the unique maximiser.

**(c)** Show that, under model conditions, $\widehat{\beta} \approx_d \mathrm{N}_p(\beta, \widehat{J}_{n,\mathrm{full}}^{-1})$, where $\widehat{J}_{n,\mathrm{full}} = J_{n,\mathrm{full}}(\widehat{\beta})$ is the observed Fisher information matrix (the Hessian matrix of minus the normalised log-likelihood function, at the ML position).

**(d)** Consider an individual, perhaps outside the dataset, with covariate vector $x_0$. Show that $x_0^{\mathrm{t}}\widehat{\beta}$ is approximately a normal $(x_0^{\mathrm{t}}\beta, x_0^{\mathrm{t}}\widehat{J}_{n,\mathrm{full}}^{-1}x_0)$, and use this to construct a confidence interval for $p(x_0) = \Pr(Y_0 = 1 \,|\, x_0)$.

**(e)** (xx repair a bit here. xx) Consider the important special case of a single $x_i$ recorded for $Y_i$, where we write the model equation as $p_i = H(a + bx_i)$, corresponding to 2-size vectors $(1, x_i)^{\mathrm{t}}$ in the more general notation used above. Show that $(\widehat{a}, \widehat{b})$ are the solutions to $\sum_{i=1}^n (y_i - p_i) = 0$ and $\sum_{i=1}^n (y_i - p_i)x_i = 0$, and that

$$J_n(a,b) = \sum_{i=1}^n \frac{\exp(a+bx_i)}{\{1+\exp(a+bx_i)\}^2} \begin{pmatrix} 1 & x_i \\ x_i & x_i^2 \end{pmatrix}.$$

**(f)** The logistic regression model uses the $H$ transform, which is symmetric around zero. A more flexible model takes $p_i = H(x_i^{\mathrm{t}}\beta)^\kappa$, allowing skewness in the underlying distribution for the latent variables. Write down the log-likelihood function, explain that standard theory applies for the ML estimator $(\widehat{\beta}, \widehat{\kappa})$. (xx point perhaps to illustration, Polish girls story, where $\kappa$ is significantly larger than 1. xx)

**Ex. 5.41** *Probit regression.* Consider again data $(x_i, Y_i)$, with covariates $x_i$ and 0-1 outcomes $Y_i$. For the logistic regression of Ex. 5.40 we modelled the 1-probabilities as $p_i = \Pr(Z_i \leq x_i^{\mathrm{t}}\beta) = H(x_i^{\mathrm{t}}\beta)$, with $H$ the logistic c.d.f. Clearly there are alternatives, and the more famous one among these is *probit regression*, which takes the $Z_i$ to be standard normal, i.e. $p_i = \Phi(x_i^{\mathrm{t}}\beta)$.

probit regression, from probability of unit

**(a)** Explain that the log-likelihood becomes $\ell_n(\beta) = \sum_{i=1}^n [y_i \log p_i(\beta) + (1-y_i)\log\{1 - p_i(\beta)\}]$. The ML estimator $\widehat{\beta}$ is its maximiser.

**(b)** Explain also that standard regression likelihood theory applies, with $\widehat{\beta} \approx_d \mathrm{N}_p(\beta, \widehat{J}_n^{-1})$, and give a recipe for computing the variance matrix.

**(c)** To compare analyses from logistic and probit regressions, we may scale the normal above to have the same variance $\tau^2 = \pi^2/3$ as the logistic. Explain that this leads to an equivalent model, though with scaled parameters, namely $p_i = \Pr(Z_i' \leq x_i^{\mathrm{t}}\beta) = \Phi(x_i^{\mathrm{t}}\beta/\tau)$.

**(d)** Simulate a simple dataset of size $n = 1000$ pairs, with $x_i$ taken $\mathrm{N}(0, \kappa^2)$ with $\kappa = 2$ and true probabilities $p_i = H(a_0 + b_0 x_i)$ with $a_0 = -0.33$ and $b_0 = 0.77$. The litt point with $\kappa = 2$ is to secure a fair range for the $x_i$, in view of the probabilities. (i) Estimate $(a, b)$ in the logistic regression model $p_i = H(a + bx_i)$; and then (ii) estimate $(a, b)$ in the probit model $p_i = \Phi((a + bx_i)/\tau)$. Draw the two estimated regression curves, say $H(\widehat{a} + \widehat{b}x)$ and $\Phi((\widetilde{a} + \widetilde{b}x)/\tau)$. Comment on what you find.

**Ex. 5.42** *Poisson regression.* Consider independent count data $y_1, \ldots, y_n$, influenced by covariate vectors $x_1, \ldots, x_n$. The Poisson regression model, in its standard form, takes $Y_i \sim \mathrm{Pois}(\mu_i)$, with $\mu_i = \exp(x_i^{\mathrm{t}}\beta)$.

**(a)** Show that the log-likelihood function becomes

$$\ell_n(\beta) = \sum_{i=1}^n \{-\mu_i + y_i \log(\mu_i)\} = \sum_{i=1}^n \{y_i x_i^{\mathrm{t}}\beta - \exp(x_i^{\mathrm{t}}\beta)\},$$

and that the equations $\sum_{i=1}^n \{y_i - \mu_i(\beta)\}x_i = 0$ define the ML estimators.

**(b)** Show that $-\partial^2 \ell_n(\beta)/\partial\beta\,\partial\beta^{\rm t} = \sum_{i=1}^n \mu_i x_i x_i^{\rm t}$, leading to the observed Fisher information matrix $\widehat{J}_{n,\rm full} = \sum_{i=1}^n \widehat{\mu}_i x_i x_i^{\rm t}$, where $\widehat{\mu}_i = \exp(x_i^{\rm t}\widehat{\beta})$.

**(c)** For a case with covariate vector $x_0$, estimate the associated expected count $\mu_0 = \exp(x_0^{\rm t}\beta)$, and construct a confidence interval.

**(d)** Sometimes there is overdispersion, compared to how the counts $y_i$ should behave under Poisson conditions. Such overdispersion could e.g. reflect 'hidden covariates' not taken on board in the model. A method handling overdispersion is to take $Y_i \,|\, \mu \sim \text{Pois}(\mu_i)$, but modelling potential extra randomness via $\mu_i \sim \text{Gam}(\exp(x_i^{\rm t}\beta)/\tau, 1/\tau)$. This is an application of Poisson-gamma mixtures from Ex. 1.27. Show that $\mu_i$ has mean $\exp(x_i^{\rm t}\beta)$ and variance $\tau\exp(x_i^{\rm t}\beta)$; $\tau$ small means getting back to ordinary Poisson regression. Show altso that $Y_i$ has mean $\exp(x_i^{\rm t}\beta)$ variance $(1+\tau)\exp(x_i^{\rm t}\beta)$. Furthermore, work out that an expression for the log-likelihood function $\ell_n(\beta,\tau)$ is

$$\sum_{i=1}^n \{(a_i/\tau)\log(1/\tau) - \log\Gamma(a_i/\tau) + \log\Gamma(a_i/\tau + y_i) - (a_i/\tau + y_i)\log(1/\tau+1)\},$$

in which $a_i = \exp(x_i^{\rm t}\beta)$. In applications it is often fruitful to profile out the $\beta$, and studying $\ell_{n,\rm prof}(\tau) = \max_{\rm all\ \beta} \ell_n(\beta,\tau)$. (xx pointer to Story #26. xx)

**Ex. 5.43** *Exponential regression.* (xx to be polished. xx) Consider regression data $(x_i, y_i)$ for $i = 1,\ldots,n$, for nonnegative outcomes $y_i$, which we now model as $Y_i \sim \text{Expo}(\theta_i)$ with $\theta_i = \exp(x_i^{\rm t}\beta)$.

**(a)** Show that the log-likelihood may be written $\ell_n(\beta) = \sum_{i=1}^n \{x_i^{\rm t}\beta - \exp(x_i^{\rm t}\beta)y_i\}$, and the the ML estimator $\widehat{\beta}$ is the solution to $\sum_{i=1}^n x_i\{1 - \exp(x_i^{\rm t}\beta)y_i\} = 0$.

**(b)** Show next that the normalised expected Fisher information matrix takes the simple form $J_n = n^{-1}\sum_{i=1}^n x_i x_i^{\rm t}$. Explain from this that $\widehat{\beta} \approx_d \text{N}_p(\beta, n^{-1}J_n^{-1})$, under model conditions.

**(c)** Conservative analysis, i.e. not necessarily trusting model conditions, leads to a more general approximation for the variance matrix, with $n^{-1}J_n^{-1}K_nJ_n^{-1}$. Identify the $K_n$ and give a consistent estimator.

**(d)** (xx perhaps pointed to illustration in a story. xx) can also do hazard regression with $h_i(t) = h_0(t)\exp(x_i^{\rm t}\beta)$. then with $h_i(t) = h_0(t,\theta)\exp(x_i^{\rm t}\beta)$.

**Ex. 5.44** *GLM regression.* (xx to be polished. xx) basic expofamily model $f(y,\theta,\kappa) = \exp\{\theta T(y) + \kappa U(y) - k(\theta,\kappa)\}h(y)$, here with one-dimensional $\theta,\kappa$. now regression with $\theta_i = x_i^{\rm t}\beta$.

$$\ell_n(\beta,\kappa) = \sum_{i=1}^n \{x_i^{\rm t}\beta T(y_i) + \kappa U(y_i) - k(x_i^{\rm t}\beta,\kappa)\}.$$

then ML $\widehat{\beta},\widehat{\kappa}$, and more.

**(a)** (xx example. xx)

**(b)** (xx somewhere: Tore-Nils GLLM, generalised linear-lnear regression, linear in two parameters. xx)

**Ex. 5.45** *A heteroscedastic linear regression model.* (xx edit and clean. xx) In various linear regression type applications for $(x_i, y_i)$ data the linear mean assumption can be reasonable, whereas the variance might not be taken constant across covariates. Consider

therefore the model with independent $Y_i \,|\, (x_i, w_i) \sim \mathrm{N}(x_i^{\mathrm{t}}\beta, \sigma_i^2)$, for $i = 1, \ldots, n$, with covariate vectors $x_i$ of length $p$ and variance related covariates $w_i$ of length $q$, influencing $\sigma_i = \sigma \exp(\gamma^{\mathrm{t}} w_i)$. These $w_i$ could be a subset of the $x_i$ or functions thereof. It is convenient to normalise these such that $\bar{w} = n^{-1} \sum_{i=1}^n w_i = 0$, which also means that $\sigma$ is the standard deviation for an average individual, with $w_i$ equal to $\bar{w}$.

**(a)** (xx log-likelihood. score function. $J_n$ and $\widehat{J}_n$. approximations. pointers. xx) Show that the log-likelihood function can be written

$$\ell_n(\beta, \gamma) = -n \log \sigma - \tfrac{1}{2}(1/\sigma^2) Q(\beta, \gamma), \quad \text{with } Q(\beta, \gamma) = \sum_{i=1}^n \frac{(y_i - x_i^{\mathrm{t}}\beta)^2}{\exp(2w_i^{\mathrm{t}}\gamma)}.$$

Show that minimising $Q(\beta, \gamma)$ over $\beta$, for fixed $\gamma$, takes place for

$$\widehat{\beta}(\gamma) = \Big\{ \sum_{i=1}^n \frac{x_i x_i^{\mathrm{t}}}{\exp(2\gamma^{\mathrm{t}} w_i)} \Big\}^{-1} \sum_{i=1}^n \frac{x_i y_i}{\exp(2\gamma^{\mathrm{t}} w_i)}.$$

Demonstrate that this leads to the profiled log-likelihood $\ell_{n,\mathrm{prof}}(\gamma) = -n \log \widehat{\sigma}(\gamma) - \frac{1}{2} n$, where $\widehat{\sigma}(\gamma)^2 = Q_0(\gamma)/n$, with $Q_0(\gamma) = Q(\widehat{\beta}(\gamma), \gamma)$ the minimum sum of squares. Deduce from this that a recipe for finding the ML estimators consists in (i) minimising $Q_0(\gamma)$ over $\gamma$, yielding $\widehat{\gamma}$; (ii) reading off $\widehat{\beta} = \widehat{\beta}(\widehat{\gamma})$ and $\widehat{\sigma} = \widehat{\sigma}(\widehat{\gamma})$.

**(b)** (xx calibrate this with Wilks things. xx) Is it worthwhile, turning from classic linear regression, to include the extra layer of variance heterogeneity sophistication? Show that the log-likelihood-ratio test becomes that of comparing $D_n = 2n \log\{\widehat{\sigma}(0)/\widehat{\sigma}(\widehat{\gamma})\}$ to the $\chi^2_q$, in which $\widehat{\sigma}(0)^2 = Q_0(0)/n$ is the standard estimator for $\sigma^2$ under variance constancy.

**(c)** For the $p + q + 1$-parameter model, with parameters $\beta, \gamma, \sigma$, show that the score function becomes

$$u(y_i \,|\, x_i) = \begin{pmatrix} (1/\sigma^2)(y_i - x_i^{\mathrm{t}}\beta) x_i / \exp(2\gamma^{\mathrm{t}} w_i) \\ -w_i + (1/\sigma^2)(y_i - x_i^{\mathrm{t}}\beta)^2 w_i / \exp(2\gamma^{\mathrm{t}} w_i) \\ -1/\sigma + (1/\sigma^3)(y_i - x_i^{\mathrm{t}}\beta)^2 / \exp(2\gamma^{\mathrm{t}} w_i) \end{pmatrix} = \begin{pmatrix} (1/\sigma)\varepsilon_i x_i / \exp(\gamma^{\mathrm{t}} w_i) \\ (1/\sigma)(\varepsilon_i^2 - 1) w_i \\ (1/\sigma)(\varepsilon_i^2 - 1) \end{pmatrix},$$

in terms of $\varepsilon_i = (y_i - x_i^{\mathrm{t}}\beta)/\{\sigma \exp(\gamma^{\mathrm{t}} w_i)\}$. Show from this that the normalised Fisher information matrix becomes $J_n = (1/\sigma_0^2)\mathrm{diag}(\Sigma_n(\gamma_0), M_n, 2)$, at the true parameters $(\beta_0, \gamma_0, \sigma_0)$, in terms of $\Sigma_n(\gamma) = n^{-1} \sum_{i=1}^n x_i x_i^{\mathrm{t}} / \exp(2\gamma^{\mathrm{t}} w_i)$ and $M_n = n^{-1} \sum_{i=1}^n w_i w_i^{\mathrm{t}}$.

**(d)** (xx check if $\widehat{J}_{n,\mathrm{full}}$ has these off-diagonal zeroes, or if it only holds for the information calculus. spell out nice behaviour for ML estimators. $\widehat{\gamma} \approx_d \mathrm{N}_q(\gamma, (\sigma^2/n) M_n^{-1})$. xx)

**(e)** (xx round off. confidence for $\mu(x_0, w_0)$, $x_0^{\mathrm{t}}\widehat{\beta} \pm 1.96\, \widehat{\sigma} \exp(\widehat{\gamma}^{\mathrm{t}} w_0)$. pointer to Story #25. xx)

**Ex. 5.46** *Nonlinear regression.* (xx calibrate this with both classic linear regression and what we've said with genereal regression models. xx) Consider in general terms the model with independent $y_i \sim \mathrm{N}(m_i(\beta), \sigma^2)$ for $i = 1, \ldots, n$, wherethe means $m_i(\beta)$ are perhaps nonlinear functions of an appropriate vector prarameter $\beta$, involving also covariates.

**(a)** Show that the log-likelihood function becomes $-n \log \sigma - \frac{1}{2} Q_n(\beta)/\sigma^2$, with $Q_n(\beta) = \sum_{i=1}^n \{y_i - m_i(\beta)\}^2$. Show that the ML estimator for $\beta$ is the minimiser of $Q_n$, and that $\widehat{\sigma}^2 = Q_n(\widehat{\beta})/n$.

**(b)** Show that the normalised Fisher information matrix becomes

$$J_n(\beta, \sigma) = \frac{1}{\sigma^2} \begin{pmatrix} \Sigma_n & 0 \\ 0 & 2 \end{pmatrix}, \quad \text{with } \Sigma_n = n^{-1} \sum_{i=1}^n m_i^*(\beta) m_i^*(\beta)^{\mathrm{t}},$$

in which $m_i^*(\beta) = \partial m_i(\beta)/\partial\beta$.

**(c)** Deduce that $\sqrt{n}(\widehat{\beta}-\beta) \to_d \mathrm{N}(0, \sigma^2\Sigma^{-1})$, under Lindeberg type conditions, where $\Sigma$ is the limit of $\Sigma_n$. (xx more here. also the case with different normalisation, for the time series cyclic thing. xx)

**Ex. 5.47** *Extending theory and methods to regression setups, II.* (xx smooth and polish. regression outside model conditions. careful with notion, point to previous for $u$, $i$. then $\theta_{0,n}$, $J_n$, $K_n$, sandwich. xx) For i.i.d. setups we have seen how the behaviour of ML estimators can be accurately described also when the data-generating density $g$ is not inside the parametric model $f_\theta$, via a clearly defined least false parameter $\theta_0$ and the two matrices $J$ and $K$, see Ex. 5.10. Under model conditions, the two are equal. Here we extend these notions and results to regression models. The setting is partly as in Ex. 5.37, with a parametric model $f(y_i \,|\, x_i, \theta)$ for $Y_i \,|\, x_i$, but notions and results need to be lifted to the agnostic state of affairs when the real density $g(y_i \,|\, x_i)$ is not necessarily inside the model.

**(a)** For given $x$, there is a KL distance

$$\mathrm{KL}_x(g(\cdot \,|\, x), f(\cdot \,|\, x, \theta)) = \int g(y \,|\, x) \log \frac{g(y \,|\, x)}{f(y \,|\, x, \theta)} \,\mathrm{d}y.$$

Consider the overall KL distance, weighted over the covariate vectors in the sample,

$$\mathrm{KL} = n^{-1} \sum_{i=1}^n \mathrm{KL}_{x_i} = \int\int g(y \,|\, x) \log \frac{g(y \,|\, x)}{f(y \,|\, x, \theta)} \,\mathrm{d}y \,\mathrm{d}R_n(x),$$

with $R_n$ the empirical distribution over these covariate vectors. Show that this is a divergence, i.e. nonnegative and equal to zero only if $g(y \,|\, x) = f(y \,|\, x, \theta)$ for all $y$ and all $x_1, \ldots, x_n$. Let $\theta_{0,n}$ be the least false parameter, minimising this KL, for the given covariate vectors, and show that this is the same as the maximiser of

$$M_n(\theta) = n^{-1} \sum_{i=1}^n \int g(y \,|\, x_i) \log f(y \,|\, x_i, \theta) \,\mathrm{d}y.$$

Explain also that $M_n(\theta) \to M(\theta)$, under ergodic conditions, with this limit involving the distribution $R$ over covariate space, using $\int h \,\mathrm{d}R_n \to \int h \,\mathrm{d}R$. Letting $\theta_0$ be the maximiser of this limit $M(\theta)$, show also that $\theta_{0,n} \to \theta_0$, under weak conditions.

**(b)** The following arguments support the notion that the ML estimator $\widehat{\theta}$ aims at this least false $\theta_{0,n}$. With $\ell_n(\theta) = \sum_{i=1}^n \log f(y_i \,|\, x_i, \theta)$ the log-likelihood function, explain that $n^{-1}\ell_n(\theta)$ has mean $M_n(\theta)$. the mean of $n^{-1}\ell_n(\theta)$ is $M_n(\theta)$, and that its variance goes to zero. Now show that $\widehat{\theta} - \theta_{0,n} \to_{\mathrm{pr}} 0$.

**(c)** As in the earlier and simpler case of (5.4) it is fruitful to work with the log-likelihood function process. Show that

$$A_n(s) = \ell_n(\theta_{0,n} + s/\sqrt{n}) - \ell_n(\theta_{0,n}) = U_n^{\mathrm{t}} s - \tfrac{1}{2} s^{\mathrm{t}} J_n s + r_n(s),$$

now with $U_n = n^{-1/2} \sum_{i=1}^n u(Y_i \,|\, x_i, \theta_{0,n})$ and $J_n = -n^{-1} \sum_{i=1}^n i(Y_i \,|\, x_i, \theta_{0,n})$. Let $K(x_i) = \mathrm{Var}\, u(Y_i \,|\, x_i, \theta_{0,n})$. Under Lindeberg type conditions, show that $U_n \to_d U \sim \mathrm{N}_p(0, K)$, with $K$ the limit of $n^{-1} \sum_{i=1}^n K(x_i)$. Explain that $A_n(s) \to_d A(s) = U^{\mathrm{t}} s - \frac{1}{2} s^{\mathrm{t}} J s$.

**(d)** (xx then read off limits. proof of pudding lies in eating, i.e. applications, below. estimates $\widehat{J}$ and $\widehat{K}$. xx) $\widehat{J} = -n^{-1}\partial^2 \ell_n(\widehat{\theta})/\partial\theta\,\partial\theta^{\mathrm{t}}$, for estimating $K$, use the estimated scores $\widehat{u}_i = u(Y_i \,|\, x_i, \widehat{\theta})$, noting that these sum to zero. then use $\widehat{K} = n^{-1}\sum_{i=1}^n \widehat{u}_i \widehat{u}_i^{\mathrm{t}}$. sandwich $\widehat{\Sigma} = \widehat{J}^{-1}\widehat{K}\widehat{J}^{-1}$.

**Ex. 5.48** *Linear regression: agnostic analysis.* (xx to be repaired and cleaned and calibrated with next. xx) When a regression model is used, without being fully correct, the general theory of Ex. 5.47 explains (i) how the ML estimators implicitly aim for the best parametric approximation, defined via Kullback–Leibler divergences weighted over the available covariate vectors, and (ii) how the approximate distributions are affected. Here we consider such consequences for the classic linear normal regression model. For a dataset of pairs $(x_i, Y_i)$, assume merely that $\mathrm{E}\, Y_i = m(x_i)$, for some smooth $m(x)$ function, with i.i.d. errors $\varepsilon_i$ with mean zero and variance $\sigma_0^2$. The linear model approximates this via the standard $Y_i \,|\, x_i \sim \mathrm{N}(x_i^{\mathrm{t}}\beta, \sigma^2)$, for which we have found the ML estimators $\widehat{\beta}$ and $\widehat{\sigma}$ in Ex. 5.39.

**(a)** Explain that minimising the KL distance, as per Ex. 5.47, is the same as maximising

$$n^{-1}\sum_{i=1}^n \{-\log\sigma - \tfrac{1}{2}\frac{1}{\sigma^2}\mathrm{E}\,(Y_i - x_i^{\mathrm{t}}\beta)^2\} = -\log\sigma - \tfrac{1}{2}\frac{1}{\sigma^2} n^{-1}\sum_{i=1}^n [\{m(x_i) - x_i^{\mathrm{t}}\beta\}^2 + \sigma_0^2].$$

Argue from this that the least false parameter $\beta_{0,n}$ is the one minimising $Q_n(\beta) = n^{-1}\sum_{i=1}^n \{m(x_i) - x_i^{\mathrm{t}}\beta\}^2$, and show that this means $\beta_{0,n} = \Sigma_n^{-1} n^{-1}\sum_{i=1}^n m(x_i)x_i$, where $\Sigma_n = n^{-1}\sum_{i=1}^n x_i x_i^{\mathrm{t}}$. If the model is perfect, there is a true $\beta_{\mathrm{true}}$ for which $\mathrm{E}\,(Y_i \,|\, x_i) = x_i^{\mathrm{t}}\beta_{\mathrm{true}}$, in which case $\beta_{0,n} = \beta_{\mathrm{true}}$. Show next that the least false $\sigma_{0,n}$ is determined by $\sigma_{0,n}^2 = Q_{0,n} + \sigma_0^2$, where $Q_{0,n}$ is the minimum of $Q_n(\beta)$. If the linear mean is a good model, the $Q_{0,n}$ is small and the $\sigma_{0,n}$ aimed at by the ML estimator $\widehat{\sigma}$ is not much bigger than $\sigma_0$. If the linear mean is not a good approximation to the real $m(x)$, then the $\widehat{\sigma}$ implicitly picks up both data variability and the difference between $x_i^{\mathrm{t}}\beta$ and the real $m(x_i)$.

**(b)** For the two matrices $J$ and $K$ determining the behaviour of the ML estimators we have found in Ex. 5.39 that $\widehat{J} = (1/\widehat{\sigma}^2)\,\mathrm{diag}(\Sigma_n, 2)$, and that the score vectors, computed at the least false values, can be written $u(y_i \,|\, x_i, \theta_0) = (1/\sigma_{0,n})(x_i z_i, z_i^2 - 1)^{\mathrm{t}}$, in which $z_i = (y_i - x_i^{\mathrm{t}}\beta_{0,n})/\sigma_{0,n}$. Note that we for the estimated standardised residuals $\widehat{z}_i = (y_i - x_i^{\mathrm{t}}\widehat{\beta})/\widehat{\sigma}$ have $\sum_{i=1}^n \widehat{z}_i x_i = 0$ and $n^{-1}\sum_{i=1}^n \widehat{z}_i^2 = 1$. For the $K$ matrix, and with notation from Ex. 5.47, show that

$$n^{-1}\sum_{i=1}^n K(x_i) = n^{-1}\sum_{i=1}^n \frac{1}{\sigma_{0,n}^2}\mathrm{Var}\begin{pmatrix} z_i x_i \\ z_i^2 - 1\end{pmatrix}.$$

(xx round this off. two substories, depending on what is assumed for $z_i$. with $y_i = m(x_i) + \sigma_0\varepsilon_i$, the $z_i$ are not i.i.d., but we may cope with that too, via the right $\widehat{K}$. simplification if we postulate $m(x_i) = x_i^{\mathrm{t}}\beta_0$, then involving only $\widehat{\gamma}_3$ and $\widehat{\gamma}_4$ for the $\widehat{z}_i$. xx) a somewhat messy model-robust approach would use

$$\widehat{K} = \frac{1}{\widehat{\sigma}^2} n^{-1}\sum_{i=1}^n \begin{pmatrix} \widehat{z}_i^2 x_i x_i^{\mathrm{t}} & \widehat{z}_i^3 x_i \\ \widehat{z}_i^3 x_i^{\mathrm{t}} & \widehat{z}_i^4 - 1\end{pmatrix}.$$

**(c)**

**Ex. 5.49** *Logistic regression, Poisson regression: agnostic analysis.* (xx to be repaired and cleaned. xx) When the regression model used does not necessarily hold up, ML

estimators aim for the relevant least false parameters, the general theory of Ex. 5.47 shows that $\widehat{\theta} \approx_d \mathrm{N}(\theta_{0,n}, \widehat{\Sigma}/n)$, for the relevant least false parameter and with estimated sandwich matrix $\widehat{\Sigma}_n = \widehat{J}^{-1}\widehat{K}\widehat{J}^{-1}$. Under model conditions, matters simplify, the underlying $J_n$ and $K_n$ matrices are equal, and we have the standard result $\widehat{\theta} \approx_d \mathrm{N}(\theta_0, \widehat{J}^{-1}/n)$, used extensively in statistical software packages for a range of regression models. Here we check what the model agnostic setup leads to, for securing model robust inference, for standard logistic and Poission regression models; see also the agnostic analysis for the linear regression model in Ex. 5.48.

**(a)** Consider logistic regression (xx point back xx), with $p_i = H(x_i^{\mathrm{t}}\beta)$: show that $u(y_i \,|\, x_i, \beta) = (y_i - p_i)x_i$, and that

$$\widehat{J}_n = n^{-1}\sum_{i=1}^n \widehat{p}_i(1-\widehat{p}_i)x_i x_i^{\mathrm{t}}, \quad \widehat{K}_n = n^{-1}\sum_{i=1}^n (y_i - \widehat{p}_i)^2 x_i x_i^{\mathrm{t}}.$$

**(b)** Then look at Poisson regression (xx point back xx), with $\mu_i = \exp(x_i^{\mathrm{t}}\beta)$: show that $u(y_i \,|\, x_i, \beta) = (y_i - \mu_i)x_i$, with

$$\widehat{J}_n = n^{-1}\sum_{i=1}^n \widehat{\mu}_i x_i x_i^{\mathrm{t}}, \quad \widehat{K}_n = n^{-1}\sum_{i=1}^n (y_i - \widehat{\mu}_i)^2 x_i x_i^{\mathrm{t}}.$$

If there is overdispersion, with $(y_i - \widehat{\mu}_i)^2$ tending to be bigger than $\widehat{\mu}_i$, this is picked up here, with sandwich matrix bigger than $\widehat{J}^{-1}$. The potential error of applying straightforward Poisson regression, perhaps via standard packages, is that variability is underestimated, with confidence intervals becoming too narrow. For an illustration of this, see Story #26.

**(c)** With the Poisson-gamma overdispersion regression model studied in Ex. 5.42, where $Y_i \,|\, x_i$ has mean $\exp(x_i^{\mathrm{t}}\beta)$ and variance $(1+\tau)\exp(x_i^{\mathrm{t}}\beta)$, show that $\widehat{K}_n \to_{\mathrm{pr}} (1+\tau)J$, in terms of the limit $J$ of $\widehat{J}_n$. Deduce that if $\widehat{\beta}$ is the ML estimator computed for the standard Poisson regression model, then $\sqrt{n}(\widehat{\beta} - \beta_{0,n}) \to_d \mathrm{N}_p(0, (1+\tau)J^{-1})$.

### The size of the hidden box

**Ex. 5.50** *How many unseen are there?* Suppose a multinomial experiment has been carried out, involving $N$ individuals, sorted in boxes $0, 1, \ldots, k$, with probabilities $p_0, p_1, \ldots, p_k$ summing to one. The full set of observations should have been $(N_0, N_1, \ldots, N_k)$, counting the numbers in these boxes, but here we have in mind situations where the zero-box count $N_0$ has not been observed. The task is to estimate $N_0$, and hence $N = N_0 + R$, with $R = N_1 + \cdots + N_k$ the total observed count in boxes $1, \ldots, k$. The setup of Ex. 5.52, involving boxes 00, 01, 10, 11, is a special case. Methods developed below are being used in Stories #66 and #67 to estimate how many persons were killed in Srebrenica 1995 and in Guatemala, 1978–1996, based on a Venn diagram constructed from respectively two and three lists of murdered people.

**(a)** Assume a smooth parametric model is used for the probabilities, with $p_j = p_j(\theta)$ for $j = 0, 1, \ldots, k$, with $\theta$ of dimension $r \le k-1$, admitting two continuous derivatives. Show that the log-likelihood may be written

$$\ell(N, \theta) = \log(N!) - \log((N-R)!) + (N-R)\log p_0(\theta) + \sum_{j=1}^k N_j \log p_j(\theta).$$

Explain that this also leads to a profiled log-likelihood in $N$,

$$\ell_{\rm prof}(N) = \max\{\ell(N,\theta)\colon \text{all } \theta\} = \ell(N,\widehat\theta_N).$$

Maximising this function gives the ML estimators $(\widehat N,\widehat\theta)$.

**(b)** We do assume that the parametric model holds, with true parameter $\theta_0$ an inner point in the parameter space; write also $N_0$ for the true $N$. Large-sample results will be reached below in the framework of growing $N_0$. It will be seen that $\widehat N$ and $\widehat\theta$ will converge at different rates, so it is helpful to write $N = N_0\rho$ and $\widehat\rho = \widehat N/N_0$. Explain that this leads to the approximate log-likelihood $\ell(\rho,\theta)$ equal to

$$N_0\rho\log(N_0\rho) - (N_0\rho - R)\log(N_0\rho - R) + (N_0\rho - R)\log p_0(\theta) + \sum_{j=1}^k N_j \log p_j(\theta).$$

**(c)** From basic multinormal limits for multinomials, as with Ex. 2.49, show that there is joint convergence $N_0^{1/2}\{N_j/N - p_j(\theta_0)\} \to_d A_j$, a zero-mean multinormal vector with $\text{cov}(A_j,A_\ell) = p_j(\theta_0)\{\delta_{j,\ell} - p_\ell(\theta_0)\}$; note that $A_0 + \cdots + A_k = 0$. With $\widetilde p_0 = 1 - R/N_0$, show that $N_0^{1/2}\{\widetilde p_0 - p_0(\theta_0)\} \to_d A_0$.

**(d)** Letting $\psi_j(\theta) = \partial \log p_j(\theta)/\partial\theta$, explain that $\sum_{j=0}^k p_j(\theta)\psi_j(\theta) = 0$. Below, write for crispness $p_j$ and $\psi_j$ for $p_j(\theta_0)$ and $\psi_j(\theta_0)$. Put up formuale for the first-order partial derivatives of the log-likelihood $\ell(\rho,\theta)$. Show that when evaluated at $(N_0,\theta_0)$, these become

$$\begin{aligned} U_0 &= N_0(-\log\widetilde p_0 + \log p_0),\\ V_0 &= N_0\Big\{(\widetilde p_0 - p_0)\psi_0 + \sum_{j=1}^k (N_j/N_0 - p_j)\psi_j\Big\}. \end{aligned}$$

**(e)** A key quantity is $M = \sum_{j=0}^k p_j\psi_j\psi_j^{\rm t}$. Show that there is convergence in distribution

$$\frac{1}{N_0^{1/2}}\begin{pmatrix}U_0\\V_0\end{pmatrix} \to_d \begin{pmatrix}U\\V\end{pmatrix} = \begin{pmatrix}-(1/p_0)A_0\\ \sum_{j=0}^k A_j\psi_j\end{pmatrix} \sim {\rm N}_{r+1}(0,J), \text{ with } J = \begin{pmatrix}(1-p_0)/p_0 & -\psi_0^{\rm t}\\ -\psi_0 & M\end{pmatrix}.$$

This is the Fisher information matrix for this reparametrised model, with parameters $(\rho,\theta)$, evaluated at $(1,\theta_0)$. Verify via derivatives and limits that $J$ is also the limit in probability of $-N_0^{-1}H_0$, where $H_0$ is the $(r+1)\times(r+1)$ Hessian matrix of second order partial derivatives of $\ell(\rho,\theta)$, evaluated at $(1,\theta_0)$.

**(f)** Show from this that $N_0^{1/2}(\widehat N/N_0 - 1) = (\widehat N - N_0)/N_0^{1/2}$ tends in distribution to ${\rm N}(0,\tau^2)$, with

$$\tau^2 = \frac{1}{(1-p_0)/p_0 - \psi_0^{\rm t}M^{-1}\psi_0} = \frac{p_0}{1-p_0-p_0\delta}, \quad \text{where } \delta = \psi_0^{\rm t}M^{-1}\psi_0.$$

Though the main concern might be estimating $N_0$ and $N = N_0 + R$, show also that $N_0^{1/2}(\widehat\theta - \theta_0) \to_d {\rm N}_r(0,K)$, with

$$K = (M - (1-p_0)\psi_0\psi_0^{\rm t}/p_0)^{-1} = M^{-1} + dM^{-1}\psi_0\psi_0^{\rm t}M^{-1},$$

where we write $d = r_0/(1-r_0\delta) = p_0/(1-p_0-p_0\delta)$ and $r_0 = p_0/(1-p_0)$. Explain that had $N_0$ been observed too, the result would have beem ${\rm N}_r(0,M^{-1})$, so $dM^{-1}\psi_0\psi_0^{\rm t}M^{-1}$ is the extra amount of uncertainty when estimating $\theta$, caused by $N_0$ not being counted.

**(g)** Show via Wilks theory for profiled log-likelihood functions, as in Ex. 5.13, that

$$D(N_0) = 2\{\ell_{\text{prof,max}} - \ell_{\text{prof}}(N_0)\} \to_d \chi^2_1.$$

This leads to confidence intervals, e.g. $\{N_0 \colon D(N_0) \le 1.96^2\}$, and a full confidence curve $\text{cc}(N) = \Gamma_1(D(N_0))$, as per Ch. 8, which we can read off directly from the profiled log-likelihood, without needing the perhaps cumbersome estimation of the $J$ matrix.

**(h)** For the above results we are assuming that the parametric model $p_j(\theta)$ holds, also for the null box, which cannot really be verified or tested from $N_1, \ldots, N_k$ data alone, without other assumptions again, or perhaps follow-up data. Show in general terms that the likelihood can be written

$$\begin{aligned} L(N,\theta) &= \frac{N!}{N_0!\,N_1!\cdots N_k!} p_0^{N_0} p_1^{N_1} \cdots p_k^{N_k} \\ &= \frac{N!}{R!\,(N-R)!} p_0^{N-R}(1-p_0)^R \frac{R!}{N_1!\cdots N_k!} q_1^{N_1}\cdots q_k^{N_k}, \end{aligned}$$

splitting into a binomial $R \sim \text{binom}(N, 1-p_0)$ and a multinomial $(R, q_1, \ldots, q_k)$ for what is actually observed, with $q_j = p_j/(1-p_0)$. We may infer the $q_j$ from the $p_j$ but not $p_0$ from $q_1, \ldots, q_k$. Show also the ML estimator $\widehat\theta$ can be found from the multinomial $(N_1, \ldots, N_k)$ given $R$, followed by $\widehat N = R/\{1 - p_0(\widehat\theta)\}$.

**Ex. 5.51** *Hardy-Weinberg when seeing only two of the three categories.* (xx perhaps a Simulated Story. xx) We visit Hardy-Weinberg island, in clear equilibritum, where individuals may be sorted into categories 0, 1, 2, with probabilities $(1-\theta)^2, 2\theta(1-\theta), \theta^2$. Suppose however that when accountants arrive, all the zero box people go into hiding, so from the trinomial $(N_0, N_1, N_2)$ only $N_1, N_2$ are counted. How can we estimate $N_0$, and the size of the population $N = N_0 + N_1 + N_2$ investigated?

**(a)** Show that given $R = N_1 + N_2$, $N_2$ is a binomial $(R, q_2)$, with $q_2 = p_1/(p_1+p_2) = \theta/(2-\theta)$. Show that the ML estimators are

$$\widehat\theta = \frac{2N_2}{N_1 + 2N_2}, \quad \widehat N = \frac{R}{1-(1-\widehat\theta)^2} = \frac{R}{1-\{N_1/(N_1+2N_2)\}^2}.$$

**(b)** We may find confidence intervals via the methods of Ex. 5.50. Explain that the log-likelihood is

$$\begin{aligned} \ell(N,\theta) &= \log\frac{(N!)}{(N-R)!} + 2(N-R)\log(1-\theta) + N_1\{\log\theta + \log(1-\theta)\} + 2N_2\log\theta \\ &= \log(N!) - \log((N-R)!) + (N_1+2N_2)\log\theta + (2(N-R)+N_1)\log(1-\theta), \end{aligned}$$

and that this leads to the profiled log-likelihood

$$\ell_{\text{prof}}(N) = \log(N!) - \log((N-R)!) + (N_1+2N_2)\log\widehat\theta_N + (2(N-R)+N_1)\log(1-\widehat\theta_N),$$

in which $\widehat\theta_N = (N_1+2N_2)/N$.

**(c)** To find the limiting variance $\tau^2$ of $(\widehat N - N)/N^{1/2}$, we need $\psi_0 = -2/(1-\theta)$, $\psi_1 = 1/\theta - 1/(1-\theta)$, $\psi_2 = 2/\theta$. Verify that $\sum_{0,1,2} p_j\psi_j = 0$, and show that

$$M = \sum_{0,1,2} p_j\psi_j^2 = 8 + 8\frac{(\theta-\frac12)^2}{\theta(1-\theta)} = \frac{2}{\theta(1-\theta)}.$$

Derive from this that $\delta = \psi_0^2/M = 2\theta/(1-\theta)$, and that the formula for $\tau^2$ of Ex. 5.50 simplifies to $(1-\theta)^2/\theta^2$. It is hence considerably harder to estimate $N$ with good precision if $\theta$ is small, with a high proportion $N_0/N$ in hiding.

**(d)** (xx round off. variance of $\widehat{N}/N$. may ask for a few cc($N$). xx)

**Ex. 5.52** *Capture-recapture.* (xx nils thinks this exercise goes away, with the material taken care of inside Story #66. xx) The methods developed in Ex. 5.50 for hidden boxes in multinomial setups apply in particular to the estimation of closed populations, via capture-recapture. The case dealt with in the Srebrenica Story #66 involves boxes 00, 01, 10, 11, leading to the Petersen (1896) estimator, is a special case, and we now apply better and more versatile methods. The fish population is $\{1, \ldots, N\}$, and the task is to estimate $N$ based on capture, mark, release, recapture.

**(a)** With capture probabiities $p$ and $q$ for two independent rounds, this leads to a Venn diagram of $N_{0,0}, N_{0,1}, N_{1,0}, N_{1,1}$, with $N_{1,1}$ the number of fish caught in both rounds, etc. The number $N_{0,0}$ of fish caught in neither of the two rounds is not observed. Show that the this amounts to a mutlinomial setup with probabilities $p_{0,0} = (1-p)(1-q)$, $p_{0,1} = (1-p)q$, $p_{1,0} = p(1-q)$, $p_{1,1} = pq$. Show that the log-likelihood function may be written

$$\begin{aligned}\ell(N,p,q) \;=\; &\log(N!) - \log((N-R)!) + (N-R)\log p_{0,0} \\ &+ N_{0,1}\log p_{0,1} + N_{1,0}\log p_{1,0} + N_{1,1}\log p_{1,1}.\end{aligned}$$

Simplify and find an expression for the profiled $\ell_{\rm prof}(N) = \ell(N, \widehat{p}_N, \widehat{q}_N)$.

**(b)** The theory of Ex. 5.50 applies, both for reading off confidence intervals from $D(N) = 2\{\ell_{\rm prof}(\widehat{N}) - \ell_{\rm prof}(N)\}$ and for assessing the variance of $\widehat{N}$. To this end, find the four log-probability derivatives $\psi_{i,j}(p,q)$, verify that $\sum_{i,j} p_{i,j}\psi_{i,j} = 0$, and show that $M = \sum_{i,j} p_{i,j}\psi_{i,j}\psi_{i,j}^{\rm t}$ is the diagonal matrix with elements $1/(p(1-p)), 1/(q(1-q))$. From this find also

$$\delta = \psi_{0,0}^{\rm t} M^{-1}\psi_{0,0} = p/(1-p) + q/(1-q),$$

and show that the limit variance for $N^{1/2}(\widehat{N}/N - 1)$ is $\tau^2 = (1-p)(1-q)/(pq)$. This is indeed equal to the limit variance found for $N^{1/2}(N^*/N - 1)$ in Story #66 for the Petersen estimator $N^*$. This is no coincidence, show that $N^*$ and $\widehat{N}$ is at most 1 apart. Note that these estimates and confidence intervals are found not only without knowing $N_{0,0}$, but also not knowing the probabilites $p$ and $q$ of fish being caught in the two rounds.

**(c)** We now go on to three rounds of capture-mark-release, with probabilities $p, q, r$. In the multinomial setup, with $2^3 = 8$ counts $N_{i,j,k}$, show that $p_{0,0,0} = (1-p)(1-q)(1-r)$, and so on, up to $p_{1,1,1} = pqr$ the probability of a fish being caught in each of the threee rounds. Again, the theory of Ex. 5.50 applies, so it is a matter of working out the details for this setup. Find an explicit formula for the profiled log-likelihood function $\ell_{\rm prof}(N) = \ell(N, \widehat{p}_N, \widehat{q}_N, \widehat{r}_N)$, amenable for implementation and for reading off confidence intervals for $N$ via the Wilks theorem. Set up formulae for the eight $\psi_{i,j,\ell}$; verify that $\sum_{i,j,\ell} p_{i,j,\ell}\psi_{i,j,\ell} = 0$; and show that $M = \sum_{i,j,\ell} p_{i,j,\ell}\psi_{i,j,\ell}\psi_{i,j,\ell}^{\rm t}$ is diagonal, with elements $\{p(1-p)\}^{-1}$, $\{q(1-q)\}^{-1}$, $\{r(1-r)\}^{-1}$. Explain that also leads to

$$\delta = \psi_{0,0,0}^{\rm t} M^{-1}\psi_{0,0,0} = p/(1-p) + q/(1-q) + r/(1-r).$$

Conclude that $(\widehat{N} - N)/N^{1/2} \to_d {\rm N}(0, \tau^2)$, with

$$\tau^2 = \frac{(1-p)(1-q)(1-r)}{1-(1-p)(1-q)(1-r)(1+\delta)} = \frac{(1-p)(1-q)(1-r)}{pq+pr+qr-2pqr}.$$

Discuss how the variance of $\widehat{N}$ is influenced by say small, moderate, and higher values of $p, q, r$. In various applications, from fishery sciences to estimating the number of bank acccount cheaters, these probabilities might tend to be small.

**(d)** Generalise the above to the case of four independent rounds of capture-mark-release.

**Further material and leftovers (as of 30-June-2025)**

## Notes and pointers

(xx some pointers to more, including Empirical Likelihood, Hjort et al. (2009, 2018). xx)

(xx Bhattacharyya, 1946, lower bounds more general than CR. xx)

(xx for the 1-para vs. 5-para kladd to be pushed to solutions follows. to be fixed. xx) we know $(1-\rho^2)^2$ variance for $R_n$ Ch2. so ratio of variances should be

$$r = \frac{(1-\rho^2)^2}{1+\rho^2}\frac{1}{(1-\rho^2)^2}$$

check with

$$\log f(x,y,\rho) = -\tfrac{1}{2}\log(1-\rho^2) - \tfrac{1}{2}(x^2+y^2-2\rho xy)/(1-\rho^2),$$

score function

$$u(x,y,\rho) = \frac{1}{(1-\rho^2)^2}\{\rho - \rho^2 + (1+\rho^2)xy - \rho(x^2+y^2)\},$$

where we may check that the mean is zero. We find $\mathrm{Var}\, XY = 1+\rho^2$, $\mathrm{Var}\,(X^2+Y^2) = 4(1+\rho^2)$, $\mathrm{cov}(XY, X^2+Y^2) = 4\rho$, and this leads to Fisher information $J(\rho) = (1+\rho^2)/(1-\rho^2)^2$.

(xx make sure we have something on board with BHHJ for regression models, placed after ML for regression. xx)

[xx CR bound: In its simplest form, the inequality goes back to Cramér (1946) and Rao (1945). xx]

(xx M-estimators, Z-estimators, Huber, minimum divergence, point to Basu et al. (2011), more on minimum divergence. two-stage estimators, WalkerHjort24. xx)

[xx least false: a term invented by Hjort, Hjort believes, see Hjort (1986b, 1992), and now used somewhat frequently in the literature. xx]

Read more about risk functions in DeGroot (1970).

For Ex. 6.6, point to Hjort and Jones (1996), Schweder and Hjort (2016), and WalkerHjort24.

[xx check and calibrate what's here and what's in Ch. 8, regarding CD things. xx]

(xx we see that ML matches CR bounds, under model conditions. point to Hajek convolution theorem, and other characterisations. perhaps Hodges superefficient thing too. xx)

(xx point to Hjort (2008), re ML and least false etc. xx)

# I.6

# Minimum divergence

The previous Ch. 5 has developed a versatile machinery for likelihood analysis in parametric models, with estimators, approximate distributions, deviances, Wilks theorems, etc. We have also seen the key role played by the Kullback–Leibler divergence $\mathrm{KL}(g, f_\theta)$. In the present chapter the scope is widened to allow for other natural divergences, leading then to classes of minimum divergence estimators. These might have favourable properties not shared by the ML ones, regarding e.g. robustness, both with respect to potential outliers and datapoints being outside the normal scope of the model, and to model specification itself. We discuss in particular the Basu-Harris-Hjort-Jones estimation scheme, associated with a particulat BHHJ divergence. The apparatus works for i.i.d. data, regression models, and beyond. Also included below is the useful tool of influence functions for estimators and functionals.

*Key words:* BHHJ estimators, Huber estimator, influence functions, minimum divergence, regression models, robustness

Our start setup for the chapter is that of i.i.d. data $Y_1, \ldots, Y_n$ from a data-generating distribution with c.d.f. $G$ and density $g$, and that a parametric model $f_\theta(y) = f(y, \theta)$ is put forward as an approximation to the $g$. Suppose in general terms that $h(y, \theta)$ is such that $H(\theta) = \mathrm{E}_G\, h(Y, \theta) = \int g(y) h(y, \theta)\, \mathrm{d}y$ has a unique minimum $\theta_0$, to be thought of as providing the best parametric approximation to $g$. Then the *minimum divergence estimator* $\widehat{\theta}$ is the minimiser of the *divergence criterion function*

$$H_n(\theta) = n^{-1} \sum_{i=1}^{n} h(y_i, \theta). \tag{6.1}$$

The chief special case is that of $h(y, \theta) = -\log f(y, \theta)$, for which $H_n(\theta)$ is minus the normalised log-likelihood function $\ell_n(\theta)$, and $\widehat{\theta}$ hence being the ML estimator. Thus the maximum likelihood machinery dealt with in Ch. 5 is a special case. The ML estimator, seen here to minimise minus the log-likelihood, aims for $\theta_0$, the maximiser of $\int g \log f_\theta\, \mathrm{d}y$, i.e. the minimiser of the KL divergence $\mathrm{KL}(g, f_\theta)$.

We might of course pedantically distinguish beetween 'minimum divergence' and 'minimum criterion function' estimators, but it is common practice to use the former for classes of minimisers of functions of the type above, even when the $h(y, \theta)$ criterium function is not immediately or necessarily related to a divergence. In many cases one may start with a bona fide divergence, say $d(g, f_\theta)$, measuring the distance from $g$ to

its approximation $f_\theta$, and then discover that a suitable version of (6.1) matches that. The ML is again a case in point, as the KL divergence may be written $\mathrm{KL}(g, f_\theta) = \int g \log g \,\mathrm{d}y - \int g \log f_\theta \,\mathrm{d}y$. The first term does not involve the parameter, so maximising an smpirical version of $\int \log f_\theta \,\mathrm{d}G(y)$ does the minimum divergence job. This is indeed the ML procedure.

The object of the present follow-up chapter is to widen the scope, exhibiting other natural divergences. As long as minimising such a divergence, perhaps with some extra efforts, can be translated to a minimisation of a criterion function, as above, then the general machinery developed below will handle the central inference questions, regarding estimation, confidence, testing, limit distributions, etc. We shall also discover that what we learned in the likelihood chapter can be lifted, largely without laborious efforts, to these more general cases. Regression setups will also be worked through, with parameters constructed to minimise suitable $\sum_{i=1}^n h(y_i \,|\, x_i, \theta)$ criteria functions. In particular, there will be robust alternatives to linear regression and other regression models, where large or outlying datapoints will be properly downplayed.

### Divergences and minimising them

**Ex. 6.1** *Minimising $L_2$ distance.* Suppose i.i.d. data $Y_1, \ldots, Y_n$ come from some data generating density $g$ which we wish to approximate with some parametrically modelled $f_\theta$. The $L_2$ distance between $g$ and $f_\theta$ is

$$D(g, f_\theta) = \int (g - f_\theta)^2 \,\mathrm{d}y = \int f_\theta^2 \,\mathrm{d}y - 2 \int g f_\theta \,\mathrm{d}y + a(g),$$

where $a(g)$ does not depend on $\theta$.

**(a)** Use this to motivate what we may call the minimum $L_2$ estimator $\widehat{\theta}$, the minimiser of $D_n(\theta) = \int f_\theta^2 \,\mathrm{d}y - 2n^{-1} \sum_{i=1}^n f(Y_i, \theta)$. Operationally, this is easiest when there is a closed-form formula for $\int f_\theta^2 \,\mathrm{d}y$, but numerical minimisation might be carried out even without this. Simulate 100 detapoints from a $\mathrm{Gam}(a, b)$, where you choose $(a, b)$ as you wish, and estimate these using this method.

**(b)** Carry out a similar simple experiment with 100 datapoints drawn from a normal, i.e. estimate the mean and standard deviation using the minimum $L_2$ method. Then do the following variation: push one of the 100 datapoints far away from the others, which will cause the traditional estimates of $\mu$ and $\sigma$ to be off; show however that the minimum $L_2$ estimates are far less affected. – The minimum $L_2$ method is a special case of the BHHJ method, see Ex. 6.4.

**Ex. 6.2** *Estimating means via minimum divergence.* Suppose again i.i.d. data $Y_1, \ldots, Y_n$ stem from some data generating density $g$. The point to convey here is that key parameters can be seen as minimisers of natural distance functions, or divergences, leading to recipes for estimating them.

**(a)** For any distribution $G$, let $\xi(G)$ be the minimiser of $H(\xi) = \mathrm{E}_G (Y - \xi)^2$, with estimator the minimiser $\widehat{\xi}$ of the empirical version $H_n(\xi) = n^{-1} \sum_{i=1}^n (Y_i - \xi)^2$. Show, for this simple illustration of minimum divergence function methods, that $\xi(G)$ is the mean $\mathrm{E}_G Y$ and that $\widehat{\xi} = \bar{Y}$, the sample mean.

**(b)** For characterising and then estimating both mean and standard deviation, consider $h(y, \xi, \sigma) = (y - \xi)^2 + \{y^2 - (\xi^2 + \sigma^2)\}^2$, and let $(\xi, \sigma)$ be the minimiser of the distance function $H(\xi, \sigma) = \mathrm{E}_G \, h(Y, \xi, \sigma)$. Show that $\xi(G)$, $\sigma(G)$ are the mean and standard

deviation, and work out expressions for $\widehat{\xi}$, $\widehat{\sigma}$, the minimisers of the empirical distance function $H_n(\xi,\sigma) = n^{-1}\sum_{i=1}^n h(Y_i,\xi,\sigma)$.

**(c)** For positive $Y_i$ data, to be fitted to the Gam$(a,b)$, consider the function $h(y,a,b) = (y-a/b)^2 + (y^2 - (a/b^2 + a^2/b^2))^2$, constructed in view of $a/b$ and $a/b^2$ being the formulae for the mean and variance for a Gamma. Explain that the empirical version of the distance function $H(a,b) = \mathrm{E}_G\, h(Y,a,b)$ becomes $H_n(a,b) = n^{-1}\sum_{i=1}^n\{(Y_i-a/b)^2 + (Y_i^2 - (a/b^2+a^2/b^2)\}^2$. Show that the minimum divergence function estimators $(\widehat{a},\widehat{b})$ are equivalent to the moment estimators, worked with in Ex. 3.27, i.e. the solution to the two equations $\bar{Y} = a/b$ and $\widehat{\sigma}^2 = a/b^2$.

**(d)** Consider $h_0(x) = x\arctan x - \frac{1}{2}\log(1+x^2)$, and define $\theta_0$ as the minimiser of $\mathrm{E}_G\, h_0(Y-\theta)$. Show that $\widehat{\theta}$, the minimiser of $H_n(\theta) = n^{-1}\sum_{i=1}^n h_0(Y_i-\theta)$, is also the unique solution to $\sum_{i=1}^n \arctan(Y_i-\theta) = 0$. (xx so connection from minimum divergence estimator to M estimator. round off. xx)

**Ex. 6.3** *Minimum divergence function estimators: general setup.* IN: MODIFY: We learn from Ex. 6.1–6.2 that classes of estimators may be formed via minimisation of suitable empirical distance functions; these estimate the corresponding minimisers of distance functions operating on the underlying distributions. We call these *minimum divergence function estimators* or *minimum distance estimators*. In general terms, for observations $Y_1,\ldots,Y_n$ from some distribution $G$, consider a parameter $\theta_0 = \theta(G)$ defined as the minimiser of the function $H(\theta) = \mathrm{E}_G\, h(Y,\theta)$, for a suitable $h(y,\theta)$. It is assumed that $\theta_0$ thus defined, which may also be multidimensional, is the unique minimiser. The empirical version of $H(\theta)$ is $H_n(\theta) = n^{-1}\sum_{i=1}^n h(Y_i,\theta)$, so a natural estimator for $\theta_0$ is $\widehat{\theta} = \mathrm{argmin}(H_n)$. In fact many important estimators are of this or related types, perhaps minimising somewhat more complicated random functions, as we shall see in this chapter. In exercises below we shall develop clear results for how the minimum divergence function estimators behave, under sets of natural assumptions, but the present exercise is meant to illustrate the basic construction via different types of examples.

minimum divergence function estimators

**(a)** Explain that $H_n(\theta)$ can be written $\int h(y,\theta)\,\mathrm{d}G_n(y)$, with $G_n$ the empirical distribution, having mass $1/n$ at each datapoint; see Ex. 3.9. Explain why $H_n(\theta) \to_{\mathrm{pr}} H(\theta)$ for each $\theta$, and find the limit distribution of $\sqrt{n}\{H_n(\theta) - H(\theta)\}$. What we need, tended to in several exercises to follow, are conditions under which $\widehat{\theta} = \mathrm{argmin}(H_n)$ tends to $\theta_0 = \mathrm{argmin}(H)$, along with a limit distribution.

**(b)** For one-dimensional $Y_i$, work through the details of $h(y,\theta) = (y-\theta)^2$. Then consider $h(y,\theta) = [\exp\{c(y-\theta)\} - 1 - c(y-\theta)]/c^2$, with $c$ a balance parameter. Draw 100 datapoints from a normal $\mathrm{N}(\theta,1)$, with $\theta$ of your choice, and estimate $\theta$ in this minimum $H_n$ fashion, for a few values of the balance parameter $c$. Show that $c$ close to zero corresponds to the mean.

**(c)** Let generally $h(y,\theta) = \{r(y)-\theta\}^{\mathrm{t}} V\{r(y)-\theta\}$, for some $r(y) = (r_1(y),\ldots,r_p(y))$ and a symmetric positive definite matrix $V$. Show that $\theta_0 = \mathrm{E}_G\, r(Y)$ and that $\widehat{\theta} = n^{-1}\sum_{i=1}^n r(Y_i)$.

**(d)** Consider $h(y,\xi,\tau) = p(\tau) + \frac{1}{2}(y-\xi)^2/\tau^2$, where $p(\tau)$ is a smooth increasing function of $\tau > 0$. Find a recipe for computing the estimates $(\widehat{\xi},\widehat{\tau})$ associated with the distance function $n^{-1}\sum_{i=1}^n h(Y_i,\xi,\tau)$. Check in particular the case of $p(\tau) = \log\tau$.

**(e)** There are connections here to the moment matching estimation method, as worked with in Ex. 3.26. For a one-parameter model first, with $\mathrm{E}\,Y = m(\theta)$, consider $h(y,\theta) = \{y - m(\theta)\}^2$. Show that the implied best parameter is for $m(\theta_0) = \mathrm{E}\,Y$, or $\theta_0 = m^{-1}(\mathrm{E}\,Y)$,

and that the minimum divergence function method leads to solving $m(\theta) = \bar{Y}$. Generalise to the case of there being two parameters in the model, and then to the general vector parameter case.

### BHHJ, maximum weighted likelihood, minimum quantile difference

**Ex. 6.4** *The BHHJ density power divergence method.* Here we set up the basics for the so-called *density power divergence method*. It involves raising the density function $f(y, \theta)$ to some power $a$, as one of its ingredients. In the literature it is sometimes called the *BHHJ divergence method*, We shall see in later exercises that it amounts to a robust modification of the ML method.

from its inventors Basu, Harris, Hjort, Jones

**(a)** For a density $g$, in what follows to be seen as the true underlying data-generating model, consider measuring the distance to an approximate $f_\theta(y) = f(y, \theta)$ density as

$$d_a(g, f_\theta) = \int \Big\{ f_\theta^{1+a} - \Big(1 + \frac{1}{a}\Big) g f_\theta^a + \frac{1}{a} g^{1+a} \Big\} \, \mathrm{d}y, \tag{6.2}$$

with $a$ a positive tuning parameter. Show that $d_a(g, f_\theta) \geq 0$, and that the distance is zero only when $g = f_\theta$ a.e. Note that this for $a = 1$ is the same as the $L_2$ distance of Ex. 6.1.

**(b)** Use Taylor expansion for $f_\theta^a$ and $g^a$ for small $a$, to demonstrate that the integrand in (6.2) may be written

$$f_\theta - g + g \log(g/f_\theta) - a(g - f_\theta) \log f_\theta + \tfrac{1}{2} a g \{ (\log g)^2 - (\log f_\theta)^2 \} + O(a^2).$$

Hence show that for $a$ small, we have $d_a(g, f_\theta) = \mathrm{KL}(g, f_\theta) + O(a)$, with the Kullback–Leibler distance $\int g \log(g/f_\theta) \, \mathrm{d}y$, assuming that the functions $g \log f_\theta$, $f_\theta \log f_\theta$, $g(\log f_\theta)^2$, $g(\log g)^2$ have finite integrals.

**(c)** Suppose $Y_1, \ldots, Y_n$ are i.i.d. from some unknown $g$, and that we wish to estimate $\theta$ by making the distance $d_a(g, f_\theta)$ small. The point is now that the third term of $d_a$ does not depend on $\theta$, and that we may accurately estimate the two first, using

$$H_{n,a}(\theta) = \int f(y, \theta)^{1+a} \, \mathrm{d}y - (1 + 1/a) \, n^{-1} \sum_{i=1}^n f(y_i, \theta)^a. \tag{6.3}$$

Show that this is also coming out of the general minimum divergence function apparatus of Ex. 6.3, with $h(y, \theta) = \int f_\theta^{1+a} \, \mathrm{d}y - (1 + 1/a) f(y, \theta)^a$. Show indeed that $H_n(\theta)$ is an unbiased estimator of the two first terms of $d_a(g, f_\theta)$, and give an expression for its variance. We call the minimiser $\widehat{\theta}$ of $H_n(\theta)$ the BHHJ estimator.

the BHHJ estimator

**(d)** Return to the ML estimator illustration of Ex. 5.3, with $n = 100$ points drawn from the uniform, fitted to the $\theta y^{\theta - 1}$ density. For some values of $a$, compute and plot the $H_{n,a}$ function, finding also the BHHJ estimate $\widehat{\theta}_a$. After having computed this for a grid of $a$, plot the resulting $\widehat{\theta}_a$ as a function of $a$, and comment.

**Ex. 6.5** *Integrals for BHHJ estimation.* Using the BHHJ method of Ex. 6.4 for estimating the parameters of models $f(y, \theta)$, we are very much helped, algorithmically and numerically, by having formulae for the term $A(a) = \int f_\theta^{1+a} \, \mathrm{d}y$.

**(a)** For the normal $\mathrm{N}(\xi, \sigma^2)$, show that

$$A(a) = \int f(y, \xi, \sigma)^{1+a} \, \mathrm{d}y = (2\pi)^{-a/2} (1 + a)^{-1/2} \sigma^{-a}.$$

Generalise to the multinormal case of $\mathrm{N}_p(\xi, \Sigma)$, with

$$A(a) = \int f(y, \xi, \Sigma)^{1+a} \, \mathrm{d}y = (2\pi)^{-ap/2}(1+a)^{-p/2}|\Sigma|^{-a/2}.$$

**(b)** For the gamma model with density proportional to $y^{\alpha-1}\exp\{-\beta(y-y_0)\}$ for $y \geq y_0$, we find

$$\int_{y_0}^{\infty} f(y, \alpha, \beta)^{1+a} \, \mathrm{d}y = \frac{\Gamma(\alpha + a\alpha - a)}{\Gamma(\alpha)^{1+a}} \frac{\beta^a}{(1+a)^{\alpha+a\alpha-a}}.$$

**(c)** For the log-normal model, where $Y$ is such that $\log Y \sim \mathrm{N}(\xi, \sigma^2)$: Show that

$$\int_0^{\infty} f(y, \xi, \sigma)^{1+a} \, \mathrm{d}y = (2\pi)^{-a/2}\sigma^{-a}(1+a)^{-1/2}\exp\{-a\xi + \tfrac{1}{2}a^2\sigma^2/(1+a)\}.$$

**(d)** For the Weibull, with c.d.f. $1 - \exp\{-(y/\alpha)^\beta\}$ for $y \geq 0$, show that

$$\int_0^{\infty} f(y, \alpha, \beta)^{1+a} \, \mathrm{d}y = \Big(\frac{\beta}{\alpha}\Big)^a \frac{\Gamma(1 + a - a/\beta)}{(1+a)^{1+a-a/\beta}}.$$

**(e)** For robust estimation of the three-parameter $t$ distribution, consider the density $f(y, \xi, \sigma, \nu) = g_\nu((y-\xi)/\sigma)(1/\sigma)$ for $Y = \xi + \sigma t_\nu$, with $g_\nu$ the $t$ density with $\nu$ degrees of freedom. Find

$$\begin{aligned} \int f^{1+a} \, \mathrm{d}y &= \frac{1}{\sigma^a} \int g_\nu^{1+a} \, \mathrm{d}x \\ &= \frac{1}{\sigma^a} \frac{\Gamma(((1+a)\nu + a)/2)}{\Gamma((1+a)(\nu+1)/2)} \Big\{ \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)} \Big\}^{1+a} \frac{1}{(\nu\,\pi)^{a/2}}. \end{aligned}$$

**Ex. 6.6** *Maximum weighted likelihood estimation.* Another fruitful generalisation of the ML method, in addition to the BHHJ method of Ex. 6.4, starts from adding a weight function to the Kullback–Leibler $\mathrm{KL}(g, f)$ divergence.

**(a)** Show that $\mathrm{KL}(g, f)$ may be written $\int \{g \log(g/f) - (g - f)\} \, \mathrm{d}y$, that the integrand is nonnegative, and is zero only if $g = f$ a.e. With any nonnegative weight function $w(y)$ on the sample space, deduce that

$$\mathrm{KL}_w(g, f) = \int w(y) \Big[ g(y) \{ \log \frac{g(y)}{f(y)} - \{g(y) - f(y)\} \Big] \, \mathrm{d}y$$

is a divergence, i.e. nonnegative, and is zero only if $g = f$ on the support set of $w$. In particular, for $g$ fixed and parametric $f(\cdot, \theta)$, show that $\mathrm{KL}_w(g, f_\theta)$ can be expressed as $\int w(-g \log f_\theta + f_\theta)$ plus other terms not depending on $\theta$.

**(b)** For data $Y_1, \ldots, Y_n$ from a generating density $g$, to be approximated with a parametric $f(y, \theta)$, explain that the general minimum divergence function method leads to maximisation of

$$\ell_{n,w}(\theta) = \sum_{i=1}^n w(y_i) \log f(Y_i, \theta) - nB(\theta),$$

with $B(\theta) = \int w f_\theta \, \mathrm{d}y$. We call the maximiser $\widehat{\theta}_w$ the maximum weighted likelihood estimator, associated with weight function $w(y)$. The classic ML estimator then corresponds to constant weight function $w(y) = 1$. Properties of $\widehat{\theta}_w$ are derived in Ex. 6.12.

**(c)** For an illustration, access the birthweight dataset for Oslo children, reported on in Story #20. For the boys and the girls, fit normal distributions, using maximum weighted likelihood, with weight function $w(y)$ being one on $[2.5, 4.5]$ kg and zero outside. Plot the estimated densities in a diagram, This requires programming a function for optimisation. Compare with curves estimated via full ML.

**Ex. 6.7** *Quantile matching estimators.* Suppose i.i.d. $Y_1, \ldots, Y_n$ are observed from some smooth density $g$, and that a parametric model $f_\theta$ is to be fitted. One may than compare the sample quantiles, say $Q_{n,1}, \ldots, Q_{n,k}$ at levels $q_1 < \cdots < q_k$, to those derived from the model, i.e. $F^{-1}(q_j, \theta)$.

**(a)** Simulated a normal dataset in your computer, say of size $n = 333$. For the nine deciles $1/10, \ldots, 9/10$, compute the sample deciles $Q_{n,j}$, and explain why minimising criterion functions of the type $\sum_{j=1}^k w_j\{Q_{n,j} - F^{-1}(q_j, \theta)\}^2$ will be sensible estimation procedure. The $w_j$ are positive weights, which can be taken equal or not, to reflect higher imoportance. Fit the normal distribution to your data in this way, using $F^{-1}(q_j, \xi, \sigma) = \xi + \sigma\Phi^{-1}(q_j)$. Experiment a bit with other quantiles and weights.

**(b)** Simulate another dataset, this time from the exponential distribution. Find the quantile function $F^{-1}(q, \theta)$. Estimate the $\theta$ by minimising $\sum_{j=1}^k w_j\{Q_{n,j} - F^{-1}(q_j, \theta)\}^2$, this time with $w_j = 1 + q_j$, i.e. weights becoming more important for growing quantiles. Play with your code a bit and comment on your findings.

### The minimum divergence process and its limit

**Ex. 6.8** *Minimum divergence function estimators: a limit process and main heuristics.* IN: MODIFY: After having motivated and worked through particular instances of the minimum divergence function estimators, we now return to the general case, aiming to demonstrate both limiting normality and other associated results, finding recipes for large-sample approximations in the process. The aim of the present exercise is to go through the basic ideas, involving also two main heuristics. These then apply to large classes of estimators and associatead minimum divergence function minima. To make these heuristics precise we need more conditions and details, to which we return in further exercises below. In particular, results derived here and in a few of the following exercises will apply immediately to ML estimators, BHHJ estimators, MWL estimators, via their definitions in Ex. 5.3, 6.4, 6.6.

So $\theta_0 = \theta_0(G)$ is the minimiser of $H(\theta) = \mathrm{E}_G\, h(y, \theta)$, and $\widehat{\theta}$ is its estimator, the minimiser of $H_n(\theta) = \int h(y, \theta)\, \mathrm{d}G_n(y)$, with $G_n$ the empirical distribution for an observed i.i.d. sample $Y_1, \ldots, Y_n$ from $G$. To present the basic ideas and main heuristics we shall start with these regularity conditions: (i) The true $\theta_0 = \theta_0(G)$ is an inner point in its parameter space inside $\mathbb{R}^p$. (ii) The $h(y, \theta)$ is smooth in $\theta$, in a neighbourhood around $\theta_0$, with at least two derivatives, say $h'(y, \theta)$ (a vector, with components $h'_a$), $h''(y, \theta)$ (a matrix, with components $h''_{a,b}$). (iii) The matrix $J = \mathrm{E}_G\, h''(Y, \theta_0)$ is finite and positive definite. (iv) The first derivative $h'(Y, \theta_0)$ has finite variance matrix $K$.

**(a)** A key idea is to work with the following random function. Write

$$\begin{aligned} A_n(s) &= n\{H_n(\theta_0 + s/\sqrt{n}) - H_n(\theta_0)\} \\ &= \sum_{i=1}^n \{h(Y_i, \theta_0 + s/\sqrt{n}) - h(Y_i, \theta_0)\} = U_n^{\mathrm{t}} s + \tfrac{1}{2} s^{\mathrm{t}} J_n s + r_n(s), \end{aligned}$$

with $U_n = (1/\sqrt{n})\sum_{i=1}^n h'(Y_i,\theta_0)$, $J_n = n^{-1}\sum_{i=1}^n h''(Y_i,\theta_0)$. Show that $A_n(s)$ is properly defined for all large enough $n$, that $U_n$ has mean zero and tends to $U \sim \mathrm{N}_p(0,K)$, and that $J_n \to_{\rm pr} J$.

**(b)** It is also clear from the Taylor expansion argument that with moderate further regularity, the remainder term $r_n(s)$ will go to zero in probability, for each $s$. There is then convergence $A_n(s) \to_d A(s) = U^{\rm t}s + \frac12 s^{\rm t}Js$, for each $s$. Heuristic One is then to go from $A_n \to_d A$ to $\operargmin(A_n) \to_d \operargmin(A)$. Explain that this then leads to $\sqrt{n}(\widehat\theta - \theta_0) \to_d -J^{-1}U$, which is a $\mathrm{N}_p(0,J^{-1}KJ^{-1})$. We call this limit distribution variance matrix $J^{-1}KJ^{-1}$ the sandwich matrix.

the sandwich matrix

**(c)** Similarly, Heuristic Two is to go from $A_n \to_d A$ to $\min(A_n) \to_d \min(A)$. Argue that this entails $W_n = 2n\{H_n(\theta_0) - H_n(\widehat\theta)\} \to_d W = U^{\rm t}J^{-1}U$. Show for this quadratic form that its mean and variance are $\mathrm{Tr}(J^{-1}K)$ and $2\,\mathrm{Tr}(J^{-1}KJ^{-1}K)$.

**(d)** (xx give an easy example, to identify $J$ and $K$. xx)

**Ex. 6.9** *Minimum divergence function estimators: from heuristics to proofs.* IN: MODIFY: (xx nils and emil check details carefully. repair and edit. xx) The general setup is as with the previous Ex. 6.8, with minimum divergence function estimator $\widehat\theta$ minimising $H_n(\theta) = n^{-1}\sum_{i=1}^n h(Y_i,\theta)$, with observations coming from $G$. In addition to regulatity conditions (i)–(iv) given there, we postulate (v), that also third derivatives of $h(y,\theta)$ exist, say $h'''_{a,b,c}(y,\theta)$, and that these have finite means in a neighbourhood around $\theta_0$. We use the process $A_n(s) = n\{H_n(\theta_0 + s/\sqrt{n}) - H_n(\theta_0)\}$ from the previous exercise.

**(a)** Explain first that the minimiser of $A_n$ is $\alpha_n = \sqrt{n}(\widehat\theta - \theta_0)$, where we shall also study the overall minimum $A_{n,\min} = A_n(\alpha_n)$ below. Let $B_n(s) = U_n^{\rm t}s + \frac12 s^{\rm t}J_n s$ be the quadratic approximation to $A_n$, with minimiser $\beta_n$ and overall minimum $B_{n,\min} = \min\{B_n(s)\colon \text{all } s\}$. Show that

$$\begin{aligned}\beta_n &= -J_n^{-1}U_n \to_d -J^{-1}U \sim \mathrm{N}_p(0,J^{-1}KJ^{-1}),\\ B_{n,\min} &= -\tfrac12 U_n^{\rm t}J_n^{-1}U_n \to_d -\tfrac12 U^{\rm t}J^{-1}U.\end{aligned}$$

So things are simple and clear for the quadratic approximation $B_n$; we need to show that the same results obtain for the real thing, the $A_n$.

**(b)** Supposing $|r_n(s)| \le \delta$ for all $s$ in a subset $S$, show from $A_n = B_n + r_n$ that

$$|\min_{s\in S} A_n(s) - \min_{s\in S} B_n(s)| \le \delta.$$

We next establish that $\alpha_n$ cannot be far away. Show that when $\|s\| \ge cn^{1/8}$, $B_n(s) \ge D_n n^{1/4}$, with $D_n$ positive and bounded in probability; and that when $\|s\| \le cn^{1/8}$, then $|r_n(s)| \le E_n/n^{1/8}$, with $E_n$ also bounded in probability. Show from this that $\alpha_n = O_{\rm pr}(n^{1/8})$, and that $A_{n,\min} - B_{n,\min} \to_{\rm pr} 0$.

**(c)** Then consider $B_n$ a certain distance away from the minimum. For given small $\varepsilon$, show that for $v$ with $\|v\| \ge \varepsilon$, we have

$$B_n(\beta_n + v) = B_{n,\min} + \tfrac12 v^{\rm t}J_n v \ge B_{n,\min} + \tfrac12 j_n\varepsilon^2,$$

where $j_n$ is the smallest eigenvalue of $J_n$. Show then that the event $\Omega_n$, where $A_n(\beta_n + v) \ge B_{n,\min} + \frac14 j_n\varepsilon^2$ for all $v$ with $\|v\| \ge \varepsilon$, must have $\Pr(\Omega_n) \to 1$. Prove from these established statements that $\alpha_n - \beta_n$ must tend to zero in probability. Conclude that

$$\begin{aligned}\sqrt{n}(\widehat\theta - \theta_0) &\to_d -J^{-1}U \sim \mathrm{N}_p(0,J^{-1}KJ^{-1}),\\ 2n\{H_n(\theta_0) - H_n(\widehat\theta)\} &\to_d W = U^{\rm t}J^{-1}U.\end{aligned} \tag{6.4}$$

**Ex. 6.10** *The $J$, $K$, and sandwich matrices for the BHHJ method.* IN: (xx place inside prose here that we also need estimation of these matrices, to which we return in Ex. 6.16. xx) When fitting a parametric family $f(y,\theta)$ to observations, we have seen in Ex. 6.8-6.9 that the behaviour of minimum divergence function estimators is characteriserd by (i) the implied least false parameter $\theta_0 = \theta_0(G)$ and then (ii) the crucial matrices $J$ and $K$, leading to the sandwich matrix $\Sigma = J^{-1}KJ^{-1}$. As above $G$, with density $g$, is the data generating mechanism. Here we consider the BHHJ estimation method Ex. 6.4, with the estimator $\widehat{\theta}$ minimising $H_{n,a}(\theta)$ of (6.3). Again, $a$ is a positive finetuning parameter, and for small $a$ the method is close to ML estimation. For the following points, we assume regularity conditions (i)-(v) are in place, from Ex. 6.8-6.9, and in particular that there is a unique minimiser $\theta_0 = \theta_{0,a} = \theta_{0,a}(G)$ of the $d_a(g, f_\theta)$ of (6.2).

**(a)** Explain that the theory developed in these earlier exercises implies that $\widehat{\theta} \to_{\rm pr} \theta_0$. In the following points, write for simplicity $f_0(y) = f(y,\theta_0)$, $u_0(y) = u(y,\theta_0)$, $i_0(y) = i(y,\theta_0)$. Show also that $\theta_0$ is characterised as the solution to $\int f_\theta^{1+a} u_\theta \,{\rm d}y = \int g f_\theta^a u_\theta \,{\rm d}y$. For this vector, evaluated at $\theta_0$, we write

$$\xi_a = \int g f_0^a u_0 \,{\rm d}y = \int f_0^{1+a} u_0 \,{\rm d}y.$$

**(b)** As seen in Ex. 6.4, the BHHJ method corresponds to using $h(y,\theta) = \int f_\theta^{1+a} \,{\rm d}y - (1+1/a) f(y,\theta)^a$ in the general minimum divergence function setup. Find the derivative $h'(y,\theta)$ and deduce that

$$K_a = (1+a)^2 \Big\{ \int g f_0^{2a} u_0 u_0^{\rm t} - \xi_a \xi_a^{\rm t} \Big\},$$

(xx find $K_a$. with simplified at model. xx)

**(c)** (xx check this carefully. also: when estimating $J_a$, we don't need such a formula, as we get it as the Hessian of the minimisation. xx) Find then an expression for the second derivative $h''(y,\theta)$, and derive a formula for the $J$ matrix:

$$J_a = (1+a) \int \{ f_0^{1+a} u_0 u_0^{\rm t} + (f_0^{1+a} - g f_0^a)(i_0 + a u_0 u_0^{\rm t}) \} \,{\rm d}y.$$

**Ex. 6.11** *BHHJ precision under model conditions.* IN: The BHHJ method is designed to do well when the parametric model used is not perfect. We may however also study its precision when the model actually holds. For matrices studied in Ex. 6.10 there are simplified expressions under model conditions, i.e. when $g(y) = f(y,\theta_0)$ for all $y$.

**(a)** Under such model conditions, and with $f_0, u_0$ notation from Ex. 6.10, show that

$$J_a = (1+a) \int f_0^{1+a} u_0 u_0^{\rm t} \,{\rm d}y, \quad K_a = (1+a)^2 \Big( \int f_0^{1+2a} u_0 u_0^{\rm t} \,{\rm d}y - \xi_a \xi_a^{\rm t} \Big),$$

with a consequent simplified sandwich matrix $\Sigma_a = J_a^{-1} K_a J_a^{-1}$. Show that for $a \to 0$, $\Sigma_a \to J_0^{-1}$, with $J_0 = J(\theta_0)$ the Fisher information matrix for the model.

**(b)** We may now check the loss of efficiency of the BHHJ method, with tuning parameter $a$, compared to the ML method, under model conditions, by comparing $J_a^{-1} K_a J_a^{-1}$ to the inverse Fisher information matrix. Carry out the relevant computations for estimation $\theta$ in the exponential $\theta \exp(-\theta y)$ model. For the case of the $\mathrm{N}(\xi, \sigma^2)$ model, check with Story #97. For small $a$, there is significant gain in robustness at a low cost in efficiency.

**(c)** For the following, assume

$$A = \int f_0 \log f_0 \, u_0 u_0^{\mathrm{t}} \, \mathrm{d}y, \quad B = \int f_0 (\log f_0)^2 \, u_0 u_0^{\mathrm{t}} \, \mathrm{d}y, \quad c = \int f_0 \log f_0 \, u_0 \, \mathrm{d}y$$

are finite; $A$ and $B$ are $p \times p$ matrices and $c$ a $p \times 1$ vector. Below we carry out Taylor expansions to order $a^2$, and use $\doteq$ to indicate the consequent approximations for small $a$. Show that to this order

$$J_a \doteq (1+a)(J_0 + aA + \tfrac{1}{2}a^2 B), \quad K_a \doteq (1+a)^2 \{J_0 + 2aA + a^2(2B - cc^{\mathrm{t}})\}.$$

As a useful interlude, consider $p \times p$ symmetric matrices $M$ and $\varepsilon$, where $M$ is positive definite and $\varepsilon$ is smaller, in the technical sense that the eigenvalues of $M^{-1}\varepsilon$ are smaller than 1 in absolute value. Show that

$$(M+\varepsilon)^{-1} = M^{-1} - M^{-1}\varepsilon M^{-1} + M^{-1}\varepsilon M^{-1}\varepsilon M^{-1} + \cdots .$$

Use this to reach

$$(J_0 + aA + \tfrac{1}{2}a^2 B)^{-1} \doteq J_0^{-1} - aJ_0^{-1}AJ_0^{-1} + a^2(J_0^{-1}AJ_0^{-1}AJ_0^{-1} - \tfrac{1}{2}J_0^{-1}BJ_0^{-1}).$$

It is then a matter of bureaucratic algebra to reach an informative approximation to the sandwich matrix

$$\Sigma_a \doteq (J_0 + aA + \tfrac{1}{2}a^2 B)^{-1}\{J_0 + 2aA + a^2(2B - cc^{\mathrm{t}})\}(J_0 + aA + \tfrac{1}{2}a^2 B)^{-1}.$$

Show that this leads to $\Sigma_a \doteq J_0^{-1} + a^2 D$, with $D = J_0^{-1}(B - cc^{\mathrm{t}} - AJ_0^{-1}A)J_0^{-1}$. This indicates indeed that for small $a$, the efficiency loss is small.

**Ex. 6.12** *The $J$, $K$, and sandwich matrices for the maximum weighted likelihood method.* (xx check all this. xx) IN: Just as we for the BHHJ estimation method worked out the basics for $J_a$, $K_a$, $J_a^{-1}K_aJ_a^{-1}$ in Ex. 6.10, we here tend to their parallels for the maximum weighted likelihood method of Ex. 6.6. For a parametric model $f(y,\theta)$, and with a given weight function $w(y)$, it consists in maximising the weighted log-likelihood function $\ell_{n,w}(\theta) = \sum_{i=1}^n w(y_i)\log f(Y_i,\theta) - nB(\theta)$, with $B(\theta) = \int w(y) f(y,\theta)\,\mathrm{d}y$. We assume the regularity conditions of Ex. 6.9 are in force.

**(a)** When observations $Y_1,\ldots,Y_n$ stem from $G$, with density $g$, show that the MWL estimator $\widehat{\theta}$ tends to $\theta_0 = \theta_0(G)$, the minimiser of the weighted KL divergence $\mathrm{KL}_w(g, f_\theta)$. When $g$ is not inside the parametric family, this least false parameter value depends also on the weight function. Show also that $\theta_0$ is characterised as the solution to $\int w(y)\{g(y) - f(y,\theta_0)\}u(y,\theta_0)\,\mathrm{d}y = 0$. As for Ex. 6.10 we write $f_0, u_0, i_0$ for the functions $f(y,\theta_0), u(y,\theta_0), i(y,\theta_0)$, and $\xi_0 = \int w g u_0 \,\mathrm{d}y = \int w f_0 u_0 \,\mathrm{d}y$.

**(b)** Explain that maximising $\ell_{n,w}(\theta)$ is the same as minimising $n^{-1}\sum_{i=1}^n h(Y_i,\theta)$, with $h(y,\theta) = w(y)\log f(y,\theta) - B(\theta)$, or solving $n^{-1}\sum_{i=1}^n h'(Y_i,\theta) = 0$, with $h'(y,\theta) = w(y)u(y,\theta) - \xi(\theta)$, where $\xi(\theta) = \partial B(\theta)/\partial\theta = \int w(y) f(y,\theta) u(y,\theta)\,\mathrm{d}y$. Show that

$$K_w = \int w^2 g u_0 u_0^{\mathrm{t}} \,\mathrm{d}y - \xi_0\xi_0^{\mathrm{t}}.$$

Under model conditions, we have $\xi_0 = 0$, and the above simplies to $K_w = \int w^2 f_0 u_0 u_0^{\mathrm{t}}\,\mathrm{d}y$.

**(c)** Then show that $J_w = -\int w g i_0 \,\mathrm{d}y + \partial^2 B(\theta_0)/\partial\theta\,\partial\theta^{\mathrm{t}}$. Under model conditions, $J_w = -\int w f_0 i_0 \,\mathrm{d}y$.

**(d)** (xx find sandwich, and compare with $J^{-1}$ under model. xx)

**Ex. 6.13** *Maximum weighted likelihood for multinomial models.* IN: The weighted likelihood ideas of Ex. 6.6, 6.12 can also be used to estimate parameters in models for multinomial probabilities, allowing different weights of importance for different outcomes. used in the Platon Story #45

**(a)** Show first that $p \log p/q - (p-q)$ is always nonnegative, for $p, q$ in $(0,1)$. Now consider $(Y_1, \ldots, Y_k)$ being multinomial with count $n$ and probability vector $p = (p_1, \ldots, p_k)$, as with Ex. 1.5. Then

$$d_w(p, p_\theta) = \sum_{j=1}^k w_j\{p_j \log(p_j/p_{j,\theta}) - (p_j - p_{j,\theta})\},$$

is a proper nonnegative divergence, from $p$ to some $p_\theta$, as long as the weights $w_1, \ldots, w_k$ are nonnegative. With equal weights, show that this is the Kullback–Leibler divergence $\mathrm{KL}(p, p_\theta)$. If $p_\theta = (p_1(\theta), \ldots, p_k(\theta))$ is some postulated model, with $\theta$ of dimension $r$, say, show that the minimum divergence method amounts to maximising the weighted and modified log-likelihood

$$\ell_{n,w}(\theta) = n\sum_{j=1}^k w_j\{\widehat{p}_j \log p_j(\theta) - p_j(\theta)\} = \sum_{j=1}^k w_j Y_j \log p_j(\theta) - n\sum_{j=1}^k w_j p_j(\theta),$$

with $\widehat{p}_j = Y_j/n$. This generalises the usual ML method, which corresponds to equal weights.

**(b)** Let $p = (p_1, \ldots, p_k)$ denote the true probability vector. Explain that the weighted likelihood estimator $\widehat{\theta}_w$, maximising $\ell_{n,w}(\theta)$, tends in probability to the least false parameter maximising $\sum_{j=1}^k w_j\{p_j \log p_j(\theta) - p_j(\theta)\}$, i.e. to the minimiser of the weighted KL distance $d_w(p, p_\theta)$. Writing $u_j(\theta) = \partial \log p_j(\theta)/\partial\theta$, show that the least false parameter is also the solution to $\sum_{j=1}^k w_j\{p_j - p_j(\theta)\}u_j(\theta) = 0$, assumed here to be unique. Show furthermore that

$$U_n = n^{-1/2}\frac{\partial \ell_{n,w}(\theta_0)}{\partial\theta} = \sum_{j=1}^k w_j\sqrt{n}\{\widehat{p}_j - p_j(\theta_0)\}u_j(\theta_0) \to_d U = \sum_{j=1}^k w_j Z_j u_j(\theta_0),$$

using results and notation from the multinomial CLT worked with in Ex. 2.49. Deduce that $U \sim \mathrm{N}_r(0, K_w)$, with

$$K_w = \sum_{j=1}^k w_j^2 p_j u_j(\theta_0)u_j(\theta_0)^{\mathrm{t}} - \xi_0\xi_0^{\mathrm{t}}, \quad \text{with } \xi_0 = \sum_{j=1}^k w_j p_j u_j(\theta_0).$$

**(c)** Next demonstrate that the normalised Hessian matrix $\widehat{J}_{n,w} = -n^{-1}\partial^2\ell_{n,w}(\widehat{\theta})/\partial\theta\partial\theta^{\mathrm{t}}$ tends in probability to a well-defined $J_w$. In fact, writing $i_j(\theta) = \partial^2 \log p_j(\theta)/\partial\theta\partial\theta^{\mathrm{t}}$, show that $J_w = J_w(\theta_0)$, where $J_w(\theta)$ is

$$-\frac{\partial^2}{\partial\theta\partial\theta^{\mathrm{t}}}\sum_{j=1}^k w_j\{p_j \log p_j(\theta) - p_j(\theta)\} = \sum_{j=1}^k w_j\big[p_j(\theta)u_j(\theta)u_j(\theta)^{\mathrm{t}} - \{p_j - p_j(\theta)\}i_j(\theta)\big],$$

with the expression simplifying under model conditions. Use this in conjunction with the general minimum divergence theory to establish that $\sqrt{n}(\widehat{\theta}_w - \theta_0) \to_d J_w^{-1}U \sim \mathrm{N}_r(0, \Sigma_w)$, with the sandwich matrix $\Sigma_w = J_w^{-1}K_w J_w^{-1}$.

**(d)** (xx a simple example here, with pointer to Story #45. xx)

## Profiling minimum divergence

**Ex. 6.14** *Profiling a minimum divergence function, I.* IN: MODIFY: For data $Y_1, \ldots, Y_n$ from some distribution $G$, we have in Ex. 6.3 considered estimating a parameter $\theta_0 = \text{argmin}(H)$, for $H(\theta) = \text{E}_G\, h(Y, \theta)$, by minimising the distance function $H_n(\theta) = n^{-1} \sum_{i=1}^n h(Y_i, \theta)$. For a focus parameter $\mu = k(\theta)$, a smooth function of $\theta = (\theta_1, \ldots, \theta_p)$, it is useful to work with the associated profile function

profiling a distance function

$$H_{n,\text{prof}}(\mu) = \min\{H_n(\theta) \colon k(\theta) = \mu\}.$$

**(a)** As an introductory illustration, consider estimating the parameters $(a, b)$ of a Gamma distribution via minimum $L_2$, as in Ex. 6.1. Simulate 100 datapoints from a gamma; compute $(\widehat{a}, \widehat{b})$; and compute and display also the profile function $H_{n,\text{prof}}(\mu)$ for the mean parameter $\mu = a/b$.

**(b)** From the full model, with ensuing minimum divergence function estimator $\widehat{\theta}$, show that the consequent $\widehat{\mu} = k(\widehat{\theta})$ becomes normal. With setup and notation as in Ex. 6.9, prove indeed that $\sqrt{n}(\widehat{\mu} - \mu_0) \to_d c^{\text{t}} J^{-1} U$, with $c = \partial k(\theta_0)/\partial\theta$, and that the limit is a zero-mean normal with variance $c^{\text{t}} J^{-1} K J^{-1} c$. Show also that this $\widehat{\mu}$ is identical to the minimiser of the profile function.

**(c)** In other words we already know the basic story for any focus parameter estimator $\widehat{\mu}$, thanks to the delta method. It is however fruiful to work with representations and approximations stemming from examining the associated profile function. With methods from Ex. 6.9, show that

$$n\{H_n(\widehat{\theta} + s/\sqrt{n}) - H_n(\widehat{\theta})\} = \tfrac{1}{2} s^{\text{t}} J_n s + r_n(s), \quad \text{with } r_n(s) = O_{\text{pr}}(\|s\|^3/\sqrt{n}).$$

For the profiling, therefore, we must minimise this expression over all $s$ such that $k(\theta) = k(\widehat{\theta} + s/\sqrt{n}) = \mu$. With $k(\theta) = \widehat{\mu} + c_n^{\text{t}} s/\sqrt{n} + O_{\text{pr}}(\|s\|^2/n)$, here writing $c_n = \partial k(\widehat{\theta})/\partial\theta$, the essence is to minimise $\frac{1}{2} s^{\text{t}} J_n s$ under $c_n^{\text{t}} s = \sqrt{n}(\mu - \widehat{\mu}) = x_n$, say. Appealing to Ex. 5.12, show that this minimum becomes $\frac{1}{2} x_n^2 / c_n^{\text{t}} J_n^{-1} c_n = n(\widehat{\mu} - \mu)^2 / c_n^{\text{t}} J_n^{-1} c_n$. Fill in more details to prove that with $\mu_0 = k(\theta_0)$ the true parameter in question,

$$2n\{H_{n,\text{prof}}(\mu_0) - H_{n,\text{prof}}(\widehat{\mu})\} = \frac{n(\widehat{\mu} - \mu_0)^2}{c_n^{\text{t}} J_n^{-1} c_n} + o_{\text{pr}}(1) \to_d \frac{(c^{\text{t}} J^{-1} U)^2}{c^{\text{t}} J^{-1} c} \sim \kappa \chi_1^2,$$

with $\kappa = c^{\text{t}} J^{-1} K J^{-1} c / c^{\text{t}} J^{-1} c$.

**(d)** (xx explain that this often leads to better approximations than the direct limiting normality thing. then an illustration of this. and pointer to Wilks. and pointer to regression versions of these methods and results; the $Y_i$ need not at all be i.i.d. xx)

**Ex. 6.15** *ML and minimum divergence function estimators in practice.* IN: MODIFY: In previous exercises we have learned that $\widehat{\theta}$, the minimiser of $H_n(\theta) = n^{-1} \sum_{i=1}^n h(Y_i, \theta)$, is a natural estimator for $\theta_0$, the minimiser of $\text{E}_G\, h(Y, \theta)$, and that its distribution approaches normality. In order to use such results in practice, for testing, setting confidence intervals, etc., we need to estimate the two crucal matrices $J = \text{E}_G\, h''(Y, \theta_0)$ and $K = \text{Var}_G\, h'(Y, \theta_0)$. These comments apply in particular to ML estimation, with $J = -\text{E}_G\, i(Y, \theta_0)$ and $K = \text{Var}_G\, u(Y, \theta_0)$, see Ex. 5.10.

**(a)** Consider first, in general terms, some function $p(y, \theta)$ with finite mean in a neighbourhood of the true $\theta_0$, with $\widehat{\theta}$ an estimator of $\theta_0$. Explain first that $p_n = n^{-1} \sum_{i=1}^n p(Y_i, \theta_0)$ tends to $p_0 = \text{E}_G\, p(Y, \theta_0)$. The best we may do for estimating $p_0$ in practice is $\widehat{p}_n =$

$n^{-1}\sum_{i=1}^n p(Y_i,\widehat\theta)$. Show that if $|p(y,\theta_0+\varepsilon)-p(y,\theta_0)|\le M(y)\|\varepsilon\|$, for all small $\|\varepsilon\|$, for some function $M(y)$ with finite mean, then indeed $\widehat p_n\to_{\rm pr} p_0$.

**(b)** Give conditions under which the natural estimators

$$\widehat J=n^{-1}\sum_{i=1}^n h''(Y_i,\widehat\theta)\quad\text{and}\quad \widehat K=n^{-1}\sum_{i=1}^n h'(Y_i,\widehat\theta)h'(Y_i,\widehat\theta)^{\rm t}$$

are consistent for $J$ and $K$. Deduce that when such hold, the empirical sandwich matrix $\widehat\Sigma=\widehat J^{-1}\widehat K\widehat J^{-1}$ is consistent for $\Sigma=J^{-1}KJ^{-1}$. Note also that $\widehat J=H_n''(\widehat\theta)$ is the Hessian matrix of the criterion function $H_n$, often computed directly when using numerical minimisation methods for finding $\widehat\theta$ in the first place. (xx nils emil small note: where do we say that $A_n\to_{\rm pr}A$ and $B_n\to_{\rm pr}B$ for matrices implies $A_nB_n\to_{\rm pr}AB$, etc.? xx)

**(c)** Explain how confidence intervals for the components of $\theta$ may be read off from this. More generally, for any focus parameter $\mu=k(\theta)$, with estimator $\widehat\mu=k(\widehat\theta)$, show that $\widehat\mu\pm1.96\,\widehat\kappa/\sqrt n$ is an approximate 95 percent interval for $\mu$, where $\widehat\kappa^2=\widehat c^{\rm t}\widehat J^{-1}\widehat K\widehat J^{-1}\widehat c$, and $\widehat c=\partial k(\widehat\theta)/\partial\theta$.

**(d)** The case of ML estimation is a special case of the setup above, see Ex. 5.10. Show that the estimated matrices become

$$\widehat J=-n^{-1}\sum_{i=1}^n\frac{\partial^2\log f(Y_i,\widehat\theta)}{\partial\theta\partial\theta^{\rm t}}\quad\text{and}\quad\widehat K=n^{-1}\sum_{i=1}^n\widehat u_i\widehat u_i^{\rm t},$$

where $\widehat u_i=u(Y_i,\widehat\theta)$. Note also that $\widehat J=-n^{-1}\ell_n''(\widehat\theta)$, the normalised Hessian matrix of the log-likelihood function computed at the ML position.

**(e)** To illustrate how the above machinery works in practice, simulate 100 points from the standard normal, and then estimate the two normal parameters $(\xi,\sigma)$ via minimum $L_2$, as in Ex. 6.3. Explain that this means minimising the empirical distance function

$$H_n(\xi,\sigma)=\int f(y,\xi,\sigma)^2\,{\rm d}y-\frac2n\sum_{i=1}^n f(y_i,\xi,\sigma)=\frac{\frac12/\pi^{1/2}}{\sigma}-\frac2n\sum_{i=1}^n\frac1\sigma\phi\Big(\frac{y_i-\xi}{\sigma}\Big).$$

Carry out this minimisation using e.g. `nlm` in R, a non-linear minimisation algorithm, which finds both $(\widehat\xi,\widehat\sigma)$ and the Hessian $\widehat J$. Compute also $\widehat K$, and find confidence intervals for $\xi$, for $\sigma$, and for $p(y_0)=\Pr(Y\ge y_0)$, with say $y_0=1.00$.

**(f)** Change one or two of your simulated datapoints to somewhat far-off values, e.g. $y_{99}=d$ and $y_{100}=d$, with $d=5.00$ (which indeed is really far off for the standard normal). Observe what then happens to the ordinary ML estimators, and compare with what happens with the minimum $L_2$ estimators. The point is the the minimum $L_2$ method is much more robust than the ML method.

**(g)** For the ML method, we know from the Bartlett identity that the two matrices $J$ and $K$ are equal under model conditions. For various models we would then have two different estimators, with the same aim, and perhaps different precision; this does not matter for the first-order large-sample theory, since consistency is what matter for $\widehat J$ and $\widehat K$, and specifically for the estimated sandwich matrix $\widehat\Sigma=\widehat J^{-1}\widehat K\widehat J^{-1}$. For illustration, consider $Y_i$ from the Pois$(\theta)$ model. Show that $J=1/\theta_0$ and $K=\tau_0^2/\theta_0^2$, in terms of the true mean and variance $\theta_0$ and $\tau_0^2$ of the underlying distribution. The recipes above lead to $\widehat J=1/\bar Y$ and $\widehat K=V_n/\bar Y^2$, in terms of sample mean and sample variance. Under model conditions, both aim for $1/\theta_0$. Show also that the consequent estimated sandwich

becomes $\widehat{\Sigma} = V_n$. The general recipes hence lead to two somewhat different confidence intervals for $\theta$, namely $\widehat{\theta} \pm z_0\widehat{\kappa}/\sqrt{n}$, with either $\widehat{\kappa} = \bar{Y}^{1/2}$ of $\widehat{\kappa} = V_n^{1/2}$. Argue that both are valid, with both aiming for the same quantity under Poisson conditions, whereas the second option might be preferred if there might be overdispersion.

**(h)** (xx profiling too, to illustrate, for $p(y_0)$. again handled by the general theory. intervals need $\kappa = c^{\rm t}J^{-1}KJ^{-1}c/c^{\rm t}J^{-1}c$. xx)

**Ex. 6.16** *BHHJ analysis in practice.* IN: We have seen how the behaviour of the model robust BHHJ estimators, defined in Ex. 6.4, is described via the two crucial matrices $J_a$ and $K_a$ worked with in Ex. 6.10. Here we go into the details of their estimation. The setting is having i.i.d. data $Y_1, \ldots, Y_n$ from some density $g$, fitted to a parametric $f(y, \theta)$ by minimising $H_{n,a}(\theta)$ of (6.3). We let $\widehat{\theta}$ be this BHHJ estimator, computed for the given balance parameter $a$. Below, we write $\widehat{u}_i = u(Y_i, \widehat{\theta})$ and $\widehat{f}_i = f(Y_i, \widehat{\theta})$.

**(a)** Show that $\widehat{\xi}_a = n^{-1}\sum_{i=1}^n \widehat{f}_i^a \widehat{u}_i$ is equal to $\int f(y,\widehat{\theta})^{1+a} u(y,\widehat{\theta})\,{\rm d}y$, and that it is consistent for the $\xi_a$ defined in Ex. 6.10.

**(b)** Show next that $\widehat{J}_a = H''_{n,a}(\widehat{\theta})$, the Hessian matrix associated with minimisation of the criterion function, is consistent for the $J_a$ matrix.

**(c)** Show that

$$\widehat{K}_a = (1+a)^2\Big\{n^{-1}\sum_{i=1}^n \widehat{f}_i^{2a}\widehat{u}_i\widehat{u}_i^{\rm t} - \widehat{\xi}_a\widehat{\xi}_a^{\rm t}\Big\} = (1+a)^2\, n^{-1}\sum_{i=1}^n (\widehat{f}_i^a\widehat{u}_i - \widehat{\xi}_a)(\widehat{f}_i^a\widehat{u}_i - \widehat{\xi}_a)^{\rm t}$$

is consistent for $K_a$.

### Divergences in regression models

(xx fill in two or three exercises, BHHJ regression. xx)

### Influence functions

**Ex. 6.17** *Influence functions.* IN: For a distribution function $F$, consider some associated parameter, say $\theta = T(F)$, with $T$ the appropriate functional mapping the distribution to the parameter value in question. Examples include the mean, the standard deviation, the skewness, the interquartiale range, a threshold probability. The *influence function* for $\theta = T(F)$ is a very useful quantity, as we shall see. It is defined as

$$\mathrm{IF}(F, y) = \lim_{\varepsilon\to 0}(1/\varepsilon)\{T((1-\varepsilon)F + \varepsilon\delta(y)) - T(F)\}. \tag{6.5}$$

Here $\delta(y)$ is the measure putting full mass 1 at the point $y$, and $(1-\varepsilon)F + \varepsilon\delta(y)$ the consequent mixture distribution. A variable $Y_\varepsilon$ drawn from this mixture is from $F$ with probability $1-\varepsilon$ and is equal to $y$ with probability $\varepsilon$.

**(a)** Consider $\theta(F) = \mathrm{E}_G\, h(Y) = \int h(y)\,{\rm d}F(y)$, the mean of $h(Y)$. Show that $\mathrm{IF}(F, y) = h(y) - \theta(F)$. In particular, the influence is bounded when $h$ is, but unbounded e.g. in the case of the plain mean $h(y) = y$, which signifies a potential lack of robustness of this mean parameter functional $\theta = \mathrm{E}_G\, Y$.

**(b)** Then consider the class of smooth functions of means. For mean type parameters $\gamma_1 = \mathrm{E}_G\, h_1(Y), \ldots, \gamma_k = \mathrm{E}_G\, h_k(Y)$, let $\theta = T(F) = A(\gamma_1(F), \ldots, \gamma_k(F))$, where

$A(u_1,\ldots,u_k)$ is smooth in a neighbourhood of $(\gamma_1(F),\ldots,\gamma_k(F))$. Show that this parameter has influence function

$$\begin{aligned}\mathrm{IF}(F,y) &= c_1(F)\,\mathrm{IF}_{\gamma_1}(F,y)+\cdots+c_k(F)\,\mathrm{IF}_{\gamma_k}(F,y)\\ &= c_1(F)\{h_1(y)-\gamma_1(F)\}+\cdots+c_k(F)\{h_k(y)-\gamma_k(F)\},\end{aligned}$$

with $c_j(F)$ is the partial derivative $\partial A(u_1,\ldots,u_k)/\partial u_j$, evaluated at $(\gamma_1(F),\ldots,\gamma_k(F))$.

**(c)** Writing $\mu_F=\mathrm{E}_G\,Y$ for the mean, show for the variance parameter $\sigma_F^2=\mathrm{E}_G\,Y^2-\mu_F^2$ that its influence function becomes

$$\mathrm{IF}(F,y)=-2\,\mu_F(y-\mu_F)+y^2-\mathrm{E}_G\,Y^2=(y-\mu_F)^2-\sigma_F^2.$$

Then for the standard deviation parameter $\sigma(F)$ itself, show that its influence function becomes

$$\mathrm{IF}_\sigma(F,y)=\tfrac12(1/\sigma_F)\{(y-\mu_F)^2-\sigma_F^2\}.$$

**(d)** For a given parametric family $f(y,\theta)$, consider the ML functional $T(F)$, mapping a given $F$ with density $f$ to the least parameter value $\theta_0=\theta_0(F)$, the minimiser of the information distance $\mathrm{KL}(f,f(\cdot,\theta))$, or the maximiser of $\int\log f(y,\theta)\,\mathrm{d}F(y)$. With $F_n$ the empirical distribution of the data, placing probability $1/n$ on each data point, cf. Ex. 3.9, Show that $T(F_n)$ is the ML estimator, and that its influence function becomes $\mathrm{IF}(F,y)=J^{-1}u(y,\theta_0)$.

**Ex. 6.18** *Influence for quantiles.* IN: Assume the c.d.f. $F$ has a smooth density $f$.

**(a)** Consider $\mu=F^{-1}(\frac12)$, the median. Show that the influence function $\mathrm{IF}(F,y)$ becomes $-\frac12/f(\mu)$ for $y\le\mu$ and $\frac12/f(\mu)$ for $y>\mu$. Verify that the influence function has mean zero and variance $\frac14/f(\mu)^2$.

**(b)** More generally, let $\mu_q=F^{-1}(q)$, for some quantile level $q\in(0,1)$. Show that $\mathrm{IF}(F,y)$ is $-(1-q)/f(\mu_q)$ for $y\le\mu_q$ and $q/f(\mu_q)$ for $y>\mu_q$. Comment on the fact that these are bounded.

**(c)** Then consider a smooth function of quantiles, say $\gamma(F)=A(Q_1(F),\ldots,Q_k(F))$, where $Q_j(F)=F^{-1}(q_j)$. Show that its influence function is $\mathrm{IF}(F,y)=c_1(F)\,\mathrm{IF}_1(F,y)+\cdots+c_k(F)\,\mathrm{IF}_k(F,y)$, in terms of $\mathrm{IF}_j(F,y)$ the influence function of $F^{-1}(q_j)$ calculated as above, and where $c_j(F)=\partial A(Q_1,\ldots,Q_k)/\partial Q_j$ computed at $(F^{-1}(q_1),\ldots,F^{-1}(q_k))$. As an illustration, find and graph the influence function for $\gamma(F)=F^{-1}(0.90)-F^{-1}(0.10)$.

**Ex. 6.19** *Robust estimation of location.* Consider data $Y_1,\ldots,Y_n$ from a symmetric density on the line, with unknown location parameter $\xi$. We may write the density as $f(y,\xi)=f_0(y-\xi)$, with $f_0$ being symmetric around zero, but otherwise taken unknown. With $c$ a positive fine-tuning parameter, the Huber estimator $\widehat\xi_c$ is the minimiser of $H_{n,c}(\xi)=n^{-1}\sum_{i=1}^n H_c(y_i-\xi)$, for the particular function

Huber estimator, Huber function

$$H_c(u)=\begin{cases}\frac12u^2 & \text{if } |u|\le c,\\ c(|u|-\frac12c) & \text{if } |u|>c.\end{cases}$$

**(a)** Show that $H_{n,c}(\xi)$ is monotone and smooth, and explain that $\widehat\xi_c$ is also the unique solution to the equation $\sum_{i=1}^n\psi_c(y_i-\xi)=0$, with $\psi_c(u)$ being $u$ inside $[-c,c]$, then $-c$ to the left and $c$ to the right of that window. Invent a dataset and construct an algorithm that finds $\widehat\xi$, and experiment a little with different $c$. Explain that the estimator for big $c$ is equal to the sample mean $\bar y$, and for small $c$ equal to the sample median $M_n$.

**(b)** Note that the general methods and results of Ex. 6.8 and 6.9 do not work here, since the criterion function $H_c(u)$ is continuous, but not with a smooth derivative. Show that $H_c(u)$ is convex, however, so the convex minimisation methods of Ex. 5.30 apply. Work first through the details of the following:

$$H_c(y_i-\xi_0-\varepsilon)-H_c(y_i-\xi_0)=\begin{cases}-(y_i-\xi_0)\varepsilon+\frac12 s^2 & \text{if } y_i-\xi_0\in(-c,c),\\ -\varepsilon & \text{if } y_i-\xi_0>c,\\ \varepsilon & \text{if } y_i-\xi_0<-c,\end{cases}$$

for small $\varepsilon$. There is a minor problem when $y_i-\xi_0$ is close to $-c$ or $c$, but these cases do not disturb the main argument. Use this to reach the representation

$$\begin{aligned}A_{n,c}(s)&=nH_{n,c}(\xi_0+s/\sqrt{n})-nH_{n,c}(\xi_0)\\&=\sum_{i=1}^n\{H_c(y_i-\xi_0-s/\sqrt{n})-H_c(y_i-\xi_0)\}=-D_{n,c}s+\tfrac12 J_{n,c}s^2+r_{n,c}(s),\end{aligned}$$

in which $D_{n,c}=n^{-1/2}\sum_{i=1}^n\psi_c(y_i-\xi_0)$, $J_{n,c}=n^{-1}\sum_{i=1}^n I\{|Y_i-\xi_0|<c\}$, and the remainder term goes pointwise to zero in probability. Explain that $\psi_c(Y_i-\xi_0)$ has mean zero, under the symmetry assumption, so the CLT applies, with $D_{n,c}\to_d D_c\sim\mathrm{N}(0,\kappa_c^2)$, where $\kappa_c^2=\int\psi_c(u)^2f_0(u)\,\mathrm{d}u$.

**(c)** Deduce from the above, partly as per Ex. 5.30, that $A_n(s)\to_d A(s)=-D_cs+\frac12 J_cs^2$, with $J_c=\Pr(|Y-\xi_0|\le c)=2F_0(c)-1$. Explain from this again that this entails (i) $\sqrt{n}(\widehat{\xi}_c-\xi_0)\to_d D_c/J_c$, a normal with variance $v_\tau^2=\kappa_c^2/J_c^2$, and (ii)

$$2n\{H_{n,c}(\xi_0)-H_{n,c}(\widehat{\xi}_c)\}\to_d D_c^2/J_c\sim v_\tau^2\chi_1^2.$$

**(d)** estimating $\tau_c$ and $J_c$ and $v_\tau=\kappa_c/J_c$. compute for the normal case. new default value $c_{1.05}=1.023$, making intervals 1.05 bigger than the optimal size, under normality.

**(e)** do the influence function. do the 'Huber model' that we invent, where Huber is the ML. two parameters, even three, can read off $\ell_{n,\mathrm{prof}}(c)$. point to nice Story #98

**Ex. 6.20** *An estimator represented via its influence function.* IN: Consider an i.i.d. sequence $Y_1,Y_2,\ldots$ from $F$, with $\theta=T(F)$ a parameter of interest. It may be estimated nonparametrically using $\widehat{\theta}=T(F_n)$, with $F_n$ the empirical distribution. Here we work towards a representation of $\widehat{\theta}-\theta=T(F_n)-T(F)$ in terms of the influence function.

**(a)** Consider the case of $\theta=A(\gamma(F))$, where $\gamma(F)=\mathrm{E}_G\,h(Y)=\int h\,\mathrm{d}F$. Show that $\widehat{\theta}=T(F_n)$ is equal to $A(\bar{h})$, with $\bar{h}=\int h\,\mathrm{d}F_n=n^{-1}\sum_{i=1}^n h(y_i)$. Assuming $A(u)$ smooth, with two derivatives, show that

$$\widehat{\theta}=A(\gamma_0)+A'(\gamma_0)(\bar{h}-\gamma_0)+\tfrac12 A''(\gamma_0)(\bar{h}-\gamma_0)^2+o_{\mathrm{pr}}(1/n),$$

with $\gamma_0=\gamma(F)$. Deduce that $\mathrm{E}\,\widehat{\theta}=\theta+\frac12 A''(\gamma_0)\tau^2/n+o(1/n)$, in terms of $\tau^2=\mathrm{Var}\,h(Y_i)$, and that $\widehat{\theta}-\theta=n^{-1}\sum_{i=1}^n\mathrm{IF}(F,y_i)+b/n+o_{\mathrm{pr}}(1/\sqrt{n})$, where $b=\frac12 A''(\gamma_0)\tau^2$.

**(b)** Generalise to the case of $T(F)=A(\gamma_1(F),\ldots,\gamma_p(F))$ being a smooth function of several means, as studied also in Ex. 6.17, with $\gamma_j(F)=\int h_j\,\mathrm{d}F$. Show that $\mathrm{E}\,\widehat{\theta}=\theta+b/n+o(1/n)$, with $b=\frac12\mathrm{Tr}(A''(\gamma_0)K)$, with $K$ the variance matrix of $(h_1(Y_i),\ldots,h_p(Y_i))^{\mathrm{t}}$, and $A''(\gamma_0)$ the second order derivative matrix of $A$, computed at $\gamma_0$. Show that

$$\widehat{\theta}-\theta=T(F_n)-T(F)=n^{-1}\sum_{i=1}^n\mathrm{IF}(F,Y_i)+b/n+\varepsilon_n,\tag{6.6}$$

with $\varepsilon_n=o_{\mathrm{pr}}(1/\sqrt{n})$, i.e. small enough to have $\sqrt{n}\varepsilon_n\to_{\mathrm{pr}}0$. Show

**(c)** The powerful representation (6.6) actually holds quite generally, as long as $T(F)$ is a moderately smooth functional (xx find refs, Shao (1991), Jullum and Hjort (2017) xx), though with no easy general formula for the bias component $b/n$. Deduce that $\sqrt{n}(\widehat{\theta}-\theta)$ has the limit distribution $\mathrm{N}(0,\kappa^2)$, with $\kappa^2$ the variance of $\mathrm{IF}(Y_i,\theta)$, and with the bias part $b/n$ disappearing in this normal limit.

**(d)** Use the above to find the limit distribution of $\sqrt{n}(\widehat{\sigma}-\sigma)$. This gives a new and partly simpler proof of things proved in Ex. 2.48.

**Ex. 6.21** *Influence functions for BHHJ and for weighted likelihood estimators.* IN: For some smooth parametric model $f_\theta(y)=f(y,\theta)$, consider the BHHJ estimation method of Ex. 6.4 and 6.10. The setup involves the data-generating density $g$, and for the given tuning parameter $a$, the defining parameter is $\theta_0=\theta_{0,a}$, the minimiser of $\int f_\theta^{1+a}\,\mathrm{d}y-(1+1/a)\int g f_\theta^a\,\mathrm{d}y$, also the solution to $\int(f_\theta^{1+a}-gf_\theta^a)u_\theta\,\mathrm{d}y=0$. Here $u_\theta(y)=u(y,\theta)$ is the score function, and we will also need the information function $i_\theta(y)=i(y,\theta)$ below. As in Ex. 6.10 we use $f_0,u_0,i_0$ for these functions at $\theta_0$.

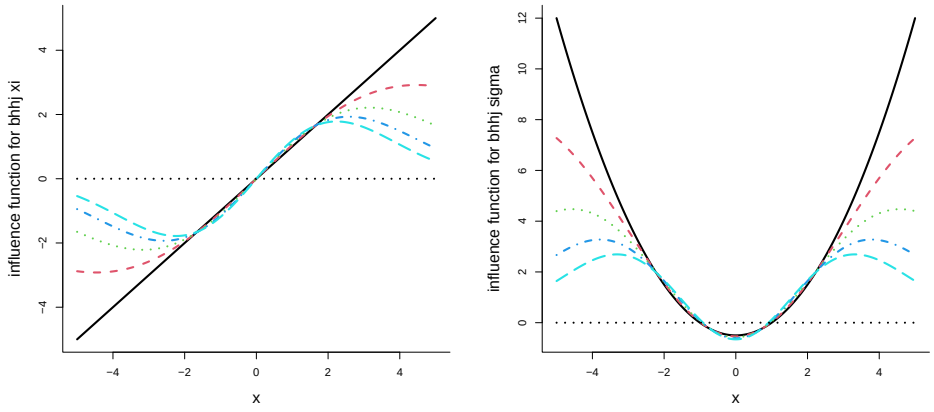



Figure 6.1: *Influence functions for the BHHJ estimator, for the normal $(\xi,\sigma^2)$ model, for values $a=0,0.05,0.10,0.15,0.20$; the full black curves are for the ML case $a=0$. Left panel: for $\widehat{\xi}_a$; right panel: for $\widehat{\sigma}_a$.*

**(a)** Explain that to find the influence function $\mathrm{IF}_a(G,y)$, we need for a small $\varepsilon$ to assess the solution $\theta_\varepsilon=\theta_0+\delta$ to the equation

$$\int f_{\theta_0+\delta}^{1+a}u_{\theta_0+\delta}\,\mathrm{d}y=(1-\varepsilon)\int gf_{\theta_0+\delta}^a u_{\theta_0+\delta}\,\mathrm{d}y+\varepsilon f(y,\theta_0+\delta)^a u(y,\theta_0+\delta).$$

The programme is now for the given small $\varepsilon$ to carry out first order Taylor expansion for $\delta$ small and then solve for $\delta$. Some details are as follows: show first

$$f_{\theta_0+\delta}^a\doteq f_0^a(1+a\delta^{\mathrm{t}}u_0),\quad f_{\theta_0+\delta}^{1+a}\doteq f_0^{1+a}\{1+(1+a)\delta^{\mathrm{t}}u_0\},\quad u_{\theta_0+\delta}\doteq u_0+i_0\delta.$$

Again with $\xi_a=\int f_0^{1+a}u_0\,\mathrm{d}u=\int gf_0^a u_0\,\mathrm{d}y$, use this to write the left and right hand sides of the equation as

$$\begin{aligned}L&\doteq\xi_a+\Big\{(1+a)\int f_0^{1+a}u_0u_0^{\mathrm{t}}\,\mathrm{d}y+\int f_0^{1+a}i_0\,\mathrm{d}y\Big\}\delta,\\ R&\doteq\xi_a+\Big\{\int gf_0^a(u_0u_0^{\mathrm{t}}+i_0)\,\mathrm{d}y\Big\}\delta+\varepsilon\{f(y,\theta_0)^a u(y,\theta_0)-\xi_a\}.\end{aligned}$$

Conclude that $\mathrm{IF}_a(G,y) = M_a^{-1}\{f_0(y)^a u_0(y) - \xi_a\}$, with the matrix

$$M_a = (1+a)\int f_0^{1+a} u_0 u_0^{\mathrm{t}}\,\mathrm{d}y + \int f_0^{1+a} i_0\,\mathrm{d}y - \int g f_0^a (u_0 u_0^{\mathrm{t}} + i_0)\,\mathrm{d}y.$$

Check with the $J_a$ matrix of Ex. 6.10 that in fact $M_a = J_a/(1+a)$, so that

$$\mathrm{IF}_a(G,y) = (1+a)J_a^{-1}\{f(y,\theta_0)^a u(y,\theta_0) - \xi_a\}.$$

Explain that as $a \to 0$, the limit is $J_0^{-1}u(y,\theta_0)$, the influence function for the ML estimation method. Show also that under conditions, $M_a$ simplifies to $\int f_0^{1+a} u_0 u_0^{\mathrm{t}}\,\mathrm{d}y$.

**(b)** Consider the case of the normal $\mathrm{N}(\xi,\sigma^2)$. Under model conditions, find formulae for and graph the influence functions $\mathrm{IF}_{1,a}(y)$ and $\mathrm{IF}_{2,a}(y)$, for the $\xi$ and the $\sigma$, at the standard position $(\xi,\sigma) = (0,1)$, for $a$ equal to 0, 0.05, 0.10, 0.15, 0.20. Construct a version of Figure 6.1. Note that the BHHJ influence functions mimic those for the ML (i.e. for $a = 0$) for the main expected data range, but that they sensibly redescend down to zero for values far away, via density downweighting. Show in particular that the influence functions are bounded, for each $a > 0$. For illustration of how the BHHJ successfully deals with outliers (without needing to identify them), and for seeing that very little efficiency is lost for small $a$, see Story #97. Formulae needed here include these, which you should prove, in addition to $\xi_{1,a} = 0$:

$$\begin{aligned}
\xi_{2,a} &= -\sigma^{-a}(2\pi)^{-a/2}a/(1+a)^{3/2},\\
M_{11,a} &= (1/\sigma^2)\sigma^{-a}(2\pi)^{-a/2}/(1+a)^{3/2},\\
M_{22,a} &= (1/\sigma^2)\sigma^{-a}(2\pi)^{-a/2}\{3/(1+a)^2 - 2/(1+a) + 1\}.
\end{aligned}$$

**(c)** Carry out similar analysis for the Gamma $(\alpha,\beta)$ model.

**(d)** Consider then the MWL method of Ex. 6.6 and 6.12. For a given weight function $w(y)$, the parameter aimed at for the MWL estimator is the maximiser $\theta_0 = \theta_{0,w}$ of $\int w(g\log f_\theta - f_\theta)$. We need $J_w = \int w f_0 u_0 u_0^{\mathrm{t}}\,\mathrm{d}y + \int w(f_0 - g)i_0\,\mathrm{d}y$ and $\xi_w = \int w f_0 u_0\,\mathrm{d}y = \int w g u_0\,\mathrm{d}y$, computed at $\theta_0$. Show that the influence function becomes

$$\mathrm{IF}(G,y) = J_w^{-1}\{w(y)u(y,\theta_0) - \xi_w\}.$$

## Leftovers and further material

## Notes and pointers

(xx various. xx) BHHJ and JHHB. Huber. The Basu book. Robustness. Influence functions here and there. Mutatis mutandis.

# I.7

# Bayesian inference and computation

In frequentist parametric inference, there is a fixed underlying true parameter value, say $\theta_0$, and methods aim at estimating this value, perhaps along with confidence regions or testing. Bayesian inference is radically different, conceptually and operationally. It starts with a prior distribution for the model parameter $\theta$, and proceeds via Bayes theorems to produce the posterior distribution, of the full $\theta$ or of relevant focus parameters. Thus 'not knowing $\theta$ well' is expressed in terms of probability distributions. This chapter goes through these concepts and operations, including also computational schemes to simulate outcomes from the posterior distributions. One prominent class of such schemes amounts to setting up a Markov Chain Monte Carlo algorithm for the given problem, where the stationary distribution for the chain is precisely the required posterior. This makes Bayesian inference possible in a host of complicated setups, without needing to rely on mathematically feasible formulae.

*Key words:* Bayes solutions, Bernshteĭn–von Mises theorems, conjugate priors, Jeffreys and vip-Jeffreys prior, loss functions, MCMC, prior to posterior distributions

The Bayesian paradigm is to formulate uncertainty about model parameters through probability distributions. If the pre-data uncertainty is a prior density $\pi(\theta)$, this is updated to the post-data posterior density $\pi(\theta \,|\, \text{data})$, via the Bayes theorems.

Consider for illustration and clarification the classical coin flipping experiment, with $\theta$ the probability of 'head up'. With $n$ independent flips we have $Y \sim \text{binom}(n, \theta)$. The frequentist postulates that there is an underlying true $\theta_0$, uses perhaps the estimator $\widehat{\theta} = Y/n$, reaches the 95 percent interval $I_n = \widehat{\theta} \pm 1.96\,\{\widehat{\theta}(1-\widehat{\theta})\}/\sqrt{n}$, etc. The key property here is $\Pr_{\theta_0}(\theta_0 \in I_n) \doteq 0.95$; so $I_n$ is a random interval, covering the true $\theta_0$ in 95 percent of actual cases. The Bayesian viewpoint is strikingly different, starting with a prior density $\pi(\theta)$ to reflect what might be considered understanding of $\theta$ before the first flip. Post flipping, the Bayesian has reached $\pi(\theta \,|\, y) \propto \pi(\theta) f(y, n, \theta)$, with the binomial likelihood. This may e.g. be used to construct a 95 percent posterior interval $J_n$ for $\theta$, with $\Pr(\theta \in J_n \,|\, y) = 0.95$. The Bayesian is then not interested in 'independent repeated experiments', but just in the data at hand. She is also allowed the statistical luxury of putting prior knowledge into the analysis; if it can be considered known that $\theta$ must be close to 0.50, with values outside $[0.40, 0.60]$ less likely than 2 percent, that can effectively be utilised in Bayesian analysis, but not so easily in frequentist analysis.

In this chapter we go through the basics of such constructions and methods, conceptually and operationally. We also uncover conditions under which the frequentist and

Bayesian might actually (approximately) agree, in their final inference statements. The two 95 percent intervals $I_n$ and $J_n$ in the previous paragraph will e.g. tend to be very similar, at least with increasing $n$.

An attractive feature of Bayesian analysis is that the answer, to a sufficiently well-posed inference question, is crystal clear, without having to study competing methods, carrying out performance and comparison analyses, etc. Essentially, if you give a Bayesian (i) data, (ii) a model, (iii) a list of possible actions, (iv) a prior for the unknown parameters, and (v) a loss function, there is a Master Recipe for the very best action.

Modern Bayesian statistics has flourished since around 1980, partly through computer power and algorithms, making calculations possible that would have been too hard for previous generations. The operational goal is often to be able to generate samples from the posterior distribution, and we give methods for accomplishing this, including Markov Chain Monte Carlo (MCMC).

### The Bayesian Master Recipe, with examples and applications

**Ex. 7.1** *Poisson data with gamma priors.* This exercise illustrates the basic prior to posterior updating mechanism in a simple Poisson setting. Suppose $Y_1, Y_2, \ldots$ are i.i.d. Poisson with unknown mean $\theta$.

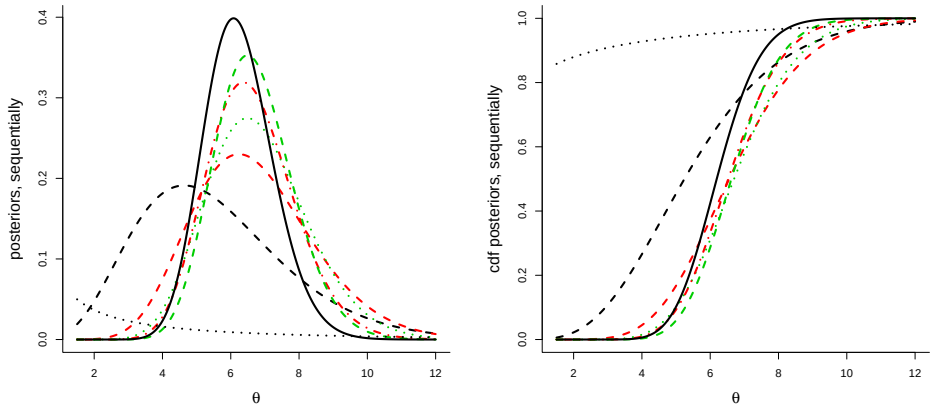



Figure 7.1: *Seven curves are displayed, corresponding to the* Gam$(0.1, 0.1)$ *intial prior for the Poisson parameter $\theta$, along with the six first updates following each of the observations 6, 8, 7, 6, 7, 4, 11, 8, 6, 3. The distributions become tighter as more data come in. The smooth black curve is after six data points, the most tight distribution so far. Left panel: the densities; right panel: the c.d.f.s.*

**(a)** Recall definition and properties of the Gamma distribution from Ex. 1.9. In the present Bayesian context, let $\theta \sim \text{Gam}(a, b)$. The prior mean and variance are $a/b = \theta_0$ and $a/b^2 = \theta_0/b$. In particular, low and high values of $b$ signify high and low variability, respectively. Explain how $(a, b)$ may be set from values of prior mean and prior variance. To exemplify, if these are $(5.5, 7.7)$, find $(a, b)$.

**(b)** With a single observation $Y$, which is Pois$(\theta)$ given $\theta$, show that $\theta \,|\, y \sim \text{Gam}(a + y, b + 1)$.

**(c)** Then suppose there are repeated observations $y_1, \ldots, y_n$, being i.i.d. $\sim$ Pois($\theta$) for given $\theta$. Use the above result repeatedly, e.g. interpreting $p(\theta \,|\, y_1)$ as the new prior before observing $y_2$, etc., to show that $\theta \,|\, y_1, \ldots, y_n \sim \text{Gam}(a + y_1 + \cdots + y_n, b + n)$. Also derive this result directly, i.e. without necessarily thinking about the data having emerged sequentially.

**(d)** Suppose the prior used is a rather flat Gam(0.1, 0.1) and that the Poisson data are 6, 8, 7, 6, 7, 4, 11, 8, 6, 3. Reconstruct a version of Figure 7.1 in your computer, plotting the six first posterior densities $p(\theta \,|\, \text{data}_j)$ (left panel), where $\text{data}_j$ is $y_1, \ldots, y_j$, along with the prior density; in the right panel we have the corresponding posterior cumulatives. Complement with another figure also including updated densities 7, 8, 9, 10, for the four last observations, and comment. Also compute the ten Bayes estimates $\widehat{\theta}_j = \text{E}(\theta \,|\, \text{data}_j)$ and the posterior standard deviations, for $j = 1, \ldots, 10$.

**(e)** The mathematics turned out to be rather uncomplicated in this situation, since the Gamma continuous density matches the Poisson discrete density so nicely. Suppose instead that the initial prior for $\theta$ is a uniform over $[5.0, 11.0]$. Try to compute posterior distributions, Bayes estimates and posterior standard deviations also in this case, and compare with what you found above. Note also that we in this exercise cared about and managed to reach answers for the posterior distributions $\theta \,|\, y_1, \ldots, y_n$, without needing to deal with the also implied marginal distribution for $(y_1, \ldots, y_m)$. These are not Poisson any longer, with the variability in $\theta$ taken into account; see Ex. 7.29.

**Ex. 7.2** *The Bayesian Master Recipe.* The general setup is as follows. We have data $y_{\text{obs}}$, seen as the outcome of a random $Y$ over the sample space $\mathcal{Y}$, generated from a model $f_\theta(y) = f(y, \theta)$, with $\theta$ having a prior distribution $\pi(\theta)\,\text{d}\theta$ over its parameter space $\Theta$. There is a decision $a$ to be reached, with $a$ belonging to an appropriate action space $\mathcal{A}$, along with a loss function $L(\theta, a)$, a measure of the consequences of decision $a$ if the truth is $\theta$. Whereas the frequentist can attempt different methods for deciding $\widehat{a} = \widehat{a}(Y)$, then compare risk functions, etc., there is a unique optimal strategy for the Bayesian.

this is the Bayes theorem

**(a)** Show that the posterior density of $\theta$, that is, the distribution of the parameter given the data, takes the form

$$\pi(\theta \,|\, y) = \frac{f_\theta(y)\pi(\theta)}{m(y)}, \tag{7.1}$$

where $m(y)$ is the required integration constant $\int_\Theta f_\theta(y)\pi(\theta)\,\text{d}\theta$. This is *Bayes' theorem*, and we typically write $\pi(\theta \,|\, y) \propto \pi(\theta) f_\theta(y)$, which reads 'posterior is proportional to prior times likelihood'. Show also that the *marginal distribution* of the data $y$ is $m(y)$.

**(b)** A decision $a = \widehat{a}(y)$ needs to be reached, as a function of the data. The *Bayes risk* for such a decision is the associated expected loss, $\text{BR}(a, \pi) = \text{E}\, L(\theta, \widehat{a}(Y))$, involving randomness on two levels; $\theta$ has a prior, and $\widehat{a}(Y) \,|\, \theta$ is random. Show that it may be expressed in two informative ways:

$$\text{BR}(a, \pi) = \begin{cases} \text{E}\,\{\text{E}\, L(\theta, \widehat{a}(Y)) \,|\, \theta\} = \int_\Theta R(\theta, \widehat{a})\pi(\theta)\,\text{d}\theta, \\ \text{E}\,\{\text{E}\, L(\theta, \widehat{a}(Y)) \,|\, Y\} = \int_{\mathcal{Y}} \{\int_\Theta L(\widehat{a}(y), \theta)\pi(\theta \,|\, y)\,\text{d}\theta\}\, m(y)\,\text{d}y. \end{cases} \tag{7.2}$$

The first expression involves the frequentist risk function $R(\theta, \widehat{a}) = \text{E}_\theta\, L(\theta, \widehat{a}(Y))$, then averaged with respect to the prior. The inner expectation of the second expression is $\text{E}_\pi\,\{L(\theta, \widehat{a}(y)) \,|\, Y = y\}$, that is, the expected loss given data.

the Bayesian Master Recipe

**(c)** Show then that the optimal Bayes strategy, the one minimising the Bayes risk, sometimes called the Bayes solution, is achieved by using

$$\widehat{a} = \text{argmin}_{a\in\mathcal{A}} \mathrm{E}_\theta \{L(\theta, a) \,|\, y_{\text{obs}}\} = \text{the value of } a \text{ minimising } \mathrm{E}_\theta \{L(\theta, a) \,|\, y_{\text{obs}}\}.$$

Here $\mathrm{E}_\theta \{L(\theta, a) \,|\, y_{\text{obs}}\}$ is the expected posterior loss. The function $g$ is evaluated and minimised over all $a$, for the given data $y = y_{\text{obs}}$. This is the Bayes recipe. Note that using the recipe in practice only concerns the observed data $y_{\text{obs}}$, and that one does not need to evaluate its risk function. The *minimum Bayes risk* $\text{MBR}(\pi) = \text{BR}(\widehat{a}, \pi)$ is the resulting Bayes risk when using this optimal strategy.

minimum Bayes risk

**(d)** Above results have been presented and reached in terms of prior and posterior densities, $\pi(\theta)$ and $\pi(\theta \,|\, y)$, partly for notational convenience. Show that the arguments go through also for more general priors; these may in particular be mixtures over continuous and discrete measures.

**Ex. 7.3** *Some loss functions and their associated Bayes rules.* The Master Recipe of Ex. 7.2 is completely general, and can be applied in new and complicated situations, as long as we have data, a model, a prior for the unknowns, and a loss function. In the Bayesian setup finding or evaluation of the posterior distribution of the parameters is always important, carrying separate weight, but if clear decisions are needed one needs also the loss function, say $L(\theta, a)$. Here we go through a short list of commonly used loss functions.

**(a)** For estimating a one-dimensional $\theta$, with squared error loss $L(\theta, a) = (a - \theta)^2$, show that the Bayes estimator is $\widehat{\theta}_B = \mathrm{E}\,(\theta \,|\, y)$, the posterior mean. Show also that the minimum Bayes risk, with this loss function, can be expressed as $\text{MBR}(\pi) = \mathrm{E}\,\text{Var}\,(\theta \,|\, Y)$.

**(b)** If the loss function is $L(\theta, a) = w(\theta)(a - \theta)^2$, show that the Bayes estimator is

$$\widehat{\theta}_B = \mathrm{E}\,\{w(\theta)\theta \,|\, \text{data}\}/\mathrm{E}\,\{w(\theta) \,|\, \text{data}\}.$$

In particular, when estimating a positive $\theta$ using loss $(a - \theta)^2/\theta$, show that the Bayes estimator is $1/\mathrm{E}\,\{(1/\theta) \,|\, \text{data}\}$, the inverse of the inverse mean.

**(c)** Consider the natural absolute loss function, $L(\theta, a) = |a - \theta|$. Show that the Bayes solution becomes the posterior median, i.e. $\widehat{\theta}_B = G^{-1}(\frac{1}{2} \,|\, \text{data})$, where $G(\theta \,|\, \text{data})$ is the posterior c.d.f.

**(d)** A differently spirited loss function is $L_\varepsilon(\theta, a) = I(|a - \theta| > \varepsilon)$, for some small $\varepsilon$; zero loss if you're very close, but then a fixed positive loss if you err with more than $\varepsilon$. Show that the Bayes solution is the $a$ which makes the probability $\Pr(\theta \in [a - \varepsilon, a + \varepsilon] \,|\, \text{data})$ as big as possible. Explain that this for small $\varepsilon$ means finding the mode of the posterior density.

**(e)** Suppose one needs the joint estimation of several parameters, say all of $\theta = (\theta_1, \ldots, \theta_p)$, via the loss function $L(\theta, a) = (a - \theta)^{\mathrm{t}} M (a - \theta)$, for an appropriate full-rank symmetric matrix $M$. Show that the Bayes solution again is the posterior mean, but now for the full vector, i.e. $\mathrm{E}\,(\theta \,|\, \text{data})$. In particular, the Bayes solution does not depend on the $M$ matrix, though the actual posterior expected loss, and the Bayes risk, do.

**(f)** In the previous subquestions the framework has been that of estimating a one-dimensional $\theta$. Check that you understand how these results and insights, for the Bayes solutions, change when the situation is changed to that of estimating a *focus parameter*, say $\mu = g(\theta_1, \ldots, \theta_p)$, a function of the full model parameter.

**Ex. 7.4** *How many streetcars in San Francisco?* The streetcars in this city are numbered $1, \ldots, N$. You observe $Y = 203$ and wonder what $N$ is.

**(a)** Supposing $Y$ has the uniform distribution on $1, \ldots, N$, set up the likelihood, and identify the ML estimate. With the prior $\pi(N)$ proportional to $1/N$, say for $N = 1, 2, \ldots, N_{\max}$ for a suitably high $N_{\max}$, find the posterior distribution, along with its mean and median.

**(b)** Suppose you after having seen no. 203 also see nos. 157, 222. Update your posterior, and again find the mean and median, qua updated estimates.

**(c)** The Bayesian setup allows any choice for the prior, so think for a minute and construct your own $\pi_{\text{you}}(N)$. Do the updating, for the data 203, 157, 222, and compare with the posterior found above. What is your best estimate of $N$, if your loss function is of 0-1 type, with $L(N, \widehat{N})$ being 0 if $\widehat{N} = N$ and 1 if $\widehat{N} \neq N$?

**Ex. 7.5** *The linex loss function.* When estimating a one-dimensional $\theta$ with a $\widetilde{\theta}$, the most traditional loss function is that of squared error, $(\widetilde{\theta} - \theta)^2$, which in particular is symmetric, treating over- and underestimation as equally important. A more flexible loss function is the so-called linex loss, with

$$L_c(\theta, \widetilde{\theta}) = \exp\{c(\widetilde{\theta} - \theta)\} - 1 - c(\widetilde{\theta} - \theta).$$

The $c$ is fine-tuning loss parameter, for the statistician to set, balancing over- against underestimation. Note that both positive and negative values of $c$ are allowed here. There is more on the linex loss in Ex. 9.21, 9.22, but here we deal with the basics.

**(a)** Show that the $L_c$ is always nonnegative. Show that $c > 0$ means penalising overestimation more than underestimation, and vice versa for $c < 0$. For small $|c|$, show that $L_c(\theta, \widetilde{\theta}) \doteq \frac{1}{2}c^2(\widetilde{\theta} - \theta)^2$, getting back to squared error loss. The constant in front is immaterial for evaluating and comparing loss and risk, and one may use $L_c^*(\theta, \widetilde{\theta}) = L_c(\theta, \widetilde{\theta})/(\frac{1}{2}c^2)$ to have a smoother transition to the $c = 0$ case of squared error loss.

**(b)** Show that the expected loss given data can be expressed as

$$\begin{aligned} \mathrm{E}\,\{L_c(\theta, t) \,|\, \text{data}\} &= \mathrm{E}\,[\exp\{c(t - \theta)\} - 1 - c(t - \theta) \,|\, \text{data}] \\ &= \exp(ct)M(-c) - 1 - c(t - \widehat{\xi}), \end{aligned}$$

where $\widehat{\xi}$ is the posterior mean and $M(-c) = \mathrm{E}\,\{\exp(-c\theta) \,|\, \text{data}\}$, the m.g.f. of $\theta$ given data, computed at $-c$.

**(c)** Show that this is minimised for the $t_0$ where $\exp(ct_0)M(-c) = 1$, or $ct_0 + \log M(-c) = 0$, so that the Bayes estimator becomes $\widehat{\theta}_B = -(1/c)\log M(-c)$. This may be computed numerically, perhaps by simulation, in cases where no clear formula exists for $M(-c)$. Show also that the expected posterior loss, using the Bayes solution, is

$$\min \mathrm{E}\,\{L_c(\theta, t) \,|\, \text{data}\} = -c(t_0 - \widehat{\xi}) = \log M(-c) + c\widehat{\xi}.$$

**(d)** Using approximations for m.g.f.s close to zero to show that $M(-c) \doteq 1 - c\widehat{\xi} + \frac{1}{2}c^2(\widehat{\xi}^2 + \widehat{\sigma})$, with $\widehat{\xi}$ and $\widehat{\sigma}^2$ the posterior mean and variance. Deduce that $\widehat{\theta}_B \doteq \widehat{\xi} - \frac{1}{2}c\widehat{\sigma}^2$.

**(e)** In situations where the posterior is based on a sample of size $n$, the posterior mean $\widehat{\xi}_n$ stays stable wheras the posterior variance $\widehat{\sigma}^2$ goes down with speed $1/n$, i.e. as $\widehat{\sigma}_0^2/n$, for the relevant $\widehat{\sigma}_0^2$. In such cases, $\widehat{\xi}_n - \frac{1}{2}c\widehat{\sigma}_0^2/n$ becomes the approximation to the Bayes linex estimator $\widehat{\theta}_B$. Find in fact the exact Bayes linex estimator, for the case of $Y_1, \ldots, Y_n$ being i.i.d. $\mathrm{N}(\theta, 1)$, with a $\mathrm{N}(0, \tau^2)$ prior for $\theta$; here you may very well work from scratch, though it is useful to see the updating results from Ex. 7.17(b).

**Ex. 7.6** *A Bayesian take on hypothesis testing.* Assume the model parameter $\theta$ is either in $\Theta_0$, which we may call the null hypothesis, or not, i.e. in its complement $\Theta_0^c$. Suppose also that the statistician needs to make a decision, either to reject the null, or to accept it. This is the basic framework of hypothesis testing, see Ch. 4, but we now consider the problem from a Bayesian viewpoint.

**(a)** The decision space is $\{\text{accept}, \text{reject}\}$. For the loss function, take $L(\theta, \text{accept})$ equal to 0 or $L_0$, if $\theta$ is inside or outside $\Theta_0$, and $L(\theta, \text{reject})$ equal to 0 or $L_1$, if $\theta$ is outside or inside $\Theta_0$. Show that

$$\mathrm{E}\,\{L(\theta, \text{accept}) \,|\, \text{data}\} = L_0\, p(\text{data}), \quad \mathrm{E}\,\{L(\theta, \text{reject}) \,|\, \text{data}\} = L_1\,\{1 - p(\text{data})\},$$

where $p(\text{data}) = \Pr(\theta \in \Theta_0^c \,|\, \text{data})$, the probability that the null is wrong, as measured by the Bayesian posterior distribution.

**(b)** Deduce that one should reject the null when the probability $p(\text{data})$ for its falseness is sufficiently overwhelming, namely when $p(\text{data}) \geq L_1/(L_0 + L_1)$. – If this threshold is 0.95, for example, show that this corresponds to $L_1/L_0 = 19$. Briefly discuss ways of assigning losses $L_0$ and $L_1$.

**(c)** Suppose now there is a 'doubt option', refraining from deciding on the either-or dichotomy. The setting is as above, with losses $L_0$ and $L_1$, but the action list is $\{\text{accept}, \text{reject}, \text{doubt}\}$, with a fixed loss $L_d$ for deciding to be in doubt. Explain that the expected losses given the data are $L_0 p(\text{data})$, $L_1\{1 - p(\text{data})\}$, $L_d$, and spell out what the Bayes solution is.

**Ex. 7.7** *Which subset does the model parameter belong to?* Consider a setup with data from a model with model parameter $\theta$ inside its region $\Theta$. Suppose you need to take one of five different possible decisions, $D_1, \ldots, D_5$, and that these are related to where the underlying parameter $\theta$ is positioned; if $\theta \in \Theta_j$ the best decision would be $D_j$, for $j = 1, \ldots, 5$. Here the $\Theta_j$ are disjoint and their union is the full parameter region.

**(a)** Suppose the loss function $L(\theta, D_j)$ is 0 if $\theta \in \Theta_j$ and 100 if $\theta \notin \Theta_j$. Show that $\mathrm{E}\,\{L(\theta, D_j) \,|\, \text{data}\} = 100\{1 - p_j(\text{data})\}$, where $p_j(\text{data}) = \Pr(\theta \in \Theta_j \,|\, \text{data})$. Hence show that the optimal Bayes strategy is to take the decision associated with the highest posterior probability $p_j(\text{data})$.

**(b)** Assume there in addition is a 'doubt option', associated with a doubt cost $L_d = 10$; this could e.g. mean planning for getting further data. With decision space $\{D_1, \ldots, D_5, \text{doubt}\}$, what is now the Bayesian strategy?

**(c)** Generalise the previous setup, and results, to the case where the costs associated with reaching the wrong decision are not equally balanced, say $L(\theta, D_j) = c_{i,j}$, if $\theta \in \Theta_i$, for $i = 1, \ldots, 5$, with $c_{i,i} = 0$ but the other $c_{i,j}$ positive.

### Binomial and beta, multinomial and Dirichlet

**Ex. 7.8** *The binomial-beta setup.* Let $Y$ given $p$ be a binomial $(n, p)$, and for $p$ take a $\text{Beta}(a, b)$ prior, see Ex. 1.22. There we worked with the marginal distribution of $Y$, and looked at certain properties, but here our aims are Bayesian.

**(a)** Show that $p \,|\, y \sim \text{Beta}(a+y, b+n-y)$. This is the main and always crucial updating step, getting from the prior to the posterior. In the present case the step is an easy one, since there is only one unknown parameter, and since the product of the prior and the likelihood takes an easy form. Give a description of the posterior also for the not quite so standard case where the prior for $p$ is uniform on $[0.30, 0.70]$.

prior updated to posterior

**(b)** Going back to the Beta$(a, b)$ prior again, show that the Bayes estimator, under squared error loss, is $\widehat{p}_B = (1 - w_n)p_0 + w_n y/n$, where $p_0 = a/(a + b)$ is the prior mean and $w_n = n/(a + b + n)$. For the case of a uniform prior, show that this leads to $(y+1)/(n+2)$. Compute the risk functions $r(p) = \mathrm{E}_p\,(\widehat{p}-p)^2$, for the classic frequentist $Y/n$ and for the this $(Y+1)/(n+2)$, and find the interval where the latter is better than the former.

**(c)** It is helpful to parametrise $(a, b)$ as $(cp_0, c(1 - p_0))$, with prior mean $p_0$ and prior variance $p_0(1 - p_0)/(c + 1)$. Explain that for the Bayes estimator and the posterior variance we then have

$$\widehat{p}_B = \mathrm{E}\,(p \,|\, y) = \frac{cp_0 + y}{c + n}, \quad \mathrm{Var}\,(\theta \,|\, y) = \frac{\widehat{p}_B(1 - p_B)}{n + c + 1}.$$

Use results from Ex. 1.22 to show that with its marginal distribution, $\mathrm{Var}\,Y = np_0(1 - p_0)(c + n)/(c + 1)$.

**(d)** Then use $\mathrm{Var}\,p = \mathrm{E}\,\mathrm{Var}\,(p \,|\, Y) + \mathrm{Var}\,\mathrm{E}\,(p \,|\, Y)$ to find that with the Beta $(cp_0, c(1-p_0))$ prior, the minimum Bayes risk is MBR $= p_0(1 - p_0)c/\{(c + 1)(c + n)\}$. Show that the maximum of the minimum Bayes risk happens for the prior Beta$(\frac{1}{2}\sqrt{n}, \frac{1}{2}\sqrt{n})$, with value $\frac{1}{4}/(\sqrt{n} + 1)^2$, and consequent Bayes estimator

$$\widehat{p}_B = \frac{\frac{1}{2}\sqrt{n} + Y}{\sqrt{n} + n} = \frac{1}{\sqrt{n} + 1}\frac{1}{2} + \frac{\sqrt{n}}{\sqrt{n} + 1}\frac{Y}{n}. \tag{7.3}$$

This is the *least favourable prior*; explain that if Nature chooses $p$ from this distribution, then this is the worst of all Beta scenarios for the statistician. The $\widehat{p}_B$ estimator has a corresponding *minimax* property, no competing estimator has a smaller maximum risk; see material on this in Ch. 9.

**(e)** It is useful to see the minimum Bayes risk in two ways, via (7.2), i.e. also via the risk function of the Bayes solution. Show that this risk function becomes

$$r_B(p) = \mathrm{E}_p\,(\widehat{p}_B - p)^2 = \frac{1}{(c + n)^2}\{np(1 - p) + c^2(p - p_0)^2\}.$$

Then take the mean of this with respect to the prior to find the MBR formula above again. For the least favourable prior, show that the risk function is constant, equal to $\frac{1}{4}/(\sqrt{n} + 1)^2$.

**(f)** If $Y$ is from the binomial $(n, p_{\mathrm{true}})$ model, show that $Y/n$ and the Bayes estimator $\widehat{p}_B$ are large-sample equivalent, with $\sqrt{n}(Y/n - \widehat{p}_B) \to_{\mathrm{pr}} 0$, as long as $c = a + b$ stays fixed or bounded. Deduce that they have the same limit distribution. This does not hold for the least favourable prior, however, as its parameters $(\frac{1}{2}\sqrt{n}, \frac{1}{2}\sqrt{n})$ then move with sample size; find the limit distribution of $\sqrt{n}(\widehat{p}_B - p_{\mathrm{true}})$ for this case.

**Ex. 7.9** *Binomial-beta with weighted loss function.* One's best decisions depend on how the consequences are to be evaluated. Consider as in Ex. 7.8 the $Y \sim \mathrm{binom}(n, p)$ situation, but now with the weighted loss function $L(p, \widehat{p}) = (\widehat{p} - p)^2/\{p(1 - p)\}$.

**(a)** Discuss briefly what such a loss function entails, compared to the traditional squared error loss, regarding situations where the $p$ probabilities are close to the boundaries. For the ML estimator $Y/n$, show that the risk function is the constant $1/n$.

**(b)** With $p$ having any prior, show that the Bayes estimate is

$$\widehat{p}_B = \mathrm{E}\left(\frac{1}{1 - p} \,|\, y\right)/\mathrm{E}\left(\frac{1}{p(1 - p)} \,|\, y\right).$$

If $p$ has a Beta$(a,b)$ disribution, show that the formula for $\mathrm{E}\,\{1/(1-p)\}/\mathrm{E}\,1/\{p(1-p)\}$ becomes $(a-1)/(a+b-2)$, provided $a>1$ and $a+b>2$. With the uniform prior, show that this again leads to $\widehat{p}_B = y/n$, first for cases $y = 1,\ldots,n-1$, then also for cases $y=0$ and $y=n$, where some extra care is required for the two boundary cases.

**Ex. 7.10** *Bayes estimators are biased.* Excpet for a few particular cases, perhaps involving special loss functions, the Bayes estimator is not unbiased; it stems from a prior, which is making its inherently biased presence felt. To be specific, suppose a one-dimensional $\theta$ is to be estimated with quadratic loss $(a-\theta)^2$, based on observations $Y$, and that one starts with a proper prior for the $\theta$. We assume, for the arguments to go through smoothly, that $\theta$ has a finite prior variance.

**(a)** We have seen in Ex. 7.3 that the Bayes estimator is $\widehat{\theta}_B(y) = \mathrm{E}\,(\theta\,|\,y)$. Suppose it is unbiased, $\mathrm{E}_\theta\,\widehat{\theta}_B(Y) = \theta$ for all $\theta$. Go through the details of the following arguments, where the expectation operator refers to the joint distribution of $(\theta, Y)$:

$$\mathrm{E}\,(\widehat{\theta}_B-\theta)^2 = \mathrm{E}\,\theta^2 - \mathrm{E}\,\theta\widehat{\theta}_B - \mathrm{E}\,\widehat{\theta}_B\theta + \mathrm{E}\,\widehat{\theta}_B^2 = \mathrm{E}\,\theta^2 - \mathrm{E}\,\theta^2 - \mathrm{E}\,\widehat{\theta}_B^2 + \mathrm{E}\,\widehat{\theta}_B^2 = 0.$$

Conclude that unbiasedness can only take place in the trivial case where $\widehat{\theta}_B = \theta$ with probability 1, which would mean $\theta$ is known.

**(b)** There are various situations where the posterior mean estimator is close to being unbiased, but by the argument above there will be some remaining bias. In the binomial setup of Ex. 7.8, the Bayes estimator is $(c\theta_0+Y)/(c+n)$, which becomes the familiar unbiased $Y/n$ for $c\to 0$; this limiting case, corresponding to a prior proportional to $\theta^{-1}(1-\theta)^{-1}$, is not a proper prior (finite integral), however. We learn that posterior mean estimators computed under certain improper priors (infinite integral) might be unbiased. Note that the arguments used in (a) involve double expectation rules, which presuppose a proper prior.

**(c)** Explain why deriving the Bayes estimator $Y/n$ in Ex. 7.9 does not contradict what is found above.

**Ex. 7.11** *The multinomial-Dirichlet setup.* Here we extend the setup and result of binomial-Beta, to the case of three or more categories. We start with $Y=(Y_1,\ldots,Y_k)$ which for given $p=(p_1,\ldots,p_k)$ is a multinomial $(n,p_1,\ldots,p_k)$. For $p$ we take the Dir$(a_1,\ldots,a_k)$ prior. For details regarding the multinomial and the Dirichlet, see Ex. 1.5 and 1.20.

**(a)** Show the important and useful result that $(p_1,\ldots,p_k)\,|\,(y_1,\ldots,y_k) \sim \mathrm{Dir}(a_1+y_1,\ldots,a_k+y_k)$.

**(b)** Show that the Bayes estimator under squared error loss becomes

$$\widehat{p}_{i,B} = \mathrm{E}\,(p_i\,|\,\mathrm{data}) = \frac{a_i+y_i}{a+n} = (1-w_n)p_{0,i} + w_n(y_i/n)$$

for $i=1,\ldots,k$, with prior means $p_{0,i}=a_i/a$, with $a=a_1+\cdots+a_k$, and weight $w_n=n/(a+n)$. Find also the posterior variance and posterior correlation between $p_i$ and $p_j$.

**(c)** Find the Bayes estimators, under quadratic loss, for the geometric mean parameters $(p_1p_2)^{1/2}$ and $(p_1\cdots p_k)^{1/k}$.

**Ex. 7.12** *Minimax estimation of multinomial parameters.* We shall consider estimation in the general multinomial model, but for concreteness start with the trinomial case.

Consider therefore $(X, Y, Z)$ trinomial $(n, p, q, r)$, with $p + q + r = 1$. We work with the loss function

$$L((p, q, r), (a, b, c)) = (a - p)^2 + (b - q)^2 + (c - r)^2$$

for estimating $(p, q, r)$ jointly.

**(a)** Show that using a Dirichlet $(cp_0, cq_0, cr_0)$ for $(p, q, r)$ leads to Bayes estimators $\widehat{p} = (cp_0 + x)/(c + n)$, $\widehat{q} = (cq_0 + y)/(c + n)$, $\widehat{r} = (cr_0 + z)/(c + n)$, and that the associated risk function becomes

$$\frac{1}{(c+n)^2}\{np(1-p) + c^2(p-p_0)^2 + nq(1-q) + c^2(q-q_0)^2 + nr(1-r) + c^2(r-r_0)^2\}.$$

**(b)** Show that the minimum Bayes risk, for such a prior, becomes

$$\text{MBR} = \frac{c}{(c+1)(c+n)}\{p_0(1-p_0) + q_0(1-q_0) + r_0(1-r_0)\}.$$

Show further that this is maximised over $c$, for $c = \sqrt{n}$, and then over all prior means for $(p_0, q_0, r_0) = (1/3, 1/3, 1/3)$. Verify that the maximised minimum Bayes risk becomes $(2/3)/(\sqrt{n}+1)^2$. This invites using the $\text{Dir}(\sqrt{n}(1/3, 1/3, 1/3))$ prior, with ensuing estimators $(\widehat{p}, \widehat{q}, \widehat{r})$ equal to

$$\frac{1/3}{\sqrt{n}+1} + \frac{\sqrt{n}}{\sqrt{n}+1}\frac{x}{n}, \frac{1/3}{\sqrt{n}+1} + \frac{\sqrt{n}}{\sqrt{n}+1}\frac{y}{n}, \frac{1/3}{\sqrt{n}+1} + \frac{\sqrt{n}}{\sqrt{n}+1}\frac{z}{n}.$$

**(c)** Show that this estimation scheme has constant risk function, equal to $(2/3)/(\sqrt{n}+1)^2$, i.e. identical to the maximised minimum Bayes risk. As for the binomial parallel (7.3), this $(\widehat{p}, \widehat{q}, \widehat{r})$ method has a nice minimax property, see Ex. 9.13.

**(d)** Generalise the above, from the trinomial to the general multinomial setting.

**Ex. 7.13** *Gott würfelt nicht.* For the multinomial-Dirichlet setup of Ex. 7.11, we reached the posterior characterisation $p \,|\, \text{data} \sim \text{Dir}(a_1 + y_1, \ldots, a_k + y_k)$. The importance of this lies in the easy usefulness of simulations, where the posterior distribution of any functions of $(p_1, \ldots, p_k)$ may be read off.

**(a)** Explain how you may simulate e.g. $10^5$ vectors $p = (p_1, \ldots, p_k)$ from the posterior distribution, using the characterisation from Ex. 1.20. Concretely, show that one may use $p_1 = G_1/G, \ldots, p_k = G_k/G$, with independent $G_1 \sim \text{Gam}(a_1 + y_1), \ldots, G_k \sim \text{Gam}(a_k + y_k)$, and sum $G = G_1 + \cdots + G_k$.

**(b)** Suppose you throw a certain and perhaps not entirely standard die 40 times and have counts $(4, 6, 6, 7, 8, 9)$ of outcomes 1, 2, 3, 4, 5, 6. Use either of the priors (i) 'flat', $\text{Dir}(1, 1, 1, 1, 1, 1)$; (ii) 'symmetric and more confident', $\text{Dir}(4, 4, 4, 4, 4, 4)$; (iii) 'unwilling to guess', $\text{Dir}(0.1, 0.1, 0.1, 0.1, 0.1, 0.1)$, for the probabilities $(p_1, \ldots, p_6)$, to assess the posterior distribution of each of the following quantities:

$$\alpha = \frac{p_6}{p_1},\ \beta = (1/6)\sum_{j=1}^{6}(p_j - 1/6)^2,\ \gamma = (1/6)\sum_{j=1}^{6}|p_j - 1/6|,\ \delta = \frac{(p_4p_5p_6)^{1/3}}{(p_1p_2p_3)^{1/3}}.$$

For each of $\alpha, \beta, \gamma, \delta$, and for each of the priors, give the 0.05, 0.50, 0.95 quantile points, from $10^5$ simulations from the posterior distributions. This leads to *credibility intervals*, different in spirit and interpretation from the frequentist *confidence intervals*, as per

credibility and confidence

(3.3). Carry out the necessary large-sample approximation, with the delta method for the ML estimator $\widehat{\delta} = (\widehat{p}_4\widehat{p}_5\widehat{p}_6)^{1/3}/(\widehat{p}_1\widehat{p}_2\widehat{p}_3)^{1/3}$, to find both

$$\Pr_{\text{true}}(\delta_{\text{true}} \in \widehat{\delta} \pm 1.645\,\widehat{\text{sd}}/\sqrt{n}) \quad \text{and} \quad \Pr_B(\delta \in [\text{low}, \text{up}] \,|\, \text{data}) = 0.90,$$

with the latter to be evaluated for each of the three priors, and for emphasis writing 'true' for probabilities under a fixed but unknown parameter vector and 'B' for Bayes. Explain these differences in perspectives. Note the relative ease with which credibility intervals may be read off from posterior simulations, without needing separate mathematical efforts like the delta method. You should also plot the posterior densities, for each of the four quantities noting the extent to which the prior influences the results.

**(c)** For the case of $\alpha = p_6/p_1$, exact numerical simulation is possible, without simulation. Do this, and compare with the answers reached via simulation.

**(d)** The above priors are slightly artificial in this context, since they do not allow the explicit possibility that the die in question is plain boring utterly simply a correct one, i.e. that $p = p_0 = (1/6, \ldots, 1/6)$. The priors used hence do not give us the possibility to admit that perhaps $\rho = 1$, $\alpha = 0$, $\beta = 0$, $\gamma = 1$, after all. This motivates using a mixture prior which allows a positive chance for $p = p_0$. Redo therefore the Bayesian analysis above, with the same $(2, 5, 3, 7, 5, 8)$ data, for the prior $\frac{1}{2}\,\delta(p_0) + \frac{1}{2}\,\text{Dir}(1, 1, 1, 1, 1, 1)$. Here $\delta(p_0)$ is the 'degenerate prior' that puts unit point mass at position $p_0$. Compute in particular the posterior probability that $p = p_0$, and display the posterior distributions of $\rho, \alpha, \beta, \gamma$.

**Ex. 7.14** *Beta-binomial and Dirichlet-multinomial: normal approximations.* (xx do this. seeing the natural similarities. xx) Consider again the Beta-binomial setup of Ex. 7.8, where $Y \,|\, \theta \sim \text{binom}(n, p)$ with a $\text{Beta}(a, b)$ prior for $p$. We shall look into certain mirror properties, for the frequentist and the Bayesian inference methods for $p$. Below let $p^* = Y/n$ be the ordinary estimator for $p$.

precursor to the Bernshteĭn–von Mises theorems of Ex. 7.32

**(a)** As a separate lemma, used below, show that if $X_c \sim \text{Beta}(cp_0, c(1-p_0))$, then $c^{1/2}(X_c - p_0) \to_d \text{N}(0, p_0(1-p_0))$. You may use the Scheffé lemma, see Ex. 2.6, or utilise the representation $X_c = U_c/(U_c + V_c)$, where $U_c \sim \text{Gam}(cp_0, 1)$ and $V_c \sim \text{Gam}(c(1-p_0), 1)$ are independent, along with limiting normality of Gamma variables; confirm that this representation follows from details of Ex. 1.20.

**(b)** (xx sketching solution, to be pushed to Solutions. xx) $\sqrt{c}(U_c/c - p_0) \to_d A$, $\sqrt{c}(V_c/c - (1-p_0)) \to_d B$, with variances $p_0$ and $1-p_0$; derivatives of $u/(u+v)$ are $(u+v-u)/(u+v)^2$ and $-u/(u+v)^2$; evaluated at $(p_0, 1-p_0)$ become $1-p_0$ and $-p_0$; giving by the delta method

to be pushed to Solutions, not here

$$\sqrt{c}\Big(\frac{U_c/c}{U_c/c + V_c/c} - \frac{p_0}{p_0 + 1 - p_0}\Big) \to_d (1-p_0)A - p_0 B,$$

which is a zero-mean normal with variance $(1-p_0)^2 p_0 + p_0^2(1-p_0) = p_0(1-p_0)$.

**(c)** When there is a true $p_{\text{true}}$ at work for the data $Y$, show from Ch. 2, or perhaps via likelihood results of Ch. 5, that $U_n = \sqrt{n}(p^* - p_{\text{true}}) \to_d U \sim \text{N}(0, p_{\text{true}}(1-p_{\text{true}}))$. Then work with the normalised posterior distribution for $U_{n,B} = \sqrt{n}(p - \widehat{p}_B)$, and show that it has the very same normal limit distribution.

**(d)** Show that $\widehat{p}_B$ and $p^*$ are close enough to deduce also that $\sqrt{n}(p - p^*) \,|\, \text{data} \to_d U$. This is the Bayesian mirror result to the frequentist one. Explain that the frequentist

confidence intervals and Bayesian crediblity intervals, here of the type

$$p^* \pm 1.645\,\{p^*(1-p^*)\}/\sqrt{n},\ [\mathrm{Be}^{-1}(0.05, a+y, b+n-y), \mathrm{Be}^{-1}(0.95, a+y, b+n-y)],$$

become first-order equivalent, for growing $n$. (xx perhaps one more sentence. Bayes interval uses $y_{\rm obs}$, but here we work out properties when $Y$ is random, from binom$(n, p_{\rm true})$. xx)

**(e)** The close-to-normal lemma for the Dirichlet, generalising the above for the Beta, is as follows. If $(X_{c,1}, \ldots, X_{c,k}) \sim \mathrm{Dir}(cp_{0,1}, \ldots, cp_{0,k})$, then the vector $c^{1/2}(X_{c,1} - p_{0,1}, \ldots, X_{c,k} - p_{0,k})$ tends to a zero-mean multinormal $(U_1, \ldots, U_k)$ with covariances $\mathrm{cov}(U_j, U_\ell) = p_{0,j}(\delta_{j,\ell} - p_{0,\ell})$. Note that this matches the multinomial covariance structure; see Ex. 2.49. To prove this you may use the gamma ratio representation $X_{c,1} = V_{c,1}/V_c, \ldots, X_{c,k} = V_{c,k}/V_c$, in terms of independent $V_{c,j} \sim \mathrm{Gam}(cp_{0,j}, 1)$ with sum $V_c$, and then apply the delta method. Then use this for posterior distribution in Dirichlet-multinomial set up of Ex. 7.11.

### Sampling from the posterior via Markov Chain Monte Carlo

**Ex. 7.15** *MCMC, I: simulating from a given distribution.* Suppose we need to simulat realisations from a given and possibly complicated density $f(x)$ on some domain. The Metropolis algorithm works as follows. The task is to form a long chain $x_1, x_2, \ldots$ in your computer, where its limiting distribution is precisely the given $f$. After having generated $x^{\rm old}$, decide on $x^{\rm new}$ by (i) deciding on a proposal, $x^{\rm prop}$, drawn from a suitable distribution symmetric in $(x^{\rm old}, x^{\rm prop})$, and then (ii) accepting this proposal with probability $p_{\rm accept} = \min(1, f(x^{\rm prop})/f(x^{\rm old}))$. It is to be noted that any factors of proportionality, as with $f(x) = cf_0(x)$ with a perhaps complicated $c$, cancel out and disappear. In algorithmic terms,

$$x_{\rm new} = (1 - {\rm ok})\, x^{\rm old} + {\rm ok}\, x^{\rm new}, \quad \text{where ok} = I(\text{accept}) = I(u \le p_{\rm accept}),$$

MCMC

writing $u$ for a simulated uniform. Markov chain theory of Ch. 13, see Ex. 13.18 and 13.20, secures that this scheme works, in the sense that the chain converges in distribution to that of $\pi$. Here, handling distributions with densities, explain that we have both $\Pr(x_n \in A) \to \int_A f(x)\,{\rm d}x$ for all sets $A$, and $\bar g_n = n^{-1}\sum_{i=1}^n g(x_i) \to_{\rm pr} g_0 = \mathrm{E}_f\, g(X) = \int gf\,{\rm d}x$ for each bounded $g$.

called ergodicity

**(a)** Set up such a scheme to generate outcomes from the standard normal (ignoring the existence of simpler direct algorithms). Start at any $x_0$, then draw proposals $x^{\rm prop} \sim \mathrm{unif}[x^{\rm old} - a, x^{\rm old} + a]$, with acceptance probabilities set up via the general scheme above. Run the chain for a suitably long time and check with a fine histogram that the distribution matches the normal. Keep track of the acceptance probabilities and the overall acceptance rate. The theory works and says that the chain will converge to the standard normal, for any positive fine-turning parameter $a$; explain however in which ways too small or too large values of $a$ will be ineffective.

**(b)** Consider a somewhat harder challenge, simulating realisations from the density $f$ of $0.05\,\mathrm{N}(-2,1) + 0.90\,\mathrm{N}(0,1) + 0.05\,\mathrm{N}(2,1)$ via MCMC. Set up a chain that converges to this $f$ in distribution; check that the output produces a fine histogram matching this $f$.

**(c)** The algorithm works also in higher dimension. Set up a chain that produces outcomes $(x, y, z)$ from the model on the unit cube with density $f(x, y, z) = 1 + \theta(x - \frac12)(y - \frac12)(z - \frac12)$,

with $\theta = 3.45$. From the output, read off the correlations, and the probability that $Z \geq (XY)^{1/2}$.

**(d)** With such MCMC output one typically reads off key quanties, like means, quantiles, probabilities for certain sets, etc. The key here is that sample averages converge, even though the chains being sampled do not produce i.i.d. output. Check with Ex. 3.42 to explain that with such a chain $x_j$, with $x_m \to_d x$ in distribution, we have $m^{-1}\sum_{j=1}^m h(x_j)$ converging to $h_0 = \int h(x)f(x)\,\mathrm{d}x$, for each function $h$, though perhaps with higher sampling variability than if the $x_j$ had been pure i.i.d. Explain also that a histogram, say based on $10^4$ values of $x_j$, is a bona fide estimator for the underlying density $f(x)$. These matters of autocorrelated chain output do not concern us much as long as computers give us long chains.

**Ex. 7.16** *MCMC, II: simulating from the posterior.* The Metropolis algorithm of Ex. 7.15 is generic, and finds extensive use in Bayesian statistics, where the object is to sample from the posterior distribution, given the data at hand.

**(a)** With the posterior density written as $\pi(\theta \,|\, \text{data}) = c\,\pi(\theta)L(\theta)$, in terms of prior times likelihood and a proportionaly constant $c$, show that the Metropolis algorithm can be described as follows. Form a long Markov chain $\theta_1, \theta_2, \ldots$ in your computer, starting anywhere in the parameter region. After having generated $\theta^{\text{old}}$, (i) get hold of a proposal $\theta^{\text{prop}}$, from a proposal distribution symmetric in $(\theta^{\text{old}}, \theta^{\text{prop}})$, and then (ii) accept it with probability $p_{\text{accept}} = \min(1, r)$, with ratio

$$r = \frac{\pi(\theta^{\text{prop}} \,|\, \text{data})}{\pi(\theta^{\text{old}} \,|\, \text{data})} = \frac{\pi(\theta^{\text{prop}})L(\theta_{\text{prop}})}{\pi(\theta^{\text{old}})L(\theta^{\text{old}})} = \frac{\pi(\theta^{\text{prop}})}{\pi(\theta^{\text{old}})} \exp\{\ell(\theta^{\text{prop}}) - \ell(\theta^{\text{old}})\}.$$

Note that the proportionality constant $c$ has disappeared, so for both the prior and the likelihood we merely need to give these up to a proportionality factor.

**(b)** In the Poisson-gamma setup of Ex. 7.1 we could find the posterior distribution directly. Suppose however that the prior is outside the gammas, say uniform on $[4.0, 8.0]$. Set up an MCMC to sample from the posterior, given data 6, 8, 7, 6, 7, 4, 11. Compute also the mean, median, standard deviation.

**(c)** As an illustration, generate first $y_1, \ldots, y_n$ from the uniform distribution on the unit interval. For these data, consider the log-linear density model $f(y, \theta_1, \theta_2) = \exp\{\theta_1(y - \frac{1}{2}) + \theta_2(y - \frac{1}{2})^2 - k(\theta_1, \theta_2)\}$, briefly studied in Ex. 1.58, and use independent zero-mean priors for $\theta_1, \theta_2$, with standard deviations $1.0, 0.5$. Set up an MCMC for the posterior of $(\theta_1, \theta_2)$. Plot say 25 densities drawn from the posteriori model in this way.

### Bayesian analysis for normal models

**Ex. 7.17** *The normal prior and posterior with normal data.* Here we go through the basic steps and results for situations with normal data and normal priors for unknown mean parameters. More elaborate constructions and technical issues are needed when there in addition are unknown parameters in the variance and covariance structure, to be pursued in Ex. 7.19, 7.30.

**(a)** There are things to think through and to learn from, by working through this very simple setup first. (i) For a single observation $Y$ assume it comes from the $\mathrm{N}(\xi, \sigma^2)$, and take $\sigma$ as known; (ii) for the unknown mean $\xi$ assume it comes from the prior $\mathrm{N}(\xi_0, \tau_0^2)$, with specified prior parameters $\xi_0, \tau_0$. Show that this leads to a binormal joint

distribution for parameter and observation,

$$\begin{pmatrix}\xi\\ Y\end{pmatrix} \sim \mathrm{N}_2(\begin{pmatrix}\xi_0\\ \xi_0\end{pmatrix}, \begin{pmatrix}\tau_0^2 & \tau_0^2\\ \tau_0^2 & \tau_0^2+\sigma^2\end{pmatrix}).$$

**(b)** Use general conditioning results from Ex. 1.45 to infer that

$$\xi \,|\, y \sim \mathrm{N}(\xi_0 + w(y-\xi_0), w\sigma^2), \quad \text{with } w = \tau^2/(\tau^2+\sigma^2).$$

So $w$ and $1-w$ are the weights given to the data-based estimate $y$ and the prior guess $\xi_0$, respectively. Also, $w$ is the reduction factor with which the variance of the prior-free estimator $Y$, from $\sigma^2$ to $w\sigma^2$.

**(c)** An easy but important extension is to the case of a full sample $Y_1, \ldots, Y_n$ from the $\mathrm{N}(\xi, \sigma^2)$ distribution, independent given the $\xi$, again with $\sigma$ taken known and the normal prior $\mathrm{N}(\xi_0, \tau_0^2)$ for the unknown mean. Show that $\bar{Y} = (1/n)\sum_{i=1}^n Y_i$ is sufficient (xx xref here xx), and that

$$\begin{pmatrix}\xi\\ \bar{Y}\end{pmatrix} \sim \mathrm{N}_2(\begin{pmatrix}\xi_0\\ \xi_0\end{pmatrix}, \begin{pmatrix}\tau_0^2 & \tau_0^2\\ \tau_0^2 & \tau_0^2+\sigma^2/n\end{pmatrix}).$$

Show from this that

$$\xi \,|\, \text{data} \sim \mathrm{N}(\xi_0 + w_n(\bar{y}-\xi_0), w_n\sigma^2/n), \quad \text{with } w_n = \frac{\tau^2}{\tau^2+\sigma^2/n} = \frac{n\tau^2}{n\tau^2+\sigma^2}.$$

Again, $w_n$ is both the weight given to the data-based estimate and the factor with which the neutral estimator's variance is reduced, from $\sigma^2/n$ to $w_n\sigma^2/n$. Note that $w_n \to 1$; discuss how this may be seen as 'the data wash out the prior'.

**(d)** Discuss the case of a 'flat prior', where $\tau$ is taken large, for $\xi \sim \mathrm{N}(\xi_0, \tau^2)$. Show indeed that this leads to the intuitively plausibe $\xi \,|\, \text{data} \sim \mathrm{N}(\bar{y}, \sigma^2/n)$.

**(e)** In addition to having a coherent updating machine, changing the prior to the posterior, for each new data point, the Bayesian structure implies positive dependence among the observations. From $\mathrm{E}\,(Y_i \,|\, \xi) = \xi$, $\mathrm{Var}\,(Y_i \,|\, \xi) = \sigma^2$, $\mathrm{E}\,(Y_iY_j \,|\, \xi) = \xi^2$, show that

$$\mathrm{E}\,Y_i = \xi_0, \quad \mathrm{Var}\,Y_i = \sigma^2+\tau_0^2, \quad \mathrm{cov}(Y_i, Y_j) = \tau_0^2, \quad \mathrm{corr}(Y_i, Y_j) = \frac{\tau_0^2}{\sigma^2+\tau_0^2},$$

Show also that $\mathrm{Var}\,\bar{Y}_n = \sigma^2/n + \tau_0^2$, and discuss what this means for large $n$.

**(f)** Prove first the convenient formula

$$v(\xi-\xi_0)^2 + n(\xi-\bar{y})^2 = (v+n)(\xi-\xi^*)^2 + d_n(\bar{y}-\xi_0)^2,$$

where

$$\xi^* = \frac{v\xi_0+n\bar{y}}{v+n} \quad \text{and} \quad d_n = \frac{vn}{v+n} = (v^{-1}+n^{-1})^{-1},$$

which also may be written and interpreted via $1/d_n = 1/v + 1/n$.

**(g)** Show that with any prior $p(\xi)$ for $\xi$, the marginal density of $(Y_1, \ldots, Y_n)$ can be written

$$\bar{f}(y_1, \ldots, y_n) = \int (2\pi)^{-n/2}\sigma^{-n} \exp\Big\{-\tfrac{1}{2}\frac{1}{\sigma^2}\sum_{i=1}^n (y_i-\xi)^2\Big\} p(\xi)\,\mathrm{d}\xi.$$

check all of this with care

For the case of $\mathrm{N}(\xi_0, \tau_0^2)$ worked with above, let first $Q_0 = \sum_{i=1}^n (y_i - \bar{y})^2$, and verify that $\sum_{i=1}^n (y_i - \xi)^2 = Q_0 + n(\xi - \bar{y})^2$. Writing for mathematical convenience $1/\tau^2 = v/\sigma^2$, and $1/d_n = 1/n + 1/v$, show that

$$\bar{f}(y_1, \ldots, y_n) = \frac{1}{(2\pi)^{n/2}} \frac{1}{\sigma^n} \exp(-\tfrac{1}{2} Q_0/\sigma^2) \Big( \frac{v}{v+n} \Big)^{1/2} \exp\{-\tfrac{1}{2} d_n (\bar{y} - \xi_0)^2/\sigma^2\}.$$

**(h)** With the marginal density seen as a function of the two fine-tuning parameters $(\xi_0, \tau_0^2)$, find the maximum marginal likelihood estimators $\widehat{\xi}_0$ and $\widehat{\sigma}$.

**(i)** We record two matrix identities here, as they will come in handy both here and on later occasions. For a square and invertible $A$, show that

$$(A + xx^{\mathrm{t}})^{-1} = A^{-1} - cA^{-1} x x^{\mathrm{t}} A^{-1}, \quad \text{with } c = 1/(1 + x^{\mathrm{t}} A^{-1} x);$$

also, that $|A + xx^{\mathrm{t}}| = |A|(1 + x^{\mathrm{t}} A^{-1} x)$.

**(j)** Argue directly that $Y = (Y_1, \ldots, Y_n)^{\mathrm{t}}$ must be multinormal, with $Y \sim \mathrm{N}_n(\xi_0 \mathbf{1}, \sigma^2 I_n + \tau_0^2 \mathbf{1}\mathbf{1}^{\mathrm{t}})$, with $\mathbf{1}$ the vector $(1, \ldots, 1)^{\mathrm{t}}$; the variance matrix has $\sigma^2 + \tau_0^2$ on the diagonal and $\tau_0^2$ outside. Show that this agrees with the marginal density formula reached above.

**Ex. 7.18** *When underestimation doesn't matter so much.* (xx nils rant; we include if turning out well. xx) Suppose $\theta$ is to be estimated, where underestimation is less important than overestimation. One version of this is to use the loss function $L(\theta, a) = L_0(a - \theta)$, with $L_0(u) = u^2$ for $u \geq -c$, but then $L_0(u) = c^2$, for all $u \leq -c$, for a relevant negative threshold $-c$. Thus a big error to the left doesn't matter more than a moderate error to the left.

**(a)** For any random $\theta$, with density $f$, c.d.f. $F$, $E_1(x) = \int_{-\infty}^x \theta f(\theta)\,\mathrm{d}\theta$, $E_2(x) = \int_{-\infty}^x \theta^2 f(\theta)\,\mathrm{d}\theta$, show that

$$\begin{aligned} R(a) = \mathrm{E}\, L_0(a - \theta) &= \int_{-\infty}^{a+c} (a - \theta)^2 f(\theta)\,\mathrm{d}\theta + c^2 \int_{a+c}^{\infty} f(\theta)\,\mathrm{d}\theta \\ &= a^2 F(a+c) - 2aE_1(a+c) + E_2(a+c) + c^2(1 - F(a+c)), \end{aligned}$$

and that its derivative simplifies to

$$R'(a) = 2aF(a+c) - 2E_1(a+c).$$

Check that the solution $a_0$ to setting this to zero is equal to the mean of $\theta$ in the case of $-c$ being far to the left.

**(b)** Explain that the Bayes estimator, for a given dataset, is the solution $\widehat{a}$ to the equation $aF(a + c \,|\, \mathrm{data}) = \mathrm{E}_1(a + c \,|\, \mathrm{data})$, featuring the posterior c.d.f. $\Pr(\theta \leq a + c \,|\, \mathrm{data})$ and $E_1(a + c \,|\, \mathrm{data}) = \int_{-\infty}^{a+c} v\pi(v \,|\, \mathrm{data})\,\mathrm{d}v$.

**(c)** For the case of $y_1, \ldots, y_n$ i.i.d. from the $\mathrm{N}(\xi, 1)$, with a flat prior for $\xi$, show first that $\xi \,|\, \mathrm{data} \sim \mathrm{N}(\bar{y}, 1/n)$, which may be represented as $\xi = \bar{y} + N/\sqrt{n}$, with a standard normal $N$. Show then that the equation for finding the Bayes estimator is

$$(\widehat{a} - \bar{y})\Phi(\sqrt{n}(\widehat{a} - \bar{y} + c)) = -\phi(\sqrt{n}(\widehat{a} - \bar{y} + c))/\sqrt{n}.$$

Explain that this means that the Bayes estimator pushes the ML estimator $\bar{y}$ to the left, with an amount diminishing with sample size $n$.

**(d)** (xx round off. an illustration. xx)

**Ex. 7.19** *The gamma-normal prior and posterior.* Let data $y_1, \ldots, y_n$ for given parameters $\xi$ and $\sigma$ be i.i.d. $\mathrm{N}(\xi, \sigma^2)$. We have seen in Ex. 7.17 that when $\sigma$ may be taken as a known quantity, the canonical class of priors for $\xi$ is the normal one. When both parameters are unknown, however, as in most practical encounters, a more elaborate analysis is called for.

**(a)** Show that the likelihood function may be written as being proportional to

$$L_n(\xi, \sigma) = \exp\Big[-n \log \sigma - \tfrac{1}{2} \frac{1}{\sigma^2} \{Q_0 + n(\xi - \bar{y})^2\}\Big],$$

where $\bar{y} = (1/n) \sum_{i=1}^n y_i$ and $Q_0 = \sum_{i=1}^n (y_i - \bar{y})^2$.

**(b)** With *any* given prior $p(\xi, \sigma)$, explain how you may set up a Metropolis type MCMC to draw samples from the posterior distribution. Try this out in practice, using the prior that takes $\xi$ and $\log \sigma$ independent and uniform on say $[-5, 5]$ and $[-10, 10]$, with data that you simulate for the occasion from a $\mathrm{N}(2.345, 1.234^2)$, with $n = 25$. Note that this approach does not need more mathematical algebra as such, apart from the likelihood function above (along with basic programming skills).

**(c)** There is however a popular and convenient conjugate class of priors for which posterior distributions become particularly clear, with the appropriate algebraic efforts. These in particular involve placing a Gamma prior on the inverse variance $\lambda = 1/\sigma^2$. Say that $(\lambda, \xi)$ has the gamma-normal distribution with parameters $(a, b, \xi_0, v)$, and write this as $(\lambda, \xi) \sim \mathrm{GN}(a, b, \xi_0, v)$, provided $\lambda = 1/\sigma^2 \sim \mathrm{Gam}(a, b)$ and $\xi \,|\, \sigma \sim \mathrm{N}(\xi_0, \sigma^2/v)$. Show that the prior can be expressed as

$$p(\lambda, \xi) \propto \lambda^{a-1} \lambda^{1/2} \exp\big[-\lambda\{b + \tfrac{1}{2} v(\xi - \xi_0)^2\}\big].$$

What is the unconditional prior variance of $\xi$?

**(d)** Using the identity from Ex. 7.17(f), show that if the prior is $(\lambda, \xi) \sim \mathrm{GN}(a, b, \xi_0, v)$, then

$$(\lambda, \xi) \,|\, \text{data} \sim \mathrm{GN}(a + \tfrac{1}{2}n, b + \tfrac{1}{2}Q_0 + \tfrac{1}{2}d_n(\bar{y} - \xi_0)^2, \xi^*, v + n).$$

**(e)** The special case of a 'flat prior' for $\xi$, corresponding to letting $v \to 0$ above, is particularly easy to deal with. Show that then

$$(\lambda, \xi) \,|\, \text{data} \sim \mathrm{GN}(a + \tfrac{1}{2}n, b + \tfrac{1}{2}Q_0, \bar{y}, n).$$

Find the posterior mean of $\sigma^2$ under this prior.

**(f)** For an illustration, consider the cigarette consumption data $x_1, \ldots, x_n$, from 5.B, for $n = 44$ US states (actually, 43 states plus the District of Columbia), from the early 1960ies, taken here to form a sample from $\mathrm{N}(\xi, \sigma^2)$. The $x_i$ is the consumption in state $i$, in hundreds per year, ranging then from 14.0 to 42.4 (which translates to from 3.84 per day to 11.61 per day, in the adult population). Fix the prior that takes (i) $\lambda = 1/\sigma^2 \sim \mathrm{Gam}(a, b)$, with $(a, b)$ taken to correspond to 0.10 and 0.90 prior quantiles for $\sigma$ being 1.0 and 10.0; (ii) given $\sigma$, $\xi$ is a $\mathrm{N}(20.0, \sigma^2/v)$ with $v = 0.50$. Find the precise posterior distribution for $(\lambda, \xi)$. Give posterior means and 90 percent credibility intervals for $\xi$, $\sigma$, and for the probability $p = \Pr(X \geq 40)$.

**(g)** For the same data and setup as in the previous point, carry out a second Bayesian analysis, with the simpler prior that corresponds to $v \to 0$. Comment briefly on the differences in results.

**Ex. 7.20** *The gamma-multinormal prior for linear regression models.* The aim of the present exercise is to generalise the gamma-normal conjugate prior class above to the linear-normal regression model. The model is the very classical one of Ex. 3.34, where

$$y_i = x_{i,1}\beta_1 + \cdots + x_{i,k}\beta_k + \varepsilon_i = x_i^{\mathrm{t}}\beta + \varepsilon_i \quad \text{for } i = 1, \ldots, n,$$

with the $\varepsilon_i$ taken i.i.d. $\mathrm{N}(0, \sigma^2)$. Write $X$ for the $n \times k$ matrix of covariates (explanatory variables), with $x_i = (x_{i,1}, \ldots, x_{i,k})$ as its $i$th row, and use $y$ and $\varepsilon$ to indicate the vectors of $y_i$ and $\varepsilon_i$. Then $y = X\beta + \varepsilon \sim \mathrm{N}_n(X\beta, \sigma^2 I_n)$ is a concise way to write the full model.

**(a)** Show that the likelihood function may be written as being proportional to

$$L_n(\beta, \sigma) = \sigma^{-n} \exp\Big[-\tfrac{1}{2}\frac{1}{\sigma^2}\{Q_0 + n(\beta - \widehat{\beta})^{\mathrm{t}} M_n (\beta - \widehat{\beta})\}\Big],$$

in which

$$\Sigma_n = (1/n)X^{\mathrm{t}}X = n^{-1}\sum_{i=1}^n x_i x_i^{\mathrm{t}} \quad \text{and} \quad \widehat{\beta} = (X^{\mathrm{t}}X)^{-1}X^{\mathrm{t}}y = \Sigma_n^{-1} n^{-1} \sum_{i=1}^n x_i y_i.$$

Also,

$$Q(\beta) = \|y - X\beta\|^2 = Q_0 + n(\beta - \widehat{\beta})^{\mathrm{t}} \Sigma_n (\beta - \widehat{\beta}),$$

with $Q_0 = \sum_{i=1}^n (y_i - x_i^{\mathrm{t}}\widehat{\beta})^2$ the minimum value of $Q$ over all $\beta$. Note that $\widehat{\beta}$ is the classical least squares estimator (and the ML estimator), which in the frequentist framework is unbiased with variance matrix equal to $\sigma^2(X^{\mathrm{t}}X)^{-1} = (\sigma^2/n)\Sigma_n^{-1}$. This is the basis of all classical methods related to the widely popular linear regression model.

**(b)** Let $p(\beta, \sigma)$ be *any* prior for the $(k+1)$-dimensional parameter of the model. Set up formulae for a Metropolis type MCMC algorithm for drawing samples from the posterior distribution of $(\beta, \sigma)$.

**(c)** In spite of the possibility of solving problems via MCMC (or perhaps acceptance-rejection sampling), as with the previous exercise it is very much worthwhile setting up explicit formulae for the case of a certain canonical prior class. Write $(\lambda, \beta) \sim \mathrm{GN}_k(a, b, \beta_0, \Sigma_0)$ to indicate the gamma-normal prior where

$$\lambda = 1/\sigma^2 \sim \mathrm{Gam}(a, b) \quad \text{and} \quad \beta \,|\, \sigma \sim \mathrm{N}_k(\beta_0, \sigma^2 \Sigma_0^{-1}).$$

Show that this prior may be expressed as

$$p(\lambda, \beta) \propto \lambda^{a-1} \lambda^{k/2} \exp\big[-\lambda\{b + \tfrac{1}{2}(\beta - \beta_0)^{\mathrm{t}} \Sigma_0 (\beta - \beta_0)\}\big].$$

**(d)** When multiplying the prior with the likelihood it is convenient to use the following linear algebra identity about quadratic forms, which you should prove first. For symmetric and invertible matrices $A$ and $B$, and for any vectors $a, b, x$ of the appropriate dimension,

$$(x-a)^{\mathrm{t}}A(x-a) + (x-b)^{\mathrm{t}}B(x-b) = (x-\xi)^{\mathrm{t}}(A+B)(x-\xi) + (b-a)^{\mathrm{t}}D(b-a),$$

where $\xi = (A+B)^{-1}(Aa + Bb)$ (a weighted average of $a$ and $b$) and $D$ is a matrix for which several equivalent formulae may be used:

$$\begin{aligned} D &= A(A+B)^{-1}B = B(A+B)^{-1}A \\ &= A - A(A+B)^{-1}A = B - B(A+B)^{-1}B = (A^{-1} + B^{-1})^{-1}. \end{aligned}$$

**(e)** Prove that if $(\lambda, \beta)$ has the $\mathrm{GN}_k(a, b, \beta_0, \Sigma_0)$ prior, then

$$(\lambda, \beta) \,|\, \text{data} \sim \mathrm{GN}_k(a + \tfrac{1}{2}n, b + \tfrac{1}{2}Q_0 + \tfrac{1}{2}(\widehat{\beta} - \beta_0)^{\mathrm{t}} D_n (\widehat{\beta} - \beta_0), \beta^*, \Sigma_0 + n\Sigma_n),$$

where

$$\beta^* = (\Sigma_0 + n\Sigma_n)^{-1}(\Sigma_0\beta_0 + n\Sigma_n\widehat{\beta}) \quad \text{and} \quad D_n = \Sigma_0(\Sigma_0 + n\Sigma_n)^{-1}n\Sigma_n.$$

This characterisation makes it easy to simulate a large number of $(\beta, \sigma)$ from the posterior distribution and hence to carry out Bayesian inference for any parameter of quantity of interest.

**(f)** Note the algebraic simplifications that result when the $\Sigma_0$ in the prior is chosen as being proportional to the covariate sample variance matrix, i.e. $\Sigma_0 = c_0\Sigma_n$. Show that then

$$\beta^* = (c_0\beta_0 + n\widehat{\beta})/(c_0 + n) \quad \text{and} \quad D_n = c_0 n/(c_0 + n).$$

In this connection $c_0$ has a natural interpretation as 'prior sample size'.

**(g)** A special case of the above, leading to simpler results, is that where $\beta$ has a flat, non-informative prior, corresponding to very large prior variances, i.e. to $\Sigma_0 \to 0$. Show that with such a prior,

$$(\lambda, \beta) \,|\, \text{data} \sim \mathrm{GN}_k(a + \tfrac{1}{2}n, b + \tfrac{1}{2}Q_0, \widehat{\beta}, n\Sigma_n).$$

The prior is improper (infinite integral), but the posterior is proper as long as $\widehat{\beta}$ exists, which requires $X^{\mathrm{t}}X$ to have full rank, which again means at least $k$ linearly independent covariate vectors, and, in particular, $n \geq k$.

**(h)** (xx repair this. do bladder cancer rates, since linear is ok, whereas lung cancer is more quadratic, done in Story #16. xx) Go again to the dataset 5.B, for illustration and for flexing your operational muscles. For $y$ use the bladder cancer column of deaths per 100,000 inhabitants and for $x$ use the number of cigarettes sold per capita. Your task is to carry out Bayesian analysis within the linear regression model $y_i = \beta_0 + \beta_1(x_i - \bar{x}) + \varepsilon_i$ for $i = 1, \ldots, 44$, with $\varepsilon_i$ taken i.i.d. $\mathrm{N}(0, \sigma^2)$. Specifically, we wish point estimates along with 95 percent credibility intervals for (i) each of the three parameters $\beta_0, \beta_1, \sigma$; (ii) the probability that $y \geq 25.0$, for a state with cigarette consumption $x = 35.0$; (iii) the bladder cancer death rates $y_{45}$ and $y_{46}$, per 100,000 inhabitants, for states with cigarette consumption rates $x_{45} = 10.0$ (low) and $x_{46} = 50.0$ (high). You are to carry out such inference with two priors (xx nils repairs this xx): (1) First, the informative one which takes $1/\sigma^2$ a gamma with 0.10 and 0.90 quantiles for $\sigma$ equal to 1.0 and 5.0, and $\beta_0$ and $\beta_1$ as independent normals $(5.0, 0.5\,\sigma^2)$ and $(0.0, (2.0\,\sigma)^2)$, given $\sigma$. (2) Then, the simpler and partly non-informative one that takes a flat prior for $(\beta_0, \beta_1)$ and the less informative one for $\sigma$ that uses 0.10 and 0.90 prior quantiles 0.5 and 10.0. Finally, compare your results from those arrived at using classical frequentist methods.

**Ex. 7.21** *Mixture priors.* Suppose data $Y$ come from a model $f(y, \theta)$, where different priors $\pi_1(\theta), \ldots, \pi_k(\theta)$ can be used, each leading to posterior distributions $\pi_1(\theta \,|\, y), \ldots, \pi_k(\theta \,|\, y)$.

**(a)** For each of these possible priors (and hence possible posteriors), show that there is a representation $f_j(y, \theta) = \pi_j(\theta) f(y, \theta) = \pi_j(\theta \,|\, y)\bar{f}_j(y)$, where $\bar{f}_j(y) = \int f(y, \theta)\pi_j(\theta)\,\mathrm{d}\theta$ is the marginal density of $Y$, associated with the $\pi_j(\theta)$ prior.

**(b)** Suppose now that a full mixture prior is assigned to $\theta$, of the type $\pi(\theta) = p_1\pi_1(\theta) + \cdots + p_k\pi_k(\theta)$, with probabilities $p_1, \ldots, p_k$ summing to 1. Explain that this can be interpreted as $\theta$ being drawn from prior $j$ with probability $p_j$. Show also that the marginal distribution of $Y$ can be expressed as $\bar f(y) = \sum_{j=1}^k p_j \bar f_j(y)$.

**(c)** Then show that the posterior distribution for $\theta$ becomes

$$\pi(\theta \,|\, y) = p_1^*\pi_1(\theta \,|\, y) + \cdots + p_k^*\pi_k(\theta \,|\, y),$$

with revised prior probabilities $p_j^* = p_j \bar f_j(y) / \sum_{j'=1}^k p_{j'} \bar f_{j'}(y)$ for the different types of priors.

**(d)** Suppose $Y \sim \text{binom}(n, \theta)$, and that the prior used for $\theta$ is $0.15\,\text{Beta}(2, 10) + 0.70\,\text{Beta}(15, 15) + 0.15\,\text{Beta}(10, 2)$. Draw this prior in a plot. With $n = 100$, compute the posterior probabilities $p_1^*, p_2^*, p_3^*$, and draw the posterior distribution for $\theta$, along with the prior, for each of the cases $y = 12$, $y = 48$, $y = 91$.

**(e)** (xx revisit the Gott würfelt nicht, Ex. 7.13. do a mixture prior, perhaps $0.50\,\text{Dir}(1, 1, 1, 1, 1, 1) + 0.50\,\text{Dir}(s, s, s, s, s, s)$, where $s$ is quite big, reflecting the possibility that the die is perfectly fair with probabilities equal to or very close to $(1/6, 1/6, 1/6, 1/6, 1/6, 1/6)$. xx)

**(f)** Generalise the above to the situation where $\pi(\theta) = \int \pi_\alpha(\theta)\,\mathrm{d}G(\alpha)$ is a mixture of $\pi_\alpha(\theta)$ priors, with $\mathrm{d}G(\alpha)$ a fully general probability measure over the space of hyperparameter $\alpha$. The $\alpha$ coud be a parameter belonging to a finite set, matching the setup above, or a full continuous mixture. Show that the posterior can be represented as $\pi(\theta \,|\, y) = \int \pi_\alpha(\theta \,|\, y)\,\mathrm{d}G(\alpha \,|\, \text{data})$, where $\mathrm{d}G(\alpha \,|\, \text{data})$ is the posterior for the hyperparameter, and $\pi_\alpha(\theta \,|\, y)$ is the posterior for $\theta$ in the setup where $\alpha$ is fixed and known.

### The Jeffreys and vip-Jeffrey priors

**Ex. 7.22** *The Jeffreys prior: the basics.* Consider a smooth parametric model $f(y, \theta)$, with score function $u(y, \theta) = \partial \log f(y, \theta)/\partial\theta$ and finite Fisher information matrix $J(\theta) = \text{Var}_\theta\, u(Y, \theta)$; see Ex. 5.6. The *Jeffreys prior* is defined as $\pi_{\text{jeff}}(\theta) \propto |J(\theta)|^{1/2}$. If this has a finite integral, over the parameter domain, the proportionality constant is fixed to make the integral 1, but we explicitly allow also improper priors here, i.e. positive functions with infinite integral. It will be seen that in most cases, the passing from $\pi_{\text{jeff}}(\theta)$ to $\pi_{\text{jeff}}(\theta \,|\, \text{data})$ via the Bayes theorem (7.1) gives a proper posterior.

**(a)** Consider a smooth 1-1 reparametrisation, from $\theta = (\theta_1, \ldots, \theta_p)$ to $\phi = (\phi_1, \ldots, \phi_p)$, with $\theta = h(\phi)$ and $\phi = h^{-1}(\theta)$. Use Ex. 1.12 to show that if $\theta$ has the Jeffreys prior, then $\phi$ has density $\pi^*(\phi) = \pi_{\text{jeff}}(h(\phi))|G(\phi)|$ with the gradient matrix $G(\phi) = \partial h(\phi)/\partial\phi$.

**(b)** We have seen in Ex. 5.9 that the Fisher information matrix on the $\phi$ scale is $J^*(\phi) = G(\phi)J(h(\phi))G(\phi)^{\text{t}}$. Deduce that the Jeffreys prior constructed on the $\phi$ scale becomes $\pi^*_{\text{jeff}}(\phi) = \pi_{\text{jeff}}(h(\phi))|G(\phi)|$. But this is the same as $\pi^*(\phi)$ above. Explain that this demonstrates that the Jeffreys prior is invariant under smooth parameter transformations. Three Bayesian statisticians, with the same data and the same model, will reach the same conclusions, even if they use three different parametrisations, as long as they use the Jeffreys prior construction in their respective model formulations.

invariance w.r.t. parametrisation

**Ex. 7.23** *The Jeffreys prior for normal models.* We go through the Fisher information and Jeffreys calculus for normal models.

**(a)** For the normal $\text{N}(\xi, \sigma^2)$ case, with $\sigma$ known, show that the Jeffreys prior for $\xi$ is flat on the full lne. With data $y_1, \ldots, y_n$, show that the Bayes rule gives the nice and

interpretable posterior $\xi \,|\, \text{data} \sim \mathrm{N}(\bar{y}, \sigma^2/n)$; in particular, even for $n = 1$ the posterior is proper though the prior isn't. Generalise this to the linear regression model, wit $y_i \sim \mathrm{N}(x_i^{\mathrm{t}}\beta, \sigma^2)$, with $x_i$ of dimension $p$, and again with $\sigma$ known. Show that Jeffreys says that the $\beta$ prior is flat on the full $\mathbb{R}^p$. Find the posterior for $\beta$, and explain that this is proper as long as there are at least $p$ linarly independent $x_i$ in the data; in particular, one needs $n \geq p$.

**(b)** Matters are more delicate for the normal models when $\sigma$ is unknown. For the single $\mathrm{N}(\xi, \sigma^2)$ model, show that $\pi_{\text{jeff}}(\xi, \sigma) \propto 1/\sigma^2$, i.e. flat for $\xi$ but $1/\sigma^2$ for $\sigma$. Explain that the posterior needs at least two datapoints in order to become proper. For the linear regression setup, show that $\pi_{\text{jeff}}(\beta, \sigma) \propto 1/\sigma^{p+1}$. Connect this to the gamma-normal setup of Ex. 7.20, and identify the proper postserior $\pi(\beta, \sigma \,|\, \text{data})$.

**Ex. 7.24** *The Jeffreys prior in certain models.* Also in various other models we may derive clear formula for the Fisher information matrix, and hence also for the Jeffreys prior $|J(\theta)|^{1/2}$.

**(a)** For the binomial $(n, p)$ model, show that the Jeffreys prior is the Beta$(\frac{1}{2}, \frac{1}{2})$. Compare this with the geometric, with $f(y, p) = (1-p)^{y-1}p$, for which you should find $\pi_{\text{jeff}}(p) \propto p^{-1}(1-p)^{-1/2}$, which can be thought of as a Beta$(\varepsilon, \frac{1}{2})$ with a very small $\varepsilon$. Show that this prior is improper, but that the posterior becomes proper as long as there are $n \geq 1$ observations.

**(b)** Then consider the trinomial $(X, Y, Z)$, with probabilities proportional to $p^x q^y (1-p-q)^z$, with $x+y+z = n$. Find the $2 \times 2$ Fisher information matrix and its determinant, and show that the Jeffreys prior is proportional to $p^{-1/2}q^{-1/2}(1-p-q)^{-1/2}$, which is the Dirichlet $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ for $(p, q, r)$. Generalise to the multinomial $(n, p_1, \ldots, p_k)$ case, with $p_1 + \cdots + p_k = 1$.

**(c)** For the Poisson with parameter $\theta$, show that the Jeffreys prior is $1/\theta^{1/2}$.

**(d)** For the Gamma $(a, b)$, show that the Jeffreys prior takes the form $\pi(a, b) = \pi_1(a)\pi_2(b)$, with $\pi_1(a) \propto \{a\psi'(a) - 1\}^{1/2}$ and $\pi_2(b) \propto 1/b$.

**(e)** Then work through the details for the Weibull model, with $F(t, a, b) = 1 - \exp\{-(t/a)^b\}$, see Ex. 1.60. You should find the Fisher information matrix

$$J(a, b) = \begin{pmatrix} b^2/a^2 & -(1-\gamma_e)/a \\ -(1-\gamma_e)/a & d/b^2 \end{pmatrix},$$

with $\gamma_e \doteq 0.5772$ the Euler constant and $d = 3 - \gamma_e^2 - \pi^2/6 \doteq 1.0219$. Deduce that $\pi_{\text{jeff}}(a, b) \propto 1/a$, i.e. flat in $b$.

**Ex. 7.25** *A simple model for deviations from uniformity.* This exercise illustrates how we can carry out Bayesian analysis for almost any given one-parameter model, via simple numerical techniques; bigger models need bigger tools, e.g. the MCMC ones of Ex. 7.16, if needed coupled with numerical evaluation of the Fisher matrix. Consider then the model $f(y, \theta) = 1 + \theta(y - \frac{1}{2})$ for $y \in [0, 1]$.

**(a)** Show that this indeed defines a bona fide model, for $\theta \in [-2, 2]$, and with c.d.f. $F(y, \theta) = y + \frac{1}{2}\theta(y^2 - y)$. Show that the Fisher information is

$$J(\theta) = \int_{-1/2}^{1/2} \frac{x^2}{1+\theta x}\,\mathrm{d}x = [\log \frac{1+\frac{1}{2}\theta}{1-\frac{1}{2}\theta} - \theta]/\theta^3.$$

(xx check this numerically. xx) Compute and display the Jeffreys prior.

**(b)** Take $\theta_{\rm true} = 0.333$, simulate say $n = 100$ points from the model, and give a graph for the log-likelihood function. Compute the ML and an approximate 90 percent interval for $\theta$ via the methods of Chapter 5. Then, with a uniform prior on $[-2, 2]$, compute and display the posterior distribution for $\theta$.

**(c)** Using a fine grid, e.g. with grid length 0.0001, sample say $10^5$ points from the posterior distribution. From these, provide $0.05, 0.50, 0.95$ quantiles. (xx just a bit more; round off; point away. xx)

**Ex. 7.26** *Bayes with Jeffreys in more challenging waters.* Carrying out Bayes analysis with the Jeffreys prior becomes harder when there is no simple formula for the latter. Consider a setup with given densities $f_0$ and $f_1$, but unknown mixture proportion $p$ for data coming from $f = (1-p)f_0 + pf_1$.

**(a)** Show that the Fisher information becomes

$$J(p) = \int \frac{\{f_1(x) - f_0(x)\}^2}{(1-p)f_0(x) + pf_1(x)}\,{\rm d}x.$$

For a few choices of $f_0$ and $f_1$, compute and plot $\pi_{\rm jeff}(p)$ numerically. Comment in particular on the case where $f_0$ and $f_1$ are normal with the same variance.

**(b)** Let for concreteness $f_0$ and $f_1$ be $\mathrm{N}(\xi_0, 1)$ and $\mathrm{N}(\xi_1, 1)$, with $(\xi_0, \xi_1) = (0, 1)$, though your code should work for any other pair of densities. Generate $n = 25$ points from the mixture density, using say $p_{\rm true} = 0.33$. Your task is to simulate realisations from $\pi(p \,|\, {\rm data})$. For this use MCMC, see Ex. 7.16, which for the acceptance probabiity computation requires numerical evaluation of $\pi_{\rm jeff}(p^{\rm prop})/\pi_{\rm jeff}(p^{\rm old})$. The point is that this is entirely possible, even without having an easier formula for the Fisher information.

**(c)** For the one-dimensional case, as above, it suffices to being able to evaluate one-dimensional integrals. This is more challenging in the vector case. Explain how you can still get hold of numerical approximations to $\pi_{\rm jeff}(a, b)$, say, in a two-parameter model, by simulating a number of score vectors at each position $(a, b)$ in the evaluation of the acceptance probabiity step for the MCMC. Try such a scheme for the case of data sampled from $f = (1-p)f_0 + pf_1$, this time with $p = 0.33$ known and fixed, but with unknown means for $\mathrm{N}(\xi_0, 1)$ and $\mathrm{N}(\xi_1, 1)$.

**Ex. 7.27** *The vip-Jeffreys combo prior.* Bayesian analysis for a model requires full specification of a prior for its full parameter vector $\theta$, of dimension say $p$. In many situations there is prior information available regarding a sub-parameter, with a separaty elicited prior $\pi_{\rm vip}(\mu)$ for that perhaps very important parameter, but otherwise not for other aspects of the full model parameter. Parametrising $\theta$ into $(\mu, \gamma)$, with ensuing Fisher iformation matrix we propose using

the vip-Jeffreys combo

$$\pi_{\rm combo}(\mu, \gamma) = \pi_{\rm vip}(\mu)\pi_{\rm jeff}(\gamma \,|\, \mu) = \pi_{\rm vip}(\mu)|J(\gamma \,|\, \mu)|^{1/2}.$$

With the full $p \times p$ Fisher information matrix $J(\mu, \gamma)$, the $J(\gamma \,|\, \mu)$ is the $(p-1) \times (p-1)$ lower right corner matrix, which is also the Fisher matrix for the model $f(y, \mu, \gamma)$ when $\mu$ is held fixed.

**(a)** Two normals means, information on the difference. $\pi_{\rm vip}(\delta) \cdot 1$.

**(b)** Two exponentials, with information about the ratio.

**(c)** Gamma $(a, b)$, with vip prior for the mean $\xi = a/b$: parametrise as $(c\xi, c)$. check with Ex. 5.17 that the Jeffreys priro for $c$, for given $\xi$, is

$${\rm jeff}(c \,|\, \xi) \propto \{\xi^2 \psi'(c\xi) - \xi/c\}^{1/2}.$$

(xx ask to plot. xx)

**(d)** Normal, with information on $\xi$, or on $\sigma$.

**(e)** Fieller problem: goes almost away, with a vip prior om $\rho = a/b$.

**(f)** (xx tighten. xx) Binormal, with information on $\rho$. First the Jeffreys in the 5-para model: the square-root of

$$|J(\theta)| = \frac{8}{(1-\rho^2)^3\sigma_1^4\sigma_2^4}.$$

First do the calculus and algebra to show that $J(\xi_1, \xi_2, \sigma_1, \sigma_2 \,|\, \rho)$ matrix has blocks

$$\frac{1}{1-\rho^2}\begin{pmatrix} 1/\sigma_1^2 & -\rho/(\sigma_1\sigma_2) \\ -\rho/(\sigma_1\sigma_2) & 1/\sigma_2^2 \end{pmatrix} \quad \text{and} \quad 2\begin{pmatrix} 1/\sigma_1^2 & \rho/(\sigma_1\sigma_2) \\ \rho/(\sigma_1\sigma_2) & 1/\sigma_2^2 \end{pmatrix},$$

with zero correlations between these. Deduce from this that the vip-Jeffreys combo prior for the five parameters becomes

$$\pi_{\text{combo}} = \pi_{\text{vip}}(\rho)\frac{(1+\rho^2)^{1/2}}{1-\rho^2}\frac{1}{\sigma_1^2\sigma_2^2}.$$

Hmm ... so when is

$$\pi_{\text{vip}}(\rho)\frac{(1+\rho^2)^{1/2}}{1-\rho^2} = 1/(1-\rho^2)^{3/2}\,?$$

**(g)** Linear regression $\mathrm{N}(a+bx, \sigma^2)$, with information on $b$. Show first that Jeffreys is $1/\sigma^3$. Then that the vip-Jeffreys is $\pi_{\text{vip}}(b)/\sigma^2$.

**Ex. 7.28** *Parameter invariance for the vip-Jeffreys prior.* (xx we do this properly. xx)

### Marginal distributions and Bernshteĭn–von Mises theorems

**Ex. 7.29** *The marginal distribution.* Suppose we have data $y_1, \ldots, y_n$ from a model $f(y, \theta)$, with a prior $\pi(\theta)$. Most of the time Bayesians care about the posterior distribution, but on occasion, also in connection with bigger setups, one needs the marginal distribution, which is $\bar{f}(y_1, \ldots, y_n) = \int L_n(\theta)\pi(\theta)\,\mathrm{d}\theta$, in terms of the likelihood function $L_n(\theta)$. In various setups there will be a clear formula for this $\bar{f}$, as seen below; see also Ex. 7.31 for a very useful approximation method for more complex cases.

**(a)** Let $y_1, \ldots, y_n$ be independent Bernoulli variables with $\Pr(y_i = 1 \,|\, \theta) = \theta$, and let $\theta \sim \mathrm{Beta}(a, b)$. Writing $z = \sum_{i=1}^n y_i$ for the number of 1s, show that

$$\bar{f}(y_1, \ldots, y_n) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\frac{\Gamma(a+z)\Gamma(b+n-z)}{\Gamma(a+b+n)}.$$

**(b)** Let then $y_1, \ldots, y_n$ be independent $\mathrm{Pois}(\theta)$, with a $\mathrm{Gam}(a, b)$ prior, as with Ex. 7.1. Show that

$$\bar{f}(y_1, \ldots, y_n) = \frac{b^a}{\Gamma(a)}\frac{\Gamma(a+n\bar{y})}{(b+n)^{a+n\bar{y}}}\frac{1}{y_1!\cdots y_n!}.$$

**(c)** Consider i.i.d. data $y_i \sim \mathrm{N}(\xi, \sigma^2)$, with known $\sigma$ and a normal prior $\xi \sim \mathrm{N}(\xi_0, \sigma_0^2)$ for $\xi$. Find the marginal distribution (xx give a formula here xx).

**(d)** (xx do also $\mathrm{N}(x_i^{\mathrm{t}}\beta, \sigma^2)$ with $\beta \sim \mathrm{N}(\beta_0, \Sigma_0)$. find the marginal. check with other exercises. xx)

**(e)** (xx then also for the gamma-normal; give a formula for $\bar{f}(y_1,\ldots,y_n)$. xx)

**(f)** (xx something to point to empirical Bayes. calibrate carefully with loss-risk Ch. 9. can already point to Stein things. and to mixtures, where these $\bar{f}(y)$ turn up as ingredients. xx)

**Ex. 7.30** *The gamma-normal induced marginal model.* (xx edit intro sentences. $Y_1,\ldots,Y_n$ are i.i.d. from the $\mathrm{N}(\xi,\sigma^2)$, given these two parameters. xx) For the direct Bayesian use one only needs the prior to posterior computation, in this case from the initial $\mathrm{GN}(a,b,\xi_0,v)$ to the updated GN given in Ex. 7.19, and one somehow bypasses the marginal density $\bar{f}(y_1,\ldots,y_n)$ of the data, the likelihood with the parameters $(\xi,\sigma)$ integrated out according to the prior. On occasion this marginal distribution is important, however, and also finds use as a model in its own right, for positively dependent data.

**(a)** (xx first things with $\xi\sim\mathrm{N}(\xi_0,\tau_0^2)$, known $\sigma$. two ways of computing and seeing $\bar{f}(y_1,\ldots,y_n)$. do marginal moments and correlations. xx)

**(b)** xx

**(c)** Then the $\mathrm{GN}(a,b,\xi_0,v)$ gamma-normal prior for $(\lambda,\xi)$, as with Ex. 7.19. Show first that the likelihood times the prior, $L_n(\lambda,\xi)p(\lambda,\xi)$, can be expressed as

$$\frac{\lambda^{n/2}}{(2\pi)^{n/2}}\exp[-\tfrac12\lambda\{Q_0+n(\xi-\bar y)^2\}]\frac{b^a}{\Gamma(a)}\lambda^{a-1}(v\lambda)^{1/2}\exp[-\lambda\{b+\tfrac12 v(\xi-\xi_0)^2\}].$$

Then integrate out the $\xi$ to get

$$\frac{1}{(2\pi)^{n/2}}\frac{b^a}{\Gamma(a)}\lambda^{a+n/2-1}\Big(\frac{v}{v+n}\Big)^{1/2}\exp\{-\lambda(b+\tfrac12 Q_0+\tfrac12 d_n(\bar y-\xi_0)^2\},$$

with $d_n=(1/v+1/n)^{-1}$ as per Ex. 7.19. Show then that this leads to the marginal density being

$$\bar f(y_1,\ldots,y_n)=(2\pi)^{-n/2}\Big(\frac{v}{v+n}\Big)^{1/2}\frac{b^a}{\Gamma(a)}\frac{\Gamma(a+n/2)}{\{b+\frac12 Q_0+\frac12 d_n(\bar y-\xi_0)^2\}^{a+n/2}}.$$

**Ex. 7.31** *Approximating the marginal distribution.* In the setup of Ex. 7.29, we go through a useful type of Laplace approximation for the marginal.

**(a)** Writing as usual $\ell_n(\theta)$ for the log-likelihood, with maximum value $\ell_{n,\max}=\ell_n(\widehat\theta)$, in terms of the ML estimator, show that

$$\bar f(y_1,\ldots,y_n)=\exp(\ell_{n,\max})\int\exp\{\ell_n(\theta)-\ell_n(\widehat\theta)\}\pi(\theta)\,\mathrm{d}\theta.$$

With $J_n=-(1/n)\partial^2\ell_n(\widehat\theta)/\partial\theta\partial\theta^{\mathrm t}$ the normalised observed information matrix, of dimension say $p\times p$, show that the marginal can be approximated with

$$\begin{aligned}\bar f&\doteq\exp(\ell_{n,\max})\int\exp\{-\tfrac12 n(\theta-\widehat\theta)^{\mathrm t}J_n(\theta-\widehat\theta)\}\pi(\theta)\,\mathrm{d}\theta\\&=\exp(\ell_{n,\max})\int\exp(-\tfrac12 s^{\mathrm t}J_n s)\pi(\widehat\theta+s/\sqrt n)\,\mathrm{d}s/n^{p/2}\\&\doteq\exp(\ell_{n,\max})\pi(\widehat\theta)(2\pi)^{p/2}|J_n|^{-1/2}/n^{p/2}.\end{aligned}$$

**(b)** (xx a couple of things here. check how successful the approximation is in two setups. the formula

$$\log\bar f\doteq\ell_{n,\max}-\tfrac12 p\log n+\log\pi(\widehat\theta)-\tfrac12\log|J_n|+\tfrac12 p\log(2\pi),$$

with its two leading terms, lead to the BIC in Ch. 12. xx)

**Ex. 7.32** *Bernshteĭn–von Mises approximations.* Suppose observations $Y_1, \ldots, Y_n$ are i.i.d. from a density $f(y, \theta)$, with $\pi(\theta)$ a prior for the model parameter, of dimension say $p$. The posterior density can of course be quite complicated, perhaps necessitating numerical efforts, or simulation, for its evaluation. Remarkably, there are generic and simple normal approximations, however. For an informative illustration and precursor for the general results below, check Ex. 7.14.

**(a)** Show that the posterior density $\pi_n(\theta) = \pi(\theta \,|\, \text{data})$ is proportional to $\pi(\theta) \exp\{\ell_n(\theta)\}$, with $\ell_n(\theta) = \sum_{i=1}^n \log f(y_i, \theta)$ the log-likelihood function.

**(b)** Let $\widehat{\theta}$ be the ML estimator, and $J_n = -(1/n)\partial^2 \ell_n(\widehat{\theta})/\partial\theta\partial\theta^{\mathrm{t}}$ the normalised observed information, as per likelihood theory of Ch. 5. Show that the density of $Z_n = \sqrt{n}(\theta - \widehat{\theta})$ is $g_n(z) = \pi_n(\widehat{\theta} + z/\sqrt{n})(1/n^{p/2})$, and that it can be approximated as

$$g_n(z) \propto \pi(\widehat{\theta} + z/\sqrt{n}) \exp\{\ell_n(\widehat{\theta} + z/\sqrt{n}) - \ell_n(\widehat{\theta})\} \doteq \pi(\widehat{\theta} + z/\sqrt{n}) \exp(-\tfrac{1}{2} z^{\mathrm{t}} J_n z).$$

**(c)** Suppose then that the data really are i.i.d. from the model, with an underlying $\theta_{\text{true}}$. In particular, then $\widehat{\theta} \to_{\text{pr}} \theta_{\text{true}}$ and $J_n \to_{\text{pr}} J = J(\theta_{\text{true}})$, by the general likelihood theory of Ch. 5. If $\pi(\theta)$ is continuous in a neighbourhood around $\theta_{\text{true}}$, show that $g_n(z)$ tends to the density of a $\mathrm{N}_p(0, J^{-1})$, in probability. In concrete terms, show

$$D_n = \int |g_n(z) - \phi_p(z, 0, J^{-1})| \, \mathrm{d}z \to_{\text{pr}} 0.$$

This is one of several versions of *Bernshteĭn–von Mises (BvM theorems.* These are Bayesian mirror versions of the classical maximum likelihood asymptotics results in the frequentist camp:

sometimes called the Bayesian Central Limit Theorem

$$\begin{aligned} \sqrt{n}(\widehat{\theta} - \theta_{\text{true}}) &\to_d \mathrm{N}(0, J^{-1}), \\ \sqrt{n}(\theta - \widehat{\theta}) \,|\, \text{data} &\to_d \mathrm{N}(0, J^{-1}), \text{ in probability.} \end{aligned}$$

**(d)** Go back to the Beta-binomial and Dirichlet-multinomial setups of Ex. 7.14 and verify that results reached there are special cases of the BvM theorem above. For another clear illustration, consider the case where $Y_1, \ldots, Y_n \,|\, \theta$ are i.i.d. Pois$(\theta)$ and $\theta \sim \mathrm{Gam}(a_0, b_0)$. Check what the BvM theorem says in this situation. Choose also $n$ and $(a_0, b_0)$ for some brief numerical investigations.

**(e)** Explain and discuss that we are learning here, quite generally, (i) that the effect of the prior is being washed out as the data volume increases, and (ii) that the posterior distributions, even for different start priors, are close to $\mathrm{N}_p(\widehat{\theta}, \widehat{J}_{\text{full}}^{-1})$, with $\widehat{J}_{\text{full}} = -\partial^2 \ell_{n,\widehat{\theta}}/\partial\theta\,\partial\theta^{\mathrm{t}}$ the observed Fisher information matrix. Argue that this invites the 'Lazy Bayesian' pragmatic view, that one may bypass the prior elicitation and jump directly to the $\theta \,|\, \text{data} \approx_d \mathrm{N}_p(\widehat{\theta}, \widehat{J}_{\text{full}}^{-1})$, at least in cases with moderate or large datasets compared to the parameter dimension.

the Lazy Bayesian

### Multiparameter and hierarchical setups

**Ex. 7.33** *Multiparameter inference* Consider independent $Y_i \sim \mathrm{N}(\theta_i, 1)$ for $i = 1, \ldots, k$, where the task is to estimate the ensemble of means, with loss function $L = \sum_{i=1}^k (a_i - \theta_i)^2$.

**(a)** The standard estimator is $Y_i$, simply; show e.g. that this is the ML estimator. Show that the risk function $R(\theta, Y)$ is constant, equal to $k$. Show also that with this loss

function, the Bayes solution, after having started with some prior for $(\theta_1,\ldots,\theta_k)$, are the posterior means, $\mathrm{E}\,(\theta_i\,|\,\mathrm{data})$.

**(b)** Then let $\theta_i$ have the prior $\mathrm{N}(0,\tau^2)$, independently of the others, for $i=1,\ldots,k$. Show as with 7.17 that $\theta_i\,|\,\mathrm{data}\sim\mathrm{N}(\rho y_i,\rho^2)$, with $\rho=\tau^2/(\tau^2+1)$. For the Bayes estimator $\widehat{\theta}_i(\tau)=\rho Y_i$, show that the risk function is $\sum_{i=1}^k\{\rho^2+(1-\rho)^2\theta_i^2\}$ and that the Bayes risk is $k\rho$.

**(c)** The $\rho Y_i$ strategy requires of the Bayesian that $\tau$ is specified. When this is a too tall order, a natural method is to use the data to estimate $\rho$, leading to the *empirical Bayes strategy* $\theta_i^*=\widehat{\theta}_i(\widehat{\rho})=\widehat{\rho}Y_i$. With the $\theta_i\sim\mathrm{N}(0,\tau^2)$ prior, explain that $Y_i$, marginally, is a $\mathrm{N}(0,\tau^2+1)$, and that $V=\sum_{i=1}^k Y_i^2\sim(\tau^2+1)\chi^2_k$. Since $V/k$ estimates $\tau^2+1$, argue that $\widehat{\rho}=\max(1-k/V,0)$, and relatives, are sensible estimators of $\rho$. Technically, this is not a bona fide Bayes method, as $\widehat{\rho}Y_i$ is not the posterior mean stemming from a prior on $(\theta_1,\ldots,\theta_k)$, but its frequentist performance may be excellent. In particular, versions of this empirical Bayes scheme will succeed in having risk functions below the standard risk level $k$, in all of the parameter space (provided $k\geq 3$, actually). This is the so-called Stein phenomenon; see connections from the present exercise to Ex. 9.23. empirical Bayes

**(d)** A full Bayesian solution, without resorting to the empirical Bayes method above, is to give a prior for $\tau$, or equivalently the $\rho$, along with $\theta_i\,|\,\tau\sim\mathrm{N}(0,\tau^2)$. Show that the joint density of the $2k+1$ quantities in the game, i.e. $\rho,\theta_1,\ldots,\theta_k,y_1,\ldots,y_k$, may be written

$$\pi(\rho)\prod_{i=1}^k\Big\{\frac{1}{\tau}\phi\Big(\frac{\theta_i}{\tau}\Big)\phi(y_i-\theta_i)\Big\}=\pi(\rho)\prod_{i=1}^k\Big\{\frac{1}{\rho^{1/2}}\phi\Big(\frac{\theta_i-\rho y_i}{\rho^{1/2}}\Big)(1-\rho)^{1/2}\phi((1-\rho)^{1/2}y_i)\Big\}.$$

Integrate out the $\theta_i$ to show that $(\rho,y_1,\ldots,y_k)$ follow the joint density proportional to $\pi(\rho)(1-\rho)^{k/2}\exp\{-\frac12(1-\rho)v)\}$. Show that the Jeffreys prior is proportional to $1/(1-\rho)$, with which the posterior for $\rho$ becomes $\pi(\rho\,|\,\mathrm{data})\propto(1-\rho)^{k/2-1}\exp\{-\frac12(1-\rho)v\}$ on the unit interval.

**(e)** Explain that posterior simulation may be carried out via (i) drawing $\rho$ from its posterior, e.g. via one-dimensional MCMC, and then (ii) drawing $\theta_1,\ldots,\theta_k$ independently from their $\mathrm{N}(\rho y_i,\rho)$. For the loss function $L$ given above the Bayes solutions are the posterior means; show that these are $\widetilde{\theta}_i=\widetilde{\rho}y_i$, with $\widetilde{\rho}=\mathrm{E}\,(\rho\,|\,\mathrm{data})$. This can easily be computed numerically.

**(f)** As long as one may carry out posterior simulations, one may also find Bayes solutions with respecte to more complicated loss functions than the quadratic one. For illustration, suppose the observations are

```
-1.277 -1.254 -0.791 -0.148 0.178 0.294 0.847 1.229 1.720 1.749
```

and set up a scheme to simulate say $10^5$ realisations of $(\theta_1,\ldots,\theta_k)$ given these datapoints, from the prior described above. Then find the Bayes estimators with the loss function $\sum_{i=1}^k|\widehat{\theta}_i-\theta_i|$. Check what happens if data point no. 10 is pushed some steps and then farther to the right.

**Ex. 7.34** *Multiparameter inference, II.* (xx nils rant. need polish and finish. xx) the natural real thing: independent $Y_i\sim\mathrm{N}(\theta_i,\sigma^2)$ for $i=1,\ldots,k$, with unknown means and the same unknown variance. the object is estimating all the means, with loss $L=\sum_{i=1}^k(\widehat{\theta}_i-\theta_i)^2/\sigma^2$. direct estimator is $Y_i$, with risk $k$. with any prior construction, the

Bayes solution is $\mathrm{E}\,(\theta_i \,|\, \text{data})$. natural prior: $\theta_i \sim \mathrm{N}(\theta_0, \tau^2)$. then

$$\theta_i \,|\, y_i \sim \mathrm{N}((1-\rho)\theta_0 + \rho y_i, \rho\sigma^2), \quad \text{with } \rho = \tau^2/(\sigma^2+\tau^2).$$

here we are at a higher level of complexity compared to Ex. 7.34; there we knew $\theta_0$ and $\sigma$, and could concentrate on a prior for $\rho$. for empirical Bayes, note that marginally, $Y_i \sim \mathrm{N}(\theta_0, \sigma^2+\tau^2)$. the sample mean is $\widehat{\theta}_0 = \bar{Y}$, and $V = \sum_{i=1}^k (Y_i - \bar{Y})^2 \sim (\sigma^2+\tau^2)\chi^2_{k-1}$, leading to an estimator for $\sigma^2+\tau^2$. the Bayes estimator is

$$\widehat{\theta_i}) = \mathrm{E}\,\{(1-\rho)\theta_0 + \rho y_i \,|\, \text{data}\}.$$

we need posterior for $(\theta_0, \rho)$, but perhaps need a prior for the full triple $(\sigma, \theta_0, \tau)$. we may write up

$$\begin{aligned} f &= \pi(\sigma,\theta_0,\tau) \prod_{i=1}^k \Big\{ \frac{1}{\tau}\phi\Big(\frac{\theta_i-\theta_0}{\tau}\Big) \frac{1}{\sigma}\phi\Big(\frac{y_i-\theta_i}{\sigma}\Big) \Big\} \\ &= \pi(\sigma,\theta_0,\tau) \prod_{i=1}^k \Big\{ \frac{1}{\rho^{1/2}\sigma}\phi\Big(\frac{\theta_i-(1-\rho)\theta_i-\rho y_i}{\rho^{1/2}\sigma}\Big) \frac{1}{(\sigma^2+\tau^2)^{1/2}}\phi\Big(\frac{y_i-\theta_0}{(\sigma^2+\tau^2)^{1/2}}\Big) \Big\}. \end{aligned}$$

integrating out the $\theta_i$ we have

$$\pi(\sigma,\theta_0,\tau) \prod_{i=1}^k \frac{1}{(\sigma^2+\tau^2)^{1/2}}\phi\Big(\frac{y_i-\theta_0}{(\sigma^2+\tau^2)^{1/2}}\Big).$$

## Notes and pointers

point to Ch. 14 for various extensions, quote Lindley 1972, Bayesians had been 'embarrassingly silent' about nonparametrics.

(xx mention Varian (1975); Zellner (1986); Claeskens and Hjort (2008a) for the linex loss; but this should perhaps be in Ch. 9. mention Stigler (1983), who uses Bayes's Theorem to help him spot a candidate for an earlier discoverer of Bayes' Theorem. mention MCMC revolution since the 1990ies. point to Story #41. xx)

(xx on the Jeffreys prior. the argument used is slightly shaky since the transformation formula is used for possibly improper densities. point to Bernardo-Berger. can push in the nils-emil combo prior; informative on some components, non-informative on the rest. Cencov papers re Jeffreys being only solution to invariance demand. xx)

(xx hierarchical Bayes, with or without empirical Bayes. point to Efron etc. xx)

# I.8

# Confidence distributions, confidence curves, combining information sources

With $\mu$ a focus parameter, a function of the full parameter vector $\theta$, the Bayesian setup gives a posterior distribution. This requires the conceptually and practically difficult task of defining a prior for the full $\theta$, however. Confidence distributions (CDs) are a frequentist parallel, yielding post-data distributions for such focus parameters, without any prior. In this chapter we develop theory for CDs and confidence curves, and also find ways of combining CDs across different information sources. Constructing and computing CDs is not an easy or automatic task, but we develop and illustrate several recipes. For the exponential family class, we derive optimal CDs, with their own clear recipes.

*Key words:* boundary constraints, combining CDs, confidence conversion, confidence curves, confidence distributions, exponential family, meta-analysis, II-CC-FF, prediction CDs, t-bootstrapping

Confidence distributions and confidence curves are fruitful statistical inference summaries. Suppose in general terms that data $y$ stem from a model $f(y, \theta)$, with model parameter $\theta = (\theta_1, \ldots, \theta_p)$, and that $\mu = \mu(\theta_1, \ldots, \theta_p)$ is a parameter of particular interest. A *confidence distribution* for $\mu$, a CD, for short, is a function $C(\mu, y)$, such that (i) it is a c.d.f. in $\mu$, for each dataset $y$, and (ii) the distribution of $U = C(\mu_0, Y)$ is uniform at the true value $\mu_0 = \mu(\theta_0)$. In other words,

$$\Pr_{\theta_0}\{a \le C(\mu_0, Y) \le b\} = b - a \quad \text{for each } a, b \in [0, 1].$$

Assuming this random c.d.f. has a unique inverse, then, we have

$$\Pr_{\theta}\{C^{-1}(0.05, Y) \le \mu \le C^{-1}(0.95, Y)\} = 0.90, \tag{8.1}$$

and of course similarly for other choices of quantiles. This is by definition making $[C^{-1}(0.05, y_{\rm obs}), C^{-1}(0.95, y_{\rm obs})]$ a 90 percent confidence interval for the focus parameter $\mu$. The CD concept is hence related to and an extension of the confidence intervals, see Ch. 4. The *confidence curve* is a related summary graph, most often computed from the CD via $\mathrm{cc}(\mu, y) = |1 - 2\,C(\mu, y)|$. It has the practical property that $\{\mu\colon \mathrm{cc}(\mu, y) \le 0.90\}$ give the 90 percent interval directly; similarly, all intervals at any desired confidence level can be read off from the confidence curve. We sometimes write simply $C(\mu)$ and $\mathrm{cc}(\mu)$, omitting the data argument, when clear from the context what the data are, in $C(\mu) = C(\mu, \mathrm{data})$.

Construction a CD is not always an easy or automatic task, but we develop several practical recipes, some of which are based on approximate normality, or on more general methods of likelihood theory. Just as tests have detection power, also CDs have power, and theory is developed below to find optimal CDs in classes of situations. This is partly parallelling the optimal testing methodology of Ch. 4. All in all we develop and illustrate the following recipes: (i) Via the c.d.f. of an estimator; (ii) normal approximation; (iii) based on a pivot; (iv) deviance and Wilks theorem; (v) t-bootstrapping; (vi) the optimal CD via conditional distributions, if inside the exponential family.

The CDs are post-data graphical summaries of the level of uncertainty for any focus parameter, and can be seen as frequentist parallels to the Bayesian posterior distributions; here there is no prior, however. We illustrate this 'clear data-only based posteriors without priors' aspect of the CDs through theory and applications (xx perhaps point to a few Stories xx).

Combining different information sources is a broad statistical theme, going back to the first meta-analysis concepts and methods of Karl Pearson just after 1900 (Simpson and Pearson, 1904). The more familiar meta-analysis methods aim at combining independent estimators for the same quantity, or for providing a broader population assessment of similar but not identical parameters. CDs are useful for such endeavours, and we provide methods for combining sources more general than the traditional ones.

### Recipes One, Two, Three for constructing CDs

**Ex. 8.1** *The probability transform.* Some of the following facts are related to various operations for confidence distributions and confidence curves

**(a)** Suppose $X$ has a continuous and increasing cumulative distribution function $F$, i.e. $F(x) = \Pr(X \le x)$. Show that $U = F(X)$ is uniform on the unit interval. Any continuously distributed random variable can hence be transformed to uniformity, via this *probability transform*.

**(b)** Show that also $U_2 = 1 - F(X)$ and $U_3 = |1 - 2\,F(X)|$ have uniform distributions.

**(c)** Simulate a million copies of $x_i \sim \mathrm{N}(0,1)$, and check the histogram of $\Gamma_1(x_i^2)$, where $\Gamma_\nu$ is the cumulative distribution function of a $\chi^2_\nu$. Comment on what you find.

**Ex. 8.2** *Recipe One: via the c.d.f. of an estimator.* Suppose $\theta$ is a one-dimensional parameter, for which we need a CD, after having observed data $y_{\rm obs}$. If there is an estimator $\widehat\theta$, with a distribution depending only on this $\theta$, there is a clear recipe.

**(a)** Assume therefore that $\widehat\theta$ has a continuous distribution function $K_\theta(x) = \Pr_\theta(\widehat\theta \le x)$; its distribution is here required to depend only on $\theta$, not on other aspects of the underlying model employed. Consider Recipe One, the construction

$$C(\theta, y_{\rm obs}) = \Pr_\theta(\widehat\theta \ge \widehat\theta_{\rm obs}) = 1 - K_\theta(\widehat\theta_{\rm obs}),$$

a curve that can be computed and plotted post-data, where $\widehat\theta_{\rm obs} = \widehat\theta(y_{\rm obs})$ is the observed estimate. Show that it has the property that the random $C(\theta, Y)$ is uniformly distributed, for each fixed $\theta$.

**(b)** To illustrate, go through the details for the case of using $\widehat\theta = 1/\bar Y$, with i.i.d. observations $Y_1, \ldots, Y_n$ from the exponential $\theta \exp(-\theta y)$. Show first that $2\theta Y_i \sim \chi^2_2$, and derive $K_\theta(x) = 1 - \Gamma_{2n}(2n\theta/x)$, with $\Gamma_{2n}$ the c.d.f. of the $\chi^2_{2n}$. Simulate data and plot the CD $C(\theta, y_{\rm obs}) = \Gamma_{2n}(2n\theta/\widehat\theta_{\rm obs})$. From the CD, find a 95 percent interval for $\theta$.

**(c)** Assume $X_1, \ldots, X_m$ are i.i.d. Expo($\theta_1$) and that $Y_1, \ldots, Y_n$ are i.i.d. Expo($\theta_2$). Find the distribution of the estimator $\widehat{\rho} = \widehat{\theta}_1/\widehat{\theta}_2$ for the ratio $\rho = \theta_1/\theta_2$, and derive the associated CD.

**(d)** Generate $n = 25$ datapoints from the double exponential density $f(y, \theta) = \frac{1}{2}\exp(-|y-\theta)$, using your favourite true $\theta_0$. Compute and display the CD for $\theta$ based on the median $M_n$.

**(e)** For a simpler and more fundamental illustration, suppose $\widehat{\theta}$ has a normal distribution centred at $\theta$, with a known variance, say $\widehat{\theta} \sim \mathrm{N}(\theta, \kappa^2)$. Show that Recipe One gives $C(\theta) = \Phi((\theta - \widehat{\theta})/\kappa)$. Check that the famous 95 percent interval $\widehat{\theta} \pm 1.96\,\kappa$ agrees with this.

**Ex. 8.3** *Confidence distribution and confidence curve for the normal standard deviation.* The confidence distribution $C$ and the confidence curve cc are close cousins, and they do not need to be both displayed for each new statistical application. Here is a simple illustration. You observe the $n = 6$ data points 4.09, 6.37, 6.87, 7.86, 8.28, 13.13 from a normal distribution and wish to assess the underlying spread parameter, the standard deviation $\sigma$.

**(a)** For the empirical variance, use $\widehat{\sigma}^2 \sim \sigma^2\chi^2_m/m$, with $m = n - 1$, to build the CD

$$C(\sigma, y_{\mathrm{obs}}) = \Pr_\sigma(\widehat{\sigma} \geq \widehat{\sigma}_{\mathrm{obs}}) = 1 - \Gamma_m(m\widehat{\sigma}^2_{\mathrm{obs}}/\sigma^2).$$

Here $y_{\mathrm{obs}}$ represents the observed data, and $\widehat{\sigma}_{\mathrm{obs}}$ the observed point estimate. Show that $C(\sigma, Y) \sim$ unif, where $Y$ represents a random data set $Y_1, \ldots, Y_n$, from the $\sigma$ in question. In particular, the distribution of $C(\sigma, Y)$ does not depend on $\sigma$. Make a graph, also of the associated confidence curve

the confidence curve

$$\mathrm{cc}(\sigma, y_{\mathrm{obs}}) = |1 - 2\,C(\sigma, y_{\mathrm{obs}})| = |1 - 2\,\Gamma_m(m\widehat{\sigma}^2_{\mathrm{obs}}/\sigma^2)|.$$

Compute the *median confidence estimate* $\widehat{\sigma}_{0.50} = C^{-1}(0.50, y_{\mathrm{obs}})$ and the natural 90 percent confidence interval $[C^{-1}(0.05, y_{\mathrm{obs}}), C^{-1}(0.95, y_{\mathrm{obs}})]$. Find and display also the *confidence density* $c(\sigma, y_{\mathrm{obs}})$, the derivative of the CD.

**(b)** Compute also the *confidence density* $c(\sigma, y_{\mathrm{obs}})$ associated with the CD. Compute furthermore its mode, say $\sigma^*$, and briefly assess its properties as an estimator of $\sigma$.

**(c)** A Bayesian approach to the same problem, i.e. finding a posterior distribution for $\sigma$, is to start with a prior $\pi(\sigma)$ and then compute $\pi(\sigma \,|\, y_{\mathrm{obs}}) \propto \pi(\sigma)g(\widehat{\sigma}, \sigma)$, where $g(\widehat{\sigma}, \sigma)$ is the likelihood, here the density function for $\widehat{\sigma}$ as a function of $\sigma$. When does such a Bayesian approach agree with the confidence density?

**(d)** Suppose there are two independent normal samples, with standard deviations $\sigma_1$ and $\sigma_2$. Construct a CD for $\rho = \sigma_1/\sigma_2$. Invent a second simple small dataset, to complement the first dataset given above, and then compute and display the confidence curve $\mathrm{cc}(\rho, \mathrm{data})$.

**Ex. 8.4** *A CD for mean by standard deviation.* As in Ex. 3.39, consider i.i.d. $Y_1, \ldots, Y_n$ from a normal $(\xi, \sigma^2)$, with inference needed for $\kappa = \xi/\sigma$. The natural estimator is $\widehat{\kappa} = \widehat{\xi}/\widehat{\sigma}$, with the usual sample mean and sample standard deviation. As we saw in the exercise pointed to, $X = \sqrt{n}\widehat{\kappa} \sim t_m(\sqrt{n}\kappa)$, with $m = n - 1$ degrees of freedom for the noncentral $t$.

**(a)** We start with a stylised version of the problem, with a single $T \sim t_m(a)$, with known $m$ and unknown $a$. Show that $C(a, t_{\mathrm{obs}}) = \Pr_a(T \geq t_{\mathrm{obs}}) = 1 - G_m(t_{\mathrm{obs}}, a)$ is a CD for

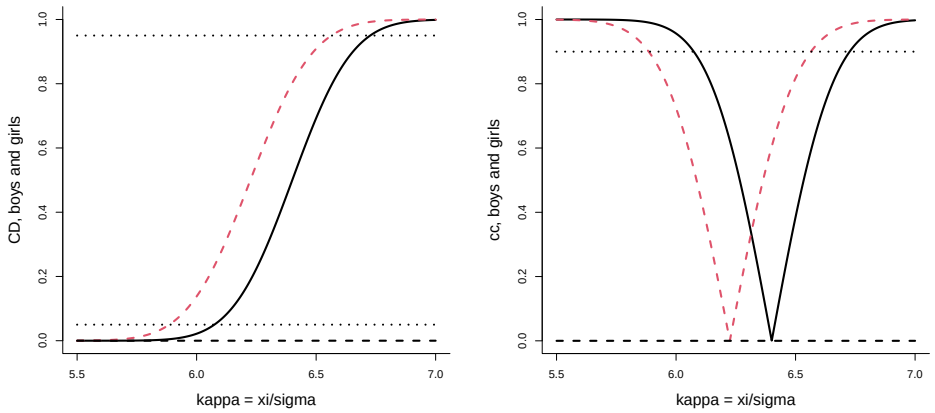



Figure 8.1: *For 548 boys and 480 girls born in Oslo, 2001–2008, CDs (left panel) and confidence curves (right panel), for the birthweight ratio $\kappa = \xi/\sigma$ (full curves for boys, slanted curves for girls). One may read off 0.05 and 0.95 confidence points from the CDs, or the 0.90 confidence interval directly from the ccs; 6.075 to 6.726 for boys, 5.886 to 6.564 for girls. Median confidence estimates are 6.400 and 6.225.*

$a$, in terms of the c.d.f. for the $t_m(a)$ distribution. The associated confidence curve is then $\mathrm{cc}(a, t_{\mathrm{obs}}) = |1 - 2\,G_m(t_{\mathrm{obs}}, a)|$.

**(b)** Going back to the normal sample, show that the natural CD for $\kappa = \xi/\sigma$ becomes

$$C(\kappa) = \Pr_\kappa(\widehat{\kappa} \geq \widehat{\kappa}_{\mathrm{obs}}) = 1 - G_m(\sqrt{n}\widehat{\kappa}_{\mathrm{obs}}, \sqrt{n}\kappa).$$

**(c)** For an illustration, we use the data from Story #20, with birthweights for 548 boys and 480 girls born in Oslo, 2001–2008. The required sufficient information $(\widehat{\xi}, \widehat{\sigma})$ is $(3.669, 0.573)$ for the boys and $(3.494, 0.561)$ for the girls. Compute and display the CDs and confidence curves, as with Figure 8.1. Find also 99 percent confidence intervals for the $\kappa$ parameter in the two groups.

**(d)** Consider a linear regression setup, with $y_i \sim \mathrm{N}(x_i^{\mathrm{t}}\beta, \sigma^2)$ for $i = 1, \ldots, n$. For two covariate vectors $x_a$ and $x_b$, examine the mean difference $\mathrm{E}\,(Y_b \,|\, x_b) - \mathrm{E}\,(Y_a \,|\, x_a) = (x_b - x_a)^{\mathrm{t}}\beta$, along with $\kappa = (x_b - x_a)^{\mathrm{t}}\beta/\sigma$, a natural quantity informing us about the essential statistical difference between the two. Construct a CD for this parameter. (xx now brief notes on solution. xx) writing $d = x_b - x_a$, note first that $Y_b - Y_a \sim \mathrm{N}(d^{\mathrm{t}}\beta, 2\sigma^2)$, so $p = \Pr(Y_b > Y_a) = \Phi(\kappa/\sqrt{2})$, with $\kappa = d^{\mathrm{t}}\beta/\sigma$. we wish a CD for this $\kappa$ and hence for $p$. for this, $Y_b - Y_a - d^{\mathrm{t}}\widehat{\beta}$ is zero-mean normal with variance $(2 + \tau_n^2/n)\sigma^2$, where $\tau_n^2 = d^{\mathrm{t}}\Sigma_n^{-1}d$, with $\Sigma_n = n^{-1}\sum_{i=1}^n x_i x_i^{\mathrm{t}}$, as per usual linear regression theory of Ch3. via this we have a CD for $Y_b - Y_a$, for large $n$ close to simply $Y_b - Y_a \sim \mathrm{N}(d^{\mathrm{t}}\beta, 2\sigma^2)$. also, with $\widehat{\kappa} = d^{\mathrm{t}}\widehat{\beta}/\widehat{\sigma}$, we have

$$\sqrt{n}\widehat{\kappa}/\tau_n = \frac{\sqrt{n}d^{\mathrm{t}}\widehat{\beta}/(\sigma\tau_n)}{\widehat{\sigma}/\sigma} = \frac{\mathrm{N}(\sqrt{n}\kappa/\tau_n, 1)}{(\chi^2_m/m)^{1/2}}$$

this then leads to the clear CD

$$C(\kappa) = \Pr_\kappa(\sqrt{n}\widehat{\kappa} \geq \sqrt{n}\kappa_{\mathrm{obs}}) = 1 - G_m(\sqrt{n}\widehat{\kappa}_{\mathrm{obs}}/\tau_n, \sqrt{n}\kappa/\tau_n).$$

this should be done for New Haven, regarding $p = \Pr(Y_b > Y_a)$.

**Ex. 8.5** *A CD for the Mahalanobis distance between populations.* (xx nils rant; may need an easy application to illustrate. For two multinormal populations, say $\mathrm{N}_p(\xi_1, \Sigma)$ and $\mathrm{N}_p(\xi_2, \Sigma)$, the Mahalanobis distance between them is $\delta = \{(\xi_2 - \xi_1)^{\mathrm{t}}\Sigma^{-1}(\xi_2 - \xi_1)\}$. check Reiser and Faraggi, Biometrics 1997, and 'confidence intervals for the Mahalanobis distance', Reiser 2001. also point to other parameters, smooth functions of $\delta$. so we solve many problems, precisely. xx)

**(a)** The natural estimator is $\widehat{\delta}^2 = (\bar{Y}_2 - \bar{Y}_1)^{\mathrm{t}}\widehat{\Sigma}^{-1}(\bar{Y}_2 - \bar{Y}_1)$, with $\widehat{\Sigma} = c_1\widehat{\Sigma}_1 + c_2\widehat{\Sigma}_2$, with $c_1 = (m-1)/(m+n-2)$ and $c_2 = (n-1)/(m+n-2)$. Then (xx via other things xx) show that

$$D^2 = \frac{mn}{m+n}\frac{m+n+p-1}{m+n-2}\widehat{\delta}^2 \sim F_{p,m+n-p-1}(\lambda),$$

with excentre parameter $\lambda = \{mn/(m+n)\}\delta^2$.

**(b)** (xx with the above result is should be easy to construct a clear CF for $\delta$. xx)

**(c)**

**Ex. 8.6** *Computing a CD with simulation and isotonic repair.* (xx to be polished. we use this is Story #62 and perhaps in other places, where simulations are expensive. xx) Suppose one observes $y_1, \ldots, y_n$ from the one-parameter Weibull distribution with c.d.f. $F(y, b) = 1 - \exp(-y^b)$, with sample size $n = 25$, and computes the data mean $\bar{y}_{\mathrm{obs}} = 1.313$.

**(a)** Though we do not actually need this in the CD computations here, find an estimate of $b$ based on $\mathrm{E}\, Y_i = \Gamma(1 + 1/b)$; see Ex. 1.60. Show that $C(b) = \Pr_b(\bar{Y} \le \bar{y}_{\mathrm{obs}})$ is a CD for $b$.

**(b)** The practical obstacle here is that $\bar{Y}$ does not have a simple distribution. But we're saved by simulation. Show that the simulation recipe $Y_i^* = V_i^{1/b}$ produces outcomes from the weibull $F(y, b)$, where the $V_i$ are unit exponential. For a grid of $b$ values, e.g. from 0.20 to 1.20, compute the simulation based $C^*(b)$, the proportion of $B$ cases where the simulated $\bar{Y}^*$ is below $\bar{y}_{\mathrm{obs}}$. Compute also the confidence curve $\mathrm{cc}^*(b) = |1 - 2\,C^*(b)|$. For this simple example it is easy to accomplish this with a high $B$, say $10^5$, to make $C^*(b)$ and $\mathrm{cc}^*(b)$ smooth and very close to the real $C(b)$ and $\mathrm{cc}(b)$; for this illustration, however, make the simulation size as relatively small as $B = 100$, and plot the curves, as in Figure 8.2.

**(c)** We learn that with a low or moderate simulation size $B$, the $C^*(b)$ and $\mathrm{cc}^*(b)$ will be wiggly. We can do better, using the prior knowledge that $C(b)$ is increasing. There are several repair mechanisms, which from the potentially wiggly $C^*(b)$ create a monotonically increasing curve. A simple scheme is so-called *isotonic regression*, the details of which we do need to get into here. Supposing you have first created `bval` and `Cval` in your `R` session, you may use `Cvaliso=isoreg(bval,Cval)$yf`, which repairs your $C^*(b)$ and $\mathrm{cc}^*(b)$ to ensure monotonicity. Produce versions of Figure 8.2, left and right panels.

isotonic regression

**(d)** (xx round off. explain salient points about generalisability. we need reduction to one-parameter situation. xx)

**Ex. 8.7** *An extension of Recipe One.* In Ex. 8.2 we saw that the simple construction $C(\theta, y) = \Pr_\theta(\widehat{\theta} \ge \widehat{\theta}_{\mathrm{obs}})$ gives a CD, in the case of one-dimensional setups with a well-defined estimator $\widehat{\theta}$.

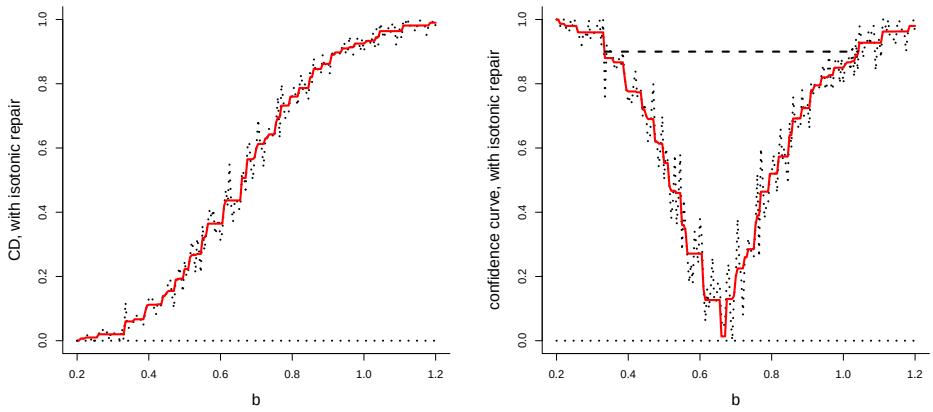



Figure 8.2: *Simulation based confidence distribution* $C^*(b)$ *and confidence curve* $\text{cc}^*(b)$ *for the Weibull parameter* $b$*, based on the observed sample mean* $\bar{y}_{\text{obs}} = 1.313$ *for* $n = 25$ *data points, along with isotonic repairs. The simulation size here is the low* $B = 100$*.*

**(a)** When working with estimators, finetuning efforts are often exuded to trim away biases, getting the scaling right, etc. In a sense this is not needed here, when constructing the CD. Show that if $\widehat{\alpha} = g(\widehat{\theta})$, with any smooth increasing $g$, the recipe $C^*(\theta) = \text{Pr}_\theta(\widehat{\alpha} \geq \widehat{\alpha}_{\text{obs}})$ gives precisely the same CD as without the $g$ transformation.

**(b)** So this CD recipe relies merely on having an informative statistic, say $Z$, with a distribution stochastically increasing in $\theta$; it does not really have to be an estimator for that parameter. Show that $C(\theta, y) = \text{Pr}_\theta(Z \geq z_{\text{obs}})$ is a bona fide CD.

**(c)** Show also that the construction works, if there are other parameters at play too, as long as the distribution of the chosen $Z$ only depends on $\theta$. Go through the details for the case of the $Y_i$ being $\text{N}(\mu, \sigma^2)$, with $Z = \sum_{i=1}^n (Y_i - \bar{Y})^2$, and also for $Z' = \sum_{i=1}^n |Y_i - M_n|$, where $M_n$ is the empirical median. Compute, display, compare both CDs, based on $Z$ and on $Z'$, for the simple dataset of Ex. 8.3 (with $n = 6$). For the $Z$ case, there is a formula, but for the $Z'$ case you would need simulation, for a grid of $\sigma$ values; see Ex. 8.6.

**(d)** (xx one more example, where there is a $Z$ carrying information, but not qua estimator. xx)

**Ex. 8.8** *Recipe Two: the normal approximation CD.* Applying Recipe One of Ex. 8.2 to the case of the estimator having a normal distribution leads as we saw there to a clear CD, provided the variance is known. But this is at least approximately so, for large classes of situations, as we've seen in Chs. 2 and 5.

**(a)** Suppose in general terms that $\widehat{\theta}$ estimates $\theta$, and that its distribution is approximately a $\text{N}(\theta, \kappa^2)$. Explain that $C(\theta) = \Phi((\theta - \widehat{\theta})/\kappa)$ then is an approximate CD for $\theta$. More formally, if $(\widehat{\theta} - \theta_0)/\widehat{\kappa} \to_d \text{N}(0, 1)$, at the true parameter $\theta_0$, show that $C(\theta, Y) = \Phi((\theta - \widehat{\theta})/\widehat{\kappa})$ has the property that it converges in distribution to the uniform, at $\theta_0$. In typical applications of these arguments, there is a $\sqrt{n}$ scaling in terms of an underlying sample size, with $\sqrt{n}(\widehat{\theta} - \theta_0) \to_d \text{N}(0, \tau^2)$, say, and $\widehat{\kappa} = \widehat{\tau}/\sqrt{n}$, with $\widehat{\tau} \to_{\text{pr}} \tau$. So this is Recipe Two, the normal approximation CD, most typically of this type $\Phi(\sqrt{n}(\theta - \widehat{\theta})/\widehat{\tau})$.

**(b)** Simulate a moderate or small dataset from a normal distribution. Compute and display two (approximate) CDs for the mean parameter $\xi$, (i) using the data mean, (ii) using the data median.

**(c)** We have seen in Chs. 2 and 5 that approximate normality is highly common, for large classes of estimators, typically along with consistent estimators for the variances. In particular, the delta method implies approximate normality of smooth functions of background estimators, see Ex. 2.53, making in its turn approximate normality CDs easily available. For a simple illustration, suppose you throw your nearest die, which has probability $p$ of giving a '6', until you get your first '6'. You carry out this geometric experiment $n = 10$ times, giving you the counts $Y_1, \ldots, Y_n$ equal to 1, 2, 17, 18, 20, 4, 3, 1, 15, 3. Use the normal approximation for $\bar{Y}$ to give an approximate CD for $p$. You may also compare this to what one achieves working with the exact distribution of $\bar{Y}$.

**(d)** (xx point to logistic and poisson regression, with delta method. estimate $\beta$ and also $p = H(x_0^{\mathrm{t}}\beta)$. xx)

**Ex. 8.9** *Recipe Three: from a pivot to a CD.* (xx check that we're not repetitive regarding pivot. xx) Suppose in general terms that $\mu$ is some parameter of interest, in a model for observations $Y$, and that a function $A = \mathrm{piv}(\mu, Y)$ of the parameter and the data has the property that its distribution does not depend on the model parameters (in particular, therefore, not on $\mu$, which might itself be a function of other model parameters). We call $A$ a pivot, in more pedantic detail a pivot for the parameter $\mu$.

**(a)** With $Y_1, \ldots, Y_n$ independent from the normal $(\mu, \sigma^2)$, let $R_n = \sum_{i=1}^n |Y_i - \bar{Y}|$ with the sample mean $\bar{Y}$. Show that $(\bar{Y} - \mu)/R_n$ is a pivot. Invent yet another pivot involving $\mu$, with a different denominator.

**(b)** With two normal samples, say $X_1, \ldots, X_m$ from $\mathrm{N}(\mu_1, \sigma_1^2)$ and $Y_1, \ldots, Y_n$ from $\mathrm{N}(\mu_2, \sigma_2^2)$, suppose $\rho = \sigma_1/\sigma_2$ is in focus. Show that $(V_1/V_2)/\rho$ is a pivot for $\rho$, where $V_1$ and $V_2$ are the interquartile ranges for the two datasets.

**(c)** Consider $Y_1, \ldots, Y_n$ from the Cauchy model with density $(1/\pi)/\{1 + (y-\theta)^2\}$. Show that $R_n - \theta$ is a pivot, where $R_n = \frac{1}{2}(Q_{n,0.10} + Q_{n,0.90})$ is the average of the 0.10 and 0.90 quantiles.

**(d)** Back to the generalities, consider a pivot $A = \mathrm{piv}(\mu, Y)$ for $\mu$ in some model, increasing in $\mu$. Assume the situation is continuous, not discrete, so that the pivot's distribution function $K$ is continuous. Show that $C(\mu, y_{\mathrm{obs}}) = 1 - K(\mathrm{piv}(\mu, y_{\mathrm{obs}}))$ is a proper CD for $\mu$.

**(e)** In clean cases we may derive the precise distribution for the pivot in question, but the CD recipe given above may be used also in more complicated setups, as long as $A = \mathrm{piv}(\mu, Y)$ may be simulated. Make an illustration of this, with the ratio of standard deviations above. Suppose two normal datasets, both of size $n = 100$, lead to interquartile ranges $V_{1,\mathrm{obs}} = 4.44$ and $V_{2,\mathrm{obs}} = 3.33$. Construct and display $C(\rho)$ and $\mathrm{cc}(\rho)$.

**(f)** (xx make the point that various constructions, involving large-sample approximations to the normal and to chisquares, lead to *approximate pivots*, and then again to approximate CDs and ccs. in particular, Method One, with $\Phi((\mu - \widehat{\mu})/\widehat{\kappa})$ and Method Two, with $\Gamma_1(D(\mu))$, can be seen via approximate pivots. also Method Three, construction of a t type ratio and then bootstrapping. xx)

**Ex. 8.10** *CDs from the t pivot.* In Ex. 8.2 we saw that the simple construction $C(\theta, y) = \Pr_\theta(\widehat{\theta} \geq \widehat{\theta}_{\mathrm{obs}})$ gives a CD, in the case of one-dimensional setups with a well-defined

estimator $\widehat{\theta}$.

**(a)** For a normal sample from $\mathrm{N}(\mu, \sigma^2)$, we see that several $\Pr_{\mu,\sigma}(Z \geq z_{\rm obs})$ schemes work, in that the $Z$ in question has a distribution depending on $\sigma$, but not $\mu$. Attempt to work with $C^*(\mu, y) = \Pr_{\mu,\sigma}(\bar{Y} \geq \bar{y}_{\rm obs})$ – and explain that it will not really work (unless $\sigma$ is known).

**(b)** But of course there *are* natural CD constructions for $\mu$ here. What is needed is a *pivot*, say $A = \mathrm{piv}(\mu, y)$, a function binding the focus parameter and data together in a way which makes its distribution not depend on the parameters. Study indeed

$$t_n = t_n(\mu, Y) = (\bar{Y} - \mu)/(\widehat{\sigma}/\sqrt{n}),$$

with $\widehat{\sigma}^2 = (n-1)^{-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ the classical empirical variance. Pretend that you in all your cleverness have not seen this $t_n$ before, and are unaware of its relation to a t distribution – but show that the distribution of $t_n$, call it $K_n$, does not depend on $(\mu, \sigma)$.

**(c)** Then show that $C(\mu, y_{\rm obs}) = K_n(t_n(\mu, y_{\rm obs}))$ is a CD for $\mu$. Even if you do not see the connection to the classic t of Student (1908), see Ex. 1.51, you may still carry through this, by simulating $B = 10^5$ realisations of $t_n$, and use

$$C(\mu, y_{\rm obs}) = K_n^*(t_n(\mu, y_{\rm obs})) = \frac{1}{B} \sum_{j=1}^B I\{t_{n,j} \leq t_n(\mu, y_{\rm obs})\}.$$

Show however that by all means $K_n$ is a $t_m$, with $m = n - 1$, so the canonical CD for $\mu$ is and remains $C(\mu, y_{\rm obs}) = G_m(\sqrt{n}(\mu - \bar{y}_{\rm obs})/\widehat{\sigma}_{\rm obs})$, with $G_m$ the c.d.f. for the $t_m$.

### Recipes Four, Five, Six for constructing CDs

**Ex. 8.11** *Recipe Four: confidence curves via Wilks theorems.* Consider data from a parametric model, leading to the log-likelihood function $\ell_n(\theta)$, and that there is a focus parameter $\mu = g(\theta)$. We have seen likelihood profiling and Wilks theorems in Ch. 5, and know that the deviance $D_n(\mu) = 2\{\ell_{\max} - \ell_{\rm prof}(\mu)\}$ has the property that $D_n(\mu_0) \to_d \chi^2_1$ at the true value $\mu_0 = g(\theta_0)$; see Ex. 5.13.

**(a)** Recipe Four, utilising the Wilks theorems, is to form $\mathrm{cc}(\mu, y) = \Gamma_1(D_n(\mu))$, with $\Gamma_1$ the c.d.f. for the $\chi^2_1$. Show that $\Pr_{\theta_0}(\mathrm{cc}(\mu_0, Y) \leq \alpha) \to \alpha$ for each $\alpha$, and explain that this makes $\mathrm{cc}(\mu, y)$ and approximate confidence curve.

**(b)** For an illustration, consider the model $F(y, \theta) = y^\theta$ for observations on $[0, 1]$, where $\theta$ is an unknown positive parameter. Write down the log-likelihood function and find a formula for the maximum likelihood (ML) estimator $\widehat{\theta}$. Use also theory of Ch. 5 to write down a normal approximation to the distribution of $\widehat{\theta}$.

**(c)** Consider the data set

```
0.013 0.054 0.234 0.286 0.332 0.507 0.703 0.763 0.772 0.920
```

Estimate $\theta$ and compute the CD $C(\theta) = \Pr_\theta(\widehat{\theta} \geq \widehat{\theta}_{\rm obs})$, along with the confidence curve $\mathrm{cc}(\theta) = |1 - 2\,C(\theta)|$, (i) using simulations, (ii) using exact probability calculus. Reproduce a version of Figure 8.3.

**(d)** Supplement these two curves with approximations based (i) on the normal approximation for $\widehat{\theta}$ and (ii) on the chi-squared approximation for the deviance.

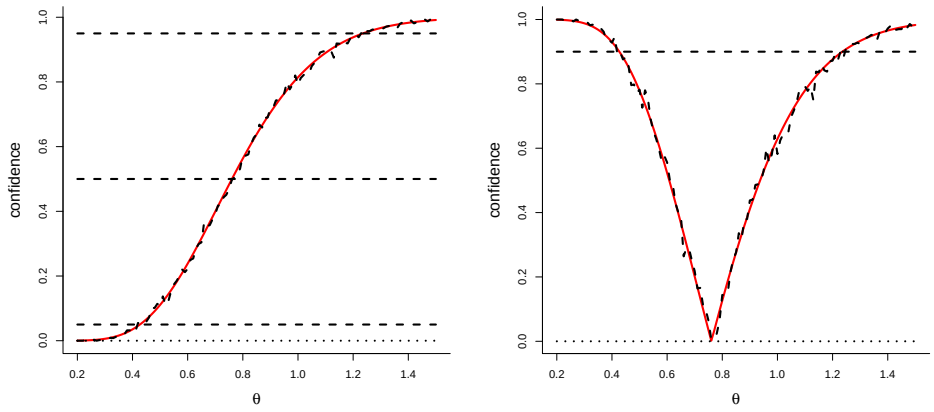



Figure 8.3: *For the simple data example of Ex. 8.11: Left panel: confidence distribution $C(\theta)$, via simulations (black and wiggly curve) and via exact calculations (red and smooth curve); right panel: the two versions of the associated confidence curve* $\mathrm{cc}(\theta)$. *From these we read off the median confidence estimate* $\widehat{\theta}_{0.50} = 0.76$, *and the 90 percent confidence interval* $[0.43, 1.24]$.

**(e)** (xx somewhere, if not here, then separately: from CD to cc, and from cc to CD, with $C(\mu) = \frac{1}{2} - \frac{1}{2}\mathrm{cc}(\mu)$ for $\mu \le \widehat{\mu}_{0.50}$ and $\frac{1}{2} + \frac{1}{2}\mathrm{cc}(\mu)$ for $\mu \ge \widehat{\mu}_{0.50}$. particularly useful with these deviance based ccs. xx)

**Ex. 8.12** *Median age for men and women in Roman Era Egypt.* In Story #49 we work with a rare dataset of lifetimes for 82 men and 59 women in Roman Era Egypt, a century B.C. For the present illustration of constructing confidence curves via log-likelihood profiling, take these lifetimes $T_i$ to have arisen from Weibull distributions, with c.d.f.s $F_m(t) = 1 - \exp\{-(t/a_m)^{b_m}\}$ for men and $F_w(t) = 1 - \exp\{-(t/a_w)^{b_w}\}$ for women.

**(a)** For a Weibull, with parameters $(a, b)$, show that the $q$ level quantile becomes $\mu(q) = F^{-1}(q) = ac(q)^{1/b}$, with $c(q) = -\log(1-q)$. In particular, the median is $a(\log 2)^{1/b}$.

**(b)** For the men and women separately, write down the log-likelihood functions $\ell_m(a_m, b_m)$ and $\ell_w(a_w, b_w)$, and then carry out profiling to compute the deviance functions, for the medians $\mu_m$ and $\mu_w$. Use Recipe Four, from Ex. 8.11, to construct the confidence curves, and make a version of Figure 8.4, left panel; also, read off 90 percent confidence intervals.

**(c)** It appears indeed that men had longer lives than women, in Egypt 2100 years ago; see Story #49 for more discussion and details. For this exercise, carry out log-likelihood profiling, for the median difference $d = \mu_m - \mu_w$, from the four-parameter model with the two Weibulls. Compute the deviance function and then the confidence curve $\mathrm{cc}(\cdot)$, leading to Figure 8.4, right panel. Read off a 90 percent interval for $d$.

**(d)** Constructing the confidence curves above involved the somewhat laborious computation of log-likelihood profiles. Compare these curves with the easier ones, computationally speaking, based on normal approximations.

**Ex. 8.13** *The Wilks confidence recipe for the normal model.* Consider the familiar setup with i.i.d. data $Y_1, \ldots, Y_n$ from the $\mathrm{N}(\xi, \sigma^2)$ model. We have handled the log-likelihood

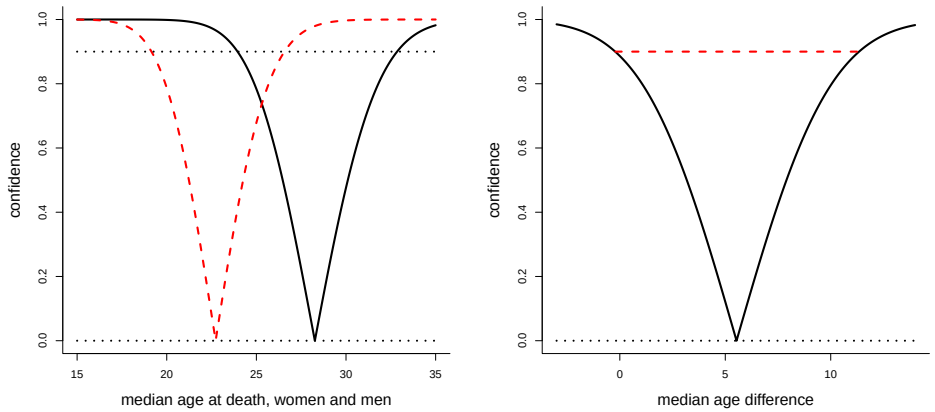



Figure 8.4: *Lifetimes in Roman Era Egypt, a century B.C.: Left panel: confidence curves for the median age $\mu_w$ and $\mu_m$, for women and men, via log-likelihood-profiling from fitting separate Weibull distributions, with 90 percent confidence intervals* $[19.95, 26.56]$ *and* $[23.95, 32.84]$. *Right panel: confidence curve for the median age difference $d = \mu_m - \mu_w$, via log-likelihood profiling, and 90 percent confidence interval* $[-0.21, 11.34]$.

profiles and accompanying Wilks theorems approximations in Ex. 5.21, and can apply results for constructing confidence curves. In the following, let as usual $\widehat{\xi}$ and $\widehat{\sigma}^2$ be the sample mean and sample variance.

**(a)** Starting with the mean parameter $\xi$, explain first that $C(\xi, \text{data}) = G_m(\sqrt{n}(\xi - \bar{Y})/\widehat{\sigma})$ is the natural exact CD, with $G_m$ the c.d.f. for the $t_m$, with $m = n - 1$ degrees of freedom. Then use the exercise pointed to to show that the Wilks recipe leads to

$$\text{cc}_w(\xi) = \Gamma_1\big(n \log(1 + n^{-1} K_n(\xi))\big),$$

with $K_n(\xi) = n(\xi - \widehat{\xi})^2/\widehat{\sigma}^2$.

**(b)** Then considering $\sigma$, explain that the natural exact CD is $C(\sigma, \text{data}) = 1 - \Gamma_m(m\widehat{\sigma}^2/\sigma^2)$, with $\Gamma_m$ the c.d.f. for $\chi^2_m$. Show that the Wilks recipe leads to

$$\text{cc}_w(\sigma) = \Gamma_1(nh(\widehat{\sigma}^2/\sigma^2)),$$

with $h(x) = x - \log x - 1$.

**(c)** (xx then compare the exact with the Wilks. approximation is good even for moderately low $n$. ask perhaps for doing this with the $n = 64$ word-length averages of Story #38. xx)

**Ex. 8.14** *Recipe Five: CDs via approximate pivots and t-bootstrapping.* Consider a parametric model $f(y, \theta)$ for data, with a model parameter of length say $p$. Suppose there is a focus parameter $\mu = g(\theta)$, estimated as $\widehat{\mu} = g(\widehat{\theta})$, for which we need a CD.

**(a)** Suppose first that $\widehat{\mu} - \mu$ has some distribution, say $G_0$, not depending on $\theta$. Under this simple pivotal assumption for 'estimator minus estimand', show that

$$C(\mu) = \Pr_\theta(\widehat{\mu} \geq \widehat{\mu}_{\text{obs}}) = 1 - G_0(\widehat{\mu}_{\text{obs}} - \mu).$$

As indicated in e.g. Ex. 8.6, this can be computed even without knowing the form of $G_0$, through simulation of many realisations of $\widehat{\mu}^* - \widehat{\mu}$, where the $\widehat{\mu}^*$ is computed from

a dataset drawn from the estimated distribution at $\widehat{\theta}$. Explain that this recipe works, even if $G_0$ is nonsymmetric and not centred well at zero. Simulating the distribution of $\widehat{\mu} - \mu$ at different points in the $\theta$ parameter space may also be helpful for checking the assumption needed for the $C(\mu)$ constructed here to be a clear CD.

**(b)** For a special and simpler case, if $\widehat{\mu} \sim \mathrm{N}(\mu, \kappa^2)$, to a good approximation, with known or well estimated $\kappa$, show that the general recipe above leads to

$$C(\mu) = \Pr_\theta(\mu + \kappa N \geq \widehat{\mu}_{\mathrm{obs}}) = \Pr(N \geq (\widehat{\mu} - \mu)/\kappa) = \mu((\mu - \widehat{\mu})/\kappa),$$

and argue that this is really a CD. Note that this requires $Z = (\widehat{\mu} - \mu)/\kappa$ having distribution equal to or close to a standard normal, regardless of where $\theta$ is in its parameter space.

**(c)** Often the standard deviation in the normal approximation is not that sharply estimated from the available data. Consider a Student type ratio $t = (\widehat{\mu} - \mu)/\widehat{\kappa}$, with some appropriate scale estimator $\widehat{\kappa}$. Assume first that $t$ really is a pivot, i.e. that its distribution $G$ is independent or nearly independent of where $\theta$ is in its parameter space. Show that

$$C(\mu) = \Pr_\theta((\widehat{\mu} - \mu)/\widehat{\kappa} \geq (\widehat{\mu}_{\mathrm{obs}} - \mu)/\widehat{\kappa}_{\mathrm{obs}}) = 1 - G((\widehat{\mu}_{\mathrm{obs}} - \mu)/\widehat{\kappa}_{\mathrm{obs}})$$

is a CD. If $G$ is not known, or too difficult to derive, use simulations, of $t^* = (\widehat{\mu}^* - \widehat{\mu}_{\mathrm{obs}})/\widehat{\kappa}^*$, via datasets simulated at position $\widehat{\theta}_{\mathrm{obs}}$. We call this a CD computed from t-bootstrapping.

**(d)** The previous recipe works well if $t = (\widehat{\mu} - \mu)/\widehat{\kappa}$ is close to pivotal, i.e. its distribution is nearly constant over the parameter region. In other cases we may take the t-bootstrapping argument one step further. Write for emphasis $\theta = (\mu, \gamma)$, perhaps in a reparametrisation, where $\mu$ is in focus and $\gamma$ is of length $p-1$. The $t$ has some distribution, depending on $\theta$, and we write $\Pr_\theta(t \leq u) = G(u, \mu, \gamma)$. Show that

$$H(\mu, \gamma) = \Pr_{\mu,\gamma}((\widehat{\mu} - \mu)/\widehat{\kappa} \geq (\widehat{\mu}_{\mathrm{obs}} - \mu)/\widehat{\kappa}_{\mathrm{obs}}) = 1 - G((\widehat{\mu}_{\mathrm{obs}} - \mu)/\widehat{\kappa}_{\mathrm{obs}}, \mu, \gamma).$$

This is not necessarily a CD, in the strict sense, as this probability may depend not only on $\mu$ but also on aspects of $\gamma$. Often the distribution of $t$ is approximately the same, though, in a neighbourhood around the true value. Argue that this leads to

$$C^*(\mu) = 1 - \widehat{G}((\widehat{\mu}_{\mathrm{obs}} - \mu)/\widehat{\kappa}_{\mathrm{obs}}, \mu) \quad \text{where } \widehat{G}(u, \mu) = G(u, \mu, \widehat{\gamma}).$$

Such an estimated distribution can be computed via bootstrapping, i.e. simulated datasets at position $\widehat{\theta}$ in the parameter space. With $B$ such simulated datasets, leading to simulated values $\widehat{\theta}^*, \widehat{\mu}^*, \widehat{\kappa}^*$, and hence $t^* = (\widehat{\mu}^* - \widehat{\mu}_{\mathrm{obs}})/\widehat{\kappa}^*$.

**Ex. 8.15** *Recipe Six: CDs in exponential families.* We have worked with the general exponential family in previous chapters, see Ex. 1.56. In particular we learned in Ex. 4.33 that there are uniformly optimal tests, for individual parameters in such models. The same holds in the present framework of CDs. Suppose data stem from a model of the form $f(y, a, b) = \exp\{aU(y) + b^{\mathrm{t}}V(y)\}h(y)$, with $U$ one-dimensional and $V$ of dimension say $p$. The optimal recipe for $a$ is

$$C^*(a) = \Pr_a\{U(Y) \geq u_{\mathrm{obs}} \,|\, V(Y) = v_{\mathrm{obs}}\}.$$

This construction is actually optimal, in a power risk function sense we come back to in Ex. 8.35, but we can already start working with this definition and see how it applies in various situations.

**(a)** Verify from arguments in Ex. 4.33 that $C^*(a)$ indeed depends only on $a$, not on $b$.

**(b)** For an illustration, consider the pair of exponentials of Ex. 4.28. To avoid confusion with the parametrisation, use now $X \sim \text{Expo}(\theta)$, $Y \sim \text{Expo}(\theta+\delta)$, with sum $Z = X+Y$. Show that the joint density is indeed of exponential form, and that the recipe leads to

$$C^*(\delta) = \Pr_\delta(Y \le y_{\text{obs}} \,|\, Z = z_{\text{obs}}) = \frac{1-\exp(-\delta y_{\text{obs}})}{1-\exp(-\delta z_{\text{obs}})}.$$

Find the positive confidence pointmass at $\delta = 0$.

**(c)** Suppose there are $m$ independent pairs of such exponentials, with $X_i \sim \text{Expo}(\theta_i)$, $Y_i \sim \text{Expo}(\theta_i+\delta)$, and sums $Z_i = X_i + Y_i$. We need a CD for the difference parameter $\delta$. Show that the joint density of the $2m$ variables is on the exponential form, and that the resulting CD must be of the form

$$C^*(\delta) = \Pr_\delta(U \le u_{\text{obs}} \,|\, Z_1 = z_{1,\text{obs}}, \ldots, z_{m,\text{obs}}),$$

with $U = \sum_{i=1}^m Y_i$. There is no clear formula for this conditional distribution, but show that $Y_i \,|\, z_i$ has density $\delta\exp(-\delta y_i)/\{1-\exp(-\delta z_i)\}$ for $y_i \in [0, z_i]$. To show how the CD can be computed, via simulations, suppose as in Ex. 4.28 that the data are the three pairs $(0.927, 0.819), (1.479, 0.408), (3.780, 1.311)$. In that exercise we worked with the optimal test for $\delta = 0$ vs. $\delta > 0$, and needed only the null distribution of $U$ given the three sums $z_1, z_2, z_3$, i.e. where $\delta = 0$. Now we need to tabulate this conditional distribution also for each $\delta > 0$, however.

**(d)**

**Ex. 8.16** *Optimal CD for a bivariate model.* (xx spell out. we do get the natural answer $C(a) = \Pr_a(\widehat{a} \ge \widehat{a}_{\text{obs}} \,|\, \widehat{b})$. then multivariate. xx)

**Ex. 8.17** *Bayesian posteriors as approximate CDs.* Consider a setup with data $y$ from a model with parameter $\theta = (\theta_1, \ldots, \theta_p)$, and with $\mu = \mu(\theta_1, \ldots, \theta_p)$ a focus parameter. A CD for $\mu$ has the property $\Pr_\theta\{C^{-1}(0.05, Y) \le \mu \le C^{-1}(0.95, Y)\} = 0.90$, etc., as with (8.1), thus delivering confidence intervals with the right coverage. This is also akin to how Bayesian posterior distributions are used. If a Bayesian prior for $\theta$ leads to a posterior for $\theta$, and hence for a cumulative $B(\mu \,|\, y_{\text{obs}})$, then the Bayesian can read off $[B^{-1}(0.05, y_{\text{obs}}) \le \mu \le B^{-1}(0.95, y_{\text{obs}})]$, as with Ex. 7.13. A question of interest and relevance also in Bayesian contexts is whether such intervals make sense also in the frequentist sense.

a Bernshteĭn–von Mises theorem for Bayesian posteriors and CDs

**(a)** For a simple start example, consider $Y_1, \ldots, Y_n$ which given $\theta$ are i.i.d. from the Pois($\theta$), and with a prior $\theta \sim \text{Gam}(a, b)$; see Ex. (xx suitable exercise Ch 6 xx). Show that the posterior cumulative for $\theta$ becomes $B_n(\theta \,|\, \text{data}) = G(\theta, a+n\bar{y}_{\text{obs}}, b+n)$, in terms for the cumulative Gamma, and with $\bar{y}_{\text{obs}}$ the observed data average. Let $\widehat{\theta}_B$ and $\widehat{\tau}_B$ be the posterior mean and standard deviation. Assume now that data $Y_1, Y_2, \ldots$ come from the Pois($\theta_0$), for a certain $\theta_0$. Show that

$$\begin{aligned} B_n(\theta_0 \,|\, Y_1, \ldots, Y_n) &= \Pr(\theta \le \theta_0 \,|\, Y_1, \ldots, Y_n) = G(\theta_0, a + n\bar{y}, b+n) \\ &\doteq \Phi((\theta_0 - \widehat{\theta}_B)/\widehat{\tau}_B) \to_d \Phi(\text{N}(0,1)) \sim \text{unif}, \end{aligned}$$

with probability 1.

**(b)** For a similar adventure, start with the Beta($a, b$) prior for a binomial probability $\theta$. Show that the posterior cumulative for $\theta$ becomes $B_n(\theta \,|\, \text{data}) = \text{Be}(\theta, a+y, b+n-y)$,

in terms of the Beta cumulative. Assuming that $Y_n$ is really from a binomial with some true $\theta_0$, show that

$$\begin{aligned} B_n(\theta_0 \,|\, Y_n) &= \Pr(\theta \le \theta_0 \,|\, Y_n) = \mathrm{Be}(\theta_0, a+Y_n, b+n-Y_n) \\ &\doteq \Phi((\theta_0 - \widehat{\theta}_B)/\widehat{\tau}_B) \to_d \Phi(\mathrm{N}(0,1)) \sim \mathrm{unif}, \end{aligned}$$

with probability 1. Discuss how these findings relate to those of Ex. 7.14.

**(c)** Consider then the general parametric situation, supposing $Y_1, \ldots, Y_n$ being i.i.d. from the density $f(y,\theta)$, with $\mu = \mu(\theta)$ a one-dimensional focus parameter. Let $\theta_0$ and $\mu_0 = \mu(\theta_0)$ denote the true parameters. As with Recipe Two, see Ex. 8.8, there is convergence to the standard normal of the standardised $\sqrt{n}(\widehat{\mu}_{\mathrm{ml}} - \mu_0)/\widehat{\kappa}$, say, with the ML estimator for $\mu$ and $\widehat{\kappa}^2$ consistently estimating $\kappa^2 = c^{\mathrm{t}} J^{-1} c$ (point back to delta method things in Ch2 xx). Now use the Bernshteĭn–von Mises theorem setup of Ex. 7.32 to explain that $\sqrt{n}(\mu - \widehat{\mu}_{\mathrm{ml}}) \,|\, \mathrm{data}$ tends in distribution to $\mathrm{N}(0, \kappa^2)$. The Bayesian posterior has a c.d.f. $B_n(\mu \,|\, \mathrm{data})$. Show that at the true value,

$$\begin{aligned} B_n(\mu_0 \,|\, \mathrm{data}) &= \Pr\big(\sqrt{n}(\mu - \widehat{\mu}_{\mathrm{ml}})/\widehat{\kappa} \le \sqrt{n}(\mu_0 - \widehat{\mu}_{\mathrm{ml}})/\widehat{\kappa} \,|\, \mathrm{data}\big) \\ &= \mu(\sqrt{n}(\mu_0 - \widehat{\mu}_{\mathrm{ml}})/\widehat{\kappa}) + \varepsilon_n, \end{aligned}$$

with $\varepsilon_n \to_{\mathrm{pr}} 0$. Deduce that $B_n(\mu)$ is a CD in the large-sample sense. In this sense we may think of any sensible Bayesian posterior distribution, in regular parametric models, as Recipe Seven for creating a CD. (xx a little more here. xx)

**Ex. 8.18** *Confidence distribution for the ratio of explained variation.* (xx to be polished. used in Story #5. xx) For the classic linear regression model $Y_i = x_i^{\mathrm{t}}\beta + \varepsilon_i$, with i.i.d. $\mathrm{N}(0,\sigma^2)$ noise terms, theory was developed in Ex. 4.38 to estimate the fraction of the variance explained via the covariates. The statistic $R^2$ given there can be seen as an estimator of the explained variation ratio $\rho = \beta^{\mathrm{t}}\Sigma_n\beta/(\beta^{\mathrm{t}}\Sigma_n\beta + \sigma^2)$, with notation from that exercise. Here we construct full CDs for such parameters.

**(a)** (xx one covariate at a time. note that $\sigma$ changes value and interpretaion, depending on whichi covariates are used in the linear regression equation. xx) $\rho = M_n\beta^2/(M_n\beta^2 + \sigma^2)$, with $M_n = (1/n)\sum_{i=1}^n (x_i - \bar{x})^2$. then $F = nM_n\widehat{\beta}^2/\widehat{\sigma}^2 \sim F(1, m, n\rho/(1-\rho))$.

**(b)** (xx then general. xx) Explain that

$$C(\rho) = \Pr_\rho(\widehat{\lambda} \ge \widehat{\lambda}_{\mathrm{obs}}) = \Pr_\rho(F \ge n\widehat{\lambda}_{\mathrm{obs}}/p) = 1 - F(n\widehat{\lambda}_{\mathrm{obs}}/p, p, m, n\rho/(1-\rho))$$

becomes a CD for $\rho$.

**(c)** (xx one more thing. xx)

### CDs for quantiles

**Ex. 8.19** *CDs for quantiles.* Let $Y_1, \ldots, Y_n$ be independent observations from a smooth density $f$, with c.d.f. $F$. How can we construct CDs for its quantiles, the $\mu_q = F^{-1}(q)$? We wish such a CD to be nonparametric, without further assumptions on the $f$. We go through the main ideas for the case of the median $\mu = F^{-1}(\frac{1}{2})$, before extending methods and results to a general quantile $q \in (0,1)$. (xx a bit more prose here; several methods; some better than others in terms of precision and coverage; we draw but briefly on density estimators, point to Ch3 exercise. nils needs to check Price and Bonett (2001, 2002). xx)

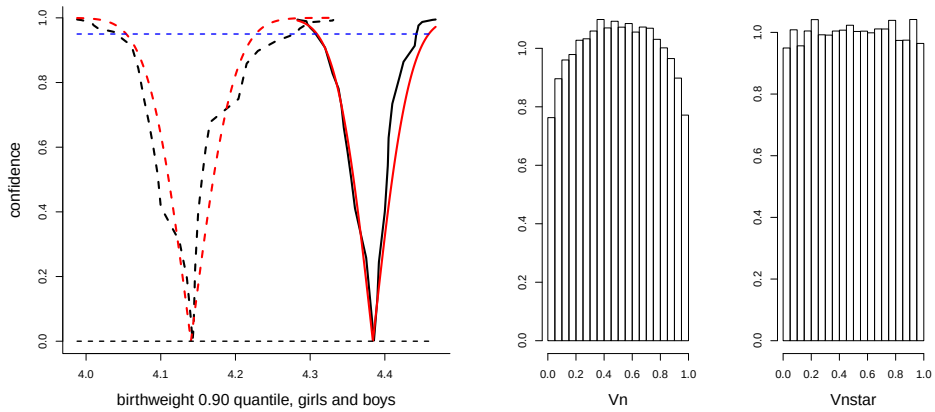



Figure 8.5: *Left panel: for Ex. 8.19, confidence curves* $\mathrm{cc}(\mu_{0.90})$*, for the 0.90 quantiles of the birthweight distributions for girls (to the left) and boys (to the right). The black curves use the Beta method* $\mathrm{cc}_n^*(\mu, \mathrm{data})$*, with linear interpolation, whereas the slanted curves use the large-sample approximation. The former yields more accurate coverage than the latter. 95 percent intervals for the two 0.90 quaniles are indicated via the blue horizontal line. Right panel: for Ex. 8.20, for the case of* $f$ *being standard normal,* $n = 100$*,* $q = 0.50$*: histograms of* $V_n$ *and* $V_n^*$ *based on* $\mathrm{sim} = 10^5$ *simulations. Also in other cases the* $V_n^*$ *is much closer to uniformity than is* $V_n$*.*

**(a)** It is not difficult to construct a first-order correct CD via large-sample results reached in Chapter 3, see in particular Ex. 3.21. With $M_n$ the sample median, we have $\sqrt{n}(M_n - \mu) \to_d \mathrm{N}(0, \frac{1}{4}/f(\mu)^2)$. Show that as long as $\widehat{\tau}$ is a consistent estimator for $f(\mu)$, then $\sqrt{n}(M_n - \mu)/(\frac{1}{2}/\widehat{\tau}) \to_d \mathrm{N}(0,1)$, and that this leads to the approximate CD

approximate CD for quantiles

$$C_n(\mu, y) = \mu\big(\sqrt{n}(\mu - M_n)/(\tfrac{1}{2}/\widehat{\tau})\big).$$

One of several choices is to take $\widehat{\tau} = \widehat{f}(M_n)$, with $\widehat{f}(y) = n^{-1}\sum_{i=1}^n h^{-1}K(h^{-1}(y_i - y))$ a kernel density estimator, with some kernel function $K$ and bandwidth $h$. The best size for this fine-tuning parameter is of the type $h = c/n^{1/5}$, as seen in Ex. 3.43, and a classic rule of thumb which we typically might resort to here is to take $h = 1.059\,\widehat{\sigma}/n^{1/2}$, with $\widehat{\sigma}$ the empirical standard deviation. Show that the confidence intervals from this CD take the form $M_n \pm z_0(\frac{1}{2}/\widehat{\tau})/\sqrt{n}$, with $z_0$ the relevant normal quantile, like 1.96 for intended 95 percent intervals.

**(b)** A different idea starts out as follows. For the ordered observations $Y_{(1)} < \cdots < Y_{(n)}$, show that

$$\Pr_f(\mu \le Y_{(i)}) = \Pr(\tfrac{1}{2} \le U_{(i)}) = 1 - \mathrm{Be}(\tfrac{1}{2}, i, n - i + 1) \quad \text{for } i = 1, \ldots, n.$$

Here $U_{(i)} = F(Y_{(i)})$; these form an ordered sample from the standard uniform, and we saw in Ex. 3.21 that they have Beta distributions. The $\mathrm{Be}(x, a, b)$ is the c.d.f. of a $\mathrm{Beta}(a, b)$. Define a full CD for $\mu$, say $C_n^*(\mu, \mathrm{data})$, via linear interpolation between the $C_n^*(y_{(i)}, \mathrm{data}) = 1 - \mathrm{Be}(\frac{1}{2}, i, n - i + 1)$ points. This also yields a confidence curve $\mathrm{cc}_n^*(\mu, \mathrm{data}) = |1 - 2\,C_n^*(\mu, \mathrm{data})|$.

**(c)** Extend the two methods above, constructed there to deal with the median, to a general quantile $\mu_q = F^{-1}(q)$. For the first CD, use $\sqrt{n}(Q_{n,q} - \mu_q) \to_d \mathrm{N}(0, q(1-q)/f(\mu_q)^2)$, with $Q_{n,q} = F_n^{-1}(q)$ the empirical $q$ quantile, and estimate $\tau_q = f(\mu_q)$ via $\widehat{f}(F_n^{-1}(q))$. For the second CD, show first that $\Pr_f(\mu_q \le Y_{(i)}) = 1 - \mathrm{Be}(q, i, n-i+1)$, and use linear interpolation:

$$C_n^*(\mu_q, \text{data}) = \text{interpolation with } 1 - \mathrm{Be}(q, i, n-i+1) \text{ at } y_{(i)}, \tag{8.2}$$

the Beta method CD for quantiles

for $i = 1, \ldots, n$. For $\mu_q$ inside $(y_{(i)}, y_{(i+1)})$, therefore, the CD value is interpolation between $1 - \mathrm{Be}(q, i, n-i+1)$ and $1 - \mathrm{Be}(\mu_1, i+1, n-i)$. We call this *the Beta method CD* for quantiles.

**(d)** (xx a bit more. for birthweights oslo boys and girls, compute, display, and interpret the confidence curves for the 0.90 quantile, using both of the CD methods. Reproduce a version of Figure 8.5. point to Story #20. xx)

**Ex. 8.20** *CDs for quantiles: how well do they work?* In Ex. 8.19 we found two nonparametric CD recipes, for any quantile $\mu_q = F^{-1}(q)$. Here we investigate how well they work, in terms of actual coverage probabilities for confidence intervals. For the methods $C_n(\mu_q, \text{data})$ and $C_n^*(\mu_q, \text{data})$, define

$$V_n = C_n(\mu_{q,\text{true}}, Y_1, \ldots, Y_n) \quad \text{and} \quad V_n^* = C_n^*(\mu_{q,\text{true}}, Y_1, \ldots, Y_n),$$

with $Y_1, \ldots, Y_n$ drawn from the density in question. Accurate coverage, at all levels, means that the distribution of these two random CDs, at the true value, should be close to the uniform.

**(a)** To check the precision of these two CDs, carry out a simple simulation experiment. Take $f$ equal to the standard normal, with $\mu_q = \Phi^{-1}(q)$ to be estimated with uncertainty; use $\widehat{\tau} = \widehat{f}(Q_{n,q})$ as above, with $K$ the standard normal kernel and bandwidth $h = 1.059\,\widehat{\sigma}/n^{1/5}$ (which is optimal for the normal case, see Ex. 3.43), using the ordinary standard deviation from the data; and then simulate say $\text{sim} = 10^5$ values of $V_n = C_n(\mu_{q,\text{true}}, Y_1, \ldots, Y_n)$ and $V_n^* = C_n^*(\mu_{q,\text{true}}, Y_1, \ldots, Y_n)$. Check, perhaps for $n = 50, 100, 500, 1000$, how close the distributions of $V_n$ and $V_n^*$ are to the uniform. – For computing the $C_n^*$, and hence for executing that part of the simulation experiment, the `approx` algorithm of `R` is handy, carrying out linear approximation between the two values $1 - \mathrm{Be}(\frac{1}{2}, i, n-i+1)$ and $1 - \mathrm{Be}(\frac{1}{2}, i+1, n-i)$, for any $\mu$ inside the $[Y_{(i)}, Y_{(i+1)}]$ interval.

**(b)** Conduct a few similar simulation experiments, to see how close $V_n$ and $V_m n^*$ are to uniformity, with different density $f$, quantile $\mu_q$, sample size $n$.

**(c)** To assess 'closeness to uniformity' more accurately, use the monitoring processes of Ex. 3.9. For each of your simulation experiments, in addition to displaying histograms of $V_n$ and $V_n^*$, compute and display the functions $Z_{\text{sim}}(t) = (\text{sim})^{1/2}\{G_{\text{sim}}(t) - t\}$ and $Z_{\text{sim}}^*(t) = (\text{sim})^{1/2}\{G_{\text{sim}}^*(t) - t\}$, where $G_{\text{sim}}$ and $G_{\text{sim}}^*$ are the empirical distribution functions of the $V_n$ and $V_n^*$. Compute also $D_{\text{sim}} = \max_t |Z_{\text{sim}}(t)|$ and $D_{\text{sim}}^* = \max_t |Z_{\text{sim}}^*(t)|$. It will transpire (i) that $V_n$ often is not particularly close to uniformity, unless $n$ is rather large; (ii) that $V_n^*$ is often so close to uniformity, even for moderate $n$ and $q$ near 0 or 1, that we cannot see that the distribution is not uniform, even with $10^5$ simulated values.

**(d)** xx

**Ex. 8.21** *Large-sample equivalence for two CDs for quantiles.* (xx to come. details for

why the two CDs are large-sample equivalent. harder to show clearly that the 2nd is better than the 1st. nils thinks that it is, though, as of 30-June-2025. xx)

### CDs in some nonregular setups

**Ex. 8.22** *The length problem.* (xx might make a Satellite Collision Story, based on Cunen et al. (2020b). contrasting CD with Bayes. xx) There are several situations, of varying degrees of complexity, where the heart of the matter is, or can be transformed to, the following: with $Y$ having the $\mathrm{N}_p(\theta, \Sigma)$ distribution, with unknown mean vector and known or partly known variance matrix, reach inference for the length $\rho = \|\theta\| = (\theta_1^2 + \cdots + \theta_p^2)^{1/2}$. See e.g. Cunen et al. (2020b) for an application involving the computation and real-time monitoring of the probability that two satellites will collide.

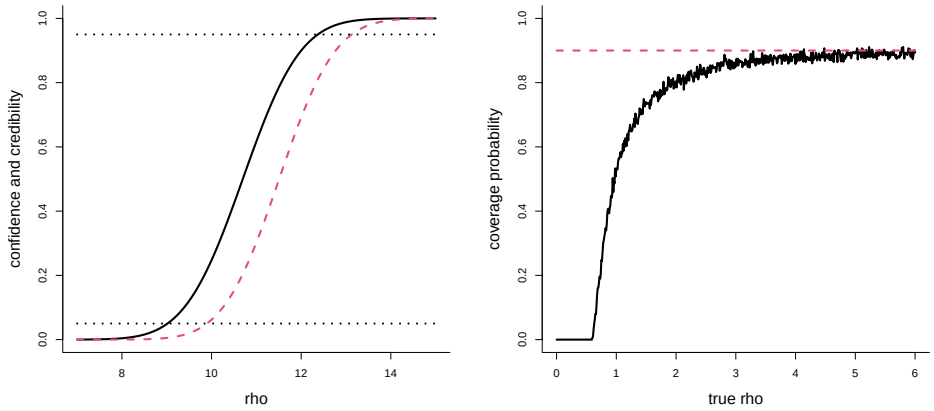



Figure 8.6: *Left panel: for dimension $p = 10$ and observed $\widehat{\rho} = 11.11$, the correct CD (full curve, with correct 90 percent confidence* $[9.02, 12.37]$*) and the Bayes cumulative posterior (slanted curve, with 90 percent credibility* $[9.89, 13.13]$*). Right panel: as a function of the true $\rho$, for dimension $p = 3$, the curve gives the actual coverage probability of the Bayesian 90 percent credibility interval, based on the posterior stemming from a flat prior for the $\theta$.*

**(a)** Take first $\Sigma = I_p$, so $y \sim \mathrm{N}_p(\theta, I_p)$, which means independent $Y_i \sim \mathrm{N}(\theta_i, 1)$ for $i = 1, \ldots, p$. Show that the ML estimator of $\rho$ is $\widehat{\rho} = \|Y\|$. Show also that $\widehat{\rho}^2 \sim \chi^2_p(\rho^2)$, the noncentral chi-squared.

**(b)** Deduce that $\widehat{\rho}^2$ is overshooting its target $\rho^2$, with mean and variance $p + \rho^2$ and $2p + 4\rho^2$. Find also an expression for $\mathrm{E}\,\widehat{\rho}$, and show that it overshoots $\rho$.

**(c)** Show that the natural CD becomes $C(\rho, y) = 1 - \Gamma_p(\widehat{\rho}^2, \rho^2)$.

this CD gives the correct coverage

**(d)** A typical Bayesian analysis would start with a flat prior for $\theta_1, \ldots, \theta_p$. Show that $\theta \,|\, y \sim \mathrm{N}_p(y, I)$, and that this entails $\rho^2 \,|\, y \sim \chi^2_p(\widehat{\rho}^2)$. For $p = 10$ and $\widehat{\rho} = 11.11$, compute and draw both the CD and the Bayesian posterior distribution,

$$C(\rho, y) = 1 - \Gamma_p(\widehat{\rho}^2, \rho^2) \quad \text{and} \quad B(\rho, y) = \Gamma_p(\rho^2, \widehat{\rho}^2),$$

constructing a version of Figure 8.6, left panel. Comment on what you find.

**(e)** Simulate to illustrate that the CD by construction works, producing confidence intervals with the correct coverage; $U_C = C(\rho_0, Y) \sim$ unif when data stem from the model, at position $\rho_0$. show however that the Bayesian posterior distribution here risks being very far from producing intervals with the right coverage; $U_B = B(\rho_0, Y)$ is very far from being uniform. Construct a version of Figure 8.6, right panel, for the too low coverage probability of the Bayesian 90 percent credibility interval; the CD based intervals have exact coverage. (xx link to Bernshteĭn–von Mises things in Ch. 7; here we're outside BvM terrain. more on why and how. xx)

**(f)** More generally, with $Y_1, \ldots, Y_n$ being i.i.d. from the $\mathrm{N}_p(\theta, \sigma^2 I_p)$, with $\sigma$ known, show first that $\bar{Y} \sim \mathrm{N}_p(\theta, (\sigma^2/n) I_p)$. Then show that $\widehat{\rho} = \|\bar{y}\|$ is the ML estimator, with distribution given by $n\widehat{\rho}^2/\sigma^2 \sim \chi^2_p(n\rho^2/\sigma^2)$. On the Bayesian side, show that a flat prior for $\theta$ leads to $\theta \,|\, \text{data} \sim \mathrm{N}_p(y, (\sigma^2/n) I_p)$. Show that these statements lead to these generalisations of the above

$$\begin{aligned} C_n(\rho, y) &= 1 - \Gamma_p(n\widehat{\rho}^2/\sigma^2, n\rho^2/\sigma^2), \\ B_n(\rho \,|\, y) &= \Gamma_p(n\rho^2/\sigma^2, (n/\sigma^2)\widehat{\rho}^2). \end{aligned}$$

**(g)** (xx a bit more, regarding BvM, which holds for fixed $p$ and $\rho$, with growing $n$. but misleading picture for finite $n$. do something to see interplay with $n$ and $p$. xx) (xx also, briefly, to the case of $Y \sim \mathrm{N}_p(\theta, \sigma^2 I_p)$, with $\sigma$ estimated via an independent $\widehat{\sigma}^2 \sim \sigma^2 \chi^2_m/m$. xx)

**Ex. 8.23** *The Fieller problem: ratio of normal means.* (xx edit and clean. xx) In various situations one meets problems related to ratios of normal means, so to speak directly or indirectly. These become difficult when the nominator is small, where e.g. delta methods do not work well. Below we construct clean confidence curves for such ratio parameters. (xx point to Stories #33 New Haven and #34 on Oslo climate. xx)

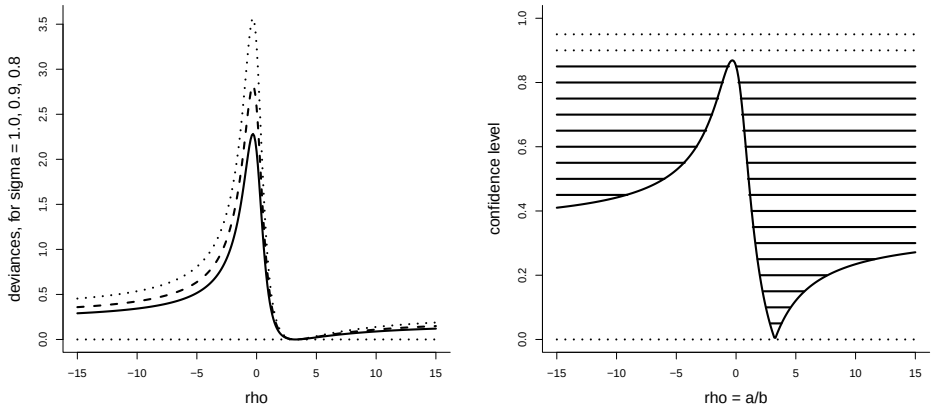



Figure 8.7: *Ratio $\rho = a/b$ of normal means, with $(\widehat{a}, \widehat{b}) = (1.444, 0.444)$ and $\widehat{\rho} = 3.252$: Left panel: the deviance function, for $\sigma$ equal to 1.0, 0.9, 0.8. Right panel: for $\sigma = 1$, the* cc$(\rho)$*. For confidence levels above 0.87, the interval is the full real line; xxxx.*

**(a)** Consider first the prototype setup for such questions, where $\widehat{a} \sim \mathrm{N}(a, 1)$ and $\widehat{b} \sim \mathrm{N}(b, 1)$ are independent, with inference required for $\rho = a/b$. Show first that the log-likelihood is a simple $\ell(a, b) = -\frac{1}{2} Q(a, b)$, with $Q(a, b) = (a - \widehat{a})^2 + (b - \widehat{b})^2$. Then derive

the following formula for the profiled version,

$$\ell_{\rm prof}(\rho) = -\tfrac12 \min\{Q(a,b)\colon a/b=\rho\} = -\tfrac12(\widehat a-\widehat b\rho)^2/(1+\rho^2),$$

and verify that the deviance function

$$D(\rho) = 2\{\ell_{\max} - \ell_{\rm prof}(\rho)\} = (\widehat a-\widehat b\rho)^2/(1+\rho^2) \quad \text{is an exact } \chi^2_1.$$

**(b)** Generalising also to the case of $\widehat a \sim {\rm N}(a,\sigma^2)$ and $\widehat b \sim {\rm N}(b,\sigma^2)$, with $\sigma$ known, explain that

$$\text{cc}(\rho) = \Gamma_1\Big(\frac{1}{\sigma^2}\frac{(\widehat a-\widehat b\rho)^2}{1+\rho^2}\Big)$$

is a clear confidence curve for the $\rho = a/b$ ratio. Construct versions of Figure 8.7, based on having observed $(\widehat a,\widehat b) = (1.444, 0.444)$, with $\widehat\rho = 3.252$. The left panel has deviance functions, for $\sigma = 1.0, 0.9, 0.8$, and the right panel displays the confidence curve $\text{cc}(\rho)$, for $\sigma = 1$. Comment on the unusual confidence regions, for different confidence levels $\alpha$: these are bounded intervals for small $\alpha$; right-sided intervals for slightly bigger $\alpha$; the union of two infinite intervals for the next levels; and equal to the full real line at level 86.9 and above. All of this is caused by $\widehat b$ being close to zero. You may play with your code to see how different values of the numerator $\widehat b$ influence the $\text{cc}(\rho)$.

**(c)** (xx fix this, perhaps with an illustration, to be placed alongside the length problem one. xx) Consider now the classic linear regression setup, where $y_i = a + b(x_i-\bar x) + \varepsilon_i$, with the $\varepsilon_i$ being i.i.d. ${\rm N}(0,\sigma^2)$; see Ex. 3.33. Where, or when, will the line reach some threshold level $c$ of interest? Argue that the answer is at $x_0 = \bar x + (c-a)/b$, for which the point estimate is $\widehat x_0 = \bar x + (c-\widehat a)/\widehat b$, in terms of the usual estimators $(\widehat a,\widehat b)$. With $M_n = \sum_{i=1}^n (x_i-\bar x)^2$, show that

$$Z_n(x_0) = \frac{\widehat a + \widehat b(x_0-\bar x) - c}{\widehat\sigma(1/n + (x_0-\bar x)^2/M_n)^{1/2}} \sim t_{\rm df}, \quad \text{with df} = n-2.$$

Use this to show that

$$\text{cc}(x_0) = F_{1,\rm df}(Z_n(x_0)^2) = F_{1,\rm df}\Big(\frac{\{\widehat a+\widehat b(x_0-\bar x)-c\}^2}{\widehat\sigma^2(1/n+(x_0-\bar x)^2/M_n)}\Big)$$

is a genuine confidence curve for the $x_0$ parameter. For $|x_0|$ becoming large, show that $\text{cc}(x_0) \to F_{1,\rm df}(M_n\widehat b^2/\widehat\sigma^2)$. Explain that this is $1-p$, with the p-value for testing $b=0$ vs. $b\neq 0$. When $t = M_n^{1/2}\widehat b/\widehat\sigma$ is not large, therefore, the $\text{cc}(x_0)$ is bounded below 1, with the implication that confidence intervals include $\pm\infty$. Discuss this finding.

**(d)** (xx bioassay. point to application. xx) When is $a+bx \geq c$, in other regression setups? Assume $(\widehat a,\widehat b)$ is approximately a binormal, with means $(a,b)$ and variance matrix estimated at $\Sigma$. The target is again $x_0 = (c-a)/b$. Explain that

$$Z(x_0) = \frac{\widehat a + \widehat b x_0 - c}{\{(1,x_0)\Sigma(1,x_0)^{\rm t}\}^{1/2}} \quad \text{is approximately a standard normal,}$$

leading to the confidence curve $\text{cc}(x_0) = \Gamma_1(Z(x_0)^2)$.

**Ex. 8.24** *CDs and posterior distributions with boundary constraints.* Here we learn about construction of CDs when there is a boundary condition on the focus parameter.

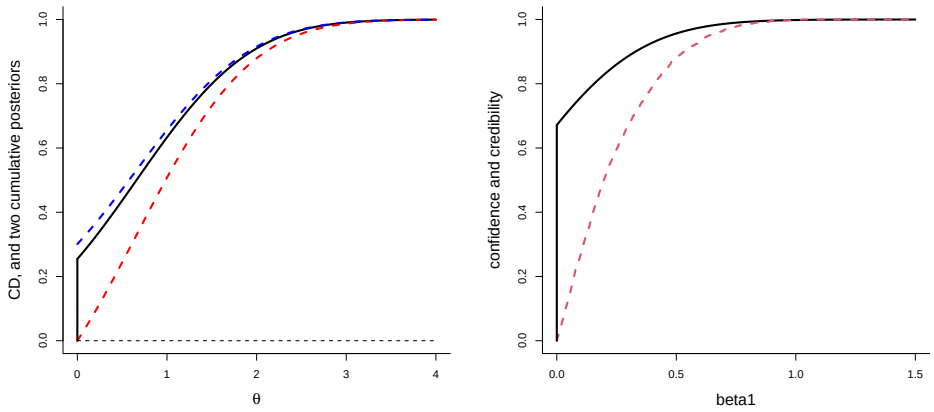



Figure 8.8: *Left panel: for Ex. 8.24, with* $y_{\text{obs}} = 0.66$ *for the* $\mathrm{N}(\theta, 1)$ *model, the black curve is the natural CD, with positive point mass 0.255 at zero. The red and the blue curves are Bayesian posterior distributions, for the flat prior on the halfline, and for the mixture prior with* $\frac{1}{2}$ *at zero and* $\frac{1}{2}$ *flat on the halfline, respectively. Right panel: for the linear regression setup of Ex. 8.25, with constraint* $\beta_1 \geq 0$*, the CD (full line) and the Bayes cumulative posterior (slanted line), with flat priors on* $\beta_0, \beta_1, \log \sigma$*.*

This is sometimes an easy task, involving a natural positive post-data probability on the boundary point. We also compare with Bayesian procedures. Matters may of course be extended and generalised in several directions here, but for simplicity and conciseness we study a very simple prototype situation: $y$ is $\mathrm{N}(\theta, 1)$, and $\theta \geq 0$ a priori.

**(a)** Before we come to the parameter constraint, we deal with the more normal situation where there is no a priori constraint. The classical CD is then $C(\theta, y) = \Phi(\theta - y)$. Show that the Bayesian starting with a flat prior for $\theta$ finds the posterior distribution $\theta \,|\, y \sim \mathrm{N}(y, 1)$, with cumulative $B(\theta \,|\, y) = \Phi(\theta - y)$, i.e. identical to the canonical CD. – The point below will partly be that this is *not* the same for the constrained problem.

**(b)** For the remaining points here, assume indeed that $\theta \geq 0$ a priori. Argue that the canonical CD should be $C(\theta, y) = \Phi(\theta - y)$ for $\theta \geq 0$. Its point mass at zero is $\Phi(-y)$. Graph the CD, for the three cases $y_{\text{obs}}$ equal to $-0.22, 0.66, 1.99$.

**(c)** One Bayesian approach in this situation, where $\theta \geq 0$ a priori, is to let $\theta$ be flat on $[0, \infty)$. Show that then

$$\theta \,|\, y \sim \frac{\phi(\theta - y)}{\int_0^\infty \phi(\theta - y)\,\mathrm{d}\theta} = \frac{\phi(\theta - y)}{\Phi(y)} \quad \text{for } \theta \geq 0,$$

and that the cumulative posterior distribution becomes

$$B(\theta \,|\, y) = \frac{\Phi(\theta - y) - \Phi(-y)}{1 - \Phi(-y)} = \frac{\Phi(\theta - y) - \Phi(-y)}{\Phi(y)} \quad \text{for } \theta \geq 0.$$

For the three cases of $y_{\text{obs}}$ given above, graph the CD along with the Bayesian $B(\theta \,|\, y_{\text{obs}})$, and comment on what you find.

**(d)** In general terms, for the case of $y \,|\, \theta \sim \mathrm{N}(\theta, 1)$, let $\theta$ have the mixture prior distribution $p_0\pi_0 + p_1\pi_1$, with the sub-priors $\pi_0$ and $\pi_1$ having their individual posteriors

$\pi_0(\theta\,|\,y)$ and $\pi_1(\theta\,|\,y)$. Show that the posterior has a natural mixture form,

$$\theta\,|\,y \sim p_0^*(y)\pi_0(\theta\,|\,y) + p_1^*(y)\pi_1(\theta\,|\,y),$$

where

$$p_0(y) = \frac{p_0 f_0(y)}{p_0 f_0(y) + p_1 f_1(y)} \quad \text{and} \quad p_1(y) = \frac{p_1 f_1(y)}{p_0 f_0(y) + p_1 f_1(y)},$$

and with $f_0(y) = \int \phi(y-\theta)\pi_0(\theta)\,\mathrm{d}\theta$ and $f_1(y) = \int \phi(y-\theta)\pi_1(\theta)\,\mathrm{d}\theta$ the marginal densities following from the two priors. (This structure generalises to general mixture priors in general models, though that does not concern us just now.)

**(e)** For the prior $p_0\pi_0 + p_1\pi_1$, with $\pi_0$ a unit pointmass at zero and $\pi_1$ a flat prior on the halfline, show that $f_0(y) = \phi(y)$ and $f_1(y) = \Phi(y)$. With a 50-50 mixture, show hence that

$$p_0(y) = \frac{\phi(y)}{\phi(y) + \Phi(y)} \quad \text{and} \quad p_1(y) = \frac{\Phi(y)}{\phi(y) + \Phi(y)}.$$

Draw curves of these two posterior probabilities, one for the zero-point and the other for the halfline-based part, as $y$ goes from say $-5$ to 5. Show that the posterior cumulative distribution becomes $B^*(\theta\,|\,y) = p_0(y) + p_1(y)B(\theta\,|\,y)$ for $\theta \geq 0$. In particular, there is a pointmass $p_0(y)$ at zero. Construct a version of Figure 8.8.

**(f)** Show that there is no choice of $(p_0, p_1)$ which makes the Bayesian cumulative posterior $B^*(\theta\,|\,y)$ agree with the CD $C(\theta, y)$. Devise a method for selection $(p_0, p_1)$ such that the distance between $B^*(\theta\,|\,y)$ and $C(\theta, y)$ is small, for a relevant range of $\theta$ and possible observed $y_{\rm obs}$.

**(g)** Generalise the formulae above to the case of $y_1, \ldots, y_n$ i.i.d. $\mathrm{N}(\theta, \sigma^2)$, with known $\sigma$.

**Ex. 8.25** *CDs for regression parameters with boundary constraints.* The setup and insights of Ex. 8.24 carry over to other models, when there are constraints on one or more parameters. To illustrate this, consider the ordinary linear regression model with $Y_i = \beta_0 + \beta_1(x_i - \bar{x}) + \varepsilon_i$ for $i = 1, \ldots, n$, with the $\varepsilon_i$ being i.i.d. $\mathrm{N}(0, \sigma^2)$. Analysis methods for this model are classic (xx point Ch3 xx), as long as there are no constraints on the parameters. Suppose however that it is a priori known that $\beta_1 \geq 0$.

**(a)** From $\widehat{\beta}_1 \sim \mathrm{N}(\beta_1, \sigma^2/M_n)$, with $M_n = \sum_{i=1}^n (x_i - \bar{x})^2$, show that $C_n(\beta_1) = G_{\rm df}(M_n^{1/2}(\beta_1 - \widehat{\beta}_1)/\widehat{\sigma})$ is a proper CD, in that $C_n(\beta_1, \text{data}) \sim$ unif, for each $\beta_1 \geq 0$. Here $G_{\rm df}$ is the c.d.f. for the $t_{\rm df}$, and $\mathrm{df} = n - 2$. The point-mass at zero is $G_{\rm df}(-M_n^{1/2}\widehat{\beta}_1/\widehat{\sigma})$. To illustrate this, we have simulated $(x_i, y_i)$ from a setup with normal data and $(\beta_0, \beta_1, \sigma) = (1, 0, 1)$, yielding uncorrelated binormal data, with estimates $(1.042, -0.173, 0.994)$, and $M_n = 6.520$. From this, compute and display this CD, as with Figure 8.8, right panel.

**(b)** The slanted curve in that figure's right panel is the c.d.f. for a Bayesian's $\beta_1\,|\,\text{data}$ distribution. Specifically, take flat priors on $(\beta_0, \beta_1, \log\sigma)$, actively using the constraint $\beta_1 \geq 0$, which requires MCMC tools from Ch. 7. Complete a construction of the figure in question along these lines. We learn that the Bayes solution is not a good one here, not quite able to discover that $\beta_1$ is close to zero (or actually zero, as here). The 90 percent credibility interval is $[0.019, 0.636]$, compared to the correct 90 percent confidence interval $[0, 0.329]$, and with a clear chance of $C(0) = 0.671$ of $\beta_1$ being zero. (xx could ask for a better prior, $\frac{1}{2}$-$\frac{1}{2}$ for zero or a flat prior, will do better. xx)

**(c)** The same phenomenon surfaces in more general regression setups. In multiple linear regression $y_i = x_i^{\mathrm{t}}\beta + \varepsilon_i$, with a constraint of the type $\beta_j \geq 0$, for a particular $j$, give the CD for that parameter. (xx round off. point to Story #62. connect to theme of Ch. 9. xx)

**Ex. 8.26** *CDs in the truncated exponential model.* Here we consider a model sometimes called the truncated exponential model. We start with its simplest form, with data $Y_1, \ldots, Y_n$ i.i.d. from the density $\exp\{-(y-a)\}$ for $y \geq a$. The $a$ is the unknown start point for the distribution.

**(a)** Show that the ML estimator is equal to $U_n = \min_{i \leq n} Y_i$, the smallest data point. Show that $n(U_n - a)$ has a unit exponential distribution. Build from this a natural CD for $a$.

**(b)** Construct a predictive CD for the next sample point $Y_{n+1}$. Illustrate by computing and displaying the confidence curve for the text sample point, after having observed the six data points `3.735, 3.338, 10.634, 3.839, 5.667, 5.808`.

**(c)** Then consider the more realistic two-parameter version of the model, with density

$$f(y_i, a, b) = (1/b)\exp\{-(y_i - a)/b\} \quad \text{for } y_i \geq a,$$

with $a$ being the unknown start-point and $b$ a scale parameter. Show that the ML estimators become $\widehat{a} = U_n$ and $\widehat{b} = (1/n)\sum_{i=1}^n (Y_i - U_n)$, again with $U_n$ being the smallest observation.

**(d)** Construct accurate CDs and confidence curves for $a$, for $b$, and for the next datapoint $Y_{n+1}$. If some of your formulae cannot be given very explicit mathematical forms, this is ok, as long as numerical solutions can be found via numerical integration or simulation. Give approximations for these CDs for large sample sizes $n$.

**(e)** Ignoring these large-sample approximations, compute and display confidence curves for $a$, $b$, $Y_{n+1}$ with the simple $n = 6$ dataset above.

**Ex. 8.27** *CD inference for the exponential rate, with censored data.* The lifelength distribution for a certain type of technical components is considered exponential, i.e. with density $\theta\exp(-\theta t)$ for $t > 0$, on a priori grounds. To arrive at a point estimate and a confidence curve for $\theta$, the firm producing these components sets in motion the simple experiment where $n$ such items are set to work, under controlled natural conditions. One cannot wait until all components have died out, however, and the firm needs to report what can be said about the lifelength distribution, via $\theta$, a certain time $t_0$ after project start.

**(a)** With data of the form observed $t_i$ for the $N$ of the items which have died within $t_0$, and the information $t_i > t_0$ for the $n - N$ which are still alive and well, show that the combined likelihood function may be expressed as

$$\theta^N \exp\Big[-\theta\Big\{\sum_{t_i \leq t_0} t_i + (n-N)t_0\Big\}\Big].$$

**(b)** Show that the ML estimator is

$$\widehat{\theta} = N/R = N/\Big\{\sum_{t_i \leq t_0} t_i + (n-N)t_0\Big\}.$$

With increasing sample size, and fixed $t_0$, find expressions for the probability limits of $N/n$ and $R/n$, and show that $\widehat{\theta}$ is consistent.

**(c)** Show in fact that there is a limiting normal distribution here, with $\sqrt{n}(\widehat{\theta}-\theta) \to_d \mathrm{N}(0, \tau(t_0,\theta)^2)$, and attempt to find an explicit (though not necessarily quick and simple) formula for the limit variance.

**(d)** Explain why the construction $C_n(\theta) = \Pr_\theta(\widehat{\theta} \geq \widehat{\theta}_{\rm obs})$ yields a CD, and also how it can be computed in practice.

**(e)** Suppose the experiment described involves $n = 20$ such items, and that the life-lengths for the $N = 11$ of these that conk out before the deadline of $t_0 = 2.00$ years are

`0.528 0.743 0.869 1.180 0.602 0.133 0.327 1.115 0.117 0.208 1.808`

Compute and display perhaps as many as three (exact or approximate) confidence curves for $\theta$, for this little experiment: the one described in (c); one based on the normal approximation to the distribution of the ML estimator; and a t-bootstrap based version. Comment on your findings.

**Ex. 8.28** *Estimating $n$ based on observing the first $r$.* Suppose $Y_1, \ldots, Y_n$ are i.i.d., from some known distribution with density $f$ and cumulative $F$, but that one only observes the first $r$ order statistics, $Y_{(1)} < \cdots < Y_{(r)}$. Can we estimate $n$? Such nonstandard problems turn up in various context, from estimating the size of a vocabulary to the number of unseen species. In this exercise we consider the special case of the unit exponential distribution, where the $Y_i$ can be seen as waiting times, so the question may be phrased as how long time do we need to wait, until we've seen all items, when we have used a certain time to observe the first $r$.

**(a)** Let then $Y_1, \ldots, Y_n$ be i.i.d. from the unit exponential, and assume $Y_{(1)} < \cdots < Y_{(r)}$ are observed, with unknown $n$. Observing these $r$ first data points is equivalent to observing the spacings $D_1 = Y_{(1)}$, $D_2 = Y_{(2)} - Y_{(1)}$, up to $D_r = Y_{(r)} - Y_{(r-1)}$. Use Ex. 1.15 to show that the joint distribution of these $f$ first spacings may be written

$$\begin{aligned} g_r(d_1,\ldots,d_r) &= n(n-1)\cdots(n-r+1) \\ &\quad \exp\big[-\{n(d_1+\cdots+d_r) - d_2 - 2d_3 - \cdots (r-1)d_r\}\big], \end{aligned}$$

and deduce from this that $Y_{(r)}$ is sufficient for $n$.

**(b)** With $F(x) = 1-\exp(-x)$, show that $F(Y_{(r)})$ has a Beta distribution with parameters $(r, n-r+1)$.

**(c)** Show that the optimal CD for $n$, based on having observed the smallest $r$ datapoints, is

$$C_r(n) = \Pr_n(Y_{(r)} \leq Y_{(r),\rm obs}) = \mathrm{Be}(F(Y_{(r),\rm obs}), r, n-r+1).$$

**(d)** (xx an example or two. suppose $Y_{(r),\rm obs} = 0.348$ with $r = 33$. estimate $n$. see nils com87a or thereabouts. give normal approximation. but these are not good for $r/n$ close to zero or one. can we characterise ML estimator. xx)

**Ex. 8.29** *(xx another discrete model thing. xx)* (xx to come. xx)

### Prediction with confidence

**Ex. 8.30** *What's the next data point?* The confidence distributions and curves worked with so far have been for focus parameters in various statistical models. The concepts carry over to unknown or not-yet-observed random variables, however. When $Y_0$ is to be predicted, based on data $Y$ and a relevant model, we say that $C(y_0, y)$ is a prediction

confidence disribution for $Y_0$, provided (i) $C(y_0, y_{\rm obs})$ is a c.d.f., for any dataset $y_{\rm obs}$, and (ii) $C(Y_0, Y) \sim$ unif. Analogous comments and definitions apply to the prediction confidene curve $\mathrm{cc}(y_0, y_{\rm obs})$, whether such a cc is constructed directly or via $\mathrm{cc}(y_0, y_{\rm obs}) = |1 - 2\,C(y_0, y_{\rm obs})|$.

prediction CDs

**(a)** Suppose $Y_1, \ldots, Y_n, Y_{n+1}$ are i.i.d. from the normal $(\xi, \sigma^2)$, with the first $n$ observed but not yet the $Y_{n+1}$ of tomorrow. Assume first that $\sigma$ is known. From $Y_{n+1} - \bar{Y} \sim \mathrm{N}(0, \sigma^2(1 + 1/n)$, show that $C(y_{n+1}, y_{\rm obs}) = \Phi(\{n/(n+1)\}^{1/2}(y_{n+1} - \bar{y}_{\rm obs})/\sigma)$ is a prediction CD for $Y_{n+1}$. When $\sigma$ is not known, and needs to be estimated via the familiar empirical standard deviation, show that

$$C(y_0, \mathrm{data}) = G_m(\{n/(n+1)\}^{1/2}(y - \bar{y}_{\rm obs})/\widehat{\sigma})$$

is a prediction CD, with $G_m$ being the t c.d.f., here with $m = n - 1$ degrees of freedom. Explain that for $n$ large, the prediction CD becomes close to the ideal $\Phi((y - \xi)/\sigma)$; one cannot do better than this. Tomorrow's randomness does not disappear, even when the parameters of the model are perfectly known.

**(b)** Suppose $Y_1, \ldots, Y_n$ have been observed, i.i.d. from $\mathrm{Expo}(\theta)$, with ML estimator $\widehat{\theta} = 1/\bar{Y}$. What will the next $Y'$ be like, and more generally the next $T_m = Y_1' + \cdots + Y_m'$, when these new $Y_j'$ are from the same distribution? Using $2\theta Y_i \sim \chi^2_2$, show that

$$\widehat{\theta}\, T_m \sim \theta \frac{2n}{\chi^2_{2n}} \frac{m}{\theta} \frac{\chi^2_{2m}}{2m} \sim m\, F_{2n,2m},$$

in terms for the F with degrees of freedom $(2n, 2m)$, and construct a prediction CD for $T_m$ based on this. An illustration is given in Story #43.

**(c)** Consider $Y_1, \ldots, Y_{n+1}$ being i.i.d. from the uniform on $[0, \theta]$. Find a prediction CD for tomorrow's $Y_{n+1}$, based on $M_n = \max_{i \le n} Y_i$.

**Ex. 8.31** *The next binomial.*

**Ex. 8.32** *What is $Y$, for a new $x$?* Consider the linear regression model, with $Y_i = a + b(x_i - \bar{x}) + \varepsilon_i$ for $i = 1, \ldots, n$, with the $\varepsilon_i$ being i.i.d. $\mathrm{N}(0, \sigma^2)$. For a new $x_0$, perhaps for an individual not part of the dataset, what will $Y_0$ be? From $Y_0 = a + b(x_0 - \bar{x}) + \varepsilon_0$, show that

$$t(Y_0, \mathrm{data}) = \frac{Y_0 - \widehat{a} - \widehat{b}(x_0 - \bar{x})}{\widehat{\sigma}\{1 + 1/n + (x_0 - \bar{x})^2/M_n\}^{1/2}} \sim t_{n-2},$$

and explain that $C(y_0, \mathrm{data}) = G_{n-2}(t(y_0, \mathrm{data}))$ becomes a prediction CD for $Y_0$. (xx a bit more. point to illustration. easiest to predict for $x_0$ in the vicinity of $\bar{x}$. xx)

**(a)** For the general linear regression setup, with $Y_i = x_i^{\rm t}\beta + \varepsilon_i$ for $i = 1, \ldots, n$, for covariate vectors $x_i$ of length $p$, show that

$$C(y_0, x_0, \mathrm{data}) = G_m\Big(\frac{y_0 - x_0^{\rm t}\widehat{\beta}}{\widehat{\sigma}(1 + x_0^{\rm t}\Sigma_n^{-1}x_0/n)^{1/2}}\Big)$$

is a prediction CD for $Y_0$, with $\Sigma_n = n^{-1}\sum_{i=1}^n x_i x_i^{\rm t}$ and degrees of freedom $m = n - p$.

**(b)** (xx round off, point to some story. xx)

**Ex. 8.33** *Prediction CD vs. Bayes prediction.* (xx well, to be cleaned. xx)

$$\bar{f}_B(y) = \int f(y, \theta)\pi(\theta \,|\, \mathrm{data}), \quad \bar{f}(y) = \int f(y, \theta)\, \mathrm{d}C(\theta \,|\, \mathrm{data}),$$

with the appropriate $C(\theta \,|\, \mathrm{data})$ .

### Risk functions for CDs

**Ex. 8.34** *Risk functions for CDs.* This exercise looks into risk functions for and hence comparisons between CDs, in simple prototype situations where calculations are easier than for general cases. We start out with $Y_1, \ldots, Y_n$ being i.i.d. from the $\mathrm{N}(\theta, 1)$ model. For a CD $C_n(\theta, y)$, where $y$ denotes the full dataset, the risk function used is

$$\mathrm{risk}_n(C_n, \theta) = \mathrm{E}_\theta \int (\theta' - \theta)^2 \, \mathrm{d}C_n(\theta', Y) = \mathrm{E}_\theta(\theta_{\mathrm{cd}} - \theta)^2,$$

where $\theta_{\mathrm{cd}}$ is the result of a two-stage random process: data $Y$ lead to the CD $C_n(\theta, Y)$, and then $\theta_{\mathrm{cd}}$ is drawn from this distribution.

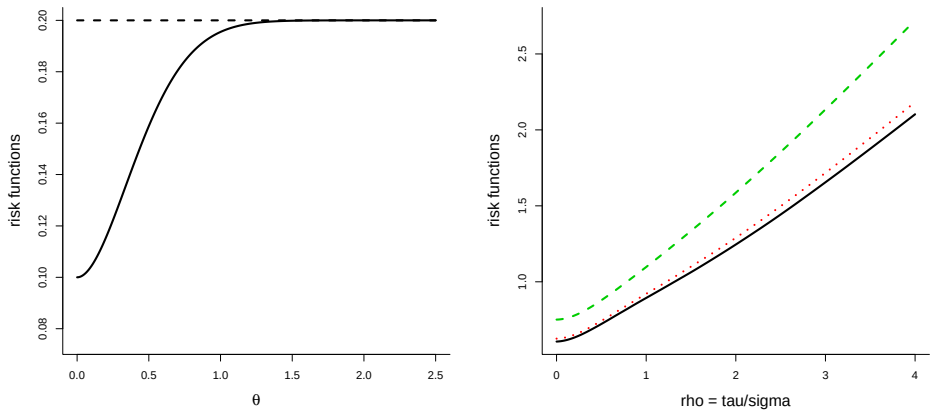



Figure 8.9: *Left panel: for Ex. 8.34, risk function for the CD, in the setup with $Y_1, \ldots, Y_n$ being i.i.d. from $\mathrm{N}(\theta, 1)$, with $n = 10$, but with the restriction $\theta \geq 0$. It starts out at $1/n$ and then grows to the $2/n$ risk of the unrestricted case as $\theta$ grows. Right panel: for Ex. 8.36, for the variance component model, with $p = 4$ and $\sigma = 1$, risk functions $r(C, \tau)$ for three CDs for $\tau$. The one based on $Z = \sum_{i=1}^p Y_i^2$ is best, closely followed by the one using $A = \sum_{i=1}^p |Y_i|$, whereas the one using the range $R = \max Y_i - \min Y_i$ does worse.*

**(a)** Show that the natural CD based on the observed sample mean $\bar{y}_{\mathrm{obs}} = n^{-1} \sum_{i=1}^n y_i$ is $C_n(\theta, y_{\mathrm{obs}}) = \Phi(\sqrt{n}(\theta - \bar{y}_{\mathrm{obs}}))$. Prove that its risk function is $\mathrm{risk}_n(C_n, \theta) = 2/n$.

**(b)** More generally, assume $\theta^*$ is some unbiased estimator of $\theta$, with finite variance $\tau_n^2$, with the property that $\widehat{\theta}^* - \theta$ has a distribution $H_n$ symmetric around zero. Show that the associated CD becomes $C_n^*(\theta, y_{\mathrm{obs}}) = H_n(\theta - \theta^*_{\mathrm{obs}})$, and show that its risk function becomes $2\tau_n^2$. The case of $\bar{Y}$ corresponds to $2/n$. Find the risk function for the case of the median based CD, with say $n = 10$, as for Figure 8.9, left panel.

**(c)** (xx fix this: Relate the above results to the optimality theorem for CDs, in certain situations, from CLP's Chapter 5. xx)

**(d)** Now we change gears a bit, by putting the a priori assumption $\theta \geq 0$ on the table. Show that the ML estimator becomes $\widehat{\theta} = \max(0, \bar{y})$, i.e. the sample mean truncated, if necessary, to zero. Argue that this leads to the natural CD

$$\widetilde{C}_n(\theta, y) = \Phi(\sqrt{n}(\theta - \bar{y}_{\mathrm{obs}})), \quad \text{for } \theta \geq 0,$$

in particular having a positive point-mass at zero.

**(e)** With $\theta_{\rm cd}$ drawn from this CD, for given data, show that it may be expressed as $\max(0, \bar{y} + N/\sqrt{n})$, with $N$ a standard normal. Show next that in the two-stage setup, with random data followed by $\theta_{\rm cd}$ drawn from the $\widetilde{C}_n$ CD, we have $\theta_{\rm cd} - \theta = \max(0, \theta + (N + N')/\sqrt{n}) - \theta$, with $N'$ another and independent standard normal. Use this to show that the $\text{risk}_n(\widetilde{C}_n, \theta)$ can be expressed as

$$\begin{aligned} r_n &= \int \big[\{\max(0, \theta + (2/n)^{1/2}x)\} - \theta\big]^2 \phi(x)\,{\rm d}x \\ &= \theta^2 \Phi(-(\tfrac{1}{2}n)^{1/2}\theta) + (2/n)\{-(\tfrac{1}{2}n)^{1/2}\theta\phi((\tfrac{1}{2}n)^{1/2}\theta) + 1 - \Phi(-(\tfrac{1}{2}n)^{1/2}\theta)\}. \end{aligned}$$

Compute and display the risk functions for $\widetilde{C}_n$ and $C_n$, for say $n = 10$, constructing a version of Figure 8.9, left panel. Comment on what we learn from this.

**(f)** (xx not fully sure about this one. xx) There are various other estimators and CDs worth considering in this $\theta \geq 0$ setting. To simplify matters, take $n = 1$, and consider the Bayes estimator $\widehat{\theta}_B$, the conditional mean of $\theta \,|\, y$, with a flat prior on $(0, \infty)$. Show in fact that $\widehat{\theta}_B = y + \phi(y)/\Phi(y)$, and verify that this is positive even when $y$ is negative. Work out an expression for the naturally associated CD $C_B(\theta) = \Pr_\theta(\widehat{\theta}_B \geq \widehat{\theta}_{B,\rm obs})$, and comment.

**Ex. 8.35** *Optimal CDs in exponential family models.* (xx relate to Ex. 8.15. spell out NP things to demonstrate optimality. xx)

**Ex. 8.36** *Risk functions for three CDs in a variance components model.* Consider the simple variance component model with independent observations $y_i \sim \text{N}(0, \sigma^2 + \tau^2)$ for $i = 1, \ldots, p$, with $\sigma$ known and $\tau$ the unknown parameter of interest; see Schweder and Hjort (2016, Example 4.1 and Exercise 5.8). The aim here is first to construct CDs based on (i) $Z = \sum_{i=1}^p y_i^2$, (ii) $A = \sum_{i=1}^p |y_i|$, and (iii) the range $R = \max y_i - \min y_i$; and then to compute and compare their risk functions. These are defined as

$$\text{risk}(C, \tau) = \text{E}_\tau\, |\tau_{\rm cd} - \tau| = \text{E}_\tau \int |\tau_{\rm cd} - \tau|\,{\rm d}C(\tau_{\rm cd}, Y),$$

with $\tau_{\rm cd}$ a random draw from the $C(\tau, Y)$ distribution, and with $Y$ itself denoting a dataset drawn from the distribution indexed by $\tau$.

**(a)** Show that the natural CDs, based on $Z$, $A$, $R$ respectively, are

$$\begin{aligned} C_Z(\tau, \text{data}) &= 1 - \Gamma_p(Z_{\rm obs}/(\sigma^2 + \tau^2)), \\ C_A(\tau, \text{data}) &= 1 - G_p(A_{\rm obs}/(\sigma^2 + \tau^2)^{1/2}), \\ C_R(\tau, \text{data}) &= 1 - H_p(R_{\rm obs}/(\sigma^2 + \tau^2)^{1/2}). \end{aligned}$$

Here $\Gamma_p$ is the cumulative distribution function of $Z_0 = \sum_{i=1}^p N_i^2$, with the $N_i$ being i.i.d. and standard normal, which means $Z_0 \sim \chi^2_p$. Similarly, $G_p$ and $H_p$ are the cumulative distribution functions of $A_0 = \sum_{i=1}^p |N_i|$ and of $R_0 = \max N_i - \min N_i$, respectively.

**(b)** Show that a random draw $\tau_{\rm cd}$ from the first of these, i.e. $C_Z$, for a given dataset, can be represented as $\tau_{\rm cd} = (Z_{\rm obs}/K - \sigma^2)_+^{1/2}$, where $x_+$ is notation for the truncated-to-zero quantity $\max(x, 0)$, and where $K \sim \chi^2_p$. In the situation where data are random, from the model at position $\tau$, deduce that

$$\tau_{\rm cd} - \tau = \{(\sigma^2 + \tau^2)K_0/K - \sigma^2\}_+^{1/2} - \tau = \sigma\big[\{(1 + \rho^2)K_0/K - 1\}_+^{1/2} - \rho\big],$$

where $\rho = \tau/\sigma$, and $K_0, K$ are two independent draws from the $\chi^2_p$. In other words, $F = K_0/K \sim F_{p,p}$, a $F$ distribution with degrees of freedom $(p, p)$. Use this to compute

the risk function risk$(C_Z, \tau)$, for $p = 4$ and $\sigma = 1$; this is the lowest of the three risk functions of Figure 8.9, right panel.

**(c)** Then consider the $C_A$ option. Show that a random draw from an observed $C_A(\tau, \text{data})$ can be written $\tau_{\text{cd}} = \{(A_{\text{obs}}/A)^2 - \sigma^2\}_+^{1/2}$. Deduce that for random data behind the CD, we have the representation

$$\tau_{\text{cd}} - \tau = \{(\sigma^2 + \tau^2)(A_0/A)^2 - \sigma^2\}_+^{1/2} - \tau = \sigma\big[\{(1+\rho^2)(A_0/A)^2 - 1\}_+^{1/2} - \rho\big],$$

with $A$ and $A_0$ two independent draws from the $G_p$ distribution. Use this to compute risk$(C_A, \tau)$. There is no simple expression for the density of $A_0/A$, so use simulation.

**(d)** Carry out similar analysis for the third CD, based on the range $R$. Construct a version of Figure 8.9, right panel.

**(e)** Use your programme to explore the three risk functions for other values of $p$.

### Meta-analysis, combining confidence distributions

(xx need more sorting and polish. xx)

**Ex. 8.37** *CD and cc for binomial probabilities.* Suppose $y$ is observed from a binomial $(n, \theta)$. The task is to construct a CD and a cc for $\theta$.

**(a)** Show that the standard normal approximations for $y$ (xx give pointer here to large-sample chapter) lead to

$$C_a(\theta, y) = \Phi\Big(\frac{n\theta - y}{\{n\theta(1-\theta)\}^{1/2}}\Big) \quad \text{and} \quad C_b(\theta, y) = \Phi\Big(\frac{\sqrt{n}(\theta - \widehat{\theta})}{\{\widehat{\theta}(1-\widehat{\theta})\}^{1/2}}\Big),$$

with $\widehat{\theta} = y/n$ the standard estimator for $\theta$.

**(b)** The recipe of Ex. 8.7 does not quite work here since $y$ has a discrete distribution. This invites the half-correction method

$$C(\theta, y) = \Pr_\theta(y > y_{\text{obs}}) + \tfrac{1}{2}\Pr_\theta(y = y_{\text{obs}}).$$

For say $n = 20$ and $y = 12$, compute and display this CD, along with (i) the same CD but without the half-correction, (ii) the two simple normal approximations above. Try other combinations of $(n, y)$, and demonstrate that they are approximately equal for moderate to large $n$.

**(c)** To investigate the basic CD property, take $n = 20$ and $\theta_{\text{true}} = 0.33$. Simulate a large number of $C(\theta_0, y)$, $C_a(\theta_0, y)$, $C_b(\theta_0, y)$, to check for their approximate uniform distribution. Try other values of $(n, \theta_0)$, and summarise your findings.

**Ex. 8.38** *CD and cc for comparing binomials.* In Ex. 8.37 we learn how to construct CDs for separate binomial parameters. Consider now the $2 \times 2$ table setup with two binomials, as with Ex. 4.31, say $y_0 \sim \text{binom}(m_0, p_0)$ and $y_1 \sim \text{binom}(m_1, p_1)$. How do we reach precise inference for the extent to which $p_0$ and $p_1$ differ?

**(a)** We first use the logistic transform $p_0 = H(\theta_0)$ and $p_1 = H(\theta_0 + \gamma)$, with $H(u) = \exp(u)/\{1 + \exp(u)\}$. Show that

$$\gamma = \log\frac{p_1/(1-p_1)}{p_0/(1-p_0)} = \log\frac{p_1}{1-p_1} - \log\frac{p_0}{1-p_0},$$

the log-odds difference. Write up the likelihood function for the observed $(Y_0, Y_1)$ to deduce (xx via the optimal CD exercise xx) that the optimal CD for $\gamma$ takes the form

$$C(\gamma) = \Pr_\gamma(Y_1 > y_{1,\text{obs}} \,|\, Z = z_{\text{obs}}) + \tfrac{1}{2}\Pr_\gamma(Y_1 = y_{1,\text{obs}} \,|\, Z = z_{\text{obs}}),$$

with $Z = Y_0 + Y_1$. The conditional distribution in question is the eccentric hypergeometric, found in Ex. 4.31. (xx do simple example here. this CD is used in both Stories #1 and #11. we use Ex. 4.31. xx)

**Ex. 8.39** *Meta-analysis for Lidocain data.* The following data table is from Normand (1999), and pertains to prophylactic use of lidocaine after a heart attack. The aim is to evaluate mortality from prophylactic use of lidocaine in acute myocardial infarction. We view the data here as pairs of binomials, with $y_{1,i} \sim \text{binom}(m_{i,1}, p_{1,i})$ and $y_{1,0} \sim \text{binom}(m_{i,0}, p_{1,0})$.

| m1 | m0 | y1 | y0 |
|---|---|---|---|
| 39 | 43 | 2 | 1 |
| 44 | 44 | 4 | 4 |
| 107 | 110 | 6 | 4 |
| 103 | 100 | 7 | 5 |
| 110 | 106 | 7 | 3 |
| 154 | 146 | 11 | 4 |

**(a)** Write the probabilities in logistic fashion, i.e. $p_{i,0} = H(\theta_{i,0})$ and $p_{i,1} = H(\theta_{i,0} + \gamma_i)$, with $H(u) = \exp(u)/\{1 + \exp(u)\}$. Show that

$$\gamma_i = H^{-1}(p_{i,1}) - H^{-1}(p_{0,i}) = \log \frac{p_{i,1}}{1 - p_{i,1}} \Big/ \frac{p_{0,i}}{1 - p_{i,0}},$$

the log-odds difference. Construct and display the optimal CD for the $\gamma_i$, and also for the odds ratio $\rho_i = \exp(\gamma_i)$, for each of the six studies.

**(b)** Assume then that the log-odds parameter $\gamma$ is the same, across studies, so that the six binomial data pairs relate to seven parameters. Find the optimal CD for this $\gamma$, and for the common odds ratio $\rho = \exp(\gamma)$. Translate the CDs to confidence curves, and display the six + one curves in a diagram. How would you conclude?

**Ex. 8.40** *Comparing Poisson parameters.* (xx ranting on a bit, to be edited. point to Story #12, application to sucide attempts rates. xx) Suppose $Y_0 \sim \text{Pois}(m_0\theta_0)$ and $Y_1 \sim \text{Pois}(m_1\theta_1)$. In what precise way is $\theta_1$ different from $\theta_0$? Writing $\gamma = \theta_1/\theta_0$, show that the likelihood is proportional to $\exp\{-\theta_0(m_0 + m_1\gamma)\}\theta_0^{y_0+y_1}\gamma^{y_1}$. Explain that the optimality recipe tells us inference should be made based on the distribution of $Y_1 \,|\, (Z = z)$, where $Z = Y_0 + Y_1$. Show that $Y_1 \,|\, (Z = z)$ has the binomial distribution $(z, m_1\gamma/(m_0 + m_1\gamma))$. Show how this leads to the optimal CD

$$\begin{aligned} C(\gamma) &= \Pr_\gamma(Y_1 > y_{1,\text{obs}} \,|\, z) + \tfrac{1}{2}\Pr_\gamma(Y_1 = y_{1,\text{obs}} \,|\, z) \\ &= 1 - B_z(y_{1,\text{obs}}, m_1\gamma/(m_0 + m_1\gamma)) + \tfrac{1}{2}b_z(y_{1,\text{obs}}, m_1\gamma/(m_0 + m_1\gamma)). \end{aligned}$$

(xx point to Story #12, for $y_0 = 1$, $y_1 = 7$, for the patient years $m_0$, $m_1$, in Aursnes et al. (2005). compare with their Bayes gamma priors, both informative and less informative. xx)

**Ex. 8.41** *Basic meta-analysis.* (xx to come, and calibrated with later stuff. xx) There is a very wide literature on combining information, with different names and labels, including meta-analysis, data fusion, etc. This exercise looks into some of the more basic versions, and where CDs will be helpful in later extensions below.

**(a)** Suppose $y_j \sim \mathrm{N}(\mu, \sigma_j^2)$, for $j = 1, \ldots, k$ independent sources, with the same focus parameter $\mu$, and with variances taken to be known or well estimated. Consider the linear combination estimator $\widehat{\mu} = \sum_{j=1}^k a_j y_j$. Show that it is unbiased, provided $\sum_{j=1}^k a_j = 1$, and find its variance. Show that the best choice, yielding minimal variance among the unbiased ones, is $a_j \propto 1/\sigma_j^2$, leading to

$$\widehat{\mu} = \frac{\sum_{j=1}^k y_j/\sigma_j^2}{\sum_{j=1}^k 1/\sigma_j^2}.$$

Show indeed that $\widehat{\mu} \sim \mathrm{N}(\mu, \kappa^2)$, with this minimal variance being $\kappa^2 = (\sum_{j=1}^k 1/\sigma_j^2)^{-1}$. Comment on what this leads to for the case where the $\sigma_j$ are equal.

**(b)** In various settings there is a need to generalise the setting above to one where $y_j \,|\, \mu_j \sim \mathrm{N}(\mu_j, \sigma_j^2)$, with these individual mean parameters not being equal, but having their own distribution, say $\mu_j \sim \mathrm{N}(\mu_0, \tau^2)$. The task is then to reach inference for both the overall mean $\mu_0$ and the spread $\tau$ among the $\mu_j$. Show that $y_j \sim \mathrm{N}(\mu_0, \sigma_j^2 + \tau^2)$, and that the log-likelihood function becomes

$$\ell(\mu_0, \tau) = -\tfrac{1}{2} \sum_{j=1}^k \Big\{ \log(\sigma_j^2 + \tau^2) + \frac{(y_j - \mu_0)^2}{\sigma_j^2 + \tau^2} \Big\}.$$

**(c)** Considering the spread parameter $\tau$ first, show that the profiled log-likelihood can be written

$$\ell_{\mathrm{prof}}(\tau) = -\tfrac{1}{2} \sum_{j=1}^k \Big[ \log(\sigma_j^2 + \tau^2) + \frac{\{y_j - \widehat{\mu}_0(\tau)\}^2}{\sigma_j^2 + \tau^2} \Big], \quad \text{for } \tau \geq 0,$$

in which

$$\widehat{\mu}_0(\tau) = \frac{\sum_{j=1}^k y_j/(\sigma_j^2 + \tau^2)}{\sum_{j=1}^k 1/(\sigma_j^2 + \tau^2)}$$

is the best linear combination estimator for $\mu_0$, for the fixed $\tau$ under inspection.

**(d)** (xx fix this, needs to be clearer. xx) Consider this profiled log-likelihood as a function of $\gamma = \tau^2$, rather than of $\tau$, and show that its derivative at zero is

$$D = \tfrac{1}{2} \sum_{j=1}^k \frac{1}{\sigma_j^2} \Big\{ \frac{(y_j - \widetilde{\mu}_0)^2}{\sigma_j^2} - 1 \Big\}.$$

Here $\widetilde{\mu}_0 = \widehat{\mu}(0)$. A small $D$ means that the $y_j$ data have a low spread, and vice versa. Show that if $D \leq 0$, then $\widehat{\tau}_{\mathrm{ml}} = 0$, and if that $D > 0$, then $\widehat{\tau}_{\mathrm{ml}}$ is positive.

**(e)** (xx fix this, needs to be clearer. xx) Show next that

$$Q(\tau) = \sum_{j=1}^k \frac{\{y_j - \widehat{\mu}(\tau)\}^2}{\sigma_j^2 + \tau^2} \sim \chi^2_{k-1}.$$

**(f)** (xx work with $\ell_{\mathrm{prof}}(\tau)$. partly from CLP. derivative at zero. xx) By maximising over $\mu_0$, for each given $\tau$, show that

$$\ell_{\mathrm{prof}}(\tau) = -\tfrac{1}{2} \sum_{j=1}^k \Big[ \log(\sigma_j^2 + \tau^2) + \frac{\{y_j - \widehat{\mu}(\tau)\}^2}{\sigma_j^2 + \tau^2} \Big].$$

**Ex. 8.42** *Combining CDs for the same parameter.* (xx a few exercises here. first for the same parameter, basic. then to CLP Ch 13 settings, then CDs to likelihood; then II-CC-FF. xx) Suppose that $C_1(\mu), \ldots, C_k(\mu)$ are independent CDs for the same parameter $\mu$, perhaps based on different sets of data. How can these be properly combined?

**(a)** Show that $N_j = \phi^{-1}(C_j(\mu_{\rm true}))$ is standard normal, at the true position in the parameter space underlying the $C_j$. With $w_1, \ldots, w_k$ numbers such that $\sum_{j=1}^k w_j^2 = 1$, show that

$$\bar{C}(\mu) = \Phi\Big(\sum_{j=1}^k w_j \Phi^{-1}(C_j(\mu))\Big)$$

is a proper combination CD for $\mu$.

**(b)** (xx point back to Ex. 8.41. xx)

**(c)**

**Ex. 8.43** *The problem of the Nile.* (xx starting a rant and will see how it goes. xx) Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$ be independent and exponential, the $X_i$ with parameter $\theta$, the $Y_i$ with parameter $1/\theta$.

**(a)** Show that the log-likelihoods, for the $X_i$ and for the $Y_i$ parts, become

$$\ell_1(\theta) = n(\log\theta - \theta\bar{X}), \quad \ell_2(\theta) = n(-\log\theta - \bar{Y}/\theta),$$

with separate ML estimators $\widehat{\theta}_1 = 1/\bar{X}$ and $\widehat{\theta}_2 = \bar{Y}$. Show further that we may represent these as $\widehat{\theta}_1 \sim \theta A$, $\widehat{\theta}_2 \sim \theta B$, where $A = 2n/K$ and $B = L/(2n)$, with $K$ and $L$ being $\chi^2_{2n}$.

**(b)** Show that the canonical CDs for $\theta$, based on the two parts, are

$$C_1(\theta) = \Gamma_{2n}(2n\theta/\widehat{\theta}_1), \quad C_2(\theta) = 1 - \Gamma_{2n}(2n\widehat{\theta}_2/\theta).$$

The combination recipe hence leads to

$$\bar{C}(\theta) = \Phi\big((1/\sqrt{2})\{\Phi^{-1}(C_1(\theta)) + \Phi^{-1}(C_2(\theta))\}\big).$$

**(c)** The full log-likelihood becomes $\ell(\theta) = -n(\theta\bar{X} + \bar{Y}/\theta)$. Show that this is maximised for

$$\widehat{\theta}_{\rm ml} = (\bar{Y}/\bar{X})^{1/2} = (\widehat{\theta}_1\widehat{\theta}_2)^{1/2} \sim \theta F^{1/2},$$

where $F = A/B = L/K \sim F_{2n,2n}$. Show that the associated with this ML estimator is $C_{\rm ml}(\theta) = 1 - F_{2n,2n}(\widehat{\theta}^2_{\rm ml}/\theta^2)$.

**(d)** Run some simple experiments, where you for a few values of $n$ simulate data and then plot the four CDs, and the four confidence curves (the two separate ones, the combination one, and the ML based one).

**(e)** (xx a bit more. ML etc. work, even though the situation is slightly irregular. $(\bar{X}, \bar{Y})$ is sufficient, but not complete. we need two informants for the single parameter. xx)

**Ex. 8.44** *From confidence to likelihood.* (xx to come. with illustrations. normal conversion. $\ell(\mu) = -\frac{1}{2}\Gamma_1^{-1}({\rm cc}_j(\mu))$. xx) Consider a model with parameter vector $\theta$ and focus parameter $\mu = \mu(\theta)$. Above we have developed methodology effectively constructing a confidence curve from the log-likelihood function, via Wilks theorems and related tools. Here we address the topic of conversion; constructing a log-likelihood function from the confidence distribution.

normal conversion

**(a)** Consider first the simple case of $\widehat{\mu} \sim \mathrm{N}(\mu, \sigma^2)$, with known $\sigma$. Explain that the log-likelihood is $-\frac{1}{2}(\mu - \widehat{\mu})^2/\sigma^2$, with confidence curve $\mathrm{cc}(\mu) = \Gamma_1((\mu - \widehat{\mu})^2/\sigma^2)$. Show from this again that the *normal conversion*, $\ell_{\mathrm{conv}}(\mu) = -\frac{1}{2}\Gamma_1^{-1}(\mathrm{cc}(\mu))$, succeeds in precisely retrieving the log-likelihood function.

**(b)** The key idea is now to carry out such normal conversion on any given confidence curve of relevance. From such a $\mathrm{cc}_n(\mu)$, carry out the normal conversion above, with

$$\ell_{\mathrm{conv},n}(\mu) = -\tfrac{1}{2}\Gamma_1^{-1}(\mathrm{cc}_n(\mu)).$$

When a statistician is given this log-likelihood function, perhaps via summaries of other work for the focus parameter in question, show that she eaches the deviance function $D_{\mathrm{conv},n}(\mu) = \Gamma_1^{-1}(cc_n(\mu))$. When the confidence curve is well calibrated, in the sense of having distribution $cc_n(\mu_0)$ close to a uniform, at the true value, show that $D_{\mathrm{conv},n}(\mu_0)$ is close to a $\chi^2_1$. Also explain that the Wilks theorem recipe is consistent with this; the same confidence intervals are generated by recipes (i) the given $\mathrm{cc}(\mu)$ and (ii) the conversion.

**(c)** (xx two examples, in situations where there is an exact cc and log-likelihood, to see that it works well, also with a bit of skewness. xx)

**Ex. 8.45** *II-CC-FF: Independent Inspection, Confidence Conversion, Focused Fusion.* (xx to come. using Cunen and Hjort (2022). point to Bayesian updating being part of this, but allows user keeping only prior for the focus parameter. aim to demonstrate iiccff in Story #12. xx) Story three sources.

**Ex. 8.46** *Private attributes.* (xx to be checked with care. xx) The probability $\psi$ of cheating at exams might be hard to estimate, but a bit of randomisation might grant anonymity and yield valid estimates. Suppose there are three cards, two with the statement 'I did cheat', and 'I did not cheat' on the third. Students are asked to draw one of the three cards randomly and answer either *true* or *false* to the drawn statement, without revealing it.

**(a)** Show that the probability of *true* is $(1 + \psi)/3$. Assume a binomial model for the number of students answering *true*, and devise a CD for $\psi$.

**(b)** Assume 1000 students go through the simple post-exam exercise above (anonymously). Find and display CDs for $\psi$ for the cases of respectively 300, 350, 400 out of the 1000 answered *true*.

## Notes and pointers

[xx A few remarks. xx]

ToDo: (i) complete the optimality for CDs in expofamilies. (ii) give the argument that testing ought to have alternatives in mind, and CDs are useful for assessing this. calibrate that with Ch4.

we point to Schweder and Hjort (2016, Example 3.11), Fisher (1930), Xie and Singh (2013), Hjort and Schweder (2018), Cunen and Hjort (2022), Singh et al. (2005), ...

for the conversion, from confidence to likelihood, point both to CLP, to Cunen and Hjort (2022), but find also the Efron.

(xx repair. we point to Story #62. xx) There's a notable discrepancy between the frequentist Schweder-Hjort CD and the Bayesian posterior distribution associated with

a flat prior on the $[0, \infty)$ interval, in cases where the $y_{\text{obs}}$ is close to, or perhaps even to the left of, the boundary point.

# I.9

# Loss, risk, performance, optimality

Statistics is a mathematical formalisation of how to make good decisions under uncertainty. One source of uncertainty is that the true state of nature, say $\theta$, is not known when we are to make our decisions, and since the utility or loss of a decision depends on $\theta$, we need to be clear about how bad it is when our decision is off. This is the role of loss functions.Decisions are based on data, and it is not without consequence how we use the data: Some procedures for going from data to a decision are better than others. Therefore, it makes sense to see how a certain procedure performs on average. This is the role played by risk functions. Risk functions average out the data, but they still depend on $\theta$, so the risk function of various decision procedures are often difficult to order. Some might be preferable for certain values of $\theta$, while others might be better for other values of $\theta$. This chapter introduces criteria that let us, nevertheless, say something about how good a decision procedure is. A decision procedure is said to be *admissible* if there is no other that does better, in terms of the risk function, whatever the truth or future may be. One says that a decision procedure is *minimax* if it is the the best one in the most unfortunate situation. In proving that certain decision procedures are admissible or minimax, Bayesian thinking is an essential tool. This includes the concepte of Bayes risk, where not only the data is average out, but also the various possible states of nature are averaged out.

*Key words:* admissibility, Rao–Blackwell, loss functions, minimaxity, multiparameter estimation, UMVU estimators, risk functions

A statistical decision, as most decisions in life when you think about it, is a function whose domain is what we observe, and whose range are all possible decisions we can make in a given setting: I look out the window and see grey clouds, and choose to take my umbrella with me when I go out. I see that you are smiling and I think that you are happy. Formally, there is a probability space $(\Omega, \mathcal{F}, \Pr)$ in the background, and the data are realistions of a random element $X\colon \Omega \to \mathcal{X}$. A decision procedure is then a function $\widehat{a}\colon \mathcal{X} \to \mathcal{A}$, where $\mathcal{A}$ is the collection of all possible decisions or actions we might take in a given setting, called the *action space*. How wise our choice of $\widehat{a}$ is or turns out to be depends on the true distribution $P_{\theta_0}$ of $X$. This distribution belongs to a family $\{P_\theta\colon \theta \in \Theta\}$ of distributions on the measurable space $(\mathcal{X}, \mathcal{B})$. A *loss function* $L\colon \Theta \times \mathcal{A} \to [0, \infty)$ measures 'how much' we lose by choosing action $a$ when the true state of nature is $\theta$, and as the name indicates, the less loss the better.

loss function

To be concrete, consider the point estimation problem with data $X_1, \ldots, X_n$ independent $\mathrm{N}(\theta, 1)$, where $\theta$ is an unknown parameter to be estimated. A common and mathe-

matically tractable loss function is then the squared error loss function $L(\theta, a) = (a-\theta)^2$. For another classical example, suppose we wish to test $H_0 \colon \theta \leq 0$ versus $H_A \colon \theta > 0$. The action space is $\mathcal{A} = \{\text{keep } H_0, \text{ reject } H_0\}$, and a natural loss function is

$$L(\theta, a) = \begin{array}{c|cc} & \theta \leq 0 & \theta > 0 \\ \hline \text{keep } H_0 & 0 & L_2 \\ \text{reject } H_0 & L_1 & 0 \end{array}$$

With this loss function, we lose $L_1 > 0$ when rejecting a true null hypothesis, and $L_2$ if we fail to reject a null hypothesis that is false. In statistical jargon, failing to reject a null hypothesis is called a Type I error, while when we fail to reject a null hypothesis that is false, we commit a Type II error.

The general setup so far is essentially as for what we built with the Bayesian framework of Ch. 7, see Ex. 7.2, where one in addition to the elements above pushes in one more ingredient, that of a prior distribution for the $\theta$. We do comment more below, and in various exercises, on Bayesian components and connections, but the main spirit of the present chapter is the classical or frequentist one, with no priors. Here different decision procedures are compared by the loss they incur for each value of $\theta$, that is, by their *risk function* the risk function

$$R(\theta, \widehat{a}) = \mathrm{E}_\theta \, L(\theta, \widehat{a}(X)) = \int_{\mathcal{X}} L(\theta, \widehat{a}(x)) \, \mathrm{d}P_\theta(x). \tag{9.1}$$

For each decision procedure $\widehat{a}$, the risk function $R(\theta, \widehat{a})$ is a function of $\theta$, and the second argument $\widehat{a}$ only indicates what procedure we are computing the risk of. For two decision procedures $\widehat{a}_1$ and $\widehat{a}_2$ their respective risk functions may cross, that is $R(\theta, \widehat{a}_1) < R(\theta, \widehat{a}_2)$ for some $\theta$, while $R(\theta, \widehat{a}_1) > R(\theta, \widehat{a}_2)$ for other values of $\theta$. This entails that comparison of risk functions only provides a partial ordering of decision procedures, and as such does not point clearly at a best decision procedure. What is clear, however, is that if $R(\theta, \widehat{a}_1) \leq R(\theta, \widehat{a}_2)$ for all $\theta$, with strict inequality for at least one value of $\theta$, then $\widehat{a}_2$ should be discarded from the competition: We say that $\widehat{a}_1$ *dominates* $\widehat{a}_2$, and $\widehat{a}_2$ is said to be *inadmissible*. A decision procedure that is not dominated by any other decision procedure is *admissible*. Clearly, inadmissible decisions procedures should be discarded, but admissibility in and of itself does not tell us much about a decision procedure. admissibility

One fix to the problem of crossing risk functions is to limit the search of a best decision procedure to certain subclasses of decision procedures. In the search for a 'best' estimator, for example, it is common to restrict ones search to the class of *unbiased estimators*. An estimator $\widehat{\theta}$ is unbiased for the parameter $\theta$ if $\mathrm{E}_\theta \, \widehat{\theta} = \theta$ for all $\theta$ in a certain parameter space $\Theta$.

Another method by which to compare decision procedures is the the *minimax principle*. According to this principle, the estimator with the best performance in the worst possible scenario ought to be chosen. We say that a decision rule $a^*$ is *minimax* if it minimises the maximum risk, that is, if $\sup_{\theta \in \Theta} R(\theta, a^*) \leq \sup_{\theta \in \Theta} R(\theta, \widehat{a})$ for all decision procedures $\widehat{a}$. minimax

If you are a Bayesian and venture into the Ch. 7 terrain of constructing prior distributions $\pi(\theta)$ over the parameter space $\Theta$, the problem of risk functions only being partially ordered can be circumvented. As studied in that chapter, what you are interested in then is the *Bayes risk* of the decision procedures you are comparing, that Bayes risk

is

$$\mathrm{BR}(\widehat{a}, \pi) = \mathrm{E}_\pi \, R(\theta, \widehat{a}) = \int_\Theta R(\theta, \widehat{a}) \, \pi(\theta) \, \mathrm{d}\theta.$$

For a given decision procedure $\widehat{a}$ and prior $\pi$, the Bayes risk is a number (in contrast to the risk function which is a function of the unknown truth), and under certain conditions that we will explored in Ch. 7 and will explore further below, there will be a unique decision procedure $\widehat{a}$ that, for a given prior $\pi$, minimises $\mathrm{BR}(\widehat{a}, \pi)$. This decision procedure is the *Bayes solution*, see Ex. 7.2. As we shall see in this chapter, Bayes solutions are, despite the name, of high importance also for the frequentists.

### Decisions and risk functions

**Ex. 9.1** *Coin tossing.* To get a feeling for some of the basic challenges and concepts concerning the comparison of various estimators, we start out with the emblematic problem of estimating the probability of heads in $n = 10$ independent tosses of a coin. Let $Y_1, \ldots, Y_n$ be independent Bernoulli random variables with expectation $\theta$.

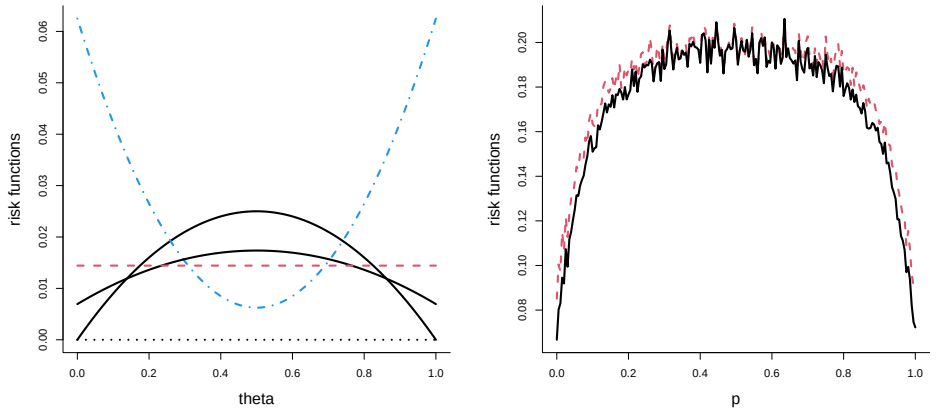



Figure 9.1: *Left panel: for Ex. 9.1. risk functions for four estimators of* $\theta = \Pr(\text{heads})$*: the ML (going down to zero at the ends); the Bayes estimator* $(Y + 1)/(n + 2)$ *(dotted line); the 50-50 between 0.50 and* $Y/n$ *(going up at the ends); and the minimax estimator (flat). Right panel: for Ex. 9.2, risk fucntions for estimation methods A and B, for the case of* $n = 100$ *and distance 1 between the two normals in the mixture, based on 1000 simulations for each grid point. The ML method is just slightly better than the simpler moment matching method, for the given loss function.*

**(a)** Sketch the risk function of the maximum likelihood (ML) estimator $\bar{Y}_n = n^{-1} \sum_{i=1}^n Y_i$ under squared error loss $L(\theta, \bar{Y}_n) = (\delta - \bar{Y}_n)^2$, as for Figure 9.1. Recall that $n = 10$.

**(b)** Suppose we have some intuition about where on the unit interval the probability $\theta$ might be located, close to a value $0 < \theta_0 < 1$, say. One way in which such a prior hunch might be employed is by taking as our estimate a convex combination of the ML estimator and $\theta_0$, that is $\widehat{a} = w\bar{Y}_n + (1-w)\theta_0$, for some user defined weight $w \in [0, 1]$. For $w = 1/2$ and $\theta_0 = 1/2$, sketch the risk function of this estimator, i.e. one of the four risk functions of Figure 9.1. Suppose that your task is to estimate the probability of giving

as was Pierre-Simon Laplace's in 1781 or so

birth to a boy. Which of the two above estimators do you prefer, the ML estimator, or $\widehat{a}$ with $w = 1/2$ and $\theta_0 = 1/2$?

**(c)** Based on the risk functions you sketched in (a) and (b), we see that the estimators $\widehat{a}$ and $\bar{Y}_n$ are difficult to compare, unless we have strong prior feelings about where the true value is. As mentioned in the introduction, a fix to this problem of comparison is to limit our search to the class of estimators that are unbiased for what we are estimating. Look back at Ex. 5.7 and explain why, when the search is restricted to the class of unbiased estimators, the ML estimator is the clear winner.

**(d)** Consider the estimator from Ex. (b) with $\theta_0 = 1/2$. Find a weight function $w = w(n)$ such that the risk function $R(\theta, \widehat{a})$ is constant. For $n = 10$, plot the risk function of this estimator (that is, the flat risk of Figure 9.1). Suppose you have absolutely no idea whatsoever about where in the unit interval $\theta$ may be located. Which of your three estimators of $\theta$ do you prefer? In Ex. 9.12 we learn that this constant risk estimator is indeed minimax.

**Ex. 9.2** *Estimating the mixture proportion.* Suppose i.i.d. observations $X_1, \ldots, X_n$ come from a mixture of known densities $f_0$ and $f_1$, i.e. from $f_p = (1-p)f_0 + pf_1$, but with unknown fraction $p$. The task is to estimate this parameter, yielding the estimated density as $\widehat{f} = (1-\widehat{p})f_0 + \widehat{p}f_1$. On this occasion we evaluate the risk based on the loss function

$$L(p, \widehat{p}) = \sqrt{n} \max |\widehat{f}(x) - f_p(x)|,$$

as opposed to simpler ones like squared error. The $\sqrt{n}$ factor is mostly cosmetic, to make the risks stable in size for growing $n$. Below we compute and display risk functions for the case of $f_0 = \mathrm{N}(0,1)$ and $f_1 = N(d,1)$, with the illustration depending on the value of $d$. For concreteness, take $d = 1$ and $n = 100$ below. Can data tell us the fraction of types 0 and 1, even with the densities are not far from each other?

**(a)** Estimator A is to match the sample mean $\bar{X}$ to the model mean. Show that this is solved by the equation $\bar{X} = (1-p)\xi_0 + p\xi_1$, with $\xi_0$ and $\xi_1$ the means under $f_0$ and $f_1$. Estimator B is the ML one. Show that this is the maximiser of $\ell_n(p) = \sum_{i=1}^n \log((1-p)f_0(X_i) + pf_1(X_i))$, which may be found after a numerical search over a fine grid of $p$, or via another numerical optimiser.

**(b)** Now compute the two risk functions, via simulations; for each $p$, simulate a high number of $\widehat{p}$ and hence a high number of losses, after which the risk is their average. Explain why this is and adequate risk finding procedure. Construct a version of Figure 9.1, right panel. When running your code you may also read off the simpler mean squared error risks, i.e. $\mathrm{E}_p\,(\widehat{p} - p)^2$, though the point is here that we can handle much more complicated loss functions.

**(c)** While we're at it, find also the limiting distributions of $\sqrt{n}(\widehat{p} - p)$, for the simple and the log-likelihood method, and compare their variances. One learns that the ML wins, with relatively speaking clearer amounts near the endpoints 0 and 1, compared to the risk results above. So the loss function matters, and with the one above the two estimation methods fare almost equally well.

**(d)** For any smooth increasing $h(x)$, show that $\mathrm{E}_p h(X) = (1-p)h_0 + ph_1$, with $h_0$ and $h_1$ the mean values under $f_0$ and $f_1$. Construct the relevant moment matching estimator for $p$. For such an $h(x)$, of your choice, compute the risk function and compare to the two of Figure 9.1, right panel.

**Ex. 9.3** *How many cards in the hat?* There is an unknown number $N$ of cards in a hat, numbered $1, \ldots, N$. You draw $X_1, \ldots, X_n$ cards, with $n = 5$, with replacement, and your task is to estimate $N$, with loss function $|\widehat{N} - N|/N = |\widehat{N}/N - 1|$.

**(a)** Show that $X_i$ has mean $(N+1)/2$. Explain that the moment matching estimator is $\widehat{N} = 2\bar{X} - 1$. Another estimator is $N^* = \max_{i \le n} X_i$. Compute and display their risk functions, say for $N = 10, \ldots, 100$. To compute these you may resort to simulation, as with Ex. 9.2. One learns that $N^*$ does better than $\widehat{N}$, regardless of the correct $N$.

**(b)** Now attempt to be more clever than $N^*$, perhaps using $N^* + c_n$ for a suitable $c_n$. For $n = 5$, compute and display also this third risk function. Do you succeed in being uniformly more clever than $N^*$?

**(c)** Now redo your analyses and risk calculations, when the $n$ cards are drawn from the $N$-hat without replacement.

**Ex. 9.4** *Decide on small, moderate, or big.* A crucial parameter $\theta$ is being assessed, in order to place it in categories small, $\theta \in B_1 = [0, 2]$; moderate, $\theta \in B_2 = (2, 4)$; or big, $\theta \in B_3 = [4, \infty)$. An action $a \in \{1, 2, 3\}$ is taken, based on data, corresponding to declaring $\theta$ to be small, moderate, or big. The potential losses incurred, i.e. $L(\theta, a)$, follow this table, with specified positive losses $\ell_{i,j}$ in cases where the action taken is not the correct one, i.e. when $j \neq i$:

| | action 1 | action 2 | action 3 |
|---|---|---|---|
| $\theta \in B_1$ | 0 | $l_{1,2}$ | $l_{1,3}$ |
| $\theta \in B_2$ | $l_{2,1}$ | 0 | $l_{2,3}$ |
| $\theta \in B_3$ | $l_{3,1}$ | $l_{3,2}$ | 0 |

**(a)** Assume for simplicity that all errors are equally unfortunate, with all $l_{i,j} = 1$ for $j \neq i$. Suppose that a single $Y \sim \text{Gam}(\theta, 1)$ is observed. Since its mean is $\theta$ it makes sense to take $\widehat{a}(Y)$ equal to 1, 2, 3, according to whether $Y$ lands in the relevant boxes. Show that the risk function becomes

$$R(\theta) = \begin{cases} 1 - G(2, \theta) & \text{if } \theta \in B_1, \\ G(2, \theta) + 1 - G(4, \theta) & \text{if } \theta \in B_2, \\ G(4, \theta) & \text{if } \theta \in B_3, \end{cases}$$

writing $G(y, \theta)$ for the c.d.f. of the $\text{Gam}(\theta, 1)$. Compute and display.

**(b)** It might be more clever to take decisions 1, 2, 3 based on $Y$ landing in $B_1^* = [0, 2-c]$, $B_2^* = (2-c, 4+c)$, $B_3^* = [4+c, \infty)$, for a suitable wise $c$. Compute and plot a few associated risk functions, to see whether this might be more clever than using $c = 0$. By tracking these risk functions numerically, find indeed the value of $c$ which gives lowest possible max-risk in this class of procedures (the answer is at 0.22).

**(c)** Generalise the situation above to the case where one observes $\bar{Y}$, the sample mean of $Y_1, \ldots, Y_m$ being i.i.d. from the $\text{Gam}(\theta, 1)$. Compute and plot risk functions for e.g. $m = 10$.

### UMVU estimators, Rao–Blackwell, and Lehmann–Scheffé

**Ex. 9.5** *Uniformly minimum variance unbiased estimators.* What is variably called a *best unbiased estimator*, the *uniformly minimum variance unbiased estimator*, or the *UMVU estimator*, is defined as follows: An estimator $\widehat{a}$ is the UMVU estimator for $g(\theta)$

UMVU estimator

if it is unbiased for $g(\theta)$, and $\mathrm{Var}_\theta\,\widehat{a} \le \mathrm{Var}_\theta\,\tilde{a}$ for all $\theta$, for any other unbiased estimator $\tilde{a}$. When it is feasible (see Ex. 5.6 and 5.7–5.8), the easiest way of establishing that an unbiased estimator is an UMVU estimator is to verify that it achieves the Cramér–Rao lower bound.

**(a)** Let us look at a few examples of the Cramér–Rao approach: (i) Suppose $Y \sim \mathrm{N}(\theta, 1)$, and show that $Y$ is UMVU for $\theta$. (ii) Suppose $Y$ has an exponential distribution with mean $\theta$. Show that $Y$ is UMVU for $\theta$. (iii) Let $Y_i = \beta_0 + \beta_1 x_i + \sigma\varepsilon_i$ for $i = 1, \ldots, n$, where the covariates $x_1, \ldots, x_n$ are fixed numbers, and the $\varepsilon_1, \ldots, \varepsilon_n$ are independent standard normal random variables. Show that the least squares estimator for $(\beta_0, \beta_1)$ is the UMVU estimator.

**(b)** Let $Y_1, \ldots, Y_n$ be i.i.d. $\mathrm{N}(\mu, \sigma^2)$. It is immediate from (a) that $\bar{Y}_n = n^{-1}\sum_{i=1}^n Y_i$ is UMVU for $\mu$. Show that the estimator $\widehat{\sigma}_n^2 = \sum_{i=1}^n (Y_i - \bar{Y}_n)^2/(n-1)$ is *not* UMVU for $\sigma^2$. In Ex. 9.6 we will see that there is no unbiased estimator of $\sigma^2$ attaining the Cramér–Rao lower bound.

**Ex. 9.6** *Cramér–Rao and Cauchy–Schwarz.* The proof of the Cramér–Rao inequality that we met in Ex. 5.7–5.8 is a clever application of the Cauchy–Schwarz inequality, introduced in Ex. 1.30.

**(a)** Let $X$ and $Y$ be two square integrable random variables with expectation zero. Show that $|\mathrm{E}\,XY| = \{\mathrm{Var}\,(X)\mathrm{Var}\,(Y)\}^{1/2}$ if and only if $X$ and $Y$ are linearly related, $Y = a + bX$, for example.

**(b)** Explain why the Cramér–Rao inequality is an equality if and only if the estimator $\delta(y)$ and the score function are linearly related, that is

$$u(y,\theta) = \partial \log f(y,\theta)/\partial\theta = a(\theta) + b(\theta)\delta(y), \quad \text{for all } \theta,$$

for some functions $a(\theta)$ and $b(\theta)$. Solve the differential equation above for $f(y,\theta)$, and state what this entails for estimators and distributions when it comes to possible attainment of the Cramér–Rao lower bound.

**(c)** In Ex. 9.5(b) we saw that with $Y_1, \ldots, Y_n$ i.i.d. $\mathrm{N}(\mu, \sigma^2)$, the unbiased estimator $\widehat{\sigma}_n^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2/(n-1)$ for $\sigma^2$ does not attain the Cramér–Rao lower bound. Use the result from (b) to argue that the Cramér–Rao lower bound may only be attained when $\mu$ is known.

**Ex. 9.7** *Sufficiency and Rao–Blackwell.* Suppose that $Y$ has a distribution from a family $\{P_\theta : \theta \in \Theta\}$ of distributions. Recall from Ex. 4.18 that $T = T(Y)$ is a sufficient statistic for this family distributions if the conditional distribution of $Y$ given $T$ does not depend on $\theta$. The Rao–Blackwell theorem says that any estimator can be improved upon by conditioning on a sufficient statistic. This is an important result, as it tells us that in our search for a best estimator, we need only consider those estimators that are functions of a sufficient statistic.

**(a)** Let $\widetilde{a} = \widetilde{a}(Y)$ be an unbiased estimator for $g(\theta)$, and suppose that $T = T(Y)$ is sufficient for $\theta$. Consider the estimator given by $\widehat{a} = \mathrm{E}_\theta\,(\widetilde{a}\,|\,T)$. First, explain why $\widehat{a}$ is an estimator for $g(\theta)$; second, show that $\widehat{a}$ is unbiased; and third, show that $\mathrm{Var}_\theta\,\widehat{a} \le \mathrm{Var}_\theta\,\widetilde{a}$ for all $\theta$. You have now proven the Rao–Blackwell theorem for unbiased estimators. In Ex. 9.11 we look at this theorem in a more general decision theoretic framework. Rao–Blackwell theorem

**(b)** Suppose that $Y_1, \ldots, Y_n$ are i.i.d. uniforms on $[0, \theta]$, with $\theta > 0$ an unknown param-

eter. Show that the estimators

$$\widetilde{\theta} = \frac{2}{n}\sum_{i=1}^{n} Y_i \quad \text{and} \quad \widehat{\theta} = \frac{n+1}{n}\max_{i\le n} Y_i$$

are both unbiased for $\theta$. There are (at least) two ways of showing that $\widehat{\theta}$ is a better estimator than $\widetilde{\theta}$. One involves computing the variances of both estimators; the second uses the Rao–Blackwell theorem. For the Rao–Blackwell approach, use the results from Ex. 3.17 to establish that $\mathrm{E}\,(\widetilde{\theta} \mid \max_{i\le n} Y_i) = \widehat{\theta}$ almost surely. Try both methods.

**(c)** Let $Y_1, \ldots, Y_n$ be i.i.d. Pois($\lambda$). We seek to estimate $\theta = \Pr(Y_1 = 0) = \exp(-\lambda)$. Find the ML estimator for $\theta$, say $\widehat{\theta}$, and show that $\mathrm{E}\,\widehat{\theta} = \theta\{1 + O(1/n)\}$, meaning the the ML estimator for $\theta$ is biased. Next, find the Cramér–Rao lower bound for unbiased estimators of $\theta$. Look back at Ex. 9.6 and consider whether this lower bound can be attained. Another estimation strategy is to estimate $\theta = \Pr(Y_1 = 0)$ by the share of zero counts, that is $\widetilde{\theta} = n^{-1}\sum_{i=1}^{n} I\{Y_i = 0\}$. This estimator is clearly unbiased for $\theta$, why is it not best unbiased? Finally, with the aid of the sufficient statistic $T(Y) = \sum_{i=1}^{n} Y_i$ we Rao–Blackwellise the estimator $\widetilde{\theta}$. Derive an expression for this Rao–Blackwellised estimator, $\widetilde{\theta}_{\mathrm{rb}}$, say. Does $\widetilde{\theta}_{\mathrm{rb}}$ attain the Cramér–Rao lower bound?

**Ex. 9.8** *Best unbiased, completeness, and Lehmann–Scheffé.* From the Rao–Blackwell theorem we know that any candidate for being an UMVU estimator must be a function of a sufficient statistic. In this exercise we first establish that in our search for the UMVU estimator, we are only looking for one estimator, i.e., the UMVU estimator is unique. Second, it is shown that an estimator is best unbiased if and only if it is uncorrelated with all unbiased estimators of zero. This yields a characterisation of the best unbiased estimators, albeit one of limited utility as it is, without further conditions, hard to describe all unbiased estimators of zero. Finally, completeness – a condition on the distribution of the data – is introduced, ensuring that the only unbiased estimator of zero is zero itself.

**(a)** Suppose that both $\widetilde{a}$ and $\widehat{a}$ are UMVU for $g(\theta)$. Define the estimator $\delta = \frac{1}{2}\widetilde{a} + \frac{1}{2}\widehat{a}$, and show that $\mathrm{Var}_\theta\,\delta \le \mathrm{Var}_\theta\,\widetilde{a}$. But since $\widetilde{a}$ is an UMVU estimator and $\delta$ is unbiased (check it), it must be the case that $\mathrm{Var}_\theta\,\delta = \mathrm{Var}_\theta\,\widetilde{a}$. Look back at the results in Ex. 9.6, and establish that if $\widetilde{a}$ and $\widehat{a}$ are both UMVU, then $\widetilde{a} = \widehat{a}$ almost surely, for all $\theta$.

**(b)** Suppose that the estimator $\widehat{a}$ is a function of a sufficient statistic and is unbiased for $g(\theta)$. How may we improve upon this estimator? Well, the family of estimators $\widetilde{a}_b = \widehat{a} + b\varepsilon$ as $b$ ranges of the real numbers, and $\varepsilon$ is some mean zero random variable, constitute a class of unbiased estimators. Show that if $\mathrm{cov}_\theta(\widehat{a}, \varepsilon) \neq 0$ for some $\theta$, then $b$ may be chosen so that $\mathrm{Var}_\theta\,\widetilde{a}_b < \mathrm{Var}_\theta\,\widehat{a}$ for some value(s) of $\theta$, in which case $\widehat{a}$ is not best unbiased. Prove the converse, namely that if $\widehat{a}$ is unbiased and $\mathrm{cov}_\theta(\widehat{a}, \varepsilon) = 0$ for all $\theta$ and all mean zero random variables $\varepsilon$, then $\widehat{a}$ is UMVU. In summary, an estimator is best unbiased if and only if it is uncorrelated with all unbiased estimators of zero.

characterisation of the UMVU estimator

**(c)** It is in general no easy task to show that an unbiased estimator, or more generally, a statistic $S = S(Y)$ say, is uncorrelated with all unbiased estimators of zero. Since the correlation between any random variable and zero is zero, the task would be much easier if we knew of the distribution of $S$ that the only unbiased estimator of zero, is zero itself. That is, if for any measurable function $h$ and for any $\theta$, $\mathrm{E}_\theta\,h(S) = 0$ implies that $h(S) = 0$ almost surely. Recall from Ex. 4.24 that a family of distributions with this property is called *complete*. Alternatively, we just say that the statistic $S(Y)$ is complete.

completeness

Now, suppose that $T = T(Y)$ is sufficient and complete for $\theta$. Let $\widehat{a} = \widehat{a}(T)$ be unbiased for $g(\theta)$. Prove the Lehmann–Scheffé theorem, that is, show that the estimator $\widehat{a}$ is the unique UMVU estimator for $g(\theta)$. Lehmann–Scheffé theorem

**(d)** Return to Ex. 9.7(b), and show that the estimator $\widehat{\theta}$ is the UMVU estimator.

**(e)** The completeness requirement in the Lehmann–Scheffé theorem was motivated by the characterisation in (b), saying that an estimator is UMVU if and only if it is uncorrelated with all unbiased estimators of zero. A perhaps more illuminating motivation comes from the fact, proven in Ex. 9.7(a), that a best estimator must be based on a sufficient statistic. Intuitively, by getting rid of information irrelevant to the estimation problem at hand, we reduce the variance of our estimation procedure. Taking this intuition to its logical conclusion, we deduce that a best estimator must be based on a statistic achieving the maximum amount of data compression, while still retaining all the information in the data about the parameter we seek to estimate. In other words, a best estimator must be based on a minimal sufficient statistic. Recall from Ex. 4.22 that a statistic $S$ is minimal sufficient if for any sufficient statistic $T$ there exists a measurable function $g$ so that $S = g(T)$. Suppose that $\delta$ is an unbiased estimator and form the estimators $\widetilde{a} = \mathrm{E}_\theta\,(\delta \,|\, T)$ and $\widehat{a} = \mathrm{E}_\theta\,(\delta \,|\, S)$, where $T$ is sufficient and $S$ is minimal sufficient. Show that then $\mathrm{Var}_\theta\,\widehat{a} \leq \mathrm{Var}_\theta\,\widetilde{a}$ for all $\theta$. Next, suppose that $T$ is sufficient *and* complete. Use the Lehmann–Scheffé theorem and (a) to conclude that $\widehat{a} = \widetilde{a}$ almost surely, for each $\theta$. In view of this equality, it may not come as a surprise that if $T$ is sufficient and complete, then $T$ is minimal sufficient. A fact we will prove in (g).

**(f)** Here is a toy example illustrating some of the points made in (e). Let $X_1$ and $X_2$ be independent Bernoulli($\theta$) random variables and consider the estimator $\widehat{\theta} = (X_1 + X_2)/2$ and the estimator $\widetilde{\theta}$ defined by

$$\widetilde{\theta}(x_1, x_2) = \begin{cases} 1, & (x_1, x_2) = (1,1), \\ 2/3, & (x_1, x_2) = (1,0), \\ 1/3, & (x_1, x_2) = (0,1), \\ 0, & (x_1, x_2) = (0,0). \end{cases}$$

Explain why both $\widehat{\theta}$ and $\widetilde{\theta}$ are sufficient for $\theta$, and that they are both unbiased for $\theta$. Show that the variance of $\widetilde{\theta}$ exceeds the variance of $\widehat{\theta}$ for all values of $\theta \in (0,1)$, and explain how this relates to minimal sufficiency.

**(g)** The result quoted at the end of (e) is Bahadur's theorem: If $T$ is sufficient and complete, then $T$ is minimal sufficient. To prove this, let $S$ be another sufficient statistic, and assume, with out loss of generality, that $T$ and $S$ are real valued. We must show that there is a function $g$ such that $T = g(S)$. If so happens that $T = \mathrm{E}_\theta\,(T \,|\, S)$, then we have found a function $g$: It is $g(s) = \mathrm{E}_\theta\,(T \,|\, S = s)$, and $g$ does not depend on $\theta$ since $S$ is sufficient. Let us therefore prove that $T$ equals $\mathrm{E}_\theta\,(T \,|\, S)$ almost surely for all $\theta$. To this end, assume that $T$ has finite variance, and define $g(S) = \mathrm{E}_\theta\,(T \,|\, S)$ and $h(T) = \mathrm{E}_\theta\,\{g(S) \,|\, T\}$. Now, use the tower property of conditional expectation a couple of times and the fact that $T$ is complete to show that $T = h(T)$ almost surely, for all $\theta$. Next, combine the above with the variance decomposition formula to obtain Bahadur's theorem

$$\mathrm{Var}_\theta\, g(S) = \mathrm{E}_\theta\,\mathrm{Var}_\theta(g(S) \,|\, T) + \mathrm{E}_\theta\,\mathrm{Var}_\theta(T \,|\, S) + \mathrm{Var}_\theta\, g(S),$$

from which we conclude that $T = \mathrm{E}_\theta\,(T \,|\, S)$ almost surely for all $\theta$. To get rid of the finite variance assumption on $T$, replace $T$ by $f(T) = 1/\{1 + \exp(-T)\}$ (which clearly has finite variance) throughout the proof, to conclude that $T = f^{-1}(g(S))$.

**Ex. 9.9** *A uniform mean.* Let $Y_1, \ldots, Y_n$ be i.i.d. unif$(a, b)$. We are to estimate the mean $\mu = (a + b)/2$.

**(a)** Show that $(\min_{i \le n} Y_i, \max_{i \le n} Y_i)$ is sufficient and complete.

**(b)** Propose an unbiased estimator for $\mu$, and find its variance.

**Ex. 9.10** *A weird unbiased estimator.* Limiting our search for a best estimator to the class of unbiased estimators lacks the decision theoretic foundations that the principle of minimising expected loss enjoys. More on this in the Notes&Pointers. Sometimes, the search for unbiasedness might lead us astray. Let $Y$ be a random variable with density

$$f(y, \lambda) = \frac{\lambda^y \exp(-\lambda)}{y!\{1 - \exp(-\lambda)\}}, \quad \text{for } y = 1, 2, \ldots,$$

with $\lambda > 0$. This is a Poisson distribution truncated at zero, and the probability of being truncated is $\theta$.

**(a)** Show that $\widehat{\theta} = (-1)^{Y+1}$ is the unique unbiased estimator for $\theta = \exp(-\lambda)$.

**(b)** Find an expression for the risk function of $\widehat{\theta}$.

**(c)** Propose an estimator with uniformly smaller risk than $\widehat{\theta}$.

**Ex. 9.11** *More Rao–Blackwellisation.* In Ex. 1.30 we proved Jensen's inequality, saying that $g(\mathrm{E}\, X) \le \mathrm{E}\, g(X)$ for convex functions $g$. Jensen's inequality extends to conditional expectations, as we show in Ex. C.11. In this exercise we consider loss functions $L(\theta, \delta)$ that are convex in $\delta$ for all $\theta$. Think of your favourite loss function, and you will realise that this is a quite natural requirement.

**(a)** Let $\widetilde{\theta}$ be an estimator of $\theta$, and suppose that $T = T(X)$ is sufficient for $\theta$. If we Rao–Blackwellise $\widetilde{\theta}$, that is, define $\widehat{\theta} = \mathrm{E}_\theta\, (\widetilde{\theta} \,|\, T)$ we obtain a new estimator (as shown in Ex. 9.7). Suppose that the loss function $L(\theta, a)$ is convex in $a$ for all $\theta$, and show that

$$R(\theta, \widehat{\theta}) = \mathrm{E}_\theta\, L(\theta, \widehat{\theta}) \le \mathrm{E}_\theta\, L(\theta, \widetilde{\theta}) = R(\theta, \widetilde{\theta}), \quad \text{for all } \theta.$$

**(b)** Under what conditions is the inequality in (a) a strict inequality for all $\theta$?

**(c)** Suppose that $\mathrm{E}_\theta\, \widehat{\theta}(T) = h(\theta)$ for all $\theta$, and that $T$ is complete. Show that, in the class of estimators $\{\delta \colon \mathrm{E}_\theta\, \delta = h(\theta), \text{ for all } \theta\}$, the estimator $\widehat{a}$ is the unique estimator minimising the risk.

**(d)** [xx Let $L(\theta, \delta) = (\delta - \theta)^2$ and specialise to UMVU estimator xx]

**(e)** Suppose $L(\theta, a)$ is convex in $a$ for each $\theta$, and that $a_\pi$ is the unique Bayes solution under the prior $\pi$. Show that $a_\pi$ must be a function of a sufficient statistic.

## Minimaxity

**Ex. 9.12** *Tools for minimaxity.* An estimator is minimax if it minimises the risk in the worst possible scenario. That is, $\widehat{a}$ is minimax if $\sup_\theta R(\theta, \widehat{a}) \le \sup_\theta R(\theta, \widetilde{a})$ for all estimators $\widetilde{a}$. In this exercise we develop some techniques for verifying that an estimator is minimax.

**(a)** Consider any prior $\pi$ and any decision procedure $\widehat{a}$, with consequent risk function $R(\theta, \widehat{a})$ and Bayes risk $\mathrm{BR}(\widehat{a}, \pi)$. Writing $R_{\max}(\widehat{a}) = \sup_{\theta \in \Theta} R(\theta, \widehat{a})$ for the maximum risk, show that $\mathrm{BR}(\widehat{a}, \pi) \le R_{\max}(\widehat{a})$, and that $\mathrm{MBR}(\pi) \le R_{\max}(\widehat{a})$.

**(b)** Let $\widehat{a}_\pi$ be a Bayes solution with respect to the prior distribution $\pi$, and suppose that

$$\mathrm{BR}(\widehat{a}_\pi, \pi) = R_{\max}(\widehat{a}_\pi), \tag{9.2}$$

that is, $\mathrm{MBR}(\pi) = R_{\max}(\widehat{a}_\pi)$. Show that $\widehat{a}_\pi$ is minimax.

**(c)** Show that if $\widehat{a}_\pi$ satisfies (9.2) and is the unique Bayes solution with respect to $\pi$, then $\widehat{a}_\pi$ is the unique minimax procedure.

**(d)** Show that if a Bayes solution has constant risk, then it is minimax.

**(e)** Show that if an estimator has constant risk and is admissible, it is minimax.

**Ex. 9.13** *Minimax estimators for binomial and multinomial parameters.* In Ex. 7.8 and 7.12 we worked through certain Bayesian priors and their associated posteriors and posterior means, along with results for their risk functions. Those efforts now lead quickly to the additional quality marks for the derived estimators: they are minimax. The points below merely spell this out.

**(a)** First, consider $Y \sim \mathrm{binom}(n, p)$, with $p$ to be estimated with quadratic loss. Show that the Bayes estimator with the $\mathrm{Beta}(\frac{1}{2}\sqrt{n}, \frac{1}{2}\sqrt{n})$ prior is $p^* = (\frac{1}{2} + Y/\sqrt{n})/(\sqrt{n}+1)$, that its risk function is the constant $\frac{1}{4}/(\sqrt{n}+1)^2$, and that it is minimax. Show also that it is fact admissible. For which interval of $p$ is the minimax risk smaller than $R(p, Y/n)$ for the usual default estimator?

**(b)** Then consider a multinomial $(Y_1, \ldots, Y_k)$, with parameters $(n, p_1, \ldots, p_k)$, with sums $\sum_{j=1}^k Y_j = n$ and $\sum_{j=1}^k p_j = 1$, i.e. there are $k-1$ free parameters for the $k$ categories. With joint estimation of the probabilities, with loss function $\sum_{j=1}^k (a_j - p_j)^2$, start with the Dirichlet prior $\sqrt{n}(1/k, \ldots, 1/k)$, show that the posterior mean estimator takes the form

$$p_j^* = \frac{1/k}{\sqrt{n}+1} + \frac{\sqrt{n}}{\sqrt{n}+1}\frac{Y_j}{n} \quad \text{for } j = 1, \ldots, k.$$

and verify that its risk function is contant $(1-1/k)/(\sqrt{n}+1)^2$. Explain that it is minimax and admissible. In which region of $(p_1, \ldots, p_k)$ is the minimax estimator better than the ML estimator, i.e. $(Y_1/n, \ldots, Y_k/n)$?

**(c)** Explain the implications of this minimax find when it comes to constructing histograms for continuous data, see Ex. 1.6 for pertinent details, say with equal binwidths.

**Ex. 9.14** *Minimaxity and sequences of priors.* In Ex. 9.12(c) we assumed that the equality in (9.2) is attained. A prior distribution that succeeds in attaining this equality is, for natural reasons, called a *least favourable* prior distribution. If no such prior distribution exists, we cannot use the conclusion of that exercise to prove minimaxity. Consider independent $X_1, \ldots, X_n \,|\, \theta$ from $\mathrm{N}(\theta, 1)$. It seems reasonable that a least favourable prior for $\theta$ should spread its mass evenly out on the real line, that is

$$\int_a^{a+c} \pi(\theta)\,\mathrm{d}\theta = \int_b^{b+c} \pi(\theta)\,\mathrm{d}\theta, \qquad \text{for all } a, b \in \mathbb{R} \text{ and } c > 0.$$

The measure $\pi(\theta)\mathrm{d}\theta$ satisfying this equality is Lebesgue measure on $\mathbb{R}$, and is not a probability distribution, and so Ex. 9.12(c) is not immediately applicable. To fix this, the idea is to approximate an improper distribution (i.e., a measure not integrating to one) with proper ones. In the case of the normals, one may try $\theta \sim \pi_k(\theta)$, where $\pi_k(\theta)$ is the density of a uniform distribution over $[-k, k]$, then let $k$ grow. improper priors

**(a)** Let $\widehat{a}$ be an estimator, $(\pi_k)_{k\geq 1}$ a sequence of prior distributions, and for each $k \geq 1$ let $\widehat{a}_{\pi_k}$ be the Bayes solution for $\pi_k$. Suppose that

$$\lim_{k\to\infty} \mathrm{BR}(\widehat{a}_{\pi_k}, \pi_k) = R_{\max}(\widehat{a}) < \infty.$$

Show that $\widehat{a}$ is then minimax.

**(b)** Let $X_1, \ldots, X_n$ be independent $\mathrm{N}(\xi, \sigma^2)$. We are to estimate $\xi$ under the squared error loss $L(\xi, \widehat{\xi}) = (\widehat{\xi} - \xi)^2$. Consider the sequence of priors $\mu \sim \mathrm{N}(0, \tau_k)$ for $k = 1, 2, \ldots$, and show that the estimator $\widehat{\xi} = \bar{X}_n$ is minimax.

**(c)** Let $X_1, \ldots, X_n$ be independent $\mathrm{Pois}(\theta)$. We want to estimate $\theta$ under the weighted loss function $L(\theta, \widehat{\theta}) = \theta^{-1}(\widehat{\theta} - \theta)^2$. Use Gamma priors to show that $\widehat{\theta} = \bar{X}_n$ is minimax.

### Admissibility and Bayes

**Ex. 9.15** *Tools for admissibility.* If we have at hand an estimator $\widehat{a}$, the most convenient way of showing that $\widehat{a}$ is admissible is to show that it is the Bayes solution for some prior. In fact, it is *almost* true that an estimator is admissible if and only if it is Bayes. We will get to the cases where this implication fails, but as a rule of thumb it is pretty safe.

**(a)** Suppose that $X \sim f_\theta$, where $\theta \in \Theta = \{\theta_1, \ldots, \theta_k\}$ for some finite $k \geq 2$. Consider the estimator $\widehat{a}_\pi$ that is Bayes for the prior $\pi = \{\pi_1, \ldots, \pi_k\}$, where $\pi_j$ is the prior mass given to $\theta_j$. Show that if $\pi_j > 0$ for $j = 1, \ldots, k$, then $\delta_\pi$ is admissible.

**(b)** Why does the conclusion in (a) fail if $\pi_j = 0$ for one or more $j$?

**(c)** Show that if an estimator is unique minimax, it is admissible.

**(d)** Show that if a Bayes solution is *unique*, then it is admissible.

**(e)** To clarify what the uniqueness of the Bayes solution refers to, let's look at an example where there are several Bayes solutions. Let $X \sim \mathrm{unif}(0, \theta)$ and suppose that we want to estimate $\theta$ under the squared error loss function $L(\theta, \delta) = (\delta - \theta)^2$. Suppose $\theta$ is given the prior distribution that is uniform on $(0, c)$. Find the posterior distribution $\theta \,|\, (X = x)$ and derive at least two different Bayes solutions (there are uncountably many). [xx comment on this exercise, more relevant in BNP, point to Nils' 1976-proof and the mistake made by Lehmann. Could also mention Lindley and his so-called Cromwell's rule xx].

**(f)** Recall that a parameter value $\theta$ is in the *support* of $\pi$ (a probability density in our notation) if it is contained in the set $\{\theta \in \Theta \colon \pi(\theta) > 0\}$. Suppose that the support of the prior $\pi$ is $\Theta$, and that the risk function $R(\theta, \delta)$ is continuous in $\theta$ for all estimators $\delta$. Show that if $\widehat{a}_\pi$ is Bayes with respect to $\pi$ and have finite Bayes risk, then $\widehat{a}_\pi$ is admissible.

**Ex. 9.16** *Generalised Bayes.* (xx need a bit more nils-emil Ch6-Ch8 calibration; some of this might be in Ch6. xx) Suppose that in some experiment involving data from a normal distribution with expectation $\theta$, you have no idea whatsoever about where on the real line $\theta$ might be located. A natural 'prior' is therefore $\pi(\theta) \propto 1$ (or just take it equal to one) that spreads the 'probability' mass uniformly over the real line. Now, $\pi(\theta) \propto 1$ corresponds to Lebesgue measure on the real line, and is not a probability measure because its integral is infinite. The fact that $\pi(\theta)$ does not integrate to one does not, however, stop us from using it to derive estimators using the Bayes theorem, or a stretched version thereof. Priors that are not probability distributions are called improper priors.

**(a)** Suppose $Y_1, \ldots, Y_n$ are i.i.d. $\mathrm{N}(\theta, \sigma^2)$, with known $\sigma$ but unknown mean. Suppose $\theta$ is given the improper prior $\pi(\theta) = 1$. With $\bar{Y}$ as usual the sample mean, show that $\pi(\theta \,|\, y_1, \ldots, y_n)$ is the normal density $\mathrm{N}(\bar{y}, \sigma^2/n)$, noting that this is a real density, as soon as $n \geq 1$ datum is observed, even though the prior is improper. A generalised Bayes estimator is the estimator $\delta$ minimising the posterior expected loss $\mathrm{E}\{L(\theta, \delta) \,|\, \mathrm{data}\}$.

With squared error loss, show that the generalised Bayes estimator for $\theta$ is $\bar{y}$.

**(b)** The estimator $\widehat{\theta} = \bar{Y}$ you found in (a) is generalised Bayes; why is this not enough to conclude that it is admissible?

**(c)** (xx decide on balance Ch6 Ch8 here. xx) Let us generalise the above to normal data from $\mathrm{N}(\theta, \sigma^2)$ with both parameters unknown. It is mathematically convenient to turn to the parametrisation $\lambda = 1/\sigma^2$. With $(\theta, \lambda)$ having the improper prior $\pi(\theta, \lambda) = 1/\lambda$, and writing $Q_0 = \sum_{i=1}^n (y_i - \bar{y})^2$, show that the posterior becomes

$$\begin{aligned}\pi_n(\theta, \lambda) &\propto \lambda^{n/2-1} \exp\{-\tfrac{1}{2}\lambda\{Q_0 + n(\theta - \bar{y})^2\} \\ &\propto \lambda^{1/2} \exp\{-\tfrac{1}{2}\lambda(\theta - \bar{y})^2\}\lambda^{(n-1)/2-1} \exp(-\tfrac{1}{2}\lambda Q_0).\end{aligned}$$

Translate this to saying (i) $(1/\sigma^2) \,|\, \text{data} \sim \mathrm{Gam}(\frac{1}{2}(n-1), \frac{1}{2}Q_0)$; (ii) $\theta \,|\, (\text{data}, \sigma) \sim \mathrm{N}(\bar{y}, \sigma^2/n)$. Show that the posterior mean of $1/\sigma^2$ is $1/\widehat{\sigma}^2$ with the familiar empirical variance $Q_0/(n-1)$.

**Ex. 9.17** *The normal sample mean is admissible, in dimension 1.* We tend to general sample size $n$ below, but start wout considering the prototype setup with a single $Y$ from the normal $(\theta, 1)$. The natural estimatoris $\bar{Y}$ itself. Here we show that it is admissible, under squared error loss, but the arguments used do not work for dimension $k \geq 2$.

**(a)** We have seen in Ex. 9.14 that $\bar{Y}$ is minimax, with constant risk function 1. Explain via Ex. 7.10 that it is not a pure Bayes estimator.

**(b)** We employ the prior $\pi_\tau \sim \mathrm{N}(0, \tau^2)$ for $\theta$, with increasing $\tau$, resempling the flat prior under which we have seen that $\bar{Y}$ is generalised Bayes. Show that the Bayes estimator is $\widehat{\theta}_\tau = \rho Y$, with $\rho = \tau^2/(1+\tau^2)$, with minimum Bayes risk equal to this $\rho$. You may redo that argument leading from this to minimaxity of of $Y$.

**(c)** Assume now that $Y$ is not admissible. Explain that there must be a competitor $\widetilde{\theta}$ with risk at most 1 and with $R(\theta, \widetilde{\theta}) \leq 1 - \varepsilon$, for some small $\varepsilon$, for some interval $[a, b]$. Deduce from this that

$$\frac{\tau^2}{\tau^2+1} \leq \int R(\theta, \widetilde{\theta})\pi_\tau(\theta)\,\mathrm{d}\theta \leq 1 - \varepsilon \int_{[a,b]} \phi(\theta/\tau)(1/\tau)\,\mathrm{d}\theta,$$

and that

$$\varepsilon(1/\tau) \int_{[a,b]} \phi(\theta/\tau)\,\mathrm{d}\theta \leq 1/(\tau^2+1) \quad \text{for all } \tau.$$

Show that this leads to a contradiction: $Y$ is admissible.

**(d)** Generalise the above to any sample size; with $Y_1, \ldots, Y_n$ being i.i.d. $\mathrm{N}(\theta, \sigma^2)$, with given $\sigma$, then the sample mean $\bar{Y}$ is minimax and amissible.

**(e)** Consider the $k$-dimensional situation, with independent components $Y_i \sim \mathrm{N}(\theta_i, 1)$ for $i = 1, \ldots, k$, and with the vector of means to be estimated with loss $\sum_{i=1}^k (a_i - \theta_i)^2$. Using a sequence of independent priors $\mathrm{N}(0, \tau^2)$, for $i = 1, \ldots, k$, show that $Y = (Y_1, \ldots, Y_k)$ is minimax, with constant rate $k$. Then try the argument above, and explain that it does not work, for $k \geq 2$. We lean indeed in Ex. 9.23 that $Y$ is inadmissible, for $k \geq 3$; there are competitors which uniformly dominate $Y$.

**Ex. 9.18** *Blyth's method.* We saw in Ex. 9.17 that admissibility may be established via contradiction. Here we formalise that arguemt, sometimes called Blyth's method. When it works, it works, often in a case by case modus; we saw for the case of a multidimensional

normal mean that the Blyth method only worked for the one-dimensional case. In general, say that $\delta$ is a *limiting Bayes* estimator if there is a sequence $\{\pi_k\}_k$ of possibly improper priors such that the corresponding Bayes estimators $\delta_{\pi_k}$ converge almost surely to $\delta$.

**(a)** Let $\widehat{a}$ be an estimator. Suppose $\Theta \subset \mathbb{R}^p$ is open, and that $R(\theta, a)$ is continuous in $\theta$ for all $a$. Let $(\pi_k)_{k\geq 1}$ be a sequence of (possibly improper) prior distributions such that $\mathrm{BR}(\widehat{a}, \pi_k) < \infty$ for all $k$, and for any open set $\Theta_0 \subset \Theta$,

$$\frac{\mathrm{BR}(\widehat{a}, \pi_k) - \mathrm{BR}(\widehat{a}_{\pi_k}, \pi_k)}{\int_{\Theta_0} \pi_k(\theta)\,\mathrm{d}\theta} \to 0, \quad \text{as } k \to \infty,$$

where $\widehat{a}_{\pi_k}$ is the Bayes solution under $\pi_k$. Show that $\widehat{a}$ is admissible. This is Blyth's method, which may be quoted, rather imprecisely, as saying that that limits of Bayes estimators are admissible.

**(b)** Check arguments from Ex. 9.17 again, to see these general arguments in action, leading to admissibility for normal means in dimension 1 but not in higher dimension.

**(c)** Consder $Y \sim \mathrm{Pois}(\theta)$ with loss function $(a-\theta)^2/\theta$. Show that the estimator $Y$ has constant risk. Use Gamma priors $(1, b)$, with growing $b$, to show that $Y$ is minimax and admissible. Attempt to see how far the argument goes for the case of $k \geq 2$ Poisson variables, with loss function $\sum_{i=1}^k (a_i - \theta_i)^2/\theta_i$; see Ex. 9.25.

**Ex. 9.19** *Bernoulli mean with weighted risk.* Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli$(\theta)$. We wish to estimate $\theta$, and we are particularly interested in precise estimates of very small and very large values of $\theta$. Therefore, we'll work with the loss function

$$L(\theta, \delta) = (\delta - \theta)^2/\{\theta(1-\theta\}.$$

**(a)** Compute the risk function of the ML estimator. What's noticeable about this risk function?

**(b)** We now take a Bayesian point of view and give $\theta$ a Beta$(a\theta', a(1-\theta'))$ prior distribution. Compute the expectation and variance of this prior.

**(c)** With the prior introduced in (b), find the posterior distribution $\pi(\theta \,|\, x_1, \ldots, x_n)$. Find also the Bayes solution $\delta_\pi$, i.e., the minimiser of the Bayes risk $\mathrm{BR}(\delta, \theta) = \int R(\delta, \theta)\pi(\theta)\,\mathrm{d}\theta$. [xx introduce Bayes risk earlier xx].

**(d)** Tweak the parameters of the Beta prior distribution, so that the Bayes solution you found above equals the ML estimator from (a). What desirable properties does the ML estimator possess?

**Ex. 9.20** *Estimating the standard deviation.* Suppose $X_1, \ldots, X_n$ are i.i.d. from $\mathrm{N}(0, \sigma^2)$. We are to estimate $\sigma$ under the loss function $L(\sigma, \delta) = (\delta - \sigma)^2/\sigma$.

**(a)** Find the ML estimator, say $\widehat{\sigma}_{\mathrm{ml}}$, and show that its risk function is

$$R(\sigma, \widehat{\sigma}) = \sigma\{(n-1)/n + b_n^2 + (b_n - 1)^2\}, \quad \text{where} \quad b_n = \sqrt{\frac{2}{n}}\frac{\Gamma(n/2)}{\Gamma((n-1)/2)}.$$

You may now use Stirling's formula $\Gamma(z) = (2\pi/z)^{1/2}(z/e)^z$ to show that $b_n \to 1$, and $R(\sigma, \widehat{\sigma}) \to 2\sigma$ as $n \to \infty$, as we already knew from ML-theory (see Ex. xx in Chapter 5).

**(b)** Consider the prior distribution $\sigma \sim \pi(\sigma)$, whose density is

$$\pi(\sigma) \propto (1/\sigma)^{a+1} \exp(-b/\sigma^2).$$

with $a > 1$ and $b > 0$. Find the prior expectation of $\sigma$. Find also the prior expectation of $1/\sigma$.

**(c)** Find the posterior distribution $\sigma \,|\, x_1, \ldots, x_n$, and derive the Bayes solution under the loss function above.

**(d)** Show that the ML estimator is inadmissible by exhibiting an estimator, say $\delta^\star$, with uniformly smaller risk. *Hint*: Consider $\delta_\alpha = \alpha\widehat{\sigma}_{\rm ml}$.

**(e)** Is $\delta^\star$ admissible? *Hint*: Use Blyth's method.

**Ex. 9.21** *Bayes and minimax normal estimation with the linex loss.* We worked out some basic properties of the linex loss function $\exp\{c(t-\theta)\} - 1 - c(t-\theta)$ in Ex. 7.5. Here we use the Bayesian machinery to find a minimax estimator for the normal mean.

**(a)** Consider the simple prototype setup where a single $X$ has the $\mathrm{N}(\theta, 1)$ distribution. Show that the estimator $X + d$ has risk function

$$r_c(\theta) = \mathrm{E}_\theta\,[\exp\{c(X+d-\theta)\} - 1 - c(X+d-\theta)] = \exp(cd + \tfrac{1}{2}c^2) - 1 - cd,$$

constant in $\theta$, and that the best estimator of this sort is $\theta^* = X - \frac{1}{2}c$. Show also that the risk achieved, by this estimator, is $\frac{1}{2}c^2$.

**(b)** Now consider Bayes estimation, with the prior $\theta \sim \mathrm{N}(0, \tau^2)$. Show via Ex. 7.17 that $(\theta \,|\, x) \sim \mathrm{N}(\rho x, \rho)$, with $\rho = \tau^2/(\tau^2+1)$. Show, perhaps via expressing $\theta \,|\, x$ as $\rho x + \rho^{1/2} N$ with $N$ a standard normal, that the posterior expected loss is

$$\mathrm{E}\,\{L_c(\theta, t) \,|\, x\} = \exp\{c(t-\rho x) + \tfrac{1}{2}\rho c^2\} - 1 - c(t-\rho x).$$

Deduce that the Bayes estimator is $\widehat{\theta}_B = \rho x - \frac{1}{2}\rho c$, and that the posterior expected loss is $\mathrm{E}\,\{L_c(\theta, \widehat{\theta}_B) \,|\, x\} = \frac{1}{2}\rho c^2$, independent of $x$.

**(c)** Show that $\theta^* = X - \frac{1}{2}c$ is minimax. Show also, via Blyth's method, that it is in fact admissible.

**(d)** Generalise the above to the case of a full sample $X_1, \ldots, X_n$ from $\mathrm{N}(\theta, 1)$. Find the Bayes estimator and associated minimum Bayes risk, for the $\mathrm{N}(0, \tau^2)$ prior, and prove that $\theta^* = \bar{X} - \frac{1}{2}c/n$ is minimax. What is its minimax risk?

**(e)** Find the distribution of $Z_n = \sqrt{n}(\theta^* - \theta)$, and comment on its limit, (i) when the loss-skewness parameter $c$ is fixed, (ii) when $c = \sqrt{n}$.

**(f)** (xx perhaps another example with linex loss. and something where we see certain arguments lead to a choice of $c$. xx)

**Ex. 9.22** *More on the linex loss.* For the linex loss, studied initially in Ex. 7.5 and then in Ex. 9.21 for the normal case, we now find out more.

**(a)** Suppose $Y_1, \ldots, Y_n$ are i.i.d. from the $\mathrm{Pois}(\theta)$, with prior $\theta \sim \mathrm{Gam}(a, b)$, as with Ex. 7.1. Show that the Bayes estimator with the linex loss is $\widehat{\theta}_B = (1/c)(a + n\bar{y}) \log\{1 + c/(b+n)\}$. Verify that when $c \to 0$, we retrieve the posterior means of Ex. 7.1.

**(b)** (xx one more case with a clear formula. perhaps $\sigma$ in normal. xx)

**(c)** As we know from Ex. 7.32, an approximation to the posterior distribution is $\theta \,|\, \mathrm{data} \sim \mathrm{N}(\widehat{\theta}_{\rm ml}, \widehat{\sigma}^2/n)$, in terms of the ML estimate and estimated inverse Fisher information. Deduce that $M_n(-c) \doteq \exp(-c\widehat{\theta}_{\rm ml} + \frac{1}{2}c^2\widehat{\sigma}^2/n)$, in the notation above, and that this leads to the approximation $\widehat{\theta}_B = \widehat{\theta}_{\rm ml} - \frac{1}{2}c\widehat{\sigma}^2/n$ for the Bayes estimator under linex loss. Show also that the posterior expected loss is approximately $\frac{1}{2}c^2\widehat{\sigma}^2/n$.

**(d)** (xx an example where we can check the approximation with the exact Bayes estimator, e.g. with Poisson and gamma. xx)

## Multiparameter estimation and the Stein phenomenon

**Ex. 9.23** *The James–Stein estimator.* When estimating the price of apples in Oslo, the height of women in Bergen, and the unemployment rate in Trondheim, it is sometimes advantageous to use information about apples is Oslo and women in Bergen to say something about the unemployment rate in Trondheim. The point is that when estimating an ensemble of unrelated quantities, we can sometimes do better by borrowing information across unrelated things. This phenomenon is known as Stein's paradox or the Stein effect; (i) it might be clever to combine these independent estimators, gaining precision for each case; and (ii) even in cases where not much is gained, it will never be worse than not-combining-at-all. In the present exercise we'll look at Stein's 1956–1961 result, a result that initiated a whole field of statistical research known by the labels of shrinkage, ensemble, small area estimation. So let $Y_i \sim \mathrm{N}(\theta_i, 1)$ be independent for $i = 1, \ldots, k$. We are to estimate $\theta_1, \ldots, \theta_k$ under the combined loss function

$$L(\theta, a) = \sum_{i=1}^{k} (a_i - \theta_i)^2.$$

We actually studied several aspects of this situation in Ex. 7.33, from Bayesian and empirical Bayesian perspectives, and succeeded in finding good ensemble procedures. Presently the focus is more on risk functions and uniform dominance.

**(a)** The standard approach is to use $Y_i$ as an estimator of $\theta_i$. This $Y_i$ is the ML estimator, it is admissible, it is the UMVU estimator, etc. For this standard estimator $Y = (Y_1, \ldots, Y_k)$, show that its risk function is the constant $k$.

**(b)** For a single $Y \sim \mathrm{N}(\theta, 1)$, show using integration by parts that under very mild conditions on the function $b(y)$, one has $\mathrm{E}_\theta\,(Y - \theta)b(Y) = \mathrm{E}_\theta\, b'(Y)$m where $b'$ is the derivative of $b$.

**(c)** Let now $b(y) = (b_1(y), \ldots, b_k(y))$. Generalise what you found in (b) to

$$\mathrm{E}_\theta\,(Y_i - \theta_i)b_i(Y) = \mathrm{E}_\theta\, b_{i,i}(Y), \quad \text{where } b_{i,i}(y) = \partial b_i(y)/\partial y_i.$$

**(d)** What you found in (b) and (c) is known as Stein's lemma. We shall use this to construct an estimator that uniformly dominates $Y$. Consider a general competitor to $Y$ of the form $\delta(Y) = (\delta_1(Y), \ldots, \delta_k(Y))$, with

$$\delta_i(Y) = Y_i - b_i(Y) \quad \text{for } i = 1, \ldots, k.$$

Show that the difference in risk between $Y$ and estimators of his form can be expressed as $R(\delta, \theta) - R(Y, \theta) = \mathrm{E}_\theta\, D(Y)$, where

$$D(y) = \sum_{i=1}^{k} \{b_i(y)^2 - 2b_{i,i}(y)\}.$$

The clever thing about $R(\delta, \theta) = k + \mathrm{E}_\theta\, D(Y)$ is that $D(y)$ does not depend on the unknown $\theta_1, \ldots, \theta_k$. We can therefore try to find a data dependent function $b(y)$ such that $D(y) < 0$ for all $y$, and in such a case identify an estimator that uniformly dominates the standard estimator. This turns out to be possible for $k \geq 3$, but not for $k \leq 2$.

**(e)** Try $b_i(y) = ay_i/\|y\|^2$, with $\|y\|^2$ being the squared Euclidian norm $\sum_{i=1}^{p} y_i^2$, leading to the estimator $\delta(y) = y - b(y) = (1 - a/\|y\|^2)y$. With this choice of $b(y)$, show that

$$D(y) = \{a^2 - 2a(k-2)\}/\|y\|^2.$$

Show that this is negative for a range of $a$ values, provided $k \geq 3$. Demonstrate that the optimal $a$ is $a = k - 2$, corresponding to the estimator

$$\delta_{\mathrm{JS}}(Y) = \{1 - (k-2)/\|Y\|^2\}\, Y. \tag{9.3}$$

This estimator is known as the James–Stein estimator. Show that its risk can be expressed as

$$R(\delta_{\mathrm{JS}}, \theta) = k - (k-2)^2 \mathrm{E}_\theta\, 1/\|Y\|^2.$$

In particular, explain that the classic estimator $(Y_1, \ldots, Y_k)$ for $(\theta_1, \ldots, \theta_k)$ is inadmissible, for any $k \geq 3$. Show that the greatest reduction in risk from using $\delta_{\mathrm{JS}}$ instead of $Y$ takes place when $\theta_1 = \cdots = \theta_k = 0$, and compute the risk $R(\delta_{\mathrm{JS}}, 0)$ in this point.

**(f)** (xx more; link to Ex. 7.33. xx) We'll now make a connection to empirical Bayes procedures. Start with a prior that takes $\theta_1, \ldots, \theta_k$ independent from $\mathrm{N}(0, \tau^2)$. Show that the Bayes solution is $\delta^B = (\delta_1^B, \ldots, \delta_k^B)$, with

$$\delta_i^B(Y) = \rho\, Y_i \text{ for } i = 1, \ldots, k, \quad \text{where} \quad \rho = \tau^2/(\tau^2 + 1).$$

The empirical Bayes approach consists in estimating hyperparameters from data. Hyperparameters are those parameters set by the statistician in a pure Bayesian approach. Show that the marginal distribution of $y_1, \ldots, y_k$ is a product of $\mathrm{N}(0, 1 + \tau^2)$ distributions. Find the ML estimator of $\alpha$. Use the ML estimator to find an unbiased estimator, say $\tilde{\alpha}$, of $\alpha$. The empirical Bayes estimator is then $\delta_{\mathrm{EB}}(Y) = \tilde{\alpha} Y$. What's noticeable about this estimator?

**Ex. 9.24** *Resolving the paradox.* [xx make an exercise based on insights from Stigler (1990), perhaps?]

**Ex. 9.25** *Poisson means and inadmissibility of ML estimator.* (xx nils emil check all details. xx) To show that an estimator is inadmissible it suffices to showcase one estimator that dominates it. Let $Y_1, \ldots, Y_k$ be independent Poisson with means $\theta_1, \ldots, \theta_k$. We are to estimate the $\theta = (\theta_1, \ldots, \theta_k)$ with the weighted loss function $L(\theta, \delta) = \sum_{i=1}^k (\delta_i - \theta_i)^2/\theta_i$, where $\delta = (\delta_1, \ldots, \delta_k)$. We demonstrate below that the usual estimator, with $Y_i$ for $\theta_i$, is minimax, but inadmissible; there are competitors doing uniformly better.

**(a)** Show first that the ML estimator in this problem is $\delta_{\mathrm{ml}} = (Y_1, \ldots, Y_k)$, i.e. using $Y_i$ for $\theta_i$. Show that its risk function is the constant $k$. Via gamma independent priors $\theta_i \sim \mathrm{Gam}(1, b)$, find the Bayes estimator and the Bayes risk, and demonstrate that the ML estimator is minimax.

**(b)** Let $Z = \sum_{i=1}^k Y_i$ be the sum of these independent Poisson observations, write $\gamma = \sum_{i=1}^k \theta_i$ for the sum of the $p$ Poisson means, and define $\pi_i = \theta_i/\gamma$ for $i = 1, \ldots, k$. Show that $(Y_1, \ldots, Y_k)$ given $Z = z$ is a multinomial $(z, \pi_1, \ldots, \pi_k)$; this is valid for any $z \geq 1$. This establishes that $\mathrm{E}(Y_i \,|\, Z) = Z\pi_i$ and $\mathrm{Var}(Y_i \,|\, Z) = Z\pi_i(1 - \pi_i)$, for $i = 1, \ldots, k$, with these formulae holding trivially also for $Z = 0$.

**(c)** Prove the following little lemma. If $X \sim \mathrm{Pois}(\theta)$ and $g$ is a function such that $g(0) = 0$, then $\mathrm{E}_\theta\, g(X)/\theta = \mathrm{E}_\theta\, g(X+1)/(X+1)$.

**(d)** Consider the estimator $\delta^\star$ whose components are

$$\delta_j^\star = \{1 - \phi(Z)\} Y_j \quad \text{for } j = 1, \ldots, k.$$

The game to be played now (as with the James–Stein estimator worked with in Ex. 9.23) is to find an expression for the risk difference $R(\theta, \delta^\star) - R(\theta, \delta_{\mathrm{ml}})$ that is independent of

the unknown parameters. Using the results from (b) and (c) show indeed that the risk difference $R(\theta,\delta^\star)-R(\theta,\delta_{\rm ml})$ can be expressed as $\mathrm{E}_\gamma\, D(Z)$, with

$$D(z)=\{\phi(Z+1)^2-2\phi(Z+1)\}\{(Z+1)+(k-1)\}+2\phi(Z)Z.$$

**(e)** Suppose that the function $\phi$ is such that $\phi(z)z$ is increasing. Under this assumption, find a function $\phi$ that ensures that $D(z)<0$ for all $z\in\{0,1,2,\ldots\}$. Conclude that the ML estimator is inadmissible. (xx check with $\phi(z)=c/(p-1+z)$, e.g. $c=1$. we require $k\geq 2$. give risk expression. smallest for $\theta=0$. x)

**(f)** We have shown that $\delta_{\rm CZ}$ uniformly [xx nytt begrep xx] dominates the ML estimator, however, we do not yet now whether or not there exists and estimator that dominates $\delta_{\rm CZ}$. Show that $\delta_{\rm CZ}$ is admissible.

**Ex. 9.26** *Multiparameter inference, III.* (xx to be calibrated with clevenson-zidek Ch8. we do Bayes, empirical Bayes, full Bayes. xx) let $Y_i\sim\text{Pois}(\theta_i)$ for $i=1,\ldots,k$, aiming to estimate the ensemble of means, with loss $L=\sum_{i=1}^k(\widehat\theta_i-\theta_i)^2/\theta_i$. standard is $Y_i$, constant risk $k$. we do $\theta_i\sim\text{Gam}(a,b)$, $\widehat\theta_i(a,b)=(y_i+a-1)/(b+1)$, i think. empirical Bayes yields almost Clevenson–Zidenk. then throw in full Bayes, with prior for $(a,b)$ too.

**(a)** The default estimator would be $\widehat\theta_j=Y_j$, for $j=1,\ldots,k$. Show that its risk function is constant, equal to $k$.

**(b)** Consider then the prior which takes $\theta_1,\ldots,\theta_k$ independent from a Gamma $(a,b)$. Show that $\theta_j\,|\,\text{data}$ is a $\text{Gam}(a_j+y_j,b+1)$, find the Bayes estimator, and its risk function. (xx jotting down details here. xx)

$$\widehat\theta_j=\widehat\theta_j(a,b)=1/(\mathrm{E}\,(1/\theta_j)\,|\,y_j)=\frac{a+y_j-1}{b+1}.$$

empirical Bayes: $\mathrm{E}\,(Y_j\,|\,\theta_j)=\theta_j$, $\text{Var}\,(Y_j\,|\,\theta_j))=\theta_j$, leading to $\mathrm{E}\,Y_j=a/b$, $\text{Var}\,Y_j=a/b+a/b^2$; can then use moment matching to find $\widetilde a,\widetilde b$. this leads to empirical Bayes estimator $\theta_j^*=\widehat\theta_j(\widetilde a,\widetilde b)=(\widetilde a+y_j-1)/(\widetilde b+1)$. also:

$$\bar f(y_j,a,b)=\int_0^\infty \exp(-\theta)\theta^{y_j}/y_j!\frac{b^a}{\Gamma(a)}\theta^{a-1}\exp(-b\theta)\,\mathrm{d}\theta=\frac{b^a}{\Gamma(a)}\frac{\Gamma(a+y_j)}{(b+1)^{a+y_j}}\frac{1}{y_j!}$$

Full Bayes, a prior for $(a,b)$:

$$\pi_0(a,b)\prod_{j=1}^k g(\theta_j,a,b)f(y_j,\theta_j)=\pi_0(a,b)\prod_{j=1}^k g(\theta_j,a+y_j,b+1)\bar f(y_j,a,b)$$

which leads to $\pi(a,b\,|\,\text{data})\propto\pi_0(a,b)\prod_{j=1}^k\bar f(y_j,a,b)$. then mcmc for sampling from $(a,b,\theta_1,\ldots,\theta_k)$ given data.

**Ex. 9.27** *Multiparameter inference, IV.* Suppose one observes independent $Y_i\sim\text{unif}[0,\theta_i]$ for $i=1,\ldots,k$, aiming to estimate all parameters with sum of squares loss function $L(\theta,a)=\sum_{i=1}^k(a_i-\theta_i)^2$. Below use $\bar\theta=k^{-1}\sum_{i=1}^k\theta_i$, $z=\sum_{i=1}^k(\theta_i-\bar\theta)^2=s-k^2$, with $s=\sum_{i=1}^k\theta_i^2$.

**(a)** A direct start procedure is to use $\widetilde\theta_i=2Y_i$ for the different components. Show that is is unbiased, with combined risk function $R(\theta,\widetilde\theta)=s/3=(1/3)(z+k\bar\theta^2)$.

**(b)** Then consider independent priors $\theta_i\sim\text{Gam}(2,b)$ for the components. Shos that the posterior distributions can be characterised as $\theta_i\,|\,\text{data}\sim y_i+V_i$, where $V_i\sim\text{Expo}(b)$, leading also to the Bayes estimator $\widehat\theta_i(b)=y_i+1/b$. Find also the Bayes risk.

**(c)** Since specifying $b$ might be hard, consider empirical Bayes estimators of the type $\theta_i^* = \widehat{\theta}_i(\widehat{b}) = y_i + \widehat{1}/b$, using data to estimate $1/b$. Explain that the mean of $2Y_i$ is $2/b$, in the marginal distribution for the data, and argue that setting $\bar{Y} = 1/\widehat{b}$ is a good scheme for estimating $b$. Explain that this leads to the empirical Bayes estimator $\theta_i^* = Y_i + \bar{Y}$.

**(d)** We then study the performance of the $\theta_i^* = Y_i + \bar{Y}$ estimator, in the spirit of the James–Stein and multi-Poisson methods worked with in Ex. 9.23 and 9.25. Show that the risk function becomes

$$\sum_{i=1}^{n} \mathrm{E}_\theta\,(Y_i + \bar{Y} - \theta_i)^2 = \sum_{i=1}^{k} [(\tfrac{1}{2}\theta_i + \tfrac{1}{2}\bar{\theta} - \theta_i)^2 + \theta_i^2/12 + (1/k)^2 s/12 + 2(\theta_i^2/12)/k]$$

and that this after some algebra may be expressed as $(1/4)z + (1/2)(1 + 3/k)(z + k\bar{\theta}^2)$. Show that this empirical Bayes method performs better than the direct one in the part of the parameter space where $w(\theta) = (1/k)z(\theta)/\bar{\theta}^2 \le 1 - 1/k$, the variance to the mean squared. Show that when the $\theta_i$ actually stem from such a $\mathrm{Gam}(2, b)$ prior, the $w(\theta)$ will be around $\frac{1}{2}$. Discuss this aspect of the methods, that the empirical Bayes scheme is expected to win over the direct $2Y_i$ method, but on this occasion not in all of the parameter space.

**(e)** Above we used the empicial Bayes scheme to motivate the estimator $\theta_i^* = Y_i + \bar{Y}$. Build also a full Bayesian prior, taking a background prior $\pi_0(b)$ and then $\theta_1, \ldots, \theta_k \sim \mathrm{Gam}(2, b)$. Find the posterior distribution of $b$ and set up a simulation scheme to estimate the $\theta_i$ parameters. Compare its performance to the $Y_i + \bar{Y}$ method. (xx round off suitably. xx)

## Hypothesis testing

**Ex. 9.28** *Testing a simple hypothesis.* (xx make connection to Bayes Ch6 and Neyman-Pearson Ch4. xx) Let $Y$ have density $f(y, \theta)$ and consider the simple hypothesis $H_0\colon \theta = \theta_0$ versus the simple alternative $\theta = \theta_1$. The statistical tests $\phi$, with $\phi(x) = 1$ meaning reject $H_0$, and $\phi(x) = 0$ keep $H_0$, are to be evaluated under the loss function

$$L(\theta_0, \phi) = \begin{cases} 0, & \text{if } \phi(x) = 0, \\ K_1, & \text{if } \phi(x) = 1, \end{cases} \qquad L(\theta_1, \phi) = \begin{cases} K_2, & \text{if } \phi(x) = 0, \\ 0, & \text{if } \phi(x) = 1. \end{cases}$$

**(a)** Let $0 < \pi_0 < 1$ be your prior probability of $H_0$ being true. Derive an expression for the posterior expected loss, and show that the Bayes solution $\phi_\pi$ is of the likelihood ratio type

$$\phi_\pi(x) = \begin{cases} 1, & \text{if } f(y, \theta_1) > k_\pi f(y, \theta_0), \\ 0, & \text{if } f(y, \theta_1) < k_\pi f(y, \theta_0). \end{cases}$$

Find $k_\pi$ and relate this quantity to the level of a test.

**(b)** Let now $Y$ have the $\mathrm{N}(\theta, 1)$ distribution, for given $\theta$. We want to test $H_0\colon \theta = 0$ versus $\theta_1 = \frac{1}{2}$ using the Bayes solution when the prior is $\pi_0 = \frac{1}{2}$. Find $K_1$ and $K_2$ such that $\mathrm{E}_{\theta_0}\,\phi_\pi(Y) = 0.05$.

**(c)** Show that any Bayesian test, with a prior giving weight to both the null- and the alternative hypothesis, is the most powerful test of its size. Hence the Bayesian method has somehow rediscovered the Neyman–Pearson Lemma of Ch. 4. *Hint*: Use what you know about Bayes solutions and admissibility.

**Ex. 9.29** *Testing2.* (xx one more exercise here, on the Bayesian take on testing. a little bit on risk functions. xx)

### Estimation in a few stylised prototype problems

**Ex. 9.30** *Estimating the squared bias.* Assume $Y \sim \mathrm{N}(b,1)$, and that $\phi = b^2$ is to be estimated. There are links from the present prototype exercise about squared biases to the FIC theory of model selection in Ch. 12.

**(a)** Go through the relevant motivation for the following three estimators for the squared bias. (i) Since $Y$ estimates $b$, use $\widehat{\phi}_1 = Y^2$. (ii) Explain that $Y^2$ has expected value $b^2+1$. This invites $\widehat{\phi}_2 = Y^2 - 1$. Show also that this is the only unbiased estimator for $b^2$. (iii) The trouble with the unbiased estimator for a nonnegative quantity is that it may be negative; what is the probability that this happens? Motivate the truncated estimator $\widehat{\phi}_3 = \max(Y^2 - 1, 0)$.

**(b)** For these three estimators, find expressions for their risk functions $r_j(b) = \mathrm{E}_b\,(\widehat{\phi}_j - b^2)^2$, and plot them in a diagram. Comment on what you find.

**(c)** Compute also risk functions numerically under absolute loss, i.e. using $\mathrm{E}_b\,|\widehat{\phi}_j - b^2|$, where the better behaviour of $\widehat{\phi}_3$ shows up more clearly.

**Ex. 9.31** *Estimating a normalised distance, from a narrow to a wide model.* Consider a situation where $Y \sim \mathrm{N}(\theta,1)$ and where it is thought not unreasonable that $\theta$ is not far from zero. One may then consider estimating $\theta$ via a smooth mixture of the simplest estimator, 0, and the natural wider model estimator, $Y$. There are links from this stylised exercise to FIC and model averaging themes of Ch. 12, just as for Ex. 9.30.

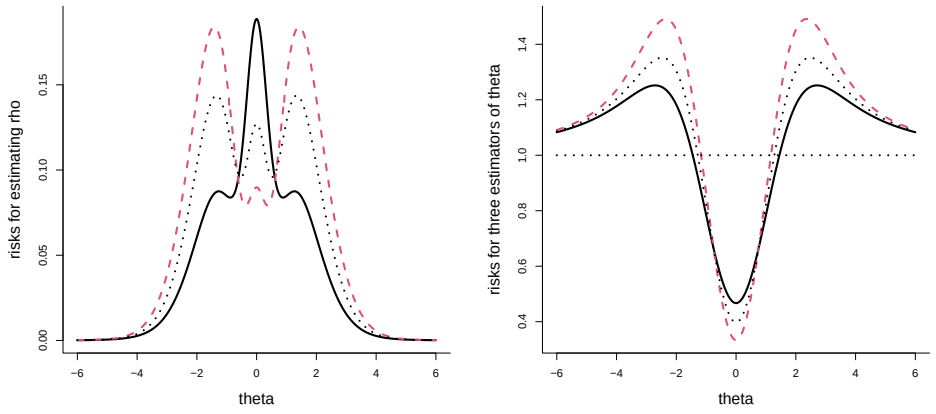



Figure 9.2: *Left panel: risk functions for the three estimators of* $\rho = \theta^2/(\theta^2+1)$ *(the first full, the second slanted, the third dotted). Right panel: risk functions for the three consequent estimators* $\widehat{\theta}_j = \widehat{\rho}_j Y$ *of* $\theta$ *(the first, the second, the rhird in increasing order of their maxima).*

**(a)** Explain that the estimator $\widehat{\theta}(a) = aY$ has mean squared error $(1-a)^2\theta^2 + a^2$. Show that this is minimised for $a_0 = \rho = \theta^2/(\theta^2+1)$, with minimum mse equal to this same $\rho$.

**(b)** There is hence a need to estimate $\rho$ well. Since $Y$ estimates $\theta$, a direct estimator is $\widehat{\rho}_1 = Y^2/(Y^2+1)$. Since $Y^2$ has mean $1+\theta^2$, however, argue that $\widehat{\rho}_2 = \max(Y^2-1,0)/Y^2$ might also work well. For good measure we include one more, between the two first, namelig $\widehat{\rho}_3 = \max(Y^2 - \frac{1}{2}, 0)/(Y^2 + c)$, with $c = \frac{1}{2}$. Compute and display the risk functions for these three estimators, under squared error loss, as with Figure 9.2, left

panel.

**(c)** Yet another estimator to be contemplated is the Bayesian solution, starting with a flat prior for $\theta$, often used in situations with an unknown normal mean. Show that $\theta \,|\, y \sim \mathrm{N}(y, 1)$, and explain that the resulting estimator for $\rho$, under squared error, is

$$\widehat{\rho}_4 = \mathrm{E}\,(\rho \,|\, Y) = \int_0^\infty \frac{x}{x+1} g_1(x, Y^2)\,\mathrm{d}x,$$

the mean of $K/(K+1)$, where $K \sim \chi^2_1(Y^2)$.

**(d)** Coming back to the intended use of estimating $\rho$, consider the two estimators of $\theta$ in the mean estimation problem,

$$\widehat{\theta}_1 = \frac{Y^2}{Y^2+1} Y, \quad \widehat{\theta}_2 = \frac{\max(Y^2-1, 0)}{Y^2} Y, \quad \widehat{\theta}_3 = \frac{\max(Y^2-\frac12, 0)}{Y^2+\frac12} Y.$$

The second estimator may be written $(1 - 1/Y^2)Y$ for $|Y| \geq 1$ and zero for $|Y| \leq 1$; similarly the third estimator is $\{(Y^2 - \frac12)/(Y^2 + \frac12)\}Y$ for $Y^2 \geq \frac12$ and zero for $Y^2 \leq \frac12$. Compute and display the risk functions of these three estimators, under squared error loss, as with Figure 9.2, right panel.

**(e)** We learn in these two situations, estimating $\rho = \theta^2/(\theta^2+1)$ and then $\theta$ via $\widehat{\rho} Y$, that there are mo clear winners. Explain that the minimax principle may be applied, leading to $\widehat{\rho}_1 Y$ being the best for the second competition. Explain further that to find a clear ranking, one solution is to apply a Bayesian prior, say a suitable $\mathrm{N}(0, \kappa^2)$ for $\theta$, here seen as a sensible weight function and not as a prior knowledge translated in to a prior, per se. With the standard normal as such a weight function, seeing values of $\theta$ outside $(-3, 3)$ as relatively unimportant, for example, compute the Bayes risks and rank the estimators. Find also the genuine Bayes solutions, for this prior, when estimating $\rho$ and $\theta$.

**Ex. 9.32** *Estimating a bounded mean.* With $Y \sim \mathrm{N}(\theta, 1)$, the task is to estimate the mean under quadratic loss, when it is a priori known that $|\theta| \leq 1$.

**(a)** Show that the ML estimator is $Y$ truncated to $[-1, 1]$. Compute and find a workable formula for its risk function

$$R(\theta) = \int_{-1}^1 (y-\theta)^2 \phi(y-\theta)\,\mathrm{d}y + \int_{-\infty}^{-1} (-1-\theta)^2 \phi(y-\theta)\,\mathrm{d}y + \int_1^\infty (1-\theta)^2 \phi(y-\theta)\,\mathrm{d}y.$$

**(b)** Then examine the Bayesian prior with $\theta$ taking values $\pm 1$ with equal probabilities. What is the marginal density $\bar{f}(y)$ for $Y$? Show that $\Pr(\theta = 1 \,|\, y)$ is $p(y) = \exp(2y)/\{1 + \exp(2y)\}$, and that this leads to the Bayes estimator

$$\widehat{\theta}_B = \mathrm{E}\,(\theta \,|\, y) = \frac{\exp(2y)-1}{\exp(2y)+1} = \frac{\sinh(Y)}{\cosh(Y)}.$$

Plot first the Bayes and ML estimators, as functions of $y$, as in Figure 9.3, left panel; then compute and display both risk functions, as in the right panel.

**(c)** Find the Bayes risk, perhaps in two ways: (i) via the risk function $R_B(\theta)$, averaged over the prior; (ii) via $\mathrm{MBR} = \mathrm{E}\,\mathrm{Var}\,(\theta \,|\, Y) = 4\,\mathrm{E}\,p(Y)\{1 - p(Y)\}$. Show that $\widehat{\theta}_B$ is a minimax estimator.

**(d)** (xx generalisations. could bother to do this. The computations above may be mimicked for the case where $\theta \in [-m, m]$ a priori, but we learn in the process that the minimax argument successfully used for $m = 1$ does not work, in cases where $m >$

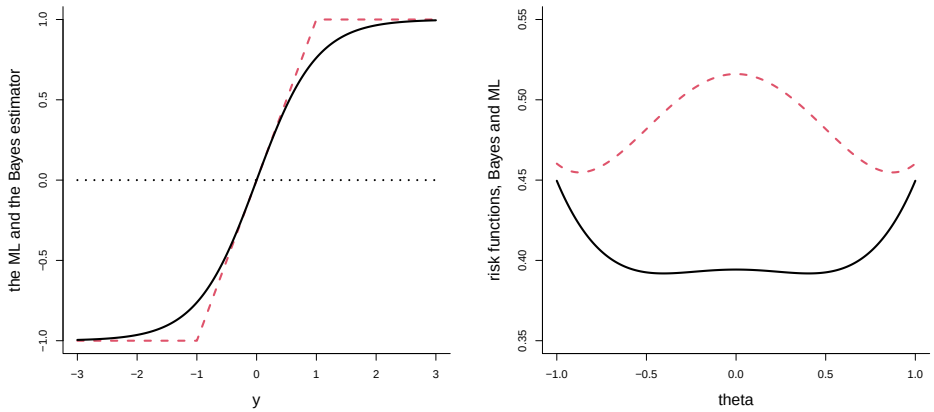



Figure 9.3: *Left panel: the Bayes (full curve) and the ML estimators (slanted), as functions of $y$. Right panel: their risk functions (with full curve for Bayes and slanted for the ML), with minimum Bayes risk 0.4496 for the two-point Bayes prior equal to the risk value at these two points.*

$m_0$, where $m_0 \approx 1.06$ (and indeed the Bayes estimator is then not minimax). Start out taking the $\theta$ prior to be symmetrically concentrated in $\{-m, m\}$. Show that the posterior probability of $\theta = m$ is equal to $p(y) = \exp(2my)/\{1 + \exp(2my)\}$, and that the posterior mean is $\mathrm{E}(\theta \,|\, y) = m \tanh(my)$. Show that the minimum Bayes risk is equal to $\mathrm{E}\, 4m^2 p(Y)\{1 - m(Y)\}$ and compute this, for values of $m$ equal to say $0.90, 1.00, 1.10$. The point is now that this minimum Bayes risk is equal to the maximum risk for $m = 0.90, 1.00$, but not for $m = 1.10$. For values $m > m_0$, the risk function starts having three modes, at $-m, 0, m$, with the one at zero becoming bigger than at the endpoints.

**Ex. 9.33** *Estimating a binormal mean with nonnegativity constraints.* (xx to be done and polished. xx) Let $(X, Y)$ be binormal with mean $(a, b)$, unit variances, and known correlation $\rho$. The task is to estimate $(a, b)$, with the natural quadratic loss function

$$\begin{pmatrix} \widehat{a} - a \\ \widehat{b} - b \end{pmatrix}^{\mathrm{t}} \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}^{-1} \begin{pmatrix} \widehat{a} - a \\ \widehat{b} - b \end{pmatrix} = \frac{1}{1-\rho^2} \{(\widehat{a} - a)^2 + (\widehat{b} - b)^2 - 2\rho(\widehat{a} - a)(\widehat{b} - b)\},$$

when it is known a priori that $a \geq 0$, $b \geq 0$. We omit the $1/(1 - \rho^2)$ constant in the loss and risk function questions below.

**(a)** A simple direct method is to truncate $X$ and $Y$ to zero, if negative, i.e. using $\widetilde{a} = \max(X, 0)$ and $\widetilde{b} = \max(Y, 0)$. Find formulae for $\mathrm{E}_a\, \widetilde{a}$, $\mathrm{Var}_a\, \widetilde{a}$, with of course similar results for $\widetilde{b}$. Find the risk function

$$R_0(a, b) = \mathrm{E}_{a,b} \{(\widetilde{a} - a)^2 + (\widetilde{b} - b)^2 - 2\rho(\widetilde{a} - a)(\widetilde{b} - b)\}.$$

**(b)** Then consider the ML estimators $(\widehat{a}, \widehat{b})$, maximising the log-likelihood under the given constraints. Show that this, for a given observed $(x, y)$, is equivalent to minimising

$$Q(a, b) = (a - x)^2 + (b - y)^2 - 2\rho(a - x)(b - y) \quad \text{with } a \geq 0,\ b \geq 0.$$

**(c)** Though $(\widehat{a}, \widehat{b})$ may be found via numerical routines, e.g. the 'nlm in box' algorithm `nlminb` in `R`, it is helpful to find explicit formulae for the ML. Going through the four

situations, for the four quadrants into which $(x, y)$ might fall, show this: (i) For $x > 0, y > 0$, we have $(\widehat{a}, \widehat{b}) = (x, y)$; (ii) for $x > 0, y < 0$, we have $\widehat{b} = 0$ and $\widehat{a} = \max(x - \rho y, 0)$; (iii) for $x < 0, y > 0$, we have $\widehat{a} = 0$ and $\widehat{b} = \max(y - \rho x, 0)$; (iv) for $x < 0, y < 0$, we have $\widehat{a} = \max(x - \rho y, 0)$ and $\widehat{b} = \max(y - \rho x, 0)$.

**(d)** There is now a well-defined risk function for the ML method,

$$R(a, b) = \mathrm{E}_{a,b}\{(\widehat{a} - a)^2 + (\widehat{b} - b)^2 - 2\rho(\widehat{a} - a)(\widehat{b} - b)\},$$

though with no easy closed-form formula. Show that for $a \geq 2.50, b \geq 2.50$, say, the chance is so high for $(X, Y)$ to land in the positive quadrant that both $R_0$ and $R$ come close to $2(1 - \rho^2)$.

**(e)** Carry out some numerical computation and comparison of the two risk functions, via simulation of binormal $(X, Y)$, for given value of $\rho$, at grid positions $(a, b)$. (xx more. ML is often better, sometimes significantly so, but not uniformly. application to regression situation with constraints on coefficients. xx)

### Estimation of parametric densities

**Ex. 9.34** *Unbiased estimation of a parametric density.* Suppose $Y_1, \ldots, Y_n$ are i.i.d. from a parametric density $f(y, \theta)$, like the normal or the Gamma or the Beta. How can we construct an unbiased estimator of the density function itself? For this exercise, assume there is a sufficient and complete statistic, say $T = (T_1, \ldots, T_p)$.

**(a)** Let $\widehat{\theta}$ be the ML estimator. Explain why it needs to be a function of $T_1, \ldots, T_p$, and that the ML estimator of the density itself is $f(y, \widehat{\theta})$.

**(b)** For a given window $A$ (make it an interval, if the distribution is one-dimensional), a very simple estimator for $p(\theta) = \Pr(Y \in A) = \int_A f(y, \theta)\,\mathrm{d}y$ is $\widehat{p} = I(Y_1 \in A)$, using a single data point. Show that it is unbiased. This also invites the somewhat more sensible estimator $\bar{p} = n^{-1}\sum_{i=1}^n I(Y_i \in A)$, the binomial proportion of data points inside the $A$ window. Show that it is unbiased and find a formula for its variance.

**(c)** Typically this estimator can be beaten, however. Consider indeed

$$p^* = \mathrm{E}\,(\widehat{p}\,|\,T) = \Pr(Y_1 \in A\,|\,T).$$

Explain why this is actually an estimator, i.e. that it does not depend on the parameter $\theta$, that it is unbiased, and indeed the UMVU estimator for $p(\theta)$. Show furthermore that the construction $\mathrm{E}\,(\bar{p}\,|\,T)$ leads to the very same $p^*$. In particular, there is a well-defined UMVU estimator for the c.d.f. $F(y, \theta)$, with this recipe being valid for any parametric distribution (again, provided there is a sufficient and complete statistic).

**(d)** Assume the situation is regular enough for the density $f_n(y\,|\,T)$ of a $Y_i$ given $T$ to exist. Explain why it does not depend on the parameter, and that

$$p^* = \int_A f_n(y\,|\,T)\,\mathrm{d}y, \quad \text{for all windows } A.$$

Show that $f_n(y\,|\,T)$ is unbiased, and the UMVU for $f(y, \theta)$.

**(e)** Let $Y_1, \ldots, Y_n$ be i.i.d. from the exponential density $\theta\exp(-\theta y)$. Show that $T = \sum_{i=1}^n Y_i$ is complete and sufficient, and that $(Y_1/T, \ldots, Y_n/T)$ has a Dirichlet $(1, \ldots, 1)$ distribution, with each $Y_i/T$ having a Beta distribution. Show from this that the UMVU estimator for the density takes the form

$$f_n(y\,|\,T) = \mathrm{be}(y/T, 1, n-1)/T = (n-1)(1 - y/T)^{n-2} \quad \text{for } y \in [0, T].$$

Why does the fact that the estimator is zero, outside the $[0,T]$ window, not contradict its unbiasedness? Show also that $f_n(y\,|\,T)$ comes close to the ML estimator $f(y,\widehat{\theta})$, for growing $n$. Give an example where the ML estimator is unbiased, but where $f(y,\widehat{\theta})$ is not unbiased for $f(y,\theta)$.

**(f)** Above we managed to find a formula $f_n(y\,|\,T)$ for the unbiased estimator of the exponential density, via the fact that $Y_1/T$ has a Beta distribution $(1,n-1)$. Suppose, for the sake of the argument, that you did not quite have the time to do the last part of the mathematics, deriving that formula. Explain and show via an illustration in your computer that the following would work. For the given dataset (perhaps with a small $n$, to see more interesting shapes), compute $T$, and draw a million realisations of $Y_1'$ given that values of $T$. Then form a very fine histogram of these $Y_1'$. That fine histogram is a numerical approximation to the $f_n(y\,|\,T)$ you need. This argument is valid for any such parametric density estimation situation; as long as you can simulate $Y_1'$ given the sufficient and complete statistic, you have your unbiased density estimator.

**(g)** (xx may make a separate short exercise for the discrete case, umvu estimation of $f(y,\theta)$ in two cases. xx) (i) Suppose $Y_1,\ldots,Y_n$ are i.i.d. Pois$(\theta)$. With $T=\sum_{i=1}^n Y_i$, show that $Y_1\,|\,T\sim\text{binom}(T,1/n)$, leading to the UMVU estimator $f_n(y\,|\,T)=\binom{T}{y}(1/n)^y(1-1/n)^{T-y}$. Show that this is close to $\exp(-\bar{Y})(\bar{Y})^y/y!$ for $n$ growing. (ii) Consider $Y_1,\ldots,Y_a$ being geometric with $(1-p)^y p=q^y p$ for $y=0,1,\ldots$. Show that $Z=\sum_{i=1}^a Y_i$ is sufficient and complete. Use formulae from Ex. 1.28 to work out that the UMVU estimator of the point probability $q^y p$ is

$$f_a(y\,|\,Z)=\binom{Z-y+a-2}{a-2}\Big/\binom{Z+a-1}{a-1}=\frac{(Z-y+a-2)!}{(Z-y)!\,(a-2)!}\,\frac{Z!\,(a-1)!}{(Z+a-1)!}$$

for $y=0,1,\ldots,Z$. (xx note: not hypergeometric?; close to $(1-\widehat{p})^y\widehat{p}$, with $\widehat{p}=1/(1+\bar{y})$. xx)

**Ex. 9.35** *UMVU estimation of the uniform density and cumulative.* In Ex. 9.34 we learned how to construct UMVU estimators for the cumulative and densities in parametric families, as long as these is a sufficient and complete statistic. This works also for the uniform case, but with certain extra componenents and nuances. The uniform density on $[0,\theta]$ is $1/\theta$ for $y\le\theta$ and zero for $y>\theta$, and there is a subtle difference between estimating this density and estimating the parameter $1/\theta$.

**(a)** With $Y_1,\ldots,Y_n$ i.i.d. from this uniform on $[0,\theta]$, show that $M_n=\max_{i\le n}Y_i$ is sufficient and complete. Find the UMVU estimator for $\theta$ and for $1/\theta$.

**(b)** Show that the distribution of $Y_1$ given $M_n$ is a mixture, equal to $M_n$ with probability $1/n$ and uniform on $[0,M_n]$ with probability $1-1/n$. Explain that the c.d.f. is $F_n(y\,|\,M_n)=(1-1/n)(y/M_n)$ for $y<M_n$ and then 1 for $y\ge M_n$. Why is this the UMVU for the parametric c.d.f. $F(y,\theta)$?

**(c)** Then set up a clear formula for the UMVU estimator of the density, and verify that it is indeed unbiased.

**(d)** Suppose i.i.d. pairs $(X_1,Y_1),\ldots,(X_n,Y_n)$ are observed from the distribution uniform on the interior of the circle $\{(x,y)\colon x^2+y^2\le\theta^2\}$. Writing $R_i=\|(X_i,Y_i)\|$, show that the ML estimator is $\widehat{\theta}=M_n=\max_{i\le n}R_i$, find its distribution, and show that it is sufficient and complete. Then find the distribution of $(X_1,Y_1)\,|\,M_n$, and from this the UMVU estimators of the density and cumulative of $R_i$.

**Ex. 9.36** *UMVU estimation of normal densities.* Suppose $Y_1,\ldots,Y_n$ are i.i.d. N$(\xi,\sigma^2)$.

How can we estimate the underlying density $f(y) = \sigma^{-1}\phi(\sigma^{-1}(y-\xi))$ unbiasedly? For this we use the general results of 9.34. (xx the following to be polished. xx)

**(a)** For a simpler separate question first, suppose $Y_1, \ldots, Y_n$ are i.i.d. $\mathrm{N}(\xi, 1)$. Here the sample average $\bar{Y}$ is sufficient and and complete, so the UMVU estimator is $f_n(y)$, the density of $Y_1 \,|\, \bar{Y}$. Writing $Y_1 = \bar{Y} + (Y_1 - \bar{Y})$, show that $Y_i - \bar{Y}$ is independent of $\bar{Y}$. Explain that $f_n(y) = g_n(y - \bar{y})$, with $g_n$ the density of $Y_1 - \bar{Y}$. Show next that $g_n$ is zero-mean normal with variance $1 - 1/n$, leading to

$$f_n(y) = \Big(\frac{n}{n-1}\Big)^{1/2} \phi\Big(\Big(\frac{n}{n-1}\Big)^{1/2}(y - \bar{y})\Big).$$

It is of course close to the simpler ML estimator based density $\phi(y - \bar{y})$.

**(b)** Then consider a dataset $Y_i$ from $\mathrm{N}(\xi, \sigma^2)$, with sufficient and complete estimators $\bar{Y}$ and $\widehat{\sigma}^2 = \sum_{i=1}^n (Y_i - \bar{Y})^2/(n-1)$. Write $Y_i = \bar{Y} + \widehat{\sigma} W_i$, with $W_i = (Y_i - \bar{Y})/\widehat{\sigma}$. Show via the Basu lemma of Ex. 4.26 that $W_i$ is independent of $(\bar{Y}, \widehat{\sigma})$. Show from this that the UMVU estimator must be $f_n(y) = \widehat{\sigma}^{-1} h_n(\widehat{\sigma}^{-1}(y - \bar{y}))$, with $h_n(z)$ the density of $W_1$. We find a formula for this density below. Explain nevertheless how that the numerical approximation trick via simulations, from Ex. 9.34(f), can be made to work here, without having the actual density formula.

**(c)** As an introductory exercise, before returning to the residuals, suppose $U_1, \ldots, U_m$ are i.i.d. $\mathrm{N}(0,1)$, and consider $V_i = U_i/(U_1^2 + \cdots + U_m^2)^{1/2}$. Working first with their squares, show that $V_i^2$ can be written $\chi_1^2/(\chi_1^2 + \chi_{m-1}^2)$, which is a Beta; more broadly, show that $(V_1^2, \ldots, V_m^2) \sim \mathrm{Dir}(\frac{1}{2}, \ldots, \frac{1}{2})$. Show next that the density of $V_1$ becomes

$$g_m(v) = \frac{\Gamma(\frac{1}{2}m)}{\Gamma(\frac{1}{2})\Gamma(\frac{1}{2}(m-1))}(1 - v^2)^{(m-1)/2-1} \quad \text{on } (-1, 1).$$

Also relate what you find here to the following: with $U_1, \ldots, U_m$ i.i.d. $\mathrm{N}(0,1)$, the distribution of $(U_1, \ldots, U_m)$ given its length $\|U\| = r$ is a uniform on the sphere of radius $r$.

**(d)** Write $G = \sum_{i=1}^n (Y_i - \bar{Y})^2$ and $V_i = (Y_i - \bar{Y})/G^{1/2}$. From $\sum_{i=1}^n V_i^2 = 1$ note that $\sum_{i=1}^n W_i^2 \le m$, with $m = n - 1$ the degrees of freedom for $G \sim \chi_m^2$. Show that the distribution of $V_i$ does not depend on the parameters, so for simplicity set $(\xi, \sigma)$ to $(0, 1)$. With row vectors $a_1 = (1/\sqrt{n}, \ldots, 1/\sqrt{n})$ and $a_2 = \{n/(n-1)\}^{1/2}(1 - 1/n, -1/n, \ldots, -1/n)$, show that these are orthogonal and of length 1. Hence there is an orthonormal matrix $A$, with $a_1, a_2$ the two first rows. Transforming to $Z = AX$, with $Z_1, \ldots, Z_n$ i.i.d. $\mathrm{N}(0,1)$ by Ex. 1.50, show that $Z_1 = \sqrt{n}\bar{Y}$ and $Z_2 = \{n/(n-1)\}^{1/2}(Y_1 - \bar{Y})$, and that this leads to

$$V_1 = \{(n-1)/n\}^{1/2} Z_2/(Z_2^2 + \cdots + Y_n^2)^{1/2}.$$

Derive a formula for $V_1$ based on this, using results from the previous point. Conclude that the UMVU estimator for the normal density is

$$f_n(y) = \frac{1}{\widehat{\sigma}}\Big(\frac{n}{n-1}\Big)^{1/2} \frac{\Gamma(\frac{1}{2}(n-1))}{\Gamma(\frac{1}{2})\Gamma(\frac{1}{2}(n-2))}\Big\{1 - \frac{n}{n-1}\Big(\frac{y - \bar{y}}{\widehat{\sigma}}^2\Big)\Big\}^{(n-2)/2-1}$$

**(e)** (xx can attempt multinormal density, would be nice. xx) $Y_i = \bar{Y} + \widehat{\Sigma}^{1/2} W_i$, with $W_i = \widehat{\Sigma}^{-1/2}(Y_i - \bar{Y})$. independent of the sufficient statistics, has a density $h_n(w)$. which implies

$$f_n(y) = |\widehat{\Sigma}|^{-1/2} h_n(\widehat{\Sigma}^{-1/2}(y - \bar{Y})).$$

would like to see this more clearly: let $U_1, \ldots, U_m$ be i.i.d. $\mathrm{N}_p(0, I)$, with $A_m = \sum_{i=1}^m U_i U_i^{\mathrm{t}}$. Then we should understand the distribution of $V_i = U_i^{\mathrm{t}} A_m^{-1} U_i$, with $\sum_{i=1}^m V_i = p$. the distribution of $A_m^{-1} U_1 U_1^{\mathrm{t}}$ is a multivariate Beta type, related to Roy's largest root test, with 1 and $m-1$ degrees of freedom, but careful with notation and terms. also keen on the root of $V_i$, i.e. $W_i = A_m^{-1/2} U_i$, with $W_i^{\mathrm{t}} W_i = V_i$. it should have a density with a formula.

## Notes and pointers

[xx some notes and pointers here xx]

Stein: See Stein (1956); James and Stein (1961) for the original articles, and, for example Efron and Morris (1977) and Stigler (1990) for lucid presentations. Clevenson and Zidek (1975), Stoltenberg and Hjort (2020), more. point to Hudson1978, generalising Stein multiparameter things to exponential families.

(xx Dan Roy Annals. Lehmann, Blyth. Bhattacharyya bounds, generalising Cramer-Rao. BNP chapter for minimax for $\theta = \mathrm{E}_F Y_i$, and admissibility. minimax for $F$. xx)

(xx there is a literature on estimating bounded means. bickel, casella, more. xx)

# Part II

# 100 Statistical Stories

# Part III

# Appendix

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